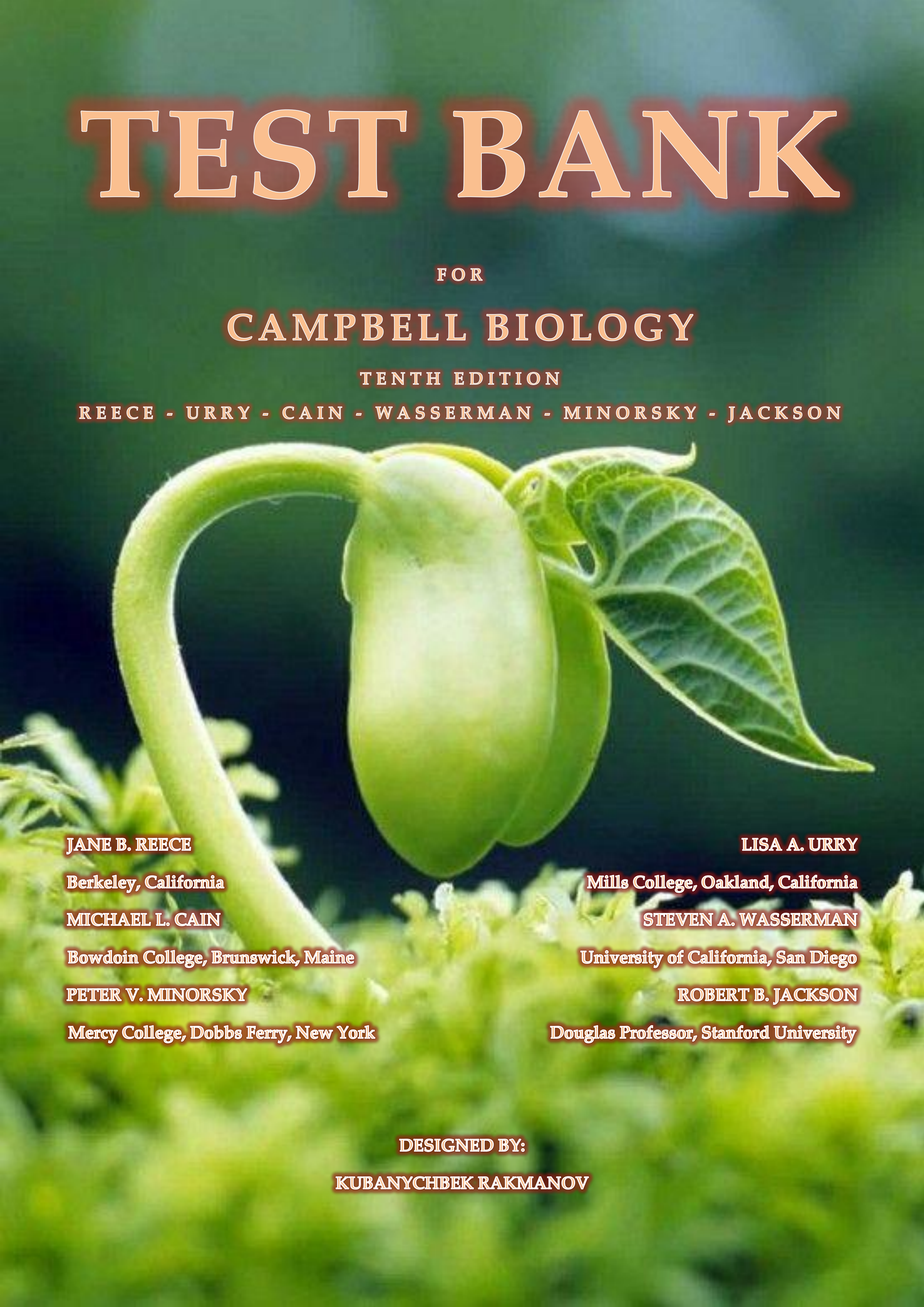

TEST BANK

FOR

CAMPBELL BIOLOGY

TENTH EDITION

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CONTENTS

CHAPTER 1:

EVOLUTION, THE THEMES OF BIOLOGY, AND SCIENTIFIC INQUIRY

1) Cells are _____.
A) Only found in pairs, because single cells cannot exist independently.
B) Limited in size to 200 and 500 micrometers in diameter.
C) Characteristic of eukaryotic but not prokaryotic organisms.
D) Characteristic of prokaryotic and eukaryotic organisms.

2) In comparison to eukaryotes, prokaryotes _____.
A) Are more structurally complex.
B) Are larger.
C) Are smaller.
D) Do not have membranes

3) Which of the following types of cells utilize deoxyribonucleic acid (DNA) as their genetic material but do not have their DNA encased within a nuclear envelope?
A) Animal.
B) Plant.
C) Archaean.
D) Fungi.

4) To understand the chemical basis of inheritance, we must understand the molecular structure of DNA. This is an example of the application of which concept to the study of biology?
A) Evolution.
B) Emergent properties.
C) Reductionism.
D) Feedback regulation.

5) A localized group of organisms that belong to the same species is called a _____.
A) Community.
B) Population.
C) Ecosystem.
D) Family.

6) Which of the following statements is FALSE regarding the complexity of biological systems?
A) An understanding of the interactions between different components within a living system is a key goal of a systems biology approach to understanding biological complexity.
B) Knowing the function of a component of a living system can provide insight into its structure and organization.
C) Understanding the chemical structure of DNA reveals how it directs the functioning of a living cell.
D) An ecosystem displays complex properties not present in the individual communities within it.

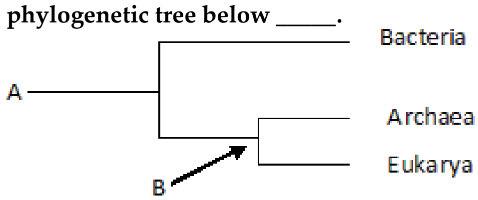
7) When a person gets dehydrated while exercising on a hot day, their pituitary gland releases ADH, a hormone that signals the kidneys to retain more water. This is an example of _____.
A) Positive feedback regulation.
B) Negative feedback regulation.
C) Chemical cycling.
D) Emergent properties.

8) Prokaryotes are classified as belonging to two different domains. What are the domains?
A) Bacteria and Eukarya.
B) Archaea and Monera.
C) Bacteria and Protista.
D) Bacteria and Archaea.

9) Which of these provides evidence of the common ancestry of all life?
A) Near universality of the genetic code.
B) Structure of the nucleus.
C) Structure of cilia.
D) Structure of chloroplasts.

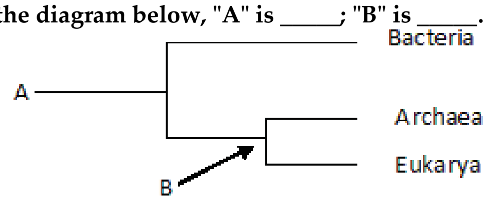
10) Which branch of biology is concerned with the naming and classifying of organisms?
A) Informatics.
B) Taxonomy.
C) Genomics.
D) Evolution.

11) The phylogenetic tree below _____.



A) Depicts the three major domains of life.
B) Includes unicellular but not multicellular life.
C) Includes unicellular and some forms of multicellular life, but not complex animals and plants.
D) Includes noncellular life-forms.

12) In the diagram below, "A" is _____; "B" is _____.



A) The most recent species to evolve on Earth; an ancestor of group "A".
B) The most recent species to evolve on Earth; the last common ancestor of Archaea and Eukarya.
C) The common ancestor of all life; the common ancestor of Bacteria and Archaea.
D) The common ancestor of all life; the last common ancestor of Archaea and Eukarya.

13) You find yourself standing next to a beautiful rose bush. Which of the following do you and the rose have in common?
A) You both are multicellular.
B) You both lack a membrane-bound nucleus.
C) You are both prokaryotic.
D) You and the rose have nothing in common.

14) Which of the following is (are) true of natural selection?
A) It requires genetic variation.
B) It results in descent with modification.
C) It involves differential reproductive success.
D) It requires genetic variation, results in descent with modification, and involves differential reproductive success.

15) Charles Darwin proposed a mechanism for descent with modification that stated that organisms of a particular species are adapted to their environment when they possess _____.
A) Non-heritable traits that enhance their survival in the local environment.
B) Non-heritable traits that enhance their survival and reproductive success in the local environment.
C) Heritable traits that enhance their survival and reproductive success in the local environment.
D) Heritable traits that decrease their survival and reproductive success in the local environment.

16) Which of the following best describes what occurred after the publication of Charles Darwin's *On the Origin of Species*?
A) The publisher was pressured to cease distribution.
B) The book was banned from schools.
C) The book was widely discussed and disseminated.
D) The book was discredited by most scientists.

17) Darwin's finches, collected from the Galápagos Islands, illustrate which of the following?
A) Mutation frequency.
B) Ancestors from different regions.
C) Adaptive radiation.
D) Vestigial anatomical structures.

18) Cotton-topped tamarins are small primates with tufts of long white hair on their heads. While studying these creatures, you notice that males with longer hair get more opportunities to mate and father more offspring. To test the hypothesis that having longer hair is adaptive in these males, you should _____.
A) Test whether other traits in these males are also adaptive.
B) Look for evidence of hair in ancestors of tamarins.
C) Determine if hair length is heritable.
D) Test whether males with shaved heads are still able to mate.

The following experiment is used for the corresponding questions (19-22).

A researcher discovered a species of moth that lays its eggs on oak trees. Eggs are laid at two distinct times of the year: early in spring when the oak trees are flowering and in midsummer when flowering is past. Caterpillars from eggs that hatch in spring feed on oak flowers and look like oak flowers. But caterpillars that hatch in summer feed on oak leaves and look like oak twigs.

How does the same population of moths produce such different-looking caterpillars on the same trees? To answer this question, the biologist caught many female moths from the same population and collected their eggs. He put at least one egg from each female into eight identical cups. The eggs hatched, and at least two larvae from each female were maintained in one of the four temperature and light conditions listed below.

In each of the four environments, one of the caterpillars was fed oak flowers, the other oak leaves. Thus, there were a total of eight treatment groups (4 environments × 2 diets).

Temperature	Day Length
Springlike	Springlike
Springlike	Summerlike
Summerlike	Springlike
Summerlike	Summerlike

19) Refer to the accompanying figure. Which one of the following is NOT a plausible hypothesis to explain the differences in caterpillar appearance observed in this population?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) The cooler temperatures of spring trigger the development of flowerlike caterpillars.
C) Differences in air pressure, due to differences in elevation, trigger the development of different types of caterpillars.
D) Differences in diet trigger the development of different types of caterpillars.

20) Refer to the accompanying figure. In every case, caterpillars that feed on oak flowers look like oak flowers. In every case, caterpillars that were raised on oak leaves looked like twigs. These results support which of the following hypotheses?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) Differences in air pressure, due to elevation, trigger the development of different types of caterpillars.
C) Differences in diet trigger the development of different types of caterpillars.
D) The differences are genetic. A female will either produce all flowerlike caterpillars or all twig-like caterpillars.

21) Refer to the accompanying figure. Recall that eggs from the same female were exposed to each of the eight treatments used. This aspect of the experimental design tested which of the following hypotheses?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) Differences in air pressure, due to elevation, trigger the development of different types of caterpillars.
C) Differences in diet trigger the development of different types of caterpillars.
D) The differences are genetic. A female will either produce all flowerlike caterpillars or all twig-like caterpillars.

22) Recall the caterpillar experiment in which caterpillars born in the spring looked like flowers, and caterpillars born in the summer looked like twigs. What is the most likely selective advantage for this difference in body shape?
A) Looking like their food sources allows the caterpillars to move through their environment more efficiently.
B) Development into the adult moth form is faster for caterpillars shaped like twigs than like flowers.
C) Looking like their food source lets the caterpillars blend into their surroundings, reducing predation.
D) Looking like their food source will increase the caterpillars' feeding efficiency; this would increase their growth rate and survival rate.

23) How does a scientific theory differ from a scientific hypothesis?
A) Theories are proposed to test scientific hypotheses.
B) Theories are usually an explanation for a more general phenomenon; hypotheses typically address more specific issues.
C) Hypotheses are usually an explanation for a more general phenomenon; theories typically address more specific issues.
D) Confirmed theories become scientific laws; hypotheses become theories.

24) A friend of yours calls to say that his car would not start this morning. He asks for your help. You say that you think the battery must be dead. If so, then jump-starting the car from a good battery will solve the problem. In doing so, you are _____.
A) Testing a theory for why the car will not start.
B) Making observations to inspire a theory for why the car will not start.
C) Stating a hypothesis and using that hypothesis to make a testable prediction.
D) Comparing multiple hypotheses for why the car will not start.

25) *Agrobacterium* infects plants and causes them to form tumors. You are asked to determine how long a plant must be exposed to these bacteria to become infected. Which of the following experiments will provide the best data to address that question?
A) Determine the survival rate of *Agrobacterium* when exposed to different concentrations of an antibiotic.
B) Measure the number of tumors formed on a plant when exposed to various concentrations of *Agrobacterium*.
C) Measure the concentration of *Agrobacterium* in different soil environments where the plants grow.
D) Measure the number of tumors formed on plants, which are exposed to *Agrobacterium* for different lengths of time.

26) *Agrobacterium* infects plants and causes them to form tumors. You determine that tumor formation requires a large amount of the plant's energy for tissue formation. How might this change the number of offspring a plant produces, and what is the most likely explanation for this change?
A) The number of offspring should increase, because in general, illness increases the reproductive output of organisms.
B) The number of offspring should increase, because the bacteria will provide energy for the plant.
C) The number of offspring should decrease, because the plant will divert energy from reproduction to tumor formation.
D) There should be no effect of infection on offspring production because energy for reproduction is independent of infection.

Use the following information when answering the corresponding questions (27-28).

In 1668, Francesco Redi performed a series of experiments on spontaneous generation. He began by putting similar pieces of meat into eight identical jars. Four jars were left open to the air, and four were sealed. He then did the same experiment with one variation: Instead of sealing four of the jars completely, he covered them with gauze (the gauze excluded the flies while allowing the meat to be exposed to air). In both experiments, he monitored the jars and recorded whether or not maggots (young flies) appeared in the meat.

27) Refer to the paragraph on Redi's experiments. What hypothesis was being tested in the initial experiment with open versus sealed jars?
A) Spontaneous generation is more likely during the long days of summer.
B) The type of meat used affects the likelihood of spontaneous generation.
C) Maggots do not arise spontaneously, but from eggs laid by adult flies.
D) Spontaneous generation can occur only if meat is exposed to air.

28) Refer to the paragraph on Redi's experiments. In both experiments, flies appeared in all of the open jars and only in the open jars. Which one of the following statements is correct?
A) The experiment was inconclusive because Redi used only one kind of meat.
B) The experiment was inconclusive because it did not run long enough.
C) The experiment supports the hypothesis that spontaneous generation occurs in rotting meat.
D) The experiment supports the hypothesis that maggots arise only from eggs laid by adult flies.

29) The best experimental design _____.
A) Includes a large sample size for each condition.
B) Includes a control.
C) Alters only one condition between the controls and the experimental condition.
D) Includes a large sample size and a control, and alters only one condition between the controls and the experimental condition.

30) In the process of science, which of these is tested?
A) A result.
B) An observation.
C) A hypothesis.
D) A control group.

31) A controlled experiment _____.
A) Is repeated many times to ensure that the results are accurate.
B) Includes at least two groups, one of which does not receive the experimental treatment.
C) Includes at least two groups, one differing from the other by two or more variables.
D) Includes one group for which the scientist controls all variables.

32) Which of the following are qualities of any good scientific hypothesis?
I. It is testable.
II. It is falsifiable.
III. It produces quantitative data.
IV. It produces results that can be replicated.
A) I only.
B) III only.
C) I and II.
D) III and IV.

33) In presenting data that result from an experiment, a group of students show that most of their measurements fall on a straight diagonal line on their graph. However, two of their data points are "outliers" and fall far to one side of the expected relationship. What should they do?
A) Do not show these points because clearly something went wrong in the experiment.
B) Average several trials, rule out the improbable results, and do not show them in the final work.
C) Show all results obtained and then try to explore the reason(s) for these outliers.
D) Change the details of the experiment until they can obtain the expected results.

34) Which of the following is the best description of a control for an experiment?
A) The control group is kept in an unchanging environment.
B) The control group is matched with the experimental group except for one experimental variable.
C) The control group is exposed to only one variable rather than several.
D) Only the experimental group is tested or measured.

35) Which of these is an example of inductive reasoning?
A) Hundreds of individuals of a species have been observed and all are photosynthetic; therefore, the species is photosynthetic.
B) These organisms live in sunny regions. Therefore, they are using photosynthesis.
C) If protists are all single-celled, then they are incapable of aggregating.
D) If two species are members of the same genus, they are more alike than each of them could be to a different genus.

36) The application of scientific knowledge for some specific purpose is known as _____.
A) Technology.
B) Deductive science.
C) Inductive science.
D) Pure science.

37) Which of the following best describes a model organism?
A) It is often pictured in textbooks and easy for students to imagine.
B) It is well studied, it is easy to grow, and results are widely applicable.
C) It is small, inexpensive to raise, and lives a long time.
D) It has been chosen for study by early biologists.

38) Why is a scientific topic best discussed by people of varying points of view, from different subdisciplines, and representing diverse cultures?
A) Robust and critical discussion between diverse groups improves scientific thinking.
B) Scientists can coordinate with others to conduct experiments in similar ways.
C) This is a way of ensuring that everyone gets the same results.
D) People need to exchange their ideas with other disciplines and cultures because everyone has a right to an opinion in science.

CHAPTER 2:

THE CHEMICAL CONTEXT OF LIFE

1) About twenty-five of the ninety-two natural elements are known to be essential to life. Which four of these twenty-five elements make up approximately 96 percent of living matter?
A) Carbon, sodium, hydrogen, nitrogen.
B) Carbon, oxygen, phosphorus, hydrogen.
C) Oxygen, hydrogen, calcium, nitrogen.
D) Carbon, hydrogen, nitrogen, oxygen.

2) Trace elements are those required by an organism in only minute quantities. Which of the following is a trace element that is required by humans and other vertebrates, but not by other organisms such as bacteria or plants?
A) Calcium.
B) Iodine.
C) Sodium.
D) Phosphorus.

3) Which of the following statements is FALSE?
A) Carbon, hydrogen, oxygen, and nitrogen are the most abundant elements of living matter.
B) Some naturally occurring elements are toxic to organisms.
C) All life requires the same essential elements.
D) Iron is needed by all humans.

4) Which of the following are compounds?
A) H_2O, O_2, and CH_4.
B) H_2O and O_2.
C) O_2 and CH_4.
D) H_2O and CH_4, but not O_2.

5) Knowing the atomic mass of an element allows inferences about which of the following?
A) The number of electrons in the element.
B) The number of protons in the element.
C) The number of protons plus neutrons in the element.
D) The number of protons plus electrons in the element.

6) In what way are elements in the same column of the periodic table the same? They have the same number of _____.
A) Protons.
B) Electrons when neutral.
C) Electrons in their valence shells when neutral.
D) Electron shells when neutral.

7) Molybdenum has an atomic number of 42. Several common isotopes exist, with mass numbers from 92-100. Therefore, which of the following can be true?
A) Molybdenum atoms can have between 50 and 58 neutrons.
B) Molybdenum atoms can have between 50 and 58 protons.
C) Molybdenum atoms can have between 50 and 58 electrons.
D) Isotopes of molybdenum have different numbers of electrons.

8) Carbon-12 is the most common isotope of carbon and has a mass number of 12. However, the average atomic mass of carbon found on a periodic table is slightly more than 12 daltons. Why?
A) The atomic mass does not include the mass of electrons.
B) Some carbon atoms in nature have an extra proton.
C) Some carbon atoms in nature have more neutrons.
D) Some carbon atoms in nature have a different valence electron distribution.

9) Which of the following best describes the relationship between the atoms described below?

Atom 1	Atom 2
$^{1}_{1}H$	$^{3}_{1}H$

A) They are isomers.
B) They are isotopes.
C) They contain 1 and 3 protons, respectively.
D) They each contain only 1 neutron.

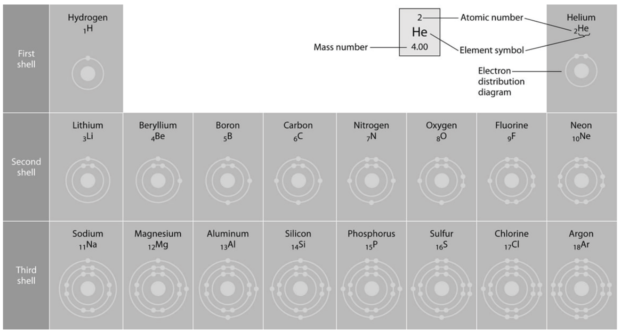
10) The atomic number of nitrogen is 7. Nitrogen-15 has a greater mass number than nitrogen-14 because the atomic nucleus of nitrogen-15 contains _____.
A) 7 neutrons.
B) 8 neutrons.
C) 8 protons.
D) 15 protons.

11) From its atomic number of 15, it is possible to predict that the phosphorus atom has _____.
A) 5 neutrons, 5 protons, and 5 electrons.
B) 15 neutrons and 15 protons.
C) 8 electrons in its outermost electron shell.
D) 15 protons and 15 electrons.

12) Fluorine has an atomic number of 9. Which of the following would you do to a neutral fluorine atom to complete its valence shell?
A) Add 1 electron.
B) Add 2 electrons.
C) Remove 1 electron.
D) Nothing. If fluorine is neutral, it has a complete valance shell.

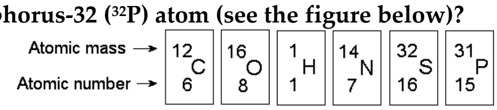
13) Magnesium has an atomic number of 12. What is the most stable charge for a magnesium ion?
A) A +1 charge.
B) A +2 charge.
C) A -1 charge.
D) A -2 charge.

14) Refer to the figure below (first three rows of the periodic table). What element has properties most similar to carbon?



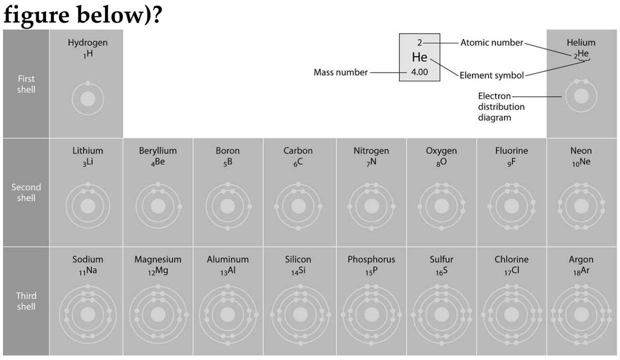
A) Boron.
B) Silicon.
C) Nitrogen.
D) Phosphorus.

15) How many neutrons are present in the nucleus of a phosphorus-32 (^{32}P) atom (see the figure below)?



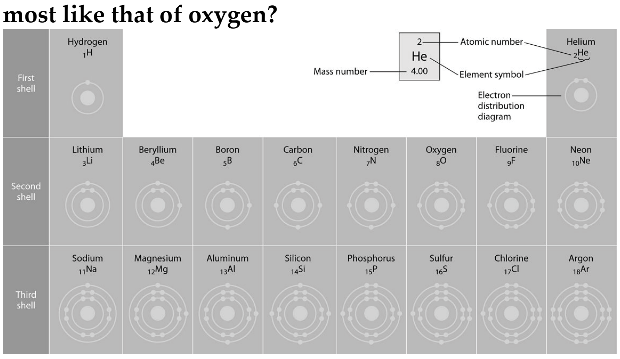
A) 15.
B) 16.
C) 17.
D) 32.

16) How many electrons will a single atom of sulfur with no charge and no bonds have in its valence shell (see the figure below)?



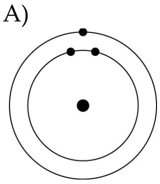
A) 6.
B) 8.
C) 16.
D) 32.

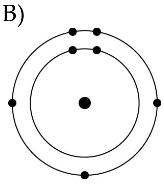
17) Based on electron configuration, which of the elements in the figure below would exhibit a chemical behavior most like that of oxygen?

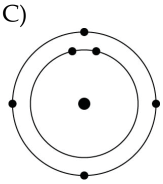


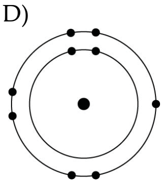
A) Carbon.
B) Nitrogen.
C) Sulfur.
D) Phosphorus.

18) Which one of the atoms shown would be most likely to form a cation with a charge of +1?

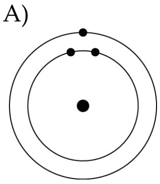
A)


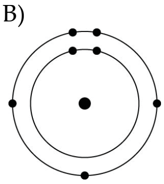
B)


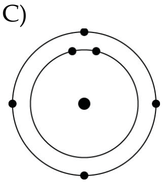
C)


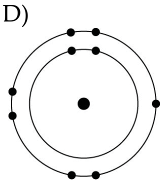
D)


19) Which one of the atoms shown would be most likely to form an anion with a charge of -1?

A)


B)


C)


D)


20) Oxygen has an atomic number of 8 and most commonly, a mass number of 16. Thus, what is the atomic mass of an oxygen atom?
A) Approximately 8 grams.
B) Approximately 8 daltons.
C) Approximately 16 grams.
D) Approximately 16 daltons.

21) If you change the number of neutrons in an atom, you create _____.
A) A cation.
B) An anion.
C) An isotope.
D) A different element.

22) Can the atomic mass of an element vary?
A) No, it is fixed. If it changes at all then you have formed a different element.
B) Yes. Adding or losing electrons will substantially change the atomic mass.
C) Yes. Adding or losing protons will change the atomic mass without forming a different element.
D) Yes. Adding or losing neutrons will change the atomic mass without forming a different element.

23) Which of the following is the best description of an atom's physical structure?
A) An atom is a solid mass of material.
B) The particles that form an atom are equidistant from each other.
C) Atoms are little bubbles of space with mass concentrated at the center of the bubble.
D) Atoms are little bubbles of space with mass concentrated on the outside surface of the bubble.

24) A salamander relies on hydrogen bonding to stick to various surfaces. Therefore, a salamander would have the greatest difficulty clinging to a _____.
A) Slightly damp surface.
B) Surface of hydrocarbons.
C) Surface of mostly carbon-oxygen bonds.
D) Surface of mostly carbon-nitrogen bonds.

25) A covalent chemical bond is one in which _____.
A) Electrons are removed from one atom and transferred to another atom so that the two atoms become oppositely charged.
B) Protons and neutrons are shared by two atoms so as to satisfy the requirements of both atoms.
C) Outer-shell electrons of two atoms are shared so as to satisfactorily fill their respective orbitals.
D) Outer-shell electrons of one atom are transferred to fill the inner electron shell of another atom.

26) What is the maximum number of covalent bonds that an oxygen atom with atomic number 8 can make with hydrogen?
A) 1.
B) 2.
C) 4.
D) 6.

27) Nitrogen (N) is more electronegative than hydrogen (H). Which of the following is a correct statement about the atoms in ammonia (NH_3)?
A) Each hydrogen atom has a partial positive charge; the nitrogen atom has a partial negative charge.
B) Ammonia has an overall positive charge.
C) Ammonia has an overall negative charge.
D) The nitrogen atom has a partial positive charge; each hydrogen atom has a partial negative charge.

28) Bonds between two atoms that are equally electronegative are _____.
A) Hydrogen bonds.
B) Polar covalent bonds.
C) Nonpolar covalent bonds.
D) Ionic bonds.

29) What results from an unequal sharing of electrons between atoms?
A) A nonpolar covalent bond.
B) A polar covalent bond.
C) An ionic bond.
D) A hydrophobic interaction.

30) A covalent bond is likely to be polar when _____.
A) One of the atoms sharing electrons is more electronegative than the other atom.
B) The two atoms sharing electrons are equally electronegative.
C) Carbon is one of the two atoms sharing electrons.
D) The two atoms sharing electrons are the same elements.

31) What is the difference between covalent bonds and ionic bonds?
A) Covalent bonds involve the sharing of pairs of electrons between atoms; ionic bonds involve the sharing of single electrons between atoms.
B) Covalent bonds involve the sharing of electrons between atoms; ionic bonds involve the electrical attraction between charged atoms.
C) Covalent bonds involve the sharing of electrons between atoms; ionic bonds involve the sharing of protons between charged atoms.
D) Covalent bonds involve the transfer of electrons between charged atoms; ionic bonds involve the sharing of electrons between atoms.

32) The atomic number of chlorine is 17. The atomic number of magnesium is 12. What is the formula for magnesium chloride?
A) MgCl.
B) $MgCl_2$.
C) Mg_2Cl.
D) $MgCl_3$.

33) How many electron pairs are shared between carbon atoms in a molecule that has the formula C_2H_4?
A) 1.
B) 2.
C) 3.
D) 4.

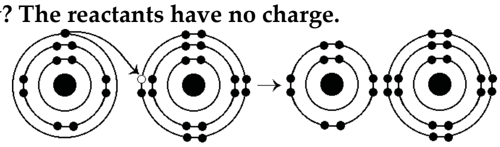
34) Which bond or interaction would be difficult to disrupt when compounds are put into water?
A) Covalent bonds between carbon atoms.
B) Hydrogen bonds.
C) Ionic bonds.
D) Ionic and hydrogen bonds.

35) Water molecules are attracted to one another by _____.
A) Nonpolar covalent bonds.
B) Ionic bonds.
C) Hydrogen bonds.
D) Hydrophobic interactions.

36) Van der Waals interactions may result when _____.
A) Electrons are not symmetrically distributed in a molecule.
B) Molecules held by ionic bonds react with water.
C) Two polar covalent bonds react.
D) A hydrogen atom loses an electron.

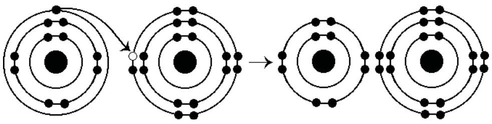
37) What is the maximum number of hydrogen atoms that can be covalently bonded in a molecule containing two carbon atoms?
A) 2.
B) 4.
C) 6.
D) 8.

38) What results from the chemical reaction illustrated below? The reactants have no charge.



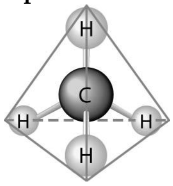
A) A cation with a net charge of +1 and an anion with a net charge of +1.
B) A cation with a net charge of -1 and an anion with a net charge of -1.
C) A cation with a net charge of -1 and an anion with a net charge of +1.
D) A cation with a net charge of +1 and an anion with a net charge of -1.

39) What is the atomic number of the cation formed in the reaction illustrated below?



A) 8
B) 10
C) 11
D) 16

40) What causes the shape of the molecule shown below?



A) The shape of the 2 *p* orbitals in the carbon atom.
B) The shape of the 1 *s* orbital in the carbon atom.
C) The shape of the *sp3* hybrid orbitals of the electrons shared between the carbon and hydrogen atoms.
D) Hydrogen bonding configurations between the carbon and hydrogen atoms.

41) How many electrons are involved in a single covalent bond?
A) One.
B) Two.
C) Three.
D) Four.

42) How many electrons are involved in a double covalent bond?
A) One.
B) Two.
C) Three.
D) Four.

43) If an atom has a charge of +1, which of the following must be true?
A) It has two more protons than neutrons.
B) It has the same number of protons as electrons.
C) It has one more electron than it does protons.
D) It has one more proton than it does electrons.

44) Elements found on the left side of the periodic table contain outer shells that are _____; these elements tend to form _____ in solution.
A) Almost empty; cations.
B) Almost empty; anions.
C) Almost full; cations.
D) Almost full; anions.

45) An atom has four electrons in its valence shell. What types of covalent bonds is it capable of forming?
A) Single, double, or triple.
B) Single and double only.
C) Single bonds only.
D) Double bonds only.

46) When are atoms most stable?
A) When they have the fewest possible valence electrons.
B) When they have the maximum number of unpaired electrons.
C) When all of the electron orbitals in the valence shell are filled.
D) When all electrons are paired.

47) When the atoms involved in a covalent bond have the same electronegativity, what type of bond results?
A) An ionic bond.
B) A hydrogen bond.
C) A nonpolar covalent bond.
D) A polar covalent bond.

48) Nitrogen (N) normally forms three covalent bonds with a valence of 5. However, ammonium has four covalent bonds, each to a different hydrogen (H) atom (H has a valence of 1). What do you predict to be the charge on ammonium?
A) +1.
B) -1.
C) +2.
D) -2.

49) You need to write down information about a molecule, but need to indicate only the type and number of atoms it contains. Which representation would work best?
A) Molecular formula.
B) Structural formula.
C) Ball-and-stick model.
D) Space-filling model.

50) You need to represent a molecule to best illustrate the relative sizes of the atoms involved and their interrelationships. Which representation would work best?
A) Molecular formula.
B) Structural formula.
C) Ball-and-stick model.
D) Space-filling model.

51) Which of the following is true for this reaction?

$$3\ H_2 + N_2 \leftrightarrow 2\ NH_3$$

A) The reaction is nonreversible.
B) Hydrogen and nitrogen are the reactants of the reverse reaction.
C) Ammonia is being formed and decomposed simultaneously.
D) Only the forward or reverse reactions can occur at one time.

52) Which of the following correctly describes all chemical equilibrium?
A) Forward and reverse reactions continue with no net effect on the concentrations of the reactants and products.
B) Concentrations of products are higher than the concentrations of the reactants.
C) There are equal concentrations of products and reactants while forward and reverse reactions continue.
D) There are equal concentrations of reactants and products, and the reactions have stopped.

53) Which of the following correctly describes a reaction that has reached chemical equilibrium?
A) The rate of the forward reaction is equal to the rate of the reverse reaction.
B) All of the reactants have been converted to the products of the reaction.
C) All of the products have been converted to the reactants of the reaction.
D) Both the forward and the reverse reactions have stopped, with no net effect on the concentration of the reactants and the products.

CHAPTER 3:

WATER AND LIFE

1) In a single molecule of water, two hydrogen atoms are bonded to a single oxygen atom by _____.
A) Hydrogen bonds.
B) Nonpolar covalent bonds.
C) Polar covalent bonds.
D) Ionic bonds.

2) The partial negative charge at one end of a water molecule is attracted to the partial positive charge of another water molecule. What is this attraction called?
A) A covalent bond.
B) A hydrogen bond.
C) An ionic bond.
D) A van der Waals interaction.

3) The partial negative charge in a molecule of water occurs because _____.
A) The oxygen atom donates an electron to each of the hydrogen atoms.
B) The electrons shared between the oxygen and hydrogen atoms spend more time around the oxygen atom nucleus than around the hydrogen atom nucleus.
C) The oxygen atom has two pairs of electrons in its valence shell that are not neutralized by hydrogen atoms.
D) One of the hydrogen atoms donates an electron to the oxygen atom.

4) Sulfur is in the same column of the periodic table as oxygen, but has electronegativity similar to carbon. Compared to water molecules, molecules of H_2S will _____.
A) Have greater cohesion to other molecules of H_2S.
B) Have a greater tendency to form hydrogen bonds with each other.
C) Have a higher capacity to absorb heat for the same change in temperature.
D) Not form hydrogen bonds with each other.

5) Water molecules can form hydrogen bonds with _____.
A) Compounds that have polar covalent bonds.
B) Oils.
C) Oxygen gas (O_2) molecules.
D) Chloride ions.

6) Which of the following is a property of liquid water? Liquid water _____.
A) Is less dense than ice.
B) Has a specific heat that is lower than that for most other substances.
C) Has a heat of vaporization that is higher than that for most other substances.
D) Is nonpolar.

7) Which of the following can be attributed to water's high specific heat?
A) Oil and water do not mix well.
B) A lake heats up more slowly than the air around it.
C) Ice floats on water.
D) Sugar dissolves in hot tea faster than in iced tea.

8) The cities of Portland, Oregon, and Minneapolis, Minnesota, are at about the same latitude, but Minneapolis has much hotter summers and much colder winters than Portland. Why?
A) They are not at the same exact latitude.
B) The ocean near Portland moderates the temperature.
C) Fresh water is more likely to freeze than salt water.
D) Minneapolis is much windier, due to its location in the middle of North America.

9) To act as an effective coolant in a car's radiator, a substance has to have the capacity to absorb a great deal of heat. You have a reference book with tables listing the physical properties of many liquids. In choosing a coolant for your car, which table would you check first?
A) pH.
B) Density at room temperature.
C) Heat of vaporization.
D) Specific heat.

10) Water has many exceptional and useful properties. Which is the rarest property among compounds?
A) Water is a solvent.
B) Solid water is less dense than liquid water.
C) Water has a high heat capacity.
D) Water has surface tension.

11) Which of the following effects can occur because of the high surface tension of water?
A) Lakes cannot freeze solid in winter, despite low temperatures.
B) A raft spider can walk across the surface of a small pond.
C) Organisms can resist temperature changes, although they give off heat due to chemical reactions.
D) Sweat can evaporate from the skin, helping to keep people from overheating.

12) Which of the following takes place as an ice cube cools a drink?
A) Molecular collisions in the drink increase.
B) Kinetic energy in the liquid water decreases.
C) A calorie of heat energy is transferred from the ice to the water of the drink.
D) The specific heat of the water in the drink decreases.

13) A dietary Calorie equals 1 kilocalorie. Which of the following statements correctly defines 1 kilocalorie? One kilocalorie equals ______.
A) 1000 calories, or the amount of heat required to raise the temperature of 1 g of water by 1°C.
B) 10,000 calories, or the amount of heat required to raise the temperature of 1 kg of water by 1°F.
C) 1000 calories, or the amount of heat required to raise the temperature of 1 kg of water by 1°C.
D) 1000 calories, or the amount of heat required to raise the temperature of 100 g of water by 100°C.

14) Which type of bond must be broken for water to vaporize?
A) Ionic bonds.
B) Polar covalent bonds.
C) Hydrogen bonds.
D) Both polar covalent bonds and hydrogen bonds.

15) Why does ice float in liquid water?
A) The high surface tension of liquid water keeps the ice on top.
B) The ionic bonds between the molecules in ice prevent the ice from sinking.
C) Stable hydrogen bonds keep water molecules of ice farther apart than water molecules of liquid water.
D) The crystalline lattice of ice causes it to be denser than liquid water.

16) Hydrophobic substances such as vegetable oil are _____.
A) Nonpolar substances that repel water molecules.
B) Nonpolar substances that have an attraction for water molecules.
C) Polar substances that repel water molecules.
D) Polar substances that have an affinity for water.

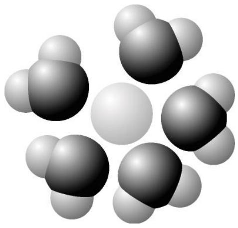
17) One mole (mol) of glucose (molecular mass = 180 daltons) is _____.
A) 180 × 1023 molecules of glucose.
B) 1 kilogram of glucose dissolved in 1 liter of solution.
C) 180 kilograms of glucose.
D) 180 grams of glucose.

18) When an ionic compound such as sodium chloride (NaCl) is placed in water, the component atoms of the NaCl crystal dissociate into individual sodium ions (Na^+) and chloride ions (Cl^-). In contrast, the atoms of covalently bonded molecules (e.g., glucose, sucrose, and glycerol) do not generally dissociate when placed in aqueous solution. Which of the following solutions would be expected to contain the greatest number of solute particles (molecules or ions)?
A) 1 liter of 0.5 *M* NaCl.
B) 1 liter of 1.0 *M* NaCl.
C) 1 liter of 1.0 *M* glucose.
D) 1 liter of 1.0 *M* NaCl and 1 liter of 1.0 *M* glucose will contain equal numbers of solute particles.

19) The molar mass of glucose is 180 grams per mole (g/mol). Which of the following procedures should you carry out to make a 1 *M* solution of glucose? Into 0.8 liter (L) of water, dissolve _____.
A) 1 g of glucose and then add more water until the total volume of the solution is 1 L.
B) 18 g of glucose and then add more water until the total volume of the solution is 1 L.
C) 180 g of glucose and then add 0.2 L more of water.
D) 180 g of glucose and then add more water until the total volume of the solution is 1 L.

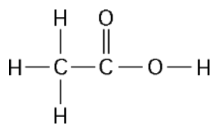
20) You have a freshly prepared 0.1 *M* glucose solution. Each liter of this solution contains how many glucose molecules?
A) 6.02×10^{23}.
B) 3.01×10^{23}.
C) 6.02×10^{24}.
D) 6.02×10^{22}.

21) Based on your knowledge of the polarity of water molecules, the solute molecule depicted here is most likely _____.



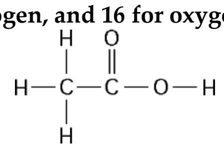
A) Positively charged.
B) Negatively charged.
C) Without charge.
D) Nonpolar.

22) One mole of the compound below would weigh how many grams? (Note: The atomic masses, in daltons, are approximately 12 for carbon, 1 for hydrogen, and 16 for oxygen.)



A) 29.
B) 30.
C) 60.
D) 150.

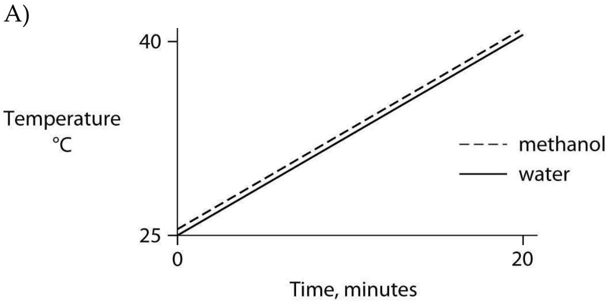
23) How many grams of the compound in the figure below are required to make 1 liter of a 0.5 *M* solution? (Note: The atomic masses, in daltons, are approximately 12 for carbon, 1 for hydrogen, and 16 for oxygen.)



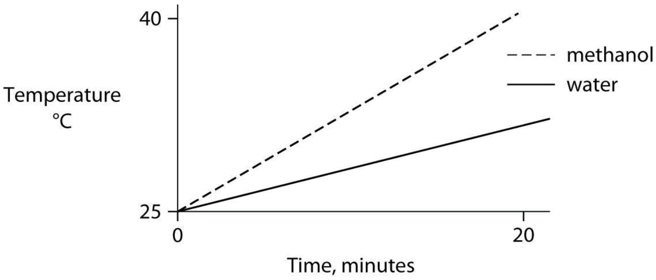
A) 29.
B) 30.
C) 60.
D) 150.

24) Identical heat lamps are arranged to shine on two identical containers, one containing water and one methanol (wood alcohol), so that each liquid absorbs the same amount of energy minute by minute. The covalent bonds of methanol molecules are nonpolar, so there are no hydrogen bonds among methanol molecules. Which of the following graphs correctly describes what will happen to the temperature of the water and the methanol?

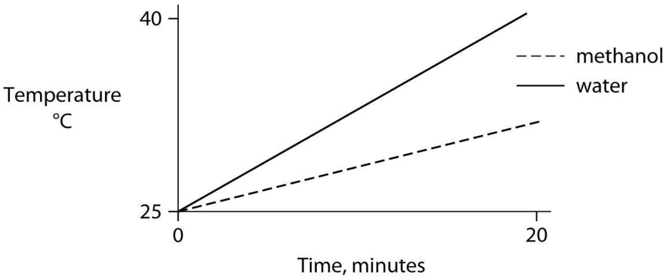
A)



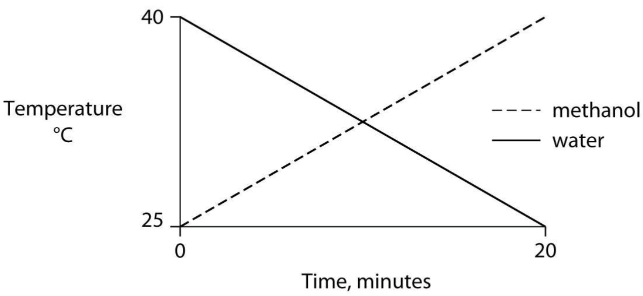
B)



C)



D)



25) You have two beakers. One contains pure water, the other contains pure methanol (wood alcohol). The covalent bonds of methanol molecules are nonpolar, so there are no hydrogen bonds among methanol molecules. You pour crystals of table salt (NaCl) into each beaker. Predict what will happen.

A) Equal amounts of NaCl crystals will dissolve in both water and methanol.
B) NaCl crystals will not dissolve in either water or methanol.
C) NaCl crystals will dissolve readily in water but will not dissolve in methanol.
D) NaCl crystals will dissolve readily in methanol but will not dissolve in water.

26) Rank, from low to high, the pH of blood, stomach acid, and urine.

A) Blood, urine, and stomach acid.
B) Stomach acid, blood, and urine.
C) Urine, blood, stomach acid.
D) Stomach acid, urine, blood.

27) A solution with a pH of 5 has how many more protons in it than a solution with a pH of 7?

A) 5 times.
B) 10 times.
C) 100 times.
D) 1000 times.

28) Consider the following reaction at equilibrium: CO_2 + H_2O ←→ H_2CO_3. What would be the effect of adding additional H_2CO_3?

A) It would drive the equilibrium dynamics to the right.
B) It would drive the equilibrium dynamics to the left.
C) Nothing would happen, because the reactants and products are in equilibrium.
D) The amounts of CO_2 and H_2O would decrease.

29) A strong acid like HCl _____.

A) Dissociates completely in an aqueous solution.
B) Increases the pH when added to an aqueous solution.
C) Reacts with strong bases to create a buffered solution.
D) Is a strong buffer at low pH.

30) Which of the following dissociates completely in solution and is considered to be a strong base (alkali)?

A) HCl.
B) NH_3.
C) H_2CO_3.
D) NaOH.

31) A 0.01 *M* solution of a substance has a pH of 2. What can you conclude about this substance?

A) It is a strong acid that dissociates completely in water.
B) It is a strong base that dissociates completely in water.
C) It is a weak acid.
D) It is a weak base.

32) A solution contains 0.0000001 (10^{-7}) moles of hydroxyl ions [OH^-] per liter. Which of the following best describes this solution?

A) Acidic: H+ acceptor.
B) Basic: H+ acceptor.
C) Acidic: H+ donor.
D) Neutral.

33) What is the pH of a solution with a hydroxyl ion (OH^-) concentration of 10^{-12} *M*?
A) pH 2.
B) pH 4.
C) pH 10.
D) pH 12.

34) Which of the following solutions would require the addition of the greatest amount of base to bring the solution to neutral pH?
A) Gastric juice at pH 2.
B) Vinegar at pH 3.
C) Black coffee at pH 5.
D) Household bleach at pH 12.

35) What is the hydrogen ion (H^+) concentration of a solution of pH 8?
A) 8 *M*.
B) 8 x 10^{-6} *M*.
C) 10^{-8} *M*.
D) 10^{-6} *M*.

36) If the pH of a solution is decreased from 9 to 8, it means that the concentration of _____.
A) H^+ has decreased to one-tenth (1/10) what it was at pH 9.
B) H^+ has doubled compared to what it was at pH 9.
C) H^+ has increased tenfold (10X) compared to what it was at pH 9.
D) OH^- has increased tenfold (10X) compared to what it was at pH 9.

37) One liter of a solution of pH 2 has how many more hydrogen ions (H^+) than 1 liter of a solution of pH 6?
A) 4 times more.
B) 40,000 times more.
C) 10,000 times more.
D) 100,000 times more.

38) Which of the following statements is true about buffer solutions?
A) They maintain a constant pH when bases are added to them but not when acids are added to them.
B) They maintain a constant pH when acids are added to them but not when bases are added to them.
C) They fluctuate in pH when either acids or bases are added to them.
D) They maintain a relatively constant pH when either acids or bases are added to them.

39) One of the buffers that contribute to pH stability in human blood is carbonic acid (H_2CO_3). Carbonic acid is a weak acid that, when placed in an aqueous solution, dissociates into a bicarbonate ion (HCO_3^-) and a hydrogen ion (H^+), as noted below.

$$H_2CO_3 \leftrightarrow HCO_3^- + H^+$$

If the pH of blood drops, one would expect _____.
A) A decrease in the concentration of H_2CO_3 and an increase in the concentration of HCO_3^-.
B) The concentration of bicarbonate ions (HCO_3^-) to increase.
C) The HCO_3^- to act as a base and remove excess H^+ by the formation of H_2CO_3.
D) The HCO_3^- to act as an acid and remove excess H^+ by the formation of H_2CO_3.

40) One of the buffers that contribute to pH stability in human blood is carbonic acid (H_2CO_3). Carbonic acid is a weak acid that, when placed in an aqueous solution, dissociates into a bicarbonate ion (HCO_3^-) and a hydrogen ion (H^+), as noted below.

$$H_2CO_3 \leftrightarrow HCO_3^- + H^+$$

If the pH of blood increases, one would expect _____.
A) A decrease in the concentration of H_2CO_3 and an increase in the concentration of HCO_3^-.
B) An increase in the concentration of H_2CO_3 and a decrease in the concentration of HCO_3^-.
C) A decrease in the concentration of HCO_3^- and an increase in the concentration of H^+.
D) An increase in the concentration of HCO_3^- and a decrease in the concentration of OH^-.

41) Assume that acid rain has lowered the pH of a particular lake to pH 4.0. What is the hydroxide ion concentration of this lake?
A) 1 × 10^{-10} mol of hydroxide ions per liter of lake water.
B) 1 × 10^{-4} mol of hydroxide ions per liter of lake water.
C) 4.0 *M* with regard to hydroxide ion concentration.
D) 4.0 × 10^{-4} mol of hydroxide ions per liter of lake water.

42) Consider two solutions: solution X has a pH of 4; solution Y has a pH of 7. From this information, we can reasonably conclude that _____.
A) Solution Y has no free hydrogen ions (H^+).
B) The concentration of hydrogen ions in solution Y is 1000 times as great as the concentration of hydrogen ions in solution X.
C) The concentration of hydrogen ions in solution X is 3 times as great as the concentration of hydrogen ions in solution Y.
D) The concentration of hydrogen ions in solution X is 1000 times as great as the concentration of hydrogen ions in solution Y.

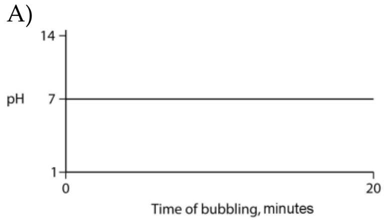
43) A beaker contains 100 milliliters (mL) of NaOH solution at pH = 13. A technician carefully pours into the beaker 10 mL of HCl at pH = 1. Which of the following statements correctly describes the result of this mixing?
A) The concentration of Na+ ions will rise.
B) The pH of the beaker's contents will increase.
C) The pH of the beaker's contents will be neutral.
D) The pH of the beaker's contents will decrease.

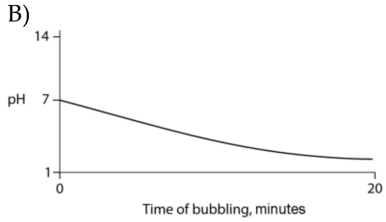
44) Increased atmospheric CO2 concentrations might have what effect on seawater?
A) Seawater will become more alkaline, and carbonate concentrations will decrease.
B) There will be no change in the pH of seawater, because carbonate will turn to bicarbonate.
C) Seawater will become more acidic, and carbonate concentrations will decrease.
D) Seawater will become more acidic, and carbonate concentrations will increase.

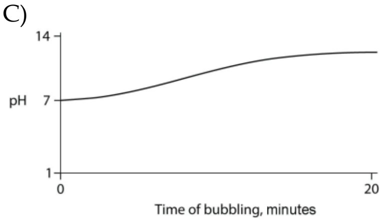
45) How would acidification of seawater affect marine organisms? Acidification of seawater would _____.
A) Increase dissolved carbonate concentrations and promote faster growth of corals and shell-building animals.
B) Decrease dissolved carbonate concentrations and promote faster growth of corals and shell-building animals.
C) Increase dissolved carbonate concentrations and hinder growth of corals and shell-building animals.
D) Decrease dissolved carbonate concentrations and hinder growth of corals and shell-building animals.

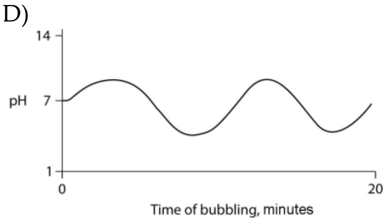
46) One idea to mitigate the effects of burning fossil fuels on atmospheric CO_2 concentrations is to pipe liquid CO_2 into the ocean at depths of 2500 feet or greater. At the high pressures at such depths, CO_2 is heavier than water. What potential effects might result from implementing such a scheme?
A) Increased carbonate concentrations in the deep waters.
B) Increased growth of corals from a change in the carbonate–bicarbonate equilibrium.
C) No effect because carbon dioxide is not soluble in water.
D) Increased acidity and decreased carbonate concentrations in the deep waters.

47) If the cytoplasm of a cell is at pH 7, and the mitochondrial matrix is at pH 8, then the concentration of H^+ ions _____.
A) Is 10 times higher in the cytoplasm than in the mitochondrial matrix.
B) Is 10 times higher in the mitochondrial matrix than in the cytoplasm.
C) In the cytoplasm is 7/8 the concentration in the mitochondrial matrix.
D) In the cytoplasm is 8/7 the concentration in the mitochondrial matrix.

48) Carbon dioxide (CO_2) is readily soluble in water, according to the equation $CO_2 + H_2O \leftrightarrow H_2CO_3$. Carbonic acid ($H_2CO_3$) is a weak acid. If CO_2 is bubbled into a beaker containing pure, freshly distilled water, which of the following graphs correctly describes the results?
A)


B)


C)


D)


49) The loss of water from a plant by transpiration cools the leaf. Movement of water in transpiration requires both adhesion to the conducting walls and wood fibers of the plant and cohesion of the molecules to each other. A scientist wanted to increase the rate of transpiration of a crop species to extend its range into warmer climates. The scientist substituted a nonpolar solution with an atomic mass similar to that of water for hydrating the plants. What do you expect the scientist's data will indicate from this experiment?
A) The rate of transpiration will be the same for both water and the nonpolar substance.
B) The rate of transpiration will be slightly lower with the nonpolar substance as the plant will not have evolved with the nonpolar compound.
C) Transpiration rates will fall to zero as nonpolar compounds do not have the properties necessary for adhesion and cohesion.
D) Transpiration rates will increase as nonpolar compounds undergo adhesion and cohesion with wood fibers more readily than water.

50) In living systems molecules involved in hydrogen bonding almost always contain either oxygen or nitrogen or both. How do you explain this phenomenon?
A) Oxygen and nitrogen are elements found in both nucleic acids and proteins.
B) Oxygen and nitrogen are elements with very high attractions for their electrons.
C) Oxygen and nitrogen are elements found in fats and carbohydrates.
D) Oxygen and nitrogen were both components of gases that made up the early atmosphere on Earth.

CHAPTER 4:

CARBON AND THE MOLECULAR DIVERSITY OF LIFE

1) The element present in all organic molecules is _____.
A) Hydrogen.
B) Oxygen.
C) Carbon.
D) Nitrogen.

2) The complexity and variety of organic molecules is due to _____.
A) The chemical versatility of carbon atoms.
B) The variety of rare elements in organic molecules.
C) The diverse bonding patterns of nitrogen.
D) Their interaction with water.

3) The experimental approach taken in current biological investigations presumes that _____.
A) Simple organic compounds can be synthesized in the laboratory from inorganic precursors, but complex organic compounds like carbohydrates and proteins can be synthesized only by living organisms.
B) A life force ultimately controls the activities of living organisms and this life force cannot be studied by physical or chemical methods.
C) Living organisms are composed of the same elements present in nonliving things, plus a few special trace elements found only in living organisms or their products.
D) Living organisms can be understood in terms of the same physical and chemical laws that can be used to explain all natural phenomena.

4) Differences among organisms are caused by differences in the _____.
A) Elemental composition from organism to organism.
B) Types and relative amounts of organic molecules synthesized by each organism.
C) Sizes of the organic molecules in each organism.
D) Types of inorganic compounds present in each organism.

5) Stanley Miller's 1953 experiments supported the hypothesis that _____.
A) Life on Earth arose from simple inorganic molecules.
B) Organic molecules can be synthesized abiotically under conditions that may have existed on early Earth.
C) Life on Earth arose from simple organic molecules, with energy from lightning and volcanoes.
D) The conditions on early Earth were conducive to the origin of life.

6) When Stanley Miller applied heat and electrical sparks to a mixture of simple inorganic compounds such as methane, hydrogen gas, ammonia, and water vapor, what compounds were produced?
A) Only simple organic compounds such as formaldehyde and cyanide.
B) Mostly hydrocarbons.
C) Only simple inorganic compounds.
D) Simple organic compounds, amino acids, and hydrocarbons.

7) Which of the following is true of carbon?
A) It forms only polar molecules.
B) It can form a maximum of three covalent bonds with other elements.
C) It is highly electronegative.
D) It can form both polar and nonpolar bonds.

8) Why is carbon so important in biology?
A) It is a common element on Earth.
B) It has very little electronegativity, making it a good electron donor.
C) It bonds to only a few other elements.
D) It can form a variety of carbon skeletons and host functional groups.

9) How many electron pairs does carbon share to complete its valence shell?
A) 2.
B) 3.
C) 4.
D) 8.

10) A carbon atom is most likely to form what kind of bond(s) with other atoms?
A) Ionic.
B) Hydrogen.
C) Covalent.
D) Ionic bonds, covalent bonds, and hydrogen bonds.

11) Why are hydrocarbons insoluble in water?
A) The majority of their bonds are polar covalent carbon-to-hydrogen linkages.
B) The majority of their bonds are nonpolar covalent carbon-to-hydrogen linkages.
C) They exhibit molecular complexity and diversity.
D) They are less dense than water.

12) Which of the following statements correctly describes cis-trans isomers?
A) They have variations in arrangement around a double bond.
B) They have an asymmetric carbon that makes them mirror images.
C) They have the same chemical properties.
D) They have different molecular formulas.

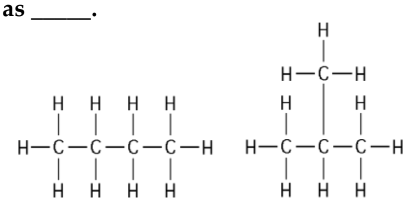
13) Research indicates that ibuprofen, a drug used to relieve inflammation and pain, is a mixture of two enantiomers; that is, molecules that _____.
A) Have identical chemical formulas but differ in the branching of their carbon skeletons.
B) Are mirror images of each other.
C) Differ in the location of their double bonds.
D) Differ in the arrangement of atoms.

14) What determines whether a carbon atom's covalent bonds to other atoms are in a tetrahedral configuration or a planar configuration?
A) The presence or absence of bonds with oxygen atoms.
B) The presence or absence of double bonds between the carbon atom and other atoms.
C) The polarity of the covalent bonds between carbon and other atoms.
D) The solvent in which the organic molecule is dissolved.

15) Compared to a hydrocarbon chain where all the carbon atoms are linked by single bonds, a hydrocarbon chain with the same number of carbon atoms, but with one or more double bonds, will _____.
A) Be more flexible in structure.
B) Be more constrained in structure.
C) Be more polar.
D) Have more hydrogen atoms.

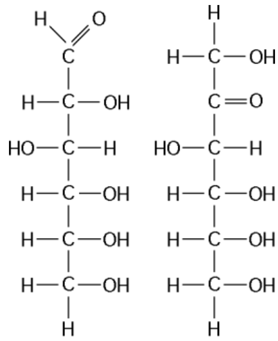
16) Organic molecules with only hydrogens and five carbon atoms cannot _____.
A) Have a branching carbon skeleton.
B) Have different combinations of double bonds between carbon atoms.
C) Have different positions of double bonds between carbon atoms.
D) Form enantiomers.

17) The two molecules shown in the figure below are best described as _____.



A) Enantiomers.
B) Structural isomers.
C) Cis-trans isomers.
D) Chain length isomers.

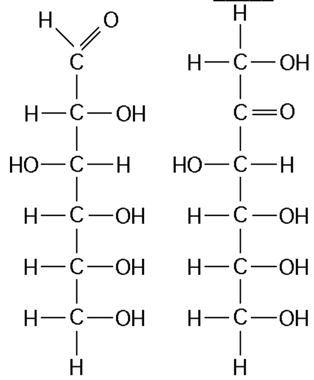
18) The figure below shows the structures of glucose and fructose. These two molecules differ in the _____.



glucose fructose

A) Number of carbon, hydrogen, and oxygen atoms.
B) Types of carbon, hydrogen, and oxygen atoms.
C) Arrangement of carbon, hydrogen, and oxygen atoms.
D) Number of oxygen atoms joined to carbon atoms by double covalent bonds.

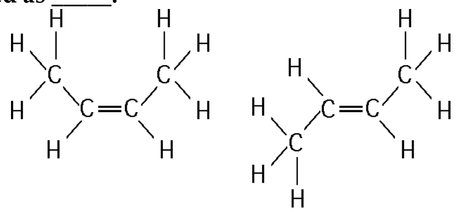
19) The figure below shows the structures of glucose and fructose. These two molecules are _____.



glucose fructose

A) Isotopes.
B) Enantiomers.
C) Cis-trans isomers.
D) Structural isomers.

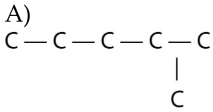
20) The two molecules shown in the figure below are best described as _____.



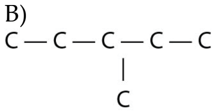
A) Enantiomers.
B) Radioactive isotopes.
C) Structural isomers.
D) Cis-trans isomers.

21) Which of the following illustrations is NOT a structural isomer of an organic compound with the molecular formula C_6H_{14}? For clarity, only the carbon skeletons are shown; hydrogen atoms that would be attached to the carbons have been omitted.

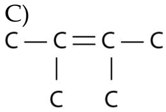
A)



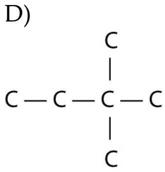
B)



C)

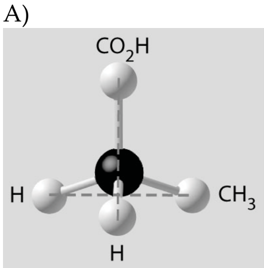


D)

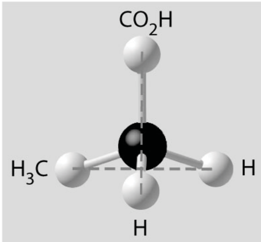


22) Which of the pairs of molecular structures shown below depict enantiomers (enantiomeric forms) of the same molecule?

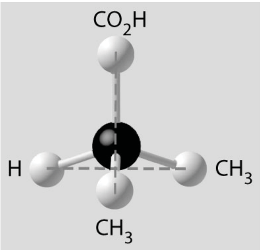
A)



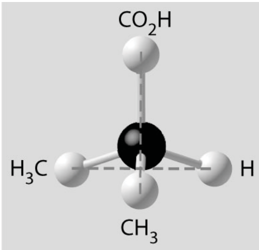
and



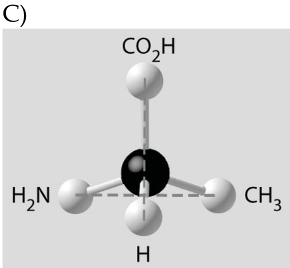
B)



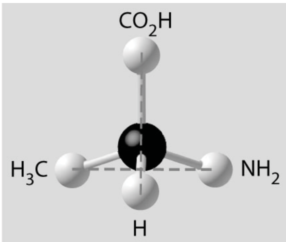
and



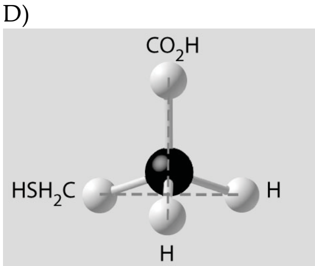
C)



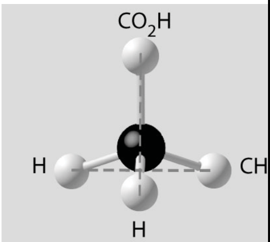
and



D)

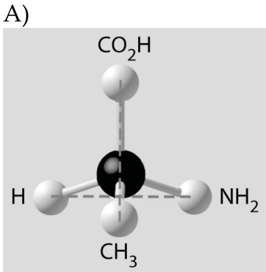


and

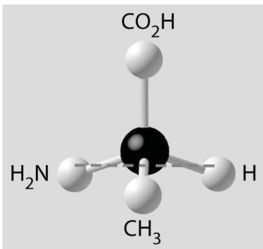


23) Which of the pairs of molecular structures shown below do NOT depict enantiomers (enantiomeric forms) of the same molecule?

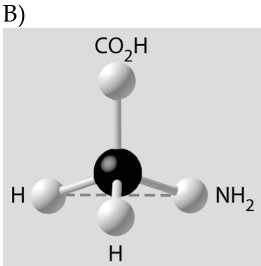
A)



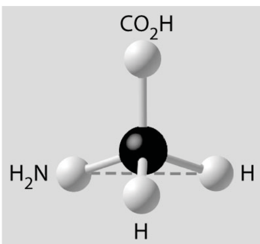
and



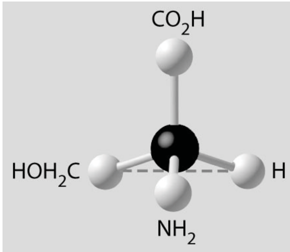
B)



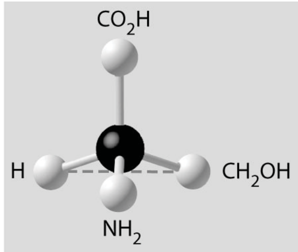
and



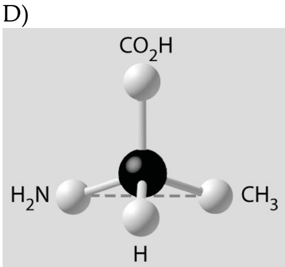
C)



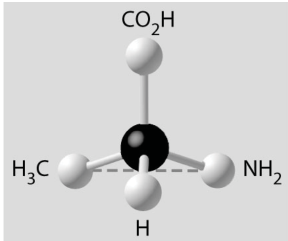
and



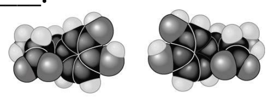
D)



and



24) Thalidomide and L-dopa, shown below, are examples of pharmaceutical drugs that occur as enantiomers or molecules that _____.



L-dopa D-dopa

A) Have identical three-dimensional shapes.
B) Are mirror images of one another.
C) Are mirror images of one another and have the same biological activity.
D) Are cis-trans isomers.

25) Which of the following molecules is polar?

C_3H_7OH C_2H_5COOH

A) C_3H_7OH and C_2H_5COOH are both polar molecules.
B) Neither C_2H_5COOH or C_3H_7OH is polar.
C) C_2H_5COOH is polar, but C_3H_7OH is not polar.
D) C_3H_7OH is not polar, but C_3H_7OH is polar.

26) Which of the functional groups below acts most like an acid in water?

A) Amino.
B) Carbonyl.
C) Carboxyl.
D) Hydroxyl.

27) A compound contains hydroxyl groups as its predominant functional group. Therefore, this compound _____.

A) Lacks an asymmetric carbon and is probably a fat or lipid.
B) Should dissolve in water.
C) Should dissolve in a nonpolar solvent.
D) Will not form hydrogen bonds with water.

28) Which two functional groups are always found in amino acids?

A) Carbonyl and amino groups.
B) Carboxyl and amino groups.
C) Amino and sulfhydryl groups.
D) Hydroxyl and carboxyl groups.

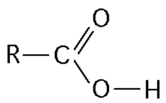
29) Amino acids are acids because they always possess which functional group?
A) Amino.
B) Carbonyl.
C) Carboxyl.
D) Phosphate.

30) A hydrocarbon skeleton is covalently bonded to an amino group at one end and a carboxyl group at the other end. When placed in water this molecule would function _____.
A) Only as an acid because of the carboxyl group.
B) Only as a base because of the amino group.
C) As an acid and a base.
D) As neither an acid nor a base.

31) Which chemical group can act as an acid?
A) Amino.
B) Carbonyl.
C) Carboxyl.
D) Methyl.

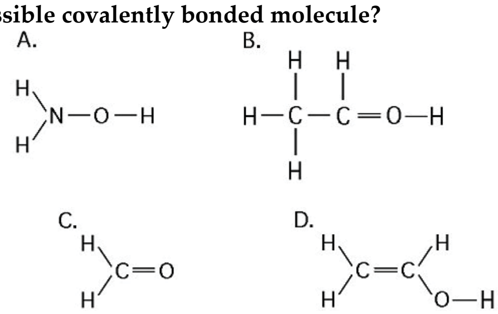
32) Testosterone and estradiol are male and female sex hormones, respectively, in many vertebrates. In what way(s) do these molecules differ from each other? Testosterone and estradiol _____.
A) Are structural isomers but have the same molecular formula.
B) Are *cis-trans* isomers but have the same molecular formula.
C) Have different functional groups attached to the same carbon skeleton.
D) Are enantiomers of the same organic molecule.

33) What is the name of the functional group shown in the figure below?



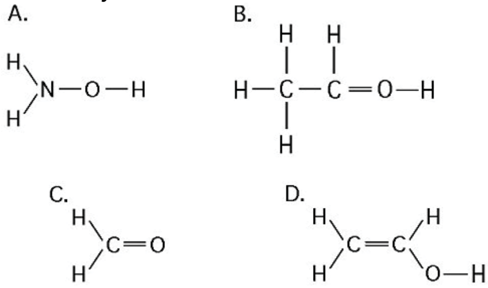
A) Carbonyl.
B) Ketone.
C) Aldehyde.
D) Carboxyl.

34) Which of the structures illustrated below is an impossible covalently bonded molecule?



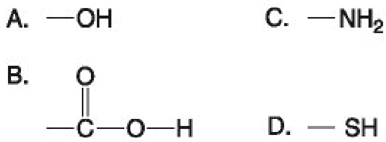
A) A.
B) B.
C) C.
D) D.

35) In which of the structures illustrated below are the atoms bonded by ionic bonds?



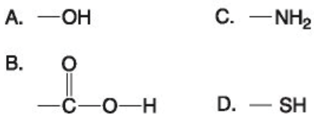
A) A.
B) B.
C) C.
D) None of the structures.

36) Which functional group shown below is characteristic of alcohols?



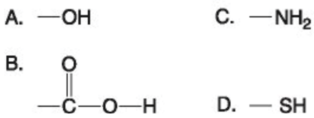
A) A.
B) B.
C) C.
D) D.

37) Which functional group(s) shown below is (are) present in all amino acids?



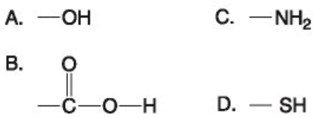
A) A and B.
B) B and D.
C) C only.
D) B and C.

38) Which of the groups shown below is a functional group that helps stabilize proteins by forming covalent cross-links within or between protein molecules?



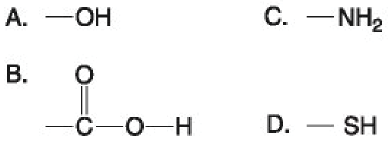
A) A.
B) B.
C) C.
D) D.

39) Which of the groups below is a carboxyl functional group?



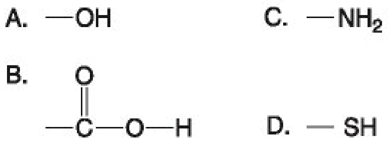
A) A.
B) B.
C) C.
D) D.

40) Which of the groups below is an acidic functional group that can dissociate and release H^+ into a solution?



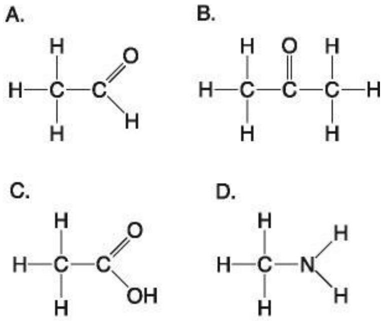
A) A.
B) B.
C) C.
D) D.

41) Which of the groups below is a basic functional group that can accept H+ and become positively charged?



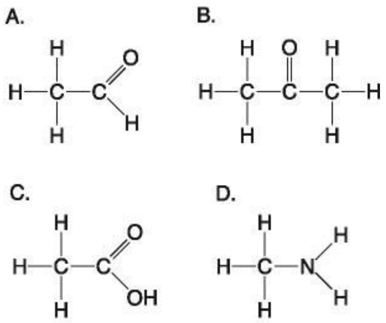
A) A.
B) B.
C) C.
D) D.

42) Which molecule shown above would have a positive charge in a cell?



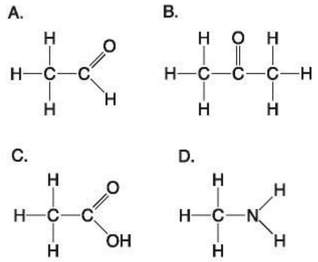
A) A.
B) B.
C) C.
D) D.

43) Which molecule(s) shown above is (are) ionized in a cell?



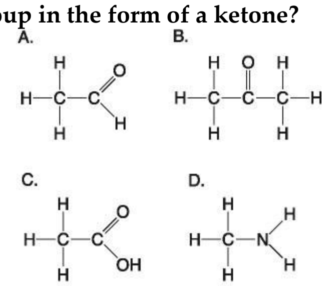
A) A.
B) B and D.
C) C and D.
D) D.

44) Which molecules shown above contain a carbonyl group?



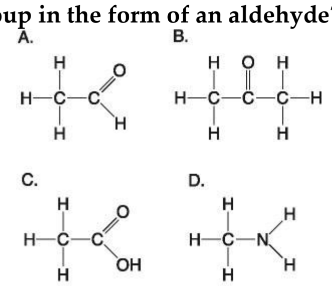
A) A and B.
B) B and C.
C) B, C, and D.
D) C and D.

45) Which molecule shown above has a carbonyl functional group in the form of a ketone?



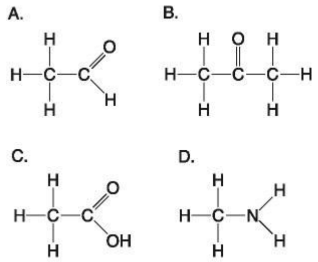
A) A.
B) B.
C) C.
D) D.

46) Which molecule shown above has a carbonyl functional group in the form of an aldehyde?



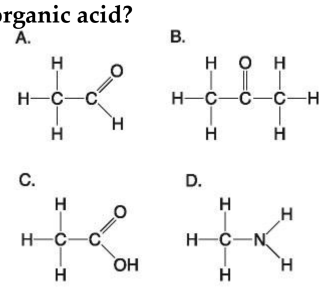
A) A.
B) B.
C) C.
D) D.

47) Which molecule shown above contains a carboxyl group?



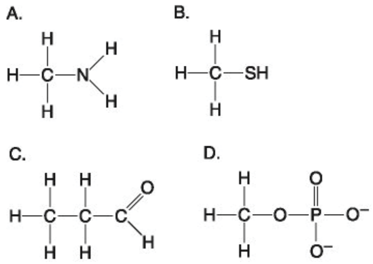
A) A.
B) B.
C) C.
D) D.

48) Which molecule shown above can increase the concentration of hydrogen ions in a solution and is, therefore, an organic acid?



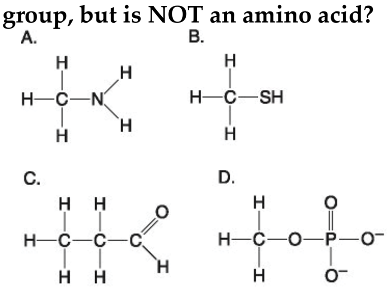
A) A.
B) B.
C) C.
D) D.

49) Which molecule shown below can form a cross linkage?



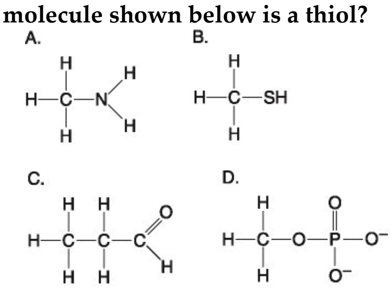
A) A.
B) B.
C) C.
D) D.

50) Which molecule shown below contains an amino functional group, but is NOT an amino acid?



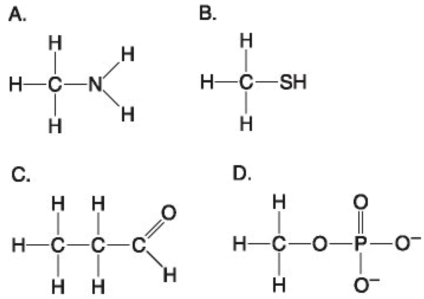
A) A.
B) B.
C) C.
D) D.

51) Which molecule shown below is a thiol?



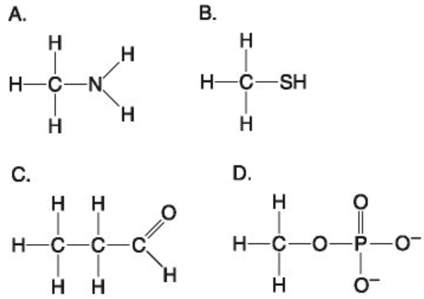
A) A.
B) B.
C) C.
D) D.

52) Which molecule shown below contains a functional group that cells use to transfer energy between organic molecules?



A) A.
B) B.
C) C.
D) D.

53) Which molecule shown below can function as a base?



A) A.
B) B.
C) C.
D) D.

54) Which of the following is a FALSE statement concerning amino groups? Amino groups _____.
A) Are basic with respect to pH.
B) Are found in amino acids.
C) Contain nitrogen.
D) Are nonpolar.

CHAPTER 5:

THE STRUCTURE AND FUNCTION OF LARGE BIOLOGICAL MOLECULES

1) Which of these classes of biological molecules does NOT include polymers?
A) Lipids.
B) Carbohydrates.
C) Proteins.
D) Nucleic acids.

2) Which of the following is NOT a polymer?
A) Glucose.
B) Starch.
C) Cellulose.
D) DNA.

3) How many molecules of water are used to completely hydrolyze a polymer that is 11 monomers long?
A) 12.
B) 11.
C) 10.
D) 9.

4) Which of the following best summarizes the relationship between dehydration and hydrolysis?
A) Dehydration reactions assemble polymers; hydrolysis reactions break polymers apart.
B) Dehydration reactions eliminate water from membranes; hydrolysis reactions add water to membranes.
C) Dehydration reactions and hydrolysis reactions assemble polymers from monomers.
D) Hydrolysis reactions create polymers and dehydration reactions create monomers.

5) The molecular formula for glucose is $C_6H_{12}O_6$. What would be the molecular formula for a molecule made by linking three glucose molecules together by dehydration reactions?
A) $C_{18}H_{36}O_{18}$.
B) $C_{18}H_{32}O_{16}$.
C) $C_6H_{10}O_5$.
D) $C_{18}H_{30}O_{15}$.

6) What is the difference between an aldose sugar and a ketose sugar?
A) The number of carbons.
B) The position of the hydroxyl groups.
C) The position of the carbonyl group.
D) One is a ring form, the other is a linear chain.

7) What is the major structural difference between starch and glycogen?
A) The types of monosaccharide subunits in the molecules.
B) The type of glycosidic linkages in the molecule.
C) Whether glucose is in the α or β form.
D) The amount of branching that occurs in the molecule.

8) Which polysaccharide is an important component in the structure of many animals and fungi?
A) Chitin.
B) Cellulose.
C) Amylopectin.
D) Amylose.

9) What does the term *insoluble fiber* refer to on food packages?
A) Cellulose.
B) Polypeptides.
C) Starch.
D) Amylopectin.

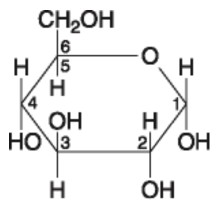
10) A molecule with the chemical formula $C_6H_{12}O_6$ is probably a _____.
A) Fatty acid.
B) Polysaccharide.
C) Nucleic acid.
D) Monosaccharide.

11) Lactose, a sugar in milk, is composed of one glucose molecule joined by a glycosidic linkage to one galactose molecule. How is lactose classified?
A) As a hexose.
B) As a monosaccharide.
C) As a disaccharide.
D) As a polysaccharide.

12) Starch and cellulose _____.
A) Are polymers of glucose.
B) Are cis-trans isomers of each other.
C) Are used for energy storage in plants.
D) Are structural components of the plant cell wall.

13) Humans can digest starch but not cellulose because _____.
A) Humans have enzymes that can hydrolyze the α-glycosidic linkages of starch but not the β-glycosidic linkages of cellulose.
B) Starch monomers are joined by covalent bonds and cellulose monomers are joined by ionic bonds.
C) The monomer of starch is glucose, while the monomer of cellulose is galactose.
D) The monomer of starch is fructose, while the monomer of cellulose is glucose.

14) The molecule shown in the accompanying figure is _____.



A) A hexose.
B) A pentose.
C) Fructose.
D) Maltose.

15) A glycosidic linkage is analogous to which of the following in proteins?
A) An amino group.
B) A peptide bond.
C) A disulfide bond.
D) A β-pleated sheet.

16) Cooking oil and gasoline (a hydrocarbon) are not amphipathic molecules because they _____.
A) Do not have a polar or charged region.
B) Do not have a nonpolar region.
C) Have hydrophobic and hydrophilic regions.
D) Are highly reduced molecules.

17) How do phospholipids interact with water molecules?
A) The polar heads avoid water; the nonpolar tails attract water (because water is polar and opposites attract).
B) Phospholipids do not interact with water because water is polar and lipids are nonpolar.
C) The polar heads interact with water; the nonpolar tails do not.
D) Phospholipids dissolve in water.

18) Phospholipids and triglycerides both _____.
A) Contain serine or some other organic compound.
B) Have three fatty acids.
C) Have a glycerol backbone.
D) Have a phosphate.

19) Which of the following is the best explanation for why vegetable oil is a liquid at room temperature while animal fats are solid?
A) Vegetable oil has more double bonds than animal fats.
B) Vegetable oil has fewer double bonds than animal fats.
C) Animal fats have no amphipathic character.
D) Vegetable oil has longer fatty-acid tails than animal fats.

20) Which of the following statements is FALSE? Saturated fats _____.
A) Are more common in animals than in plants.
B) Have many double bonds in the carbon chains of their fatty acids.
C) Usually solidify at room temperature.
D) Contain more hydrogen than unsaturated fats that consist of the same number of carbon atoms.

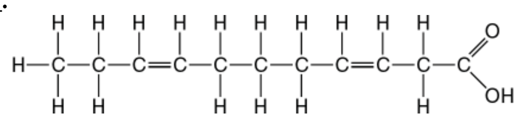
21) Lipids _____.
A) Are insoluble in water.
B) Are made from glycerol, fatty acids, and nitrogen.
C) Contain less energy than proteins and carbohydrates.
D) Are made by dehydration reactions.

22) The label on a container of margarine lists "hydrogenated vegetable oil" as the major ingredient. Hydrogenated vegetable oil _____.
A) Is solid at room temperature.
B) Has more "kinks" in the fatty acid chains.
C) Has fewer trans fatty acids.
D) Is less likely to clog arteries.

23) Saturated fatty acids _____.
A) Have double bonds between carbon atoms of the fatty acids.
B) Are the principal molecules in lard and butter.
C) Are usually liquid at room temperature.
D) Are usually produced by plants.

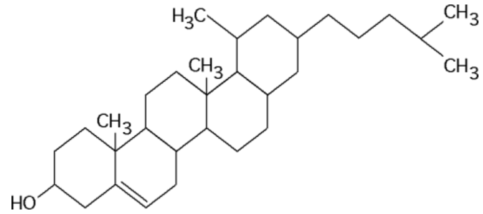
24) Steroids are considered to be lipids because they _____.
A) Are essential components of cell membranes.
B) Are not soluble in water.
C) Are made of fatty acids.
D) Contribute to atherosclerosis.

25) The molecule illustrated in the accompanying figure _____.



A) Is a saturated fatty acid.
B) Stores genetic information.
C) Will be liquid at room temperature.
D) Is a carbohydrate.

26) The molecule shown the accompanying figure is a _____.



A) Fatty acid.
B) Steroid.
C) Protein.
D) Phospholipid.

27) Which one of the following is NOT a component of each monomer used to make proteins?
A) A phosphorus atom, P.
B) An amino functional group, NH_2.
C) A side chain, R.
D) A carboxyl group, COOH.

28) What component of amino acid structure varies among different amino acids?
A) The long carbon-hydrogen tails of the molecule.
B) The presence of a central C atom.
C) The components of the R-group.
D) The glycerol molecule that forms the backbone of the amino acid.

29) You disrupt all hydrogen bonds in a protein. What level of structure will be preserved?
A) Primary structure.
B) Secondary structure.
C) Tertiary structure.
D) Quaternary structure.

30) Which of the following is the strongest evidence that protein structure and function are correlated?
A) Proteins function best at certain temperatures.
B) Proteins have four distinct levels of structure and many functions.
C) Enzymes tend to be globular in shape.
D) Denatured (unfolded) proteins do not function normally.

31) You have just sequenced a new protein found in mice and observe that sulfur-containing cysteine residues occur at regular intervals. What is the significance of this finding?
A) Cysteine residues are required for the formation of α-helices and β-pleated sheets.
B) It will be important to include cysteine in the diet of the mice.
C) Cysteine residues are involved in disulfide bridges that help form tertiary structure.
D) Cysteine causes bends, or angles, to occur in the tertiary structure of proteins.

Use the following information when answering the question (32).

Rhodopsins are light-sensitive molecules composed of a protein (opsin) and retinal (derivative of vitamin A). Opsin is a membrane protein with several α-helical segments that loop back and forth through the plasma membrane. There are two classes of rhodopsins. According to Oded Beje, one class has relatively slow dynamics (a photocycle of approximately 0.5 second) and is well suited for light detection. The second class has faster dynamics (a photocycle of approximately 0.02 seconds) and is well suited for chemiosmosis: pumping of protons or chloride ions across cell membranes. Oded Beje was the first, in September 2000, to report on a rhodopsin (proteorhodopsin) found in the domain Bacteria. [SOURCE: O. Beje et al., Science 289 (2000): 1902.]

32) Proteorhodopsin consists of a single polypeptide chain. What is the highest level of structure found in this protein?
A) Primary.
B) Secondary.
C) Tertiary.
D) Quaternary.

33) All of the following contain amino acids EXCEPT ___.
A) Hemoglobin.
B) Cholesterol.
C) Enzymes.
D) Insulin.

34) Which level of protein structure do the α-helix and the β-pleated sheet represent?
A) Primary.
B) Secondary.
C) Tertiary.
D) Quaternary.

35) The tertiary structure of a protein is the _____.
A) Order in which amino acids are joined in a polypeptide chain.
B) Unique three-dimensional shape of the fully folded polypeptide.
C) Organization of a polypeptide chain into a α-helix or β-pleated sheet.
D) Overall protein structure resulting from the aggregation of two or more polypeptide subunits.

36) The R-group, or side chain, of the amino acid serine is -CH_2-OH. The R-group, or side chain, of the amino acid leucine is -CH_2-CH-$(CH_3)_2$. Where would you expect to find these amino acids in a globular protein in aqueous solution?
A) Serine would be in the interior, and leucine would be on the exterior of the globular protein.
B) Leucine would be in the interior, and serine would be on the exterior of the globular protein.
C) Serine and leucine would both be in the interior of the globular protein.
D) Serine and leucine would both be on the exterior of the globular protein.

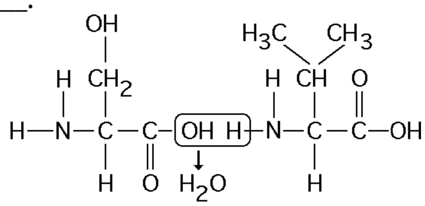
37) Misfolding of polypeptides is a serious problem in cells. Which of the following diseases are associated with an accumulation of misfolded polypeptides?
A) Alzheimer's only.
B) Parkinson's only.
C) Diabetes mellitus only.
D) Alzheimer's and Parkinson's only.

38) Changing a single amino acid in a protein consisting of 325 amino acids would _____.
A) Alter the primary structure of the protein but not its tertiary structure or function.
B) Cause the tertiary structure of the protein to unfold.
C) Always alter the biological activity or function of the protein.
D) Always alter the primary structure of the protein, sometimes alter the tertiary structure of the protein, and sometimes affect its biological activity.

39) Normal hemoglobin is a tetramer, consisting of two molecules of β hemoglobin and two molecules of α hemoglobin. In sickle-cell disease, as a result of a single amino acid change, the mutant hemoglobin tetramers associate with each other and assemble into large fibers. Based on this information alone, we can conclude that sickle-cell hemoglobin exhibits _____.
A) Only altered primary structure.
B) Only altered tertiary structure.
C) Only altered quaternary structure.
D) Altered primary structure and altered quaternary structure; the secondary and tertiary structures may or may not be altered.

40) What is the term used for a protein molecule that assists in the proper folding of other proteins?
A) Tertiary protein.
B) Chaperonin.
C) Renaturing protein.
D) Denaturing protein.

41) The chemical reaction illustrated in the accompanying figure _____.



A) Is a hydrolysis reaction.
B) Results in a peptide bond.
C) Joins two fatty acids together.
D) Links two polymers to form a monomer.

42) Nucleic acids are polymers made up of which of the following monomers?
A) Nucleotides.
B) Sugars.
C) Amino acids.
D) Nitrogenous bases.

43) Which of the following includes all of the pyrimidines found in RNA and DNA?
A) Cytosine and uracil.
B) Cytosine and thymine.
C) Cytosine, uracil, and thymine.
D) Cytosine, uracil, and guanine.

44) When nucleotides polymerize to form a nucleic acid _____.
A) A covalent bond forms between the sugar of one nucleotide and the phosphate of a second.
B) A hydrogen bond forms between the sugar of one nucleotide and the phosphate of a second.
C) Covalent bonds form between the bases of two nucleotides.
D) Hydrogen bonds form between the bases of two nucleotides.

45) Which of the following statements about the 5' end of a polynucleotide strand of RNA is correct?
A) The 5′ end has a hydroxyl group attached to the number 5 carbon of ribose.
B) The 5′ end has a phosphate group attached to the number 5 carbon of ribose.
C) The 5′ end has phosphate attached to the number 5 carbon of the nitrogenous base.
D) The 5′ end has a carboxyl group attached to the number 5 carbon of ribose.

46) One of the primary functions of RNA molecules is to _____.
A) Transmit genetic information to offspring.
B) Function in the synthesis of proteins.
C) Make a copy of itself, thus ensuring genetic continuity.
D) Act as a pattern or blueprint to form DNA.

47) If ^{14}C-labeled uracil is added to the growth medium of cells, what macromolecules will be labeled?
A) DNA.
B) RNA.
C) Both DNA and RNA.
D) Proteins.

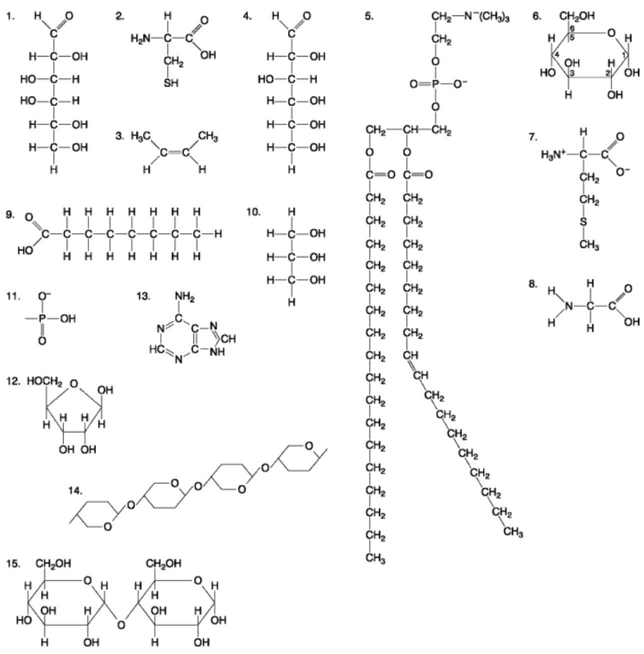
48) Which of the following descriptions best fits the class of molecules known as nucleotides?
A) A nitrogenous base and a phosphate group.
B) A nitrogenous base and a sugar.
C) A nitrogenous base, a phosphate group, and a sugar.
D) A sugar and a purine or pyrimidine.

49) If a DNA sample were composed of 10% thymine, what would be the percentage of guanine?
A) 10.
B) 40.
C) 80.
D) It is impossible to tell from the information given.

50) Which of the following statements best summarizes the differences between DNA and RNA?
A) DNA encodes hereditary information, whereas RNA does not.
B) The bases in DNA contain sugars, whereas the bases in RNA do not contain sugar.
C) DNA nucleotides contain a different sugar than RNA nucleotides.
D) DNA contains the base uracil, whereas RNA contains the base thymine.

51) If one strand of a DNA molecule has the sequence of bases 5'ATTGCA3', the other complementary strand would have the sequence _____.
A) 5'TAACGT3'.
B) 5'TGCAAT3'.
C) 3'UAACGU5'.
D) 5'UGCAAU3'.

The following questions (52-59) are based on the 15 molecules illustrated in the accompanying figure. Each molecule may be used once, more than once, or not at all.



52) Which molecule has both hydrophilic and hydrophobic properties and is found in plasma membranes?
A) 1.
B) 5.
C) 12.
D) 14.

53) Which of the following combinations of molecules illustrated could be linked to form a nucleotide?
A) 1, 2, and 11.
B) 3, 7, and 8.
C) 5, 9, and 10.
D) 11, 12, and 13.

54) Which molecule is a saturated fatty acid?
A) 1.
B) 5.
C) 8.
D) 9.

55) Which of the following molecules is a purine nitrogenous base?
A) 2.
B) 5.
C) 12.
D) 13.

56) Which of the following molecules act as building blocks (monomers) of polypeptides?
A) 1, 4, and 6.
B) 2, 7, and 8.
C) 7, 8, and 13.
D) 11, 12, and 13.

57) Which of the following pairs of molecules could be joined together by a peptide bond in a dehydration reaction?
A) 2 and 3.
B) 7 and 8.
C) 8 and 9.
D) 12 and 13.

58) A fat (or triacylglycerol) would be formed as a result of a dehydration reaction between _____.
A) One molecule of 9 and three molecules of 10.
B) Three molecules of 9 and one molecule of 10.
C) One molecule of 5 and three molecules of 9.
D) One molecule of 5 and three molecules of 10.

59) Which of the following molecules is the pentose sugar found in RNA?
A) 1.
B) 6.
C) 12.
D) 13.

60) A new organism is discovered in the forests of Costa Rica. Scientists there determine that the polypeptide sequence of hemoglobin from the new organism has 72 amino acid differences from humans, 65 differences from a gibbon, 49 differences from a rat, and 5 differences from a frog. These data suggest that the new organism is more closely related to _____.
A) Humans than to frogs.
B) Frogs than to humans.
C) Rats than to frogs.
D) Gibbons than to rats.

61) Absorbance at Various pH Levels

Time (sec)	pH 4	pH 5	pH 6	pH 7	pH 8
20	0.003	0.025	0.055	0.146	0.005
40	0.009	0.109	0.152	0.300	0.015
60	0.012	0.195	0.255	0.432	0.038
80	0.020	0.215	0.341	0.552	0.065
100	0.023	0.333	0.413	0.659	0.081
120	0.025	0.360	0.478	0.755	0.090

The table given above represents the results of an experiment where the effects of pH buffers on an enzyme found in saliva (amylase) were studied. A spectrophotometer set at 500nm was used to measure absorbance at the various pH levels every 20 sec for 2 min. The higher absorbance values would indicate greater enzyme activity. All experiments were conducted at the same temperature.
Which statement correctly identifies the result that the optimum pH for amylase function is 7?
A) The pH with the lowest absorbance values would indicate the optimum pH for amylase since this pH does not affect the structure or function of the protein.
B) The pH with the highest absorbance values would indicate the optimum pH for amylase since this pH does not affect the structure or function of the protein.

CHAPTER 6:

A TOUR OF THE CELL

1) The smallest cell structure that would most likely be visible with a standard (not super-resolution) research-grade light microscope is a _____.
A) Mitochondrion.
B) Microtubule.
C) Ribosome.
D) Microfilament.

2) The advantage of light microscopy over electron microscopy is that _____.
A) Light microscopy provides for higher magnification than electron microscopy.
B) Light microscopy provides for higher resolving power than electron microscopy.
C) Light microscopy allows one to view dynamic processes in living cells.
D) Light microscopy provides higher contrast than electron microscopy.

3) In the fractionation of homogenized cells using centrifugation, the primary factor that determines whether a specific cellular component ends up in the supernatant or the pellet is the _____.
A) Relative solubility of the component.
B) Size and weight of the component.
C) Percentage of carbohydrates in the component.
D) Presence or absence of lipids in the component.

4) What is the reason that a modern transmission electron microscope (TEM) can resolve biological images to the subnanometer level, as opposed to tens of nanometers achievable for the best super-resolution light microscope?
A) The focal length of the electron microscope is significantly longer.
B) Contrast is enhanced by staining with atoms of heavy metal.
C) Electron beams have much shorter wavelengths than visible light.
D) The electron microscope has a much greater ratio of image size to real size.

5) What technique would be most appropriate to use to observe the movements of condensed chromosomes during cell division?
A) Standard light microscopy.
B) Scanning electron microscopy.
C) Transmission electron microscopy.
D) Drifting light microscopy.

6) A newspaper ad for a local toy store indicates that an inexpensive toy microscope available for a small child is able to magnify specimens nearly as much as the more costly microscope available in your college lab. What is the primary reason for the price difference?
A) The toy microscope does not have the same fine control for focus of the specimen.
B) The toy microscope magnifies a good deal, but has low resolution and therefore poor quality images.
C) The college microscope produces greater contrast in the specimens.
D) The toy microscope usually uses a different wavelength of light source.

7) All of the following are part of a prokaryotic cell EXCEPT _____.
A) A cell wall.
B) A plasma membrane.
C) Ribosomes.
D) An endoplasmic reticulum.

8) Cell size is limited by _____.
A) The number of proteins within the plasma membrane.
B) The surface area of mitochondria in the cytoplasm.
C) Surface to volume ratios.
D) The size of the endomembrane system.

9) Which of the following is a major difference between prokaryotic cells and eukaryotic cells?
A) Prokaryotes have cells while eukaryotes do not.
B) Eukaryotic cells have more intracellular organelles than prokaryotes.
C) Prokaryotes are not able to carry out aerobic respiration, relying instead on anaerobic metabolism.
D) Prokaryotes are generally larger than eukaryotes.

10) You have a cube of modeling clay in your hands. Which of the following changes to the shape of this cube of clay will decrease its surface area relative to its volume?
A) Pinch the edges of the cube into small folds.
B) Flatten the cube into a pancake shape.
C) Round the clay up into a sphere.
D) Stretch the cube into a long, shoebox shape.

11) Prokaryotes are classified as belonging to two different domains. What are the domains?
A) Bacteria and Eukarya.
B) Bacteria and Archaea.
C) Archaea and Protista.
D) Bacteria and Protista.

12) Which structure is common to plant and animal cells?
A) Chloroplast.
B) Central vacuole.
C) Mitochondrion.
D) Centriole.

13) Which of the following is present in a prokaryotic cell?
A) Mitochondrion.
B) Ribosome.
C) Chloroplast.
D) ER.

14) In a bacterium, we will find DNA in _____.
A) A membrane-enclosed nucleus.
B) Mitochondria.
C) The nucleoid.
D) Ribosomes.

15) Which organelle or structure is absent in plant cells?
A) Mitochondria.
B) Microtubules.
C) Centrosomes.
D) Peroxisomes.

16) What is the function of the nuclear pore complex found in eukaryotes?
A) It regulates the movement of proteins and RNAs into and out of the nucleus.
B) It synthesizes the proteins required to copy DNA and make mRNA.
C) It selectively transports molecules out of the nucleus, but prevents all inbound molecules from entering the nucleus.
D) It assembles ribosomes from raw materials that are synthesized in the nucleus.

17) Which of the following macromolecules leaves the nucleus of a eukaryotic cell through pores in the nuclear membrane?
A) DNA
B) Amino acids
C) mRNA
D) Phospholipids.

18) Which of the following statements correctly describes some aspect of protein secretion from prokaryotic cells?
A) Prokaryotes cannot secrete proteins because they lack an endomembrane system.
B) The mechanism of protein secretion in prokaryotes is probably the same as that in eukaryotes.
C) Proteins secreted by prokaryotes are synthesized on ribosomes bound to the cytoplasmic surface of the plasma membrane.
D) Prokaryotes cannot secrete proteins because they lack ribosomes.

19) Large numbers of ribosomes are present in cells that specialize in producing which of the following molecules?
A) Lipids.
B) Glycogen.
C) Proteins.
D) Nucleic acids.

20) The nuclear lamina is an array of filaments on the inner side of the nuclear membrane. If a method were found that could cause the lamina to fall into disarray, what would you most likely expect to be the immediate consequence?
A) The loss of all nuclear function.
B) The inability of the nucleus to divide during cell division.
C) A change in the shape of the nucleus.
D) Failure of chromosomes to carry genetic information.

21) A cell with a predominance of free ribosomes is most likely _____.
A) Primarily producing proteins for secretion.
B) Primarily producing proteins in the cytosol.
C) Constructing an extensive cell wall or extracellular matrix.
D) Enlarging its vacuole.

22) Which organelle often takes up much of the volume of a plant cell?
A) Lysosome.
B) Vacuole.
C) Golgi apparatus.
D) Peroxisome.

23) A cell with an extensive area of smooth endoplasmic reticulum is specialized to _____.
A) Play a role in storage.
B) Synthesize large quantities of lipids.
C) Actively export protein molecules.
D) Import and export protein molecules.

24) Which structure is NOT part of the endomembrane system?
A) Nuclear envelope.
B) Chloroplast.
C) Golgi apparatus.
D) Plasma membrane.

25) The Golgi apparatus has a polarity, or sidedness, to its structure and function. Which of the following statements correctly describes this polarity?
A) Transport vesicles fuse with one side of the Golgi and leave from the opposite side.
B) Proteins in the membrane of the Golgi may be sorted and modified as they move from one side of the Golgi to the other.
C) Lipids in the membrane of the Golgi may be sorted and modified as they move from one side of the Golgi to the other.
D) All of the listed responses correctly describe polarity characteristics of the Golgi function.

26) The difference in lipid and protein composition between the membranes of the endomembrane system is largely determined by the _____.
A) Transportation of membrane lipids among the membranes of the endomembrane system by small membrane vesicles.
B) Function of the Golgi apparatus in sorting and directing membrane components.
C) Modification of the membrane components once they reach their final destination.
D) Synthesis of different lipids and proteins in each of the organelles of the endomembrane system.

27) Which structure is the site of the synthesis of proteins that may be exported from the cell?
A) Rough ER.
B) Plasmodesmata.
C) Golgi vesicles.
D) Free cytoplasmic ribosomes.

28) Tay-Sachs disease is a human genetic abnormality that results in cells accumulating and becoming clogged with very large, complex, undigested lipids. Which cellular organelle must be involved in this condition?
A) The endoplasmic reticulum.
B) The Golgi apparatus.
C) The lysosome.
D) Mitochondrion.

29) The liver is involved in detoxification of many poisons and drugs. Which of the following structures is primarily involved in this process and, therefore, abundant in liver cells?
A) Rough ER.
B) Smooth ER.
C) Golgi apparatus.
D) Nuclear envelope.

30) Which of the following produces and modifies polysaccharides that will be secreted?
A) Lysosome.
B) Mitochondrion.
C) Golgi apparatus.
D) Peroxisome.

31) What is the most likely pathway taken by a newly synthesized protein that will be secreted by a cell?
A) ER → Golgi → nucleus.
B) Golgi → ER → lysosome.
C) ER → Golgi → vesicles that fuse with plasma membrane.
D) ER → lysosomes → vesicles that fuse with plasma membrane.

32) Asbestos is a material that was once used extensively in construction. One risk from working in a building that contains asbestos is the development of asbestosis caused by the inhalation of asbestos fibers. Cells will phagocytize asbestos, but are not able to degrade it. As a result, asbestos fibers accumulate in _____.
A) Mitochondria.
B) Ribosomes.
C) Peroxisomes.
D) Lysosomes.

33) Which of the following is NOT true? Both chloroplasts and mitochondria _____.
A) Have their own DNA.
B) Have multiple membranes.
C) Are part of the endomembrane system.
D) Are capable of reproducing themselves.

34) Which organelle is the primary site of ATP synthesis in eukaryotic cells?
A) Lysosome.
B) Mitochondrion.
C) Golgi apparatus.
D) Peroxisome.

35) Thylakoids, DNA, and ribosomes are all components found in _____.
A) Chloroplasts.
B) Mitochondria.
C) Lysosomes.
D) Nuclei.

36) In a plant cell, DNA may be found _____.
A) Only in the nucleus.
B) Only in the nucleus and chloroplasts.
C) In the nucleus, mitochondria, and chloroplasts.
D) In the nucleus, mitochondria, chloroplasts, and peroxisomes.

37) In a liver cell detoxifying alcohol and some other poisons, the enzymes of the peroxisome remove hydrogen from these molecules and _____.
A) Combine the hydrogen with water molecules to generate hydrogen peroxide.
B) Use the hydrogen to break down hydrogen peroxide.
C) Transfer the hydrogen to the mitochondria.
D) Transfer the hydrogen to oxygen molecules to generate hydrogen peroxide.

38) The evolution of eukaryotic cells most likely involved _____.
A) Endosymbiosis of an aerobic bacterium in a larger host cell—the endosymbiont evolved into mitochondria.
B) Anaerobic archaea taking up residence inside a larger bacterial host cell to escape toxic oxygen—the anaerobic bacterium evolved into chloroplasts.
C) An endosymbiotic fungal cell evolving into the nucleus.
D) Acquisition of an endomembrane system and subsequent evolution of mitochondria from a portion of the Golgi.

39) Where are proteins produced other than on ribosomes free in the cytosol or ribosomes attached to the ER?
A) In the extracellular matrix.
B) In the Golgi apparatus.
C) In mitochondria.
D) In the nucleolus.

40) Suppose a cell has the following molecules and structures: enzymes, DNA, ribosomes, plasma membrane, and mitochondria. It could be a cell from _____.
A) A bacterium.
B) An animal but not a plant.
C) Nearly any eukaryotic organism.
D) A plant but not an animal.

41) Cyanide binds with at least one molecule involved in producing ATP. If a cell is exposed to cyanide, most of the cyanide will be found within the _____.
A) Mitochondria.
B) Peroxisomes.
C) Lysosomes.
D) Endoplasmic reticulum.

42) Suppose a young boy is always tired and fatigued, suffering from a metabolic disease. Which of the following organelles is most likely involved in this disease?
A) Lysosomes.
B) Golgi apparatus.
C) Ribosomes.
D) Mitochondria.

43) Motor proteins provide for molecular motion in cells by interacting with what types of cellular structures?
A) Membrane proteins of the inner nuclear envelope.
B) Free ribosomes and ribosomes attached to the ER.
C) Components of the cytoskeleton.
D) Cellulose fibers in the cell wall.

44) Which of the following contain the 9 + 2 arrangement of microtubules, consisting of nine doublets of microtubules surrounding a pair of single microtubules?
A) Motile cilia and primary (nonmotile) cilia.
B) Flagella and motile cilia.
C) Basal bodies and primary (nonmotile) cilia.
D) Centrioles and basal bodies.

45) Vinblastine, a drug that inhibits microtubule polymerization, is used to treat some forms of cancer. Cancer cells given vinblastine would be unable to _____.
A) Form cleavage furrows during cell division.
B) Migrate by amoeboid movement.
C) Separate chromosomes during cell division.
D) Maintain the shape of the nucleus.

46) Amoebae move by crawling over a surface (cell crawling), which involves _____.
A) Growth of actin filaments to form bulges in the plasma membrane.
B) Setting up microtubule extensions that vesicles can follow in the movement of cytoplasm.
C) Reinforcing the pseudopod with intermediate filaments.
D) Cytoplasmic streaming.

47) Researchers tried to explain how vesicular transport occurs in cells by attempting to assemble the transport components. They set up microtubular tracks along which vesicles could be transported, and they added vesicles and ATP (because they knew the transport process requires energy). Yet, when they put everything together, there was no movement or transport of vesicles. What were they missing?
A) An axon.
B) Contractile microfilaments.
C) Endoplasmic reticulum.
D) Motor proteins.

48) Cilia and flagella bend because of _____.
A) Conformational changes in ATP that thrust microtubules laterally.
B) A motor protein called radial spokes.
C) The quick inward movements of water by osmosis.
D) A motor protein called dynein.

49) Spherocytosis is a human blood disorder associated with a defective cytoskeletal protein in the red blood cells (RBCs). What do you suspect is the consequence of such a defect?
A) Abnormally shaped RBCs.
B) An insufficient supply of ATP in the RBCs.
C) An insufficient supply of oxygen-transporting proteins in the RBCs.
D) Adherence of RBCs to blood vessel walls, causing plaque formation.

50) Cytochalasin D is a drug that prevents actin polymerization. A cell treated with cytochalasin D will still be able to _____.
A) Divide in two.
B) Contract muscle fibers.
C) Extend pseudopodia.
D) Move vesicles within a cell.

51) Cells require which of the following to form cilia or flagella?
A) Tubulin.
B) Laminin.
C) Actin.
D) Intermediate filaments.

52) Which of the following statements about the cytoskeleton is true?
A) The cytoskeleton of eukaryotes is a static structure most resembling scaffolding used at construction sites.
B) Although microtubules are common within a cell, actin filaments are rarely found outside of the nucleus.
C) Movement of cilia and flagella is the result of motor proteins causing microtubules to move relative to each other.
D) Chemicals that block the assembly of the cytoskeleton would have little effect on a cell's response to external stimuli.

53) The cell walls of bacteria, fungi, and plant cells and the extracellular matrix of animal cells are all external to the plasma membrane. Which of the following is a characteristic common to all of these extracellular structures?
A) They must block water and small molecules to regulate the exchange of matter and energy with their environment.
B) They must provide a rigid structure that maintains an appropriate ratio of cell surface area to volume.
C) They are constructed of polymers that are synthesized in the cytoplasm and then transported out of the cell.
D) They are composed of a mixture of lipids and nucleotides.

54) A mutation that disrupts the ability of an animal cell to add polysaccharide modifications to proteins would most likely cause defects in its _____.
A) Nuclear matrix and extracellular matrix.
B) Mitochondria and Golgi apparatus.
C) Golgi apparatus and extracellular matrix.
D) Nuclear pores and secretory vesicles.

55) The extracellular matrix is thought to participate in the regulation of animal cell behavior by communicating information from the outside to the inside of the cell via which of the following?
A) Gap junctions.
B) The nucleus.
C) DNA and RNA.
D) Integrins.

56) Plasmodesmata in plant cells are most similar in function to which of the following structures in animal cells?
A) Desmosomes.
B) Gap junctions.
C) Extracellular matrix.
D) Tight junctions.

57) Ions can travel directly from the cytoplasm of one animal cell to the cytoplasm of an adjacent cell through _____.
A) Plasmodesmata.
B) Tight junctions.
C) Desmosomes.
D) Gap junctions.

58) In plant cells, the middle lamella _____.
A) Allows adjacent cells to adhere to one another.
B) Prevents dehydration of adjacent cells.
C) Maintains the plant's circulatory system.
D) Allows for gas and nutrient exchange among adjacent cells.

59) Where would you expect to find tight junctions?
A) In the epithelium of an animal's stomach.
B) Between the smooth endoplasmic reticulum and the rough endoplasmic reticulum.
C) Between plant cells in a woody plant.
D) In the plasma membrane of prokaryotes.

60) H. V. Wilson worked with sponges to gain some insight into exactly what was responsible for holding adjacent cells together. He exposed two species of differently pigmented sponges to a chemical that disrupted the cell-cell interaction (cell junctions), and the cells of the sponges dissociated. Wilson then mixed the cells of the two species and removed the chemical that caused the cells to dissociate. Wilson found that the sponges reassembled into two separate species. The cells from one species did not interact or form associations with the cells of the other species. How do you explain the results of Wilson's experiments?
A) The two species of sponge had different enzymes that functioned in the reassembly process.
B) The molecules responsible for cell-cell adhesion (cell junctions) were irreversibly destroyed during the experiment.
C) The molecules responsible for cell-cell adhesion (cell junctions) differed between the two species of sponge.
D) One cell functioned as the nucleus for each organism, thereby attracting only cells of the same pigment.

61) Gaucher disease is the most common of lipid storage diseases in humans. It is caused by a deficiency of an enzyme necessary for lipid metabolism. This leads to a collection of fatty material in organs of the body including the spleen, liver, kidneys, lungs, brain, and bone marrow.

Using your knowledge of the structure of eukaryotic cells, identify the statement below that best explains how internal membranes and the organelles of cells would be involved in Gaucher disease.
A) The mitochondria are most likely defective and do not produce adequate amounts of ATP needed for cellular respiration.
B) The rough endoplasmic reticulum contains too many ribosomes which results in an overproduction of the enzyme involved in carbohydrate catalysis.
C) The lysosomes lack sufficient amounts of enzymes necessary for the metabolism of lipids.
D) The Golgi apparatus produces vesicles with faulty membranes that leak their contents into the cytoplasm of the cell.

62) Both the volume and the surface area for three different cells were measured. These values are listed in the following table:

	Volume	Surface Area
Cell 1	9.3 μm3	26.5 μm2
Cell 2	12.2 μm3	37.1 μm2
Cell 3	17.6 μm3	40.6 μm2

Using data from the table above, select the best explanation for why that cell will be able to eliminate waste most efficiently?

A) Cell 1 since it has the smallest volume and will not produce as much waste as the other cells.

B) Cell 2 since it has the highest surface area -to-volume ratio which facilitates the exchange of materials between a cell and its environment.

C) Cell 3 since it has the largest surface area which will enable it to eliminate all of its wastes quickly.

D) Cell 3 because it is big enough to allow wastes to easily diffuse through the plasma membrane.

CHAPTER 7:

MEMBRANE STRUCTURE AND FUNCTION

1) Integral membrane proteins are _____.
A) Hydrophilic.
B) Hydrophobic.
C) Amphipathic, with at least one hydrophobic region.
D) Exposed on only one surface of the membrane.

2) You have a planar bilayer with equal amounts of saturated and unsaturated phospholipids. After testing the permeability of this membrane to glucose, you increase the proportion of unsaturated phospholipids in the bilayer. What will happen to the membrane's permeability to glucose?
A) Permeability to glucose will increase.
B) Permeability to glucose will decrease.
C) Permeability to glucose will stay the same.
D) You cannot predict the outcome. You simply have to make the measurement.

3) According to the fluid mosaic model of cell membranes, phospholipids _____.
A) Can move laterally along the plane of the membrane.
B) Frequently flip-flop from one side of the membrane to the other.
C) Occur in an uninterrupted bilayer, with membrane proteins restricted to the surface of the membrane.
D) Have hydrophilic tails in the interior of the membrane.

4) The membranes of winter wheat are able to remain fluid when it is extremely cold by _____.
A) Increasing the percentage of unsaturated phospholipids in the membrane.
B) Increasing the percentage of cholesterol molecules in the membrane.
C) Decreasing the number of hydrophobic proteins in the membrane.
D) Cotransport of glucose and hydrogen.

5) Some regions of the plasma membrane, called lipid rafts, have a higher concentration of cholesterol molecules. At higher temperatures, these regions _____.
A) Are more fluid than the surrounding membrane.
B) Are less fluid than the surrounding membrane.
C) Detach from the plasma membrane and clog arteries.
D) Have higher rates of lateral diffusion of lipids and proteins into and out of these regions.

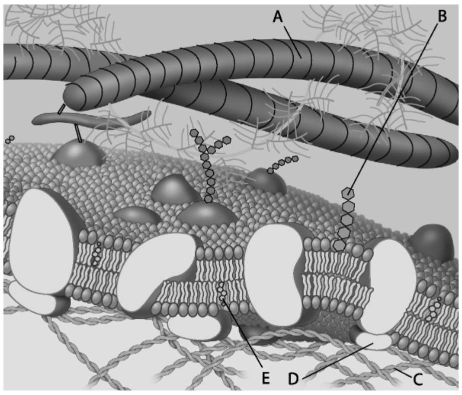
6) Singer and Nicolson's fluid mosaic model of the membrane proposed that membranes _____.
A) Are a phospholipid bilayer between two layers of hydrophilic proteins.
B) Are a single layer of phospholipids and proteins.
C) Consist of protein molecules embedded in a fluid bilayer of phospholipids.
D) Consist of a mosaic of polysaccharides and proteins.

7) An animal cell lacking oligosaccharides on the external surface of its plasma membrane would likely be impaired in which function?
A) Transporting ions against an electrochemical gradient.
B) Cell-cell recognition.
C) Attaching the plasma membrane to the cytoskeleton.
D) Establishing a diffusion barrier to charged molecules.

8) Which of these are NOT embedded in the hydrophobic portion of the lipid bilayer at all?
A) Transmembrane proteins.
B) Integral proteins.
C) Peripheral proteins.
D) All of these are embedded in the hydrophobic portion of the lipid bilayer.

9) Why are lipids and proteins free to move laterally in membranes?
A) The interior of the membrane is filled with liquid water.
B) Lipids and proteins repulse each other in the membrane.
C) Hydrophilic portions of the lipids are in the interior of the membrane.
D) There are only weak hydrophobic interactions in the interior of the membrane.

For the following questions (10-14), match the labeled component of the cell membrane in the figure with its description.



10) Which component is a peripheral protein?
A) A.
B) B.
C) C.
D) D.

11) Which component is cholesterol?
A) B.
B) C.
C) D.
D) E.

12) Which component is a protein fiber of the extracellular matrix?
A) A.
B) B.
C) C.
D) E.

13) Which component is a microfilament (actin filament) of the cytoskeleton?
A) A.
B) B.
C) C.
D) D.

14) Which component is a glycolipid?
A) A.
B) B.
C) C.
D) E.

15) Cell membranes are asymmetrical. Which of the following statements is the most likely explanation for the membrane's asymmetrical nature?
A) Since the cell membrane forms a border between one cell and another in tightly packed tissues such as epithelium, the membrane must be asymmetrical.
B) Since cell membranes communicate signals from one organism to another, the cell membranes must be asymmetrical.
C) The two sides of a cell membrane face different environments and carry out different functions.
D) Proteins only function on the cytoplasmic side of the cell membrane, which results in the membrane's asymmetrical nature.

16) In what way do the membranes of a eukaryotic cell vary?
A) Phospholipids are found only in certain membranes.
B) Certain proteins are unique to each membrane.
C) Only certain membranes of the cell are selectively permeable.
D) Some membranes have hydrophobic surfaces exposed to the cytoplasm, while others have hydrophilic surfaces facing the cytoplasm.

17) Which of the following is a reasonable explanation for why unsaturated fatty acids help keep a membrane more fluid at lower temperatures?
A) The double bonds form kinks in the fatty acid tails, preventing adjacent lipids from packing tightly.
B) Unsaturated fatty acids have a higher cholesterol content and, therefore, more cholesterol in membranes.
C) Unsaturated fatty acids are more polar than saturated fatty acids.
D) The double bonds block interaction among the hydrophilic head groups of the lipids.

18) What kinds of molecules pass through a cell membrane most easily?
A) Large and hydrophobic.
B) Small and hydrophobic.
C) Large polar.
D) Ionic.

19) Which of the following most accurately describes selective permeability?
A) An input of energy is required for transport.
B) Lipid-soluble molecules pass through a membrane.
C) There must be a concentration gradient for molecules to pass through a membrane.
D) Only certain molecules can cross a cell membrane.

20) Which of the following is a characteristic feature of a carrier protein in a plasma membrane?
A) It exhibits a specificity for a particular type of molecule.
B) It requires the expenditure of cellular energy to function.
C) It works against diffusion.
D) It has no hydrophobic regions.

21) Which of the following would likely move through the lipid bilayer of a plasma membrane most rapidly?
A) CO_2.
B) An amino acid.
C) Glucose.
D) K^+.

22) Which of the following allows water to move much faster across cell membranes?
A) The sodium-potassium pump.
B) ATP.
C) Peripheral proteins.
D) Aquaporins.

23) You are working on a team that is designing a new drug. For this drug to work, it must enter the cytoplasm of specific target cells. Which of the following would be a factor that determines whether the molecule selectively enters the target cells?
A) Hydrophobicity of the drug molecule.
B) Lack of charge on the drug molecule.
C) Similarity of the drug molecule to other molecules transported by the target cells.
D) Lipid composition of the target cells' plasma membrane.

24) Diffusion _____.
A) Is very rapid over long distances.
B) Requires an expenditure of energy by the cell.
C) Is a passive process in which molecules move from a region of higher concentration to a region of lower concentration.
D) Requires integral proteins in the cell membrane.

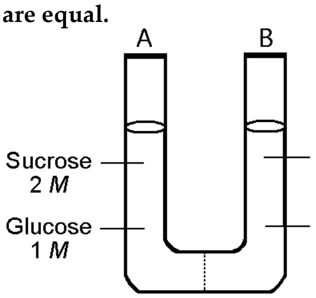
25) Which of the following processes includes all others?
A) Osmosis.
B) Facilitated diffusion.
C) Passive transport.
D) Transport of an ion down its electrochemical gradient.

26) When a cell is in equilibrium with its environment, which of the following occurs for substances that can diffuse through the cell?
A) There is random movement of substances into and out of the cell.
B) There is directed movement of substances into and out of the cell.
C) There is no movement of substances into and out of the cell.
D) All movement of molecules is directed by active transport.

27) Which of the following is true of osmosis?
A) Osmosis only takes place in red blood cells.
B) Osmosis is an energy-demanding or "active" process.
C) In osmosis, water moves across a membrane from areas of lower solute concentration to areas of higher solute concentration.
D) In osmosis, solutes move across a membrane from areas of lower water concentration to areas of higher water concentration.

Use the information and figure given below to answer the following questions (28-29).

The solutions in the two arms of this U-tube are separated by a membrane that is permeable to water and glucose but not to sucrose. Side A is half-filled with a solution of 2 *M* sucrose and 1 *M* glucose. Side B is half-filled with 1 *M* sucrose and 2 *M* glucose. Initially, the liquid levels on both sides are equal.



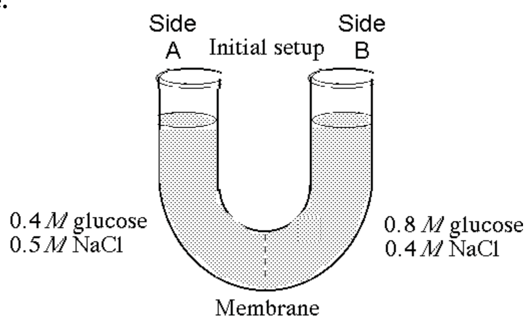
28) Refer to the figure. Initially, in terms of tonicity, the solution in side A with respect to the solution in side B is _____.
A) Hypotonic.
B) Isotonic.
C) Saturated.
D) Hypertonic.

29) Refer to the figure. After the system reaches equilibrium, what changes are observed?
A) The molarity of sucrose is higher than that of glucose on side A.
B) The water level is higher in side A than in side B.
C) The water level is unchanged.
D) The water level is higher in side B than in side A.

30) A patient was involved a serious accident and lost a large quantity of blood. In an attempt to replenish body fluids, distilled water—equal to the volume of blood lost—is added to the blood directly via one of his veins. What will be the most probable result of this transfusion?
A) The patient's red blood cells will shrivel up because the blood has become hypotonic compared to the cells.
B) The patient's red blood cells will swell and possibly burst because the blood has become hypotonic compared to the cells.
C) The patient's red blood cells will shrivel up because the blood has become hypertonic compared to the cells.
D) The patient's red blood cells will burst because the blood has become hypertonic compared to the cells.

Use the information and figure given below to answer the following questions (31-32).

The solutions in the arms of a U-tube are separated at the bottom of the tube by a selectively permeable membrane. The membrane is permeable to sodium chloride but not to glucose. Side A is filled with a solution of 0.4 *M* glucose and 0.5 *M* sodium chloride (NaCl), and side B is filled with a solution containing 0.8 *M* glucose and 0.4 *M* sodium chloride. Initially, the volume in both arms is the same.

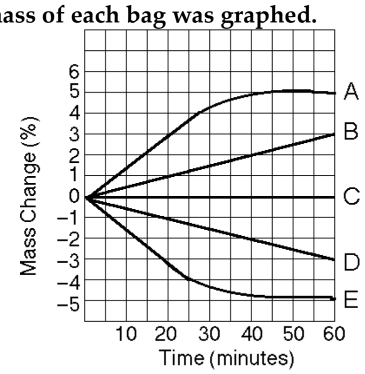


31) Refer to the figure. At the beginning of the experiment…
A) Side A is hypertonic to side B.
B) Side A is hypotonic to side B.
C) Side A is hypertonic to side B with respect to glucose.
D) Side A is hypotonic to side B with respect to NaCl.

32) Refer to the figure. If you examine side A after three days, you should find _____.
A) A decrease in the concentration of NaCl and glucose and an increase in the water level.
B) A decrease in the concentration of NaCl, an increase in water level, and no change in the concentration of glucose.
C) A decrease in the concentration of NaCl and a decrease in the water level.
D) No change in the concentration of NaCl and glucose and an increase in the water level.

Use the information and figure given below to answer the following questions (33-35).

Five dialysis bags constructed of membrane, which is permeable to water and impermeable to sucrose, were filled with various concentrations of sucrose and then placed in separate beakers containing an initial concentration of 0.6 *M* sucrose solution. At 10-minute intervals, the bags were massed (weighed) and the percent change in mass of each bag was graphed.



33) Which line in the graph represents the bag that contained a solution isotonic to the 0.6 *M* solution at the beginning of the experiment?
A) A.
B) B.
C) C.
D) D.

34) Which line in the graph represents the bag with the highest initial concentration of sucrose?
A) A.
B) B.
C) C.
D) D.

35) Which line or lines in the graph represent(s) bags that contain a solution that is hypertonic at 50 minutes?
A) A and B.
B) B.
C) D.
D) D and E.

36) Celery stalks that are immersed in fresh water for several hours become stiff. Similar stalks left in a 0.15 *M* salt solution become limp. From this we can deduce that the fresh water _____.
A) And the salt solutions are both hypertonic to the cells of the celery stalks.
B) Is hypotonic and the salt solution is hypertonic to the cells of the celery stalks.
C) Is hypertonic and the salt solution is hypotonic to the cells of the celery stalks.
D) Is isotonic and the salt solution is hypertonic to the cells of the celery stalks.

37) What will happen to a red blood cell (RBC), which has an internal ion concentration of about 0.9 percent, if it is placed into a beaker of pure water?
A) The cell would shrink because the water in the beaker is hypotonic relative to the cytoplasm of the RBC.
B) The cell would shrink because the water in the beaker is hypertonic relative to the cytoplasm of the RBC.
C) The cell would swell because the water in the beaker is hypotonic relative to the cytoplasm of the RBC.
D) The cell will remain the same size because the solution outside the cell is isotonic.

38) Which of the following statements correctly describes the normal tonicity conditions for typical plant and animal cells? The animal cell is in _____.
A) A hypotonic solution, and the plant cell is in an isotonic solution.
B) An isotonic solution, and the plant cell is in a hypertonic solution.
C) A hypertonic solution, and the plant cell is in an isotonic solution.
D) An isotonic solution, and the plant cell is in a hypotonic solution.

39) In which of the following would there be the greatest need for osmoregulation?
A) An animal connective tissue cell bathed in isotonic body fluid.
B) A salmon moving from a river into an ocean.
C) A red blood cell surrounded by plasma.
D) A plant being grown hydroponically in a watery mixture of designated nutrients.

40) When a plant cell, such as one from a rose stem, is submerged in a very hypotonic solution, what is likely to occur?
A) The cell will burst.
B) Plasmolysis will shrink the interior.
C) The cell will become flaccid.
D) The cell will become turgid.

41) A sodium-potassium pump _____.
A) Moves three potassium ions out of a cell and two sodium ions into a cell while producing an ATP for each cycle.
B) Move three sodium ions out of a cell and two potassium ions into a cell while consuming an ATP for each cycle.
C) Moves three potassium ions out of a cell and two sodium ions into a cell while consuming 2 ATP in each cycle.
D) Move three sodium ions out of a cell and two potassium ions into a cell and generates an ATP in each cycle.

42) The sodium-potassium pump is called an electrogenic pump because it _____.
A) Pumps equal quantities of Na+ and K+ across the membrane.
B) Contributes to the membrane potential.
C) Ionizes sodium and potassium atoms.
D) Is used to drive the transport of other molecules against a concentration gradient.

43) Which of the following membrane activities requires energy from ATP?
A) Facilitated diffusion of chloride ions across the membrane through a chloride channel.
B) Movement of Na+ ions from a lower concentration in a mammalian cell to a higher concentration in the extracellular fluid.
C) Movement of glucose molecules into a bacterial cell from a medium containing a higher concentration of glucose than inside the cell.
D) Movement of carbon dioxide out of a paramecium.

44) The voltage across a membrane is called the _____.
A) Chemical gradient.
B) Membrane potential.
C) Osmotic potential.
D) Electrochemical gradient.

45) Ions diffuse across membranes through specific ion channels down _____.
A) Their chemical gradients.
B) Their concentration gradients.
C) The electrical gradients.
D) Their electrochemical gradients.

46) Which of the following would increase the electrochemical gradient across a membrane?
A) A sucrose-proton cotransporter.
B) A proton pump.
C) A potassium channel.
D) Both a proton pump and a potassium channel.

47) The phosphate transport system in bacteria imports phosphate into the cell even when the concentration of phosphate outside the cell is much lower than the cytoplasmic phosphate concentration. Phosphate import depends on a pH gradient across the membrane—more acidic outside the cell than inside the cell. Phosphate transport is an example of _____.
A) Passive diffusion.
B) Facilitated diffusion.
C) Active transport.
D) Cotransport.

48) In some cells, there are many ion electrochemical gradients across the plasma membrane even though there are usually only one or two proton pumps present in the membrane. The gradients of the other ions are most likely accounted for by _____.
A) Cotransport proteins.
B) Ion channels.
C) Pores in the plasma membrane.
D) Passive diffusion across the plasma membrane.

49) Which of the following is most likely true of a protein that cotransports glucose and sodium ions into the intestinal cells of an animal?
A) Sodium and glucose compete for the same binding site in the cotransporter.
B) Glucose entering the cell down its concentration gradient provides energy for uptake of sodium ions against the electrochemical gradient.
C) Sodium ions can move down their electrochemical gradient through the cotransporter whether or not glucose is present outside the cell.
D) A substance that blocks sodium ions from binding to the cotransport protein will also block the transport of glucose.

50) Proton pumps are used in various ways by members of every domain of organisms: Bacteria, Archaea, and Eukarya. What does this most probably mean?
A) Proton gradients across a membrane were used by cells that were the common ancestor of all three domains of life.
B) The high concentration of protons in the ancient atmosphere must have necessitated a pump mechanism.
C) Cells of each domain evolved proton pumps independently when oceans became more acidic.
D) Proton pumps are necessary to all cell membranes.

51) Several epidemic microbial diseases of earlier centuries incurred high death rates because they resulted in severe dehydration due to vomiting and diarrhea. Today they are usually not fatal because we have developed which of the following?
A) Antiviral medications that are efficient and work well with most viruses.
B) Intravenous feeding techniques.
C) Medications to slow blood loss.
D) Hydrating drinks with high concentrations of salts and glucose.

52) The force driving simple diffusion is _____, while the energy source for active transport is _____.
A) The concentration gradient; ADP.
B) The concentration gradient; ATP.
C) Transmembrane pumps; electron transport.
D) Phosphorylated protein carriers; ATP.

53) An organism with a cell wall would most likely be unable to take in materials through _____.
A) Osmosis.
B) Active transport.
C) Phagocytosis.
D) Facilitated diffusion.

54) White blood cells engulf bacteria using _____.
A) Phagocytosis.
B) Pinocytosis.
C) Osmosis.
D) Receptor-mediated exocytosis.

55) Familial hypercholesterolemia is characterized by _____.
A) Defective LDL receptors on the cell membranes.
B) Poor attachment of the cholesterol to the extracellular matrix of cells.
C) A poorly formed lipid bilayer that cannot incorporate cholesterol into cell membranes.
D) Inhibition of the cholesterol active transport system in red blood cells.

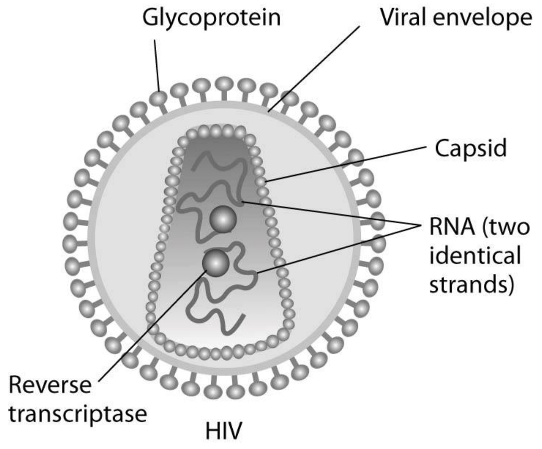
56) The difference between pinocytosis and receptor-mediated endocytosis is that _____.
A) Pinocytosis brings only water molecules into the cell, but receptor-mediated endocytosis brings in other molecules as well.
B) Pinocytosis increases the surface area of the plasma membrane, whereas receptor-mediated endocytosis decreases the plasma membrane surface area.
C) Pinocytosis is nonselective in the molecules it brings into the cell, whereas receptor-mediated endocytosis offers more selectivity.
D) Pinocytosis can concentrate substances from the extracellular fluid, but receptor-mediated endocytosis cannot.

57) In receptor-mediated endocytosis, receptor molecules initially project to the outside of the cell. Where do they end up after endocytosis?
A) On the outside of vesicles.
B) On the inside surface of the cell membrane.
C) On the inside surface of the vesicle.
D) On the outer surface of the nucleus.

58) A bacterium engulfed by a white blood cell through phagocytosis will be digested by enzymes contained in _____.
A) Lysosomes.
B) Golgi vesicles.
C) Vacuoles.
D) Secretory vesicles.

Use the paragraph and accompanying figure to answer the following questions (59-60).

Human immunodeficiency virus (HIV) infects cells that have both CD4 and CCR5 cell surface molecules. The viral nucleic acid molecules are enclosed in a protein capsid, and the protein capsid is itself contained inside an envelope consisting of a lipid bilayer membrane and viral glycoproteins. One hypothesis for viral entry into cells is that binding of HIV membrane glycoproteins to CD4 and CCR5 initiates fusion of the HIV membrane with the plasma membrane, releasing the viral capsid into the cytoplasm. An alternative hypothesis is that HIV gains entry into the cell via receptor -mediated endocytosis, and membrane fusion occurs in the endocytotic vesicle. To test these alternative hypotheses for HIV entry, researchers labeled the lipids on the HIV membrane with a red fluorescent dye.



59) In an HIV-infected cell producing HIV virus particles, the viral glycoprotein is expressed on the plasma membrane. How do the viral glycoproteins get to the plasma membrane? They are synthesized _____.
A) On ribosomes on the plasma membrane.
B) By ribosomes in the rough ER and arrive at the plasma membrane in the membrane of secretory vesicles.
C) On free cytoplasmic ribosomes and then inserted into the plasma membrane.
D) By ribosomes in the rough ER, secreted from the cell, and inserted into the plasma membrane from the outside

60) What would be observed by live-cell fluorescence microscopy immediately after HIV entry if HIV is endocytosed first, and then later fuses with the endocytotic vesicle membrane?
A) A spot of red fluorescence will be visible on the infected cell's plasma membrane, marking the site of membrane fusion and HIV entry.
B) The red fluorescent dye-labeled lipids will appear in the infected cell's interior.
C) A spot of red fluorescence will diffuse in the infected cell's cytoplasm.
D) A spot of red fluorescence will remain outside the cell after delivering the viral capsid.

CHAPTER 8:

AN INTRODUCTION TO METABOLISM

1) Which of the following is true of metabolism in its entirety in all organisms?
A) Metabolism depends on a constant supply of energy from food.
B) Metabolism uses all of an organism's resource.
C) Metabolism consists of all the energy transformation reactions in an organism.
D) Metabolism manages the increase of entropy in an organism.

2) Which of the following is an example of potential rather than kinetic energy?
A) Water rushing over Niagara Falls.
B) Light flashes emitted by a firefly.
C) A molecule of glucose.
D) A crawling beetle foraging for food.

3) Most cells cannot harness heat to perform work because _____.
A) Heat is not a form of energy.
B) Temperature is usually uniform throughout a cell.
C) Heat can never be used to do work.
D) Heat must remain constant during work.

4) Which of the following involves a decrease in entropy?
A) Condensation reactions.
B) Reactions that separate monomers.
C) Depolymerization reactions.
D) Hydrolysis reactions.

5) Which term most precisely describes the cellular process of breaking down large molecules into smaller ones?
A) Catabolism (catabolic pathways).
B) Metabolism.
C) Anabolism (anabolic pathways).
D) Dehydration.

6) Anabolic pathways _____.
A) Are usually highly spontaneous chemical reactions.
B) Consume energy to build up polymers from monomers.
C) Release energy as they degrade polymers to monomers.
D) Consume energy to decrease the entropy of the organism and its environment.

7) Which of the following is a statement of the first law of thermodynamics?
A) Energy cannot be created or destroyed.
B) The entropy of the universe is decreasing.
C) The entropy of the universe is constant.
D) Energy cannot be transferred or transformed.

8) For living organisms, which of the following is an important consequence of the first law of thermodynamics?
A) The energy content of an organism is constant.
B) The organism ultimately must obtain all of the necessary energy for life from its environment.
C) The entropy of an organism decreases with time as the organism grows in complexity.
D) Organisms grow by converting energy into organic matter.

9) Living organisms increase in complexity as they grow, resulting in a decrease in the entropy of an organism. How does this relate to the second law of thermodynamics?
A) Living organisms do not obey the second law of thermodynamics, which states that entropy must increase with time.
B) Life obeys the second law of thermodynamics because the decrease in entropy as the organism grows is exactly balanced by an increase in the entropy of the universe.
C) As a consequence of growing, organisms cause a greater increase in entropy in their environment than the decrease in entropy associated with their growth.
D) Living organisms are able to transform energy into entropy.

10) Which of the following statements is a logical consequence of the second law of thermodynamics?
A) If the entropy of a system increases, there must be a corresponding decrease in the entropy of the universe.
B) If there is an increase in the energy of a system, there must be a corresponding decrease in the energy of the rest of the universe.
C) Every chemical reaction must increase the total entropy of the universe.
D) Energy can be transferred or transformed, but it cannot be created or destroyed.

11) Which of the following statements is representative of the second law of thermodynamics?
A) Conversion of energy from one form to another is always accompanied by some gain of free energy.
B) Without an input of energy, organisms would tend toward decreasing entropy.
C) Cells require a constant input of energy to maintain their high level of organization.
D) Every energy transformation by a cell decreases the entropy of the universe.

12) Which of the following types of reactions would decrease the entropy within a cell?
A) Anabolic reactions.
B) Hydrolysis.
C) Digestion.
D) Catabolic reactions.

13) Biological evolution of life on Earth, from simple prokaryote-like cells to large, multicellular eukaryotic organisms, _____.
A) Has occurred in accordance with the laws of thermodynamics.
B) Has caused an increase in the entropy of the planet.
C) Has been made possible by expending Earth's energy resources.
D) Has occurred in accordance with the laws of thermodynamics, by expending Earth's energy resources and causing an increase in the entropy of the planet.

14) The mathematical expression for the change in free energy of a system is $\Delta G = \Delta H - T\Delta S$. Which of the following is (are) correct?
A) ΔS is the change in enthalpy, a measure of randomness.
B) ΔH is the change in entropy, the energy available to do work.
C) ΔG is the change in free energy.
D) T is the temperature in degrees Celsius.

15) A system at chemical equilibrium _____.
A) Consumes energy at a steady rate.
B) Releases energy at a steady rate.
C) Has zero kinetic energy.
D) Can do no work.

16) Which of the following is true for all exergonic reactions?
A) The products have more total energy than the reactants.
B) The reaction proceeds with a net release of free energy.
C) The reaction goes only in a forward direction: all reactants will be converted to products, but no products will be converted to reactants.
D) A net input of energy from the surroundings is required for the reactions to proceed.

17) A chemical reaction that has a positive ΔG is best described as _____.
A) Endergonic.
B) Enthalpic.
C) Spontaneous.
D) Exergonic.

18) Chemical equilibrium is relatively rare in living cells. An example of a reaction at chemical equilibrium in a cell would be _____.
A) One in which the free energy at equilibrium is higher than the energy content at any point away from equilibrium.
B) One in which the entropy change in the reaction is just balanced by an opposite entropy change in the cell's surroundings.
C) An endergonic reaction in an active metabolic pathway where the energy for that reaction is supplied only by heat from the environment.
D) A chemical reaction in which both the reactants and products are not being produced or used in any active metabolic pathway at that time in the cell.

19) Choose the pair of terms that correctly completes this sentence: Catabolism is to anabolism as _____ is to _____.
A) Exergonic; spontaneous.
B) Exergonic; endergonic.
C) Free energy; entropy.
D) Work; energy.

20) In solution, why do hydrolysis reactions occur more readily than condensation reactions?
A) Hydrolysis increases entropy and is exergonic.
B) Hydrolysis raises G, or Gibbs free energy.
C) Hydrolysis decreases entropy and is exergonic.
D) Hydrolysis increases entropy and is endergonic.

21) Why is ATP an important molecule in metabolism?
A) Its hydrolysis provides an input of free energy for exergonic reactions.
B) It provides energy coupling between exergonic and endergonic reactions.
C) Its terminal phosphate group contains a strong covalent bond that, when hydrolyzed, releases free energy.
D) Its terminal phosphate bond has higher energy than the other two phosphate bonds.

22) When 10,000 molecules of ATP are hydrolyzed to ADP and Pi in a test tube, about half as much heat is liberated as when a cell hydrolyzes the same amount of ATP. Which of the following is the best explanation for this observation?
A) Cells are open systems, but a test tube is an isolated system.
B) Cells are less efficient at heat production than nonliving systems.
C) The reaction in cells must be catalyzed by enzymes, but the reaction in a test tube does not need enzymes.
D) Reactant and product concentrations in the test tube are different from those in the cell.

23) Which of the following is most similar in structure to ATP?
A) A pentose sugar.
B) A DNA nucleotide.
C) An RNA nucleotide.
D) An amino acid with three phosphate groups attached.

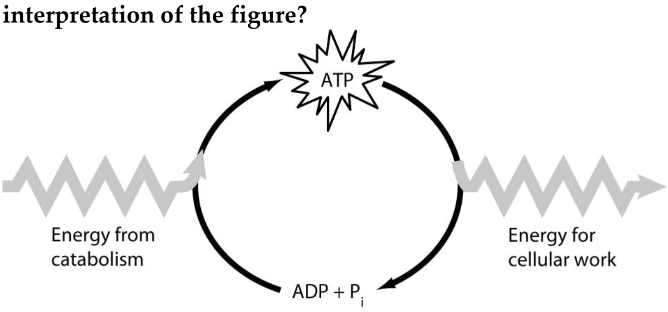
24) Catabolic pathways _____.
A) Combine molecules into more energy-rich molecules.
B) Supply energy, primarily in the form of ATP, for the cell's work.
C) Are endergonic.
D) Are spontaneous and do not need enzyme catalysis.

25) When chemical, transport, or mechanical work is done by an organism, what happens to the heat generated?
A) It is used to power yet more cellular work.
B) It is used to store energy as more ATP.
C) It is used to generate ADP from nucleotide precursors.
D) It is lost to the environment.

26) When ATP releases some energy, it also releases inorganic phosphate. What happens to the inorganic phosphate in the cell?
A) It is secreted as waste.
B) It is used only to regenerate more ATP.
C) It may be used to form a phosphorylated intermediate.
D) It enters the nucleus and affects gene expression.

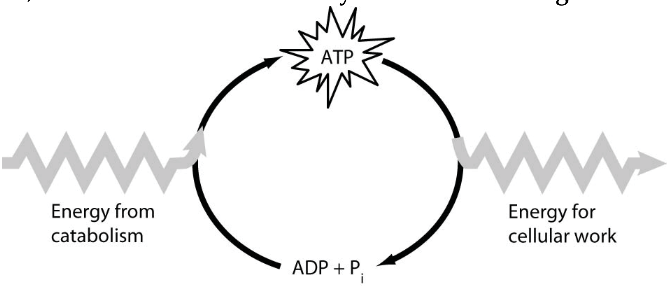
27) A number of systems for pumping ions across membranes are powered by ATP. Such ATP-powered pumps are often called ATPases, although they do not often hydrolyze ATP unless they are simultaneously transporting ions. Because small increases in calcium ions in the cytosol can trigger a number of different intracellular reactions, cells keep the cytosolic calcium concentration quite low under normal conditions, using ATP-powered calcium pumps. For example, muscle cells transport calcium from the cytosol into the membranous system called the sarcoplasmic reticulum (SR). If a resting muscle cell's cytosol has a free calcium ion concentration of 10^{-7} while the concentration in the SR is 10^{-2}, then how is the ATPase acting?
A) ATPase activity must be powering an inflow of calcium from the outside of the cell into the SR.
B) ATPase activity must be transferring Pi to the SR to enable this to occur.
C) ATPase activity must be pumping calcium from the cytosol to the SR against the concentration gradient.
D) ATPase activity must be opening a channel for the calcium ions to diffuse back into the SR along the concentration gradient.

28) Which of the following is the most correct interpretation of the figure?



A) Energy from catabolism can be used directly for performing cellular work.
B) ADP + Pi are a set of molecules that store energy for catabolism.
C) ATP is a molecule that acts as an intermediary to store energy for cellular work.
D) Pi acts as a shuttle molecule to move energy from ATP to ADP.

29) How do cells use the ATP cycle shown in the figure?



A) Cells use the cycle to recycle ADP and phosphate.
B) Cells use the cycle to recycle energy released by ATP hydrolysis.
C) Cells use the cycle to recycle ADP, phosphate, and the energy released by ATP hydrolysis.
D) Cells use the cycle primarily to generate heat.

30) Which of the following is true of enzymes?
A) Enzyme function is increased if the 3- D structure or conformation of an enzyme is altered.
B) Enzyme function is independent of physical and chemical environmental factors such as pH and temperature.
C) Enzymes increase the rate of chemical reaction by lowering activation energy barriers.
D) Enzymes increase the rate of chemical reaction by providing activation energy to the substrate.

31) Which of the following is true when comparing an uncatalyzed reaction to the same reaction with a catalyst?
A) The catalyzed reaction will be slower.
B) The catalyzed reaction will have the same ΔG.
C) The catalyzed reaction will have higher activation energy.
D) The catalyzed reaction will consume all of the catalyst.

32) The lock-and-key analogy for enzymes applies to the specificity of enzymes _____.
A) As they form their tertiary and quaternary structure.
B) Binding to their substrate.
C) Interacting with water.
D) Interacting with ions.

33) You have discovered an enzyme that can catalyze two different chemical reactions. Which of the following is most likely to be correct?
A) The enzyme contains α-helices and β-pleated sheets.
B) The enzyme is subject to competitive inhibition and allosteric regulation.
C) Two types of allosteric regulation occur: The binding of one molecule activates the enzyme, while the binding of a different molecule inhibits it.
D) Either the enzyme has two distinct active sites or the reactants involved in the two reactions are very similar in size and shape.

34) Reactants capable of interacting to form products in a chemical reaction must first overcome a thermodynamic barrier known as the reaction's _____.
A) Entropy.
B) Activation energy.
C) Equilibrium point.
D) Free-energy content.

35) During a laboratory experiment, you discover that an enzyme -catalyzed reaction has a Δ*G* of -20 kcal/mol. If you double the amount of enzyme in the reaction, what will be the Δ*G* for the new reaction?
A) -40 kcal/mol.
B) -20 kcal/mol.
C) 0 kcal/mol.
D) +20 kcal/mol.

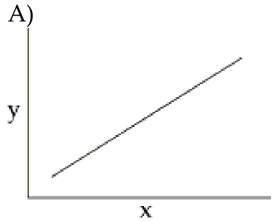
36) The active site of an enzyme is the region that _____.
A) Binds allosteric regulators of the enzyme.
B) Is involved in the catalytic reaction of the enzyme.
C) Binds noncompetitive inhibitors of the enzyme.
D) Is inhibited by the presence of a coenzyme or a cofactor.

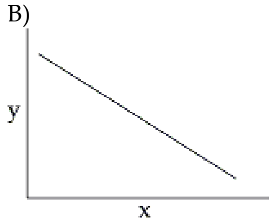
37) According to the induced fit hypothesis of enzyme catalysis, _____.
A) The binding of the substrate depends on the shape of the active site.
B) Some enzymes change their structure when activators bind to the enzyme.
C) The binding of the substrate changes the shape of the enzyme's active site.
D) The active site creates a microenvironment ideal for the reaction.

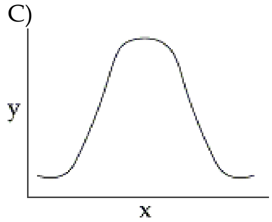
38) Increasing the substrate concentration in an enzymatic reaction could overcome which of the following?
A) The need for a coenzyme.
B) Allosteric inhibition.
C) Competitive inhibition.
D) Insufficient cofactors.

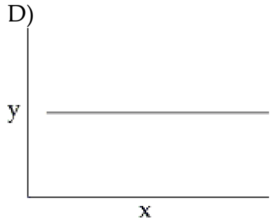
39) Zinc, an essential trace element for most organisms, is present in the active site of the enzyme carboxypeptidase. The zinc most likely functions as _____.
A) A noncompetitive inhibitor of the enzyme.
B) An allosteric activator of the enzyme.
C) A cofactor necessary for enzyme activity.
D) A coenzyme derived from a vitamin.

40) A noncompetitive inhibitor decreases the rate of an enzyme reaction by _____.
A) Binding at the active site of the enzyme.
B) Changing the shape of the enzyme's active site.
C) Changing the free energy change of the reaction.
D) Acting as a coenzyme for the reaction.

41) You collect data on the effect of pH on the function of the enzyme catalase in human cells. Which of the following graphs would you expect?
A)


B)


C)


D)


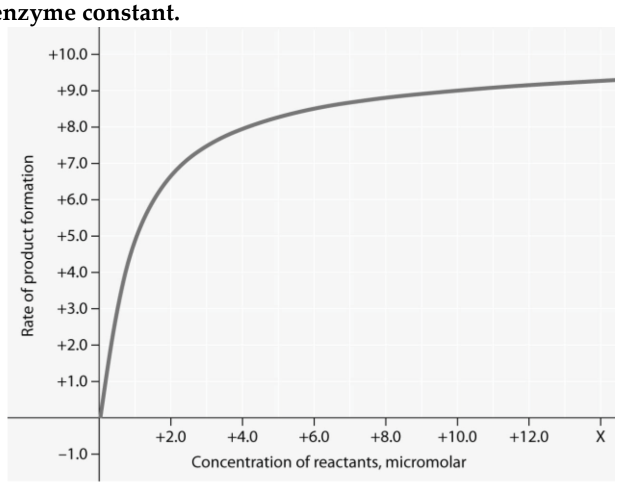
42) How might a change of one amino acid at a site, distant from the active site of an enzyme, alter an enzyme's substrate specificity?
A) By changing the enzyme's stability.
B) By changing the shape of an enzyme.
C) By changing the enzyme's pH optimum.
D) An amino acid change away from the active site cannot alter the enzyme's substrate specificity.

43) For the enzyme- catalyzed reaction shown in the figure, if the initial reactant concentration is 1.0 micromolar, which of these treatments will cause the greatest increase in the rate of the reaction?
A) Doubling the activation energy needed.
B) Cooling the reaction by 10°C.
C) Doubling the enzyme concentration.
D) Increasing the concentration of reactants to 10.0 micromolar, while reducing the concentration of enzyme by 1/2.

Use the information and figure given below to answer the following questions (28-29).

Use the information and figure given below to answer the following question (44).

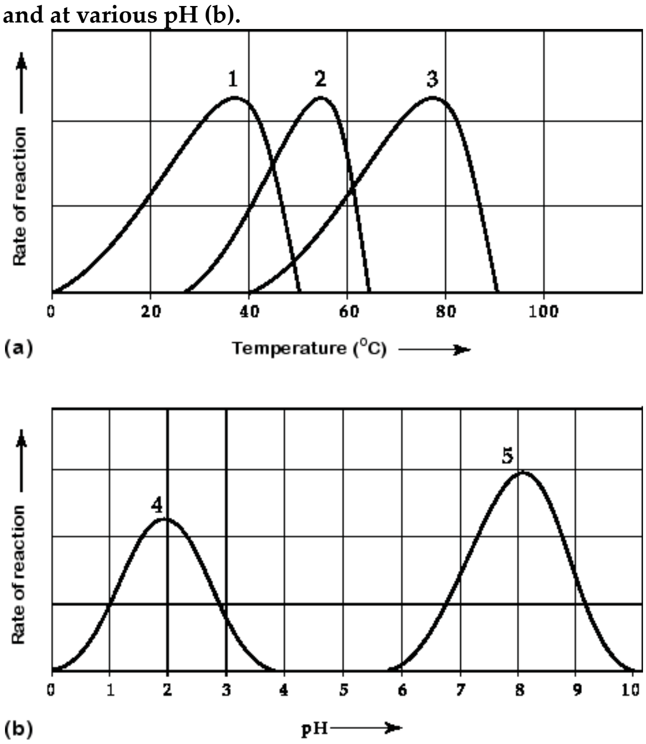
Rate of an enzyme-catalyzed reaction as a function of varying reactant concentration, with the concentration of enzyme constant.



44) In the figure, why does the reaction rate plateau at higher reactant concentrations?
A) Feedback inhibition by product occurs at high reactant concentrations.
B) Most enzyme molecules are occupied by substrate at high reactant concentrations.
C) The reaction nears equilibrium at high reactant concentrations.
D) The rate of the reverse reaction increases with reactant concentration.

Use the information and figure given below to answer the following question (45).

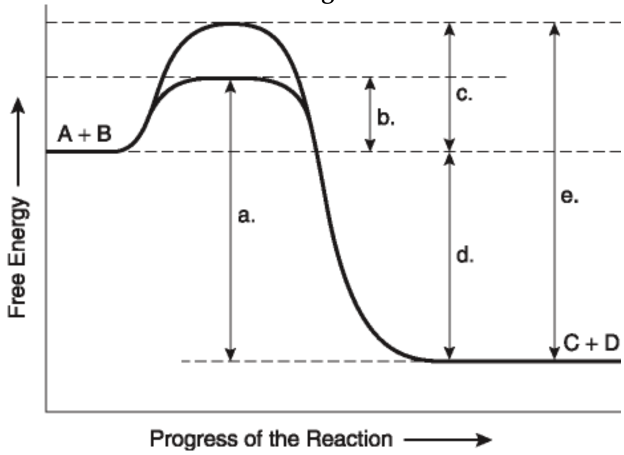
Activity of various enzymes at various temperatures (a) and at various pH (b).



45) Which curves on the graphs may represent the temperature and pH profiles of an enzyme taken from a bacterium that lives in a mildly alkaline hot springs at temperatures of 70°C or higher?
A) Curves 1 and 5.
B) Curves 2 and 5.
C) Curves 3 and 4.
D) Curves 3 and 5.

46) Which temperature and pH profile curves on the graphs were most likely generated from analysis of an enzyme from a human stomach where conditions are strongly acid?
A) Curves 1 and 4.
B) Curves 1 and 5.
C) Curves 2 and 4.
D) Curves 3 and 4.

The following questions (47-49) are based on the reaction A + B ↔ C + D shown in the figure below.



47) Which of the following terms best describes the forward reaction in the figure?
A) Endergonic, $\Delta G > 0$.
B) Exergonic, $\Delta G < 0$.
C) Endergonic, $\Delta G < 0$.
D) Exergonic, $\Delta G > 0$.

48) Which of the following in the figure would be the same in either an enzyme -catalyzed or a noncatalyzed reaction?
A) a.
B) b.
C) c.
D) d.

49) Which of the following represents the activation energy required for the enzyme-catalyzed reaction in the figure?
A) a.
B) b.
C) c.
D) d.

Use the following information to answer the questions (50-52) below.

Succinate dehydrogenase catalyzes the conversion of succinate to fumarate. The reaction is inhibited by malonic acid, which resembles succinate but cannot be acted upon by succinate dehydrogenase. Increasing the ratio of succinate to malonic acid reduces the inhibitory effect of malonic acid.

50) Based on this information, which of the following is correct?
A) Succinate dehydrogenase is the enzyme, and fumarate is the substrate.
B) Succinate dehydrogenase is the enzyme, and malonic acid is the substrate.
C) Succinate is the substrate, and fumarate is the product.
D) Fumarate is the product, and malonic acid is a noncompetitive inhibitor.

51) What is malonic acid's role with respect to succinate dehydrogenase? Malonic acid _____.
A) Is a competitive inhibitor.
B) Blocks the binding of fumarate.
C) Is a noncompetitive inhibitor.
D) Is an allosteric regulator.

52) HIV is the virus that causes AIDS. In the mid-1990s, researchers discovered an enzyme in HIV called protease. Once the enzyme's structure was known, researchers began looking for drugs that would fit into the active site and block it. If this strategy for stopping HIV infections were successful, it would be an example of what phenomenon?
A) Vaccination.
B) Denaturation.
C) Allosteric regulation.
D) Competitive inhibition.

Use the following information to answer the questions (53-54) below.

A series of enzymes catalyze the reaction X → Y → Z → A. Product A binds to the enzyme that converts X to Y at a position remote from its active site. This binding decreases the activity of the enzyme.

53) What is substance X?
A) An allosteric inhibitor.
B) A substrate.
C) An intermediate.
D) The product.

54) With respect to the enzyme that converts X to Y, substance A functions as _____.
A) An allosteric inhibitor.
B) The substrate.
C) An intermediate.
D) A competitive inhibitor.

55) The mechanism in which the end product of a metabolic pathway inhibits an earlier step in the pathway is most precisely described as _____.
A) Metabolic inhibition.
B) Feedback inhibition.
C) Allosteric inhibition.
D) Noncooperative inhibition.

56) You have isolated a previously unstudied protein, identified its complete structure in detail, and determined that it catalyzes the breakdown of a large substrate. You notice it has two binding sites. One of these is large, apparently the bonding site for the large substrate; the other is small, possibly a binding site for a regulatory molecule. What do these findings tell you about the mechanism of this protein?
A) It is probably a structural protein that is involved in cell-to-cell adhesion.
B) It is probably an enzyme that works through allosteric regulation.
C) It is probably an enzyme that works through competitive inhibition.
D) It is probably a cell membrane transport protein–like an ion channel.

57) Allosteric enzyme regulation is usually associated with _____.
A) Feedback inhibition.
B) Activating activity.
C) An enzyme with more than one subunit.
D) The need for cofactors.

58) Which of the following is an example of cooperativity?
A) The binding of an end product of a metabolic pathway to the first enzyme that acts in the pathway.
B) One enzyme in a metabolic pathway passing its product to act as a substrate for the next enzyme in the pathway.
C) A molecule binding at one unit of a tetramer, allowing faster binding at each of the other three.
D) Binding of an ATP molecule along with one of the substrate molecules in an active site.

59) Besides turning enzymes on or off, what other means does a cell use to control enzymatic activity?
A) Localization of enzymes into specific organelles or membranes.
B) Exporting enzymes out of the cell.
C) Connecting enzymes into large aggregates.
D) Hydrophobic interactions.

60) Protein kinases are enzymes that transfer the terminal phosphate from ATP to an amino acid residue on the target protein. Many are located on the plasma membrane as integral membrane proteins or peripheral membrane proteins. What purpose may be served by their plasma membrane localization?
A) ATP is more abundant near the plasma membrane.
B) They can more readily encounter and phosphorylate other membrane proteins.
C) Membrane localization lowers the activation energy of the phosphorylation reaction.
D) They flip back and forth across the membrane to access target proteins on either side.

61) Biological systems use free energy based on empirical data that all organisms require a constant energy input. The first law of thermodynamics states that energy can be neither created nor destroyed. For living organisms, which of the following statements is an important consequence of this first law?
A) The energy content of an organism is constant except for when its cells are dividing.
B) The organism must ultimately obtain all the necessary energy for life from its environment.
C) The entropy of an organism decreases with time as the organism grows in complexity.
D) Organisms are unable to transform energy from the different states in which it can exist.

62) In a biological reaction, succinate dehydrogenase catalyzes the conversion of succinate to fumarate. The reaction is inhibited by malonic acid, a substance that resembles succinate but cannot be acted upon by succinate dehydrogenase. Increasing the amount of succinate molecules to those of malonic acid reduces the inhibitory effect if malonic acid. Select the correct identification of the molecules described in the reaction.
A) Succinate dehydrogenase is the enzyme, and fumarate is the substrate in the reaction.
B) Succinate dehydrogenase is the enzyme, and malonic acid is the substrate in the reaction.
C) Succinate is the substrate, and fumarate is the product in the reaction.
D) Fumarate is the product, and malonic acid is a noncompetitive inhibitor in the reaction.

CHAPTER 9:

CELLULAR RESPIRATION AND FERMENTATION

1) Substrate-level phosphorylation occurs _____.
A) In glycolysis.
B) In the citric acid cycle.
C) In both glycolysis and the citric acid cycle.
D) During oxidative phosphorylation.

2) The molecule that functions as the reducing agent (electron donor) in a redox or oxidation-reduction reaction _____.
A) Gains electrons and gains potential energy.
B) Loses electrons and loses potential energy.
C) Gains electrons and loses potential energy.
D) Loses electrons and gains potential energy.

3) When electrons move closer to a more electronegative atom, what happens? The more electronegative atom is _____.
A) Reduced and energy is released.
B) Reduced and energy is consumed.
C) Oxidized and energy is consumed.
D) Oxidized and energy is released.

4) Which of the listed statements describes the results of the following reaction?

$$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + \text{Energy}$$

A) $C_6H_{12}O_6$ is oxidized and O_2 is reduced.
B) O_2 is oxidized and H_2O is reduced.
C) CO_2 is reduced and O_2 is oxidized.
D) O_2 is reduced and CO_2 is oxidized.

5) When a glucose molecule loses a hydrogen atom as the result of an oxidation-reduction reaction, the molecule becomes _____.
A) Hydrolyzed.
B) Oxidized.
C) Reduced.
D) An oxidizing agent.

6) When a molecule of NAD+ (nicotinamide adenine dinucleotide) gains a hydrogen atom (not a proton), the molecule becomes _____.
A) Dehydrogenated.
B) Oxidized.
C) Reduced.
D) Redoxed.

7) Which of the following statements about NAD^+ is true?
A) NAD^+ is reduced to NADH during glycolysis, pyruvate oxidation, and the citric acid cycle.
B) NAD^+ has more chemical energy than NADH.
C) NAD^+ can donate electrons for use in oxidative phosphorylation.
D) In the absence of NAD^+, glycolysis can still function.

8) The oxygen consumed during cellular respiration is involved directly in which process or event?
A) Glycolysis.
B) Accepting electrons at the end of the electron transport chain.
C) The citric acid cycle.
D) The oxidation of pyruvate to acetyl CoA.

9) Carbohydrates and fats are considered high -energy foods because they _____.
A) Have a lot of oxygen atoms.
B) Have no nitrogen in their makeup.
C) Have a lot of electrons associated with hydrogen.
D) Are easily reduced.

10) A cell has enough available ATP to meet its needs for about 30 seconds. What is likely to happen when an athlete exhausts his or her ATP supply?
A) He or she has to sit down and rest.
B) Catabolic processes are activated that generate more ATP.
C) ATP is transported into the cell from the circulatory system.
D) Other cells take over, and the muscle cells that have used up their ATP cease to function.

11) Substrate-level phosphorylation accounts for approximately what percentage of the ATP formed by the reactions of glycolysis?
A) 0%.
B) 2%.
C) 38%.
D) 100%.

12) The free energy for the oxidation of glucose to CO_2 and water is -686 kcal/mol and the free energy for the reduction of NAD^+ to NADH is +53 kcal/mol. Why are only two molecules of NADH formed during glycolysis when it appears that as many as a dozen could be formed?
A) Most of the free energy available from the oxidation of glucose is used in the production of ATP in glycolysis.
B) Glycolysis is a very inefficient reaction, with much of the energy of glucose released as heat.
C) Most of the free energy available from the oxidation of glucose remains in pyruvate, one of the products of glycolysis.
D) There is no CO_2 or water produced as products of glycolysis.

13) Starting with one molecule of glucose, the energy-containing products of glycolysis are _____.
A) 2 NAD^+, 2 pyruvate, and 2 ATP.
B) 2 NADH, 2 pyruvate, and 2 ATP.
C) 2 FADH2, 2 pyruvate, and 4 ATP.
D) 6 CO_2, 2 pyruvate, and 2 ATP.

14) In glycolysis, for each molecule of glucose oxidized to pyruvate _____.
A) Two molecules of ATP are used and two molecules of ATP are produced.
B) Two molecules of ATP are used and four molecules of ATP are produced.
C) Four molecules of ATP are used and two molecules of ATP are produced.
D) Two molecules of ATP are used and six molecules of ATP are produced.

15) Which kind of metabolic poison would most directly interfere with glycolysis?
A) An agent that reacts with oxygen and depletes its concentration in the cell.
B) An agent that binds to pyruvate and inactivates it.
C) An agent that closely mimics the structure of glucose but is not metabolized.
D) An agent that reacts with NADH and oxidizes it to NAD^+.

16) Most of the CO_2 from the catabolism of glucose is released during _____.
A) Glycolysis.
B) Electron transport.
C) Chemiosmosis.
D) The citric acid cycle.

17) Following glycolysis and the citric acid cycle, but before the electron transport chain and oxidative phosphorylation, the carbon skeleton of glucose has been broken down to CO_2 with some net gain of ATP. Most of the energy from the original glucose molecule at that point in the process, however, is in the form of _____.
A) Acetyl-CoA.
B) Glucose.
C) Pyruvate.
D) NADH.

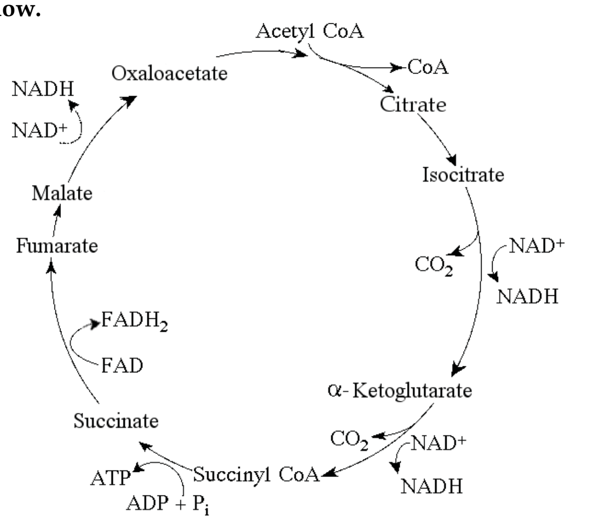
18) Which electron carrier(s) function in the citric acid cycle?
A) NAD^+ only.
B) NADH and $FADH_2$.
C) The electron transport chain.
D) ADP and ATP.

19) If you were to add one of the eight citric acid cycle intermediates to the culture medium of yeast growing in the laboratory, what do you think would happen to the rates of ATP and carbon dioxide production?
A) There would be no change in ATP production, but we would observe an increased rate of carbon dioxide production.
B) The rates of ATP production and carbon dioxide production would both increase.
C) The rate of ATP production would decrease, but the rate of carbon dioxide production would increase.
D) Rates of ATP and carbon dioxide production would probably both decrease.

20) Carbon dioxide (CO2) is released during which of the following stages of cellular respiration?
A) Glycolysis and the oxidation of pyruvate to acetyl CoA.
B) Oxidation of pyruvate to acetyl CoA and the citric acid cycle.
C) Oxidative phosphorylation and fermentation.
D) Fermentation and glycolysis.

21) If glucose is the sole energy source, what fraction of the carbon dioxide exhaled by animals is generated by the reactions of the citric acid cycle?
A) 1/6.
B) 1/3.
C) 2/3.
D) All of it.

Use the following figure to answer the questions (22-24) below.



The citric acid cycle.

22) For each mole of glucose ($C_6H_{12}O_6$) oxidized by cellular respiration, how many moles of CO_2 are released in the citric acid cycle given above?
A) 2.
B) 4.
C) 6.
D) 32.

23) If pyruvate oxidation is blocked, what will happen to the levels of oxaloacetate and citric acid in the citric acid cycle shown in figure given above?
A) Oxaloacetate will decrease and citric acid will accumulate.
B) Oxaloacetate will accumulate and citric acid will decrease.
C) Both oxaloacetate and citric acid will decrease.
D) Both oxaloacetate and citric acid will accumulate.

24) Starting with citrate, which of the following combinations of products would result from three acetyl CoA molecules entering the citric acid cycle given above?
A) 1 ATP, 2 CO_2, 3 NADH, and 1 $FADH_2$.
B) 3 ATP, 3 CO_2, 3 NADH, and 3 $FADH_2$.
C) 3 ATP, 6 CO_2, 9 NADH, and 3 $FADH_2$.
D) 38 ATP, 6 CO_2, 3 NADH, and 12 $FADH_2$.

Use the following information to answer the questions (25-26) below.

In the presence of oxygen, the three-carbon compound pyruvate can be catabolized in the citric acid cycle. First, however, the pyruvate (1) loses a carbon, which is given off as a molecule of CO_2, (2) is oxidized to form a two-carbon compound called acetate, and (3) is bonded to coenzyme A.

25) The three listed steps result in the formation of _____.
A) Acetyl CoA, O_2, and ATP.
B) Acetyl CoA, $FADH_2$, and CO_2.
C) Acetyl CoA, NADH, and CO_2.
D) Acetyl CoA, NAD^+, ATP, and CO_2.

26) Which one of the following is formed by the removal of a carbon (as CO_2) from a molecule of pyruvate?
A) Glyceraldehyde 3-phosphate.
B) Oxaloacetate.
C) Acetyl CoA.
D) Citrate.

27) Which of the following events takes place in the electron transport chain?
A) The breakdown of glucose into two pyruvate molecules.
B) The breakdown of an acetyl group to carbon dioxide.
C) The extraction of energy from high-energy electrons remaining from glycolysis and the citric acid cycle.
D) Substrate-level phosphorylation.

28) The electron transport chain _____.
A) Is a series of redox reactions.
B) Is a series of substitution reactions.
C) Is driven by ATP consumption.
D) Takes place in the cytoplasm of prokaryotic cells.

29) The chemiosmotic hypothesis is an important concept in our understanding of cellular metabolism in general because it explains _____.
A) How ATP is synthesized by a proton motive force.
B) How electron transport can fuel substrate-level phosphorylation.
C) The sequence of the electron transport chain molecules.
D) The reduction of oxygen to water in the final steps of oxidative metabolism.

30) During aerobic respiration, electrons travel downhill in which sequence?
A) Glucose → NADH → electron transport chain → oxygen.
B) Glucose → pyruvate → ATP → oxygen.
C) Glucose → ATP → electron transport chain → NADH.
D) Food → glycolysis → citric acid cycle → NADH → ATP.

31) Proteins of the electron transport chain are located ___.
A) Mitochondrial outer membrane.
B) Mitochondrial inner membrane.
C) Mitochondrial intermembrane space.
D) Mitochondrial matrix.

32) During aerobic respiration, which of the following directly donates electrons to the electron transport chain at the lowest energy level?
A) NADH.
B) ATP.
C) ADP + Pi.
D) FADH2.

33) The primary role of oxygen in cellular respiration is to _____.
A) Yield energy in the form of ATP as it is passed down the respiratory chain.
B) Act as an acceptor for electrons and hydrogen, forming water.
C) Combine with carbon, forming CO_2.
D) Combine with lactate, forming pyruvate.

34) During aerobic respiration, H_2O is formed. Where does the oxygen atom for the formation of the water come from?
A) Carbon dioxide (CO_2).
B) Glucose ($C_6H_{12}O_6$).
C) Molecular oxygen (O_2).
D) Pyruvate ($C_3H_3O_3^-$).

35) In chemiosmosis, what is the most direct source of energy that is used to convert ADP + Pi to ATP?
A) Energy released as electrons flow through the electron transport system.
B) Energy released from substrate-level phosphorylation.
C) Energy released from movement of protons through ATP synthase, down their electrochemical gradient.
D) No external source of energy is required because the reaction is exergonic.

36) Energy released by the electron transport chain is used to pump H^+ into which location in eukaryotic cells?
A) Mitochondrial outer membrane.
B) Mitochondrial inner membrane.
C) Mitochondrial intermembrane space.
D) Mitochondrial matrix.

37) When hydrogen ions are pumped from the mitochondrial matrix across the inner membrane and into the intermembrane space, the result is the _____.
A) Formation of ATP.
B) Reduction of NAD^+.
C) Creation of a proton-motive force.
D) Lowering of pH in the mitochondrial matrix.

38) Approximately how many molecules of ATP are produced from the complete oxidation of one molecule of glucose ($C_6H_{12}O_6$) in aerobic cellular respiration?
A) 2.
B) 4.
C) 18-24.
D) 30-32.

39) The synthesis of ATP by oxidative phosphorylation, using the energy released by movement of protons across the membrane down their electrochemical gradient, is an example of _____.
A) Active transport.
B) An endergonic reaction coupled to an exergonic reaction.
C) A reaction with a positive ΔG.
D) Allosteric regulation.

40) If a cell is able to synthesize 30 ATP molecules for each molecule of glucose completely oxidized to carbon dioxide and water, approximately how many ATP molecules can the cell synthesize for each molecule of pyruvate oxidized to carbon dioxide and water?
A) 0.
B) 12.
C) 14.
D) 26.

41) In liver cells, the inner mitochondrial membranes are about five times the area of the outer mitochondrial membranes. What purpose must this serve?
A) It allows for an increased rate of glycolysis.
B) It allows for an increased rate of the citric acid cycle.
C) It increases the surface for oxidative phosphorylation.
D) It increases the surface for substrate-level phosphorylation.

42) You have a friend who lost 7 kg (about 15 pounds) of fat on a regimen of strict diet and exercise. How did the fat leave his body?
A) It was released as CO_2 and H_2O.
B) It was converted to heat and then released.
C) It was converted to ATP, which weighs much less than fat.
D) It was converted to urine and eliminated from the body.

Use the following information to answer the questions below.

Exposing inner mitochondrial membranes to ultrasonic vibrations will disrupt the membranes. However, the fragments will reseal "inside out". The little vesicles that result can still transfer electrons from NADH to oxygen and synthesize ATP.

43) After the disruption, when electron transfer and ATP synthesis still occur, what must be present?
A) All of the electron transport proteins and ATP synthase.
B) All of the electron transport system and the ability to add CoA to acetyl groups.
C) The ATP synthase system.
D) The electron transport system.

44) These inside-out membrane vesicles will _____.
A) Become acidic inside the vesicles when NADH is added.
B) Become alkaline inside the vesicles when NADH is added.
C) Make ATP from ADP and Pi if transferred to a pH 4 buffered solution after incubation in a pH 7 buffered solution.
D) Hydrolyze ATP to pump protons out of the interior of the vesicle to the exterior.

45) Chemiosmotic ATP synthesis (oxidative phosphorylation) occurs in _____.
A) All cells, but only in the presence of oxygen.
B) Only eukaryotic cells, in the presence of oxygen.
C) Only in mitochondria, using either oxygen or other electron acceptors.
D) All respiring cells, both prokaryotic and eukaryotic, using either oxygen or other electron acceptors.

46) Which of the following normally occurs regardless of whether or not oxygen (O 2) is present?
A) Glycolysis.
B) Fermentation.
C) Citric acid cycle.
D) Oxidative phosphorylation (chemiosmosis).

47) Which of the following occurs in the cytosol of a eukaryotic cell?
A) Glycolysis and fermentation.
B) Fermentation and chemiosmosis.
C) Oxidation of pyruvate to acetyl CoA.
D) Citric acid cycle.

48) In the absence of oxygen, yeast cells can obtain energy by fermentation, resulting in the production of _____.
A) ATP, CO_2, and ethanol (ethyl alcohol).
B) ATP, CO_2, and lactate.
C) ATP, NADH, and pyruvate.
D) ATP, pyruvate, and acetyl CoA.

49) One function of both alcohol fermentation and lactic acid fermentation is to _____.
A) Reduce NAD^+ to NADH.
B) Reduce FAD^+ to $FADH_2$.
C) Oxidize NADH to NAD^+.
D) Reduce $FADH_2$ to FAD^+.

50) An organism is discovered that thrives in both the presence and absence of oxygen in the air. Curiously, the consumption of sugar increases as oxygen is removed from the organism's environment, even though the organism does not gain much weight. This organism _____.
A) Is a normal eukaryotic organism.
B) Is photosynthetic.
C) Is an anaerobic organism.
D) Is a facultative anaerobe.

51) Why is glycolysis considered to be one of the first metabolic pathways to have evolved?
A) It produces much less ATP than does oxidative phosphorylation.
B) It does not involve organelles or specialized structures, does not require oxygen, and is present in most organisms.
C) It is found in prokaryotic cells but not in eukaryotic cells.
D) It requires the presence of membrane-enclosed cell organelles found only in eukaryotic cells.

52) Yeast cells that have defective mitochondria incapable of respiration will be able to grow by catabolizing which of the following carbon sources for energy?
A) Glucose.
B) Proteins.
C) Fatty acids.
D) Such yeast cells will not be capable of catabolizing any food molecules, and therefore, will die.

53) What is the oxidizing agent in the following reaction?
$$\text{Pyruvate} + \text{NADH} + \text{H}^+ \rightarrow \text{Lactate} + \text{NAD}^+$$
A) NADH.
B) NAD^+.
C) Lactate.
D) Pyruvate.

54) High levels of citric acid inhibit the enzyme phosphofructokinase, a key enzyme in glycolysis. Citric acid binds to the enzyme at a different location from the active site. This is an example of _____.
A) Competitive inhibition.
B) Allosteric regulation.
C) The specificity of enzymes for their substrates.
D) Positive feedback regulation.

55) Glycolysis is active when cellular energy levels are _____; the regulatory enzyme, phosphofructokinase, is _____ by ATP.
A) Low; activated.
B) Low; inhibited.
C) High; activated.
D) High; inhibited.

56) Canine phosphofructokinase (PFK) deficiency afflicts Springer spaniels, affecting an estimated 10% of the breed. Given its critical role in glycolysis, one implication of the genetic defect resulting in PFK deficiency in dogs is _____.
A) Early embryonic mortality.
B) Elevated blood-glucose levels in the dog's blood.
C) An intolerance for exercise.
D) A reduced life span.

57) A young dog has never had much energy. He is brought to a veterinarian for help and she decides to conduct several diagnostic tests. She discovers that the dog's mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal. Of the following, which is the best explanation of the dog's condition?
A) His mitochondria lack the transport protein that moves pyruvate across the outer mitochondrial membrane.
B) His cells cannot move NADH from glycolysis into the mitochondria.
C) His cells lack the enzyme in glycolysis that forms pyruvate.
D) His cells have a defective electron transport chain, so glucose goes to lactate instead of to acetyl CoA.

58) Even though plants cells photosynthesize, they still use their mitochondria for oxidation of pyruvate. This will occur in _____.
A) Photosynthetic cells in the light, while photosynthesis occurs concurrently.
B) Cells that are storing glucose only.
C) All cells all the time.
D) Photosynthesizing cells in the light and in other tissues in the dark.

59) In respiration, beta oxidation involves the _____.
A) Oxidation of glucose.
B) Oxidation of pyruvate.
C) Regulation of glycolysis.
D) Breakdown of fatty acids.

60) Fatty acids usually have an even number of carbons in their structures. They are catabolized by a process called beta-oxidation. The end products of the metabolic pathway are acetyl groups of acetyl CoA molecules. These acetyl groups _____.
A) Directly enter the electron transport chain.
B) Directly enter the energy-yielding stages of glycolysis.
C) Are directly decarboxylated by pyruvate dehydrogenase.
D) Directly enter the citric acid cycle.

61) New biosensors, applied like a temporary tattoo to the skin, can alert serious athletes that they are about to "hit the wall" and find it difficult to continue exercising. These biosensors monitor lactate, a form of lactic acid, released in sweat during strenuous exercise.
Which of the statements below is the best explanation of why athletes would need to monitor lactate levels?
A) During aerobic respiration, muscle cells cannot produce enough lactate to fuel muscle cell contractions and muscles begin to cramp, thus athletic performance suffers.
B) During anaerobic respiration, lactate levels increase when muscles cells need more energy, however muscles cells eventually fatigue, thus athletes should modify their activities to increase aerobic respiration.
C) During aerobic respiration, muscles cells produce too much lactate which causes a rise in the pH of the muscle cells, thus athletes must consume increased amounts of sports drinks, high in electrolytes, to buffer the pH.
D) During anaerobic respiration, muscle cells receive too little oxygen and begin to convert lactate to pyruvate (pyruvic acid), thus athletes experience cramping and fatigue.

CHAPTER 10:

PHOTOSYNTHESIS

1) The process of photosynthesis probably originated _____.
A) In plants.
B) In prokaryotes.
C) In fungi.
D) Three separate times during evolution.

2) In autotrophic bacteria, where is chlorophyll located?
A) In chloroplast membranes.
B) In the ribosomes.
C) In the nucleoid.
D) In the infolded plasma membrane.

3) Plants photosynthesize _____.
A) Only in the light but respire only in the dark.
B) Only in the dark but respire only in the light.
C) Only in the light but respire in light and dark.
D) And respire only in the light.

4) Early investigators thought the oxygen produced by photosynthetic plants came from carbon dioxide. In fact, it comes from _____.
A) Water.
B) Glucose.
C) Air.
D) Electrons from NADPH.

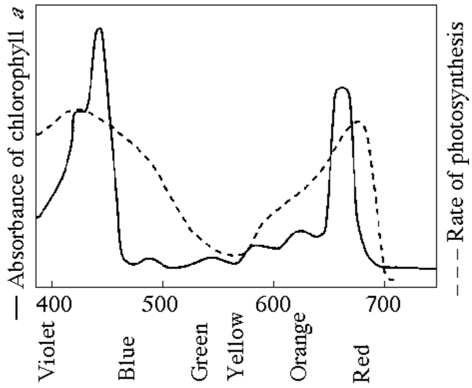
5) If photosynthesizing green algae are provided with CO_2 containing heavy oxygen (^{18}O), later analysis will show that all of the following molecules produced by the algae contain ^{18}O EXCEPT _____.
A) Glyceraldehyde 3-phosphate (G3P).
B) Glucose.
C) Ribulose bisphosphate (RuBP).
D) O_2.

6) Every ecosystem must have _____.
A) Autotrophs and heterotrophs.
B) Producers and primary consumers.
C) Photosynthesizers.
D) Autotrophs.

7) When oxygen is released as a result of photosynthesis, it is a direct by -product of _____.
A) Splitting water molecules.
B) Chemiosmosis.
C) The electron transfer system of photosystem I.
D) The electron transfer system of photosystem II.

8) Which of the following statements is a correct distinction between autotrophs and heterotrophs?
A) Cellular respiration is unique to heterotrophs.
B) Only heterotrophs have mitochondria.
C) Autotrophs, but not heterotrophs, can nourish themselves beginning with CO_2 and other nutrients that are inorganic.
D) Only heterotrophs require oxygen.

Use the following figure to answer the questions (9-10) below.



9) The figure shows the absorption spectrum for chlorophyll *a* and the action spectrum for photosynthesis. Why are they different?
A) Green and yellow wavelengths inhibit the absorption of red and blue wavelengths.
B) Oxygen given off during photosynthesis interferes with the absorption of light.
C) Other pigments absorb light in addition to chlorophyll *a*.
D) Aerobic bacteria take up oxygen, which changes the measurement of the rate of photosynthesis.

10) What wavelength of light in the figure is most effective in driving photosynthesis?
A) 420 mm.
B) 575 mm.
C) 625 mm.
D) 730 mm.

Use the following information to answer the questions (11-12) below.

Theodor W. Engelmann illuminated a filament of algae with light that passed through a prism, thus exposing different segments of algae to different wavelengths of light. He added aerobic bacteria and then noted in which areas the bacteria congregated. He noted that the largest groups were found in the areas illuminated by the red and blue light.

11) What did Engelmann conclude about the congregation of bacteria in the red and blue areas?
A) Bacteria congregated in these areas due to an increase in the temperature of the red and blue light.
B) Bacteria congregated in these areas because these areas had the most oxygen being released.
C) Bacteria are attracted to red and blue light and thus these wavelengths are more reactive than other wavelengths.
D) Bacteria congregated in these areas due to an increase in the temperature caused by an increase in photosynthesis.

12) An outcome of Engelmann's experiment was to help determine the relationship between _____.
A) Wavelengths of light and the rate of aerobic respiration.
B) Wavelengths of light and the amount of heat released.
C) Wavelengths of light and the rate of photosynthesis
D) The concentration of carbon dioxide and the rate of photosynthesis

Use the following information to answer the questions (13-15) below.

A spaceship is designed to support animal life for a multiyear voyage to the outer planets of the solar system. Plants will be grown to provide oxygen and to recycle carbon dioxide. Since the spaceship will be too far from the sun for photosynthesis, an artificial light source will be needed.

13) What wavelengths of light should be used to maximize plant growth with a minimum of energy expenditure?
A) Full-spectrum white light.
B) Green light.
C) A mixture of blue and red light.
D) UV light.

14) Suppose a plant has a unique photosynthetic pigment and the leaves of this plant appear to be reddish yellow. What wavelengths of visible light are absorbed by this pigment?
A) Red and yellow.
B) Blue and violet.
C) Green and yellow.
D) Blue, green, and red.

15) *Halobacterium* has a photosynthetic membrane that appears purple. Its photosynthetic action spectrum is the inverse of the action spectrum for green plants. (That is, the *Halobacterium* action spectrum has a peak where the green plant action spectrum has a trough.) What wavelengths of light do the *Halobacterium* photosynthetic pigments absorb?
A) Red and yellow.
B) Blue, green, and red.
C) Green and yellow.
D) Blue and red.

16) Why are there several structurally different pigments in the reaction centers of photosystems?
A) Excited electrons must pass through several pigments before they can be transferred to electron acceptors of the electron transport chain.
B) This arrangement enables the plant to absorb light energy of a variety of wavelengths.
C) They enable the plant to absorb more photons from light energy, all of which are at the same wavelength.
D) They enable the reaction center to excite electrons to a higher energy level.

17) If pigments from a particular species of plant are extracted and subjected to paper chromatography, which of the following is most likely?
A) Paper chromatography for the plant would isolate a single band of pigment that is characteristic of that particular plant.
B) Paper chromatography would separate the pigments from a particular plant into several bands.
C) The isolated pigments would be some shade of green.
D) Paper chromatography would isolate only the pigments that reflect green light.

18) In autumn, the leaves of deciduous trees change colors. This is because chlorophyll is degraded and _____.
A) Carotenoids and other pigments are still present in the leaves.
B) The degraded chlorophyll changes into many other colors.
C) Water supply to the leaves has been reduced.
D) Sugars are sent to most of the cells of the leaves.

19) What event accompanies energy absorption by chlorophyll (or other pigment molecules of the antenna complex)?
A) ATP is synthesized from the energy absorbed.
B) A carboxylation reaction of the Calvin cycle occurs.
C) Electrons are stripped from NADPH.
D) An electron is excited.

20) As electrons are passed through the system of electron carriers associated with photosystem II, they lose energy. What happens to this energy?
A) It excites electrons of the reaction center of photosystem I.
B) It is lost as heat.
C) It is used to establish and maintain a proton gradient.
D) It is used to phosphorylate NAD^+ to NADPH, the molecule that accepts electrons from photosystem I.

21) The final electron acceptor associated with photosystem I is _____.
A) Oxygen.
B) Water.
C) NADP
D) NADPH.

22) The electrons of photosystem II are excited and transferred to electron carriers. From which molecule or structure do the photosystem II replacement electrons come?
A) The electron carrier, plastocyanin.
B) Photosystem I.
C) Water.
D) Oxygen.

23) In the thylakoid membranes, the pigment molecules in a light -harvesting complex _____.
A) Split water and release oxygen from the reaction-center chlorophyll.
B) Absorb and transfer light energy to the reaction-center chlorophyll.
C) Synthesize ATP from ADP and Pi.
D) Transfer electrons to ferredoxin and then NADPH.

24) Which of the following are directly associated with photosystem I?
A) Receiving electrons from the thylakoid membrane electron transport chain.
B) Generation of molecular oxygen.
C) Extraction of hydrogen electrons from the splitting of water.
D) Passing electrons to the cytochrome complex.

25) Some photosynthetic organisms contain chloroplasts that lack photosystem II, yet are able to survive. The best way to detect the lack of photosystem II in these organisms would be to _____.
A) Determine if they have thylakoids in the chloroplasts.
B) Test for liberation of O_2 in the light.
C) Test for CO_2 fixation in the dark.
D) Do experiments to generate an action spectrum.

26) What are the products of linear electron flow?
A) Heat and fluorescence.
B) ATP and P700.
C) ATP and NADPH.
D) ADP and $NADP^+$.

27) As a research scientist, you measure the amount of ATP and NADPH consumed by the Calvin cycle in 1 hour. You find that 30,000 molecules of ATP were consumed, but only 20,000 molecules of NADPH were consumed. Where did the extra ATP molecules come from?
A) Photosystem II.
B) Photosystem I.
C) Cyclic electron flow.
D) Linear electron flow.

28) Assume a thylakoid is somehow punctured so that the interior of the thylakoid is no longer separated from the stroma. This damage will most directly affect the _____.
A) Splitting of water.
B) Flow of electrons from photosystem II to photosystem I.
C) Synthesis of ATP.
D) Reduction of $NADP^+$.

29) In a plant cell, where are the ATP synthase complexes located?
A) Thylakoid membrane only.
B) Inner mitochondrial membrane only.
C) Thylakoid membrane and inner mitochondrial membrane.
D) Thylakoid membrane and plasma membrane.

30) In mitochondria, chemiosmosis moves protons from the matrix into the intermembrane space, whereas in chloroplasts, chemiosmosis moves protons from the _____.
A) Matrix to the stroma.
B) Stroma to the thylakoid space.
C) Intermembrane space to the matrix.
D) Thylakoid space to the stroma.

31) Which of the following statements best describes the relationship between photosynthesis and respiration?
A) Respiration runs the biochemical pathways of photosynthesis in reverse.
B) Photosynthesis stores energy in complex organic molecules; respiration releases energy from complex organic molecules.
C) Photosynthesis occurs only in plants; respiration occurs only in animals.
D) Photosynthesis is catabolic; respiration is anabolic.

32) In photosynthetic cells, synthesis of ATP by the chemiosmotic mechanism occurs during _____.
A) Photosynthesis only.
B) Respiration only.
C) Photosynthesis and respiration.
D) Neither photosynthesis nor respiration.

33) Carbon dioxide is split to form oxygen gas and carbon compounds _____.
A) During photosynthesis.
B) During respiration.
C) During photosynthesis and respiration.
D) In neither photosynthesis nor respiration.

34) What is the relationship between the wavelength of light and the quantity of energy per photon?
A) They have a direct, linear relationship.
B) They are inversely related.
C) They are logarithmically related.
D) They are separate phenomena.

35) $P680^+$ is said to be the strongest biological oxidizing agent. Given its function, why is this necessary?
A) It is the receptor for the most excited electron in either photosystem of photosynthesis.
B) It is the molecule that transfers electrons to plastoquinone (Pq) of the electron transfer system.
C) It transfers its electrons to reduce $NADP^+$ to NADPH.
D) It obtains electrons from the oxygen atom in a water molecule, so it must have a stronger attraction for electrons than oxygen has.

36) Carotenoids are often found in foods that are considered to have antioxidant properties in human nutrition. What related function do they have in plants?
A) They serve as accessory pigments to increase light absorption.
B) They protect against oxidative damage from excessive light energy.
C) They shield the sensitive chromosomes of the plant from harmful ultraviolet radiation.
D) They reflect orange light and enhance red light absorption by chlorophyll.

37) In a plant, the reactions that produce molecular oxygen (O_2) take place in _____.
A) The light reactions alone.
B) The Calvin cycle alone.
C) The light reactions and the Calvin cycle.
D) Neither the light reactions nor the Calvin cycle.

38) The accumulation of free oxygen in Earth's atmosphere began with the origin of _____.
A) Life and respiratory metabolism.
B) Cyanobacteria using photosystem II.
C) Chloroplasts in photosynthetic eukaryotic algae.
D) Land plants.

39) In its mechanism, photophosphorylation is most similar to _____.
A) Substrate-level phosphorylation in glycolysis.
B) Oxidative phosphorylation in cellular respiration.
C) The Calvin cycle.
D) Reduction of $NADP^+$.

40) Which process is most directly driven by light energy?
A) Creation of a pH gradient by pumping protons across the thylakoid membrane.
B) Carbon fixation in the stroma.
C) Reduction of $NADP^+$ molecules.
D) Removal of electrons from chlorophyll molecules.

41) A gardener is concerned that her greenhouse is getting too hot from too much light and seeks to shade her plants with colored translucent plastic sheets, the color of which allows passage of only that wavelength. What color should she use to reduce overall light energy but still maximize plant growth?
A) Green.
B) Blue.
C) Orange.
D) Any color will work equally well.

42) A flask containing photosynthetic green algae and a control flask containing water with no algae are both placed under a bank of lights, which are set to cycle between 12 hours of light and 12 hours of dark. The dissolved oxygen concentrations in both flasks are monitored. Predict what the relative dissolved oxygen concentrations will be in the flask with algae compared to the control flask. The dissolved oxygen in the flask with algae will _____.
A) Always be higher.
B) Always be lower.
C) Be higher in the light, but the same in the dark.
D) Be higher in the light, but lower in the dark.

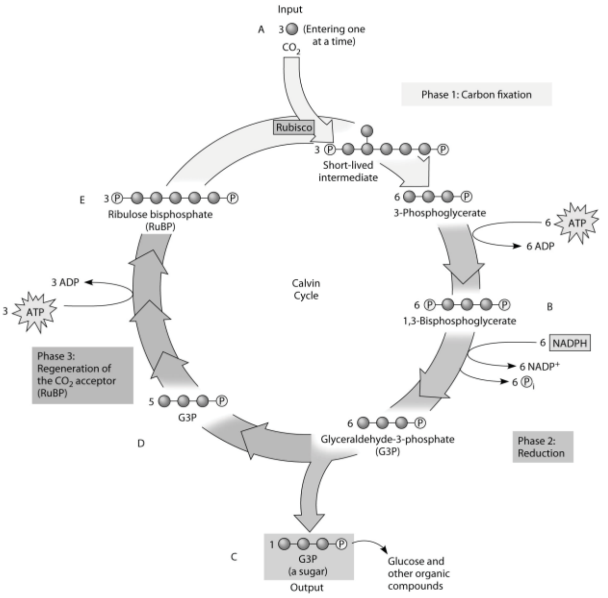
43) Which of the following are products of the light reactions of photosynthesis that are utilized in the Calvin cycle?
A) CO_2 and glucose.
B) H_2O and O_2.
C) ADP, Pi, and $NADP^+$.
D) ATP and NADPH.

44) Where does the Calvin cycle take place?
A) Stroma of the chloroplast.
B) Thylakoid membrane.
C) Interior of the thylakoid (thylakoid space).
D) Outer membrane of the chloroplast.

45) What is the primary function of the Calvin cycle?
A) Use NADPH to release carbon dioxide.
B) Split water and release oxygen.
C) Transport RuBP out of the chloroplast.
D) Synthesize simple sugars from carbon dioxide.

46) In the process of carbon fixation, RuBP attaches a CO_2 to produce a six-carbon molecule, which is then split to produce two molecules of 3-phosphoglycerate. After phosphorylation and reduction produces glyceraldehyde 3-phosphate (G3P), what more needs to happen to complete the Calvin cycle?
A) Addition of a pair of electrons from NADPH.
B) Regeneration of ATP from ADP.
C) Regeneration of RuBP.
D) Regeneration of $NADP^+$.

Use the accompanying figure below and the molecules labeled A, B, C, D, and E to answer the following questions (47-48).



47) Refer to the figure. If the carbon atom of each of the incoming CO2 molecules is labeled with a radioactive isotope of carbon, which organic molecules will be radioactively labeled after one cycle?

A) C only.
B) B, C, D, and E.
C) C, D, and E only.
D) B and C only.

48) Refer to the figure. To identify the molecule that accepts CO_2, Calvin and Benson manipulated the carbon-fixation cycle by either cutting off CO_2 or cutting off light from cultures of photosynthetic algae. They then measured the concentrations of various metabolites immediately following the manipulation. How would these experiments help identify the CO_2 acceptor?

A) The CO_2 acceptor concentration would decrease when either the CO_2 or light are cut off.
B) The CO_2 acceptor concentration would increase when either the CO_2 or light are cut off.
C) The CO_2 acceptor concentration would increase when the CO_2 is cut off, but decrease when the light is cut off.
D) The CO_2 acceptor concentration would decrease when the CO_2 is cut off, but increase when the light is cut off.

49) Which of the following sequences correctly represents the flow of electrons during photosynthesis?

A) NADPH → O_2 → CO_2.
B) H_2O → NADPH → Calvin cycle.
C) NADPH → chlorophyll → Calvin cycle.
D) NADPH → electron transport chain → O_2.

50) Which of the following does NOT occur during the Calvin cycle?

A) Oxidation of NADPH.
B) Release of oxygen.
C) Regeneration of the CO_2 acceptor.
D) Consumption of ATP.

51) What compound provides the reducing power for Calvin cycle reactions?

A) ATP.
B) NADH.
C) $NADP^+$.
D) NADPH.

52) What would be the expected effect on plants if the atmospheric CO_2 concentration was doubled?

A) All plants would experience increased rates of photosynthesis.
B) C_3 plants would have faster growth; C_4 plants would be minimally affected.
C) C_4 plants would have faster growth; C_3 plants would be minimally affected.
D) C_3 plants would have faster growth; C_4 plants would have slower growth.

53) Why are C_4 plants able to photosynthesize with no apparent photorespiration?

A) They do not participate in the Calvin cycle.
B) They use PEP carboxylase to initially fix CO_2.
C) They conserve water more efficiently.
D) They exclude oxygen from their tissues.

54) CAM plants keep stomata closed in the daytime, thus reducing loss of water. They can do this because they ____.

A) Fix CO_2 into organic acids during the night.
B) Fix CO_2 into sugars in the bundle-sheath cells.
C) Fix CO_2 into pyruvate in the mesophyll cells.
D) Use photosystem I and photosystem II at night.

55) The alternative pathways of photosynthesis using the C_4 or CAM systems are said to be compromises. Why?

A) Each one minimizes both water loss and rate of photosynthesis.
B) C_4 compromises on water loss and CAM compromises on photorespiration.
C) Both minimize photorespiration but expend more ATP during carbon fixation.
D) CAM plants allow more water loss, while C_4 plants allow less CO_2 into the plant.

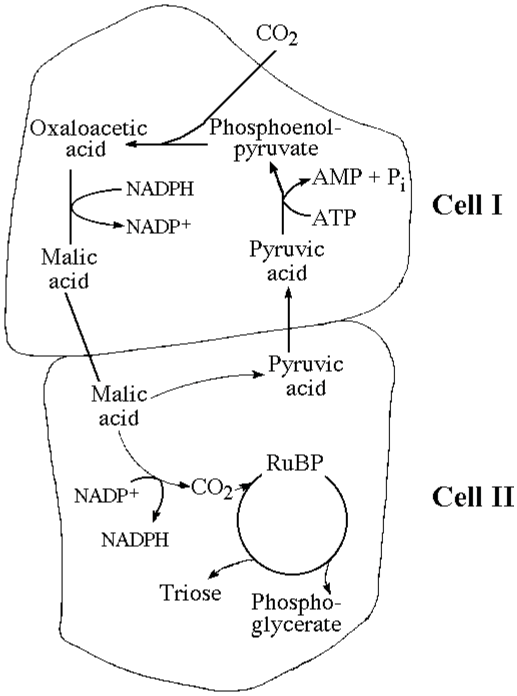
56) If plant gene alterations cause plants to be deficient in photorespiration, what would most probably occur?

A) Photosynthetic efficiency would be reduced at low light intensities.
B) Cells would carry on the Calvin cycle at a much slower rate.
C) There would be more light-induced damage to the cells.
D) Less oxygen would be produced.

57) Compared to C_3 plants, C_4 plants _____.

A) Can continue to fix CO_2 even at lower CO_2 concentrations and higher oxygen concentrations.
B) Have higher rates of photorespiration.
C) Do not use rubisco for carbon fixation.
D) Make a four-carbon compound, oxaloacetate, which is then delivered to the citric acid cycle in mitochondria.

Use the following figure to answer the questions (58-59) below.



58) Which of the following statements is true concerning of the figure given above?
A) It represents a C_4 photosynthetic system.
B) It represents an adaptation that maximizes photorespiration.
C) It represents a C_3 photosynthetic system.
D) It represents a CAM photosynthetic system.

59) Referring to the figure given above, oxygen would inhibit the CO_2 fixation reactions in _____.
A) Cell I only.
B) Cell II only.
C) Neither cell I nor cell II.
D) Both cell I and cell II.

60) Photorespiration _____.
A) Generates carbon dioxide and consumes ATP and oxygen.
B) Generates ATP and sugars and consumes oxygen and carbon dioxide.
C) Generates oxygen and consumes ATP, carbon dioxide, and sugars.
D) Consumes carbon dioxide and generates ATP, sugars, and oxygen.

61) Students conducted an experiment to determine the effect of light intensity on the rate of photosynthesis. They punched 40 leaf disks from spinach leaves and used a syringe partially filled with water to pull the gases from the leaf disks so that all leaf disks sunk to the bottom of the syringe. Ten (10) leaf disks from the syringe were placed in each of four cups and covered with 50 ml of the solutions as indicated below. All leaf disks were resting on the bottom of the cups when the experiment began. The volume of liquid in each cup and the temperature of the solutions were held constant. All cups were placed 0.5 meters from the designated light source. A large beaker of water was placed between the light and the cups to act as a heat sink to prevent a change in temperature. At the end of 10 minutes, the number of disks floating in each cup was recorded.

Trial	Grams of baking soda (CO_2) source)	Wattage of light bulb	Disks floating at 10 minutes
1	0.5	25	3
2	0.5	50	5
3	0.5	75	9
4	0	75	0

Use your knowledge of the mechanism of photosynthesis and the data presented in the chart to determine which of the statements below a correct explanation for the student's data is.
A) Cup 1 had a low rate of photosynthesis because 0.5 grams of baking soda did not provide a sufficient amount of CO_2.
B) Cup 2 had the highest rate of photosynthesis because 5 disks were floating at the end of 10 minutes using a 50 watt light bulb.
C) Cup 3 had the same rate of photosynthesis as Cup 1 because they had the same ratio of disks floating to wattage of light.
D) Cup 4 had the slowest rate of photosynthesis because it had the least baking soda.

CHAPTER 11:

CELL COMMUNICATION

1) In yeast signal transduction, a yeast cell releases a mating factor which _____.
A) Acts back on the same cell that secreted the mating factor, changing its development.
B) Passes through the membranes of neighboring cells, binds to DNA, and initiates transcription.
C) Binds to receptors on the membranes of other types of yeast cells.
D) Diffuses through the membranes of distant cells, causing them to produce factors that initiate long-distance migrations.

2) Which of the following statements about quorum sensing is FALSE? Quorum sensing _____.
A) Is cell-cell communication in eukaryotes.
B) Is species specific.
C) May result in biofilm formation.
D) Is particularly well studied because of its medical importance.

3) In the formation of biofilms, such as those forming on unbrushed teeth, cell signaling serves which function?
A) Formation of mating complexes.
B) Aggregation of bacteria that can cause cavities.
C) Secretion of substances that inhibit foreign bacteria.
D) Digestion of unwanted parasite populations.

4) Which of the following is a type of local signaling in which a cell secretes a signal molecule that affects neighboring cells?
A) Hormonal signaling.
B) Autocrine signaling.
C) Paracrine signaling.
D) Synaptic signaling.

5) Hormones are chemical substances produced in one organ that are released into the bloodstream and affect the function of a target organ. For the target organ to respond to a particular hormone, it must _____.
A) Modify its plasma membrane to alter the hormone entering the cytoplasm.
B) Be from the same cell type as the organ that produced the hormone.
C) Experience an imbalance, disrupting its normal function.
D) Have receptors that recognize and bind the hormone molecule.

6) In which of the following ways do plant hormones differ from hormones in animals?
A) Plant hormones most often travel in air as a gas.
B) Animal hormones are only local regulators.
C) Plant hormones are typically released into the soil.
D) Animal hormones are primarily for mating and embryonic development.

7) When a neuron responds to a particular neurotransmitter by opening gated ion channels, the neurotransmitter is serving as which part of the signal pathway?
A) Relay molecule.
B) Transducer.
C) Signal molecule.
D) Response molecule.

The following questions (8-9) are based on the figure given below.

8) Which of the following types of signaling is represented in the figure?
A) Autocrine.
B) Paracrine.
C) Hormonal.
D) Synaptic.

9) In the figure, the dots in the space between the two structures represent which of the following?
A) Receptor molecules.
B) Signal transducers.
C) Neurotransmitters.
D) Hormones.

10) Which observation suggested to Sutherland the involvement of a second messenger in epinephrine's effect on liver cells?
A) Receptor studies indicated that epinephrine was a ligand.
B) Glycogen breakdown was observed only when epinephrine was administered to intact cells.
C) Glycogen breakdown was observed when epinephrine and glycogen phosphorylase were combined.
D) Epinephrine was known to have different effects on different types of cells.

11) A G-protein receptor with GTP bound to it _____.
A) Is in its active state.
B) Signals a protein to maintain its shape and conformation.
C) Will use cGMP as a second messenger.
D) Directly affects gene expression.

12) Testosterone functions inside a cell by _____.
A) Acting as a signal receptor that activates tyrosine kinases.
B) Binding with a receptor protein that enters the nucleus and activates specific genes.
C) Acting as a steroid signal receptor that activates ion channel proteins.
D) Coordinating a phosphorylation cascade that increases spermatogenesis.

13) Scientists have found that extracellular matrix components may induce specific gene expression in embryonic tissues such as the liver and testes. For this to happen there must be direct communication between the extracellular matrix and the developing cells. Which kind of transmembrane protein would most likely be involved in this kind of induction?
A) Integrins.
B) Collagens.
C) Actins.
D) Fibronectins.

14) One of the major categories of receptors in the plasma membrane reacts by forming dimers, adding phosphate groups, and then activating relay proteins. Which type does this?
A) G protein-coupled receptors.
B) Ligand-gated ion channels.
C) Steroid receptors.
D) Receptor tyrosine kinases.

15) A drug designed to inhibit the response of cells to testosterone would most likely result in _____.
A) Lower cytoplasmic levels of cAMP.
B) A decrease in transcriptional activity of certain genes.
C) An increase in cytosolic calcium concentration.
D) A decrease in G protein activity.

Use this description to answer the following questions (16-17).

A major group of G protein-coupled receptors contains seven transmembrane α helixes. The amino end of the protein lies at the exterior of the plasma membrane. Loops of amino acids connect the helixes either at the exterior face or on the cytosol face of the membrane. The loop on the cytosol side between helixes 5 and 6 is usually substantially longer than the others.

16) Where would you expect to find the carboxyl end?
A) At the exterior surface.
B) At the cytosol surface.
C) Connected with the loop at H5 and H6.
D) Between the membrane layers.

17) The coupled G protein most likely interacts with this receptor _____.
A) At the NH_3 end.
B) At the COOH end.
C) Along the exterior margin.
D) At the loop between H5 and H6.

18) Binding of a signaling molecule to which type of receptor leads directly to a change in the distribution of ions on opposite sides of the membrane?
A) Receptor tyrosine kinase.
B) G protein-coupled receptor.
C) Ligand-gated ion channel.
D) Intracellular receptor.

19) Lipid-soluble signaling molecules, such as testosterone, cross the membranes of all cells but affect only target cells because _____.
A) Only target cells retain the appropriate DNA segments.
B) Intracellular receptors are present only in target cells.
C) Only target cells possess the cytosolic enzymes that transduce the testosterone.
D) Only in target cells is testosterone able to initiate the phosphorylation cascade leading to activated transcription factor.

20) If an animal cell suddenly lost the ability to produce GTP, what might happen to its signaling system?
A) It would not be able to activate and inactivate the G protein on the cytoplasmic side of the plasma membrane.
B) It would be able to carry out reception and transduction but would not be able to respond to a signal.
C) It would use ATP instead of GTP to activate and inactivate the G protein on the cytoplasmic side of the plasma membrane.
D) It would employ a transduction pathway directly from an external messenger.

21) Which of the following is true of steroid receptors?
A) The receptor molecules are themselves lipids or glycolipids.
B) The receptor may be inside the nucleus of a target cell.
C) The unbound steroid receptors are quickly recycled by lysosomes.
D) Steroid receptors are typically bound to the external surface of the nuclear membrane.

22) Particular receptor tyrosine kinases (RTKs) that promote excessive cell division are found at high levels on various cancer cells. A protein, Herceptin, has been found to bind to an RTK known as HER2. HER2 is sometimes excessive in cancer cells. This information can now be utilized in breast cancer treatment if which of the following is true?
A) If HER2, administered by injection, causes cell division.
B) If the patient's cancer cells have excessive levels of HER2.
C) If the patient's genome codes for the HER2 receptor.
D) If the patient has RTKs only in cancer cells.

23) Which of the following would be inhibited by a drug that specifically blocks the addition of phosphate groups to proteins?
A) G protein-coupled receptor binding.
B) Ligand-gated ion channel signaling.
C) Adenylyl cyclase activity.
D) Receptor tyrosine kinase activity.

24) Which of the following is characteristic of a steroid hormone action?
A) Protein phosphorylation.
B) Cell-surface receptor binding.
C) Internal receptor binding.
D) Second messenger activation.

25) The receptors for steroid hormones are located inside the cell instead of on the membrane surface like most other signal receptors. This is not a problem for steroids because _____.
A) The receptors can be readily stimulated to exit and relocate on the membrane surface.
B) Steroids do not directly affect cells but instead alter the chemistry of blood plasma.
C) Steroid hormones are lipid soluble, so they can readily diffuse through the lipid bilayer of the cell membrane.
D) Steroids must first bond to a steroid activator, forming a complex that then binds to the cell surface.

26) Not all intercellular signals require transduction. Which one of the following signals would be processed without transduction?
A) A lipid-soluble signal.
B) A signal that is weakly bound to a nucleotide.
C) A signal that binds to a receptor in the cell membrane.
D) A signal that binds to the ECM.

27) What does it mean to say that a signal is transduced?
A) The signal enters the cell directly and binds to a receptor inside.
B) The physical form of the signal changes from one form to another.
C) The signal is amplified, such than even a single molecule evokes a large response.
D) The signal triggers a sequence of phosphorylation events inside the cell.

28) Protein phosphorylation is commonly involved with all of the following EXCEPT _____.
A) Regulation of transcription by extracellular signaling molecules.
B) Enzyme activation.
C) Activation of G protein-coupled receptors.
D) Activation of protein kinase molecules.

29) In general, a signal transmitted via phosphorylation of a series of proteins _____.
A) Results in a conformational change to each protein.
B) Requires binding of a hormone to an intracellular receptor.
C) Activates a transcription event.
D) Generates ATP in the process of signal transduction.

30) Which of the following is the best explanation for the inability of a specific animal cell to reduce the Ca^{2+} concentration in its cytosol compared with the extracellular fluid?
A) Blockage of the synaptic signal.
B) Loss of transcription factors.
C) Insufficient ATP levels in the cytosol.
D) Low levels of protein kinase in the cell.

31) The toxin of *Vibrio cholerae* causes profuse diarrhea because it _____.
A) Modifies a G protein involved in regulating salt and water secretion.
B) Binds with adenylyl cyclase and triggers the formation of cAMP.
C) Signals IP_3 to act as a second messenger for the release of calcium.
D) Modifies calmodulin and activates a cascade of protein kinases.

32) Which of the following would most likely be an immediate result of a growth factor binding to its receptor?
A) Protein kinase activity.
B) Adenylyl cyclase activity.
C) Protein phosphatase activity.
D) Phosphorylase activity.

33) Adenylyl cyclase has the opposite effect of which of the following?
A) Protein kinase.
B) Protein phosphatase.
C) Phosphodiesterase.
D) Phosphorylase.

34) Caffeine is an inhibitor of phosphodiesterase. Therefore, the cells of a person who has recently consumed coffee would have increased levels of _____.
A) Phosphorylated proteins.
B) cAMP.
C) Adenylyl cyclase.
D) Activated G proteins.

35) An inhibitor of which of the following could be used to block the release of calcium from the endoplasmic reticulum?
A) Serine/threonine kinases.
B) Phosphodiesterase.
C) Phospholipase C.
D) Adenylyl cyclase.

36) Which of the following statements is true of signal molecules?
A) When signal molecules first bind to receptor tyrosine kinases, the receptors phosphorylate a number of nearby molecules.
B) In response to some G protein-mediated signals, a special type of lipid molecule associated with the plasma membrane is cleaved to form IP3 and calcium.
C) In most cases, signal molecules interact with the cell at the plasma membrane, enter the cell, and eventually enter the nucleus.
D) Protein kinase A activation is one possible result of signal molecules binding to G protein-coupled receptors.

37) Which of the following is a correct association?
A) Kinase activity and the addition of a tyrosine.
B) Phosphodiesterase activity and the removal of phosphate groups.
C) GTPase activity and hydrolysis of GTP to GDP.
D) Adenylyl cyclase activity and the conversion of cAMP to AMP.

38) Protein kinase is an enzyme that _____.
A) Functions as a second messenger molecule.
B) Serves as a receptor for various signal molecules.
C) Activates or inactivates other proteins by adding a phosphate group to them.
D) Produces second messenger molecules.

39) Viagra causes dilation of blood vessels and increased blood flow to the penis, facilitating erection. Viagra acts by inhibiting _____.
A) The hydrolysis of cGMP to GMP.
B) The hydrolysis of GTP to GDP.
C) The dephosphorylation of cGMP.
D) The removal of GMP from the cell.

40) Which of the following amino acids are most frequently phosphorylated by protein kinases in the cytoplasm during signal transduction?
A) Tyrosines.
B) Glycine and histidine.
C) Serine and threonine.
D) Glycine and glutamic acid.

41) In signal transduction, phosphatases _____.
A) Move the phosphate group of the transduction pathway to the next molecule of a series.
B) Prevent a protein kinase from being reused when there is another extracellular signal.
C) Amplify the second messengers such as cAMP.
D) Inactivate protein kinases and turn off the signal transduction.

42) If a pharmaceutical company wished to design a drug to maintain low blood sugar levels, one approach might be to design a compound _____.
A) That activates epinephrine receptors.
B) That increases cAMP production in liver cells.
C) To block G protein activity in liver cells.
D) That increases phosphorylase activity.

43) If a pharmaceutical company wished to design a drug to maintain low blood sugar levels, one approach might be to design a compound _____.
A) That mimics epinephrine and can bind to the epinephrine receptor.
B) That stimulates cAMP production in liver cells.
C) To stimulate G protein activity in liver cells.
D) That increases phosphodiesterase activity.

44) Consider this pathway: epinephrine → G protein-coupled receptor → G protein →adenylyl cyclase → cAMP. The second messenger in this pathway is _____.
A) cAMP.
B) G protein.
C) Adenylyl cyclase.
D) G protein-coupled receptor.

45) Sutherland discovered that the signaling molecule epinephrine _____.
A) Brings about a decrease in levels of cAMP as a result of bypassing the plasma membrane.
B) Causes lower blood glucose by binding to liver cells.
C) Interacts directly with glycogen phosphorylase.
D) Elevates cytosolic concentrations of cyclic AMP.

46) Which of the following is true during a typical cAMP-type signal transduction event?
A) The second messenger is the last part of the system to be activated.
B) The hormone activates the second messenger by directly binding to it.
C) The second messenger amplifies the hormonal response by attracting more hormones to the cell being affected.
D) Adenylyl cyclase is activated after the hormone binds to the cell and before phosphorylation of proteins occurs.

47) Put the steps of the process of signal transduction in the order they occur:
1. A conformational change in the signal-receptor complex activates an enzyme.
2. Protein kinases are activated.
3. A signal molecule binds to a receptor.
4. Target proteins are phosphorylated.
5. Second messenger molecules are released.
A) 1, 2, 3, 4, 5.
B) 3, 1, 2, 4, 5.
C) 3, 1, 5, 2, 4.
D) 1, 2, 5, 3, 4.

48) Transcription factors _____.
A) Regulate the synthesis of DNA in response to a signal.
B) Transcribe ATP into cAMP.
C) Control gene expression.
D) Regulate the synthesis of lipids in the cytoplasm.

49) At puberty, an adolescent female body changes in both structure and function of several organ systems, primarily under the influence of changing concentrations of estrogens and other steroid hormones. How can one hormone, such as estrogen, mediate so many effects?
A) Estrogen is produced in very large concentration by nearly every tissue of the body.
B) Each cell responds in the same way when steroids bind to the cell surface.
C) Estrogen is kept away from the surface of any cells not able to bind it at the surface.
D) Estrogen binds to specific receptors inside many kinds of cells, each with different responses.

50) Scaffolding proteins are _____.
A) Microtubular protein arrays that allow lipid-soluble hormones to get from the cell membrane to the nuclear pores.
B) Large molecules to which several relay proteins attach to facilitate cascade effects.
C) Relay proteins that orient receptors and their ligands in appropriate directions to facilitate their complexing.
D) Proteins that can reach into the nucleus of a cell to affect transcription.

51) Phosphorylation cascades involving a series of protein kinases are useful for cellular signal transduction because they _____.
A) Are species specific.
B) Always lead to the same cellular response.
C) Amplify the original signal many times.
D) Counter the harmful effects of phosphatases.

52) GTPase activity is important in the regulation of signal transduction because it _____.
A) Increases the available concentration of phosphate.
B) Decreases the amount of G protein in the membrane.
C) Hydrolyzes GTP to GDP, thus shutting down the pathway.
D) Converts cGMP to GTP.

53) Why has *C. elegans* proven to be a useful model for understanding apoptosis?
A) The animal does not naturally use apoptosis, but can be induced to do so in the laboratory.
B) The nematode undergoes a fixed and easy-to-visualize number of apoptotic events during its normal development.
C) This plant has a long-studied aging mechanism that has made understanding its death just a last stage.
D) While the organism ages, its cells die progressively until the whole organism is dead.

54) Which of the following describes the events of apoptosis?
A) The cell dies, it is lysed, its organelles are phagocytized, and its contents are recycled.
B) The cell's DNA and organelles become fragmented, the cell dies, and it is phagocytized.
C) The cell's DNA and organelles become fragmented, the cell shrinks and forms blebs, and the cell's parts are packaged in vesicles that are digested by specialized cells.
D) The cell's nucleus and organelles are lysed, then the cell enlarges and bursts.

55) If an adult person has a faulty version of the human analog to *ced-4* of the nematode, which of the following is most likely to result?
A) Activation of a developmental pathway found in the worm but not in humans.
B) A form of cancer in which there is insufficient apoptosis.
C) Formation of molecular pores in the mitochondrial outer membrane.
D) Excess skin loss.

56) Why is apoptosis potentially threatening to the healthy "neighbors" of a dying cell?
A) Cell death would usually spread from one cell to the next via paracrine signals.
B) Lysosomal enzymes exiting the dying cell would damage surrounding cells.
C) Bits of membrane from the dying cell could merge with neighboring cells and bring in foreign receptors.
D) Neighboring cells would activate immunological responses.

57) In the nematode *C. elegans*, *ced-9* prevents apoptosis in a normal cell in which of the following ways?
A) It prevents the caspase activity of *ced-3* and *ced-4.*
B) *Ced-9* remains inactive until it is signaled by *ced-3* and other caspases.
C) *Ced-9* cleaves to produce *ced-3* and *ced-4.*
D) *Ced-9* prevents blebbing by its action on the cell membrane.

58) In research on aging (both cellular aging and organismal aging), it has been found that aged cells do not progress through the cell cycle as they had previously. Which of the following, if found in cells or organisms as they age, would provide evidence that this is related to cell signaling?
A) Growth factor ligands do not bind as efficiently to receptors.
B) Hormone concentrations decrease.
C) cAMP levels change very frequently.
D) Enzymatic activity declines.

59) Apoptosis involves all but which of the following?
A) Fragmentation of the DNA.
B) Activation of cellular enzymes.
C) Lysis of the cell.
D) Digestion of cellular contents by scavenger cells.

60) An unlabeled signaling diagram for steroid hormones: Fig. 11.8 p. 213, Campbell 8e.
A steroid hormone, like estrogen, passes through the plasma membrane and binds to an intracellular protein as shown in the diagram below. This activates a signal-transduction pathway which results in an increased production of a specific protein.
Which of the following statements would explain what would occur as a result of the signal pathway represented by the diagram?
A) Transfer RNA (t-RNA) would accumulate in high levels because it is not required for protein synthesis.
B) Ribosomal RNA (r-RNA) levels would increase because ribosomes are specific for the messenger RNA (m-RNA) with which they bind during transcription of the polypeptide.
C) DNA levels would increase in the nucleus as a result of the binding of the hormone-receptor complex with the DNA.
D) Messenger RNA (m-RNA) levels would increase in order to be translated into the protein required by the cell.

61) Which of the following poses the best evidence that cell-signaling pathways evolved early in the history of life?

A) Cell-signaling pathways are seen in "primitive" cells such as bacteria and yeast.
B) Bacteria and yeast cells signal each other in a process called quorum sensing.
C) Signal transduction molecules identified in distantly related organisms are similar.
D) Most signals in all types of cells are received by cell surface receptors.
Section: 11.1

62) Cells that are infected, damaged, or have reached the end of their functional life span often undergo "programmed cell death." This controlled cell suicide is called apoptosis. Select the appropriate description of this event on a cell's life cycle.

A) Apoptosis is regulated by cell surface receptors that signal when a cell has reached its density-dependent limits.
B) During apoptosis, dying cells leak out their contents including digestive enzymes that also destroy healthy cells.
C) During apoptosis, cellular agents chop up the DNA and fragment the organelles and other cytoplasmic components of a cell.
D) Each cell organelle has protein signals that initiate the breakdown of the organelle's components which leads to cell death.

CHAPTER 12:

THE CELL CYCLE

1) In eukaryotic cells, chromosomes are composed of ___.
A) DNA and RNA.
B) DNA only.
C) DNA and proteins.
D) DNA and phospholipids.

2) What is the final result of mitosis in a human?
A) Genetically identical 2n somatic cells.
B) Genetically different 2n somatic cells.
C) Genetically identical 1n somatic cells.
D) Genetically identical 2n gamete cells.

3) Starting with a fertilized egg (zygote), a series of five cell divisions would produce how many cells?
A) 8.
B) 16.
C) 32.
D) 64.

4) If there are 20 duplicated chromosomes in a cell, how many centromeres are there?
A) 10.
B) 20.
C) 30.
D) 40.

5) Scientists isolate cells in various phases of the cell cycle. They find a group of cells that have 1*1/2 times more DNA than G_1 phase cells. The cells of this group are _____.
A) Between the G_1 and S phases in the cell cycle.
B) In the G_2 phase of the cell cycle.
C) In the M phase of the cell cycle.
D) In the S phase of the cell cycle.

6) The first gap in the cell cycle (G_1) corresponds to _____.
A) Normal growth and cell function.
B) The phase in which DNA is being replicated.
C) The beginning of mitosis.
D) The phase between DNA replication and the M phase.

7) The microtubule-organizing center found in animal cells is an identifiable structure present during all phases of the cell cycle. Specifically, it is known as the _____.
A) Microtubulere.
B) Centrosome.
C) Centromere.
D) Kinetochore.

8) In human and many other eukaryotic species' cells, the nuclear membrane has to disappear to permit _____.
A) Cytokinesis.
B) The attachment of microtubules to kinetochores.
C) The splitting of the centrosomes.
D) The disassembly of the nucleolus.

9) The mitotic spindle is a microtubular structure that is involved in _____.
A) Splitting of the cell (cytokinesis) following mitosis.
B) Triggering the compaction of chromosomes.
C) Dissolving the nuclear membrane.
D) Separation of sister chromatids.

10) Metaphase is characterized by _____.
A) Aligning of chromosomes on the equator.
B) Splitting of the centromeres.
C) Cytokinesis.
D) Separation of sister chromatids.

11) Kinetochore microtubules assist in the process of splitting centromeres by _____.
A) Using motor proteins to split the centromere at specific arginine residues.
B) Creating tension by pulling toward opposite poles.
C) Sliding past each other like actin filaments.
D) Phosphorylating the centromere, thereby changing its conformation.

12) Some cells have several nuclei per cell. How could such multinucleated cells be explained?
A) The cell underwent repeated cytokinesis but no mitosis.
B) The cell underwent repeated mitosis with simultaneous cytokinesis.
C) The cell underwent repeated mitosis, but cytokinesis did not occur.
D) The cell had multiple S phases before it entered mitosis.

13) How is plant cell cytokinesis different from animal cell cytokinesis?
A) The contractile filaments found in plant cells are structures composed of carbohydrates; the cleavage furrow in animal cells is composed of contractile phospholipids.
B) Plant cells deposit vesicles containing cell-wall building blocks on the metaphase plate; animal cells form a cleavage furrow.
C) The structural proteins of plant cells separate the two cells; in animal cells, a cell membrane separates the two daughter cells.
D) Plant cells divide after metaphase but before anaphase; animal cells divide after anaphase.

14) FtsZ is a bacterial cytoskeletal protein that forms a contractile ring involved in bacterial cytokinesis. Its function is analogous to _____.
A) The cleavage furrow of eukaryotic animal cells.
B) The cell plate of eukaryotic plant cells.
C) The mitotic spindle of eukaryotic cells.
D) The microtubule-organizing center of eukaryotic cells.

15) At which phase are centrioles beginning to move apart in animal cells?
A) Anaphase.
B) Prometaphase.
C) Metaphase.
D) Prophase.

16) If there are 20 centromeres in a cell at anaphase, how many chromosomes are there in each daughter cell following cytokinesis?
A) 10.
B) 20.
C) 40.
D) 80.

17) Taxol is an anticancer drug extracted from the Pacific yew tree. In animal cells, Taxol disrupts microtubule formation. Surprisingly, this stops mitosis. Specifically, Taxol must affect _____.
A) The structure of the mitotic spindle.
B) Anaphase.
C) Formation of the centrioles.
D) Chromatid assembly.

18) Which of the following are primarily responsible for cytokinesis in plant cells but not in animal cells?
A) Kinetochores.
B) Golgi-derived vesicles.
C) Actin and myosin.
D) Centrioles and centromeres.

19) Movement of the chromosomes during anaphase would be most affected by a drug that prevents _____.
A) Nuclear envelope breakdown.
B) Elongation of microtubules.
C) Shortening of microtubules.
D) Formation of a cleavage furrow.

20) Measurements of the amount of DNA per nucleus were taken on a large number of cells from a growing fungus. The measured DNA levels ranged from 3 to 6 picograms per nucleus. In which stage of the cell cycle did the nucleus contain 6 picograms of DNA?
A) G_1.
B) S.
C) G_2.
D) M.

21) A group of cells is assayed for DNA content immediately following mitosis and is found to have an average of 8 picograms of DNA per nucleus. How many picograms would be found at the end of S and the end of G_2?
A) 8; 8.
B) 8; 16.
C) 16; 8.
D) 16; 16.

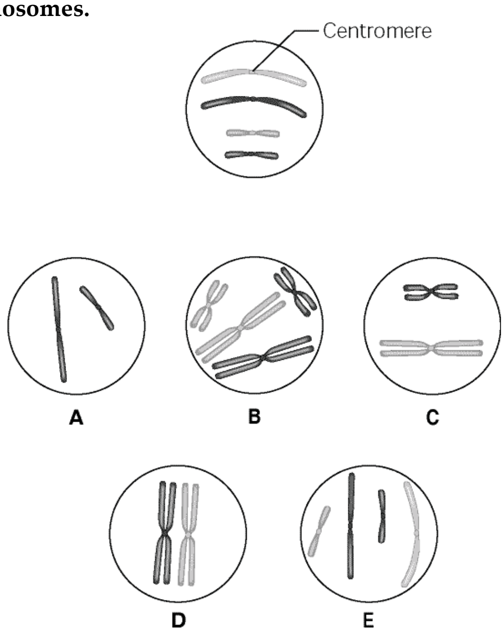
22) The beginning of anaphase is indicated by which of the following?
A) Chromatids lose their kinetochores.
B) Cohesin attaches the sister chromatids to each other.
C) Cohesin is cleaved enzymatically.
D) Spindle microtubules begin to polymerize.

23) During which phase of mitosis do the chromatids become chromosomes?
A) Telophase.
B) Anaphase.
C) Prophase.
D) Metaphase.

24) A cleavage furrow is _____.
A) A ring of vesicles forming a cell plate.
B) The separation of divided prokaryotes.
C) A groove in the plasma membrane between daughter nuclei.
D) The space that is created between two chromatids during anaphase .

Use the following information to answer the questions (25-26) below.

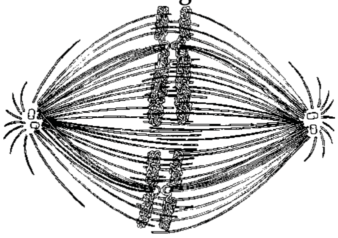
The unlettered circle at the top of the figure shows a diploid nucleus with four chromosomes that have not yet replicated. There are two pairs of homologous chromosomes, one long and the other short.
One haploid set is black, and the other is gray. The circles labeled A to E show various combinations of these chromosomes.



25) What is the correct chromosomal condition at prometaphase of mitosis?
A) B.
B) C.
C) D.
D) E.

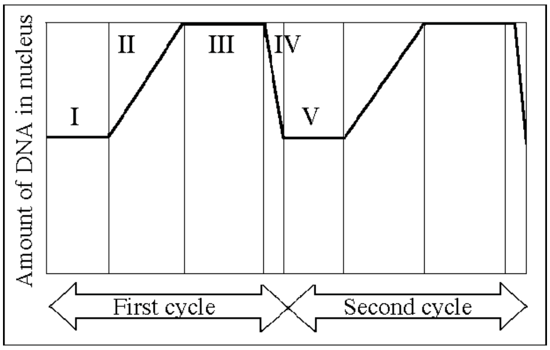
26) What is the correct chromosomal condition for one daughter nucleus at telophase of mitosis?
A) B.
B) C.
C) D.
D) E.

27) If the cell whose nuclear material is shown in the figure given below continues toward completion of mitosis, which of the following events would occur next?



A) Spindle fiber formation.
B) Nuclear envelope breakdown.
C) Formation of telophase nuclei.
D) Synthesis of chromatids.

The following questions (28-30) are based on the figure given below.



28) In the figure above, G_1 is represented by which numbered part(s) of the cycle?
A) I or V.
B) II or IV.
C) III only.
D) V only.

29) In the figure above, which number represents DNA synthesis?
A) I.
B) II.
C) III.
D) IV.

30) In the figure above, at which of the numbered regions would you expect to find cells at metaphase?
A) I and IV.
B) II only.
C) III only.
D) IV only.

Use the data in the accompanying table to answer the following question (31).
The data were obtained from a study of the length of time spent in each phase of the cell cycle by cells of three eukaryotic organisms designated beta, delta, and gamma.

Cell Type	G_1	S	G_2	M
Beta	18	24	12	16
Delta	100	1	0	0
Gamma	18	48	14	20

Minutes Spent in Cell Cycle Phases

31) Which of the following is the best conclusion that concerns the difference between the S phases for beta and gamma cell type?
A) Gamma contains more DNA than beta.
B) Beta and gamma contain the same amount of DNA.
C) Beta cells reproduce asexually.
D) Beta is a plant cell and gamma is an animal cell.

The following information applies to the questions (32-33) below.

Several organisms, primarily protists, have what are called intermediate mitotic organization.

32) These protists are intermediate in what sense?
A) They reproduce by binary fission in their early stages of development and by mitosis when they are mature.
B) They use mitotic division but only have circular chromosomes.
C) They maintain a nuclear envelope during division.
D) None of them form spindles.

33) What is the most probable hypothesis about these intermediate forms of cell division?
A) They represent a form of cell reproduction, which must have evolved completely separately from those of other organisms.
B) They rely on totally different proteins for the processes they undergo.
C) They may be more closely related to plant forms that also have unusual mitosis.
D) They show some but not all of the evolutionary steps toward complete mitosis.

Use the following information to answer the questions (34-35) below.

Nucleotides can be radiolabeled before they are incorporated into newly forming DNA and, therefore, can be assayed to track their incorporation. In a set of experiments, a student of faculty research team used labeled T nucleotides and introduced these into the culture of dividing human cells at specific times.

34) Which of the following questions might be answered by using the method described?
A) How many cells are produced by the culture per hour?
B) What is the length of the S phase of the cell cycle?
C) How many picograms of DNA are made per cell cycle?
D) When do spindle fibers attach to chromosomes?

35) The research team used their experiments to study the incorporation of labeled nucleotides into a culture of lymphocytes and found that the lymphocytes incorporated the labeled nucleotide at a significantly higher level after a pathogen was introduced into the culture. They concluded that _____.
A) The presence of the pathogen made the experiments too contaminated to trust the results.
B) Infection causes lymphocytes to divide more rapidly.
C) Infection causes cell cultures in general to reproduce more rapidly.
D) Infection causes lymphocyte cultures to skip some parts of the cell cycle.

36) Through a microscope, you can see a cell plate beginning to develop across the middle of a cell and nuclei forming on either side of the cell plate. This cell is most likely _____.
A) An animal cell in the process of cytokinesis.
B) A plant cell in the process of cytokinesis.
C) An animal cell in the S phase of the cell cycle.
D) A plant cell in metaphase.

37) Which of the following does NOT occur during mitosis?
A) Condensation of the chromosomes.
B) Replication of the DNA.
C) Spindle formation.
D) Separation of the spindle poles.

38) The drug cytochalasin B blocks the function of actin. Which of the following aspects of the cell cycle would be most disrupted by cytochalasin B?
A) Spindle formation.
B) Spindle attachment to kinetochores.
C) Cell elongation during anaphase.
D) Cleavage furrow formation and cytokinesis.

39) Motor proteins require which of the following to function in the movement of chromosomes toward the poles of the mitotic spindle?
A) Intact centromeres.
B) A microtubule-organizing center.
C) ATP as an energy source.
D) Synthesis of cohesin.

Use the data in the accompanying table to answer the following questions.

The data were obtained from a study of the length of time spent in each phase of the cell cycle by cells of three eukaryotic organisms designated beta, delta, and gamma.

Cell Type	G_1	S	G_2	M
Beta	18	24	12	16
Delta	100	1	0	0
Gamma	18	48	14	20

Minutes Spent in Cell Cycle Phases

40) The best conclusion concerning delta is that the cells _____.
A) Contain no DNA.
B) Contain no RNA.
C) Are in the G0 phase.
D) Divide in the G1 phase.

41) Neurons and some other specialized cells divide infrequently because they _____.
A) No longer have active nuclei.
B) Have entered into G_0.
C) Can no longer bind Cdk to cyclin.
D) Show a drop in MPF concentration.

42) MPF is a dimer consisting of _____.
A) A growth factor and mitotic factor.
B) ATP synthetase and a protease.
C) Cyclin and tubulin.
D) Cyclin and a cyclin-dependent kinase.

43) Cyclin-dependent kinase (Cdk) is _____.
A) Inactive, or "turned off", in the presence of cyclin.
B) Present only during the S phase of the cell cycle.
C) The enzyme that catalyzes the attachment of chromosomes to microtubules.
D) An enzyme that attaches phosphate groups to other proteins.

44) What happens if MPF (mitosis-promoting factor) is introduced into immature frog oocytes that are arrested in G_2?
A) Nothing happens.
B) The cells undergo meiosis.
C) The cells enter mitosis.
D) Cell differentiation is triggered.

45) Once a cell completes mitosis, molecular division triggers must be turned off. What happens to MPF during mitosis?
A) It is completely degraded.
B) It is exported from the cell.
C) The cyclin component of MPF is degraded.
D) The Cdk component of MPF is degraded and exported from the cell.

46) The M-phase checkpoint ensures that all chromosomes are attached to the mitotic spindle. If this does not happen, cells would most likely be arrested in _____.
A) Telophase.
B) Prophase.
C) Prometaphase.
D) Metaphase.

47) Which of the following is released by platelets in the vicinity of an injury?
A) PDGF.
B) MPF.
C) Cyclin.
D) Cdk.

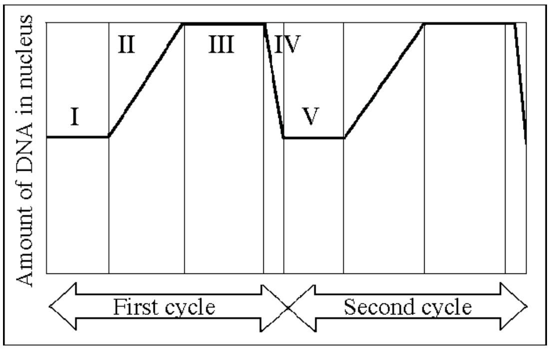
48) Which of the following is a protein synthesized at specific times during the cell cycle that associates with a kinase to form a catalytically active complex?
A) PDGF.
B) MPF.
C) Cyclin.
D) Cdk.

49) Which of the following is a protein maintained at steady levels throughout the cell cycle that requires cyclin to become catalytically active?
A) PDGF.
B) MPF.
C) Cyclin.
D) Cdk.

50) Which of the following triggers the cell's passage past the G_2 checkpoint into mitosis?
A) PDGF.
B) MPF.
C) Cyclin.
D) Cdk.

51) The cyclin component of MPF is destroyed toward the end of which phase?
A) G_1.
B) S.
C) G_2.
D) M.

The following questions are based on the accompanying figure.



52) In the figure above, MPF reaches its highest concentration during this stage:
A) I.
B) II.
C) III.
D) IV.

53) Density-dependent inhibition is explained by which of the following?
A) As cells become more numerous, they begin to squeeze against each other, restricting their size and ability to produce control factors.
B) As cells become more numerous, the cell surface proteins of one cell contact the adjoining cells and they stop dividing.
C) As cells become more numerous, the protein kinases they produce begin to compete with each other, such that the proteins produced by one cell essentially cancel those produced by its neighbor.
D) As cells become more numerous, the level of waste products increases, eventually slowing down metabolism.

54) Besides the ability of some cancer cells to overproliferate, what else could logically result in a tumor?
A) Changes in the order of cell cycle stages.
B) Lack of appropriate cell death.
C) Inability to form spindles.
D) Inability of chromosomes to meet at the metaphase plate.

55) Anchorage dependence of animal cells in vitro or in vivo depends on which of the following?
A) Attachment of spindle fibers to centrioles.
B) Response of the cell cycle controls to signals from the plasma membrane.
C) The binding of cell-surface phospholipids to those of adjoining cells.
D) Response of the cell cycle controls to the binding of cell-surface phospholipids.

56) A research team began a study of a cultured cell line. Their preliminary observations showed them that the cell line did not exhibit either density-dependent inhibition or anchorage dependence. What could they conclude right away?
A) The cells are unable to form spindle microtubules.
B) They have altered the series of cell cycle phases.
C) The cells show characteristics of tumors.
D) They were originally derived from an elderly organism.

57) For a chemotherapeutic drug to be useful for treating cancer cells, which of the following is most desirable?
A) It is safe enough to limit all apoptosis.
B) It does not alter metabolically active cells.
C) It interferes with cells entering G_0.
D) It interferes with rapidly dividing cells.

58) Cells from advanced malignant tumors often have very abnormal chromosomes and an abnormal number of chromosomes. What might explain the association between malignant tumors and chromosomal abnormalities?
A) Cancer cells are no longer density-dependent.
B) Cancer cells are no longer anchorage-dependent.
C) Cell cycle checkpoints are not in place to stop cells with chromosome abnormalities.
D) Transformation introduces new chromosomes into cells.

59) Exposure of zebrafish nuclei to meiotic cytosol resulted in phosphorylation of NEP55 and L68 proteins by cyclin-dependent kinase 2. NEP55 is a protein of the inner nuclear membrane, and L68 is a protein of the nuclear lamina. What is the most likely role of phosphorylation of these proteins in the process of mitosis?
A) They enable the attachment of the spindle microtubules to kinetochore regions of the centromere.
B) They are involved in the disassembly and dispersal of the nucleolus.
C) They are involved in the disassembly of the nuclear envelope.
D) They assist in the movement of the centrosomes to opposite sides of the nucleus.

60) Density-dependent inhibition is a phenomenon in which crowded cells stop dividing at some optimal density and location. This phenomenon involves binding of a cell-surface protein to its counterpart on an adjoining cell's surface. A growth inhibiting signal is sent to both cells, preventing them from dividing. Certain external physical factors can affect this inhibition mechanism.

Select the statement that makes a correct prediction about natural phenomena that could occur during the cell cycle to prevent cell growth.
A) As cells become more numerous, they begin to squeeze against each other, restricting their size and ability to allow cell growth.
B) As cells become more numerous, the protein kinases they produce begin to compete with each other until only one cell has the proteins necessary for growth.
C) As cells become more numerous, the amount of required growth factors and nutrients per cell becomes insufficient to allow for cell growth.
D) As cells become more numerous, more and more of them enter the synthesis part of the cell cycle and duplicate DNA to inhibit cell growth.

CHAPTER 13:

MEIOSIS AND SEXUAL LIFE CYCLES

1) If a horticulturist breeding gardenias succeeds in having a single plant with a particularly desirable set of traits, which of the following would be her most probable and efficient route to establishing a line of such plants?
A) Backtrack through her previous experiments to obtain another plant with the same traits.
B) Breed this plant with another plant with much weaker traits.
C) Clone the plant.
D) Force the plant to self-pollinate to obtain an identical one.

2) Which of the following defines a genome?
A) The complete set of an organism's polypeptides.
B) The complete set of a species' polypeptides.
C) A karyotype.
D) The complete set of an organism's gene and other DNA sequences.

3) Asexual reproduction occurs during _____.
A) Meiosis.
B) Mitosis.
C) Fertilization.
D) Chromosome exchange between organisms of different species.

4) Quaking aspen can send out underground stems for asexual reproduction. Sexual reproduction is not as common, but when it does happen, the haploid gametes have 19 chromosomes. How many chromosomes are in the cells of the underground stems?
A) 9.
B) 10.
C) 19.
D) 38.

5) Which of the following is a true statement about sexual vs. asexual reproduction?
A) Asexual reproduction, but not sexual reproduction, is characteristic of plants and fungi.
B) In sexual reproduction, individuals transmit half of their nuclear genes to each of their offspring.
C) In asexual reproduction, offspring are produced by fertilization without meiosis.
D) Asexual reproduction produces only haploid offspring.

6) At which stage of mitosis are chromosomes usually photographed in the preparation of a karyotype?
A) Prophase.
B) Metaphase.
C) Anaphase.
D) Interphase.

7) Which of the following is true of a species that has a chromosome number of $2n = 16$?
A) The species is diploid with 32 chromosomes per cell.
B) The species has 16 sets of chromosomes per cell.
C) Each diploid cell has eight homologous pairs.
D) A gamete from this species has four chromosomes.

8) Eukaryotic sexual life cycles show tremendous variation. Of the following elements, which do all sexual life cycles have in common?
I. Alternation of generations
II. Meiosis
III. Fertilization
IV. Gametes
V. Spores
A) I, II, and IV.
B) II, III, and IV.
C) II, IV, and V.
D) I, II, III, IV, and V.

9) In a plant's sexual life cycle _____.
A) Sporophytes produce gametes by meiosis.
B) Gametophytes produce gametes by mitosis.
C) Gametophytes produce gametes by meiosis.
D) Sporophytes produce gametes by mitosis.

10) Which of the following is an example of alternation of generations?
A) A grandparent and grandchild each have dark hair, but the parent has blond hair.
B) A diploid plant (sporophyte) produces, by meiosis, a spore that gives rise to a multicellular, haploid pollen grain (gametophyte).
C) A diploid animal produces gametes by meiosis, and the gametes undergo fertilization to produce a diploid zygote.
D) A haploid mushroom produces gametes by mitosis, and the gametes undergo fertilization, which is immediately followed by meiosis.

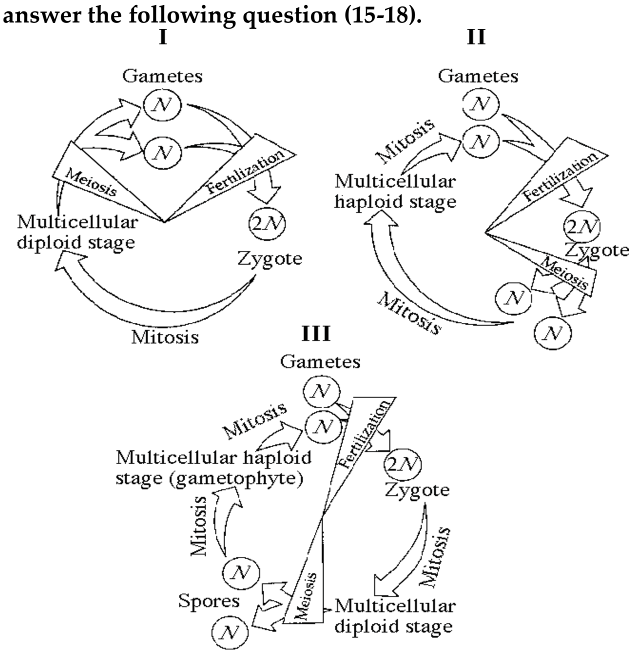
11) A given organism has 46 chromosomes in its karyotype. Therefore, we can conclude that it must _____.
A) Be human.
B) Be an animal.
C) Reproduce sexually.
D) Have gametes with 23 chromosomes.

12) A triploid cell contains sets of three homologous chromosomes. If a cell of a usually diploid species with 42 chromosomes per cell is triploid, this cell would be expected to have which of the following?
A) 63 chromosomes in 31*1/2 pairs.
B) 63 chromosomes in 21 sets of 3.
C) 63 chromosomes, each with three chromatids.
D) 21 chromosome pairs and 21 unique chromosomes.

13) Which of the following might result in a human zygote with 45 chromosomes?
A) An error in either egg or sperm meiotic anaphase.
B) Failure of the egg nucleus to be fertilized by the sperm.
C) Failure of an egg to complete meiosis II.
D) Incomplete cytokinesis during spermatogenesis after meiosis I.

14) In a human karyotype, chromosomes are arranged in 23 pairs. If we choose one of these pairs, such as pair 14, which of the following do the two chromosomes of the pair have in common?
A) Length and position of the centromere only.
B) Length, centromere position, and staining pattern only.
C) Length, centromere position, staining pattern, and traits coded for by their genes.
D) They have nothing in common except that they are X-shaped.

Refer to the life cycles illustrated in the figure below to answer the following question (15-18).



15) Which of the life cycles is typical for animals?
A) I only.
B) II only.
C) III only.
D) I and III.

16) Which of the life cycles is typical for plants and some algae?
A) I only.
B) II only.
C) III only.
D) I and III.

17) Which of the life cycles is typical for most fungi and some protists?
A) I only.
B) II only.
C) III only.
D) I and II.

18) In a life cycle such as that shown in part III of the figure above, if the zygote's chromosome number is 10, which of the following will be true?
A) The sporophyte's chromosome number per cell is 10 and the gametophyte's is 5.
B) The sporophyte's chromosome number per cell is 5 and the gametophyte's is 10.
C) The sporophyte and gametophyte each have 10 chromosomes per cell.
D) The sporophyte and gametophyte each have 5 chromosomes per cell.

19) Homologous chromosomes _____.
A) Are identical.
B) Carry information for the same traits.
C) Carry the same alleles.
D) Align on the metaphase plate in meiosis II.

20) If meiosis produces haploid cells, how is the diploid number restored for those organisms that spend most of their life cycle in the diploid state?
A) DNA replication.
B) Reverse transcription.
C) Synapsis.
D) Fertilization.

21) The human X and Y chromosomes _____.
A) Are both present in somatic cells of males and females.
B) Are the same size and have the same number of genes.
C) Include genes that determine an individual's sex.
D) Are called autosomes.

22) Which of these is a karyotype?
A) A display of all of the cell types in an organism.
B) Organized images of a cell's chromosomes.
C) The appearance of an organism.
D) A display of a cell's mitotic stages.

23) If a cell has completed meiosis I and is just beginning meiosis II, which of the following is an appropriate description of its contents?
A) It has half the amount of DNA.
B) It has half the chromosomes but twice the DNA of the originating cell.
C) It has one-fourth the DNA and one-half the chromosomes as the originating cell.
D) It is identical in content to another cell formed from the same meiosis I event.

24) The somatic cells of a privet shrub each contain 46 chromosomes. How do privet chromosomes differ from the chromosomes of humans, who also have 46?
A) Privet cells cannot reproduce sexually.
B) Privet sex cells have chromosomes that can synapse with human chromosomes in the laboratory.
C) Genes of privet chromosomes are significantly different than those in humans.
D) Privet shrubs must be metabolically more like animals than like other shrubs.

25) After telophase I of meiosis, the chromosomal makeup of each daughter cell is _____.
A) Diploid, and the chromosomes are each composed of a single chromatid.
B) Diploid, and the chromosomes are each composed of two chromatids.
C) Haploid, and the chromosomes are each composed of a single chromatid.
D) Haploid, and the chromosomes are each composed of two chromatids.

26) How do cells at the completion of meiosis compare with cells that are in prophase of meiosis I? They have _____.
A) Half the number of chromosomes and half the amount of DNA.
B) The same number of chromosomes and half the amount of DNA.
C) Half the number of chromosomes and one-fourth the amount of DNA.
D) Half the amount of cytoplasm and twice the amount of DNA.

27) Which of the following happens at the conclusion of meiosis I?
A) Homologous chromosomes of a pair are separated from each other.
B) The chromosome number per cell remains the same.
C) Sister chromatids are separated.
D) Four daughter cells are formed.

28) Sister chromatids separate from each other during _____.
A) Meiosis I only.
B) Meiosis II only.
C) Mitosis and meiosis I.
D) Mitosis and meiosis II.

29) Which of the following occurs in meiosis but not in mitosis?
A) Chromosome replication.
B) Synapsis of chromosomes.
C) Alignment of chromosomes at the equator.
D) Condensation of chromosomes.

30) When we first see chiasmata under a microscope, we know that _____.
A) Meiosis II has occurred.
B) Anaphase II has occurred.
C) Prophase I is occurring.
D) Separation of homologs has occurred.

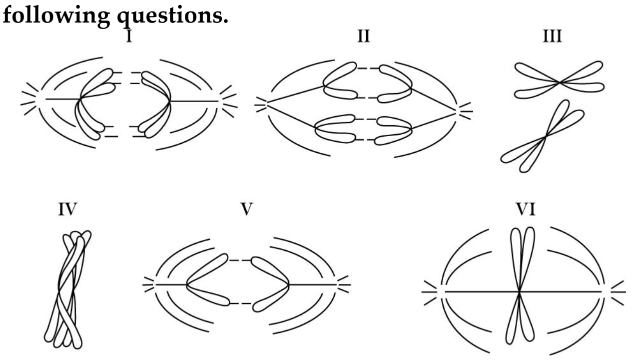
For the following questions (31-32), match the key event of meiosis with the stages listed below.

I. Prophase I	**V. Prophase II**
II. Metaphase I	**VI. Metaphase II**
III. Anaphase I	**VII. Anaphase II**
IV. Telophase I	**VIII. Telophase II**

31) Homologous chromosomes are aligned at the equator of the spindle.
A) I.
B) II.
C) IV.
D) VI.

32) Centromeres of sister chromatids disjoin and chromatids separate.
A) III.
B) IV.
C) V.
D) VII.

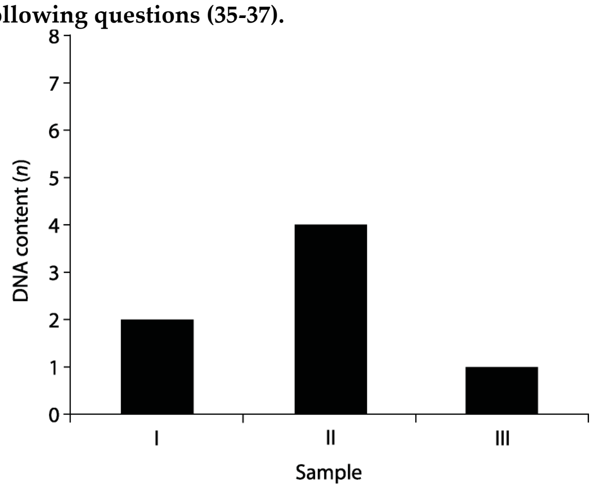
Refer to the drawings in the figure below of a single pair of homologous chromosomes as they might appear during various stages of either mitosis or meiosis, and answer the following questions.



33) Which diagram represents anaphase I of meiosis?
A) I.
B) II.
C) IV.
D) V.

34) Which diagram represents anaphase II of meiosis?
A) I.
B) III.
C) IV.
D) V.

You have isolated DNA from three different cell types of the same organism, determined the relative DNA content for each type, and plotted the results on the graph shown in the figure below. Refer to the graph to answer the following questions (35-37).



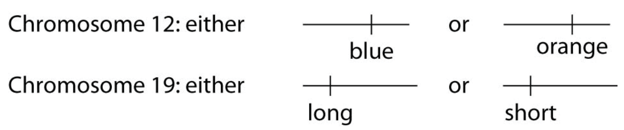
35) Which sample of DNA might be from a nerve cell arrested in G_0 of the cell cycle?
A) I.
B) II.
C) III.
D) Either I or II.

36) Which sample might represent an animal cell in the G_2 phase of the cell cycle?
A) I.
B) II.
C) III.
D) Both I and II.

37) Which sample might represent a zygote?
A) I.
B) II.
C) III.
D) Either I or II.

Refer to the information and figure below to answer the following question (38-40).

A certain (hypothetical) organism is diploid, has either blue or orange wings as the consequence of one of its genes on chromosome 12, and has either long or short antennae as the result of a second gene on chromosome 19, as shown in the figure.



38) A certain female's number 12 chromosomes both have the blue gene and number 19 chromosomes both have the long gene. As cells in her ovaries undergo meiosis, her resulting eggs (ova) may have which of the following?
A) Either two number 12 chromosomes with blue genes or two with orange genes.
B) Either two number 19 chromosomes with long genes or two with short genes.
C) Either one blue or one orange gene in addition to either one long or one short gene.
D) One chromosome 12 with one blue gene and one chromosome 19 with one long gene.

39) If a female of this species has one chromosome 12 with a blue gene and another chromosome 12 with an orange gene, and has both number 19 chromosomes with short genes, she will produce which of the following egg types?
A) Only blue short gene eggs.
B) Only orange short gene eggs.
C) One-half blue short and one-half orange short gene eggs.
D) Three-fourths blue short and one-fourth orange short gene eggs.

40) A female with a paternal set of one orange and one long gene chromosome and a maternal set comprised of one blue and one short gene chromosome is expected to produce which of the following types of eggs after meiosis?
A) All eggs will have maternal types of gene combinations.
B) All eggs will have paternal types of gene combinations.
C) Half the eggs will have maternal and half will have paternal combinations.
D) Each egg has a one-fourth chance of having either blue long, blue short, orange long, or orange short combinations.

41) Somatic cells of roundworms have four individual chromosomes per cell. How many chromosomes would you expect to find in an ovum from a roundworm?
A) Four.
B) Two.
C) Eight.
D) A diploid number.

42) Which of the following can occur by the process of meiosis but not mitosis?
A) Haploid cells fuse to form diploid cells.
B) Haploid cells multiply into more haploid cells.
C) Diploid cells form haploid cells.
D) A diploid cell combines with a haploid cell.

43) In meiosis, homologous chromosomes are separated during _____.
A) Anaphase II.
B) Prophase I.
C) Mitosis.
D) Anaphase I.

44) What is a major difference between meiosis II and mitosis in a diploid animal?
A) Homologues align on the metaphase plate in meiosis II.
B) Sister chromatids separate in mitosis, and homologues separate in meiosis II.
C) Meiosis II occurs in a haploid cell, while mitosis occurs in diploid cells.
D) Crossover takes place in meiosis II.

45) What is a major difference between mitosis and meiosis I in a diploid organism?
A) Sister chromatids separate in mitosis, while homologous pairs of chromosomes separate in meiosis I.
B) Sister chromatids separate in mitosis, while homologous pairs of chromosomes separate in meiosis II.
C) DNA replication takes place prior to mitosis, but not before meiosis I.
D) Only meiosis I results in daughter cells that contain identical genetic information.

46) Crossing over normally takes place during _____.
A) Meiosis II.
B) Meiosis I.
C) Mitosis.
D) Mitosis and meiosis II.

47) For the duration of meiosis I, each chromosome _____.
A) Is paired with a homologous chromosome.
B) Consists of two sister chromatids joined by a centromere.
C) Consists of a single strand of DNA.
D) Is joined with its homologous pair to form a synaptonemal complex.

48) Homologous pairs of chromosomes align opposite of each other at the equator of a cell during _____.
A) Mitosis metaphase.
B) Meiosis metaphase I.
C) Meiosis telophase II.
D) Meiosis metaphase II.

49) Centromeres split and sister chromatids migrate to opposite poles in meiosis _____.
A) Anaphase I.
B) Telophase I.
C) Anaphase II.
D) Telophase II.

50) Quaking aspen can send out underground stems for asexual reproduction. Sexual reproduction is not as common, but when it does happen, the haploid gametes have 19 chromosomes. How many chromosomes are in the cells of the underground stems?
A) 9.
B) 10.
C) 19.
D) 38.

51) Independent assortment of chromosomes occurs during _____.
A) Meiosis I only.
B) Meiosis II only.
C) Mitosis and meiosis I.
D) Mitosis and meiosis II.

52) For a species with a haploid number of 23 chromosomes, how many different combinations of maternal and paternal chromosomes are possible for the gametes?
A) 23.
B) 46.
C) About 1000.
D) About 8 million.

53) Independent assortment of chromosomes is a result of _____.
A) The random way each pair of homologous chromosomes lines up at the metaphase plate during meiosis I.
B) The random combinations of eggs and sperm during fertilization.
C) The random distribution of the sister chromatids to the two daughter cells during anaphase II.
D) The diverse combination of alleles that may be found within any given chromosome.

54) When homologous chromosomes cross over, what occurs?
A) Two chromatids get tangled, resulting in one re-sequencing its DNA.
B) Two sister chromatids exchange identical pieces of DNA.
C) Corresponding segments of non-sister chromatids are exchanged.
D) Maternal alleles are "corrected" to be like paternal alleles and vice versa.

55) How is natural selection related to sexual reproduction as opposed to asexual reproduction?
A) Sexual reproduction results in many new gene combinations, some of which will lead to differential reproduction.
B) Sexual reproduction results new mutations.
C) Sexual reproduction allows the greatest number of offspring to be produced.
D) Sexual reproduction utilizes far less energy than asexual.

56) The diploid number of a roundworm species is 4. Assuming there is no crossover, and random segregation of homologues during meiosis, how many different possible combinations of chromosomes might there be in the offspring (not including variety generated by crossing over)?
A) 4.
B) 8.
C) 16.
D) 64.

57) The bulldog ant has a diploid number of two chromosomes. Therefore, following meiosis, each daughter cell will have a single chromosome. Diversity in this species may be generated by mutations and _____.
A) Crossing over.
B) Independent assortment.
C) Crossing over and independent assortment.
D) Nothing else.

58) The fastest way for organisms to adapt to a changing environment involves _____.
A) Mutation.
B) Asexual reproduction.
C) Sexual reproduction.
D) Movement.

59) Imagine that there are twenty-five different species of protists living in a tide pool. Some of these species reproduce both sexually and asexually, and some of them can reproduce only asexually. The pool gradually becomes infested with disease-causing viruses and bacteria. Which species are more likely to thrive in the changing environment?
A) The sexually reproducing species.
B) The asexually reproducing species.
C) Sexually and asexually reproducing species are equally likely to thrive.
D) Nonreproductive species.

60) In eukaryotes, genetic information is passed to the next generation by processes that include mitosis or meiosis. Which of the explanations identifies the correct process and supports the claim that heritable information is passed from one generation to another?
A) During mitosis, DNA replication occurs twice within the cell cycle to insure a full set of chromosomes within each of the daughter cells produced.
B) Mitosis, followed by cytokinesis, produces daughter cells that are genetically different from the parent cell, thus insuring variation within the population.
C) In asexual reproduction, a single individual is the sole parent and passes copies of its genes to its offspring without the fusion of gametes.
D) Single-celled organisms can fuse their cells, reproducing asexually through mitosis to form new cells that are not identical to the parent cell.

61) Genetic variation leads to genetic diversity in populations and is the raw material for evolution. Biological systems have multiple processes, such as reproduction, that affect genetic variation. They are evolutionarily conserved and shared by various organisms.

Which statement best represents the connection between reproduction and evolution?
A) Plants that use sexual reproduction are rare since this type of reproduction in plants does not contribute to genetic diversity.
B) In order to increase genetic diversity for evolution in sexually reproducing organisms, mutations must occur in the zygote after fertilization.
C) Since prokaryotic organisms reproduce asexually, there is no mechanism for them to add genetic diversity for evolution.
D) Sexual reproduction increases genetic variation because random mutations can be shuffled between organisms.

CHAPTER 14:

MENDEL AND THE GENE IDEA

1) What do we mean when we use the terms monohybrid cross and dihybrid cross?
A) A monohybrid cross involves a single parent, whereas a dihybrid cross involves two parents.
B) A dihybrid cross involves organisms that are heterozygous for two characters that are being studied, and a monohybrid cross involves organisms that are heterozygous for only one character being studied.
C) A monohybrid cross is performed for one generation, whereas a dihybrid cross is performed for two generations.
D) A monohybrid cross results in a 9:3:3:1 ratio whereas a dihybrid cross gives a 3:1 ratio.

2) What was the most significant conclusion that Gregor Mendel drew from his experiments with pea plants?
A) There is considerable genetic variation in garden peas.
B) Traits are inherited in discrete units and are not the results of "blending".
C) Recessive genes occur more frequently in the F_1 generation than do dominant ones.
D) Genes are composed of DNA.

3) How many unique gametes could be produced through independent assortment by an individual with the genotype *AaBbCCDdEE?*
A) 4.
B) 8.
C) 16.
D) 64.

4) The individual with genotype *AaBbCCDdEE* can make many kinds of gametes. Which of the following is the major reason?
A) Recurrent mutations forming new alleles.
B) Crossing over during prophase I.
C) Different possible assortment of chromosomes into gametes.
D) The tendency for dominant alleles to segregate together.

5) Mendel continued some of his experiments into the F_2 or F_3 generation to _____.
A) Obtain a larger number of offspring on which to base statistics.
B) Observe whether or not a recessive trait would reappear.
C) Observe whether or not the dominant trait would reappear.
D) Distinguish which alleles were segregating.

6) Which of the following statements about independent assortment and segregation is correct?
A) The law of independent assortment requires describing two or more genes relative to one another.
B) The law of segregation requires describing two or more genes relative to one another.
C) The law of independent assortment is accounted for by observations of prophase I.
D) The law of segregation is accounted for by anaphase of mitosis.

7) A sexually reproducing animal has two unlinked genes, one for head shape (*H*) and one for tail length (*T*). Its genotype is *HhTt*. Which of the following genotypes is possible in a gamete from this organism?
A) *Hh.*
B) *HhTt.*
C) *T.*
D) *HT.*

8) Mendel accounted for the observation that traits that had disappeared in the F_1 generation reappeared in the F_2 generation by proposing that _____.
A) New mutations were frequently generated in the F_2 progeny, "reinventing" traits that had been lost in the F_1.
B) The mechanism controlling the appearance of traits was different between the F_1 and the F_2 plants.
C) Traits can be dominant or recessive and the recessive traits were obscured by the dominant ones in the F_1.
D) Members of the F_1 generation had only one allele for each trait, but members of the F_2 had two alleles for each trait.

9) The fact that all seven of the pea plant traits studied by Mendel obeyed the principle of independent assortment most probably indicates which of the following?
A) None of the traits obeyed the law of segregation.
B) The diploid number of chromosomes in the pea plants was 7.
C) All of the genes controlling the traits were located on the same chromosome.
D) All of the genes controlling the traits behaved as if they were on different chromosomes.

10) Mendel's observation of the segregation of alleles in gamete formation has its basis in which of the following phases of cell division?
A) Prophase I of meiosis.
B) Anaphase II of meiosis.
C) Metaphase II of meiosis.
D) Anaphase I of meiosis.

11) Mendel's second law of independent assortment has its basis in which of the following events of meiosis I?
A) Synapsis of homologous chromosomes.
B) Crossing over.
C) Alignment of tetrads at the equator.
D) Separation of cells at telophase.

Use the figure and the following description to answer the questions (12-14) below.

In a particular plant, leaf color is controlled by gene locus *D*. Plants with at least one allele *D* have dark green leaves, and plants with the homozygous recessive *dd* genotype have light green leaves. A true-breeding, dark-leaved plant is crossed with a light-leaved one, and the F_1 offspring is allowed to self-pollinate. The predicted outcome of the F_2 is diagrammed in the Punnett square shown in the figure, where 1, 2, 3, and 4 represent the genotypes corresponding to each box within the square.

	D	*d*
D	1	2
d	3	4

12) Which of the boxes marked 1-4 correspond to plants with dark leaves?
A) 1 only.
B) 2 and 3.
C) 4 only.
D) 1, 2, and 3.

13) Which of the boxes marked 1-4 correspond to plants with a heterozygous genotype?
A) 1.
B) 1, 2, and 3.
C) 2 and 3.
D) 2, 3, and 4.

14) Which of the boxes marked 1-4 correspond to plants that will be true-breeding?
A) 1 and 4 only.
B) 2 and 3 only.
C) 1, 2, 3, and 4.
D) 1 only.

15) Skin color in a certain species of fish is inherited via a single gene with four different alleles. How many different types of gametes would be possible in this system?
A) 2.
B) 4.
C) 8.
D) 16.

16) Why did the F_1 offspring of Mendel's classic pea cross always look like one of the two parental varieties?
A) No genes interacted to produce the parental phenotype.
B) Each allele affected phenotypic expression.
C) The traits blended together during fertilization.
D) One allele was dominant.

17) Mendel crossed yellow-seeded and green-seeded pea plants and then allowed the offspring to self-pollinate to produce an F_2 generation. The results were as follows: 6022 yellow and 2001 green (8023 total). The allele for green seeds has what relationship to the allele for yellow seeds?
A) Dominant.
B) Incomplete dominant.
C) Recessive.
D) Codominant.

18) Albinism is an autosomal (not sex-linked) recessive trait. A man and woman are both of normal pigmentation, but both have one parent who is albino (without melanin pigmentation). What is the probability that their first child will be an albino?
A) 0.
B) 1/2.
C) 1/4.
D) 1.

19) Albinism is an autosomal (not sex-linked) recessive trait. A man and woman are both of normal pigmentation and have one child out of three who is albino (without melanin pigmentation). What are the genotypes of the albino's parents?
A) One parent must be homozygous for the recessive allele; the other parent can be homozygous dominant, homozygous recessive, or heterozygous.
B) One parent must be heterozygous; the other parent can be homozygous dominant, homozygous recessive, or heterozygous.
C) Both parents must be heterozygous.
D) One parent must be homozygous dominant; the other parent must be heterozygous.

20) A black guinea pig crossed with an albino guinea pig produced twelve black offspring. When the albino was crossed with a second black animal, six blacks and six albinos were obtained. What is the best explanation for this genetic situation?
A) Albino is recessive; black is dominant.
B) Albino is dominant; black is incompletely dominant.
C) Albino and black are codominant.
D) Albino is recessive; black is codominant.

21) Gray seed color in peas is dominant to white. Assume that Mendel conducted a series of experiments where plants with gray seeds were crossed among themselves, and the following progeny were produced: 302 gray and 98 white.
(a) What is the most probable genotype of each parent?
(b) Based on your answer in (a) above, what genotypic and phenotypic ratios are expected in these progeny?
(Assume the following symbols: G = gray and g = white.)
A) (a) *GG* × *gg*; (b) genotype = 3:1, phenotypic = 1:2:1.
B) (a) *Gg* × *Gg*; (b) genotypic = 1:2:1, phenotypic = 3:1.
C) (a) *GG* × *Gg*; (b) genotypic = 1:2:1, phenotypic = 2:1.
D) (a) *gg* × *Gg*; (b) genotypic = 1:2, phenotypic = 3:1.

22) When Mendel crossed yellow-seeded and green-seeded pea plants, all the offspring were yellow seeded. When he took these F_1 yellow-seeded plants and crossed them to green-seeded plants, what genotypic ratio was expected?
A) 1:2:1.
B) 3:1.
C) 1:1.
D) 1:1:1:1.

23) Black fur in mice (*B*) is dominant to brown fur (*b*). Short tails (*T*) are dominant to long tails (*t*). What fraction of the progeny of crosses *BbTt* × *BBtt* will be expected to have black fur and long tails?
A) 1/16.
B) 3/8.
C) 1/2.
D) 9/16.

24) In certain plants, tall is dominant to short. If a heterozygous plant is crossed with a homozygous tall plant, what is the probability that the offspring will be short?
A) 1.
B) 1/2.
C) 1/4.
D) 0.

25) In the cross *AaBbCc* × *AaBbCc*, what is the probability of producing the genotype *AABBCC*?
A) 1/4.
B) 1/8.
C) 1/16.
D) 1/64.

26) Given the parents *AABBCc* × *AabbCc*, assume simple dominance for each trait and independent assortment. What proportion of the progeny will be expected to phenotypically resemble the first parent with genotype *AABBCc*?
A) 1/4.
B) 3/4.
C) 3/8.
D) 1.

27) Which of the following is the best statement of the use of the addition rule of probability?
A) The probability that two or more independent events will both occur.
B) The probability that either one of two independent events will occur.
C) The probability of producing two or more heterozygous offspring.
D) The likelihood that a trait is due to two or more meiotic events.

28) Which of the following calculations require that you utilize the addition rule?
A) Calculate the probability of black offspring from the cross *AaBb* × *AaBb*, where *B* is the symbol for black.
B) Calculate the probability of children with both cystic fibrosis and polydactyly when parents are each heterozygous for both genes.
C) Calculate the probability of each of four children having cystic fibrosis if the parents are both heterozygous.
D) Calculate the probability of a child having either sickle-cell anemia or cystic fibrosis if parents are each heterozygous for both.

29) Two true-breeding stocks of pea plants are crossed. One parent has red, axial flowers and the other has white, terminal flowers; all F_1 individuals have red, axial flowers. The genes for flower color and location assort independently. Among the F_2 offspring, what is the probability of plants with white axial flowers?
A) 9/16.
B) 1/16.
C) 3/16.
D) 1/4.

30) A man has extra digits (six fingers on each hand and six toes on each foot). His wife and their daughter have a normal number of digits. Having extra digits is a dominant trait. The couple's second child has extra digits. What is the probability that their next (third) child will have extra digits?
A) 1/2.
B) 1/16.
C) 1/8.
D) 3/4.

31) Phenylketonuria is an inherited disease caused by a recessive autosomal allele. If a woman and her husband are both carriers, what is the probability that their first child will be a phenotypically normal girl?
A) 1/4.
B) 1/16.
C) 3/16.
D) 3/8.

32) Assuming independent assortment for all gene pairs, what is the probability that the following parents, *AABbCc* × *AaBbCc*, will produce an *AaBbCc* offspring?
A) 1/2.
B) 1/16.
C) 1/8.
D) 3/4.

33) Suppose two *AaBbCc* individuals are mated. What fraction of the offspring are expected to be homozygous recessive for the three traits (genes are not linked)?
A) 1/4.
B) 1/8.
C) 1/16.
D) 1/64.

34) In cattle, roan coat color (mixed red and white hairs) occurs in the heterozygous (*Rr*) offspring of red (*RR*) and white (*rr*) homozygotes. Which of the following crosses would produce offspring in the ratio of 1 red: 2 roan: 1 white?
A) Red × white.
B) Roan × roan.
C) White × roan
D) Red × roan.

35) Which of the following describes the ability of a single allele to have multiple phenotypic effects?
A) Incomplete dominance.
B) Multiple alleles.
C) Pleiotropy.
D) Epistasis.

36) Which of the following is an example of polygenic inheritance?
A) Pink flowers in snapdragons.
B) The ABO blood group in humans.
C) White and purple flower color in peas.
D) Skin pigmentation in humans.

37) Hydrangea plants of the same genotype are planted in a large flower garden. Some of the plants produce blue flowers and others pink flowers. This can be best explained by which of the following?
A) The knowledge that multiple alleles are involved.
B) The allele for blue hydrangea being completely dominant.
C) The alleles being codominant.
D) Environmental factors such as soil pH.

38) Which of the following provides an example of epistasis?
A) Recessive genotypes for each of two genes (*aabb*) results in an albino corn snake.
B) In rabbits and many other mammals, one genotype (*ee*) prevents any fur color from developing.
C) In Drosophila (fruit flies), white eyes can be due to an X-linked gene or to a combination of other genes.
D) In cacti, there are several genes for the type of spines.

39) Radish flowers may be red, purple, or white. A cross between a red -flowered plant and a white-flowered plant yields all-purple offspring. The part of the radish we eat may be oval or long, with long being the dominant trait. If true-breeding red long radishes are crossed with true-breeding white oval radishes, the F_1 will be expected to be which of the following?
A) Red and long.
B) White and long
C) Purple and long.
D) Purple and oval.

40) Radish flowers may be red, purple, or white. A cross between a red -flowered plant and a white-flowered plant yields all-purple offspring. The flower color trait in radishes is an example of which of the following?
A) A multiple allelic system.
B) Sex linkage.
C) Codominance.
D) Incomplete dominance.

41) Skin color in a certain species of fish is inherited via a single gene with four different alleles. One fish of this type has alleles 1 and 3 (S_1S_3) and its mate has alleles 2 and 4 (S_2S_4). If each allele confers a unit of color darkness such that S_1 has one unit, S_2 has two units, and so on, then what proportion of their offspring would be expected to have five units of color?
A) 1/4.
B) 1/8.
C) 1/2.
D) 0.

42) Gene *S* controls the sharpness of spines in a type of cactus. Cactuses with the dominant allele, *S*, have sharp spines, whereas homozygous recessive ss cactuses have dull spines. At the same time, a second gene, *N*, determines whether or not cactuses have spines. Homozygous recessive nn cactuses have no spines at all. The relationship between genes *S* and *N* is an example of _____.
A) Incomplete dominance.
B) Epistasis.
C) Pleiotropy.
D) Codominance.

43) Gene *S* controls the sharpness of spines in a type of cactus. Cactuses with the dominant allele, *S*, have sharp spines, whereas homozygous recessive ss cactuses have dull spines. At the same time, a second gene, *N*, determines whether or not cactuses have spines. Homozygous recessive nn cactuses have no spines at all. A cross between a true-breeding sharp-spined cactus and a spineless cactus would produce _____.
A) All sharp-spined progeny.
B) 50% sharp-spined, 50% dull-spined progeny.
C) 25% sharp-spined, 50% dull-spined, 25% spineless progeny.
D) It is impossible to determine the phenotypes of the progeny.

44) Gene *S* controls the sharpness of spines in a type of cactus. Cactuses with the dominant allele, *S*, have sharp spines, whereas homozygous recessive ss cactuses have dull spines. At the same time, a second gene, *N*, determines whether or not cactuses have spines. Homozygous recessive *nn* cactuses have no spines at all. If doubly heterozygous *SsNn* cactuses were allowed to self-pollinate, the F_2 would segregate in which of the following ratios?
A) 3 sharp-spined:1 spineless.
B) 1 sharp-spined:2 dull-spined:1 spineless.
C) 1 sharp-spined:1 dull-spined:1 spineless.
D) 9 sharp-spined:3 dull-spined:4 spineless.

45) Feather color in budgies is determined by two different genes, *Y* and *B*, one for pigment on the outside and one for the inside of the feather. *YYBB*, *YyBB*, or *YYBb* is green; *yyBB* or *yyBb* is blue; *YYbb* or *Yybb* is yellow; and *yybb* is white. A blue budgie is crossed with a white budgie. Which of the following results is NOT possible?
A) Green offspring only.
B) Yellow offspring only.
C) Blue offspring only.
D) Green and yellow offspring.

46) Feather color in budgies is determined by two different genes, *Y* and *B*, one for pigment on the outside and one for the inside of the feather. *YYBB*, *YyBB*, or *YYBb* is green; *yyBB* or *yyBb* is blue; *YYbb* or *Yybb* is *yellow*; and *yybb* is white. Two blue budgies were crossed. Over the years, they produced twenty-two offspring, five of which were white. What are the most likely genotypes for the two blue budgies?
A) *yyBB* and *yyBB*.
B) *yyBB* and *yyBb*.
C) *yyBb* and *yyBb*.
D) *yyBb* and *yybb*.

47) A woman who has blood type A positive has a daughter who is type O positive and a son who is type B negative. Rh positive is a trait that shows simple dominance over Rh negative. Which of the following is a possible phenotype for the father?
A) A negative.
B) O negative.
C) B positive.
D) AB negative.

48) A gene for the MN blood group has codominant alleles *M* and *N*. If both children are of blood type M, which of the following is possible?
A) Each parent is either M or MN.
B) Each parent must be type M.
C) Both children are heterozygous for this gene.
D) Neither parent can have the *N* allele.

49) Marfan syndrome in humans is caused by an abnormality of the connective tissue protein fibrillin. Patients are usually very tall and thin, with long spindly fingers, and sometimes ocular problems, such as lens dislocation. Which of the following would you conclude about Marfan syndrome from this information?
A) It is recessive.
B) It is dominant.
C) It is pleiotropic.
D) It is epistatic.

50) In rabbits, the homozygous *CC* is normal, *Cc* results in deformed legs, and *cc* results in very short legs. The genotype *BB* produces black fur, *Bb* brown fur, and *bb* white fur. If a cross is made between brown rabbits with deformed legs and white rabbits with deformed legs, what percentage of the offspring would be expected to have deformed legs and white fur?
A) 25%.
B) 33%.
C) 100%.
D) 50%.

51) In humans, ABO blood types refer to glycoproteins in the membranes of red blood cells. There are three alleles for this autosomal gene: I^A, I^B, and *i*. The I^A allele codes for the A glycoprotein, The I^B allele codes for the B glycoprotein, and the *i* allele doesn't code for any membrane glycoprotein. I^A and I^B are codominant, and *i* is recessive to both I^A and I^B. People with type A blood have the genotypes I^AI^A or I^Ai, people with type B blood are I^BI^B or I^Bi, people with type AB blood are I^AI^B, and people with type O blood are *ii*. If a woman with type AB blood marries a man with type O blood, which of the following blood types could their children possibly have?
A) A and B.
B) AB and O.
C) A, B, and O.
D) A, B, AB, and O.

52) An obstetrician knows that one of her patients is a pregnant woman whose fetus is at risk for a serious disorder that is detectable biochemically in fetal cells. The obstetrician would most reasonably offer which of the following procedures to her patient?
A) Karyotyping of the woman's somatic cells.
B) X-ray.
C) Amniocentesis or CVS.
D) Blood transfusion.

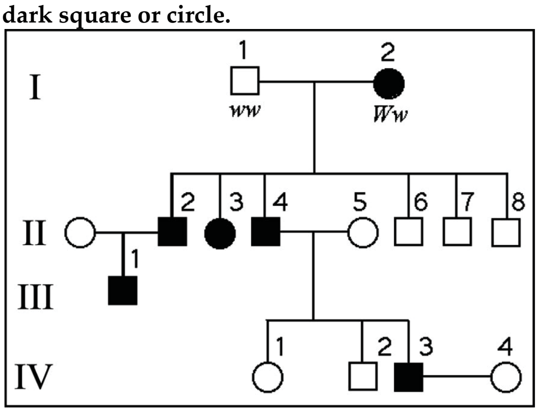
53) In some parts of Africa, the frequency of heterozygosity for the sickle-cell anemia allele is unusually high, presumably because this reduces the frequency of malaria. Such a relationship is related to which of the following?
A) Mendel's law of independent assortment.
B) Mendel's law of segregation.
C) Darwin's explanation of natural selection.
D) The malarial parasite changing the allele.

54) Phenylketonuria (PKU) is a recessive human disorder in which an individual cannot appropriately metabolize the amino acid phenylalanine. This amino acid is not naturally produced by humans. Therefore, the most efficient and effective treatment is which of the following?
A) Feed them the substrate that can be metabolized into this amino acid.
B) Regulate the diet of the affected persons to severely limit the uptake of the amino acid.
C) Feed the patients the missing enzymes in a regular cycle, such as twice per week.
D) Feed the patients an excess of the missing product.

55) Hutchinson-Gilford progeria is an exceedingly rare human genetic disorder in which there is very early senility and death, usually from coronary artery disease, at an average age of 13 years. Patients, who look very old even as children, do not live to reproduce. Which of the following represents the most likely assumption?
A) The disease is autosomal dominant.
B) The disorder will increase in frequency in successive generations within a family.
C) The disorder may be due to mutation in a single protein-coding gene.
D) Each patient will have had at least one affected grandparent or parent.

56) One of two major forms of a human condition called neurofibromatosis (NF 1) is inherited as a dominant gene, although it may range from mildly to very severely expressed. Which of the following is the best explanation for why a young, affected child is the first in her family to be diagnosed?
A) The mother carries the gene but does not express it.
B) One of the parents has a mild expression of the gene.
C) The condition skipped a generation in the family.
D) The child has one more chromosome than either of the parents.

The following questions refer to the pedigree chart in the figure below for a family, some of whose members exhibit the dominant trait, W. Affected individuals are indicated by a dark square or circle.



57) What is the genotype of individual II-5?
A) *WW*.
B) *Ww*.
C) *ww*.
D) *ww* or *Ww*.

58) What is the likelihood that the progeny of IV-3 and IV-4 will have the trait?
A) 0%.
B) 50%.
C) 75%.
D) 100%

59) What is the probability that individual III-1 is Ww?
A) 3/4.
B) 1/4.
C) 2/4.
D) 1.

The figure below shows the pedigree for a family. Dark-shaded symbols represent individuals with one of the two major types of colon cancer. Numbers under the symbols are the individual's age at the time of diagnosis. Males are represented by squares, females by circles.

60) From this pedigree, this trait seems to be inherited _____.
A) From mothers.
B) As an autosomal recessive.
C) As a result of epistasis.
D) As an autosomal dominant.

61) Which of the following statements is a correct explanation for the observation that all offspring exhibit a phenotype for a particular trait that appears to be a blend of the two parental varieties?
A) Neither of the parental genes is dominate over the other.
B) The genes for the trait are dominant in both of the parents.
C) The genes are linked and do not separate during meiosis.
D) The genes for the trait are recessive in both of the parents.

62) The pattern of inheritance (monohybrid, dihybrid, sex-linked, and genes linked on the same chromosomes) can be predicted from data if one is given the parent or offspring genotypes or phenotypes. Two organisms, with genotypes *BbDD* and *BBDd*, are mated. Assuming independent assortment of the *B*/*b* and *D*/*d* genes, determine the genotypic ratios in offspring that would occur.
A) 1/2 *BBDD*, 1/2 *bbdd*.
B) 1/4 *BBDD*, 1/4 *BbDD*, 1/4 *BBDd*, 1/4 *BbDd*.
C) 9/16 *BBDD*, 3/16 *BbDD*, 3/16 *BBDd*, 1/16 *bbdd*.
D) 1/4 *BBDD*, 1/2 *BbDd*, 1/4 *bbdd*.

CHAPTER 15:

THE CHROMOSOMAL BASIS OF INHERITANCE

1) When Thomas Hunt Morgan crossed his red-eyed F_1 generation flies to each other, the F2 generation included both red- and white-eyed flies. Remarkably, all the white-eyed flies were male. What was the explanation for this result?
A) The gene involved is on the Y chromosome.
B) The gene involved is on the X chromosome.
C) The gene involved is on an autosome, but only in males.
D) Other male-specific factors influence eye color in flies.

2) Which of the following is the meaning of the chromosome theory of inheritance as expressed in the early twentieth century?
A) Individuals inherit particular chromosomes attached to genes.
B) Mendelian genes are at specific loci on the chromosome and, in turn, segregate during meiosis.
C) No more than a single pair of chromosomes can be found in a healthy normal cell.
D) Natural selection acts on certain chromosome arrays rather than on genes.

3) Males are more often affected by sex-linked traits than females because _____.
A) Male hormones such as testosterone often alter the effects of mutations on the X chromosome.
B) Female hormones such as estrogen often compensate for the effects of mutations on the X chromosome.
C) X chromosomes in males generally have more mutations than X chromosomes in females.
D) Males are hemizygous for the X chromosome.

4) *SRY* is best described as _____.
A) A gene present on the X chromosome that triggers female development.
B) An autosomal gene that is required for the expression of genes on the Y chromosome.
C) A gene region present on the Y chromosome that triggers male development.
D) An autosomal gene that is required for the expression of genes on the X chromosome.

5) In cats, black fur color is caused by an X-linked allele; the other allele at this locus causes orange color. The heterozygote is tortoiseshell. What kinds of offspring would you expect from the cross of a black female and an orange male?
A) Tortoiseshell females; tortoiseshell males.
B) Black females; orange males.
C) Tortoiseshell females; black males.
D) Orange females; black males.

6) Red-green color blindness is a sex-linked recessive trait in humans. Two people with normal color vision have a color-blind son. What are the genotypes of the parents?
A) X^nX^n and X^nY
B) X^NX^N and X^nY
C) X^NX^N and X^NY
D) X^NX^n and X^NY

7) Cinnabar eyes are a sex-linked, recessive characteristic in fruit flies. If a female having cinnabar eyes is crossed with a wild-type male, what percentage of the F_1 males will have cinnabar eyes?
A) 0%.
B) 25%.
C) 50%.
D) 100%.

8) Normally, only female cats have the tortoiseshell phenotype because _____.
A) A male inherits only one allele of the X-linked gene controlling hair color.
B) The Y chromosome has a gene blocking orange coloration.
C) Only males can have Barr bodies.
D) Multiple crossovers on the Y chromosome prevent orange pigment production.

9) In birds, sex is determined by a *ZW* chromosome scheme. Males are *ZZ* and females are *ZW*. A recessive lethal allele that causes death of the embryo is sometimes present on the Z chromosome in pigeons. What would be the sex ratio in the offspring of a cross between a male that is heterozygous for the lethal allele and a normal female?
A) 2:1 male to female.
B) 1:2 male to female.
C) 1:1 male to female.
D) 3:1 male to female.

10) Sex determination in mammals is due to the *SRY* gene. Which of the following could allow a person with an *XX* karyotype to develop a male phenotype?
A) The loss of the *SRY* gene from an autosome.
B) Translocation of *SRY* to an X chromosome.
C) A person with an extra autosomal chromosome.
D) A person with one normal and one shortened (deleted) X.

11) In humans, clear gender differentiation occurs, not at fertilization, but after the second month of gestation. What is the first event of this differentiation?
A) Formation of testosterone in male embryos.
B) Formation of estrogens in female embryos.
C) Activation of *SRY* in male embryos and masculinization of the gonads.
D) Activation of *SRY* in females and feminization of the gonads.

12) Duchenne muscular dystrophy is a serious condition caused by a recessive allele of a gene on the human X chromosome. The patients have muscles that weaken over time because they have absent or decreased dystrophin, a muscle protein. They rarely live past their twenties. How likely is it for a woman to have this condition?
A) Women can never have this condition.
B) One-fourth of the daughters of an affected man would have this condition.
C) One-half of the daughters of an affected father and a carrier mother could have this condition.
D) Only if a woman is XXX could she have this condition.

13) All female mammals have one active X chromosome per cell instead of two. What causes this?
A) Activation of the *XIST* gene on the X chromosome that will become the Barr body.
B) Activation of the *BARR* gene on one X chromosome, which then becomes inactive.
C) Inactivation of the *XIST* gene on the X chromosome derived from the male parent.
D) Attachment of methyl (CH3) groups to the X chromosome that will remain active.

14) A man who is an achondroplastic dwarf with normal vision marries a color-blind woman of normal height. The man's father was six feet tall, and both the woman's parents were of average height. Achondroplastic dwarfism is autosomal dominant, and red-green color blindness is X-linked recessive.
How many of their daughters might be expected to be color-blind dwarfs?
A) None.
B) Half.
C) One out of four.
D) Three out of four.

15) A man who is an achondroplastic dwarf with normal vision marries a color-blind woman of normal height. The man's father was six feet tall, and both the woman's parents were of average height. Achondroplastic dwarfism is autosomal dominant, and red-green color blindness is X-linked recessive.
What proportion of their sons would be color-blind and of normal height?
A) None.
B) Half.
C) One out of four.
D) All.

16) Pseudohypertrophic muscular dystrophy is a human disorder that causes gradual deterioration of the muscles. Only boys are affected, and they are always born to phenotypically normal parents. Due to the severity of the disease, the boys die in their teens. Is this disorder likely to be caused by a dominant or recessive allele? Is its inheritance sex-linked or autosomal?
A) Dominant, sex-linked.
B) Recessive, autosomal.
C) Recessive, sex-linked.
D) Incomplete dominant, sex-linked.

17) In birds, sex is determined by a ZW chromosome scheme that is much like the typical XY scheme seen in humans and many other organisms, except that the system is reversed: Males are ZZ (similar to XX in humans) and females are ZW (similar to XY in humans). A lethal recessive allele that causes death of the embryo occurs on the Z chromosome in pigeons. What would be the sex ratio in the offspring of a cross between a male heterozygous for the lethal allele and a normal female?
A) 1:1 male to female.
B) 3:1 male to female.
C) 1:2 male to female.
D) 2:1 male to female.

18) A recessive allele on the X chromosome is responsible for red-green color blindness in humans. A woman with normal vision whose father is color blind marries a color-blind male. What is the probability that this couple's first son will be color blind?
A) 1/4.
B) 1/2.
C) 2/3.
D) 3/4.

19) A man who carries an allele of an X-linked gene will pass it on to _____.
A) All of his daughters.
B) Half of his daughters.
C) All of his sons.
D) All of his children.

20) Glucose-6-phosphate dehydrogenase deficiency (*G6PD*) is inherited as a recessive allele of an X-linked gene in humans. A woman whose father suffered from *G6PD* marries a normal man.
(a) What proportion of their sons is expected to be *G6PD*?
(b) If the husband was not normal but was *G6PD* deficient, would you change your answer in part (a)?
A) (a) 100%; (b) no.
B) (a) 1/2; (b) yes.
C) (a) 1/2; (b) no.
D) (a) zero; (b) no.

21) In a Drosophila experiment, a cross was made between homozygous wild-type females and yellow-bodied males. All of the resulting F_1s were phenotypically wild type. However, adult flies of the F_2 generation (resulting from matings of the F_1s) had the characteristics shown in the figure below. Consider the following questions:

Sex	Phenotype	Number
male	wild	123
male	yellow	116
female	wild	240

(a) Is the mutant allele for yellow body recessive or dominant?
(b) Is the yellow locus autosomal (not X-linked) or X-linked?
A) (a) recessive; (b) X-linked.
B) (a) recessive; (b) not X-linked.
C) (a) dominant; (b) X-linked.
D) (a) dominant; (b) not X-linked.

22) Sturtevant provided genetic evidence for the existence of four pairs of chromosomes in *Drosophila* in which of these ways?
A) There are four major functional classes of genes in *Drosophila.*
B) *Drosophila* genes cluster into four distinct groups of linked genes.
C) The overall number of genes in *Drosophila* is a multiple of four.
D) *Drosophila* genes have, on average, four different alleles.

23) Which of the following statements is true of linkage?
A) The closer two genes are on a chromosome, the lower the probability that a crossover will occur between them.
B) The observed frequency of recombination of two genes that are far apart from each other has a maximum value of 100%.
C) All of the traits that Mendel studied seed color, pod shape, flower color, and others are due to genes linked on the same chromosome.
D) Linked genes are found on different chromosomes.

24) How would one explain a testcross involving F_1 dihybrid flies in which more parental-type offspring than recombinant-type offspring are produced?
A) The two genes are closely linked on the same chromosome.
B) The two genes are linked but on different chromosomes.
C) Recombination did not occur in the cell during meiosis.
D) Both of the characters are controlled by more than one gene.

25) What does a frequency of recombination of 50% indicate?
A) The two genes are likely to be located on different chromosomes.
B) All of the offspring have combinations of traits that match one of the two parents.
C) The genes are located on sex chromosomes.
D) Abnormal meiosis has occurred.

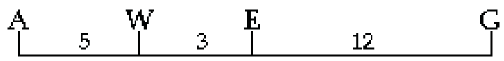
26) What is the definition of one map unit?
A) The physical distance between two linked genes.
B) A 1% frequency of recombination between two genes.
C) 1 nanometer of distance between two genes.
D) The recombination frequency between two genes assorting independently.

27) Recombination between linked genes comes about for what reason?
A) Nonrecombinant chromosomes break and then rejoin with one another.
B) Independent assortment sometimes fails.
C) Linked genes travel together at anaphase.
D) Crossovers between these genes result in chromosomal exchange.

28) What is an adaptive advantage of recombination between linked genes?
A) Recombination is required for independent assortment.
B) Recombination must occur or genes will not assort independently.
C) New allele combinations are acted upon by natural selection.
D) The forces on the cell during meiosis II results in recombination.

29) Map units on a linkage map cannot be relied upon to calculate physical distances on a chromosome for which of the following reasons?
A) The frequency of crossing over varies along the length of the chromosome.
B) The relationship between recombination frequency and map units is different in every individual.
C) Physical distances between genes change during the course of the cell cycle.
D) The gene order on the chromosomes is slightly different in every individual.

The following is a map of four genes on a chromosome.



30) Between which two genes would you expect the highest frequency of recombination?
A) *A* and *W.*
B) *E* and *G.*
C) *A* and *E.*
D) *A* and *G.*

31) In a series of mapping experiments, the recombination frequencies for four different linked genes of *Drosophila* were determined as shown in the figure below. What is the order of these genes on a chromosome map?

b = black body
cn = cinnabar eyes
rb = reduced bristles
vg = vestigial wings

Gene				
b	0			
cn	9	0		
rb	3.5	6.5	0	
vg	19	9.0	16	0
	b	*cn*	*rb*	*vg*

The numbers in the boxes are the recombination frequencies in between the genes (in percent).

A) *rb-cn-vg-b*.
B) *cn-rb-b-vg*.
C) *b-rb-cn-vg*.
D) *vg-cn-b-rb*.

Use the following information to answer the questions (32-34) below.

A plantlike organism on the planet Pandora can have three recessive genetic traits: bluish leaves, due to an allele (*a*) of gene *A*; a feathered stem, due to an allele (*b*) of gene *B*; and hollow roots due to an allele (*c*) of gene *C*. The three genes are linked and recombine as follows:

A geneticist did a testcross with an organism that had been found to be heterozygous for the three recessive traits and she was able to identify progeny of the following phenotypic distribution (+ = wild type):

Phenotypes	Leaves	Stems	Roots	Number
1	a	+	+	14
2	a	+	c	0
3	a	b	+	32
4	a	b	c	440
5	+	b	+	0
6	+	b	c	16
7	+	+	c	28
8	+	+	+	470
			Total	1000

32) Which of the following are the phenotypes of the parents in this cross?
A) 2 and 5.
B) 1 and 6.
C) 4 and 8.
D) 3 and 7.

33) Which of the progeny phenotypes will require recombination between genes *A* and *B*?
A) 1, 2, 5, and 6.
B) 1, 3, 6, and 7.
C) 2, 4, 5, and 8.
D) 2, 3, 5, and 7.

34) If recombination frequency is equal to distance in map units, what is the approximate distance between genes *A* and *B*?
A) 3 map units.
B) 6 map units.
C) 15 map units.
D) 30 map units.

35) What is the greatest benefit of having used a testcross for this experiment?
A) The homozygous recessive parents are obvious to the naked eye.
B) The homozygous parents are the only ones whose crossovers make a difference.
C) The phenotypes of the progeny reveal the allelic content of the gamete from the heterozygous parent.
D) All of the progeny will be heterozygous.

36) The greatest distance among the three genes is between *a* and *c*. What does this mean?
A) Gene c is between *a* and *b*.
B) Genes are in the order: *a—b—c*.
C) Gene *a* is not recombining with *c*.
D) Gene *a* is between *b* and *c*.

37) What is the reason that closely linked genes are typically inherited together?
A) They are located close together on the same chromosome.
B) The number of genes in a cell is greater than the number of chromosomes.
C) Alleles are paired together during meiosis.
D) Genes align that way during metaphase I of meiosis.

38) A homozygous tomato plant with red fruit and yellow flowers was crossed with a homozygous tomato plant with golden fruit and white flowers. The F_1 all had red fruit and yellow flowers. The F_1 were testcrossed by crossing them to homozygous recessive individuals and the following offspring were obtained:

Red fruit and yellow flowers = 41
Red fruit and white flowers = 7
Golden fruit and yellow flowers = 8
Golden fruit and white flowers = 44

How many map units separate these genes?
A) 17.6.
B) 15.
C) 17.1.
D) 18.1.

39) In *Drosophila melanogaster*, vestigial wings are caused by a recessive allele of a gene that is linked to a gene with a recessive allele that causes black body color. Morgan crossed black-bodied, normal-winged females and gray-bodied, vestigial-winged males. The F_1 were all gray bodied, normal winged. The F_1 females were crossed to homozygous recessive males to produce testcross progeny. Morgan calculated the map distance to be 17 map units. Which of the following is correct about the testcross progeny?
A) Black-bodied, normal-winged flies = 17% of the total.
B) Black-bodied, normal-winged flies PLUS gray-bodied, vestigial-winged flies = 17% of the total.
C) Gray-bodied, normal-winged flies PLUS black-bodied, vestigial-winged flies = 17% of the total.
D) Black-bodied, vestigial-winged flies = 17% of the total.

40) If cell X enters meiosis, and nondisjunction of one chromosome occurs in one of its daughter cells during meiosis II, what will be the result at the completion of meiosis?
A) All the gametes descended from cell X will be diploid.
B) Half of the gametes descended from cell X will be $n + 1$, and half will be $n - 1$.
C) 1/4 of the gametes descended from cell X will be $n + 1$, 1/4 will be n - 1, and 1/2 will be n.
D) Two of the four gametes descended from cell X will be haploid, and two will be diploid.

41) One possible result of chromosomal breakage is for a fragment to join a nonhomologous chromosome. What is this alteration called?
A) Deletion.
B) Inversion.
C) Translocation.
D) Duplication.

42) A nonreciprocal crossover causes which of the following products?
A) Deletion only.
B) Duplication only.
C) Nondisjunction.
D) Deletion and duplication.

43) Of the following human aneuploidies, which is the one that generally has the most severe impact on the health of the individual?
A) 47, trisomy 21.
B) 47, XXY.
C) 47, XXX.
D) 45, X.

44) A phenotypically normal prospective couple seeks genetic counseling because the man knows that he has a translocation of a portion of his chromosome 4 that has been exchanged with a portion of his chromosome 12. Although his translocation is balanced, he and his wife want to know the probability that his sperm will be abnormal. What is your prognosis regarding his sperm?
A) 1/4 will carry the two normal chromosomes, 4 and 12, 1/4 will have only the two translocation chromosomes and no normal chromosomes 4 and 12, and 1/2 will have both normal and translocated chromosomes.
B) All will carry the same translocation as the father.
C) None will carry the translocation.
D) 1/2 will be normal and the rest will have the father's translocation.

45) Abnormal chromosomes are frequently found in malignant tumors. Errors such as translocations may place a gene in close proximity to different control regions. Which of the following might then occur to make the cancer worse?
A) An increase in nondisjunction.
B) Expression of inappropriate gene products.
C) A decrease in mitotic frequency.
D) Failure of the cancer cells to multiply.

46) A couple has a child with Down syndrome. The mother is 39 years old at the time of delivery. Which of the following is the most probable cause of the child's condition?
A) The woman inherited this tendency from her parents.
B) The mother had a chromosomal duplication.
C) One member of the couple underwent nondisjunction in somatic cell production.
D) The mother most likely underwent nondisjunction during gamete production.

47) What is a syndrome?
A) A characteristic facial appearance.
B) A trait that leads to cancer at some stage in life.
C) A group of traits typically found in conjunction with a particular chromosomal aberration or gene mutation.
D) A specific characteristic that appears in conjunction with one specific aneuploidy.

48) Which of the following is known as a Philadelphia chromosome?
A) A human chromosome 22 that has had a specific translocation.
B) A human chromosome 9 that is found only in one type of cancer.
C) An animal chromosome found primarily in the mid-Atlantic area of the United States.
D) A chromosome found only in mitochondria.

49) Which of the following is generally true of aneuploidies in newborns?
A) A monosomy is more frequent than a trisomy.
B) Monosomy X is the only viable monosomy known to occur in humans.
C) Human aneuploidy usually conveys an adaptive advantage in humans.
D) An aneuploidy resulting in the deletion of a chromosome segment is less serious than a duplication.

50) A woman is found to have forty-seven chromosomes, including three X chromosomes. Which of the following describes her expected phenotype?
A) A female with masculine characteristics such as facial hair.
B) An apparent male who is sterile.
C) Healthy female of slightly above-average height.
D) A sterile female.

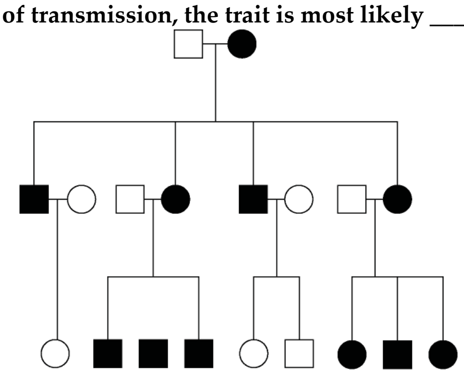
51) Which of the following is an example of monosomy?
A) Turner's syndrome.
B) Klinefelter's syndrome.
C) Down syndrome.
D) Trisomy X.

52) Genomic imprinting is generally due to the addition of methyl (—CH_3) groups to C nucleotides and chemical histone changes to silence a given gene. If this depends on the sex of the parent who transmits the gene, which of the following must be true?
A) Genes required for early development stages must not be imprinted.
B) Methylation of this kind must occur more in males than in females.
C) Methylation must be reversible in ovarian and testicular cells.
D) The imprints are transmitted only to gamete-producing cells.

53) Correns found that the inheritance of variegated color on the leaves of certain plants was determined only by the maternal parent. What phenomenon explains this pattern?
A) Ribosome structure.
B) Chloroplast inheritance.
C) Genomic imprinting.
D) Sex-linkage.

54) Mitochondrial DNA is primarily involved in coding for proteins needed for protein complexes of the electron transport chain and ATP synthase. Therefore, mutations in mitochondrial genes would most affect _____.
A) DNA synthesis in cells of the immune system.
B) The movement of oxygen into erythrocytes.
C) Generation of ATP in muscle cells.
D) The storage of urine in the urinary bladder.

55) The pedigree in the figure below shows the transmission of a trait in a particular family. Based on this pattern of transmission, the trait is most likely _____.



A) Mitochondrial.
B) Sex-linked dominant.
C) Sex-linked recessive.
D) Autosomal dominant.

56) A certain kind of snail can have a right-handed direction of shell coiling (*DD* or *Dd*) or left-handed coiling (*dd*). However, if direction of coiling is due to a protein deposited by the mother in the egg cytoplasm, then a *Dd* egg-producing snail and a *dd* sperm-producing snail will have offspring of which genotype(s) and phenotype(s)?
A) 1/2 *Dd*: 1/2 *dd*; all right-coiling.
B) All *Dd*; all right-coiling.
C) 1/2 *Dd*: 1/2 dd; half right-coiling and half left-coiling.
D) All *Dd*; half right-coiling and half left-coiling.

57) During meiosis, a defect occurs in a cell that results in the failure of microtubules, spindle fibers, to bind at the kinetochores, a protein structure on chromatids where the spindle fibers attach during cell division to pull sister chromatids apart. Which of the following is the most likely result of such a defect?
A) New microtubules with more effective binding capabilities to kinetochores will be synthesized to compensate for the defect.
B) Excessive cell divisions will occur resulting in cancerous tumors and an increase in the chromosome numbers known as polyploidy.
C) The defect will be bypassed in order to and ensure normal chromosome distribution in the new cells.
D) The resulting cells will not receive the correct number of chromosomes in the gametes, a condition known as aneuploidy.

58) Inheritance patterns cannot always be explained by Mendel's models of inheritance. If a pair of homologous chromosomes fails to separate during meiosis I, select the choice that shows the chromosome number of the four resulting gametes with respect to the normal haploid number (n)?
A) n+1; n+1; n-1; n-1
B) n+1; n-1; n; n
C) n+1; n-1; n-1; n-1
D) n+1; n+1; n; n

CHAPTER 16:

THE MOLECULAR BASIS OF INHERITANCE

1) In his transformation experiments, what did Griffith observe?
A) Mixing a heat-killed pathogenic strain of bacteria with a living nonpathogenic strain can convert some of the living cells into the pathogenic form.
B) Mixing a heat-killed nonpathogenic strain of bacteria with a living pathogenic strain makes the pathogenic strain nonpathogenic.
C) Infecting mice with nonpathogenic strains of bacteria makes them resistant to pathogenic strains.
D) Mice infected with a pathogenic strain of bacteria can spread the infection to other mice.

2) How do we describe transformation in bacteria?
A) The creation of a strand of DNA from an RNA molecule.
B) The creation of a strand of RNA from a DNA molecule.
C) The infection of cells by a phage DNA molecule.
D) Assimilation of external DNA into a cell.

3) After mixing a heat-killed, phosphorescent (light-emitting) strain of bacteria with a living, nonphosphorescent strain, you discover that some of the living cells are now phosphorescent. Which observation(s) would provide the best evidence that the ability to phosphoresce is a heritable trait?
A) Evidence that DNA was passed from the heat-killed strain to the living strain.
B) Evidence that protein passed from the heat-killed strain to the living strain.
C) Especially bright phosphorescence in the living strain.
D) Phosphorescence in descendants of the living cells.

4) In trying to determine whether DNA or protein is the genetic material, Hershey and Chase made use of which of the following facts?
A) DNA contains sulfur, whereas protein does not.
B) DNA contains phosphorus, whereas protein does not.
C) DNA contains nitrogen, whereas protein does not.
D) DNA contains purines, whereas protein includes pyrimidines.

5) Which of the following investigators were responsible for the following discovery? In DNA from any species, the amount of adenine equals the amount of thymine, and the amount of guanine equals the amount of cytosine.
A) Alfred Hershey and Martha Chase.
B) Oswald Avery, Maclyn McCarty, and Colin MacLeod.
C) Erwin Chargaff.
D) Matthew Meselson and Franklin Stahl.

6) Cytosine makes up 42% of the nucleotides in a sample of DNA from an organism. What percentage of the nucleotides in this sample will be thymine?
A) 8%.
B) 16%.
C) 42%.
D) 58%

7) It became apparent to Watson and Crick after completion of their model that the DNA molecule could carry a vast amount of hereditary information in which of the following?
A) Sequence of bases.
B) Phosphate-sugar backbones.
C) Complementary pairing of bases.
D) Side groups of nitrogenous bases.

8) In an analysis of the nucleotide composition of DNA, which of the following will be found?
A) A = C.
B) A = G and C = T.
C) A + C = G + T.
D) G + C = T + A.

9) For a science fair project, two students decided to repeat the Hershey and Chase experiment, with modifications. They decided to label the nitrogen of the DNA, rather than the phosphate. They reasoned that each nucleotide has only one phosphate and two to five nitrogens. Thus, labeling the nitrogens would provide a stronger signal than labeling the phosphates. Why won't this experiment work?
A) There is no radioactive isotope of nitrogen.
B) Radioactive nitrogen has a half-life of 100,000 years, and the material would be too dangerous for too long.
C) Although there are more nitrogens in a nucleotide, labeled phosphates actually have sixteen extra neutrons; therefore, they are more radioactive.
D) Amino acids (and thus proteins) also have nitrogen atoms; thus, the radioactivity would not distinguish between DNA and proteins.

10) Hershey and Chase set out to determine what molecule served as the unit of inheritance. They completed a series of experiments in which *E. coli* was infected by a T2 virus. Which molecular component of the T2 virus actually ended up inside the cell?
A) Protein.
B) RNA.
C) Ribosome.
D) DNA.

11) In the polymerization of DNA, a phosphodiester bond is formed between a phosphate group of the nucleotide being added and _____ of the last nucleotide.
A) The 5′ phosphate.
B) C_6.
C) The 3′ OH.
D) A nitrogen from the nitrogen-containing base.

12) Replication in prokaryotes differs from replication in eukaryotes for which of the following reasons?
A) Prokaryotic chromosomes have histones, whereas eukaryotic chromosomes do not.
B) Prokaryotic chromosomes have a single origin of replication, whereas eukaryotic chromosomes have many.
C) The rate of elongation during DNA replication is slower in prokaryotes than in eukaryotes.
D) Prokaryotes produce Okazaki fragments during DNA replication, but eukaryotes do not.

13) What is meant by the description "antiparallel" regarding the strands that make up DNA?
A) The twisting nature of DNA creates nonparallel strands.
B) The 5' to 3' direction of one strand runs counter to the 5′ to 3′ direction of the other strand.
C) Base pairings create unequal spacing between the two DNA strands.
D) One strand contains only purines and the other contains only pyrimidines.

14) Suppose you are provided with an actively dividing culture of *E. coli* bacteria to which radioactive thymine has been added. What would happen if a cell replicates once in the presence of this radioactive base?
A) One of the daughter cells, but not the other, would have radioactive DNA.
B) Neither of the two daughter cells would be radioactive.
C) All four bases of the DNA would be radioactive.
D) DNA in both daughter cells would be radioactive.

15) In *E. coli*, there is a mutation in a gene called *dnaB* that alters the helicase that normally acts at the origin. Which of the following would you expect as a result of this mutation?
A) Additional proofreading will occur.
B) No replication fork will be formed.
C) Replication will occur via RNA polymerase alone.
D) Replication will require a DNA template from another source.

16) In *E. coli*, which enzyme catalyzes the elongation of a new DNA strand in the 5′ → 3′ direction?
A) Primase.
B) DNA ligase.
C) DNA polymerase III.
D) Helicase.

17) Eukaryotic telomeres replicate differently than the rest of the chromosome. This is a consequence of which of the following?
A) The evolution of telomerase enzyme.
B) DNA polymerase that cannot replicate the leading strand template to its 5′ end.
C) Gaps left at the 5′ end of the lagging strand.
D) Gaps left at the 3′ end of the lagging strand because of the need for a primer.

18) How does the enzyme telomerase meet the challenge of replicating the ends of linear chromosomes?
A) It adds a single 5′ cap structure that resists degradation by nucleases.
B) It causes specific double-strand DNA breaks that result in blunt ends on both strands.
C) It catalyzes the lengthening of telomeres, compensating for the shortening that could occur during replication without telomerase activity.
D) It adds numerous GC pairs, which resist hydrolysis and maintain chromosome integrity.

19) The DNA of telomeres has been highly conserved throughout the evolution of eukaryotes. This most likely reflects _____.
A) The low frequency of mutations occurring in this DNA.
B) Continued evolution of telomeres.
C) That new mutations in telomeres have been advantageous.
D) A critical function of telomeres.

20) At a specific area of a chromosome, the sequence of nucleotides below is present where the chain opens to form a replication fork:

3′ C C T A G G C <u>T</u> G C A A T C C 5′

An RNA primer is formed starting at the underlined T (<u>T</u>) of the template. Which of the following represents the primer sequence?
A) 5′ G C C T A G G 3′.
B) 5′ A C G T T A G G 3′.
C) 5′ A C G U U A G G 3′.
D) 5′ G C C U A G G 3′.

21) In *E. coli*, to repair a thymine dimer by nucleotide excision repair, in which order do the necessary enzymes act?
A) Nuclease, DNA polymerase III, RNA primase.
B) Helicase, DNA polymerase I, DNA ligase.
C) DNA ligase, nuclease, helicase.
D) Nuclease, DNA polymerase I, DNA ligase.

22) In *E. coli*, what is the function of DNA polymerase III?
A) To unwind the DNA helix during replication.
B) To seal together the broken ends of DNA strands.
C) To add nucleotides to the 3′ end of a growing DNA strand.
D) To degrade damaged DNA molecules.

23) The difference between ATP and the nucleoside triphosphates used during DNA synthesis is that _____.
A) The nucleoside triphosphates have the sugar deoxyribose; ATP has the sugar ribose.
B) The nucleoside triphosphates have two phosphate groups; ATP has three phosphate groups.
C) ATP contains three high-energy bonds; the nucleoside triphosphates have two.
D) ATP is found only in human cells; the nucleoside triphosphates are found in all animal and plant cells.

24) The leading and the lagging strands differ in that _____.
A) The leading strand is synthesized in the same direction as the movement of the replication fork, and the lagging strand is synthesized in the opposite direction.
B) The leading strand is synthesized by adding nucleotides to the 3′ end of the growing strand, and the lagging strand is synthesized by adding nucleotides to the 5′ end.
C) The lagging strand is synthesized continuously, whereas the leading strand is synthesized in short fragments that are ultimately stitched together.
D) The leading strand is synthesized at twice the rate of the lagging strand.

25) A new DNA strand elongates only in the 5′ to 3′ direction because _____.
A) DNA polymerase begins adding nucleotides at the 5′ end of the template.
B) The polarity of the DNA molecule prevents addition of nucleotides at the 3′ end.
C) Replication must progress toward the replication fork.
D) DNA polymerase can add nucleotides only to the free 3′ end.

26) What is the function of topoisomerase?
A) Relieving strain in the DNA ahead of the replication fork.
B) Elongating new DNA at a replication fork by adding nucleotides to the existing chain.
C) Unwinding of the double helix.
D) Stabilizing single-stranded DNA at the replication fork.

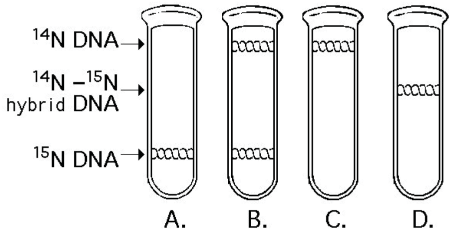
27) What is the role of DNA ligase in the elongation of the lagging strand during DNA replication?
A) It synthesizes RNA nucleotides to make a primer.
B) It joins Okazaki fragments together.
C) It unwinds the parental double helix.
D) It stabilizes the unwound parental DNA.

28) Which of the following help(s) to hold the DNA strands apart while they are being replicated?
A) Primase.
B) Ligase.
C) DNA polymerase.
D) Single-strand DNA binding proteins.

29) Individuals with the disorder xeroderma pigmentosum are hypersensitive to sunlight. This occurs because their cells cannot _____.
A) Replicate DNA.
B) Undergo mitosis.
C) Exchange DNA with other cells.
D) Repair thymine dimers.

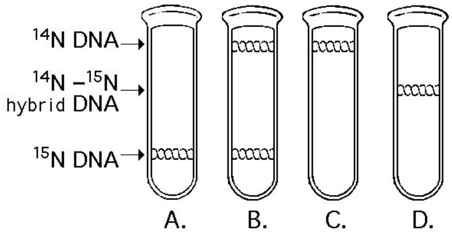
30) Which of the following would you expect of a eukaryote lacking telomerase?
A) A high probability of somatic cells becoming cancerous.
B) An inability to produce Okazaki fragments.
C) An inability to repair thymine dimers.
D) A reduction in chromosome length in gametes.

31) In the late 1950s, Meselson and Stahl grew bacteria in a medium containing "heavy" nitrogen (^{15}N) and then transferred them to a medium containing ^{14}N. Which of the results in the figure below would be expected after one round of DNA replication in the presence of ^{14}N?



A) A
B) B.
C) C.
D) D.

32) A space probe returns with a culture of a microorganism found on a distant planet. Analysis shows that it is a carbon-based life-form that has DNA. You grow the cells in 15N medium for several generations and then transfer them to 14N medium. Which pattern in the figure below would you expect if the DNA was replicated in a conservative manner?



A) A.
B) B.
C) C.
D) D.

33) After the first replication was observed in their experiments testing the nature of DNA replication, Meselson and Stahl could be confident of which of the following conclusions?
A) Replication is semi-conservative.
B) Replication is not dispersive.
C) Replication is not conservative.
D) Replication is neither dispersive nor conservative.

34) You briefly expose bacteria undergoing DNA replication to radioactively labeled nucleotides. When you centrifuge the DNA isolated from the bacteria, the DNA separates into two classes. One class of labeled DNA includes very large molecules (thousands or even millions of nucleotides long), and the other includes short stretches of DNA (several hundred to a few thousand nucleotides in length). These two classes of DNA probably represent _____.
A) Leading strands and Okazaki fragments.
B) Lagging strands and Okazaki fragments.
C) Okazaki fragments and RNA primers.
D) Leading strands and RNA primers.

35) Within a double-stranded DNA molecule, adenine forms hydrogen bonds with thymine and cytosine forms hydrogen bonds with guanine. This arrangement _____.
A) Allows variable width of the double helix.
B) Permits complementary base pairing.
C) Determines the tertiary structure of a DNA molecule.
D) Determines the type of protein produced.

36) Semiconservative replication involves a template. What is the template?
A) Single-stranded binding proteins.
B) DNA polymerase.
C) One strand of the DNA molecule.
D) An RNA molecule.

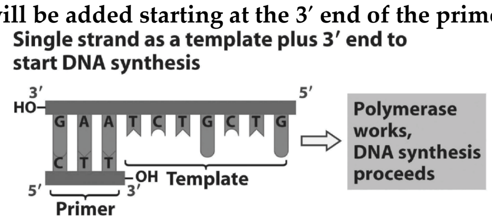
37) DNA is synthesized through a process known as _____.
A) Semiconservative replication.
B) Conservative replication.
C) Translation.
D) Transcription.

38) Who performed classic experiments that supported the semiconservative model of DNA replication?
A) Watson and Crick.
B) Meselson and Stahl.
C) Hershey and Chase.
D) Franklin and Wilkins.

39) DNA contains the template needed to copy itself, but it has no catalytic activity in cells. What catalyzes the formation of phosphodiester bonds between adjacent nucleotides in the DNA polymer being formed?
A) Ribozymes.
B) DNA polymerase.
C) ATP.
D) Deoxyribonucleotide triphosphates.

40) What provides the energy for the polymerization reactions in DNA synthesis?
A) ATP.
B) DNA polymerase.
C) Breaking the hydrogen bonds between complementary DNA strands.
D) The deoxyribonucleotide triphosphate substrates.

41) Refer to the figure given below. What bases will be added to the primer as DNA replication proceeds? The bases should appear in the new strand in the order that they will be added starting at the 3′ end of the primer.



A) C, A, G, C, A, G, A.
B) T, C, T, G, C, T, G.
C) A, G, A, C, G, A, C.
D) G, T, C, G, T, C, T.

42) What is the difference between the leading strand and the lagging strand in DNA replication?
A) The leading strand is synthesized in the 3′ → 5′ direction in a discontinuous fashion, while the lagging strand is synthesized in the 5′ → 3′ direction in a continuous fashion.
B) The leading strand is synthesized continuously in the 5′ → 3′ direction, while the lagging strand is synthesized discontinuously in the 5′ → 3′ direction.
C) The leading strand requires an RNA primer, whereas the lagging strand does not.
D) There are different DNA polymerases involved in elongation of the leading strand and the lagging strand.

43) What is a major difference between eukaryotic DNA replication and prokaryotic DNA replication?
A) Prokaryotic replication does not require a primer.
B) Prokaryotic chromosomes have a single origin of replication, while eukaryotic chromosomes have multiple origins of replication.
C) DNA replication in prokaryotic cells is conservative. DNA replication in eukaryotic cells is semi-conservative.
D) DNA polymerases of prokaryotes can add nucleotides to both 3′ and 5′ ends of DNA strands, while those of eukaryotes function only in the 5′ → 3′ direction.

44) What is a telomere?
A) The mechanism that holds two sister chromatids together.
B) DNA replication during telophase.
C) The site of origin of DNA replication.
D) The ends of linear chromosomes.

45) Telomere shortening puts a limit on the number of times a cell can divide. Research has shown that telomerase can extend the life span of cultured human cells. How might adding telomerase affect cellular aging?
A) Telomerase will speed up the rate of cell proliferation.
B) Telomerase eliminates telomere shortening and retards aging.
C) Telomerase shortens telomeres, which delays cellular aging.
D) Telomerase would have no effect on cellular aging.

46) Telomere shortening is a problem in which types of cells?
A) Only prokaryotic cells.
B) Only eukaryotic cells.
C) Cells in prokaryotes and eukaryotes.
D) Viruses.

47) Which of the following cells have reduced or very little active telomerase activity?
A) Most normal somatic cells.
B) Most normal germ cells.
C) Most cancer cells.
D) Viruses.

48) Researchers found *E. coli* that had mutation rates one hundred times higher than normal. Which of the following is the most likely cause of these results?
A) The single-stranded binding proteins were malfunctioning.
B) There were one or more mismatches in the RNA primer.
C) The proofreading mechanism of DNA polymerase was not working properly.
D) The DNA polymerase was unable to add bases to the 3′ end of the growing nucleic acid chain.

49) In a healthy cell, the rate of DNA repair is equal to the rate of DNA mutation. When the rate of repair lags behind the rate of mutation, what is a possible fate of the cell?
A) The cell can be transformed to a cancerous cell.
B) RNA may be used instead of DNA as inheritance material.
C) The cell will become embryonic.
D) DNA synthesis will continue by a new mechanism.

50) Which of the following statements describes a eukaryotic chromosome?
A) A single strand of DNA.
B) A series of nucleosomes wrapped around two DNA molecules.
C) A chromosome with different numbers of genes in different cell types of an organism.
D) A single linear molecule of double-stranded DNA plus proteins.

51) If a cell were unable to produce histone proteins, which of the following would be a likely effect?
A) There would be an increase in the amount of "satellite" DNA produced during centrifugation.
B) The cell's DNA couldn't be packed into its nucleus.
C) Spindle fibers would not form during prophase.
D) Amplification of other genes would compensate for the lack of histones.

52) Which of the following statements is true of histones?
A) Each nucleosome consists of two molecules of histone H1.
B) Histone H1 is not present in the nucleosome bead; instead, it draws the nucleosomes together.
C) The carboxyl end of each histone extends outward from the nucleosome and is called a "histone tail."
D) Histones are found in mammals, but not in other animals or in plants or fungi.

53) Why do histones bind tightly to DNA?
A) Histones are positively charged, and DNA is negatively charged.
B) Histones are negatively charged, and DNA is positively charged.
C) Both histones and DNA are strongly hydrophobic.
D) Histones are covalently linked to the DNA.

54) Which of the following represents the order of increasingly higher levels of organization of chromatin?
A) Nucleosome, 30-nm chromatin fiber, looped domain.
B) Looped domain, 30-nm chromatin fiber, and nucleosome.
C) Nucleosome, looped domain, 30-nm chromatin fiber.
D) 30-nm chromatin fiber, nucleosome, looped domain.

55) Which of the following statements describes chromatin?
A) Heterochromatin is composed of DNA, whereas euchromatin is made of DNA and RNA.
B) Both heterochromatin and euchromatin are found in the cytoplasm.
C) Heterochromatin is highly condensed, whereas euchromatin is less compact.
D) Euchromatin is not transcribed, whereas heterochromatin is transcribed.

56) Which of the following is most critical for the association between histones and DNA?
A) Histones are small proteins.
B) Histones are highly conserved (that is, histones are very similar in every eukaryote).
C) There are at least five different histone proteins in every eukaryote.
D) Histones are positively charged.

57) In *E. coli* replication the enzyme primase is used to attach a 5 to 10 base ribonucleotide strand complementary to the parental DNA strand. The RNA strand serves as a starting point for the DNA polymerase that replicates the DNA. If a mutation occurred in the primase gene, which of the following would you expect?
A) Replication would only occur on the leading strand.
B) Replication would only occur on the lagging strand.
C) Replication would not occur on either the leading or lagging strand.
D) Replication would not be affected as the enzyme primase in involved with RNA synthesis.

58) Hershey and Chase used a DNA-based virus for their work. What would the results have been if they had used an RNA virus?
A) With an RNA virus radioactive protein would have been in the final pellet.
B) With an RNA virus radioactive RNA would have been in the final pellet.
C) With an RNA virus neither sample would have had a radioactive pellet.
D) With an RNA virus the protein shell would have been radioactive in both samples.

59) The lagging strand is characterized by a series of short segments of DNA (Okazaki fragments) that will be joined together to form a finished lagging strand. The experiments that led to the discovery of Okazaki fragments gave evidence for which of the following ideas?
A) DNA polymerase is a directional enzyme that synthesizes leading and lagging strands during replication.
B) DNA is a polymer consisting of four monomers: adenine, thymine, guanine, and cytosine.
C) DNA is the genetic material.
D) Bacterial replication is fundamentally different from eukaryotic replication. The key shouldn't be way longer than the distractors.

CHAPTER 17:

GENE EXPRESSION: FROM GENE TO PROTEIN

1) Garrod hypothesized that "inborn errors of metabolism" such as alkaptonuria occur because _____.
A) Metabolic enzymes require vitamin cofactors, and affected individuals have significant nutritional deficiencies.
B) Enzymes are made of DNA, and affected individuals lack DNA polymerase.
C) Certain metabolic reactions are carried out by ribozymes, and affected individuals lack key splicing factors.
D) Genes dictate the production of specific enzymes, and affected individuals have genetic defects that cause them to lack certain enzymes.

2) A particular triplet of bases in the template strand of DNA is 5' AGT 3'. The corresponding codon for the mRNA transcribed is _____.
A) 3′ UCA 5′.
B) 3′ UGA 5′.
C) 5′ TCA 3′.
D) 3′ ACU 5′.

3) The genetic code is essentially the same for all organisms. From this, one can logically assume which of the following?
A) A gene from an organism can theoretically be expressed by any other organism.
B) DNA was the first genetic material.
C) The same codons in different organisms translate into different amino acids.
D) Different organisms have different types of amino acids.

4) The figure above shows a simple metabolic pathway. According to Beadle and Tatum's hypothesis, how many genes are necessary for this pathway?

$$A \xrightarrow{\text{enzyme } A} B \xrightarrow{\text{enzyme } B} C$$

A) 1.
B) 2.
C) 3.
D) It cannot be determined from the pathway.

5) Refer to the metabolic pathway illustrated above. If A, B, and C are all required for growth, a strain that is mutant for the gene-encoding enzyme A would be able to grow on medium supplemented with _____.
A) Nutrient A only.
B) Nutrient B only.
C) Nutrient C only.
D) Nutrients A and C.

6) Refer to the metabolic pathway illustrated above. If A, B, and C are all required for growth, a strain mutant for the gene -encoding enzyme B would be able to grow on medium supplemented with _____.
A) Nutrient A only.
B) Nutrient B only.
C) Nutrient C only.
D) Nutrients A and C.

The following questions (7-9) refer to this table of codons.

First Base	Second Base: U	Second Base: C	Second Base: A	Second Base: G	Third Base
U	UUU, UUC Phe; UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr; UAA Stop; UAG Stop	UGU, UGC Cys; UGA Stop; UGG Trp	U C A G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His; CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U C A G
A	AUU, AUC, AUA Ile; AUG Met or Start	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn; AAA, AAG Lys	AGU, AGC Ser; AGA, AGG Arg	U C A G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp; GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U C A G

7) A possible sequence of nucleotides in the template strand of DNA that would code for the polypeptide sequence phe-leu-ile-val would be _____.
A) 5′ TTG-CTA-CAG-TAG 3′.
B) 5′ AUG-CTG-CAG-TAT 3′.
C) 3′ AAA-AAT-ATA-ACA 5′.
D) 3′ AAA-GAA-TAA-CAA 5′.

8) What amino acid sequence will be generated based on the following mRNA codon sequence?

5' AUG-UCU-UCG-UUA-UCC-UUG 3′

A) met-arg-glu-arg-glu-arg.
B) met-glu-arg-arg-glu-leu.
C) met-ser-leu-ser-leu-ser.
D) met-ser-ser-leu-ser-leu.

9) Refer to the figure above. What would the anticodon be for a tRNA that transports phenylalanine to a ribosome?
A) UUU.
B) AAA.
C) TTT.
D) CCC.

10) Which of the following contradicts the one-gene, one-enzyme hypothesis?
A) A mutation in a single gene can result in a defective protein.
B) Alkaptonuria results when individuals lack a single enzyme involved in the catalysis of homogentisic acid.
C) Sickle-cell anemia results in defective hemoglobin.
D) A single antibody gene can code for different related proteins, depending on the splicing that takes place post-transcriptionally.

11) Which of the following is directly related to a single amino acid?
A) The base sequence of the tRNA.
B) The amino acetyl tRNA synthase.
C) The three-base sequence of mRNA.
D) The complementarity of DNA and RNA.

12) In the process of transcription _____.
A) DNA is replicated.
B) RNA is synthesized.
C) Proteins are synthesized.
D) mRNA attaches to ribosomes.

13) Codons are part of the molecular structure of _____.
A) A protein.
B) mRNA.
C) tRNA.
D) rRNA.

14) What does it mean when we say the genetic code is redundant?
A) A single codon can specify the addition of more than one amino acid.
B) The genetic code is different for different domains of organisms.
C) The genetic code is universal (the same for all organisms).
D) More than one codon can specify the addition of the same amino acid.

15) Once researchers identified DNA as the unit of inheritance, they asked how information was transferred from the DNA in the nucleus to the site of protein synthesis in the cytoplasm. What is the mechanism of information transfer in eukaryotes?
A) DNA from a single gene is replicated and transferred to the cytoplasm, where it serves as a template for protein synthesis.
B) Messenger RNA is transcribed from a single gene and transfers information from the DNA in the nucleus to the cytoplasm, where protein synthesis takes place.
C) Proteins transfer information from the nucleus to the ribosome, where protein synthesis takes place.
D) Transfer RNA takes information from DNA directly to a ribosome, where protein synthesis takes place.

16) According to the central dogma, what molecule should go in the blank? DNA → _____ → Proteins.
A) mtDNA.
B) rRNA.
C) mRNA.
D) tRNA.

17) Codons are three-base sequences that specify the addition of a single amino acid. How do eukaryotic codons and prokaryotic codons compare?
A) Prokaryotic codons usually contain different bases than those of eukaryotes.
B) Prokaryotic codons usually specify different amino acids than those of eukaryotes.
C) The translation of codons is mediated by tRNAs in eukaryotes, but translation requires no intermediate molecules such as tRNAs in prokaryotes.
D) Codons are a nearly universal language among all organisms.

18) Which of the following occurs in prokaryotes but not in eukaryotes?
A) Post-transcriptional splicing.
B) Concurrent transcription and translation.
C) Translation in the absence of a ribosome.
D) Gene regulation.

19) Which of the following statements best describes the termination of transcription in prokaryotes?
A) RNA polymerase transcribes through the polyadenylation signal, causing proteins to associate with the transcript and cut it free from the polymerase.
B) RNA polymerase transcribes through the terminator sequence, causing the polymerase to separate from the DNA and release the transcript.
C) Once transcription has initiated, RNA polymerase transcribes until it reaches the end of the chromosome.
D) RNA polymerase transcribes through a stop codon, causing the polymerase to stop advancing through the gene and release the mRNA.

20) In eukaryotes there are several different types of RNA polymerase. Which type is involved in transcription of mRNA for a globin protein?
A) RNA polymerase I.
B) RNA polymerase II.
C) RNA polymerase III.
D) Primase.

21) Transcription in eukaryotes requires which of the following in addition to RNA polymerase?
A) Start and stop codons.
B) Ribosomes and tRNA.
C) Several transcription factors.
D) Aminoacyl-tRNA synthetase.

22) Which of the following best describes the significance of the TATA box in eukaryotic promoters?
A) It is the recognition site for a specific transcription factor.
B) It sets the reading frame of the mRNA.
C) It is the recognition site for ribosomal binding.
D) Its significance has not yet been determined.

23) Which of the following does not occur in prokaryotic gene expression, but does occur in eukaryotic gene expression?
A) mRNA, tRNA, and rRNA are transcribed.
B) RNA polymerase binds to the promoter.
C) A cap is added to the 5′ end of the mRNA.
D) RNA polymerase requires a primer to elongate the molecule.

24) A ribozyme is _____.
A) A catalyst that uses RNA as a substrate.
B) An RNA with catalytic activity.
C) An enzyme that catalyzes the association between the large and small ribosomal subunits.
D) An enzyme that synthesizes RNA as part of the transcription process.

25) Alternative RNA splicing _____.
A) Is a mechanism for increasing the rate of translation.
B) Can allow the production of proteins of different sizes and functions from a single mRNA.
C) Can allow the production of similar proteins from different RNAs.
D) Increases the rate of transcription.

26) In the structural organization of many eukaryotic genes, individual exons may be related to which of the following?
A) The sequence of the intron that immediately precedes each exon.
B) The number of polypeptides making up the functional protein.
C) The various domains of the polypeptide product.
D) The number of start sites for transcription.

27) In an experimental situation, a student researcher inserts an mRNA molecule into a eukaryotic cell after she has removed its 5′ cap and poly-A tail. Which of the following would you expect her to find?
A) The mRNA is quickly converted into a ribosomal subunit.
B) The cell adds a new poly-A tail to the mRNA.
C) The mRNA attaches to a ribosome and is translated, but more slowly.
D) The molecule is digested by enzymes because it is not protected at the 5′ end.

Use this model of a eukaryotic transcript to answer the following question(s). E = exon and I = intron

5′ UTR E1 I1 E2 I2 E3 I3 E4 UTR 3′

28) Which components of the previous molecule will also be found in mRNA in the cytosol?
A) 5′ UTR I1 I2 I3 UTR 3′.
B) 5′ E1 E2 E3 E4 3′.
C) 5′ UTR E1 E2 E3 E4 UTR 3′.
D) 5′ E1 I1 E2 I2 E3 I3 E4 3′.

29) Which one of the following statements about RNA processing is true?
A) Exons are cut out before mRNA leaves the nucleus.
B) Ribozymes may function in RNA splicing.
C) RNA splicing can be catalyzed by tRNA.
D) A primary transcript is often much shorter than the final RNA molecule that leaves the nucleus.

30) A primary transcript in the nucleus of a eukaryotic cell is _____ the functional mRNA, while a primary transcript in a prokaryotic cell is _____ the functional mRNA.
A) The same size as; smaller than.
B) Larger than; the same size as
C) Larger than; smaller than.
D) The same size as; larger than.

31) A particular triplet of bases in the coding sequence of DNA is AAA. The anticodon on the tRNA that binds the mRNA codon is _____.
A) TTT.
B) UUA.
C) UUU.
D) AAA.

32) Accuracy in the translation of mRNA into the primary structure of a polypeptide depends on specificity in the _____.
A) Binding of ribosomes to mRNA.
B) Binding of the anticodon to small subunit of the ribosome.
C) Attachment of amino acids to rRNAs.
D) Binding of the anticodon to the codon and the attachment of amino acids to tRNAs.

33) A mutant bacterial cell has a defective aminoacyl-tRNA synthetase that attaches a lysine to tRNAs with the anticodon AAA instead of the normal phenylalanine. The consequence of this for the cell will be that _____.
A) None of the proteins in the cell will contain phenylalanine.
B) Proteins in the cell will include lysine instead of phenylalanine at amino acid positions specified by the codon UUU.
C) The cell will compensate for the defect by attaching phenylalanine to tRNAs with lysine-specifying anticodons.
D) The ribosome will skip a codon every time a UUU is encountered.

34) There are sixty-one mRNA codons that specify an amino acid, but only forty -five tRNAs. This is best explained by the fact that _____.
A) Some tRNAs have anticodons that recognize four or more different codons.
B) The rules for base pairing between the third base of a codon and tRNA are flexible.
C) Many codons are never used, so the tRNAs that recognize them are dispensable.
D) The DNA codes for all sixty-one tRNAs, but some are then destroyed.

35) Which of the following is the first event to take place in translation in eukaryotes?
A) Base pairing of activated methionine-tRNA to AUG of the messenger RNA.
B) Binding of the larger ribosomal subunit to smaller ribosomal subunits.
C) Covalent bonding between the first two amino acids.
D) The small subunit of the ribosome recognizes and attaches to the 5′ cap of mRNA.

36) A signal peptide _____.
A) Directs an mRNA molecule into the cisternal space of the ER.
B) Terminates translation of messenger RNA.
C) Helps target a protein to the ER.
D) Signals the initiation of transcription.

37) The release factor (RF) _____.
A) Binds to the stop codon in the A site in place of a tRNA.
B) Releases the amino acid from its tRNA to allow the amino acid to form a peptide bond.
C) Supplies a source of energy for termination of translation.
D) Releases the ribosome from the ER to allow polypeptides into the cytosol.

Use the following information to answer the questions (38-39) below.

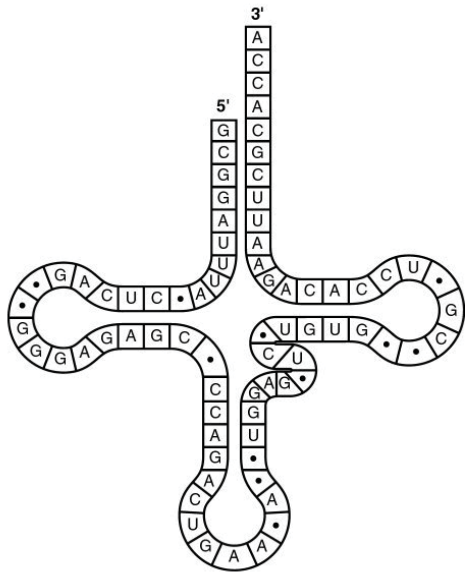
A part of an mRNA molecule with the following sequence is being read by a ribosome: 5′ CCG-ACG 3′(mRNA). The following charged transfer RNA molecules (with their anticodons shown in the 3′ to 5′ direction) are available. Two of them can correctly match the mRNA so that a dipeptide can form.

tRNA Anticodon	Amino Acid
GGC	Proline
CGU	Alanine
UGC	Threonine
CCG	Glycine
ACG	Cysteine
CGG	Alanine

38) The dipeptide that will form will be _____.
A) Cysteine-alanine.
B) Proline-threonine.
C) Glycine-cysteine.
D) Alanine-alanine.

39) The anticodon loop of the first tRNA that will complement this mRNA is _____.
A) 3′ GGC 5′.
B) 5′ GGC 3′.
C) 5′ UGC 3′.
D) 3′ UGC 5′.

Use the figure given below to answer the questions (40-42) below.



40) What type of bonding is responsible for maintaining the shape of the tRNA molecule shown in the figure above?
A) Ionic bonding between phosphates.
B) Hydrogen bonding between base pairs.
C) Van der Waals interactions between hydrogen atoms.
D) Peptide bonding between amino acids.

41) The figure above represents tRNA that recognizes and binds a particular amino acid (in this instance, phenylalanine). Which codon on the mRNA strand codes for this amino acid?
A) UGG.
B) GUG.
C) GUA.
D) UUC.

42) The tRNA shown in the figure above has its 3′ end projecting beyond its 5′ end. What will occur at this 3′ end?
A) The amino acid binds covalently.
B) The excess nucleotides (ACCA) will be cleaved off at the ribosome.
C) The small and large subunits of the ribosome will attach to it.
D) The 5′ cap of the mRNA will become covalently bound.

43) What must occure before a newly made polypeptide is secreted from a cell?
A) It must be translated by a ribosome that remains free within the cytosol.
B) Its signal sequence must target it to the ER, after which it goes to the Golgi.
C) Its signal sequence must be cleaved off before the polypeptide can enter the ER.
D) Its signal sequence must target it to the plasma membrane, where it causes exocytosis.

44) Translation requires _____.
A) mRNA, tRNA, DNA, and rRNA.
B) mRNA, DNA, and rRNA.
C) mRNA, tRNA, and rRNA.
D) mRNA, tRNA, and DNA.

45) During elongation, which site in the ribosome represents the location where a codon is being read?
A) E site.
B) P site.
C) A site.
D) The small ribosomal subunit.

46) Once a peptide has been formed between the amino acid attached to the tRNA in the P site and the amino acid associated with the tRNA in the A site, what occurs next?
A) Translocation.
B) Reading of the next codon of mRNA.
C) Initiation.
D) The codon-anticodon hydrogen bonds holding the tRNA in the A site are broken.

47) Which one of the following, if missing, would usually prevent translation from starting?
A) Exon.
B) 5′ cap.
C) AUG codon.
D) Poly-A tail.

48) Put the following events of elongation in prokaryotic translation in chronological order.
1. Binding of mRNA with small ribosomal subunit.
2. Recognition of initiation codon.
3. Complementary base pairing between initiator codon and anticodon of initiator tRNA.
4. Base pairing of the mRNA codon following the initiator codon with its complementary tRNA.
5. Attachment of the large subunit.
A) 1, 2, 3, 4, 5.
B) 2, 1, 4, 3, 5.
C) 5, 4, 3, 2, 1.
D) 1, 2, 3, 5, 4.

49) How does termination of translation take place?
A) The end of the mRNA molecule is reached.
B) A stop codon is reached.
C) The 5′ cap is reached.
D) The poly-A tail is reached.

50) Post-translational modifications of proteins may include the _____.
A) Removal of introns.
B) Addition of a 5′ cap.
C) Addition of a poly-A tail.
D) Addition of carbohydrates to form a glycoprotein.

51) Which of the following statements is true about protein synthesis in prokaryotes?
A) Extensive RNA processing is required before prokaryotic transcripts can be translated.
B) Translation can begin while transcription is still in progress.
C) Prokaryotic cells have complicated mechanisms for targeting proteins to the appropriate cellular organelles.
D) Unlike eukaryotes, prokaryotes require no initiation or elongation factors.

52) Which of the following types of mutation, resulting in an error in the mRNA just after the AUG start of translation, is likely to have the most serious effect on the polypeptide product?
A) A deletion of a codon.
B) A deletion of two nucleotides.
C) A substitution of the third nucleotide in an ACC codon.
D) A substitution of the first nucleotide of a GGG codon.

53) A nonsense mutation in a gene _____.
A) Changes an amino acid in the encoded protein.
B) Has no effect on the amino acid sequence of the encoded protein.
C) Introduces a premature stop codon into the mRNA.
D) Alters the reading frame of the mRNA.

54) Which of the following DNA mutations is most likely to damage the protein it specifies?
A) A base-pair deletion.
B) An addition of three nucleotides.
C) A substitution in the last base of a codon.
D) A codon deletion.

55) The most commonly occurring mutation in people with cystic fibrosis is a deletion of a single codon. This results in _____.
A) A base-pair substitution.
B) A frameshift mutation.
C) A polypeptide missing an amino acid.
D) A nonsense mutation.

56) Of the following, which is the most current description of a gene?
A) A unit of heredity that causes formation of a phenotypic characteristic.
B) A DNA subunit that codes for a single complete protein.
C) A DNA sequence that is expressed to form a functional product: either RNA or polypeptide.
D) A discrete unit of hereditary information that consists of a sequence of amino acids.

57) How might a single base substitution in the sequence of a gene affect the amino acid sequence of a protein encoded by the gene, and why?
A) Only a single amino acid could change, because the reading frame is unaffected.
B) The amino acid sequence would be substantially altered, because the reading frame would change with a single base substitution.
C) All amino acids following the substitution would be affected, because the reading frame would be shifted.
D) It is not possible for a single base substitution to affect protein structure, because each codon is three bases long.

58) An original section of DNA has the base sequence AGCGTTACCGT. A mutation in this DNA strand results in the base sequence AGGCGTTACCGT. This change represents _____.
A) A missense mutation.
B) A point mutation.
C) A silent mutation.
D) Frameshift mutation.

59) A single base substitution mutation is least likely to be deleterious when the base change results in _____.
A) A stop codon.
B) A codon that specifies the same amino acid as the original codon.
C) An amino acid substitution that alters the tertiary structure of the protein.
D) An amino acid substitution at the active site of an enzyme.

60) Rank the following one-base point mutations (from most likely to least likely) with respect to their likelihood of affecting the structure of the corresponding polypeptide.
1. Insertion mutation deep within an intron.
2. Substitution mutation at the third position of an exonic codon.
3. Substitution mutation at the second position of an exonic codon.
4. Deletion mutation within the first exon of the gene.
A) 1, 2, 3, 4.
B) 4, 3, 2, 1.
C) 2, 1, 4, 3.
D) 3, 1, 4, 2.

CHAPTER 18:

REGULATION OF GENE EXPRESSION

1) Which of the following is a protein produced by a regulatory gene?
A) Operon.
B) Inducer.
C) Promoter.
D) Repressor.

2) A lack of which molecule would result in a cell's inability to "turn off" genes?
A) Operon.
B) Inducer.
C) Promoter.
D) Corepressor.

3) Which of the following, when taken up by a cell, binds to a repressor so that the repressor no longer binds to the operator?
A) Inducer.
B) Promoter.
C) Repressor.
D) Corepressor.

4) Most repressor proteins are allosteric. Which of the following binds with the repressor to alter its conformation?
A) Inducer.
B) Promoter.
C) Transcription factor.
D) cAMP.

5) A mutation that inactivates a regulatory gene of a repressible operon in an E. coli cell would result in _____.
A) Continuous transcription of the structural gene controlled by that regulator.
B) Complete inhibition of transcription of the structural gene controlled by that regulator.
C) Irreversible binding of the repressor to the operator.
D) Continuous translation of the mRNA because of alteration of its structure.

6) The lactose operon is likely to be transcribed when _____.
A) There is more glucose in the cell than lactose.
B) There is glucose but no lactose in the cell.
C) The cyclic AMP and lactose levels are both high within the cell.
D) The cAMP level is high and the lactose level is low.

7) Transcription of structural genes in an inducible operon_____.
A) Occurs continuously in the cell.
B) Starts when the pathway's substrate is present.
C) Starts when the pathway's product is present.
D) Stops when the pathway's product is present.

8) For a repressible operon to be transcribed, which of the following must occur?
A) A corepressor must be present.
B) RNA polymerase and the active repressor must be present.
C) RNA polymerase must bind to the promoter, and the repressor must be inactive.
D) RNA polymerase must not occupy the promoter, and the repressor must be inactive.

9) Altering patterns of gene expression in prokaryotes would most likely serve an organism's survival by _____.
A) Organizing gene expression, so that genes are expressed in a given order.
B) Allowing each gene to be expressed an equal number of times.
C) Allowing an organism to adjust to changes in environmental conditions.
D) Allowing environmental changes to alter a prokaryote's genome.

10) In positive control of several sugar-metabolism-related operons, the catabolite activator protein (CAP) binds to DNA to stimulate transcription. What causes an increase in CAP activity in stimulating transcription?
A) An increase in glucose and an increase in cAMP.
B) A decrease in glucose and an increase in cAMP.
C) An increase in glucose and a decrease in cAMP.
D) A decrease in glucose and a decrease in the repressor.

11) There is a mutation in the repressor that results in a molecule known as a super-repressor because it represses the lac operon permanently. Which of these would characterize such a mutant?
A) It cannot bind to the operator.
B) It cannot make a functional repressor.
C) It cannot bind to the inducer.
D) It makes a repressor that binds CAP.

Use this information to answer the questions (12-14) below.

Suppose an experimenter becomes proficient with a technique that allows her to move DNA sequences within a prokaryotic genome.

12) If she moves the promoter for the *lac* operon to the region between the beta galactosidase (*lacZ*) gene and the permease (*lacY*) gene, which of the following would be likely?
A) The three structural genes will be expressed normally.
B) RNA polymerase will no longer transcribe permease.
C) The operon will still transcribe the *lacZ* and *lacY* genes, but the mRNA will not be translated.
D) Beta galactosidase will not be produced.

13) If she moves the operator to the far end of the operon, past the transacetylase (*lacA*) gene, which of the following would likely occur when the cell is exposed to lactose?
A) The inducer will no longer bind to the repressor.
B) The repressor will no longer bind to the operator.
C) The operon will never be transcribed.
D) The structural genes will be transcribed continuously.

14) If she moves the repressor gene (*lacI*), along with its promoter, to a position at some several thousand base pairs away from its normal position, we would expect the _____.
A) Repressor will no longer bind to the operator.
B) Repressor will no longer bind to the inducer.
C) *Lac* operon will be expressed continuously.
D) *Lac* operon will function normally.

15) What would occur if the repressor of an inducible operon were mutated so that it could not bind the operator?
A) Irreversible binding of the repressor to the promoter.
B) Reduced transcription of the operon's genes.
C) Continuous transcription of the operon's genes.
D) Overproduction of catabolite activator protein (CAP).

16) According to the lac operon model proposed by Jacob and Monod, what is predicted to occur if the operator is removed from the operon?
A) The *lac* operon would be transcribed continuously.
B) Only *lacZ* would be transcribed.
C) Only *lacY* would be transcribed.
D) Galactosidase permease would be produced, but would be incapable of transporting lactose.

17) The *trp* repressor blocks transcription of the *trp* operon when the repressor _____.
A) Binds to the inducer.
B) Binds to tryptophan.
C) Is not bound to tryptophan.
D) Is not bound to the operator.

18) Extracellular glucose inhibits transcription of the *lac* operon by _____.
A) Strengthening the binding of the repressor to the operator.
B) Weakening the binding of the repressor to the operator.
C) Inhibiting RNA polymerase from opening the strands of DNA to initiate transcription.
D) Reducing the levels of intracellular Camp.

19) CAP is said to be responsible for positive regulation of the *lac* operon because _____.
A) CAP binds cAMP.
B) CAP binds to the CAP-binding site.
C) CAP prevents binding of the repressor to the operator.
D) CAP bound to the CAP-binding site increases the frequency of transcription initiation.

20) Imagine that you have isolated a yeast mutant that contains histones resistant to acetylation. What phenotype do you predict for this mutant?
A) The mutant will grow rapidly.
B) The mutant will require galactose for growth.
C) The mutant will show low levels of gene expression.
D) The mutant will show high levels of gene expression.

21) The primary difference between enhancers and promoter -proximal elements is that enhancers _____.
A) Are transcription factors; promoter-proximal elements are DNA sequences.
B) Enhance transcription; promoter-proximal elements inhibit transcription.
C) Are at considerable distances from the promoter; promoter-proximal elements are close to the promoter.
D) Are DNA sequences; promoter-proximal elements are proteins.

22) The reason for differences in the sets of proteins expressed in a nerve and a pancreatic cell of the same individual is that nerve and pancreatic cells contain different _____.
A) Genes.
B) Regulatory sequences.
C) Sets of regulatory proteins.
D) Promoters.

23) Gene expression is often assayed by measuring the level of mRNA produced from a gene. If one is interested in knowing the amount of a final active gene product, a potential problem of this method is that it ignores the possibility of _____.
A) Chromatin condensation control.
B) Transcriptional control.
C) Alternative splicing.
D) Translational control.

24) Not long ago, it was believed that a count of the number of protein -coding genes would provide a count of the number of proteins produced in any given eukaryotic species. This is incorrect, largely due to the discovery of widespread _____.
A) Chromatin condensation control.
B) Transcriptional control.
C) Alternative splicing.
D) Translational control.

25) One way to detect alternative splicing of transcripts from a given gene is to _____.
A) Compare the DNA sequence of the given gene to that of a similar gene in a related organism.
B) Measure the relative rates of transcription of the given gene compared to that of a gene known to be constitutively spliced.
C) Compare the sequences of different primary transcripts made from the given gene.
D) Compare the sequences of different mRNAs made from the given gene.

26) Which of the following mechanisms is (are) used to coordinate the expression of multiple, related genes in eukaryotic cells?
A) Environmental signals enter the cell and bind directly to promoters.
B) The genes share a single common enhancer, which allows appropriate activators to turn on their transcription at the same time.
C) The genes are organized into a large operon, allowing them to be coordinately controlled as a single unit.
D) A single repressor is able to turn off many related genes.

27) DNA methylation and histone acetylation are examples of _____.
A) Genetic mutation.
B) Chromosomal rearrangements.
C) Epigenetic phenomena.
D) Translocation.

28) In eukaryotes, general transcription factors _____.

A) Bind to other proteins or to the TATA box.
B) Inhibit RNA polymerase binding to the promoter and begin transcribing.
C) Usually lead to a high level of transcription even without additional specific transcription factors.
D) Bind to sequences just after the start site of transcription.

29) Steroid hormones produce their effects in cells by _____.
A) Activating key enzymes in metabolic pathways.
B) Activating translation of certain mRNAs.
C) Promoting the degradation of specific mRNAs.
D) Binding to intracellular receptors and promoting transcription of specific genes.

30) Which of the following is most likely to have a small protein called ubiquitin attached to it?
A) A cyclin protein, that usually acts in G_1, in a cell that is in G_2.
B) A cell surface protein that requires transport from the ER.
C) An mRNA leaving the nucleus to be translated.
D) An mRNA produced by an egg cell that will be retained until after fertilization.

Use this information to answer the questions (31-32) below.

A researcher found a method she could use to manipulate and quantify phosphorylation and methylation in embryonic cells in culture.

31) In one set of experiments she succeeded in increasing acetylation of histone tails. Which of the following results would she most likely see?
A) Increased chromatin condensation.
B) Decreased chromatin condensation.
C) Decreased binding of transcription factors.
D) Inactivation of the selected genes.

32) One of her colleagues suggested she try increased methylation of C nucleotides in the DNA of promoters of a mammalian system. Which of the following results would she most likely see?
A) Decreased chromatin condensation.
B) Activation of histone tails for enzymatic function.
C) Higher levels of transcription of certain genes.
D) Inactivation of the selected genes.

33) Which method is utilized by eukaryotes to control their gene expression that is NOT used in bacteria?
A) Control of chromatin remodeling.
B) Control of RNA splicing.
C) Transcriptional control.
D) Control of both RNA splicing and chromatin remodeling.

34) The phenomenon in which RNA molecules in a cell are destroyed if they have a sequence complementary to an introduced double-stranded RNA is called _____.
A) RNA interference.
B) RNA obstruction.
C) RNA blocking.
D) RNA disposal.

35) At the beginning of this century there was a general announcement regarding the sequencing of the human genome and the genomes of many other multicellular eukaryotes. Many people were surprised that the number of protein-coding sequences was much smaller than they had expected. Which of the following could account for much of the DNA that is not coding for proteins?
A) DNA that consists of histone coding sequences.
B) DNA that is translated directly without being transcribed.
C) Non-protein-coding DNA that is transcribed into several kinds of small RNAs with biological function.
D) Non-protein-coding DNA that serves as binding sites for reverse transcriptase.

36) Among the newly discovered small noncoding RNAs, one type reestablishes methylation patterns during gamete formation and blocks expression of some transposons. These are known as _____.
A) miRNA.
B) piRNA.
C) snRNA.
D) siRNA.

37) Which of the following best describes siRNA?
A) A double-stranded RNA, one of whose strands can complement and inactivate a sequence of mRNA.
B) A single-stranded RNA that can, where it has internal complementary base pairs, fold into cloverleaf patterns.
C) A double-stranded RNA that is formed by cleavage of hairpin loops in a larger precursor.
D) A portion of rRNA that allows it to bind to several ribosomal proteins in forming large or small subunits.

Use this information to answer the question (38-39) below.

A researcher introduces double-stranded RNA into a culture of mammalian cells and can identify its location or that of its smaller subsections experimentally, using a fluorescent probe.

38) Sometime later, she finds that the introduced strand separates into single -stranded RNAs, one of which is degraded. What does this enable the remaining strand to do?
A) Attach to histones in the chromatin.
B) Bind to complementary regions of target mRNAs.
C) Activate other siRNAs in the cell.
D) Bind to noncomplementary RNA sequences.

39) When she finds that the introduced strand separates into single-stranded RNAs, what other evidence of this single-stranded RNA piece's activity can she find?
A) She can measure the degradation rate of the remaining single strand.
B) The rate of accumulation of the polypeptide encoded by the target mRNA is reduced.
C) The amount of miRNA is multiplied by its replication.
D) The cell's translation ability is entirely shut down.

40) The fact that plants can be cloned from somatic cells demonstrates that _____.
A) Differentiated cells retain all the genes of the zygote.
B) Genes are lost during differentiation.
C) The differentiated state is normally very unstable.
D) Differentiation does not occur in plants.

41) Your brother has just purchased a new plastic model airplane. He places all the parts on the table in approximately the positions in which they will be located when the model is complete. His actions are analogous to which process in development?
A) Morphogenesis.
B) Determination.
C) Differentiation.
D) Pattern formation.

42) The product of the bicoid gene in *Drosophila* provides essential information about _____.
A) The dorsal-ventral axis.
B) The left-right axis.
C) Segmentation.
D) The anterior-posterior axis.

43) If a *Drosophila* female has a homozygous mutation for a maternal effect gene, _____.
A) She will not develop past the early embryonic stage.
B) All of her offspring will show the mutant phenotype, regardless of their genotype.
C) Only her male offspring will show the mutant phenotype.
D) Only her female offspring will show the mutant phenotype.

44) Mutations in which of the following genes lead to transformations in the identity of entire body parts?
A) Segmentation genes.
B) Egg-polarity genes.
C) Homeotic genes.
D) Inducers.

45) Which of the following are maternal effect genes that control the orientation of the egg and thus the *Drosophila* embryo?
A) Homeotic genes.
B) Segmentation genes.
C) Egg-polarity genes.
D) Morphogens.

46) The bicoid gene product is normally localized to the anterior end of the embryo. If large amounts of the product were injected into the posterior end as well, which of the following would occur?
A) The embryo would grow extra wings and legs.
B) The embryo would probably show no anterior development and die.
C) Anterior structures would form in both ends of the embryo.
D) The embryo would develop normally.

47) In colorectal cancer, several genes must be mutated for a cell to develop into a cancer cell. Which of the following kinds of genes would you expect to be mutated?
A) Genes coding for enzymes that act in the colon.
B) Genes involved in control of the cell cycle.
C) Genes that are especially susceptible to mutation.
D) Genes of the bacteria, which are abundant in the colon.

48) A cell is considered to be differentiated when it _____.
A) Replicates by the process of mitosis.
B) Loses connections to the surrounding cells.
C) Produces proteins specific to a particular cell type.
D) Appears to be different from the surrounding cells.

49) When the Bicoid protein is expressed in *Drosophila*, the embryo is still syncytial (divisions between cells are not yet fully developed). This information helps to explain which observation by Nüsslein-Volhard and Wieschaus?
A) mRNA from the egg is translated into the Bicoid protein.
B) Bicoid protein diffuses throughout the embryo in a concentration gradient.
C) Bicoid protein serves as a transcription regulator.
D) Bicoid protein determines the dorsoventral axis of the embryo.

50) The protein of the *bicoid* gene in Drosophila determines the _____ of the embryo.
A) Anterior-posterior axis.
B) Anterior-lateral axis.
C) Posterior-dorsal axis.
D) Posterior-ventral axis.

51) Which of the following types of mutation would convert a proto-oncogene into an oncogene?
A) A mutation that blocks transcription of the proto-oncogene.
B) A mutation that creates an unstable proto-oncogene mRNA.
C) A mutation that greatly increases the amount of the proto-oncogene protein.
D) A deletion of most of the proto-oncogene coding sequence.

52) Proto-oncogenes _____.
A) Normally suppress tumor growth.
B) Are produced by somatic mutations induced by carcinogenic substances.
C) Stimulate normal cell growth and division.
D) Are underexpressed in cancer cells.

53) The product of the *p53* gene _____.
A) Inhibits the cell cycle.
B) Slows down the rate of DNA replication by interfering with the binding of DNA polymerase.
C) Causes cells to reduce expression of genes involved in DNA repair.
D) Allows cells to pass on mutations due to DNA damage.

54) Tumor-suppressor genes _____.
A) Are frequently overexpressed in cancerous cells.
B) Are cancer-causing genes introduced into cells by viruses.
C) Encode proteins that help prevent uncontrolled cell growth.
D) Often encode proteins that stimulate the cell cycle.

55) *BRCA1* and *BRCA2* are considered to be tumor-suppressor genes because _____.
A) Their normal products participate in repair of DNA damage.
B) The mutant forms of either one of these prevent breast cancer.
C) The normal genes make estrogen receptors.
D) They block penetration of breast cells by chemical carcinogens.

56) Forms of the Ras protein found in tumors usually cause which of the following?
A) DNA replication to stop.
B) Cell-to-cell adhesion to be nonfunctional.
C) Cell division to cease.
D) Excessive cell division.

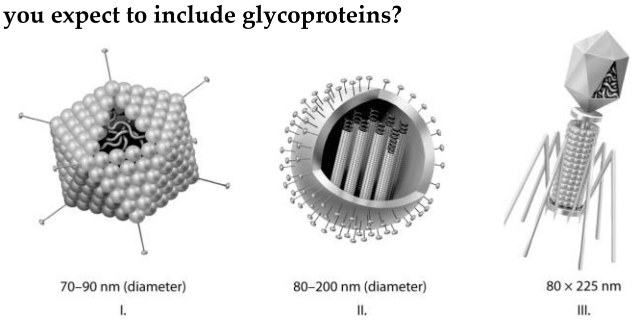
57) A genetic test to detect predisposition to cancer would likely examine the *APC* gene for involvement in which type(s) of cancer?
A) Colorectal only.
B) Lung and breast.
C) Lung only.
D) Lung and prostate.

58) In colorectal cancer, several genes must be mutated for a cell to develop into a cancer cell. Which of the following kinds of genes would you expect to be mutated?
A) Genes coding for enzymes that act in the colon.
B) Genes involved in control of the cell cycle.
C) Genes that are especially susceptible to mutation.
D) Genes of the bacteria, which are abundant in the colon.

CHAPTER 19:

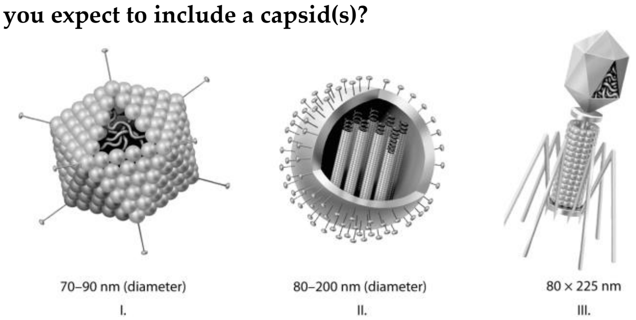
VIRUSES

1) Which of the three types of viruses shown above would you expect to include glycoproteins?



A) I only.
B) II only.
C) III only.
D) I and II only.

2) Which of the three types of viruses shown above would you expect to include a capsid(s)?



A) I only.
B) II only.
C) III only.
D) I, II, and III.

3) Which of the following supports the argument that viruses are nonliving?
A) They are not cellular.
B) Their DNA does not encode proteins.
C) They have RNA rather than DNA.
D) They do not evolve.

4) Viruses ______.
A) Manufacture their own ATP, proteins, and nucleic acids.
B) Use the host cell to copy themselves and make viral proteins.
C) Use the host cell to copy themselves and then viruses synthesize their own proteins.
D) Metabolize food and produce their own ATP.

5) What is the main structural difference between enveloped and nonenveloped viruses?
A) Enveloped viruses have their genetic material enclosed by a layer made only of protein.
B) Nonenveloped viruses have only a phospholipid membrane, while enveloped viruses have two membranes, the other one being a protein capsid.
C) Enveloped viruses have a phospholipid membrane outside their capsid, whereas nonenveloped viruses do not have a phospholipid membrane.
D) Both types of viruses have a capsid and phospholipid membrane; but in the nonenveloped virus the genetic material is between these two membranes, while in the enveloped virus the genetic material is inside both membranes.

6) The host range of a virus is determined by ______.
A) The enzymes carried by the virus.
B) Whether its nucleic acid is DNA or RNA.
C) The proteins in the host's cytoplasm.
D) The proteins on its surface and that of the host.

7) Which of the following accounts for someone who has had regular herpesvirus-mediated cold sore or genital sore flare-ups?
A) Re-infection by a closely related herpesvirus of a different strain.
B) Re-infection by the same herpesvirus strain.
C) Copies of the herpesvirus genome permanently maintained in host nuclei.
D) Copies of the herpesvirus genome permanently maintained in host cell cytoplasm.

8) In many ways, the regulation of the genes of a particular group of viruses will be similar to the regulation of the host genes. Therefore, which of the following would you expect of the genes of a bacteriophage?
A) Regulation via acetylation of histones.
B) Positive control mechanisms rather than negative.
C) Control of more than one gene in an operon.
D) Reliance on transcription activators.

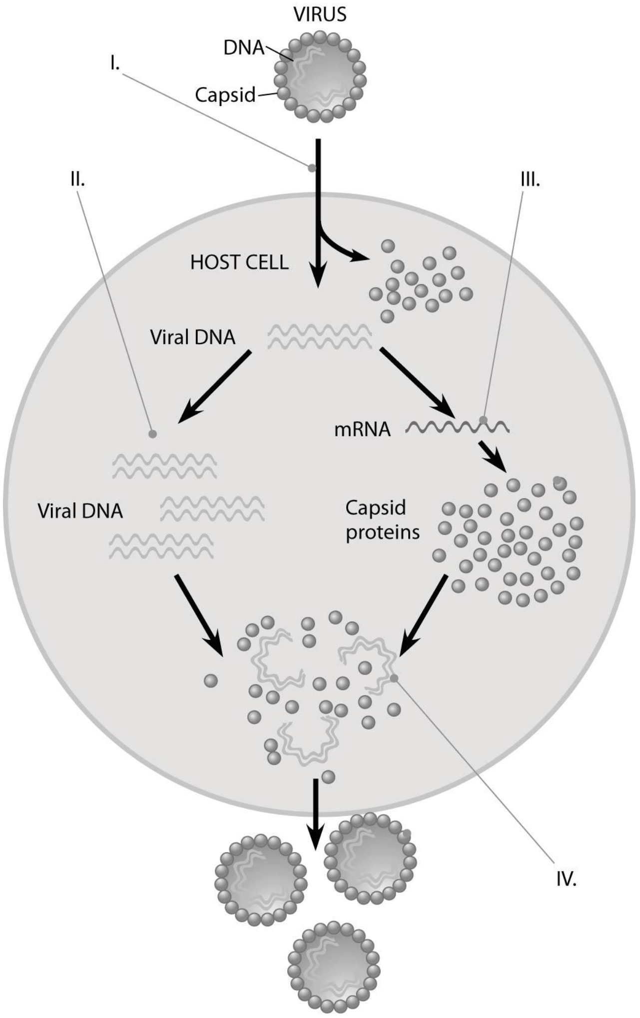
9) Which of the following characterizes of the lytic cycle?
A) Viral DNA is incorporated into the host genome.
B) The viral genome replicates without destroying the host.
C) A large number of phages are released at a time.
D) The virus-host relationship usually lasts for generations.

10) Which of the following statements describes the lysogenic cycle of lambda (λ) phage?
A) After infection, the viral genes immediately turn the host cell into a lambda-producing factory, and the host cell then lyses.
B) Most of the prophage genes are activated by the product of a particular prophage gene.
C) The phage genome replicates along with the host genome.
D) The phage DNA is copied and exits the cell as a phage.

11) Which viruses have single-stranded RNA that acts as a template for DNA synthesis?
A) Proviruses.
B) Viroids.
C) Bacteriophages.
D) Retroviruses.

12) What is the function of reverse transcriptase in retroviruses?
A) It uses viral RNA as a template for DNA synthesis.
B) It converts host cell RNA into viral DNA.
C) It translates viral RNA into proteins.
D) It uses viral RNA as a template for making complementary RNA strands.

Use the figure given below to answer the question (13-14) below.



Viral reproductive cycle

13) In the figure, at the arrow marked II, what enzyme(s) are being utilized?
A) Reverse transcriptase.
B) Viral DNA polymerase.
C) Host cell DNA polymerase.
D) Host cell RNA polymerase.

14) In the figure, when new viruses are being assembled at the point marked IV, what mediates the assembly?
A) Host cell chaperones.
B) Assembly proteins coded for by the host nucleus.
C) Assembly proteins coded for by the viral genes.
D) Nothing; they self-assemble.

Some viruses can be crystallized and their structures analyzed. One such virus is yellow mottle virus, which infects beans. This virus has a single-stranded RNA genome containing about 6300 nucleotides. Its capsid is 25-30 nm in diameter and contains 180 identical capsomeres.

15) If the yellow mottle virus begins its infection of a cell by using its genome as mRNA, which of the following would you expect to be able to measure?
A) Replication rate.
B) Transcription rate.
C) Translation rate.
D) Formation of new transcription factors.

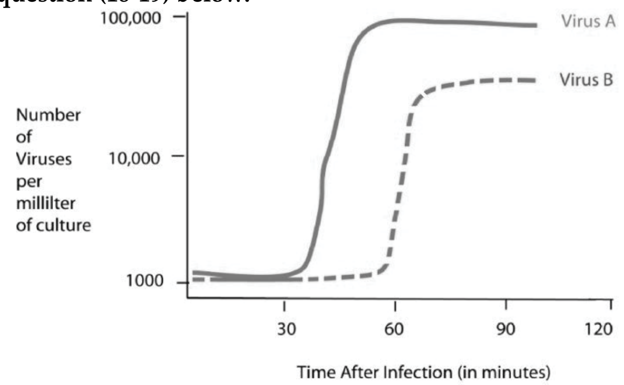
Use the following information to answer the question (16) below.

The herpes viruses are important enveloped DNA viruses that cause disease in vertebrates and in some invertebrates such as oysters. Some of the human forms are herpes simplex virus (HSV) types I and II, causing facial and genital lesions, and the varicella zoster virus (VSV), causing chicken pox and shingles. Each of these three actively infects nervous tissue. Primary infections are fairly mild, but the virus is not then cleared from the host; rather, viral genomes are maintained in cells in a latent phase. The virus can later reactivate, replicate again, and infect others.

16) In electron micrographs of HSV infection, it can be seen that the intact virus initially reacts with cell surface proteoglycans, then with specific receptors. This is later followed by viral capsids docking with nuclear pores. Afterward, the capsids go from being full to being "empty". Which of the following best fits these observations?
A) Viral capsids are needed for the cell to become infected; only the capsids enter the nucleus.
B) The viral envelope is not required for infectivity, since the envelope does not enter the nucleus.
C) Only the genetic material of the virus is involved in the cell's infectivity, and is injected like the genome of a phage.
D) The viral envelope mediates entry into the cell, the capsid mediates entry into the nuclear membrane, and the genome is all that enters the nucleus.

17) Poliovirus is an RNA virus of the picornavirus group, which uses its RNA as mRNA. At its 5′ end, the RNA genome has a viral protein (VPg) instead of a 5′ cap. This is followed by a nontranslated leader sequence, and then a single long protein -coding region (~7000 nucleotides), followed by a poly-A tail. Observations were made that used radioactive amino acid analogues. Short period use of the radioactive amino acids result in labeling of only very long proteins, while longer periods of labeling result in several different short polypeptides. What conclusion is most consistent with the results of the radioactive labeling experiment?
A) Host cell ribosomes only translate the viral code into short polypeptides.
B) The RNA is only translated into a single long polypeptide, which is then cleaved into shorter ones.
C) The RNA is translated into short polypeptides, which are subsequently assembled into large ones.
D) The large radioactive polypeptides are coded by the host, whereas the short ones are coded for by the virus.

Use the following figure and information to answer the question (18-19) below.



Cells were infected with approximately 1000 copies of either virus A or virus B at the 0 time point. At five-minute intervals, a sample of the virus and cell mixture was removed. The intact cells were removed from the sample, and the number of viruses per milliliter of culture was determined.

18) Using the data in the figure above, how long does it take for virus A to go through one lytic cycle?
A) 15 minutes
B) 30 minutes
C) 45 minutes
D) 90 minutes.

19) Using the data in the figure above, how long does it take for virus B to go through one lytic cycle?
A) 15 minutes.
B) 30 minutes.
C) 45 minutes.
D) 60 minutes.

20) The virus genome and viral proteins are assembled into virions (virus particles) during _____.
A) The lytic cycle and the lysogenic cycle in all known host organisms.
B) The lysogenic cycle only.
C) The lytic cycle only.
D) The lytic cycle in all host organisms but the lysogenic cycle only in bacteria.

21) Which of the following viruses would most likely have reverse transcriptase?
A) An RNA-based lytic virus.
B) An RNA-based lysogenic virus.
C) A DNA-based lytic virus.
D) A DNA-based lysogenic virus.

22) If a viral host cell has a mutation that interferes with the addition of carbohydrates to proteins in the Golgi, which of the following could likely result?
A) The viral envelope proteins would not be glycosylated and might not arrive at the host plasma membrane.
B) The viral capsid proteins would not be glycosylated and might not arrive at the host plasma membrane.
C) The viral core proteins would not be glycosylated and might not arrive at the host plasma membrane.
D) The virus would be unable to reproduce within the host cell.

23) HIV is inactivated in the laboratory after a few minutes of sitting at room temperature, but the flu virus is still active after sitting for several hours. What are the practical consequences of these findings?
A) HIV can be transmitted more easily from person to person than the flu virus.
B) The flu virus can be transmitted more easily from person to person than HIV.
C) This property of HIV makes it more likely to be a pandemic than the flu virus.
D) Disinfecting surfaces is more important to reduce the spread of HIV than the flu.

24) Viruses use the host's machinery to make copies of themselves. However, some human viruses require a type of replication that humans do not normally have. For example, humans normally do not have the ability to convert RNA into DNA. How can these types of viruses infect humans, when human cells cannot perform a particular role that the virus requires?
A) The virus causes mutations in the human cells, resulting in the formation of new enzymes that are capable of performing these roles.
B) The viral genome codes for specialized enzymes not in the host.
C) The virus infects only those cells and species that can perform all the replication roles necessary.
D) Viruses can stay in a quiescent state until the host cell evolves this ability.

25) The first class of drugs developed to treat AIDS, such as AZT, were known as reverse transcriptase inhibitors. They worked because they _____.
A) Targeted and destroyed the viral genome before it could be reverse transcribed into DNA.
B) Bonded to the dsDNA genome of the virus in such a way that it could not separate for replication to occur.
C) Bonded to the viral reverse transcriptase enzyme, thus preventing the virus from making a DNA copy of its RNA genome.
D) Prevented host cells from producing the enzymes used by the virus to replicate its genome.

26) A virus consisting of a single strand of RNA, which is transcribed into complementary DNA, is a _____.
A) Protease.
B) Retrovirus.
C) RNA replicase virus.
D) Nonenveloped virus.

27) Which of the following could use reverse transcriptase to transcribe its genome?
A) ssRNA.
B) dsRNA.
C) ssDNA.
D) dsDNA.

28) To make a vaccine against mumps, measles, or rabies, which type of viruses would be useful?
A) dsDNA viruses.
B) Negative-sense ssRNA viruses.
C) Positive-sense ssRNA viruses.
D) dsRNA viruses.

29) Which of the following human diseases is caused by a virus that requires reverse transcriptase to transcribe its genome inside the host cell?
A) Herpes.
B) AIDS.
C) Smallpox.
D) Influenza.

30) Why do RNA viruses appear to have higher rates of mutation?
A) RNA nucleotides are more unstable than DNA nucleotides.
B) Replication of their genomes does not involve proofreading.
C) RNA viruses can incorporate a variety of nonstandard bases.
D) RNA viruses are more sensitive to mutagens.

31) A researcher lyses a cell that contains nucleic acid molecules and capsomeres of tobacco mosaic virus (TMV). The cell contents are left in a covered test tube overnight. The next day this mixture is sprayed on tobacco plants. We expect that the plants would _____.
A) Develop some but not all of the symptoms of the TMV infection.
B) Develop the typical symptoms of TMV infection.
C) Not show any disease symptoms.
D) Become infected, but the sap from these plants would be unable to infect other plants.

32) Which of the following can be effective in preventing the onset of viral infection in humans?
A) Taking vitamins.
B) Getting vaccinated.
C) Taking antibiotics.
D) Taking drugs that inhibit transcription.

33) Viral infections in plants _____.
A) Can be controlled with antibiotics.
B) Can spread within a plant via plasmodesmata.
C) Have little effect on plant growth.
D) Are not spread by animals.

34) Which of the following represents a difference between viruses and viroids?
A) Viruses infect many types of cells, whereas viroids infect only prokaryotic cells.
B) Viruses have capsids composed of protein, whereas viroids have no capsids.
C) Viruses have genomes composed of RNA, whereas viroids have genomes composed of DNA.
D) Viruses cannot pass through plasmodesmata, whereas viroids can.

35) The difference between vertical and horizontal transmission of plant viruses is that vertical transmission is _____.
A) Transmission of a virus from a parent plant to its progeny, and horizontal transmission is one plant spreading the virus to another plant.
B) The spread of viruses from upper leaves to lower leaves of the plant, and horizontal transmission is the spread of a virus among leaves at the same general level.
C) The spread of viruses from trees and tall plants to bushes and other smaller plants, and horizontal transmission is the spread of viruses among plants of similar size.
D) The transfer of DNA from a plant of one species to a plant of a different species, and horizontal transmission is the spread of viruses among plants of the same species.

36) What are prions?
A) Mobile segments of DNA.
B) Tiny circular molecules of RNA that can infect plants.
C) Viral DNA that attaches itself to the host genome and causes disease.
D) Misfolded versions of normal protein that can cause disease.

37) A person is most likely to recover from a viral infection if the infected cells _____.
A) Can undergo normal cell division.
B) Can carry on translation, at least for a few hours.
C) Produce and release viral protein.
D) Transcribe viral mRNA.

38) Effective antiviral drugs are usually associated with which of the following properties?
A) Interference with viral replication.
B) Prevention of the host from becoming infected.
C) Removal of viral proteins.
D) Removal of viral mRNAs.

39) Which of the following best reflects what we know about how the flu virus moves between species?
A) The flu virus in a pig is mutated and replicated in alternate arrangements so that humans who eat the pig products can be infected.
B) A flu virus from a human epidemic or pandemic infects birds; the birds replicate the virus differently and then pass it back to humans.
C) An influenza virus gains new sequences of DNA from another virus, such as a herpesvirus; this enables it to be transmitted to a human host.
D) An animal such as a pig is infected with more than one virus, genetic recombination occurs, the new virus mutates, the virus is passed to a new species such as a bird, and the virus mutates again and can now be transmitted to humans.

Refer to the treatments listed below to answer the following questions (40-41).
You isolate an infectious substance capable of causing disease in plants, but you do not know whether the infectious agent is a bacterium, virus, viroid, or prion. You have four methods at your disposal to analyze the substance and determine the nature of the infectious agent.
I. Treat the substance with enzymes that destroy all nucleic acids and then determine whether the substance is still infectious.
II. Filter the substance to remove all elements smaller than what can be easily seen under a light microscope.
III. Culture the substance on nutritive medium, away from any plant cells.
IV. Treat the sample with proteases that digest all proteins and then determining whether the substance is still infectious.

40) If you already know that the infectious agent was either bacterial or viral, which method(s) listed above would allow you to distinguish between these two possibilities?
A) I
B) II
C) II or III
D) IV.

41) If you already know that the infectious agent was either a viroid or a prion, which method(s) listed above would allow you to distinguish between these two possibilities?
A) I only.
B) II only.
C) IV only.
D) Either I or IV.

Use the following information to answer the question (42) below.

The herpes viruses are important enveloped DNA viruses that cause disease in vertebrates and in some invertebrates such as oysters. Some of the human forms are herpes simplex virus (HSV) types I and II, causing facial and genital lesions, and the varicella zoster virus (VSV), causing chicken pox and shingles. Each of these three actively infects nervous tissue. Primary infections are fairly mild, but the virus is not then cleared from the host; rather, viral genomes are maintained in cells in a latent phase. The virus can later reactivate, replicate again, and infect others.

42) If scientists are trying to use what they know about HSV to devise a means of protecting other people from being infected, which of the following would have the best chance of lowering the number of new cases of infection?
A) Vaccinate of all persons with preexisting cases of HSV.
B) Interfere with new viral replication in preexisting cases of HSV.
C) Treat HSV lesions to shorten the breakout.
D) Educate people about avoiding sources of infection.

43) What is difference between an epidemic and a pandemic?
A) An epidemic is a disease; a pandemic is a treatment.
B) An epidemic is restricted to a local region; a pandemic is global.
C) An epidemic has low mortality; a pandemic has higher mortality.
D) An epidemic is caused by a bacterial infection; a pandemic is caused by a viral infection.

44) Will treating a viral infection with antibiotics affect the course of the infection?
A) No; antibiotics work by inhibiting enzymes specific to bacteria. Antibiotics have no effect on eukaryotic or virally encoded enzymes.
B) No; antibiotics do not kill viruses because viruses do not have DNA or RNA.
C) Yes; antibiotics activate the immune system, and this decreases the severity of the infection.
D) Yes; antibiotics can prevent viral entry into the cell by binding to host-receptor proteins.

45) Why do scientists consider HIV to be an emerging virus?
A) HIV infected humans long before the 1980s, but it has now mutated to a more deadly form.
B) HIV mutates rapidly making the virus very different from HIV in the early 1980s.
C) HIV suddenly became apparent and widespread in the 1980s.
D) HIV is now starting to cause diseases other than AIDS, such as rare types of cancers and pneumonias.

46) A population of viruses with similar characteristics is called a _____.
A) Strain.
B) Species.
C) Type.
D) Genome.

47) Evidence suggests that factors which contribute towards the virulence of E. coli strain O157:H7, a bacterial strain reported to cause several food poisoning deaths, are caused by genes from a virus that infects bacteria. Considering this evidence, which statement most likely explains how the O157:H7 population acquired the genetic variation that distinguishes the strain from harmless *E. coli* strains, such as those that reside in our intestines?
A) The virus entered the bacterial cell and incorporated its DNA into the bacterial genome, allowing the bacteria's cellular machinery to create new viruses.
B) Viral envelope proteins bind to receptors on the bacterial membrane, allowing the viral genetic material to enter the bacterium and become translated into proteins.
C) The virus entered the cell and acquired specific genes from the bacteria to increase the virulence of the virus.
D) The virus infected the bacterium, and allowed the bacterial population to replicate with a copy of the phage genome in each new bacterium.

48) Which of the following processes within viral replication is the greatest source of genetic variation in RNA virus populations?
A) High mutation rate due to lack of proofreading of RNA genome replication errors.
B) Transcription from the host cell RNA polymerase introduces numerous mutations.
C) Capsid proteins from the host cell can replace the viral capsid.
D) Viral RNA is translated by host cell ribosomes.

49) In 2009, a flu pandemic was believed to have originated when viral transmission occurred from pig to human, thereby earning the designation, "swine flu". Although pigs are thought to have been the breeding ground for the 2009 virus, sequences from bird, pig, and human viruses were all found within this newly identified virus. What is the most likely explanation of why this virus contained sequences from bird, pig, and human viruses?
A) The virus was descended from a common ancestor of bird, pig, and human flu viruses.
B) The infected individuals happened to be infected with all three virus types.
C) Related viruses can undergo genetic recombination if the RNA genomes mix and match during viral assembly.
D) The human was likely infected with various bacterial strains that contained all three RNA viruses.

CHAPTER 20:

DNA TOOLS AND BIOTECHNOLOGY

1) Bacterial cells protect their own DNA from restriction enzymes (endonucleases) by _____.
A) Adding methyl groups to adenines and cytosines.
B) Using DNA ligase to seal the bacterial DNA into a closed circle.
C) Adding histones to protect the double-stranded DNA.
D) Forming "sticky ends" of bacterial DNA to prevent the enzyme (endonuclease) from attaching.

2) What is the most logical sequence of steps for splicing foreign DNA into a plasmid and inserting the plasmid into a bacterium?
I. Transform bacteria with a recombinant DNA molecule.
II. Cut the plasmid DNA using restriction enzymes (endonucleases).
III. Extract plasmid DNA from bacterial cells.
IV. Hydrogen-bond the plasmid DNA to nonplasmid DNA fragments.
V. Use ligase to seal plasmid DNA to nonplasmid DNA.
A) II, III, V, IV, I.
B) III, II, IV, V, I.
C) III, IV, V, I, II.
D) IV, V, I, II, III.

3) A principal problem with inserting an unmodified mammalian gene into a plasmid and then getting that gene expressed in bacteria is that _____.
A) Prokaryotes use a different genetic code from that of eukaryotes.
B) Bacteria translate only mRNAs that have multiple messages.
C) Bacteria cannot remove eukaryotic introns.
D) Bacterial RNA polymerase cannot make RNA complementary to mammalian DNA.

4) Yeast cells are frequently used as hosts for cloning because they _____.
A) Easily form colonies.
B) Can remove exons from mRNA.
C) Do not have plasmids.
D) Are eukaryotic cells.

5) Sequencing an entire genome, such as that of C. elegans, a nematode, is most important because _____.
A) It allows researchers to use the sequence to build a "better" nematode, which is resistant to disease.
B) It allows research on a group of organisms we do not usually care much about.
C) A sequence that is found to have a particular function in the nematode is likely to have a closely related function in vertebrates.
D) A sequence that is found to have no introns in the nematode genome is likely to have acquired the introns from higher organisms.

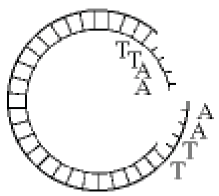
6) To introduce a particular piece of DNA into an animal cell, such as that of a mouse, you would most likely be successful with which of the following methods?
A) Electroporation followed by recombination.
B) Introducing a plasmid into the cell.
C) Infecting the mouse cell with a Ti plasmid.
D) Transcription and translation.

7) Why is it so important to be able to amplify DNA fragments when studying genes?
A) Before amplification, DNA fragments are likely to bind to RNA and no longer be able to be analyzed.
B) A gene may represent only a millionth of the cell's DNA.
C) Restriction enzymes (endonucleases) cut DNA into fragments that are too small.
D) A clone requires multiple copies of each gene per clone.

8) *Pax-6* is a gene that is involved in eye formation in many invertebrates, such as *Drosophila*. *Pax-6* is also found in vertebrates. A *Pax-6* gene from a mouse can be expressed in a fly and the protein (PAX-6) leads to a compound fly eye. This information suggests which of the following?
A) *Pax-6* genes are identical in nucleotide sequence.
B) PAX-6 proteins have identical amino acid sequences.
C) *Pax-6* is highly conserved and shows shared evolutionary ancestry.
D) PAX-6 proteins are different for formation of different kinds of eyes.

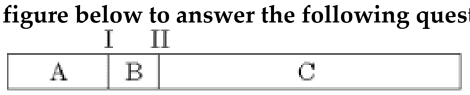
9) The reason for using Taq polymerase for PCR is that _____.
A) It is heat stable and can withstand the heating step of PCR.
B) Only minute amounts are needed for each cycle of PCR.
C) It binds more readily than other polymerases to the primers.
D) It has regions that are complementary to the primers.

Use the figure below to answer the following question.



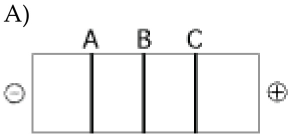
10) Which enzyme was used to produce the molecule in the figure above?
A) Ligase.
B) A restriction enzyme (endonuclease).
C) RNA polymerase.
D) DNA polymerase.

Use the figure below to answer the following question.

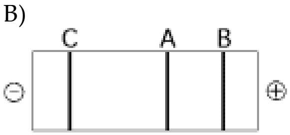


11) The segment of DNA shown in the figure above has restriction sites I and II, which create restriction fragments A, B, and C. Which of the gels produced by electrophoresis shown below best represents the separation and identity of these fragments?

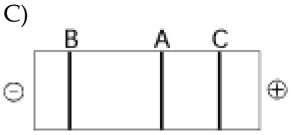
A)



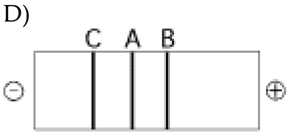
B)



C)



D)



12) A laboratory might use dideoxyribonucleotides to ___.

A) Separate DNA fragments.
B) Produce cDNA from mRNA.
C) Sequence a DNA fragment.
D) Visualize DNA expression.

13) Many identical copies of genes cloned in bacteria are produced as a result of _____.

A) Plasmid replication.
B) Bacterial cell replication.
C) Transformation.
D) Plasmid and bacterial cell replication.

14) Which of the following sequences is most likely to be cut by a restriction enzyme?

A) AATTCT TTAAGA.
B) AATATT TTATAA.
C) AAAATT TTTTAA.
D) ACTACT TGATGA.

15) What information is critical to the success of polymerase chain reaction (PCR) itself?

A) The DNA sequence of the ends of the DNA to be amplified must be known.
B) The complete DNA sequence of the DNA to be amplified must be known.
C) The sequence of restriction-enzyme recognition sites in the DNA to be amplified must be known.
D) The sequence of restriction-enzyme recognition sites in the DNA to be amplified and in the plasmid where the amplified DNA fragment will be cloned must be known.

16) Which of the following is in the correct order for one cycle of polymerase chain reaction (PCR)?

A) Denature DNA; add fresh enzyme; anneal primers; add dNTPs; extend primers.
B) Anneal primers; denature DNA; extend primers.
C) Extend primers; anneal primers; denature DNA.
D) Denature DNA; anneal primers; extend primers.

17) The final step in a Sanger DNA sequencing reaction is to run the DNA fragments on a gel. What purpose does this serve?

A) It adds ddNTP to the end of each DNA fragment.
B) It changes the length of the DNA fragments.
C) It separates DNA fragments based on their charge.
D) It separates DNA fragments generated during the sequencing reaction based on one-nucleotide differences in their size.

18) DNA sequencing has transformed our understanding of genes, genomes and evolution. Which of the following statements comparing two common sequencing techniques, the chain termination method and next generation sequencing is TRUE?

A) The chain termination method is faster and more efficient, so it is used to generate large-scale sequences, while next generation synthesis is used for routine, small-scale jobs.
B) The chain termination method employs the polymerase chain reaction, but next generation sequencing does not.
C) In the chain termination method, the order of bases is detected by fluorescently labeling each dideoxy-nucleotide in a different color, while next generation sequencing determines the order of bases by detecting the release of PPi during the formation of the phosphodiester bond.
D) Next generation sequencing employs electrophoresis, but the chain termination method does not.

19) A gene that contains introns can be made shorter (but remain functional) for genetic engineering purposes by using _____.

A) A restriction enzyme (endonuclease) to cut the gene into shorter pieces.
B) Reverse transcriptase to reconstruct the gene from its mRNA.
C) DNA polymerase to reconstruct the gene from its polypeptide product.
D) DNA ligase to put together fragments of the DNA that code for a particular polypeptide.

20) Which of the following is required to make complementary DNA (cDNA) from RNA?

A) Restriction enzymes (endonucleases).
B) Gene cloning.
C) DNA ligase.
D) Reverse transcriptase.

21) DNA microarrays have made a huge impact on genomic studies because they _____.
A) Can be used to eliminate the function of any gene in the genome.
B) Can be used to introduce entire genomes into bacterial cells.
C) Allow the expression of many or even all of the genes in the genome to be compared at once.
D) Allow physical maps of the genome to be assembled in a very short time.

22) RNAi methodology uses double-stranded pieces of RNA to trigger breakdown of a specific mRNA or inhibit its translation. For which of the following might this technique be useful?
A) To decrease the production from a harmful mutated gene.
B) To destroy an unwanted allele in a homozygous individual.
C) To form a knockout organism that will not pass the deleted sequence to its progeny.
D) To raise the concentration of a desired protein.

23) A researcher has used in vitro mutagenesis to mutate a cloned gene and then has reinserted the mutated gene into a cell. To have the mutated sequence disable the function of the gene, what must then occur?
A) Recombination resulting in replacement of the wild type with the mutated gene.
B) Use of a microarray to verify continued expression of the original gene.
C) Replication of the cloned gene using a bacterial plasmid.
D) Transcription of the cloned gene using a BAC.

24) Silencing of selected genes is often done using RNA interference (RNAi). Which of the following questions would NOT be answered with this process?
A) What is the function of gene 432 in a particular species of annelid?
B) What will happen in a particular insect's digestion if gene 173 is not able to be translated?
C) Is gene HA292 expressed in individuals for a disorder in humans?
D) Will the disabling of this gene in Drosophila and in a mouse cause similar results?

25) In large scale, genome-wide association studies in humans we look for _____.
A) Lengthy sequences that might be shared by most members of a population.
B) SNPs where one allele is found more often in persons with a particular disorder than in healthy controls.
C) SNPs where one allele is found in families with particular introns sequence.
D) SNPs where one allele is found in two or more adjacent genes.

26) For a particular microarray assay (DNA chip), cDNA has been made from the mRNAs of a dozen patients' breast tumor biopsies. The researchers will be looking for _____.
A) A particular gene that is amplified in all or most of the patient samples
B) A pattern of fluorescence that indicates which cells are over proliferating
C) A pattern shared among some or all of the samples that indicates gene expression differing from control samples
D) A group of cDNAs that match those in non-breast cancer control samples from the same population

27) Imagine that you compare two DNA sequences found in the same location on homologous chromosomes. On one of the homologs, the sequence is AACTACGA. On the other homolog, the sequence is AACTTCGA. Within a population, you discover that each of these sequences is common. These sequences _____.
A) Contain an SNP that may be useful for genetic mapping.
B) Identify a protein-coding region of a gene.
C) Cause disease.
D) Do none of the listed actions.

28) Which of the following uses labeled probes to visualize the expression of genes in whole tissues and organisms?
A) RT-PCR.
B) In situ hybridization.
C) DNA microarrays.
D) RNA interference.

29) Which of the following is most like the formation of identical twins?
A) Cell cloning.
B) Therapeutic cloning.
C) Use of adult stem cells.
D) Organismal cloning.

30) In 1997, Dolly the sheep was cloned. Which of the following processes was used?
A) Replication and dedifferentiation of adult stem cells from sheep bone marrow.
B) Separation of an early stage sheep blastula into separate cells, one of which was incubated in a surrogate ewe.
C) Fusion of an adult cell's nucleus with an enucleated sheep egg, followed by incubation in a surrogate.
D) Isolation of stem cells from a lamb embryo and production of a zygote equivalent.

31) Which of the following problems with animal cloning might result in premature death of the clones?
A) Use of pluripotent instead of totipotent stem cells.
B) Use of nuclear DNA as well as mtDNA.
C) Abnormal gene regulation due to variant methylation.
D) The indefinite replication of totipotent stem cells.

32) Reproductive cloning of human embryos is generally considered unethical. However, on the subject of therapeutic cloning there is a wider divergence of opinion. Which of the following is a likely explanation?
A) The use of adult stem cells is likely to produce more cell types than the use of embryonic stem cells.
B) Cloning to produce embryonic stem cells may lead to great medical benefits for many.
C) Cloning to produce stem cells relies on a different initial procedure than reproductive cloning.
D) A clone that lives until the blastocyst stage does not yet have human DNA.

33) Which of the following is true of embryonic stem cells but not of adult stem cells?
A) They normally differentiate into only eggs and sperm.
B) They can give rise to all cell types in the organism.
C) They can continue to reproduce for an indefinite period.
D) One aim of using them is to provide cells for repair of diseased tissue.

34) A researcher is using adult stem cells and comparing them to other adult cells from the same tissue. Which of the following is a likely finding?
A) The cells from the two sources exhibit different patterns of DNA methylation.
B) Adult stem cells have more DNA nucleotides than their counterparts.
C) The two kinds of cells have virtually identical gene expression patterns in microarrays.
D) The nonstem cells have fewer repressed genes.

35) In animals, what is the difference between reproductive cloning and therapeutic cloning?
A) Reproductive cloning uses totipotent cells, whereas therapeutic cloning does not.
B) Reproductive cloning uses embryonic stem cells, whereas therapeutic cloning does not.
C) Therapeutic cloning uses nuclei of adult cells transplanted into enucleated nonfertilized eggs.
D) Therapeutic cloning supplies cells for repair of diseased or injured organs.

36) The first cloned cat, called Carbon Copy, was a calico, but she looked significantly different from her female parent because _____.
A) The cloning was done poorly and it was likely that some contaminating cat DNA became part of Carbon Copy's genome.
B) Fur color genes in cats are determined by differential acetylation patterns.
C) Cloned animals have been found to have a higher frequency of transposon activation.
D) X inactivation in the embryo is random and produces different patterns.

37) In recent times, it has been shown that adult cells can be induced to become pluripotent stem cells (iPS). To make this conversion, what has been done to the adult cells?
A) A retrovirus is used to introduce four specific regulatory genes.
B) The adult stem cells must be fused with embryonic cells.
C) Cytoplasm from embryonic cells is injected into the adult cells.
D) The nucleus of an embryonic cell is used to replace the nucleus of an adult cell.

38) Let us suppose that someone is successful at producing induced pluripotent stem cells (iPS) for replacement of pancreatic insulin-producing cells for people with type 1 diabetes. Which of the following could still be problems?
I. The possibility that, once introduced into the patient, the iPS cells produce nonpancreatic cells.
II. The failure of the iPS cells to take up residence in the pancreas.
III. The inability of the iPS cells to respond to appropriate regulatory signals.
A) I only.
B) II only.
C) III only.
D) I, II, and III.

39) Genetically engineered plants _____.
A) Are more difficult to develop than genetically engineered animals.
B) Include transgenic rice plants that can grow in water of high salinity.
C) Are used in research but not yet in commercial agricultural production.
D) Are banned throughout the world.

40) Scientists developed a set of guidelines to address the safety of DNA technology. Which of the following is one of the adopted safety measures?
A) Microorganisms used in recombinant DNA experiments are genetically crippled to ensure that they cannot survive outside of the laboratory.
B) Genetically modified organisms are not allowed to be part of our food supply.
C) Transgenic plants are engineered so that the plant genes cannot hybridize.
D) Experiments involving HIV or other potentially dangerous viruses have been banned.

41) Which of the following is one of the technical reasons why gene therapy is problematic?
A) Most cells with an engineered gene do not produce gene product.
B) Cells with transferred genes are unlikely to replicate.
C) Transferred genes may not have appropriately controlled activity.
D) mRNA from transferred genes cannot be translated.

42) One possible use of transgenic plants is in the production of human proteins, such as vaccines. Which of the following is a possible hindrance that must be overcome?
A) Prevention of transmission of plant allergens to the vaccine recipients.
B) Prevention of vaccine-containing plants being consumed by insects.
C) Use of plant cells to translate non-plant-derived mRNA.
D) Inability of the human digestive system to accept plant-derived protein.

43) What characteristic of short tandem repeats (STRs) DNA makes it useful for DNA fingerprinting?
A) The number of repeats varies widely from person to person or animal to animal.
B) The sequence of DNA that is repeated varies significantly from individual to individual.
C) The sequence variation is acted upon differently by natural selection in different environments.
D) Every racial and ethnic group has inherited different short tandem repeats.

44) Transgenic mice are useful to human researchers because they _____.
A) Can be valuable animal models of human disease.
B) Are essential for mapping human genes.
C) Are now used in place of bacteria for cloning human genes.
D) Were instrumental in pinpointing the location of the huntingtin gene.

45) Gene therapy requires _____.
A) Knowledge and availability of the normal allele of the defective gene.
B) The ability to introduce the normal allele into the patient.
C) The ability to express the introduced gene at the correct level, time, and tissue site within the patient.
D) Knowledge and availability of the normal allele of the defective gene, an ability to introduce the normal allele into the patient, and an ability to express the introduced gene at the correct level, and time, and tissue site within the patient.

46) For applications in gene therapy, what is the most favorable characteristic of retroviruses?
A) Retroviruses have an RNA genome.
B) Retroviruses possess reverse transcriptase.
C) DNA copies of retroviral genomes become integrated into the genome of the infected cell.
D) Retroviruses mutate often.

47) Using retroviral vectors for gene therapy might increase the patient's risk of developing cancer because they might ____.
A) Introduce proteins from the virus.
B) Not express the genes that were introduced into a patient's cells.
C) Not integrate their recombinant DNA into the patient's genome.
D) Integrate recombinant DNA into the genome in ways that misregulate the expression of genes at or near the site of integration.

48) In the form of gene therapy used successfully for severe combined immunodeficiency syndrome (SCID)-X1, the genetic engineering of human cells is done by _____.
A) Injecting engineered viruses into the patient's bloodstream.
B) Injecting engineered viruses into the patient's bone marrow.
C) Treating a relative's cultured bone marrow cells with genetically engineered viruses and then injecting these cells into the patient's bone marrow.
D) Isolating the patient's bone marrow cells, infecting them with genetically engineered viruses, and injecting them back into the patient's bone marrow.

49) One predicted aspect of climate change is that climates, including precipitation and temperature, over most of the Earth will become more variable. Which of the following is a good crop genetic engineering strategy if this is true?
A) Only plant crops that are genetically engineered.
B) Genetically engineer most crops to withstand very long droughts.
C) Genetically engineer several genotypes within single crop types.
D) Genetically engineer the fastest growing crops possible.

50) Plasmids are used as vectors in plant and bacterial genetic engineering. However, there is a major difference in the fate of genes introduced into bacteria on most bacterial plasmids and into plants on tumor-inducing (Ti) plasmids. What is this difference?
A) In bacteria, genes are stably expressed; in plants, gene expression is always lost quickly.
B) Gene expression tends to decrease rapidly and unpredictably in bacteria; gene expression is much more stable in plants.
C) Bacterial plasmids are circular DNAs; Ti plasmid DNA is linear.
D) Bacterial plasmids and the genes they carry usually are not integrated into the chromosome; Ti plasmids and the genes they carry are integrated into the chromosome.

51) In an experiment, DNA from the linear form of the bacteriophage Lambda was cut into fragments using the restriction enzyme Hind III. Restriction enzymes are isolated from bacteria and cut DNA in specific locations. Hind III cuts the Lambda DNA between the adenine nucleotides on the complimentary strands in a specific sequence, as indicated in the diagram, producing eight different size fragments. These fragments are then separated with an electrical current based on size after the DNA fragments are placed in a porous gel, a process called gel electrophoresis.

Cut site
↓
5′ A A G C T T 3′
3′ T T C G A A 5′
↑
Cut site

Select an observation that best describes a correct aspect of the two processes of restriction digest and gel electrophoresis:

A) When separated on a gel, the pattern of DNA bands will be characteristic of those cut with Hind III, different restriction enzymes will not produce these same fragments.
B) The sequence AAGCTT is found eight times in the Lambda genome and the restriction enzyme Hind III finds each location.
C) If an electrical current is not used, eight separate DNA bands would be visible, but they would not be separated as much as when an electrical current is used.
D) Only the restriction enzyme Hind III can be used to cut Lambda DNA since restriction enzymes are specific to the type of DNA they can cut.

52) Organisms share many conserved core processes and features, including transcription and translation using a uniform genetic code. Scientists have used these shared processes and features in biotechnology. For example, for the process of some transformations, a plasmid is constructed when a eukaryotic gene of interest is added with an antibiotic resistant gene such beta-lactamase, which is used for ampicillin resistance. This plasmid is then inserted into a prokaryotic bacterial cell, such as *E-coli*, through a transformation process that leads to the production of the product protein from the eukaryotic organism. To culture the bacteria and obtain the protein product, the bacteria must grow.
Select the appropriate condition to determine if the plasmid has entered the *E-coli* bacterial cell.

A) Nutrient broth to which no antibiotic has been added.
B) Water to which ampicillin has been added.
C) Nutrient broth to which ampicillin has been added.
D) Nutrient broth to which other resistant bacteria have been added.

CHAPTER 21:

GENOMES AND THEIR EVOLUTION

1) What is metagenomics?
A) Genomics as applied to a species that most typifies the average phenotype of its genus.
B) The sequencing of one or two representative genes from several species.
C) The sequencing of only the most highly conserved genes in a lineage.
D) Sequencing DNA from a group of species from the same ecosystem.

2) An early step in shotgun sequencing is to _____.
A) Break genomic DNA at random sites.
B) Map the position of cloned DNA fragments.
C) Randomly select DNA primers and hybridize these to random positions of chromosomes in preparation for sequencing.
D) Making copy of a gene.

3) Using modern techniques of sequencing by synthesis and the shotgun approach, sequences are assembled into chromosomes by _____.
A) Placing them on previously generated genetic maps.
B) Cloning them into plasmid vectors.
C) Computer analysis looking for sequence overlaps.
D) Cloning them into plasmid vectors, placing them on previously generated genetic maps, followed by computer analysis looking for sequence overlaps.

4) Proteomics is defined as the _____.
A) Linkage of each gene to a particular protein.
B) Study of the full protein set encoded by a genome.
C) Totality of the functional possibilities of a single protein.
D) Study of how amino acids are ordered in a protein

5) Bioinformatics can be used to scan for short sequences that specify known mRNAs, called _____.
A) Expressed sequence tags.
B) Multigene families.
C) Proteomes.
D) Short tandem repeats.

6) What is gene annotation in bioinformatics?
A) Finding transcriptional start and stop sites, RNA splice sites, and ESTs in DNA sequences.
B) Assigning names to newly discovered genes.
C) Describing the functions of noncoding regions of the genome.
D) Matching the corresponding phenotypes of different species.

7) Bioinformatics includes _____.
I. Using computer programs to align DNA sequences.
II. Creating recombinant DNA from separate species.
III. Developing computer-based tools for genome analysis.
IV. Using mathematical tools to make sense of biological systems.
A) I and II.
B) II and III.
C) II and IV.
D) I, III, and IV.

8) After finding a new medicinal plant, a pharmaceutical company decides to determine if the plant has genes similar to those of other known medicinal plants. To do this, the company annotates the genome of the new plant to _____.
A) Determine what proteins are produced.
B) Determine what mRNA transcripts are produced.
C) Identify genes and determine their functions.
D) Identify the location of mRNA within the plant cells.

9) If the sequence of a cDNA has matches with DNA sequences in the genome, then this genomic DNA is likely to _____.
A) Code for a protein.
B) Code for an rRNA.
C) Be part of an intron.
D) Be a regulatory sequence.

10) In what sense are studies by nineteenth-century naturalists and those by early twenty-first-century genomic biologists similar?
A) Both focused on cellular evolution.
B) Both worked to understand how genes evolved.
C) Both took a reductionist approach by studying only one small part of a complex system.
D) Both focused on observing and describing what exists in their realms of investigation.

11) Which of the following techniques would be most appropriate to test the hypothesis that humans and chimps differ in the expression of a large set of shared genes?
A) DNA microarray analysis.
B) Polymerase chain reaction (PCR).
C) DNA sequencing.
D) Protein-protein interaction assays.

12) A DNA microarray is a tool that owes its existence to earlier genomics investigations. What essential contribution of genomics makes microarrays possible?
A) Recently improved RNA sequencing technologies.
B) Continuously improving methods of gene cloning.
C) More efficient techniques for cDNA synthesis.
D) Knowledge of which DNA sequences to synthesize for the array.

13) What can proteomics reveal that genomics cannot?
A) The number of genes characteristic of a species.
B) The patterns of alternative splicing.
C) The set of proteins present within a cell or tissue type.
D) The movement of transposable elements within the genome.

14) A sequence database such as GenBank could be used to do all of the following EXCEPT _____.
A) Compare cow and human insulin protein sequences.
B) Construct a tree to determine the evolutionary relationships between various bird species.
C) Search for genes in a fruit fly that are similar to a human gene.
D) Compare patterns of gene expression in cancerous and non-cancerous cells.

15) Current analysis indicates that less than 2% of the human genome codes for proteins. Based on the systems approach employed by the ENCODE project, what percentage of the genome is estimated to contain functional elements (includes functional RNAs and regulatory sequences)?
A) Less than 2%.
B) About 50%.
C) At least 80%.
D) 100%.

16) Which of the following is a representation of gene density?
A) Humans have 2900 Mb per genome.
B) *C. elegans* has ~20,000 genes.
C) Humans have ~20,000 protein-encoding genes in 2900 Mb.
D) *Fritillaria* has a genome 40 times the size of a human.

17) Why might the cricket genome have eleven times as many base pairs as that of *Drosophila melanogaster?*
A) Crickets have higher gene density.
B) *Drosophila* are more complex organisms.
C) Crickets must have more noncoding DNA.
D) Crickets must make many more proteins.

18) The comparison between the number of human genes and those of other animal species has led to many conclusions, including that _____.
A) The density of the human genome is far higher than in most other animals.
B) The number of proteins expressed by the human genome is far more than the number of its genes.
C) Most human DNA consists of genes for protein, tRNA, rRNA, and miRNA.
D) The genomes of most other organisms are significantly smaller than the human genome.

19) It is more difficult to identify eukaryotic genes than prokaryotic genes because in eukaryotes _____.
A) The proteins are larger than in prokaryotes.
B) The coding portions of genes are shorter than in prokaryotes.
C) There are no start codons.
D) There are introns.

20) If alternative splicing did NOT occur then _____.
A) The human genome would likely contain many more genes.
B) The *E. coli* genome would contain many fewer genes.
C) There would be little correlation between the complexity of organisms and genome size.
D) There would be many fewer genes devoted to metabolism in Arabidopsis and yeast.

21) A multigene family is composed of _____.
A) Multiple genes whose products must be coordinately expressed.
B) Genes whose sequences are very similar and that probably arose by duplication.
C) A gene whose exons can be spliced in a number of different ways.
D) A highly conserved gene found in a number of different species.

22) Retrotransposons _____.
A) Use an RNA molecule as an intermediate in transposition.
B) Are found only in animal cells.
C) Generally move by a cut-and-paste mechanism.
D) Contribute a significant portion of the genetic variability seen within a population of gametes.

23) In humans, the embryonic and fetal forms of hemoglobin have a higher affinity for oxygen than that of adults. This is due to _____.
A) Nonidentical genes that produce different versions of goblins during development.
B) Pseudogenes, which interfere with gene expression in adults.
C) The attachment of methyl groups to cytosine following birth, which changes the type of hemoglobin produced.
D) Histone proteins changing shape during embryonic development.

24) Sequencing eukaryotic genomes is more difficult than sequencing genomes of bacteria or archaea because of the _____.
A) Large size of eukaryotic proteins.
B) Hard-to-find proteins.
C) High proportion of G-C base pairs in eukaryotic DNA.
D) Large size of eukaryotic genomes and the large amount of eukaryotic repetitive DNA.

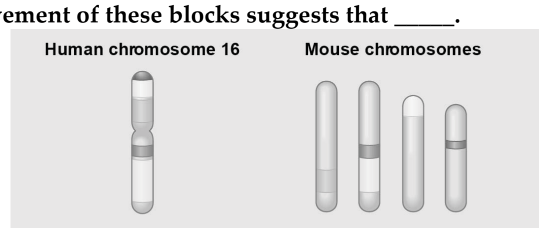
25) Because they both produce a reverse transcriptase, long interspersed nuclear elements (LINEs) as transposable elements may be related to _____.
A) Plasmids.
B) Retroviruses.
C) Poliovirus.
D) Parasitic bacteria.

26) Although transposable elements and short tandem repeats (STRs) are repetitive DNAs, they differ in that ___.
A) STRs occur within exons; transposable elements occur within introns.
B) STRs occur within introns; transposable elements occur within exons.
C) The repeated unit in STRs is clustered one after another; transposable element repeats are scattered throughout the genome.
D) The repeated unit in STRs is much larger than the repeated unit of transposable elements.

27) Which of the following can be duplicated in a genome?
A) Only DNA sequences.
B) Only entire sets of chromosomes.
C) Only entire chromosomes.
D) DNA sequences, chromosomes, or sets of chromosomes.

28) Unequal crossing over during prophase I can result in one sister chromosome with a deletion and another with a duplication. A mutated form of hemoglobin, so -called hemoglobin Lepore, exists in the human population. Hemoglobin Lepore has a deleted series of amino acids. If this mutated form was caused by unequal crossing over, what would be an expected consequence?
A) There should also be persons whose hemoglobin contains two copies of the series of amino acids that is deleted in hemoglobin Lepore.
B) Each of the genes in the hemoglobin gene family must show the same deletion.
C) The deleted gene must have undergone exon shuffling.
D) The deleted region must be located in a different area of the individual's genome.

29) The figure below shows a diagram of blocks of genes on human chromosome 16 and the locations of blocks of similar genes on four chromosomes of the mouse. The movement of these blocks suggests that _____.



A) During evolutionary time, these sequences have separated and have returned to their original positions.
B) DNA sequences within these blocks have become increasingly divergent.
C) Sequences represented have duplicated at least three times.
D) Chromosomal translocations have moved blocks of sequences to other chromosomes.

30) Humans have twenty-three pairs of chromosomes and chimps have twenty-four pairs of chromosomes. What is the most likely explanation for these differences in human and chimp genomes?
A) The common ancestor of humans and chimps had twenty -four pairs of chromosomes. After the two groups evolved, two human chromosomes fused end to end.
B) In the evolution of chimps, new adaptations resulted from additional chromosomal material.
C) At some point in evolution, human and chimp ancestors reproduced with each other.
D) Errors in mitosis resulted in an additional pair of chromosomes in chimps.

31) Exon shuffling occurs during _____.
A) Splicing of DNA.
B) DNA replication.
C) Meiotic recombination.
D) Translation.

32) When gene duplication occurs to its ultimate extent by doubling all genes in a genome, what has occurred?
A) Pseudogene creation.
B) Creation of a gene cluster.
C) Creation of a polyploid.
D) Creation of a diploid.

33) Mutations that occur in one member of a gene pair that arose from gene duplication may create _____.
A) A pseudogene.
B) A gene with a new function.
C) A gene family with two distinct but related members.
D) A pseudogene, a gene with a new function, and a gene family with two distinct but related members.

34) Based on the data in the Amino Acid Identity Table, which two members of the human globin gene family are the most divergent?
A) a1 and ß.
B) a2 and ß.
C) a1 and a2.
D) a1 and G1.

35) Fragments of DNA have been extracted from the remnants of extinct woolly mammoths, amplified, and sequenced. These can now be used to _____.
A) Introduce certain mammoth traits into relatives, such as elephants.
B) Clone live woolly mammoths.
C) Appreciate the reasons why mammoths went extinct.
D) Better understand the evolutionary relationships among members of related taxa.

36) Homeotic genes contain a homeobox sequence that is highly conserved among very diverse species. The homeobox is the code for that domain of a protein that binds to DNA in a regulatory developmental process. Therefore, you would expect that _____.
A) Homeotic genes are selectively expressed as an organism develops.
B) Homeoboxes cannot be expressed in nonhomeotic genes.
C) Homeotic genes in apes and humans are very different.
D) All organisms must have homeotic genes.

37) A recent study compared the Homo sapiens genome with that of Neanderthals. The results of the study indicated that there was a mixing of the two genomes at some period in evolutionary history. Additional data consistent with this hypothesis could be the discovery of _____.
A) Some Neanderthal sequences not found in living humans.
B) A few modern H. sapiens with some Neanderthal sequences.
C) Duplications of several Neanderthal genes on a Neanderthal chromosome.
D) Some Neanderthal chromosomes those are shorter than their counterparts in living humans.

38) Several of the different globin genes are expressed in humans, but at different times in development. What mechanism could allow for this?
A) Exon shuffling.
B) Pseudogene activation.
C) Differential translation of mRNAs.
D) Differential gene regulation over time.

39) Biologists now routinely test for homology between genes in different species. If genes are determined to be homologous, it means that they are related _____.
A) By descent from a common ancestor.
B) Because of convergent evolution.
C) By chance mutations.
D) In function but not structure.

40) A current view of how the human and chimpanzee can share most of their nucleotide sequence yet exhibit significant phenotypic differences is that many of the most important sequence differences alter _____.
A) Structural genes.
B) The number of repeated sequences.
C) Regulatory sequences.
D) Environmental factors.

41) Studies in knockout mice have demonstrated an important role of the FOXP2 transcription factor in the development of vocalizations. Recent sequence comparisons of the FOXP2 gene in Neanderthals and modern humans show that while the DNA sequence may be different, the protein sequence it codes for is identical. What might logically be inferred from this information?
A) There was a problem with the experiment because different DNA sequences cannot result in the same protein sequence.
B) The differences in DNA sequence support the hypothesis that Neanderthals were primitive beings that could only grunt.
C) Human and Neanderthal vocalizations may have been more similar than previously thought.
D) The experiments in mice demonstrating the function of the FOXP2 gene are not relevant to humans and Neanderthals because they are not primates.

42) Comparisons of DNA sequences within the human species have revealed many variations. Which of the following variations involves duplication of relatively long stretches of chromosomes, often including the duplication of protein-coding genes?
A) CNVs.
B) SNPs.
C) STRs.
D) Transposable elements.

43) A microarray is a tool used in genetic research to determine the mRNAs being produced in a particular tissue, and their relative level of expression. Known genes can therefore be assayed for their expression in different situations. One use of the technology is in cancer diagnosis and treatment. If a known gene functions as a tumor suppressor, predict which of the following pieces of evidence would be most useful in diagnosis of a cancer due to a mutation in this tumor-suppressor gene.
A) The tissue sample shows a high level of gene expression relative to a control (noncancerous) sample.
B) The tissue sample responds to treatment with a mitosis-promoting compound.
C) The mRNAs for the targeted tumor suppressor sequence are not being produced.
D) The mRNAs for cyclins and kinases show unusually high levels of expression.

CHAPTER 22:

DESCENT WITH MODIFICATION: A DARWINIAN VIEW OF LIFE

1) Darwin and Wallace's theory of evolution by natural selection was revolutionary because it _____.
A) Was the first theory to refute the ideas of special creation.
B) Proved that individuals acclimated to their environment over time.
C) Dismissed the idea that species are constant and emphasized the importance of variation and change in populations.
D) Was the first time a biologist had proposed that species changed through time.

2) Catastrophism was Cuvier's attempt to explain the existence of _____.
A) Evolution.
B) The fossil record.
C) Uniformitarianism.
D) The origin of new species.

3) With what other idea of his time was Cuvier's theory of catastrophism most in conflict?
A) The *scala naturae.*
B) The fixity of species.
C) Island biogeography.
D) Uniformitarianism.

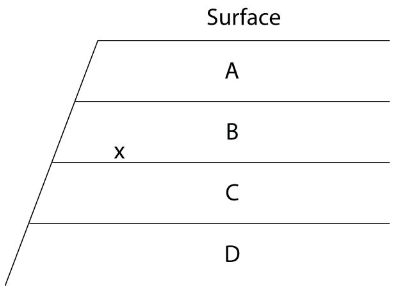
4) Prior to the work of Lyell and Darwin, the prevailing belief was that Earth is _____.
A) A few thousand years old, and populations are unchanging.
B) A few thousand years old, and populations change.
C) Millions of years old, and populations rapidly change.
D) Millions of years old, and populations are unchanging.

5) During a study session about evolution, one of your fellow students' remarks, "The giraffe stretched its neck while reaching for higher leaves; its offspring inherited longer necks as a result.". Which statement is most likely to be helpful in correcting this student's misconception?
A) Characteristics acquired during an organism's life are generally not passed on through genes.
B) Spontaneous mutations can result in the appearance of new traits.
C) Only favorable adaptations have survival value.
D) Disuse of an organ may lead to its eventual disappearance.

6) When Cuvier considered the fossils found in the vicinity of Paris, he concluded that the extinction of species _____.
A) Occurs, but that there is no evolution.
B) And the evolution of species both occurs.
C) And the evolutions of species do not occur.
D) Does not occur, but evolution does occur.

7) In the mid-1900s, the Soviet geneticist Lysenko believed that his winter wheat plants, exposed to increasingly colder temperatures, would eventually give rise to more cold-tolerant winter wheat. Lysenko's attempts in this regard were most in agreement with the ideas of _____.
A) Cuvier.
B) Lamarck.
C) Darwin.
D) Lyell.

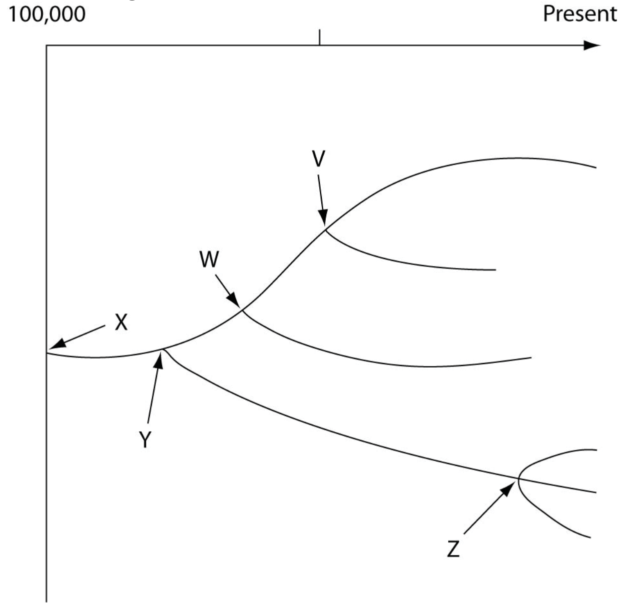
The following questions (8-9) refer to the figure below, which shows an outcrop of sedimentary rock whose strata are labeled A-D.



8) If x indicates the location of fossils of two closely related species, then fossils of their most-recent common ancestor are most likely to occur in which stratum?
A) A.
B) B.
C) C.
D) D.

9) If x indicates the fossils of two closely related species, neither of which is extinct, then their remains may be found in how many of these strata?
A) One stratum.
B) Two strata.
C) Three strata.
D) Four strata.

The following questions (10-13) refer to the evolutionary tree in the figure below.



The horizontal axis of the cladogram depicted below is a timeline that extends from 100,000 years ago to the present; the vertical axis represents nothing in particular. The labeled branch points on the tree (V-Z) represent various common ancestors. Let's say that only since 50,000 years ago has there been enough variation between the lineages depicted here to separate them into distinct species, and only the tips of the lineages on this tree represent distinct species.

10) Which pair of scientists below would probably have agreed with the process that is depicted by this tree?
A) Cuvier and Lamarck.
B) Lamarck and Wallace.
C) Aristotle and Lyell.
D) Wallace and Linnaeus.

11) The cow *Bos primigenius* (which is bred for meat and milk) has a smaller brain and larger eyes than closely related wild species of ungulates. These traits most likely arose by _____.
A) Natural selection, because these traits evolved in the population over time.
B) Natural selection, because these traits were not consciously selected by humans.
C) Artificial selection, because changes in these traits co-occurred with human selection for high milk output and high muscle content.
D) Artificial selection, because these animals differ from their close relatives and common ancestor.

12) Starting from the wild mustard *Brassica oleracea*, breeders have created the strains known as Brussels sprouts, broccoli, kale, and cabbage. Therefore, which of the following statements is correct?
A) In this wild mustard, there is enough heritable variation to permit these different varieties.
B) Heritable variation is low in wild mustard otherwise this wild strain would have different characteristics.
C) Natural selection is rare in wild populations of wild mustard.
D) In wild mustard, most of the variation is due to differences in soil or other aspects of the environment.

13) Which of the following scientists argued that variation among individuals allows evolution to occur?
A) Aristotle.
B) Lamarck.
C) Linnaeus.
D) Wallace.

14) Which of these conditions are always true of populations evolving due to natural selection?
Condition 1: The population must vary in traits that are heritable.
Condition 2: Some heritable traits must increase reproductive success.
Condition 3: Individuals pass on most traits that they acquire during their lifetime.
A) Condition 1 only.
B) Condition 2 only.
C) Conditions 1 and 2.
D) Conditions 2 and 3.

15) A farmer uses triazine herbicide to control pigweed in his field. For the first few years, the triazine works well and almost all the pigweed dies; but after several years, the farmer sees more and more pigweed. Which of these explanations best explains what happened?
A) The herbicide company lost its triazine formula and started selling poor-quality triazine.
B) Natural selection caused the pigweed to mutate, creating a new triazine-resistant species.
C) Triazine-resistant pigweed has less-efficient photosynthesis metabolism.
D) Triazine-resistant weeds were more likely to survive and reproduce.

16) After the drought of 1977, researchers on the island of Daphne Major hypothesized that medium ground finches that had large, deep beaks, survived better than those with smaller beaks because they could more easily crack and eat the tough *Tribulus cistoides* fruits. If this hypothesis is correct, what would you expect to observe if a population of these medium ground finches colonizes a nearby island where *Tribulus cistoides* is the most abundant food for the next 1000 years? Assume that (1) even the survivors of the 1977 drought sometimes had difficulty cracking the tough *T. cistoides* fruits and would eat other seeds when offered a choice; and (2) food availability is the primary limit on finch fitness on this new island.
A) Evolution of yet larger, deeper beaks over time.
B) Evolution of smaller, pointier beaks over time.
C) Random fluctuations in beak size and shape.
D) No change in beak size and shape.

17) After the drought of 1977, researchers hypothesized that on the Galápagos Island Daphne Major, medium ground finches with large, deep beaks survived better than those with smaller beaks because they could more easily crack and eat the tough Tribulus cistoides fruits. A tourist company sets up reliable feeding stations with a variety of bird seeds (different types and sizes) so that tourists can get a better look at the finches. Which of these events is now most likely to occur to finch beaks on this island?
A) Evolution of yet larger, deeper beaks over time, until all birds have relatively large, deep beaks.
B) Evolution of smaller, pointier beaks over time, until all birds have relatively small, pointy beaks.
C) Increased variation in beak size and shape over time.
D) No change in beak size and shape over time.

18) *Claytonia virginica* is a woodland spring herb with flowers that vary from white to pale pink to bright pink. Slugs prefer to eat pink-flowering over white-flowering plants (due to chemical differences between the two), and plants experiencing severe herbivory are more likely to die. The bees that pollinate this plant also prefer pink to white flowers, so that *Claytonia* with pink flowers have greater relative fruit set than *Claytonia* with white flowers. A researcher observes that the percentage of different flower colors remains stable in the study population from year to year. Given no other information, if the researcher removes all slugs from the study population, what do you expect to happen to the distribution of flower colors in the population over time?
A) The percentage of pink flowers should increase over time.
B) The percentage of white flowers should increase over time.
C) The distribution of flower colors should not change.
D) The distribution of flower colors should randomly fluctuate over time.

19) Parasitic species tend to have simple morphologies. Which of the following statements best explains this observation?
A) Parasites are lower organisms, and this is why they have simple morphologies.
B) Parasites do not live long enough to inherit acquired characteristics.
C) Simple morphologies convey some advantage in most parasites.
D) Parasites have not yet had time to progress, because they are young evolutionarily.

20) Darwin and Wallace were the first to propose _____.
A) That evolution occurs.
B) A mechanism for how evolution occurs.
C) That Earth is older than a few thousand years.
D) Natural selection as the mechanism of evolution.

21) A population of organisms will not evolve if _____.
A) All individual variation is due only to environmental factors.
B) The environment is changing at a relatively slow rate.
C) The population size is large.
D) The population lives in a habitat without competing species present.

22) Which of the following represents an idea that Darwin learned from the writings of Thomas Malthus?
A) Technological innovation in agricultural practices will permit exponential growth of the human population into the foreseeable future.
B) Populations tend to increase at a faster rate than their food supply normally allows.
C) Earth changed over the years through a series of catastrophic upheavals.
D) The environment is responsible for natural selection.

23) Given a population that contains genetic variation, what is the correct sequence of the following events under the influence of natural selection?
1. Well-adapted individuals leave more offspring than do poorly adapted individuals.
2. A change occurs in the environment.
3. Genetic frequencies within the population change.
4. Poorly adapted individuals have decreased survivorship.
A) 2 → 4 → 1 → 3.
B) 4 → 2 → 1 → 3.
C) 4 → 2 → 3 → 1.
D) 2 → 4 → 3 → 1.

24) A biologist studied a population of squirrels for fifteen years. During that time, the population was never fewer than thirty squirrels and never more than forty-five. Her data showed that over half of the squirrels born did not survive to reproduce, because of both competition for food and predation. In a single generation, 90% of the squirrels that were born lived to reproduce, and the population increased to eighty. Which inference(s) about this most recent surge in the population size might be true?
A) The amount of available food may have increased.
B) The parental generation of squirrels developed better eyesight due to improved diet; the subsequent squirrel generation inherited better eyesight.
C) The number of predators that prey upon squirrels may have decreased.
D) The amount of available food may have increased and/or the predators that prey upon squirrels may have decreased.

25) Which of the following must exist in a population before natural selection can act upon that population?
A) Genetic variation among individuals.
B) Variation among individuals caused by environmental factors.
C) Sexual reproduction.
D) The population has predators.

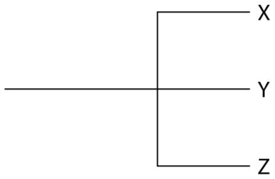
26) Which of Darwin's ideas had the strongest connection to his reading of Malthus's essay on human population growth?
A) Descent with modification.
B) Variation among individuals in a population.
C) Struggle for existence.
D) That the ancestors of the Galápagos finches had come from the South American mainland.

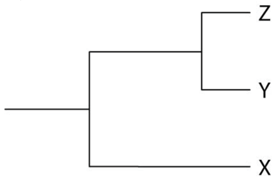
27) If Darwin had been aware of genes and their typical mode of transmission to subsequent generations, with which statement would he most likely have been in agreement?
A) If natural selection can change gene frequency in a population over generations, given enough time and genetic diversity, then natural selection can cause sufficient genetic change to produce new species from old ones.
B) If an organism's somatic cell genes change during its lifetime, making it more fit, then it will be able to pass these genes on to its offspring.
C) If an organism acquires new genes by engulfing, or being infected by, another organism, then a new genetic species will result.
D) A single mutation in a single gene in a single gamete, if inherited by future generations, will produce a new species.

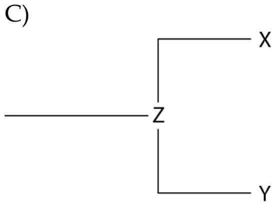
28) The role that humans play in artificial selection is to _____.
A) Determine who lives and who dies.
B) Create genetic diversity.
C) Choose which organisms reproduce.
D) Perform artificial insemination.

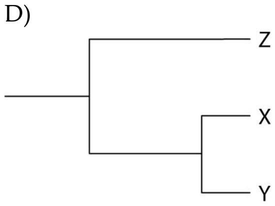
29) Currently, two extant elephant species (X and Y) are classified in the genus *Loxodonta*, and a third species (Z) is placed in the genus *Elephas*. Thus, which statement should be true?
A) Species X and Y are not related to species Z.
B) Species X and Y share a greater number of homologies with each other than either does with species Z.
C) Species X and Y share a common ancestor that is alive today.
D) Species X and Y are the result of artificial selection.

30) In a hypothetical environment, fishes called pike-cichlids are visual predators of large, adult algae-eating fish (in other words, they locate their prey by sight). The population of algae-eaters experiences predatory pressure from pike-cichlids. Which of the following is least likely to result in the algae-eater population in future generations?
A) Selection for drab coloration of the algae-eaters.
B) Selection for nocturnal algae-eaters (active only at night).
C) Selection for larger female algae-eaters, bearing broods composed of more, and larger, young.
D) Selection for algae-eaters that become sexually mature at smaller overall body sizes.

31) Currently, two of the living elephant species (X and Y) are placed in the genus *Loxodonta* and a third surviving species (Z) is placed in the genus *Elephas*. Assuming this classification reflects evolutionary relatedness, which of the following is the most accurate phylogenetic tree?
A)


B)


C)


D)


32) Cotton-topped tamarins are small primates with tufts of long white hair on their heads. While studying these creatures, you notice that males with longer hair get more opportunities to mate and father more offspring. To test the hypothesis that having longer hair is adaptive in these males, you should _____.
A) Test whether other traits in these males are also adaptive.
B) Look for evidence of hair in ancestors of tamarins.
C) Determine if hair length is heritable.
D) Test whether males with shaved heads are still able to mate.

33) Fossils of *Thrinaxodon*, a species that lived during the Triassic period, have been found in both South Africa and Antarctica. *Thrinaxodon* had a reptile-like skeleton and laid eggs, but small depressions on the front of its skull suggest it had whiskers and, therefore, fur. *Thrinaxodon* may have been warm-blooded. The fossils of *Thrinaxodon* are consistent with the hypothesis that _____.
A) Fossils found in a given area look like the modern species in that same area.
B) The environment where it lived was very warm.
C) Mammals evolved from a reptilian ancestor.
D) Antarctica and South Africa separated after *Thrinaxodon* went extinct.

34) Many crustaceans (for example, lobsters, shrimp, and crayfish) use their tails to swim, but crabs have reduced tails that curl under their shells and are not used in swimming. This is an example of _____.
A) Convergent evolution.
B) A homologous structure.
C) Natural selection.
D) A vestigial trait.

35) Which of the following, if discovered, could refute our current understanding of the pattern of evolution?
A) No fossils of soft-bodied animals.
B) A modern bird having reptile-like scales on its legs.
C) Radioactive dating of rocks showing that rocks closer to the Earth's surface are younger than lower rock strata.
D) Diverse fossils of mammals in Precambrian rock.

36) Researchers discovered that a new strain of bacteria that cause tuberculosis (*M. tuberculosis*) taken from a dead patient has a point mutation in the *rpoB* gene that codes for part of the RNA polymerase enzyme. This mutant form of RNA polymerase does not function as well as the more common form of RNA polymerase. A commonly used antibiotic called rifampin does not affect the mutant *rpoB* bacteria.
A researcher mixes M. tuberculosis with and without the *rpoB* mutation and adds the bacteria to cell cultures. Half the cell cultures contain only standard nutrients, while the other half of the cell cultures contain rifampin and the standard nutrients. After many cell generations, the researcher finds that _____.
A) Very few M. tuberculosis in the standard nutrient cell cultures carry the *rpoB* gene mutation, but almost all of the M. tuberculosis in the cell cultures with rifampin carry the *rpoB* mutation.
B) Almost all M. tuberculosis in the standard nutrient cell cultures carry the *rpoB* gene mutation, but very few of the M. tuberculosis in the cell cultures with rifampin carry the *rpoB* mutation.
C) Very few M. tuberculosis in any of the cell cultures carry the *rpoB* gene mutation.
D) Almost all of the M. tuberculosis in both types of cell cultures carry the *rpoB* mutation.

37) Scientific theories _____.
A) Are nearly the same things as hypotheses.
B) Are supported by, and make sense of, many observations.
C) Cannot be tested because the described events occurred only once.
D) Are predictions of future events.

38) DDT was once considered a "silver bullet" that would permanently eradicate insect pests. Instead, DDT is largely useless against many insects. Which of these would have prevented this evolution of DDT resistance in insect pests?
A) All habitats should have received applications of DDT at about the same time.
B) The frequency of DDT application should have been higher.
C) None of the insect pests would have genetic variations that resulted in DDT resistance.
D) DDT application should have been continual.

39) If the bacterium *Staphylococcus aureus* experiences a cost for maintaining one or more antibiotic-resistance genes, what would happen in environments that lack antibiotics?
A) These genes would be maintained in case the antibiotics appear.
B) These bacteria would be outcompeted and replaced by bacteria that have lost these genes.
C) These bacteria would try to make the cost worthwhile by locating and migrating to microenvironments where traces of antibiotics are present.
D) The number of genes conveying antibiotic resistance would increase in these bacteria.

40) Of the following anatomical structures, which is homologous to the bones in the wing of a bird?
A) Bones in the hind limb of a kangaroo.
B) Chitinous struts in the wing of a butterfly.
C) Bony rays in the tail fin of a flying fish.
D) Bones in the flipper of a whale.

41) Structures as different as human arms, bat wings, and dolphin flippers contain many of the same bones, which develop from similar embryonic tissues. These structural similarities are an example of _____.
A) Homology.
B) Convergent evolution.
C) The evolution of common structure as a result of common function.
D) The evolution of similar appearance as a result of common function.

42) Over long periods of time, many cave-dwelling organisms have lost their eyes. Tapeworms have lost their digestive systems. Whales have lost their hind limbs. How can natural selection account for these losses?
A) Natural selection cannot account for losses, but accounts only for new structures and functions.
B) Natural selection accounts for these losses by the principle of use and disuse.
C) Under particular circumstances that persisted for long periods, each of these structures presented greater costs than benefits.
D) The ancestors of these organisms experienced harmful mutations that forced them to lose these structures.

43) Which of the following evidence most strongly supports the common origin of all life on Earth? All organisms _____.
A) Require energy.
B) Use essentially the same genetic code.
C) Reproduce.
D) Show heritable variation..

44) Members of two different species possess a similar-looking structure that they use in a similar way to perform about the same function. Which of the following would suggest that the relationship more likely represents homology instead of convergent evolution?
A) The two species live at great distance from each other.
B) The two species share many proteins in common, and the nucleotide sequences that code for these proteins are almost identical.
C) The structures in adult members of both species are similar in size.
D) Both species are well adapted to their particular environments.

45) What must be true of any organ described as *vestigial?*
A) It must be analogous to some feature in an ancestor.
B) It must be homologous to some feature in an ancestor.
C) It must be both homologous and analogous to some feature in an ancestor.
D) It need be neither homologous nor analogous to some feature in an ancestor.

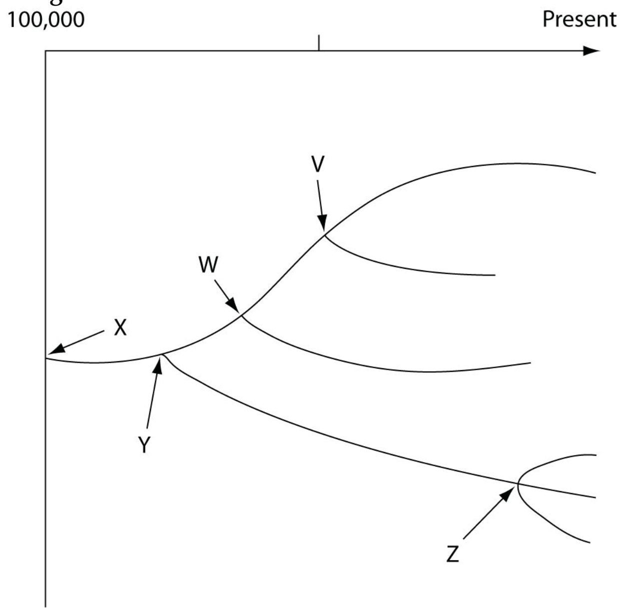
46) Pseudogenes are _____.
A) Composed of RNA, rather than DNA.
B) The same things as introns.
C) Unrelated genes that code for the same gene product.
D) Nonfunctional vestigial genes.

47) It has been observed that organisms on islands are different from, but closely related to, similar forms found on the nearest continent. This is taken as evidence that _____.
A) Island forms are descended from mainland forms.
B) Common environments are inhabited by the same organisms.
C) Island forms and mainland forms have identical gene pools.
D) The island forms and mainland forms are converging.

48) Given what we know about evolutionary biology, we expect to find the largest number of endemic species in which of the following geological features, which have existed for at least a few million years?
A) An isolated ocean island in the tropics.
B) An extensive mountain range.
C) A grassland in the center of a large continent, with extreme climatic conditions.
D) A shallow estuary on a warm-water coast.

49) The greatest number of endemic species is expected in environments that are _____.
A) Easily reached and ecologically diverse.
B) Isolated and show little ecological diversity.
C) Isolated and ecologically diverse.
D) Easily reached and show little ecological diversity.

The following questions refer to the evolutionary tree in the figure below.



The horizontal axis of the cladogram depicted below is a timeline that extends from 100,000 years ago to the present; the vertical axis represents nothing in particular. The labeled branch points on the tree (V-Z) represent various common ancestors. Let's say that only since 50,000 years ago has there been enough variation between the lineages depicted here to separate them into distinct species, and only the tips of the lineages on this tree represent distinct species.

50) How many distinct species, both living and extinct, are depicted in this tree?
A) Five.
B) Six.
C) Nine.
D) Eleven.

51) Which of the five common ancestors, labeled V-Z, is the common ancestor of the greatest number of species, both living and extinct?
A) V.
B) W.
C) Y.
D) Z.

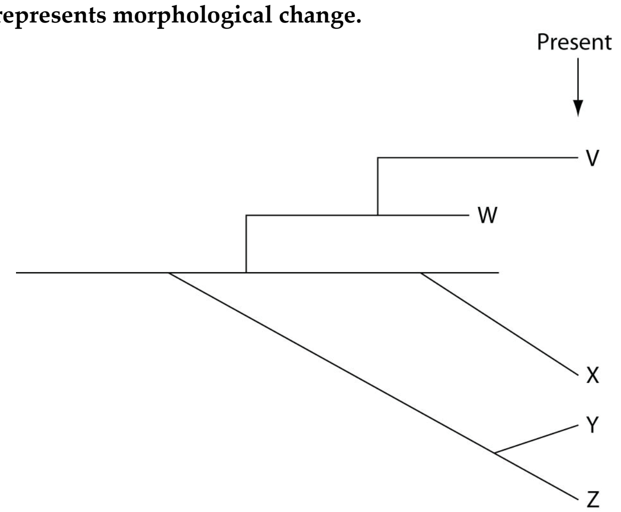
52) Which of the five species, labeled V-Z, is the common ancestor of the fewest number of species?
A) V.
B) W.
C) Y.
D) Z.

53) Evolutionary trees such as this are properly understood by scientists to be _____.
A) Theories.
B) Hypotheses.
C) Dogmas.
D) Facts.

54) About thirteen different species of finches inhabit the Galápagos Islands today, all descendants of a common ancestor from the South American mainland that arrived a few million years ago. Genetically, there are four distinct lineages, but the thirteen species are currently classified among three genera. The first lineage to diverge from the ancestral lineage was the warbler finch (genus *Certhidea*). Next to diverge was the vegetarian finch (genus *Camarhynchus*), followed by five tree finch species (also in genus *Camarhynchus*) and six ground finch species (genus *Geospiza*). If the six ground finch species have evolved most recently, then which of these is the most logical prediction?

A) They should be limited to the six islands that most recently emerged from the sea.
B) Their genomes should be more similar to each other than are the genomes of the five tree finch species.
C) They should share fewer anatomical homologies with each other than they share with the tree finches.
D) The chances of hybridization between two ground finch species should be less than the chances of hybridization between two tree finch species.

The questions (55-57) below refer to the following evolutionary tree, in which the horizontal axis represents time (present time is on the far right) and the vertical axis represents morphological change.



55) Which species is most closely related to species W?

A) V is most closely related to species W.
B) X is most closely related to species W.
C) Y and Z are equally closely related to W.
D) It is not possible to say from this tree.

56) Which of these is the extant (that is, living) species most closely related to species X?

A) V.
B) W.
C) Y.
D) Z.

57) Logically, which of these should cast the most doubt on the relationships depicted by an evolutionary tree?

A) Some of the organisms depicted by the tree had lived in different habitats.
B) The skeletal remains of the organisms depicted by the tree were incomplete (in other words, some bones were missing).
C) Transitional fossils had not been found.
D) Relationships between DNA sequences among the species did not match relationships between skeletal patterns.

CHAPTER 23:

THE EVOLUTION OF POPULATIONS

1) Which of the following is the best modern definition of evolution?
A) Descent with modification.
B) Change in the number of genes in a population over time.
C) Survival of the fittest.
D) Inheritance of acquired characters.

2) Which variable is likely to undergo the largest change in value resulting from a mutation that introduces a new allele into a population at a locus for which all individuals formerly had been fully homozygous?
A) Average heterozygosity.
B) Nucleotide variability.
C) Geographic variability.
D) Average number of loci.

3) Which statement about the beak size of finches on the island of Daphne Major during prolonged drought is true?
A) Each bird evolved a deeper, stronger beak as the drought persisted.
B) Each bird's survival was strongly influenced by the depth and strength of its beak as the drought persisted.
C) Each bird that survived the drought produced only offspring with deeper, stronger beaks than seen in the previous generation.
D) The frequency of the strong-beak alleles increased in each bird as the drought persisted.

4) Which statement about variation is true?
A) All phenotypic variation is the result of genotypic variation.
B) All genetic variation produces phenotypic variation.
C) All nucleotide variability results in neutral variation.
D) All new alleles are the result of nucleotide variability.

5) Rank the following one-base point mutations (from most likely to least likely) with respect to their likelihood of affecting the structure of the corresponding polypeptide.
1. Insertion mutation deep within an intron.
2. Substitution mutation at the third position of an exonic codon.
3. Substitution mutation at the second position of an exonic codon.
4. Deletion mutation within the first exon of the gene.
A) 1, 2, 3, 4.
B) 4, 3, 2, 1.
C) 2, 1, 4, 3.
D) 3, 1, 4, 2.

6) Genetic variation _____.
A) Is created by the direct action of natural selection.
B) Arises in response to changes in the environment.
C) Must be present in a population before natural selection can act upon the population.
D) Tends to be reduced by when diploid organisms produce gametes.

The questions (7-8) below refer to the following paragraph.

HIV's genome of RNA includes the code for reverse transcriptase (RT), an enzyme that acts early in infection to synthesize a DNA genome off of an RNA template. The HIV genome also codes for protease (PR), an enzyme that acts later in infection by cutting long viral polyproteins into smaller, functional proteins. Both RT and PR represent potential targets for antiretroviral drugs. Drugs called nucleoside analogs (NA) act against RT, whereas drugs called protease inhibitors (PI) act against PR.

7) Which of the following represents the treatment option most likely to avoid the evolution of drug-resistant HIV (assuming no drug interactions or side effects)?
A) Using a series of NAs, one at a time, and changed about once a week.
B) Using a single PI, but slowly increasing the dosage over the course of a week.
C) Using high doses of NA and a PI at the same time for a period not to exceed one day.
D) Using moderate doses of NA and two different PIs at the same time for several months.

8) Every HIV particle contains two RNA molecules. If two genes from one RNA molecule become detached and then, as a unit, get attached to one end of the other RNA molecule within a single HIV particle, which of these is true?
A) There are now fewer genes within the viral particle.
B) There are now more genes within the viral particle.
C) A point substitution mutation has occurred in the retroviral genome.
D) One of the RNA molecules has experienced gene duplication as the result of translocation.

The following experiment is used for the corresponding questions (9-11).

A researcher discovered a species of moth that lays its eggs on oak trees. Eggs are laid at two distinct times of the year: early in spring when the oak trees are flowering and in midsummer when flowering is past. Caterpillars from eggs that hatch in spring feed on oak flowers and look like oak flowers. But caterpillars that hatch in summer feed on oak leaves and look like oak twigs.

How does the same population of moths produce such different-looking caterpillars on the same trees? To answer this question, the biologist caught many female moths from the same population and collected their eggs. He put at least one egg from each female into eight identical cups. The eggs hatched, and at least two larvae from each female were maintained in one of the four temperature and light conditions listed below.

Temperature	Day Length
Springlike	Springlike
Springlike	Summerlike
Summerlike	Springlike
Summerlike	Summerlike

In each of the four environments, one of the caterpillars was fed oak flowers, the other oak leaves. Thus, there were a total of eight treatment groups (4 environments × 2 diets).

9) Refer to the accompanying figure. Which one of the following is NOT a plausible hypothesis to explain the differences in caterpillar appearance observed in this population?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) The cooler temperatures of spring trigger the development of flowerlike caterpillars.
C) Differences in air pressure, due to differences in elevation, trigger the development of different types of caterpillars.
D) Differences in diet trigger the development of different types of caterpillars.

10) Refer to the accompanying figure. In every case, caterpillars that feed on oak flowers look like oak flowers. In every case, caterpillars that were raised on oak leaves looked like twigs. These results support which of the following hypotheses?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) Differences in air pressure, due to elevation, trigger the development of different types of caterpillars.
C) Differences in diet trigger the development of different types of caterpillars.
D) The differences are genetic. A female will either produce all flowerlike caterpillars or all twig-like caterpillars.

11) Refer to the accompanying figure. Recall that eggs from the same female were exposed to each of the eight treatments used. This aspect of the experimental design tested which of the following hypotheses?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) Differences in air pressure, due to elevation, trigger the development of different types of caterpillars.
C) Differences in diet trigger the development of different types of caterpillars.
D) The differences are genetic. A female will either produce all flowerlike caterpillars or all twig-like caterpillars.

12) Cystic fibrosis is a genetic disorder in homozygous recessives that causes death during the teenage years. If 9 in 10,000 newborn babies have the disease, what are the expected frequencies of the dominant (*A1*) and recessive (*A2*) alleles according to the Hardy-Weinberg model?
A) f(*A1*) = 0.9997, f(*A2*) = 0.0003.
B) f(*A1*) = 0.9800, f(*A2*) = 0.0200.
C) f(*A1*) = 0.9700, f(*A2*) = 0.0300.
D) f(*A1*) = 0.9604, f(*A2*) = 0.0392.

13) Suppose 64% of a remote mountain village can taste phenylthiocarbamide (PTC) and must, therefore, have at least one copy of the dominant PTC taster allele. If this population conforms to Hardy-Weinberg expectations for this gene, what percentage of the population must be heterozygous for this trait?
A) 16%.
B) 32%.
C) 40%.
D) 48%.

14) For biologists studying a large flatworm population in the lab, which Hardy -Weinberg condition is most difficult to meet?
A) No selection.
B) No genetic drift.
C) No gene flow.
D) No mutation.

15) For a biologist studying a small fish population in the lab, which Hardy -Weinberg condition is easiest to meet?
A) No selection.
B) No genetic drift.
C) No gene flow.
D) No mutation.

Use the following information to answer the questions (16-17) below.

Researchers studying a small milkweed population note that some plants produce a toxin and other plants do not. They identify the gene responsible for toxin production. The dominant allele (*T*) codes for an enzyme that makes the toxin, and the recessive allele (*t*) codes for a nonfunctional enzyme that cannot produce the toxin. Heterozygotes produce an intermediate amount of toxin. The genotypes of all individuals in the population are determined (see chart) and used to determine the actual allele frequencies in the population.

Genotype Frequencies			Allele Frequencies	
TT	*Tt*	*tt*	*T*	*T*
0.56	0.28	0.16		

16) Refer to the figure above. Is this population in Hardy-Weinberg equilibrium?
A) Yes.
B) No; there are more heterozygotes than expected.
C) No; there are more homozygotes than expected.
D) More information is needed to answer this question.

17) If, on average, 46% of the loci in a species' gene pool are heterozygous, then the average homozygosity of the species should be _____.
A) 23%.
B) 46%.
C) 54%.
D) 92%.

18) The higher the proportion of loci that are "fixed" in a population, the lower are that population's _____.
A) Nucleotide variability.
B) Chromosome number.
C) Average heterozygosity.
D) Nucleotide variability and average heterozygosity.

19) Whenever diploid populations are in Hardy-Weinberg equilibrium at a particular locus, _____.
A) The allele's frequency should not change from one generation to the next.
B) Natural selection, gene flow, and genetic drift are acting equally to change an allele's frequency.
C) Two alleles are present in equal proportions.
D) Individuals within the population are evolving.

20) In the formula for determining a population's genotype frequencies, the "2" in the term $2pq$ is necessary because _____.
A) The population is diploid.
B) Heterozygotes can come about in two ways.
C) The population is doubling in number.
D) Heterozygotes have two alleles.

21) In the formula for determining a population's genotype frequencies, the "pq" in the term $2pq$ is necessary because _____.
A) The population is diploid.
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C) The population is doubling in number.
D) Heterozygotes have two alleles.

22) In a Hardy-Weinberg population with two alleles, *A* and *a*, that are in equilibrium, the frequency of the allele *a* is 0.3. What is the frequency of individuals that are homozygous for this allele?
A) 0.09.
B) 0.49.
C) 0.9.
D) 9.0.

23) In a Hardy-Weinberg population with two alleles, *A* and *a*, that are in equilibrium, the frequency of allele *a* is 0.2. What is the frequency of individuals that are heterozygous for this allele?
A) 0.020.
B) 0.04.
C) 0.16.
D) 0.32.

24) In a Hardy-Weinberg population with two alleles, *A* and *a*, that are in equilibrium, the frequency of allele *a* is 0.1. What is the frequency of individuals with *AA* genotype?
A) 0.20.
B) 0.32.
C) 0.42.
D) 0.81.

25) You sample a population of butterflies and find that 56% are heterozygous at a particular locus. What should be the frequency of the recessive allele in this population?
A) 0.08.
B) 0.09.
C) 0.70.
D) Allele frequency cannot be determined from this information.

26) In peas, a gene controls flower color such that *R* = purple and *r* = white. In an isolated pea patch, there are 36 purple-flowering plants and 64 white-flowering plants. Assuming Hardy-Weinberg equilibrium, what is the value of *q* for this population?
A) 0.36.
B) 0.64.
C) 0.75.
D) 0.80.

Use the following information to answer the questions (27-29) below.

A large population of laboratory animals has been allowed to breed randomly for a number of generations. After several generations, 25% of the animals display a recessive trait (*aa*), the same percentage as at the beginning of the breeding program. The rest of the animals show the dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.

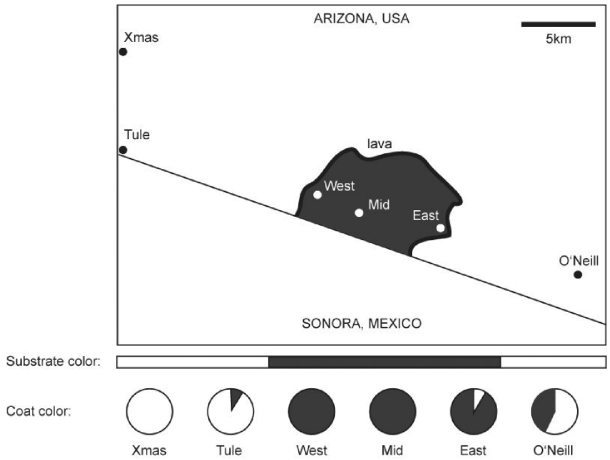
27) What is the most reasonable conclusion that can be drawn from the fact that the frequency of the recessive trait (*aa*) has not changed over time?
A) The two phenotypes are about equally adaptive under laboratory conditions.
B) The genotype *AA* is lethal.
C) There has been a high rate of mutation of allele *A* to allele *a*.
D) There has been sexual selection favoring allele *a*.

28) What is the estimated frequency of allele *A* in the gene pool?
A) 0.25.
B) 0.50.
C) 0.75.
D) 0.125.

29) What proportion of the population is probably heterozygous (*Aa*) for this trait?
A) 0.05.
B) 0.25.
C) 0.50.
D) 0.75.

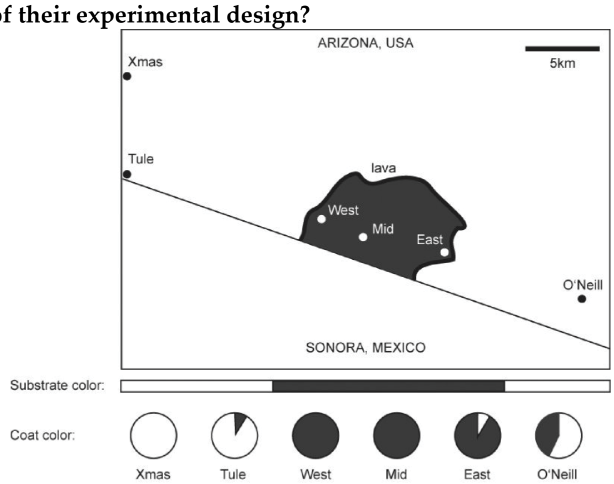
30) Mutation is the only evolutionary mechanism that _____.
A) Does little to change allele frequencies.
B) Is more important in eukaryotes than in prokaryotes.
C) Happens in all populations.
D) Has no effect on genetic variation.

31) The figure below shows the distribution of pocket-mouse coat colors in several Arizona populations found either on light-colored granite substrate or on dark volcanic rock (dark substrate). The Melanocortin-1 receptor (*Mc1r*) alleles, *D* and *d*, differ by four amino acids. Mice with *DD* and *Dd* genotypes have dark coats, whereas mice with the *dd* genotype are light colored. What sort of genotype frequencies might you expect to find in the Xmas, Mid, and O'Neill populations?



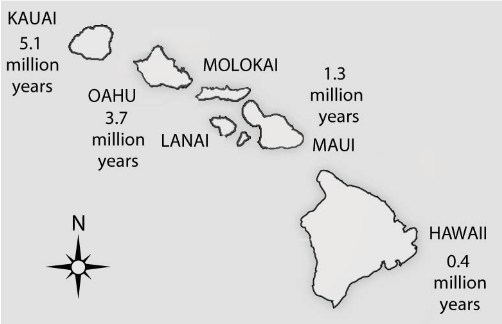
A) Xmas-high *DD* frequency; Mid-high *Dd* frequency, O'Neill-high dd frequency.
B) Xmas-high *Dd* frequency; Mid-high *DD* frequency, O'Neill-high dd frequency.
C) Xmas-high *dd* frequency; Mid-high *Dd* frequency, O'Neill-high DD frequency.
D) Xmas-high *dd* frequency; Mid-high *DD* frequency, O'Neill-high Dd frequency.

32) Refer to the figure above. In their investigation of natural selection on *Mc1r* alleles (the gene that determines coat color) in Arizona pocket mice, Hoekstra et al. determined the frequency of the *D* and *d* alleles in each population. They also determined the frequency of alleles for two neutral mitochondrial DNA genes (genes that do not affect and are not linked to coat color). Why did the researchers include the mitochondrial DNA genes as part of their experimental design?



A) Allele change for the neutral mitochondrial genes serves as an experimental group and gives information on any general background genetic difference among these populations.
B) Allele change for the neutral mitochondrial genes serves as a control and determines coat-color differences among these populations.
C) Allele change for the neutral mitochondrial genes serves as an experimental group and gives information on coat-color differences among these populations.
D) Allele change for the neutral mitochondrial genes serves as a control and gives information on any general background genetic difference among these populations.

33) Soon after the island of Hawaii rose above the sea surface (somewhat less than one million years ago), the evolution of life on this new island should have been most strongly influenced by _____.



A) A genetic bottleneck.
B) Sexual selection.
C) Habitat differentiation.
D) The founder effect.

Use the following information to answer the questions (34-35) below.

In 1983 a population of dark-eyed junco birds became established on the campus of the University of California, San Diego (UCSD), which is located many miles from the junco's normal habitat in the mixed-coniferous temperate forests in the mountains. Juncos have white outer tail feathers that the males display during aggressive interactions and during courtship displays. Males with more white in their tail are more likely to win aggressive interactions, and females prefer to mate with males with more white in their tails. Females have less white in their tails than do males, and display it less often. (Pamela J. Yeh. 2004. Rapid evolution of a sexually selected trait following population establishment in a novel habitat. *Evolution* 58[1]:166-74.)

34) Refer to the paragraph on dark-eyed junco birds. The UCSD campus male junco population tails were, on average, 36% white, whereas the tails of males from nearby mountain populations averaged 40—45% white. If this observed trait difference were due to a difference in the original colonizing population, it would most likely be due to _____.
A) Mutations in the UCSD population.
B) Gene flow between populations.
C) A genetic bottleneck.
D) A founder effect.

35) Refer to the paragraph on dark-eyed junco birds. The UCSD campus male junco population tails are about 36% white, whereas the tails of males from nearby mountain populations are about 40—45% white. The founding stock of UCSD birds was likely from the nearby mountain populations because some of those birds overwinter on the UCSD campus each year. Population sizes on the UCSD campus have been reasonably large, and there are significant habitat differences between the UCSD campus and the mountain coniferous forests; UCSD campus has a more open environment (making birds more visible) and a lower junco density (decreasing intraspecific competition) than that in the mountain forests. Given this information, which of the following evolutionary mechanisms do you think is most likely responsible for the difference between the UCSD and mountain populations?
A) Natural selection.
B) Genetic drift.
C) Gene flow.
D) Mutation.

36) The Dunkers are a religious group that moved from Germany to Pennsylvania in the mid-1700s. They do not marry with members outside their own immediate community. Today, the Dunkers are genetically unique and differ in gene frequencies, at many loci, from all other populations including those in their original homeland. Which of the following likely explains the genetic uniqueness of this population?
A) Population bottleneck and Hardy-Weinberg equilibrium.
B) Heterozygote advantage and stabilizing selection.
C) Mutation and natural selection.
D) Founder effect and genetic drift.

37) An earthquake decimates a ground-squirrel population, killing 98% of the squirrels. The surviving population happens to have broader stripes, on average, than the initial population. If broadness of stripes is genetically determined, what effect has the ground-squirrel population experienced during the earthquake?
A) Directional selection.
B) Disruptive selection.
C) A founder event.
D) A genetic bottleneck.

38) Which of the following is the most predictable outcome of increased gene flow between two populations?
A) Lower average fitness in both populations.
B) Higher average fitness in both populations.
C) Increased genetic difference between the two populations.
D) Decreased genetic difference between the two populations.

39) In 1986, a nuclear power accident in Chernobyl, USSR (now Ukraine), and led to high radiation levels for miles surrounding the plant. The high levels of radiation caused elevated mutation rates in the surviving organisms, and evolutionary biologists have been studying rodent populations in the Chernobyl area ever since. Based on your understanding of evolutionary mechanisms, which of the following most likely occurred in the rodent populations following the accident?
A) Mutations caused major changes in rodent physiology over time.
B) Mutation led to increased genetic variation.
C) Mutation caused genetic drift and decreased fitness.
D) Mutation caused the fixation of new alleles.

40) Over time, the movement of people on Earth has steadily increased. This has altered the course of human evolution by increasing _____.
A) Nonrandom mating.
B) Geographic isolation.
C) Genetic drift.
D) Gene flow.

41) You are maintaining a small population of fruit flies in the laboratory by transferring the flies to a new culture bottle after each generation. After several generations, you notice that the viability of the flies has decreased greatly. Recognizing that small population size is likely to be linked to decreased viability, the best way to reverse this trend is to _____.
A) Cross your flies with flies from another lab.
B) Reduce the number of flies that you transfer at each generation.
C) Transfer only the largest flies.
D) Change the temperature at which you rear the flies.

42) The inability of organisms to evolve anything that could be an advantage reflects _____.
A) The limits of historical constraints.
B) The inability to compromise.
C) The consequences of random mutations.
D) The consequences of inbreeding.

43) Which of the following is a fitness trade-off (compromise)?
A) In some hornbill species, the male helps seal the female in a tree with her nest until the young are ready to fledge.
B) Hummingbirds are the best pollinators of certain flowers, but bees are the best pollinators for orchids.
C) The strong, thick beak of a woodpecker helps it find insects in trees.
D) Turtle shells provide protection but are heavy and burdensome when moving.

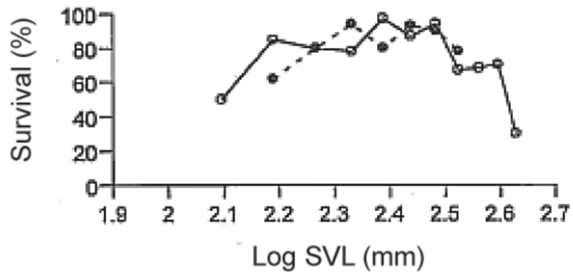
Use the following description to answer the question (44) below.

On the Bahamian island of Andros, mosquitofish populations live in various, now -isolated, freshwater ponds that were once united. Currently, some predator-rich ponds have mosquitofish that can swim in short, fast bursts; other predator-poor ponds have mosquitofish that can swim continuously for a long time. When placed together in the same body of water, the two kinds of female mosquitofish exhibit exclusive breeding preferences.

44) If one builds a canal linking a predator-rich pond to a predator-poor pond, then what type(s) of selection should subsequently be most expected among the mosquitofish in the original predator-rich pond, and what type(s) should be most expected among the mosquitofish in the formerly predator-poor pond?
A) Stabilizing selection; directional selection.
B) Stabilizing selection; stabilizing selection.
C) Less-intense directional selection; more-intense directional selection.
D) Less-intense disruptive selection; more-intense disruptive selection.

Use the following information to answer the questions (45-46) below.

Martin Wikelski and L. Michael Romero (Body size, performance and fitness in Galápagos marine iguanas, Integrative and Comparative Biology 43 [2003]:376-86) measured the snout-to-vent (anus) length of Galápagos marine iguanas and observed the percent survival of different-sized animals, all of the same age. The graph shows the log snout-vent length (SVL, a measure of overall body size) plotted against the percent survival of these different size classes for males and females.



45) Examine the figure above. What type of selection for body size appears to be occurring in these marine iguanas?
A) Directional selection.
B) Stabilizing selection.
C) Disruptive selection.
D) You cannot determine the type of selection from the above information.

46) Currently the only predators of Galápagos marine iguanas are Galápagos hawks. Iguana body size is not correlated with risk of hawk predation, although small iguanas can sprint faster than large iguanas. If predators (for example, cats) that preferably catch and eat slower iguanas are introduced to the island, iguana body size is likely to _____ in the absence of other factors; the iguanas would then be under _____ selection.
A) Increase; directional.
B) Increase; disruptive.
C) Decrease; directional.
D) Decrease; disruptive.

47) Three-spined stickleback fish (*Gasterosteus aculeatus*) show substantial heritable variation in gill-raker length related to differences in their diets. Longer gill rakers appear to function better for capturing open-water prey, while shorter gill rakers function better for capturing shallow-water prey. Which of the following types of selection is most likely to be found in a large lake (open water in the middle and shallow water around the sides) with a high density of these fish?
A) Directional selection.
B) Stabilizing selection.
C) Disruptive selection.
D) Sexual selection.

48) A biologist doing a long-term study on a wild spider population observes increased variation in silk thickness. Spider population is could be experiencing _____.
A) Directional selection.
B) Stabilizing selection.
C) Disruptive selection.
D) Genetic drift.

49) In some jacana species, males take care of the eggs and young, and females compete among themselves for territories that contain one to several males. Female jacanas are significantly larger than males. Which of these statements would you predict to be true of this bird species?
1. Male jacana fitness is primarily limited by ability to take care of eggs and raise young.
2. Female jacana fitness is limited by the number of males in her territory with which a female mates.
3. Variation in reproductive success should be greater in male jacanas than in females.
4. Variation in reproductive success should be greater in female jacanas than in males.
5. Males and females have equal variation in reproductive success.
A) 1 and 3.
B) 2 and 4.
C) 1, 2, and 4.
D) 5.

50) The restriction enzymes of bacteria protect the bacteria from successful attack by bacteriophages, whose genomes can be degraded by the restriction enzymes. The bacterial genomes are not vulnerable to these restriction enzymes because bacterial DNA is methylated. This situation selects for bacteriophages whose genomes are also methylated. As new strains of resistant bacteriophages become more prevalent, this in turn selects for bacteria whose genomes are not methylated and whose restriction enzymes instead degrade methylated DNA. The outcome of the conflict between bacteria and bacteriophage at any point in time results from _____.
A) Frequency-dependent selection.
B) Evolutionary imbalance.
C) Heterozygote advantage.
D) Neutral variation.

51) The restriction enzymes of bacteria protect the bacteria from successful attack by bacteriophages, whose genomes can be degraded by the restriction enzymes. The bacterial genomes are not vulnerable to these restriction enzymes because bacterial DNA is methylated. This situation selects for bacteriophages whose genomes are also methylated. As new strains of resistant bacteriophages become more prevalent, this in turn selects for bacteria whose genomes are not methylated and whose restriction enzymes instead degrade methylated DNA. Over the course of evolutionary time, what should occur?
A) Methylated DNA should become fixed in the gene pools of bacterial species.
B) Nonmethylated DNA should become fixed in the gene pools of bacteriophages.
C) Methylated DNA should become fixed in the gene pools of bacteriophages.
D) Methylated and nonmethylated strains should be maintained among both bacteria and bacteriophages, with ratios that vary over time.

52) Arrange the following in order from most general to most specific.
1. Natural selection.
2. Microevolution.
3. Intrasexual selection.
4. Evolution.
5. Sexual selection.
A) 4, 1, 2, 3, 5.
B) 4, 2, 1, 3, 5.
C) 4, 2, 1, 5, 3.
D) 1, 4, 2, 5, 3.

53) Adult male humans generally have deeper voices than do adult female humans, which is the direct result of higher levels of testosterone causing growth of the larynx. If the fossil records of apes and humans alike show a trend toward decreasing larynx size in adult females and increasing larynx size in adult males, then _____.
A) Sexual dimorphism was evolving over time in these species.
B) Intrasexual selection seems to have occurred in both species.
C) Stabilizing selection was occurring in these species concerning larynx size.
D) Selection was acting more directly upon genotype than upon phenotype.

54) Most Swiss starlings produce four to five eggs in each clutch. Starlings producing fewer or more than this have reduced fitness. Which of the following terms best describes this situation?
A) Directional selection.
B) Stabilizing selection.
C) Disruptive selection.
D) Sexual selection.

55) When imbalances occur in the sex ratio of sexual species that have two sexes (that is, other than a 50:50 ratio), the members of the minority sex often receive a greater proportion of care and resources from parents than do the offspring of the majority sex. This is most clearly an example of _____.

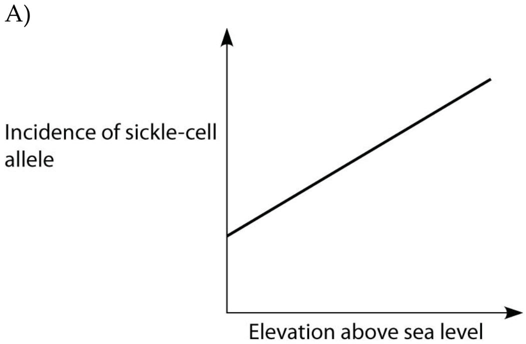
A) Sexual selection.
B) Balancing selection.
C) Stabilizing selection.
D) Frequency-dependent selection.

56) A proficient engineer can easily design skeletal structures that are more functional than those currently found in the forelimbs of such diverse mammals as horses, whales, and bats. The actual forelimbs of these mammals do not seem to be optimally arranged because _____.

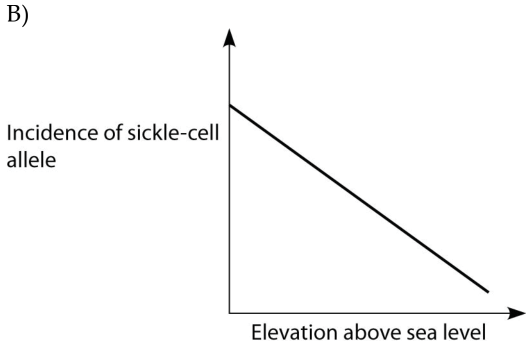
A) Natural selection has not had sufficient time to create the optimal design in each case, but will do so given enough time.
B) In many cases, phenotype is determined by genotype and the environment.
C) Though we may not consider the fit between the current skeletal arrangements and their functions excellent, we should not doubt that natural selection ultimately produces the best design.
D) Natural selection is generally limited to modifying structures that were present in previous generations and in previous species.

57) Anopheles mosquitoes, which carry the malaria parasite, cannot live above elevations of 5900 feet. In addition, oxygen availability decreases with higher altitude. Consider a hypothetical human population that is adapted to life on the slopes of Mt. Kilimanjaro in Tanzania, a country in equatorial Africa. Mt. Kilimanjaro's base is about 2600 feet above sea level and its peak is 19,341 feet above sea level. If the incidence of the sickle-cell allele in the population is plotted against altitude (feet above sea level), which of the following distributions is most likely, assuming little migration of people up or down the mountain?

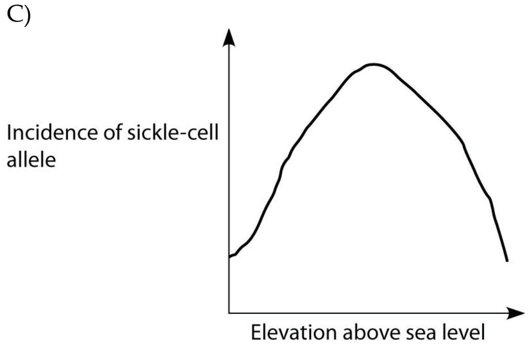
A)



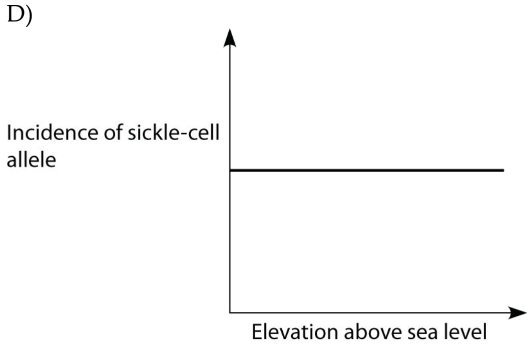
B)



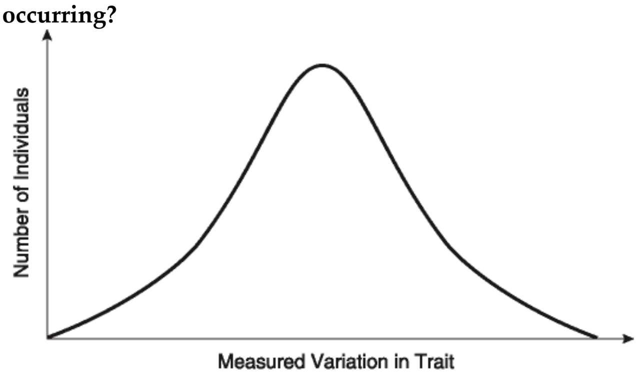
C)



D)



58) In a very large population, a quantitative trait has the following distribution pattern. If there is no gene flow, the curve shifts to the left or to the right, and the population size consequently increases over successive generations, which of the following is most likely occurring?



A) Immigration or emigration.
B) Directional selection.
C) Disruptive selection.
D) Genetic drift.

Use the following information to answer the question (59) below.

In those parts of equatorial Africa where the malaria parasite is most common, the sickle-cell allele constitutes 20% of the β hemoglobin alleles in the human gene pool.

59) In the United States, the parasite that causes malaria is not present, but African-Americans whose ancestors were from equatorial Africa are present. What should be happening to the sickle-cell allele in the United States, and what should be happening to it in equatorial Africa?
A) Stabilizing selection; disruptive selection.
B) Disruptive selection; stabilizing selection.
C) Directional selection; disruptive selection.
D) Directional selection; stabilizing selection.

60) Swine are vulnerable to infection by bird flu virus and human flu virus, which can both be present in an individual pig at the same time. When this occurs, it is possible for genes from bird flu virus and human flu virus to be combined. If the human flu virus contributes a gene for Tamiflu resistance (Tamiflu is an antiviral drug) to the new virus, and if the new virus is introduced to an environment lacking Tamiflu, then what is most likely to occur?
A) The new virus will maintain its Tamiflu-resistance gene, in case of future exposure to Tamiflu.
B) The Tamiflu-resistance gene will undergo mutations that convert it into a gene that has a useful function in this environment.
C) If the Tamiflu-resistance gene involves a cost, it will experience directional selection leading to reduction in its frequency.
D) If the Tamiflu-resistance gene confers no benefit in the current environment, and has no cost, the virus will increase in frequency.

CHAPTER 24:

THE ORIGIN OF SPECIES

Use the following information to answer the questions (1-2) below.

Two populations of birds with somewhat different coloration live on opposite sides of a peninsula. The habitat between the populations is not suitable for these birds. When birds from the two populations are brought together, they produce young whose appearance is intermediate between the two parents. These offspring will breed with each other or with birds from either parent population, and all offspring of these pairings appear intermediate to various degrees.

1) What keeps the two populations separate?
A) Temporal reproductive isolation.
B) Lack of hybrid viability.
C) Behavioral reproductive isolation.
D) Habitat isolation.

2) The two populations are _____.
A) Different subspecies, under the morphological species concept.
B) Different species, under the biological species concept.
C) Different species, under the phylogenetic species concept.
D) None of them.

3) Three populations of crickets look very similar, but the males have courtship songs that sound different. What function would this difference in song likely serve if the populations came in contact?
A) A temporal reproductive isolating mechanism.
B) A postzygotic isolating mechanism.
C) A behavioral reproductive isolating mechanism.
D) A gametic reproductive isolating mechanism.

4) Many songbirds breed in North America in the spring and summer and then migrate to Central and South America in the fall. They spend the winter in these warmer areas, where they feed and prepare for the spring migration north and another breeding season. Two hypothetical species of sparrow, A and B, overwinter together in mixed flocks in Costa Rica. In spring, species A goes to the east coast of North America, and species B goes to the west coast. What can you say about the isolating mechanisms of these two species?
A) They must have strong postzygotic isolating mechanisms to spend winter in such close proximity.
B) They must have strong prezygotic isolating mechanisms to spend winter in such close proximity.
C) Their winter habitat has no bearing on their degree of reproductive isolation.
D) Reinforcement must be occurring when they winter together.

5) The peppered moth provides a well-known example of natural selection. The light-colored form of the moth was predominant in England before the Industrial Revolution. In the mid-nineteenth century, a dark-colored form appeared. The difference is produced by a dominant allele of one gene. By about 1900, approximately 90% of the moths around industrial areas were dark colored, whereas light -colored moths were still abundant elsewhere. Apparently, birds could readily find the light moths against the soot-darkened background in industrial areas and, therefore, were eating more light moths. Recently, use of cleaner fuels has greatly reduced soot in the landscape, and the dark-colored moths have been disappearing. Should the two forms of moths be considered separate species?
A) Yes, because natural selection has affected the frequency of the two different forms.
B) Yes, because they have completely different coloration.
C) Yes, because they are reproductively isolated based on habitat.
D) No.

Use the following information to answer the question (6) below.

About 3 million years ago, the Isthmus of Panama (a narrow strip of land connecting North and South America) formed, dividing marine organisms into Pacific and Caribbean populations. Researchers have examined species of snapping shrimp on both sides of the isthmus. Based on the morphological species concept, there appeared to be seven pairs of species, with one species of each pair in the Pacific and the other in the Caribbean. The different species pairs live at somewhat different depths in the ocean. Using mitochondrial DNA sequences, the researchers estimated phylogenies and found that each of these species pairs, separated by the isthmus, were indeed each other's closest relatives. The researchers investigated mating in the lab and found that many species pairs were not very interested in courting with each other, and any that did mate almost never produced fertile offspring. (N. Knowlton, L. A. Weigt, L. A. Solorzano, D. K. Mills, and E. Bermingham. 1993. Divergence in proteins, mitochondrial DNA, and reproductive incompatibility across the Isthmus of Panama. Science 260:1629-32.)

6) Refer to the paragraph about the formation of the Isthmus of Panama. The sister populations on opposite sides of the isthmus are true species under which species concept?
A) The morphological species concept.
B) The biological species concept.
C) The phylogenetic species concept.
D) The morphological species, biological species, and phylogenetic species concepts.

7) The common edible frog of Europe is a hybrid between two species, *Rana lessonae* and *Rana ridibunda*. The hybrids were first described in 1758 and have a wide distribution, from France across central Europe to Russia. Both male and female hybrids exist, but when they mate among themselves, they are rarely successful in producing offspring. What can you infer from this information?
A) Postzygotic isolation exists between the two frog species.
B) Prezygotic isolation exists between the two frog species.
C) These two species are likely in the process of fusing back into one species.
D) The hybrids form a separate species under the biological species concept.

8) The approach to estimating phylogenetic trees is most like the approach of which species concept?
A) Morphological species concept.
B) Biological species concept.
C) Phylogenetic species concept.
D) None of them.

9) Macroevolution is _____.
A) The same as microevolution, but includes the origin of new species.
B) Evolution above the species level.
C) Defined as the evolution of microscopic organisms into organisms that can be seen with the naked eye.
D) Defined as a change in allele or gene frequency over the course of many generations.

10) Which of the various species concepts distinguishes two species based on the degree of genetic exchange between their gene pools?
A) Phylogenetic.
B) Ecological.
C) Biological.
D) Morphological.

11) There is still some controversy among biologists about whether Neanderthals should be placed within the same species as modern humans or into a separate species of their own. Most DNA sequence data analyzed so far indicate that there was probably little or no gene flow between Neanderthals and *Homo sapiens*. Which species concept is most applicable in this example?
A) Phylogenetic.
B) Ecological.
C) Morphological.
D) Biological.

12) You are confronted with a box of preserved grasshoppers of various species that are new to science and have not been described. Your assignment is to separate them into species. Which species concept will you have to use?
A) Biological.
B) Phylogenetic.
C) Ecological.
D) Morphological.

13) Dog breeders maintain the purity of breeds by keeping dogs of different breeds apart when they are fertile. This kind of isolation is most similar to which of the following reproductive isolating mechanisms?
A) Temporal isolation.
B) Behavioral isolation.
C) Habitat isolation.
D) Gametic isolation.

14) Rank the following in order from most general to most specific:
1. Gametic isolation.
2. Reproductive isolating mechanism.
3. Sperm-egg incompatibility in sea urchins.
4. Prezygotic isolating mechanism.
A) 2, 3, 1, 4.
B) 2, 4, 1, 3.
C) 4, 1, 2, 3.
D) 4, 2, 1, 3.

15) Two species of frogs belonging to the same genus occasionally mate, but the embryos stop developing after a day and then die. These two frog species separate by _____.
A) Reduced hybrid viability.
B) Hybrid breakdown.
C) Reduced hybrid fertility.
D) Gametic isolation.

16) The production of sterile mules by interbreeding between female horses (mares) and male donkeys (jacks) is an example of _____.
A) Reduced hybrid viability.
B) Hybrid breakdown.
C) Reduced hybrid fertility.
D) Mechanical isolation.

17) Dogs (*Canis lupus familiaris*) and gray wolves (*Canis lupus*) can interbreed to produce viable, fertile offspring. These species shared a common ancestor recently (in geologic time) and have a high degree of genetic similarity, although their anatomies vary widely. Judging from this evidence, which two species concepts are most likely to place dogs and wolves together into a single species?
A) Ecological and morphological.
B) Ecological and phylogenetic.
C) Biological and morphological.
D) Biological and phylogenetic.

18) Rocky Mountain juniper (*Juniperus scopulorum*) and one-seeded juniper (*J. monosperma*) have overlapping ranges. Pollen grains (which contain sperm cells) from one species are unable to germinate and make pollen tubes on female ovules (which contain egg cells) of the other species. These two juniper species are kept separate by _____.

A) Habitat isolation.
B) Temporal isolation.
C) Gametic isolation.
D) Behavioral isolation.

19) What does the biological species concept use as the primary criterion for determining species boundaries?

A) Geographic isolation.
B) Niche differences.
C) Gene flow.
D) Morphological similarity.

20) The largest unit within which gene flow can readily occur is _____.

A) A population.
B) A species.
C) The entire range of a genus.
D) The hybrid zone.

Use the following information to answer the question (21) below.

About 3 million years ago, the Isthmus of Panama (a narrow strip of land connecting North and South America) formed, dividing marine organisms into Pacific and Caribbean populations. Researchers have examined species of snapping shrimp on both sides of the isthmus. Based on the morphological species concept, there appeared to be seven pairs of species, with one species of each pair in the Pacific and the other in the Caribbean. The different species pairs live at somewhat different depths in the ocean. Using mitochondrial DNA sequences, the researchers estimated phylogenies and found that each of these species pairs, separated by the isthmus, were indeed each other's closest relatives. The researchers investigated mating in the lab and found that many species pairs were not very interested in courting with each other, and any that did mate almost never produced fertile offspring. (N. Knowlton, L. A. Weigt, L. A. Solorzano, D. K. Mills, and E. Bermingham. 1993. Divergence in proteins, mitochondrial DNA, and reproductive incompatibility across the Isthmus of Panama. Science 260:1629-32.)

21) Refer to the paragraph about the formation of the Isthmus of Panama. If the isthmus formed gradually rather than suddenly, what pattern of genetic divergence would you expect to find in these species pairs?

A) Similar percentages of difference in DNA sequence between all pairs of sister species.
B) Greater percentage of difference in DNA sequence between species that inhabit deep water than between species that inhabit shallow water.
C) Greater percentage of difference in DNA sequence between species that inhabit shallow water than between species that inhabit deep water.
D) None of them.

22) Which of the following describes the most likely order of events in allopatric speciation?

A) Genetic drift, genetic isolation, divergence.
B) Genetic isolation, divergence, genetic drift.
C) Divergence, genetic drift, genetic isolation.
D) Genetic isolation, genetic drift, divergence.

23) You want to study divergence of populations, and you need to maximize the rate of divergence to see results within the period of your grant funding. You will form a new population by taking some individuals from a source population and isolating them so the two populations cannot interbreed. What combination of characteristics would maximize your chance of seeing divergence in this study?

1. Choose a random sample of individuals to form the new population.
2. Choose individuals from one extreme to form the new population.
3. Choose a species to study that produces many offspring.
4. Choose a species to study that produces a few, large offspring.
5. Place the new population in the same type of environment as the source population.
6. Place the new population in a novel environment compared to that of the source population.

A) 1, 3, and 6.
B) 1, 4, and 6.
C) 2, 3, and 5.
D) 2, 3, and 6.

24) How are two different species most likely to evolve from one ancestral species?

A) Sympatrically, by a point mutation affecting morphology or behavior.
B) Sympatrically, due to extensive inbreeding.
C) Allopatrically, due to extensive inbreeding.
D) Allopatrically, after the ancestral species has split into two populations.

25) House finches were found only in western North America until 1939, when a few individuals were released in New York City. These individuals established a breeding population and gradually expanded their range. The western population also expanded its range somewhat eastward, and the two populations have recently come in contact. If the two forms were unable to interbreed when their expanding ranges met, it would be an example of _____.
A) Prezygotic isolation.
B) Reinforcement.
C) Allopatric speciation.
D) Sympatric speciation.

26) Most causes of speciation are relatively slow, in that they may take many generations to see changes, with the exception of _____.
A) Polyploidy.
B) Reinforcement.
C) Colonization.
D) Natural selection.

27) Two researchers experimentally formed tetraploid frogs by fertilizing diploid eggs from *Rana porosa brevipoda* with diploid sperm from *Rana nigromaculata*. When they mated these tetraploid frogs with each other, most of the offspring that survived to maturity were tetraploid, with chromosome sets of both diploid parent species. Based on these results, if this type of tetraploid formed in the wild, what would be the result? (Y. Kondo and A. Kashiwagi. 2004. Experimentally induced autotetraploidy and allotetraploidy in two Japanese pond frogs. Journal of Herpetology 38(3):381-92.)
A) The two parent species would interbreed and fuse into one species.
B) The two parent species would recognize each other as mates.
C) The tetraploids would be reproductively isolated from both parent species.
D) The tetraploids would be selected against.

28) A researcher notices that in a certain moth species, some females prefer to feed and lay eggs on domesticated *solanaceous* plants like potatoes and tomatoes. Other females prefer to feed and lay eggs on wild *solanaceous* plants like *Datura*. Both male and female moths primarily use scent to find these plants from afar. Females tend to mate where they feed, and the researcher finds a genetic basis for scent preference in these moths. Based on the above information, what might be occurring in this moth species?
A) Divergence in sympatry.
B) Divergence due to habitat fragmentation.
C) Postzygotic isolation.
D) Polyploidization.

29) Two species of tree frogs that live sympatrically in the northeastern United States differ in ploidy: *Hyla chrysoscelis* is diploid, and *Hyla versicolor* is tetraploid. The frogs are identical in appearance, but their mating calls, which females use to find mates, differ. Which difference most likely evolved first?
A) Polyploidy.
B) Difference in mating calls.
C) Polyploidy and different mating calls must have evolved at the same time.
D) None of them.

30) In a hypothetical situation, a certain species of flea feeds only on pronghorn antelopes. In the western United States, pronghorns and cattle often associate with one another in the same open rangeland. Some of these fleas develop a strong preference for cattle blood and mate only with other fleas that prefer cattle blood. The host mammal can be considered as the fleas' habitat. If this situation persists, and new species evolve, this would be an example of _____.
A) Sympatric speciation and habitat isolation.
B) Sympatric speciation and temporal isolation.
C) Allopatric speciation and habitat isolation.
D) Allopatric speciation and gametic isolation.

31) The difference between geographic isolation and habitat differentiation (isolation) is the _____.
A) Relative locations of two populations as speciation occurs.
B) Speed (tempo) at which two populations undergo speciation.
C) Amount of genetic variation that occurs among two gene pools as speciation occurs.
D) Identity of the phylogenetic kingdom or domain in which these phenomena occur.

32) Among known plant species, which of these have been the two most commonly occurring phenomena that have led to the origin of new species?
A) Allopatric speciation and sexual selection.
B) Allopatric speciation and polyploidy.
C) Sympatric speciation and sexual selection.
D) Sympatric speciation and polyploidy.

33) Beetle pollinators of a particular plant are attracted to its flowers' bright orange color. The beetles not only pollinate the flowers, but they mate while inside of the flowers. A mutant version of the plant with red flowers becomes more common with the passage of time. A particular variant of the beetle prefers the red flowers to the orange flowers. Over time, these two beetle variants diverge from each other to such an extent that interbreeding is no longer possible. What kind of speciation has occurred, and what has driven it?
A) Allopatric speciation; ecological isolation.
B) Sympatric speciation; habitat differentiation.
C) Allopatric speciation; behavioral isolation.
D) Sympatric speciation; allopolyploidy.

Use the following description to answer the question (34) below.

On the volcanic, equatorial West African island of Sao Tomé, two species of fruit fly exist. *Drosophila yakuba* inhabits the island's lowlands, and is also found on the African mainland, located about two hundred miles away. At higher elevations, and only on Sao Tomé, is found the very closely related *Drosophila santomea*. The two species can hybridize, though male hybrids are sterile. A hybrid zone exists at middle elevations, though hybrids there are greatly outnumbered by *D. santomea*. Studies of the two species' nuclear genomes reveal that *D. yakuba* on the island is more closely related to mainland *D. yakuba* than to *D. santomea* (2n = 4 in both species). Sao Tomé rose from the Atlantic Ocean about fourteen million years ago.

34) Using only the information provided in the paragraph, which of the following is the best initial hypothesis for how *D. santomea* descended from D. *yakuba?*
A) Allopolyploidy.
B) Autopolyploidy.
C) Habitat differentiation.
D) Sexual selection.

Use the following description to answer the question (35) below.

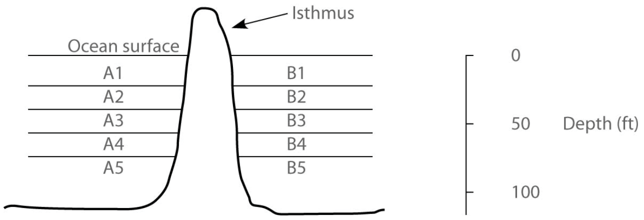
On the Bahamian island of Andros, mosquitofish populations live in various, now -isolated, freshwater ponds that were once united. Currently, some predator-rich ponds have mosquitofish that can swim in short, fast bursts; other predator-poor ponds have mosquitofish that can swim continuously for a long time. When placed together in the same body of water, the two kinds of female mosquitofish exhibit exclusive breeding preferences.

35) Which two of the following have operated to increase divergence between mosquitofish populations on Andros?
1. Improved gene flow.
2. Bottleneck effect.
3. Sexual selection.
4. Founder effect.
5. Natural selection.
A) 1 and 3.
B) 2 and 3.
C) 2 and 4.
D) 3 and 5.

Use the following description to answer the question (36-41) below.

In the ocean, on either side of the Isthmus of Panama, are thirty species of snapping shrimp; some are shallow-water species, others are adapted to deep water. There are fifteen species on the Pacific side and fifteen different species on the Atlantic side. The Isthmus of Panama started rising about ten million years ago. The oceans were completely separated by the isthmus about three million years ago.

In the following figure, the isthmus separates the Pacific Ocean on the left (side A) from the Atlantic Ocean on the right (side B). The seawater on either side of the isthmus is separated into five depth habitats (1-5), with 1 being the shallowest.



36) Why should deepwater shrimp on different sides of the isthmus have diverged from each other earlier than shallow-water shrimp?
A) They have been geographically isolated from each other for a longer time.
B) Cold temperatures, associated with deep water, have accelerated the mutation rate, resulting in faster divergence in deepwater shrimp.
C) The rise of the land bridge was accompanied by much volcanic activity. Volcanic ash contains heavy metals, which are known mutagens. Ash fall caused high levels of heavy metals in the ocean sediments underlying the deep water, resulting in accelerated mutation rates and faster divergence in deepwater shrimp.
D) Fresh water entering the ocean from the canal is both less dense and cloudier than seawater. The cloudy fresh water interferes with the ability of shallow-water shrimp to locate mating partners, which reduces the frequency of mating, thereby slowing the introduction of genetic variation.

37) In which habitat should one find snapping shrimp most closely related to shrimp that live in habitat A4?
A) A3.
B) A5.
C) B4.
D) Either A3 or A5.

38) Which of these habitats is likely to harbor the most recently diverged species?
A) A5.
B) B4.
C) A3.
D) A1.

39) Which habitats should harbor snapping shrimp species with the greatest degree of genetic divergence from each other?
A) A1 and A5.
B) A1 and B5.
C) A5 and B5.
D) Both A1/A5 and B1/B5 should have the greatest, but equal amounts of, genetic divergence.

40) Which factor is most important for explaining why there are equal numbers of snapping shrimp species on either side of the isthmus?
A) The relative shortness of time they have been separated.
B) The depth of the ocean.
C) The number of actual depth habitats between the surface and the sea floor.
D) The elevation of the isthmus above sea level.

41) The Panama Canal was completed in 1914, and its depth is about fifty feet. After 1914, snapping shrimp species from which habitats should be most likely to form hybrids as the result of the canal?
A) A5 and B5.
B) A3 and B3.
C) A1 and B1.
D) A1-A3 and B1-B3 have equal likelihoods of harboring snapping shrimp species that can hybridize.

42) Plant species A has a diploid number of 12. Plant species B has a diploid number of 16. A new species, C, arises as an allopolyploid from A and B. The diploid number for species C would probably be _____.
A) 14.
B) 16.
C) 28.
D) 56.

43) A small number of birds arrive on an island from a neighboring larger island. This small population begins to adapt to the new food plants available on the island, and their beaks begin to change. About twice a year, one or two more birds from the neighboring island arrive. These new arrivals _____.
A) Speed up the process of speciation.
B) Tend to promote adaptation to the new food plants.
C) Tend to retard adaptation to the new food plants.
D) Represent a colonizing event.

44) Male frogs give calls that attract female frogs to approach and mate. Researchers examined mating calls of closely related tree frogs in South America. If reinforcement is occurring, what would you expect if you compare the calls of the two species in zones of sympatry versus zones of allopatry?
A) Calls would be about the same in both areas.
B) Calls would be more similar in areas of sympatry.
C) Calls would be more different in areas of sympatry.
D) None of them.

45) Male frogs give calls that attract female frogs to approach and mate. Researchers examined mating calls of closely related but separate species of tree frogs in South America. What outcomes could possibly occur where the ranges of two species overlap?
I) The species will interbreed, eventually fusing over time.
II) A stable hybrid zone will form if hybrids are better adapted to the area of overlap than either parent species is.
III) Species will continue to diverge and be isolated by behavioral or genetic mechanisms.
A) I.
B) II.
C) III.
D) I, II, and III.

46) Reinforcement is most likely to occur when _____.
A) The environment is changing.
B) Hybrids have lower fitness than either parent population.
C) Prezygotic isolating mechanisms are in place.
D) Gene flow is low.

47) The phenomenon of fusion is likely to occur when, after a period of geographic isolation, two populations meet again and _____.
A) An increasing number of infertile hybrids is produced over the course of the next one hundred generations.
B) No reproduction occurs in the hybrid zone.
C) An increasing number of viable, fertile hybrids is produced over the course of the next one hundred generations.
D) A decreasing number of viable, fertile hybrids is produced over the course of the next one hundred generations.

48) A hybrid zone is properly defined as _____.
A) An area where the ranges of two closely related species overlap, but do not interbreed.
B) An area where mating occurs between members of two closely related species, producing viable offspring.
C) A zone where sterile hybrids form, kept separate by postzygotic barriers.
D) An area where members of two closely related species intermingle, but gene flow is prevented by prezygotic barriers.

49) In hybrid zones where reinforcement is occurring, we should see a decline in _____.
A) Gene flow between distinct gene pools.
B) Speciation.
C) The genetic distinctness of two gene pools.
D) Mutation rates.

50) Other than predation by introduced Nile perch, the most likely explanation for the recent decline in cichlid species diversity in Lake Victoria is _____.
A) Reinforcement.
B) Fusion.
C) Stability.
D) Polyploidy.

51) A narrow hybrid zone separates the toad species *Bombina bombina* and *Bombina variegata*. What is true of those alleles that are unique to the parental species?
A) Such alleles should be absent.
B) Their allele frequency should be nearly the same as the allele frequencies in toad populations distant from the hybrid zone.
C) The alleles' heterozygosity should be higher among the hybrid toads than in toad populations distant from the hybrid zone.
D) Their allele frequency on one edge of the hybrid zone should roughly equal their frequency on the opposite edge of the hybrid zone.

Use the following description to answer the question (52) below.

On the volcanic, equatorial West African island of Sao Tomé, two species of fruit fly exist. *Drosophila yakuba* inhabits the island's lowlands, and is also found on the African mainland, located about two hundred miles away. At higher elevations, and only on Sao Tomé, is found the very closely related *Drosophila santomea*. The two species can hybridize, though male hybrids are sterile. A hybrid zone exists at middle elevations, though hybrids there are greatly outnumbered by *D. santomea*. Studies of the two species' nuclear genomes reveal that *D. yakuba* on the island is more closely related to mainland *D. yakuba* than to *D. santomea* (2n = 4 in both species). Sao Tomé rose from the Atlantic Ocean about fourteen million years ago.

52) The observation that island *D. yakuba* are more closely related to mainland *D. yakuba* than island *D. yakuba* are to *D. santomea* is best explained by proposing that *D. santomea* _____.
A) Descended from a now-extinct, non-African fruit fly.
B) Arrived on the island before *D. yakuba*.
C) Descended from a single colony of *D. yakuba*, which had been introduced from elsewhere, with no subsequent colonization events.
D) Descended from an original colony of *D. yakuba*, of which there are no surviving members. The current island *D. yakuba* represent a second colonization event from elsewhere.

Use the following description to answer the question (53) below.

On the Bahamian island of Andros, mosquitofish populations live in various, now -isolated, freshwater ponds that were once united. Currently, some predator-rich ponds have mosquitofish that can swim in short, fast bursts; other predator-poor ponds have mosquitofish that can swim continuously for a long time. When placed together in the same body of water, the two kinds of female mosquitofish exhibit exclusive breeding preferences.

53) What is the best way to promote fusion between two related populations of mosquitofish, one of which lives in a predator-rich pond and the other of which lives in a predator-poor pond?
A) Build a canal linking the two ponds that permits free movement of mosquitofish, but not of predators.
B) Transfer only female mosquitofish from a predator-rich pond to a predator-poor pond.
C) Perform a reciprocal transfer of females between predator-rich and predator-poor ponds.
D) Remove predators from a predator-rich pond and transfer them to a predator-poor pond.

54) Suppose that a group of male pied flycatchers migrated from a region where there were no collared flycatchers to a region where both species were present. Assuming events like this are very rare, which of the following scenarios is LEAST likely?
A) Migrant pied males would produce fewer offspring than would resident pied males.
B) Pied females would rarely mate with collared males.
C) Migrant males would mate with collared females more often than with pied females.
D) The frequency of hybrid offspring would decrease.

The following question (55) refers to this hypothetical situation.

A female fly, full of fertilized eggs, is swept by high winds to an island far out to sea. She is the first fly to arrive on this island and the only fly to arrive in this way. Thousands of years later, her numerous offspring occupy the island, but none of them resembles her. There are, instead, several species, each of which eats only a certain type of food. None of the species can fly and their balancing organs (halteres) are now used in courtship displays. The male members of each species bear modified halteres that are unique in appearance to their species. Females bear vestigial halteres. The ranges of all of the daughter species overlap.

55) Fly species W, found in a certain part of the island, produces fertile offspring with species Y. Species W does not produce fertile offspring with species X or Z. If no other species can hybridize, then which of the following statements about species W and Y are true?
I) Species W and Y have genomes that are still similar enough for successful meiosis to occur in hybrid flies.
II) Species W and Y have more genetic similarity with each other than either did with the other two species.
III) Species W and Y may fuse into a single species if their hybrids remain fertile over the course of many generations.
A) Only I is correct.
B) Only II is correct.
C) Only III is correct.
D) I, II, and III are correct.

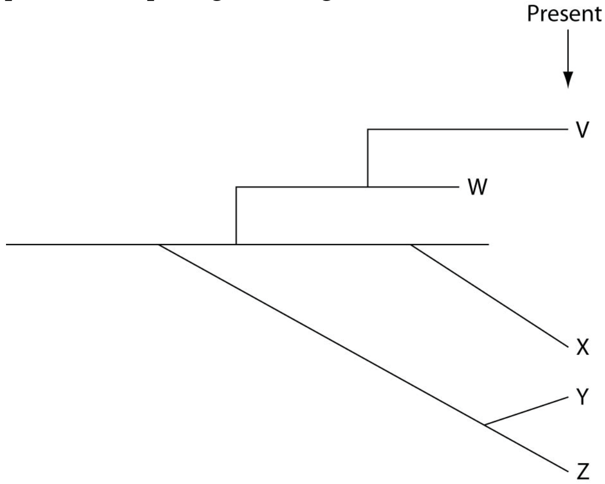
56) According to the concept of punctuated equilibrium, the "sudden" appearance of a new species in the fossil record means that _____.
A) The species is now extinct.
B) Speciation occurred in one generation.
C) Speciation occurred rapidly in geologic time.
D) The species will consequently have a relatively short existence, compared with other species.

57) According to the concept of punctuated equilibrium, _____.
A) Natural selection is unimportant as a mechanism of evolution.
B) Given enough time, most existing species will gradually give rise to new species.
C) A new species accumulates most of its unique features as it comes into existence.
D) Evolution of new species features long periods during which changes are occurring, interspersed with short periods of equilibrium, or stasis.

58) Speciation _____.
A) Occurs at such a slow pace that no one has ever observed the emergence of new species.
B) Occurs only by the accumulation of small genetic changes over vast expanses of time.
C) Must begin with the geographic isolation of a small, frontier population.
D) Can involve changes to a single gene.

59) According to the biological species concept, for speciation to occur, _____.
A) The number of chromosomes in the gene pool must change.
B) Changes to centromere location or chromosome size must occur within the gene pool.
C) Large numbers of genes that affect numerous phenotypic traits must change.
D) At least one gene, affecting at least one phenotypic trait, must change.

The question (60) below refer to the following evolutionary tree, in which the horizontal axis represents time (present time is on the far right) and the vertical axis represents morphological change.



60) Which conclusion can be drawn from this evolutionary tree?
A) A single clade (that is, a group of species that share a common ancestor) can include species that formed by gradualism and other species that formed by punctuated equilibrium.
B) A single clade (that is, a group of species that share a common ancestor) will either include species that formed by gradualism or species that formed by punctuated equilibrium.
C) Assuming that the tip of each line represents a species, there are five extant (that is, not extinct) species resulting from the earliest common ancestor.
D) Species X and Z best represent species that evolved by punctuated equilibrium.

CHAPTER 25:

THE HISTORY OF LIFE ON EARTH

1) Which of the following is the correct sequence of events in the origin of life?
I. Formation of protobionts.
II. Synthesis of organic monomers.
III. Synthesis of organic polymers.
IV. Formation of DNA-based genetic systems.
A) I, II, III, IV.
B) I, III, II, IV.
C) II, III, I, IV.
D) II, III, IV, I.

2) Which of the following is a defining characteristic that all protocells had in common?
A) The ability to synthesize enzymes.
B) A surrounding membrane or membrane-like structure.
C) RNA genes.
D) The ability to replicate RNA.

3) The first genetic material on Earth was probably _____.
A) DNA produced by reverse transcriptase from abiotically produced RNA.
B) DNA molecules whose information was transcribed to RNA and later translated in polypeptides.
C) Self-replicating RNA molecules.
D) Oligopeptides located within protobionts.

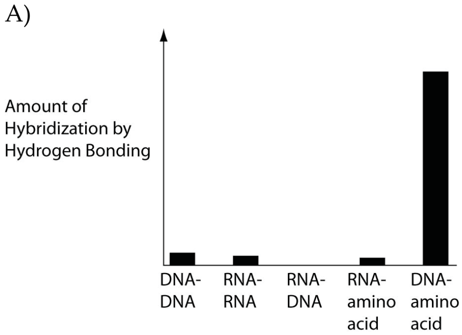
4) Several scientific laboratories across the globe are involved in research concerning the origin of life on Earth. Which of these questions is currently the most problematic and would have the greatest impact on our understanding if we were able to answer it?
A) How can amino acids, simple sugars, and nucleotides be synthesized abiotically?
B) How did RNA sequences come to carry the code for amino acid sequences?
C) How could polymers involving lipids and/or proteins form membranes in aqueous environments?
D) How can RNA molecules act as templates for the synthesis of complementary RNA molecules?

5) Which of the following steps has yet to be accomplished by scientists studying the origin of life?
A) Synthesis of small RNA polymers by ribozymes.
B) Abiotic synthesis of polypeptides.
C) Formation of molecular aggregates with selectively permeable membranes.
D) Formation of protocells that use DNA to direct the polymerization of amino acids.
Section: 25.1

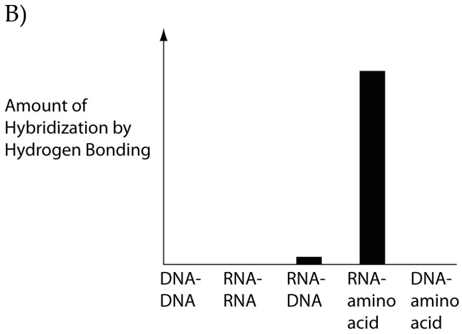
6) How were conditions on the early Earth of more than three billion years ago different from those on today's Earth? Unlike Earth today, early Earth _____.
A) Had an atmosphere rich in gases released from volcanic eruptions.
B) Had an oxidizing atmosphere.
C) Experienced little high energy radiation from the sun.
D) Had an atmosphere with significant quantities of ozone.

7) Several scientific laboratories across the globe are performing research concerning the origin of life on Earth. Which of the following are the most likely results of abiotic experiments testing the potential for hydrogen bonding between various nucleic acids and amino acids? Base your response on our current understanding of Earth's first genetic systems.

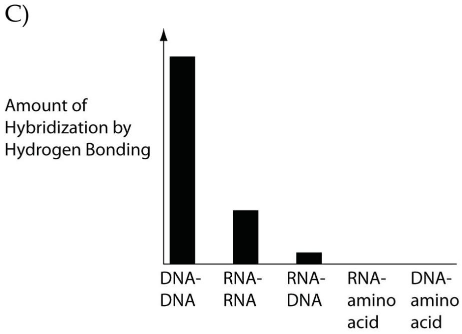
A)



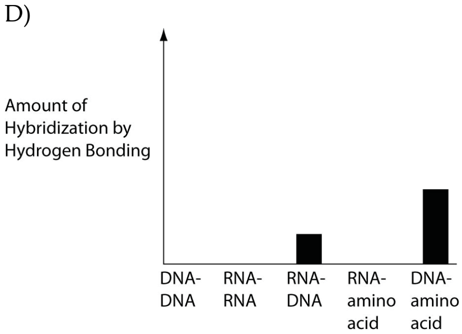
B)



C)

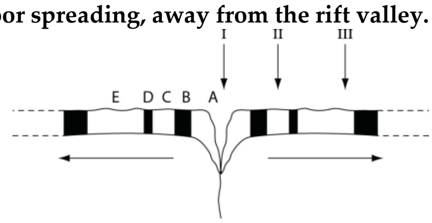


D)



The questions (8-14) below refer to the following description and figure.

The figure below represents a cross section of the sea floor through a mid-ocean rift valley, with alternating patches of black and white indicating sea floor with reversed magnetic polarities. At the arrow labeled "I", the igneous rock of the sea floor is so young that it can be accurately dated using carbon-14 dating. At the arrow labeled "III", however, the igneous rock is about one million years old, and potassium-40 dating is typically used to date such rocks. Note: The horizontal arrows indicate the direction of sea-floor spreading, away from the rift valley.



8) Which section of sea-floor crust should have the thickest layer of overlying sediment, assuming a continuous rate of sediment deposition?
A) A.
B) B.
C) D.
D) E.

9) If a particular marine organism is fossilized in the sediments immediately overlying the igneous rock at the arrow labeled "II", at which other location, labeled A-E, would a search be most likely to find more fossils of this organism?
A) B.
B) C.
C) D.
D) E.

10) Which of the following organisms would be most likely to fossilize?
A) A rare worm.
B) A common worm.
C) A rare squirrel.
D) A common squirrel.

11) Which of the following would be LEAST likely in the fossil record?
A) Burrowing species.
B) Marine-dwelling species.
C) Marsh-dwelling species.
D) Desert-dwelling species.

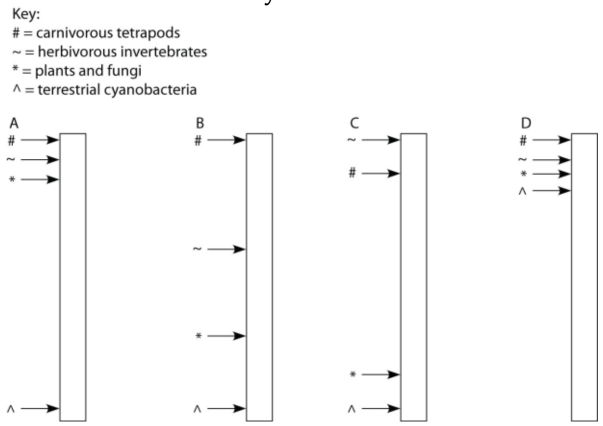
12) If the half-life of carbon-14 is about 5730 years, then a fossil that has one-sixteenth the normal proportion of carbon-14 to carbon-12 should be about how many years old?
A) 2800.
B) 11,200.
C) 16,800.
D) 22,900.

13) What is true of the fossil record of mammalian origins?
A) It shows that mammals and birds evolved from the same kind of dinosaur.
B) It includes transitional forms with progressively specialized teeth.
C) It indicates that mammals and dinosaurs did not overlap in geologic time.
D) It includes a series that shows the gradual change of scales into fur.

14) If a fossil is encased in a stratum of sedimentary rock without any strata of volcanic rock (for example, lava or ash) nearby, then it should be _____.
A) Easy to determine the absolute age of the fossil, because the radioisotopes in the sediments will not have been "reset" by the heat of the igneous rocks.
B) Easy to determine the absolute age of the fossil, because the igneous rocks will not have physically obstructed the deposition of sediment of a single age next to the fossil.
C) Difficult to determine the absolute age of the fossil, because the "marker fossils" common to igneous rock will be absent.
D) Difficult to determine the absolute age of the fossil, because radiometric dating of sedimentary rock is less accurate than that of igneous rock.

The question (15) below refer to the following paragraph.

A sediment core is removed from the floor of an inland sea. The sea has been in existence, off and on, throughout the entire time that terrestrial life has existed. Researchers wish to locate and study the terrestrial organisms fossilized in this core. The core is illustrated as a vertical column, with the top of the column representing the most recent strata and the bottom representing the time when land was first colonized by life.



15) To assign absolute dates to fossils in this sediment core, it would be most helpful if _____.
A) We knew the order in which the fossils occurred.
B) The sediments had not been affected by underwater currents during their deposition.
C) Volcanic ash layers were regularly interspersed between the sedimentary strata.
D) Fossils throughout the column were equally spaced apart.

Refer to the following information to answer the question (16) below.

Fossils of *Lystrosaurus*, a dicynodont therapsid, are most common in parts of modern-day South America, South Africa, Madagascar, India, South Australia, and Antarctica. The animal apparently lived in arid regions, and was mostly herbivorous. It originated during the mid-Permian period, survived the Permian extinction, and dwindled by the late Triassic, though there is evidence of a relict population in Australia during the Cretaceous period. Some dicynodonts had two large tusks, extending down from their upper jaws. The tusks were not used for food gathering, and in some species were limited to males. Food was gathered using an otherwise toothless beak. Judging from the fossil record in sedimentary rocks, these pig-sized organisms were the most common mammal-like reptiles of the Permian.

16) Anatomically, *Lystrosaurus* _____.
A) Would have had a lower jaw that consisted of a single pair of bones.
B) Was a tetrapod.
C) Had skin without scales, typical of modern amphibians.
D) Would have had no temporal fenestra in its skull.

17) What is thought to be the correct sequence of these events, from earliest to most recent, in the evolution of life on Earth?
1. Origin of mitochondria.
2. Origin of multicellular eukaryotes.
3. Origin of chloroplasts.
4. Origin of cyanobacteria.
5. Origin of fungal-plant symbioses.
A) 4, 3, 2, 1, 5.
B) 4, 1, 2, 3, 5.
C) 4, 1, 3, 2, 5.
D) 4, 3, 1, 5, 2.

18) Which of these observations gives the most support to the endosymbiotic theory for the origin of eukaryotic cells?
A) The existence of structural and molecular differences between the plasma membranes of prokaryotes and the internal membranes of mitochondria and chloroplasts.
B) The similarity in size between the cytosolic ribosomes of prokaryotes and the ribosomes within mitochondria and chloroplasts.
C) The size disparity between most prokaryotic cells and most eukaryotic cells.
D) The observation that some eukaryotic cells lack mitochondria.

19) Which of the following was derived from an ancestral cyanobacterium?
A) Chloroplast.
B) Mitochondrion.
C) Flagella.
D) Mitosome.

20) For many years scientists believed that almost all animal lineages burst into being during the Cambrian era (just after the end of the Precambrian super eon). However, there have been many recent findings of animal-like fossils and "trace fossils" (fossils of an animal-like organisms movement) from the late Precambrian. Which of the following best explains why it took so long to realize there was animal -like life in the Precambrian?
A) Animals from the late Precambrian had soft bodies.
B) There were many hard-shelled animals in the Cambrian.
C) The global climate was such that there was poor fossilization in the Precambrian.
D) There were very few animals during this period.

21) Which of the following showed their greatest diversity during the Mesozoic era, but were a small, less diverse group during the Paleozoic era?
A) Gymnosperms.
B) Fungi.
C) Flowering plants.
D) Mammals.

22) Which listing of geological periods is in the correct order, from oldest to most recent?
A) Cambrian, Devonian, Permian, Cretaceous.
B) Devonian, Cambrian, Permian, Cretaceous.
C) Cambrian, Permian, Devonian, Cretaceous.
D) Permian, Cambrian, Cretaceous, Devonian.

23) You are the lucky student of a wacky professor who develops a time machine. He asks if you will test it with him. You get in and there is an immediate glitch–the date readout fails so that when you land you are not sure what era you are in. Your professor begins to panic, but you see something that tells you are in the Cenozoic era. Which of the following could it be?
A) A rabbit eating a daisy.
B) A water lily floating in a pond.
C) Masses of green ferns with dragonflies hovering above them.
D) A bee pollinating a flower.

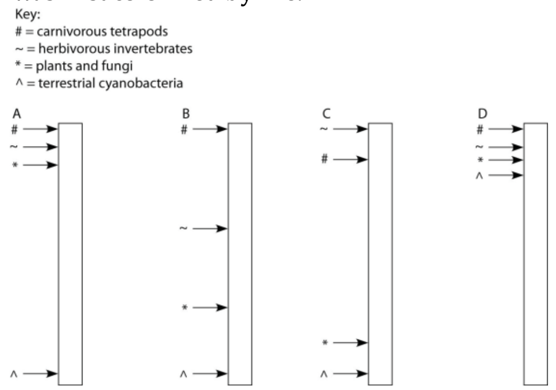
24) You are the lucky student of a wacky professor who develops a time machine. He asks if you will test it with him. You get in and there is an immediate glitch–the date readout fails so that when you land you are not sure what era you are in. As your time machine lands, you see an unusual landscape before you. As you open the door you realize you cannot breathe. You quickly shut the door, realizing you are in the _____.
A) Archaean eon.
B) Cambrian period.
C) Cenozoic era.
D) Mesozoic era.

25) An early consequence of the release of oxygen gas by plant and bacterial photosynthesis was to _____.
A) Change the atmosphere from oxidizing to reducing.
B) Make it easier to maintain reduced molecules.
C) Cause iron in ocean water and terrestrial rocks to rust (oxidize).
D) Prevent the formation of an ozone layer.

26) What is true of the Cambrian explosion?
A) There are fossils of animals in geological strata that are older than the Cambrian explosion.
B) Only the fossils of microorganisms are found in geological strata older than the Cambrian explosion.
C) The Cambrian explosion is evidence for the instantaneous creation of life on Earth.
D) The Cambrian explosion marks the appearance of filter-feeding animals in the fossil record.

27) Which of the following characteristics are expected in the first animals to have colonized land?
1. Were probably herbivores (ate Photosynthesizers).
2. Had four appendages.
3. Had the ability to resist dehydration.
4. Had lobe-finned fishes as ancestors.
5. Were invertebrates.
A) 3 only.
B) 3 and 5.
C) 1, 3, and 5.
D) 1, 2, 3, and 4.

The questions below refer to the following paragraph. A sediment core is removed from the floor of an inland sea. The sea has been in existence, off and on, throughout the entire time that terrestrial life has existed. Researchers wish to locate and study the terrestrial organisms fossilized in this core. The core is illustrated as a vertical column, with the top of the column representing the most recent strata and the bottom representing the time when land was first colonized by life.



28) If arrows indicate locations in the column where fossils of a particular type (see key above) first appear, then which core in the figure above has the most accurate arrangement of fossils?
A) Core A.
B) Core B.
C) Core C.
D) Core D.

29) Which factor most likely caused animals and plants in India to differ greatly from species in nearby Southeast Asia?
A) The climates of the two regions are similar.
B) India is in the process of separating from the rest of Asia.
C) Life in India was wiped out by ancient volcanic eruptions.
D) India was a separate continent until forty-five million years ago.

The following questions (30-31) refer to the description below.

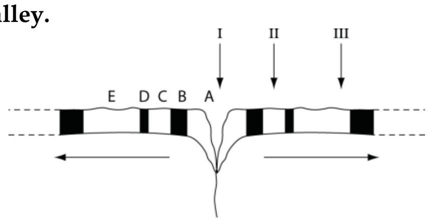
All animals with eyes or eyespots that have been studied so far share a gene in common. When mutated, the gene *Pax-6* causes lack of eyes in fruit flies, tiny eyes in mice, and missing irises (and other eye parts) in humans. The sequence of *Pax-6* in humans and mice is identical. There are so few sequence differences with fruit fly Pax-6 that the human/mouse version can cause eye formation in eyeless fruit flies, even though vertebrates and invertebrates last shared a common ancestor more than five hundred million years ago.

30) The appearance of *Pax-6* in all animals with eyes can be explained in multiple ways. Based on the information above, which explanation is most likely?
A) *Pax-6* in all of these animals is not homologous; it arose independently in many different animal phyla due to intense selective pressure favoring vision.
B) The *Pax-6* gene is really not one gene. It is many different genes that, over evolutionary time and due to convergence, have come to have a similar nucleotide sequence and function.
C) The *Pax-6* gene was an innovation of an ancestral animal of the early Cambrian period. Animals with eyes or eyespots are descendants of this ancestor.
D) The need for eyes has resulted in the separate evolution of *Pax-6* genes.

31) Fruit-fly eyes are of the compound type, which is structurally very different from the camera-type eyes of mammals. Even the camera-type eyes of mollusks, such as octopi, are structurally quite different from those of mammals. Yet, fruit flies, octopi, and mammals possess very similar versions of *Pax-6*. The fact that the same gene helps produce very different types of eyes is most likely due to _____.
A) The few differences in nucleotide sequence among the *Pax-6* genes of these organisms.
B) Variations in the number of *Pax-6* genes among these organisms.
C) The independent evolution of this gene at many different times during animal evolution.
D) Differences in the control of *Pax-6* expression among these organisms.

The questions (32-34) below refer to the following description and figure.

The figure below represents a cross section of the sea floor through a mid-ocean rift valley, with alternating patches of black and white indicating sea floor with reversed magnetic polarities. At the arrow labeled "I" (the rift valley), the igneous rock of the sea floor is so young that it can be accurately dated using carbon-14 dating. At the arrow labeled "III", however, the igneous rock is about one million years old, and potassium-40 dating is typically used to date such rocks. Note: The horizontal arrows indicate the direction of sea-floor spreading, away from the rift valley.



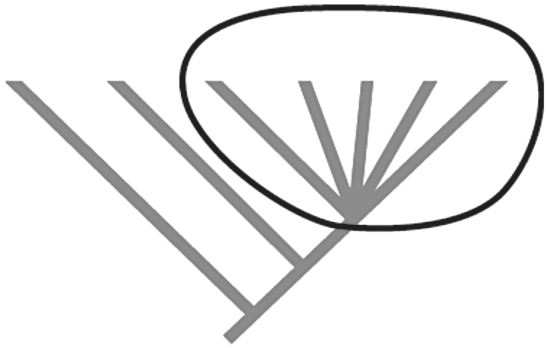
32) Assuming that the rate of sea-floor spreading was constant during the one-million-year period depicted above, on average Earth's magnetic field has undergone reversal once every _____.
A) 25,000 years.
B) 100,000 years.
C) 250,000 years.
D) 1,000,000 years.

33) How many other bands of sea-floor crust in the figure above have the same magnetic polarity as the crust that directly straddles the rift valley?
A) Two bands.
B) Four bands.
C) Five bands.
D) Eight bands.

34) Assuming that the rate of sea-floor spreading was constant during the one-million-year period depicted above, what should be the approximate age of marine fossils found in undisturbed sedimentary rock immediately overlying the igneous rock at the arrow labeled "II"?
A) 10,000 years.
B) 250,000 years.
C) 500,000 years.
D) 1,000,000 years.

35) The Permian period ended and then rapid speciation occurred as new animal and plant forms evolved. The most likely explanation for this is _____.
I) Adaptive radiation.
II) Ecological opportunity.
III) Lack of competition.
IV) Morphological innovation.
A) Just one of the above.
B) Two of the above.
C) Three of the above.
D) All of the above.

36) What does the circled part of the phylogenetic tree given below indicate?

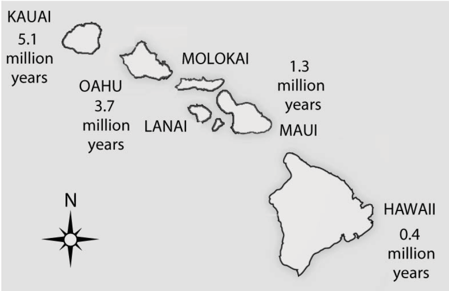


A) An adaptive radiation.
B) A mass extinction event.
C) Rapid speciation.
D) An adaptive radiation and rapid speciation.

37) A hypothetical island lies far from any other landmasses. There are many different types of plants, but only one animal, a beetle that can fly or walk from plant to plant and feeds by chewing leaves. Given that the beetle is not exploiting all of the resources on the island, which morphological change would be most likely to trigger an adaptive radiation of the beetles?
A) A change in wing shape that improves flight speed.
B) An additional segment on a pair of legs.
C) A mouthpart that can pierce fruits and seeds.
D) None of them.

38) On the basis of their morphologies, how might Linnaeus have classified the Hawaiian silverswords? Linnaeus would have _____.
A) Placed them all in the same species.
B) Classified them the same way that modern botanists do.
C) Placed more of them in different genera than modern botanists do.
D) Used evolutionary relatedness as the primary criterion for their classification.

The questions (39-41) below refer to the following figure.



39) According to the theory of sea-floor spreading, oceanic islands, such as the Hawaiian Islands depicted in the figure above, form as oceanic crustal plates move over a stationary "hot spot" in the mantle. Currently, the big island of Hawaii is thought to be over a hot spot, which is why it is the only one of the seven large islands that has active volcanoes. What should be true of the island of Hawaii?

1. Scientists in search of ongoing speciation events are more likely to find them here than on the other six large islands.
2. Its species should be more closely related to those of nearer islands than to those of farther islands.
3. It should have a rich fossil record of terrestrial organisms.
4. It should have species that are not found anywhere else on Earth.
5. On average, it should have fewer species per unit surface area than the other six islands.
A) 1, 2, and 3.
B) 1, 2, and 5.
C) 1, 2, 3, and 4.
D) 1, 2, 4, and 5.

40) Hawaii is the most southeastern of the seven largest islands and is also closest to the sea-floor spreading center from which the Pacific plate originates, which lies about 5600 km further to the southeast. Assuming equal sedimentation rates, what should be the location of the thickest sediment layer and, thus, the area with the greatest diversity of fossils above the oceanic crust?
A) Between the island of Hawaii and the sea-floor spreading center.
B) Around the base of the island of Hawaii.
C) Around the base of Kauai, the oldest of the large Hawaiian Islands.
D) Where the islands are most concentrated (highest number of islands per unit surface area).

41) Upon being formed, oceanic islands, such as the Hawaiian Islands, should feature what characteristic, leading to which phenomenon?
A) Mass extinctions, leading to bottleneck effect.
B) Major evolutionary innovations, leading to rafting to nearby continents.
C) A variety of empty ecological niches, leading to adaptive radiation.
D) Adaptive radiation, leading to founder effect.

Refer to the following information to answer the question (42) below.

Fossils of *Lystrosaurus*, a dicynodont therapsid, are most common in parts of modern-day South America, South Africa, Madagascar, India, South Australia, and Antarctica. The animal apparently lived in arid regions, and was mostly herbivorous. It originated during the mid-Permian period, survived the Permian extinction, and dwindled by the late Triassic, though there is evidence of a relict population in Australia during the Cretaceous period. Some dicynodonts had two large tusks, extending down from their upper jaws. The tusks were not used for food gathering, and in some species were limited to males. Food was gathered using an otherwise toothless beak. Judging from the fossil record in sedimentary rocks, these pig-sized organisms were the most common mammal-like reptiles of the Permian.

42) Which of the following is the most likely explanation for the modern-day distribution of dicynodont fossils? The dicynodonts were _____.
A) Carnivores that traveled widely to find prey.
B) Evenly distributed throughout all of Pangaea.
C) Most abundantly distributed throughout Gondwanaland.
D) Amphibious and able to swim long distances.

43) If an organism has a relatively large number of *Hox* genes in its genome, it most likely _____.
A) Has *Hox* genes expressed in all cells of its body.
B) Has multiple paired appendages along the length of its body.
C) Has a relatively complex anatomy.
D) Belongs to one of the first groups to evolve on Earth.

44) Bagworm moth caterpillars feed on evergreens and carry a silken case or bag around with them in which they eventually pupate. Adult female bagworm moths are larval in appearance; they lack the wings and other structures of the adult male and instead retain the appearance of a caterpillar even though they are sexually mature and can lay eggs within the bag. This is a good example of _____.
A) Allometric growth.
B) Paedomorphosis.
C) Sympatric speciation.
D) Adaptive radiation.

45) The loss of ventral spines by modern freshwater sticklebacks is due to natural selection operating on the phenotypic effects of *Pitx1* gene _____.
A) Duplication (gain in number).
B) Elimination (loss).
C) Mutation (change).
D) Silencing (loss of expression).

The following questions (46-50) refer to this hypothetical situation.

A female fly, full of fertilized eggs, is swept by high winds to an island far out to sea. She is the first fly to arrive on this island and the only fly to arrive in this way. Thousands of years later, her numerous offspring occupy the island, but none of them resembles her. There are, instead, several species, each of which eats only a certain type of food. None of the species can fly and their balancing organs (halteres) are now used in courtship displays. The male members of each species bear modified halteres that are unique in appearance to their species. Females bear vestigial halteres. The ranges of all of the daughter species overlap.

46) In each fly species, the entire body segment that gave rise to the original flight wings is missing. The mutation(s) that led to the flightless condition could have _____.
A) Duplicated all of the *Hox* genes in these flies' genomes.
B) Resulted in Paedomorphosis.
C) Altered the expression of a *Hox* gene.
D) Originated in another species.

47) A genetic change that caused a certain *Hox* gene to be expressed along the tip of a vertebrate limb bud instead of farther back helped make possible the evolution of the tetrapod limb. This type of change is illustrative of _____.
A) The influence of environment on development.
B) Paedomorphosis.
C) A change in a developmental gene or its regulation that altered the spatial organization of body parts.
D) Heterochrony.

48) The duplication of homeotic (*Hox*) genes has been significant in the evolution of animals because it _____.
A) Permitted the evolution of novel forms.
B) Caused the extinction of major groups.
C) Reduced morphological diversity into simpler forms of life.
D) Allowed animals to survive on significantly fewer calories.

49) Why would gene duplication events, such as those seen in the *Hox* gene complex, set the stage for adaptive radiation?
A) There are more copies of genes, meaning speciation had occurred by polyploidy.
B) The original gene copy is the out group, and the new gene copies are the adaptive radiation.
C) Without duplicated genes, species would be vulnerable to extinction.
D) One copy of a gene can perform the original function, while other copies are available to take on new functions.

50) All of the following events can trigger an adaptive radiation EXCEPT _____.
A) An unusual event splitting a habitat, such as a severe hurricane.
B) The evolution of a new morphological feature.
C) The extinction of competitors.
D) *Hox* gene duplication events.

The following questions refer to the description below.

All animals with eyes or eyespots that have been studied so far share a gene in common. When mutated, the gene Pax-6 causes lack of eyes in fruit flies, tiny eyes in mice, and missing irises (and other eye parts) in humans. The sequence of Pax-6 in humans and mice is identical. There are so few sequence differences with fruit fly Pax-6 that the human/mouse version can cause eye formation in eyeless fruit flies, even though vertebrates and invertebrates last shared a common ancestor more than five hundred million years ago.

51) *Pax-6* usually causes the production of a type of light-receptor pigment. In vertebrate eyes, though, a different gene (the *rh* gene family) is responsible for the light-receptor pigments of the retina. The *rh* gene, like *Pax-6*, is ancient. In the marine ragworm, for example, the *rh* gene causes production of c-opsin, which helps regulate the worm's biological clock. Which of the following most likely accounts for vertebrate vision?
A) The *Pax-6* gene mutated to become the *rh* gene among early mammals.
B) During vertebrate evolution, the *rh* gene for biological clock opsin was co-opted as a gene for visual receptor pigments.
C) In animals more ancient than ragworms, the *rh* gene(s) coded for visual receptor pigments; in lineages more recent than ragworms, *rh* has flip-flopped several times between producing biological clock opsins and visual receptor pigments.
D) *Pax-6* was lost from the mammalian genome, and replaced by the *rh* gene much later.

52) A swim bladder is a gas-filled sac that helps fish maintains buoyancy. The evolution of the swim bladder from lungs of an ancestral fish is an example of _____.
A) Exaptation.
B) Changes in Hox gene expression.
C) Paedomorphosis.
D) Adaptive radiation.

53) Which of the following is a limit of evolution that results in exaptations?
A) Natural selection and sexual selection can work at cross-purposes to each other.
B) Evolution is limited by historical constraints.
C) Adaptations are often compromises.
D) Chance events affect the evolutionary history of populations in environments that can change unpredictably.

54) Insect wings may have begun to evolve as lateral extensions of the body that were used as heat dissipaters for thermoregulation. When they had become sufficiently large, these extensions became useful for gliding through the air. Additional selection refined them as flight-producing wings. If this hypothesis is correct, modern insect wings would be an example of _____.
A) The loss of *Hox* genes in the evolution of new form.
B) Mutations.
C) An exaptation.
D) An adaptive radiation.

55) If one organ is an exaptation of another organ, then these two organs _____.
A) Are homologous.
B) Are undergoing convergent evolution.
C) Are found in the same species.
D) Have the same function.

56) Many species of snakes lay eggs. However, in the forests of northern Minnesota, where growing seasons are short, only live-bearing snake species are present. This trend toward species that have live birth in a particular environment is an example of _____.
A) An exaptation.
B) Sexual selection.
C) Species selection.
D) Goal direction in evolution.

57) The existence of evolutionary trends, such as increasing body sizes among horse species, is evidence that _____.
A) A larger volume-to-surface area ratio is adaptive in many mammals.
B) Evolution generally progresses toward some goal.
C) Evolution tends toward increased complexity or increased size.
D) In particular environments, similar adaptations can be beneficial to more than one species.

The following questions (58-59) are based on the observation that several dozen different proteins comprise the prokaryotic flagellum and its attachment to the prokaryotic cell, producing a highly complex structure.

58) If the complex protein assemblage of the prokaryotic flagellum arose by the same general processes as those of the complex eyes of mollusks (such as squids and octopi), then _____.
A) Natural selection cannot account for the rise of the prokaryotic flagellum.
B) Ancestral versions of this protein assemblage were either less functional or had different functions than modern prokaryotic flagella.
C) Neither eyes nor flagella could have arisen by evolution because both are too complex to have evolved.
D) The need for more complex structure must have driven evolution.

59) Certain proteins of the complex motor that drives bacterial flagella are modified versions of proteins that had previously belonged to plasma membrane pumps. This evidence supports the claim that _____.
A) Natural selection produces organs that will be needed in future environments.
B) The motors of bacterial flagella must have originated in other organisms.
C) Natural selection can produce new structures by coupling together parts of other structures.
D) Bacteria that possess flagella must have lost the ability to pump certain chemicals across their plasma membranes.

The following question (60) refer to this hypothetical situation.

A female fly, full of fertilized eggs, is swept by high winds to an island far out to sea. She is the first fly to arrive on this island and the only fly to arrive in this way. Thousands of years later, her numerous offspring occupy the island, but none of them resembles her. There are, instead, several species, each of which eats only a certain type of food. None of the species can fly and their balancing organs (halteres) are now used in courtship displays. The male members of each species bear modified halteres that are unique in appearance to their species. Females bear vestigial halteres. The ranges of all of the daughter species overlap.

60) Which of these fly organs, as they exist in the description above, best illustrates an exaptation?
A) Vestigial halteres.
B) Halteres.
C) Mouthparts.
D) Eggs.

61) One hypothesis that has been proposed regarding the origin of life suggests that life evolved from an "RNA world" to today's "DNA world." Considering the properties of RNA and DNA molecules, which questions would direct an investigation of the most insightful test(s) of this hypothesis?
A) Is it likely that RNA molecules functioned as ribozymes to synthesize DNA from amino acids, and that this role was reversed when DNA became the information source?
B) Is it likely that simple, yet stable RNA molecules served as a primitive precursor to a less stable DNA molecule that was more capable of storing more information?
C) Could RNA have provided a template for DNA assembly, thereby enabling a more stable molecule that is replicated more accurately?
D) Since ribozymes could freely enter and leave the vesicles, could these molecules have brought external DNA into the cell as a less stable, but more reliable storage molecule of double-stranded DNA?

62) Endosymbiosis is an evolutionary theory that explains the origin of eukaryotes and suggests a specific order in which this might have occurred. Ancestral cells engulfed and then began to use the metabolic processes of the smaller cells. Based on shared core processes and features, which statement most accurately describes the order, the theory and the evolutionary implications for all organisms within domain Eukarya?

A) Ancestral heterotrophic eukaryotes most likely engulfed both a heterotrophic and an autotrophic prokaryote, whereas ancestral photosynthetic eukaryotes probably provided the host cell for the first mitochondria. Over time, natural selection favored these relationships and these cells became ancestors of all eukaryotes.

B) All ancestral eukaryotes would have most likely consumed a nucleus -like prokaryote that eventually became the eukaryotic nucleus. These new eukaryotic cells would have had an advantage over prokaryotic cells by acquiring a nuclear command center for regulating cellular activities.

C) As carbon dioxide levels were increasing over time, natural selection would have favored organisms that acquired a photosynthetic prokaryote to convert carbon dioxide into sugars. These would have likely been the first eukaryotic cells. At which point, these ancestral cells engulfed mitochondria-like prokaryotes that would have provided an even greater advantage for cells in this environment.

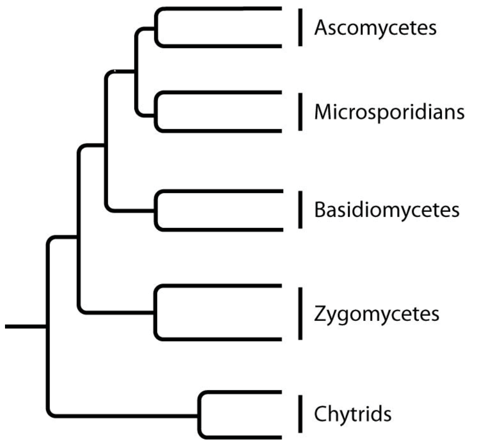
D) As Earth was becoming more aerobic, mitochondria would have provided an advantage to host cells by converting "toxic" oxygen into energy for heterotrophic cells. Since mitochondria are found in all eukaryotes, these combinations likely evolved first. Photosynthetic eukaryotes probably acquired an autotrophic prokaryote, which developed an advantageous symbiotic relationship with the host cell.

CHAPTER 26:

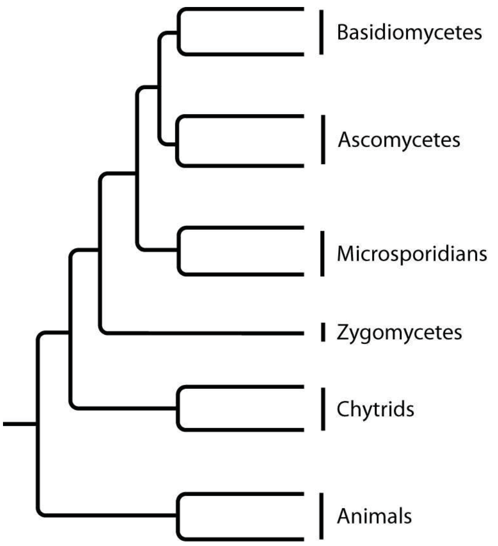
PHYLOGENY AND THE TREE OF LIFE

The questions (1-3) below refer to the following phylogenetic trees.

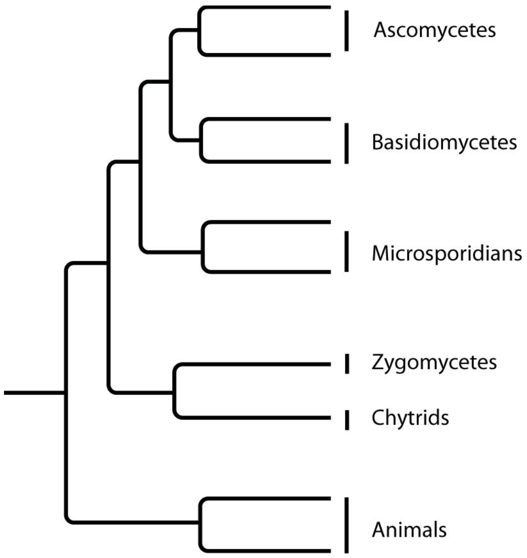
I.



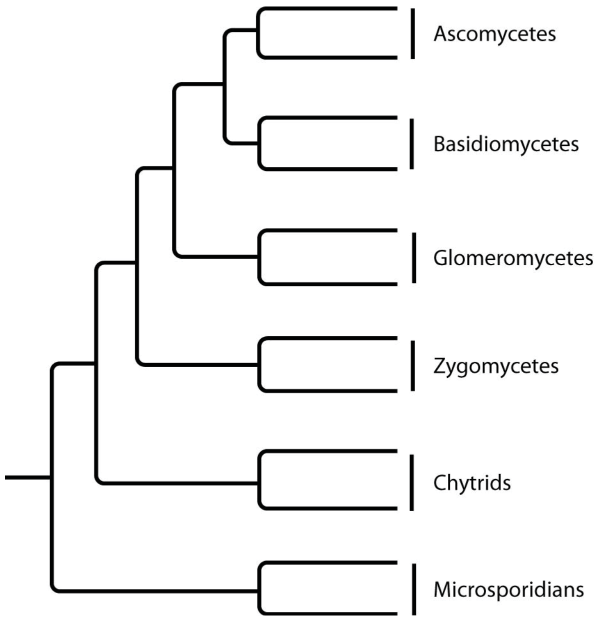
II.



III.



IV.

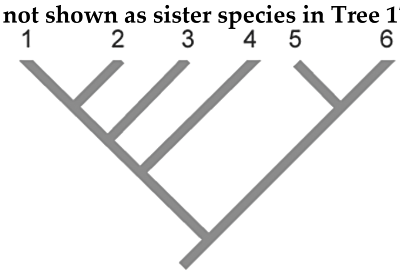


1) Which tree depicts the microsporidians as a sister group of the ascomycetes?
A) I.
B) II.
C) III.
D) IV.

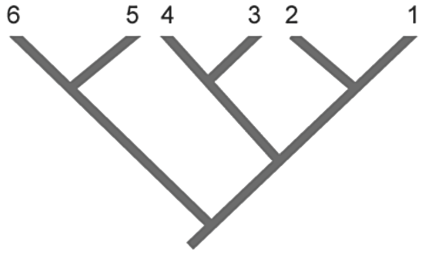
2) Which tree depicts the closest relationship between zygomycetes and chytrids?
A) I.
B) II.
C) III.
D) IV.

3) Which tree depicts the microsporidians as a sister group of the fungi, rather than as a fungus?
A) I.
B) II.
C) III.
D) I.

4) In the phylogenetic trees below, numbers represent species and the same species are shown in both trees. Which two species are represented as sister species in Tree 2 but are not shown as sister species in Tree 1?



Tree 1



Tree 2

A) 1 and 2.
B) 2 and 3.
C) 3 and 4.
D) 4 and 5.

5) The legless condition that is observed in several groups of extant reptiles is the result of _____.
A) Their common ancestor having been legless.
B) A shared adaptation to an arboreal (living in trees) lifestyle.
C) Several instances of the legless condition arising independently of each other.
D) Individual lizards adapting to a fossorial (living in burrows) lifestyle during their lifetimes.

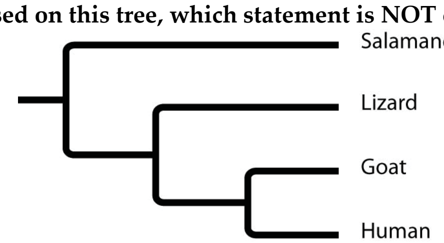
6) The various taxonomic levels (for example, phyla, genera, classes) of the hierarchical classification system differ from each other on the basis of _____.
A) How widely the organisms assigned to each are distributed throughout the environment.
B) Their inclusiveness.
C) The relative genome sizes of the organisms assigned to each.
D) Morphological characters those are applicable to all organisms.

7) If organisms A, B, and C belong to the same class but to different orders and if organisms C, D, and E belong to the same order but to different families, which of the following pairs of organisms would be expected to show the greatest degree of structural homology?
A) A and D.
B) B and D.
C) B and C.
D) D and E.

8) Linnaeus believed that species remained fixed in the form in which they had been created. Linnaeus would have been uncomfortable with _____.
A) A hierarchical classification scheme.
B) Taxonomy.
C) Phylogenies.
D) Nested, increasingly inclusive categories of organisms.

9) The best classification system is that which most closely_____.
A) Unites organisms that possess similar morphologies.
B) Conforms to traditional, Linnaean taxonomic practices.
C) Reflects evolutionary history.
D) Reflects the basic separation of prokaryotes from eukaryotes.

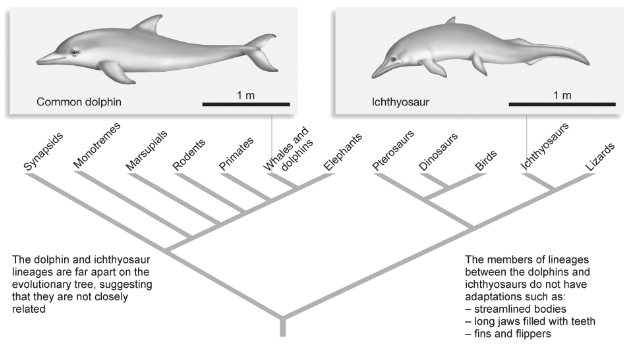
10) Based on this tree, which statement is NOT correct?



A) The salamander lineage is a basal taxon.
B) Salamanders are a sister group to the group containing lizards, goats, and humans.
C) Salamanders are as closely related to goats as to humans.
D) Lizards are more closely related to salamanders than to humans.

11) We know the streamlined bodies shown in the figure below are examples of homoplasy. If the following groups also had streamlined bodies, which of the groupings would give the most support to this body type being homologous?

Analogous traits: Similarities result from convergent evolution.



A) Lizards and elephants.
B) Pterosaurs, dinosaurs, and birds.
C) Synapsids, monotremes, marsupials, rodents, and primates.
D) Lizards, pterosaurs, dinosaurs, birds, synapsids, monotremes, marsupials, rodents, and primates.

12) Some beetles and flies have antler-like structures on their heads, much like male deer do. The existence of antlers in beetle, fly, and deer species with strong male-male competition is an example of _____.
A) Convergent evolution.
B) A synapomorphy.
C) Homology.
D) Parsimony.

13) The term *homoplasy* is most applicable to which of the following features?
A) The legless condition found in various lineages of extant lizards.
B) The five-digit condition of human hands and bat wings.
C) The fur that covers Australian moles and North American moles.
D) The bones of bat forelimbs and the bones of bird forelimbs.

14) If, someday, an Archaean cell is discovered whose rRNA sequence is more similar to that of humans than the sequence of mouse rRNA is to that of humans, the best explanation for this apparent discrepancy would be _____.
A) Homology.
B) Homoplasy.
C) Common ancestry.
D) Retro-evolution by humans.

15) Which of the following pairs are the best examples of homologous structures?
A) Bones in the bat wing and bones in the human forelimb.
B) Owl wing and hornet wing.
C) Bat wing and bird wing.
D) Eyelessness in the Australian mole and eyelessness in the North American mole.

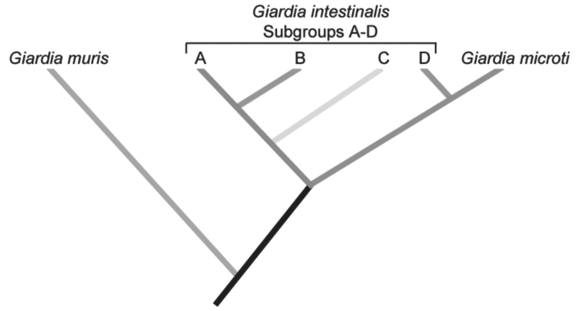
16) Some molecular data place the giant panda in the bear family (*Ursidae*) but place the lesser panda in the raccoon family (*Procyonidae*). If the molecular data best reflect the evolutionary history of these two groups, then the morphological similarities of these two species is most likely due to _____.
A) The inheritance of acquired characteristics.
B) Sexual selection.
C) Possession of analogous (convergent) traits.
D) Possession of shared primitive characters.

17) The importance of computers and of computer software to modern cladistics is most closely linked to advances in _____.
A) Light microscopy.
B) Radiometric dating.
C) Fossil discovery techniques.
D) Molecular genetics.

18) The common ancestors of birds and mammals were very early (stem) reptiles, which almost certainly possessed three-chambered hearts (two atria, one ventricle). Birds and mammals, however, are alike in having four-chambered hearts (two atria, two ventricles). The four-chambered hearts of birds and mammals are best described as _____.
A) Structural homologies.
B) Vestiges.
C) Homoplasies.
D) The result of shared ancestry.

Use the following information to answer the questions (19-20) below.

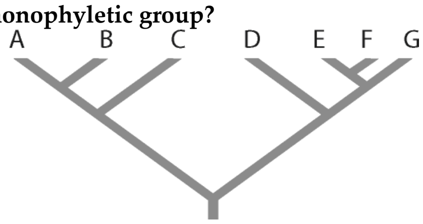
***Giardia intestinalis* can cause disease in several different mammalian species, including humans. *Giardia organisms (G. intestinalis)* that infect humans are similar morphologically to those that infect other mammals; thus they have been considered a single species. However, *G. intestinalis* has been divided into different subgroups based on their host and a few other characteristics.**
In 1999, a DNA sequence comparison study tested the hypothesis that these subgroups actually constitute different species. The following phylogenetic tree was constructed from the sequence comparison of rRNA from several subgroups of *G. intestinalis* and a few other morphologically distinct species of *Giardia*. The researchers concluded that the subgroups of *Giardia* are sufficiently different from one another genetically that they could be considered different species. (T. Monis et al. 1999. Molecular systematics of the parasitic protozoan *Giardia intestinalis*. Mol. Biol. Evol. 16[9]:1135-44.)



19) According to the phylogenetic tree in the figure above, G. intestinalis constitutes a group _____.
A) Paraphyletic.
B) Monophyletic.
C) Polyphyletic.
D) None of them.

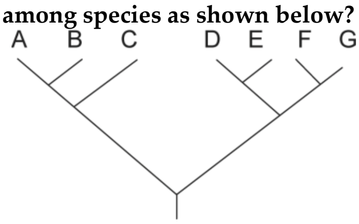
20) By examining the phylogenetic tree diagrammed in the figure above, what conclusion can you draw about the species *G. microti?*
A) It evolved before *G. intestinalis.*
B) It is more closely related to *G. muris* than to *G. intestinalis.*
C) It should not be labeled a species distinct from *G. intestinalis.*
D) It is part of a monophyletic group that also includes *G. intestinalis.*

21) Refer to the figure below. Which of the following forms a monophyletic group?

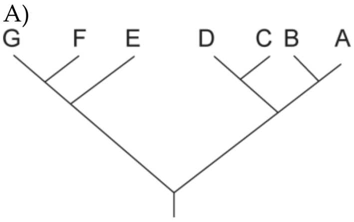


A) A, B, C, D.
B) C and D.
C) D, E, and F.
D) E, F, and G.

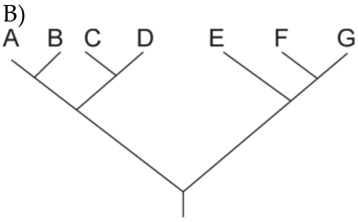
22) Which of the following trees, if any, depicts the same relationship among species as shown below?



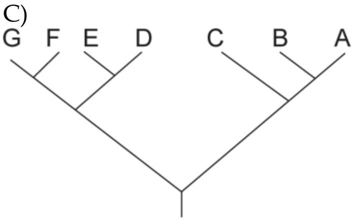
A)



B)



C)



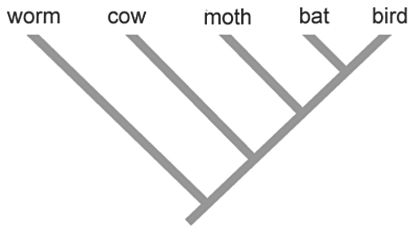
D) None of them.

23) Which of the following would be useful in creating a phylogenetic tree of a taxon?
I) Morphological data from fossil species.
II) Genetic sequences from living species.
III) Behavioral data from living species.
A) I.
B) II.
C) III.
D) I, II, and III.

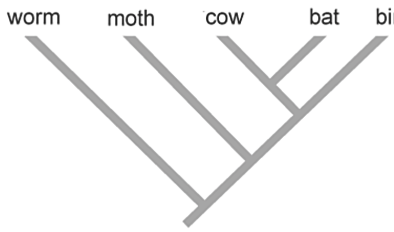
24) Your professor wants you to construct a phylogenetic tree of orchids. She gives you tissue from seven orchid species and one lily. What is the most likely reason she gave you the lily?
A) To serve as an out group.
B) To see if the lily is a cryptic orchid species.
C) To see if the lily and the orchids show all the same shared derived characters (synapomorphies).
D) To demonstrate likely homoplasies.

25) Which of the following statements best describes the rationale for applying the principle of parsimony in constructing phylogenetic trees?
A) Parsimony allows the researcher to "root" the tree.
B) Similarity due to common ancestry should be more common than similarity due to convergent evolution.
C) The molecular clock validates the principle of parsimony.
D) The outgroup roots the tree, allowing the principle of parsimony to be applied.

26) Applying the principle of parsimony to the trait "ability to fly", which of the two phylogenetic trees above is better?



Tree 1



Tree 2

A) Tree 1.
B) Tree 2.
C) Both trees are equally parsimonious.
D) Since the trees show different evolutionary relationships, you cannot determine which is more parsimonious.

27) Which of the following statements is true about a phylogeny, as represented by a phylogenetic tree?
I) Descendant groups (branches) from the same node do not necessarily share any derived characters.
II) A monophyletic group can be properly based on convergent features.
III) The ancestral group often has all the synapomorphies of the descendant species.
A) Only I
B) Only II
C) Only III
D) None of them.

28) Given that phylogenies are based on shared derived characteristics, which of the following traits is useful in generating a phylogeny of species W, X, Y, and Z?

	Species W	Species X	Species Y	Species Z
Trait 1	A	A	A	A
Trait 2	A	A	B	B
Trait 3	A	B	C	D

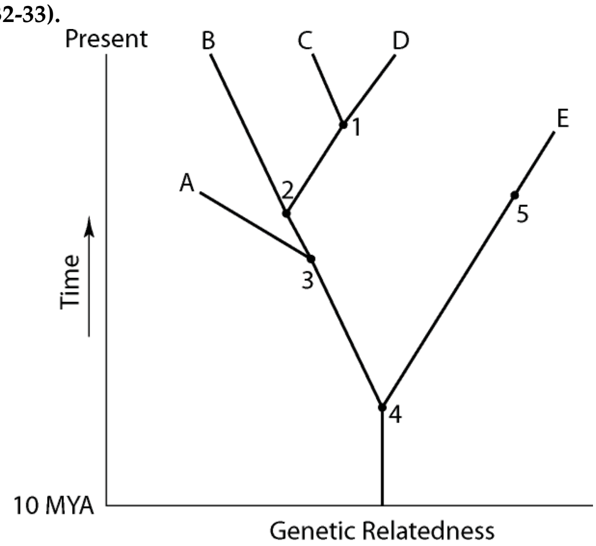
A) Trait 1.
B) Trait 2.
C) Trait 3.
D) Traits 1, 2, and 3.

29) Which of the following is (are) problematic when the goal is to construct phylogenies that accurately reflect evolutionary history?
A) Polyphyletic taxa.
B) Paraphyletic taxa.
C) Monophyletic taxa.
D) Polyphyletic taxa and paraphyletic taxa.

30) Phylogenetic trees constructed from evidence from molecular systematics are based on similarities in _____.
A) Morphology.
B) The pattern of embryological development.
C) Biochemical pathways.
D) Mutations to homologous genes.

31) There is some evidence that reptiles called cynodonts may have had whisker -like hairs around their mouths. If true, then hair is a shared _____.
A) Derived character of mammals, even if cynodonts continue to be classified as reptiles.
B) Derived character of the amniote clade and not of the mammal clade.
C) Ancestral character of the amniote clade, but only if cynodonts are reclassified as mammals.
D) Derived character of mammals, but only if cynodonts are reclassified as mammals.

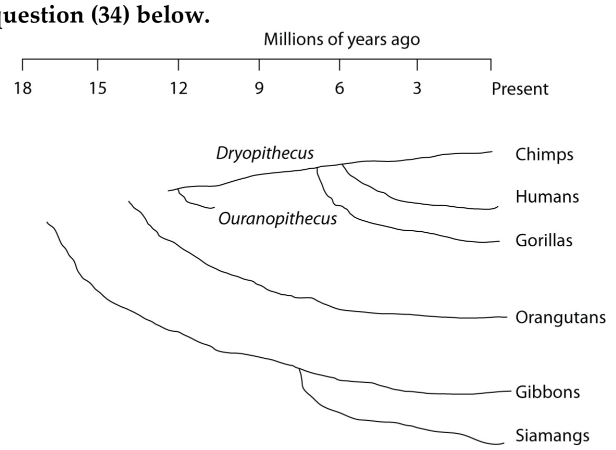
Use the figure below to answer the following questions (32-33).



32) Which extinct species should be the best candidate to serve as the outgroup for the clade whose common ancestor occurs at position 2 in the figure above?
A) A.
B) B.
C) C.
D) E.

33) If the figure above is an accurate depiction of relatedness, then which of the following should be correct?
1. The entire tree is based on maximum parsimony.
2. If all species depicted here make up a taxon, this taxon is monophyletic.
3. The last common ancestor of species B and C occurred more recently than the last common ancestor of species D and E.
4. Species A is the direct ancestor of both species B and species C.
5. The species present at position 3 is ancestral to C, D, and E.
A) 1 and 3.
B) 3 and 4.
C) 2, 3, and 4.
D) 1, 2, and 3.

Use the following figure and description to answer the question (34) below.



Humans, chimpanzees, gorillas, and orangutans are members of a clade called the great apes, which shared a common ancestor about fifteen million years ago. Gibbons and siamangs comprise a clade called the lesser apes. Tree-branch lengths indicate elapsed time.

34) Assuming chimps and gorillas are humans' closest relatives, removing humans from the great ape clade and placing them in a different clade has the effect of making the phylogenetic tree of the great apes _____.

A) Polyphyletic.

B) Paraphyletic.

C) Monophyletic.

D) Into a new order.

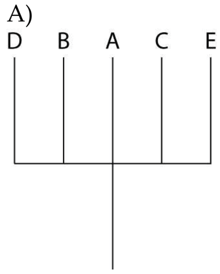
The questions below refer to the following table, which compares the % sequence homology of four different parts (two introns and two exons) of a gene that is found in five different eukaryotic species. Each part is numbered to indicate its distance from the promoter (for example, Intron I is the one closest to the promoter). The data reported for species A were obtained by comparing DNA from one member of species A to another member of species A.

% Sequence Homology

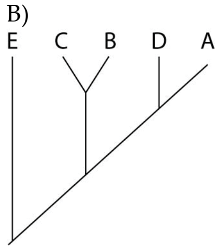
Species	Intron I	Exon I	Intron VI	Exon V
A	100%	100%	100%	100%
B	98%	99%	82%	96%
C	98%	99%	89%	96%
D	99%	99%	92%	97%
E	98%	99%	80%	94%

35) Based on the tabular data, and assuming that time advances vertically, which phylogenetic tree is the most likely depiction of the evolutionary relationships among these five species?

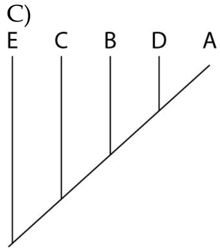
A)



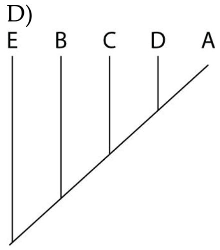
B)



C)



D)



36) Regarding these sequence homology data, the principle of maximum parsimony would be applicable in

.

A) Distinguishing introns from exons.

B) Determining degree of sequence homology.

C) Selecting appropriate genes for comparison among species.

D) Inferring evolutionary relatedness from the number of sequence differences.

37) In a comparison of birds and mammals, having four limbs is _____.

A) A shared ancestral character.

B) A shared derived character.

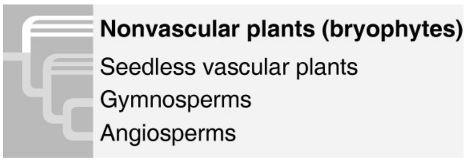
C) A character useful for distinguishing birds from mammals.

D) An example of analogy rather than homology.

38) To apply parsimony to constructing a phylogenetic tree, _____.
A) Choose the tree that assumes all evolutionary changes are equally probable.
B) Choose the tree in which the branch points are based on as many shared derived characters as possible.
C) Choose the tree that represents the fewest evolutionary changes, either in DNA sequences or morphology.
D) Choose the tree with the fewest branch points.

39) If you were using cladistics to build a phylogenetic tree of cats, which of the following would be the best outgroup?
A) Lion.
B) Domestic cat.
C) Wolf.
D) Leopard.

40) Based upon the figure below, the phylogenetic tree __.



1. Depicts uncertainty about whether the bryophytes or the vascular plants evolved first.
2. Depicts an evolutionary hypothesis.
3. Includes polytomies.
4. Shows those members of the phylum Pterophyta are the closest living relatives to the seed plants.
5. Indicates that seeds are a shared ancestral character of all vascular plants.
A) 1 and 2.
B) 2 and 3.
C) 1, 2, and 3.
D) 1, 2, and 4.

41) Which value(s) would be required to calculate how long ago the most recent ancestor of ungulates lived?
I) The number of base pairs that differ among species in a certain genetic sequence.
II) The total number of base pairs in the genetic sequence examined.
III) The age of a fossil ancestor for calibration.
A) I.
B) II.
C) III.
D) I, II, and III.

42) Concerning growth in genome size over evolutionary time, which of these is LEAST associated with the others?
A) Orthologous genes.
B) Gene duplications.
C) Paralogous genes.
D) Gene families.

43) Eukaryotes that are not closely related and that do not share many anatomical similarities can still be placed together on the same phylogenetic tree by comparing their _____.
A) Plasmids.
B) Mitochondrial genomes.
C) Homologous genes that are poorly conserved.
D) Homologous genes that are highly conserved.

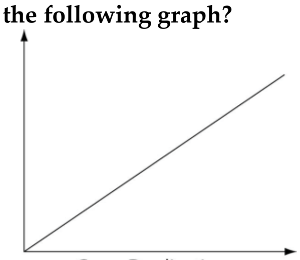
44) A phylogenetic tree constructed using sequence differences in mitochondrial DNA would be most valid for discerning the evolutionary relatedness of _____.
A) Archaeans and bacteria.
B) Fungi and animals.
C) Chimpanzees and humans.
D) Sharks and dolphins.

45) The lakes of northern Minnesota are home to many similar species of damselflies of the genus Enallagma. These species have apparently undergone speciation from ancestral stock since the last glacial retreat about ten thousand years ago. Sequencing which of the following would probably be most useful in sorting out evolutionary relationships among these closely related species?
A) Conserved regions of nuclear DNA.
B) Mitochondrial DNA.
C) Amino acids in proteins.
D) Ribosomal RNA.

46) Which statement represents the best explanation for the observation that the nuclear DNA of wolves and domestic dogs has a very high degree of sequence homology? Dogs and wolves _____.
A) Have very similar morphologies.
B) Belong to the same order.
C) Are both members of the order Carnivora.
D) Share a very recent common ancestor.

47) The reason that paralogous genes can diverge from each other within the same gene pool, whereas orthologous genes diverge only after gene pools are isolated from each other, is that _____.
A) Having multiple copies of genes is essential for the occurrence of sympatric speciation in the wild.
B) Paralogous genes can occur only in diploid species; thus, they are absent from most prokaryotes.
C) Polyploidy is a necessary precondition for the occurrence of sympatric speciation in the wild.
D) Having an extra copy of a gene permits modifications to the copy without loss of the original gene product.

48) Which of the following items is LEAST likely to form a simple linear relationship with the number of gene-duplication events, when placed as the label on the vertical axis of the following graph?



A) Number of genes.
B) Number of DNA base pairs.
C) Genome size.
D) Phenotypic complexity.

49) The most important feature that permits a gene to act as a molecular clock is _____.
A) A large number of base pairs.
B) Being acted upon by natural selection.
C) A reliable average rate of mutation.
D) A recent origin by a gene-duplication event.

50) Neutral theory proposes that _____.
A) Molecular clocks are more reliable when the surrounding pH is close to 7.0.
B) Most mutations of highly conserved DNA sequences should have no functional effect.
C) DNA is less susceptible to mutation when it codes for amino acid sequences whose side groups (or R-groups) have a neutral pH.
D) A significant proportion of mutations are not acted upon by natural selection.

51) Which of the following would, if it had acted upon a gene, prevent this gene from acting as a reliable molecular clock?
A) Neutral mutations.
B) Genetic drift.
C) Mutations within introns.
D) Natural selection.

The question (52) below refer to the following table, which compares the % sequence homology of four different parts (two introns and two exons) of a gene that is found in five different eukaryotic species. Each part is numbered to indicate its distance from the promoter (for example, Intron I is the one closest to the promoter). The data reported for species A were obtained by comparing DNA from one member of species A to another member of species A.

% Sequence Homology

Species	Intron I	Exon I	Intron VI	Exon V
A	100%	100%	100%	100%
B	98%	99%	82%	96%
C	98%	99%	89%	96%
D	99%	99%	92%	97%
E	98%	99%	80%	94%

52) Which of these four gene parts should allow the construction of the most accurate phylogenetic tree, assuming that this is the only part of the gene that has acted as a reliable molecular clock?
A) Intron I.
B) Exon I.
C) Intron VI.
D) Exon V.

53) Based on cladistics, which eukaryotic kingdom is polyphyletic and, therefore, unacceptable?
A) Plantae.
B) Fungi.
C) Animalia.
D) Protista.

54) Which eukaryotic kingdom includes members that are the result of endosymbiosis that included an ancient aerobic bacterium and an ancient cyanobacterium?
A) Plantae.
B) Fungi.
C) Animalia.
D) Protista.

55) A large proportion of Archaeans are extremophiles, so called because they inhabit extreme environments with high acidity, salinity, and/or temperature. Such environments are thought to have been much more common on the primitive Earth. Thus, modern extremophiles survive only in places that their ancestors became adapted to long ago. Which of the following is, consequently, a valid statement about modern extremophiles, assuming that their habitats have remained relatively unchanged?
A) Among themselves, they should share relatively few ancestral traits, especially those that enabled ancestral forms to adapt to extreme conditions.
B) On a phylogenetic tree whose branch lengths are proportional to the amount of genetic change, the branches of the extremophiles should be shorter than the non-extremophilic archaeans.
C) They should contain genes that originated in eukaryotes that are the hosts for numerous species of bacteria.
D) They should currently be undergoing a high level of horizontal gene transfer with non-extremophilic archaeans.

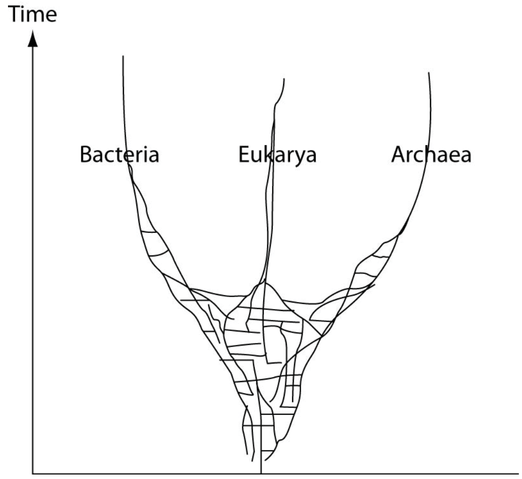
Use the following information to answer the question (56) below.

Paulinella chromatophora **is one of the few cercozoans that is autotrophic, carrying out aerobic photosynthesis with its two elongated "chromatophores." The chromatophores are contained within vesicles of the host cell, and each is derived from a cyanobacterium, though not the same type of cyanobacterium that gave rise to the chloroplasts of algae and plants.**

56) A crucial photosynthetic gene of the cyanobacterium that gave rise to the chromatophore is called *psaE*. This gene is present in the nuclear genome of the cercozoan, but is not in the genome of the chromatophore. This is evidence of _____.
A) Reciprocal mutations in the chromatophore and nuclear genomes.
B) Horizontal gene transfer from bacterium to eukaryotes.
C) Genetic recombination involving a protist and an archaean.
D) Transduction by a phage that infects both prokaryotes and eukaryotes.

57) What kind of evidence has recently made it necessary to assign the prokaryotes to either of two different domains, rather than assigning all prokaryotes to the same kingdom?
A) mtDNA.
B) rRNA genes.
C) Morphological.
D) Ecological.

The following questions refer to this phylogenetic tree, depicting the origins of life and of the three domains. Horizontal lines indicate instances of gene or genome transfer.



A possible phylogenetic tree for the three domains of life.

58) If the early history of life on Earth is accurately depicted by the figure above, then which statement is most in agreement with the hypothesis proposed by this tree?
A) The last universal common ancestor of all extant species is one individual species.
B) The origin of the three domains appears as a polytomy.
C) Archaean genomes should not contain genes that originated in bacteria, and vice versa.
D) Eukaryotes are more closely related to archaeans than to bacteria.

59) Which of these processes can be included among those responsible for the horizontal components of the figure above?
A) Endosymbiosis.
B) Mitosis.
C) Binary fission.
D) Point mutations.

60) Which portion of the figure above may ultimately be better depicted as a "ring"?
A) The bacterial lineage.
B) The archaean lineage.
C) The eukaryotic lineage.
D) The trunk of the tree.

CHAPTER 27:

BACTERIA AND ARCHAEA

1) The predatory bacterium *Bdellovibrio bacteriophorus* drills into a prey bacterium and, once inside, digests it. In an attack upon a Gram-negative bacterium that has a slimy cell covering, what is the correct sequence of structures penetrated by B. bacteriophorus on its way to the prey's cytoplasm?
1. Membrane composed mostly of lipopolysaccharide.
2. Membrane composed mostly of phospholipids.
3. Peptidoglycan.
4. Capsule.
A) 2, 4, 3, 1.
B) 1, 3, 4, 2.
C) 1, 4, 3, 2.
D) 4, 1, 3, 2.

2) Jams, jellies, preserves, honey, and other foods with high sugar content hardly ever become contaminated by bacteria, even when the food containers are left open at room temperature. This is because bacteria that encounter such an environment _____.
A) Undergo death as a result of water loss from the cell.
B) Are unable to metabolize the glucose or fructose, and thus starve to death.
C) Are obligate anaerobes.
D) Are unable to swim through these thick and viscous materials.

Use the information in the following paragraph to answer the question (3-6) below.

A hypothetical bacterium swims among human intestinal contents until it finds a suitable location on the intestinal lining. It adheres to the intestinal lining using a feature that also protects it from phagocytes, bacteriophages, and dehydration. Fecal matter from a human in whose intestine this bacterium lives can spread the bacterium, even after being mixed with water and boiled. The bacterium is not susceptible to the penicillin family of antibiotics. It contains no plasmids and relatively little peptidoglycan.

3) This bacterium's ability to survive in a human who is taking penicillin pills may be due to the presence of _____.
1. Penicillin-resistance genes
2. A secretory system that removes penicillin from the cell
3. A Gram-positive cell wall
4. A Gram-negative cell wall
5. An endospore
A) 1 or 5.
B) 2 or 3.
C) 4 or 5.
D) 2, 4, or 5.

4) Adherence to the intestinal lining by this bacterium is due to its possession of _____.
A) Fimbriae.
B) Pili.
C) A capsule.
D) A flagellum.

5) What should be true of the cell wall of this bacterium?
A) Its innermost layer is composed of a phospholipid bilayer.
B) After it has been subjected to Gram staining, the cell should remain purple.
C) It has an outer membrane of lipopolysaccharide.
D) It is mostly composed of a complex, cross-linked polysaccharide.

6) In which feature(s) should one be able to locate a complete chromosome of this bacterium?
1. Nucleolus.
2. Prophage.
3. Endospore.
4. Nucleoid.
A) 4 only.
B) 1 and 3.
C) 2 and 3.
D) 3 and 4.

The following table depicts characteristics of five prokaryotic species (A -E). Use the information in the table to answer the questions (7-9) below

Trait	Species A	Species B	Species C	Species D	Species E
Plasmid	R	None	R	F	None
Gram Staining Results	Variable	Variable	Negative	Negative	Negative
Nutritional Mode	Chemo hetero-troph	Chemo auto-troph	Chemo hetero-troph	Chemo hetero-troph	Photo autotroph
Specialized Metabolic Pathways	Aerobic methanotroph (obtains carbon and energy from methane)	Anaerobic methanogen	Anaerobic butanolic fermentation	Anaerobic lactic acid fermentation	Anaerobic nitrogen fixation and aerobic photosystems I and II
Other Features	Fimbriae	Internal membranes	Flagellum	Pili	Thylakoids

7) Which two species should have much more phospholipid, in the form of bilayers, in their cytoplasms than most other bacteria?
A) Species A and B.
B) Species A and C.
C) Species B and E.
D) Species C and D.

8) Which species are capable of directed movement?
A) Species A.
B) Species B.
C) Species C.
D) Species D.

9) How many of these species probably have a cell wall that consists partly of an outer membrane of lipopolysaccharide?
A) Only one species.
B) Two species.
C) Three species.
D) Four species.

10) Which of the following observations about flagella is true and is consistent with the scientific conclusion that the flagella from protists and bacteria evolved independently?
A) The flagella of both protists and bacteria are made of the same protein, but the configuration is different.
B) The mechanics of movement and protein structure are the same in these flagella, but there are significant genetic differences.
C) Although the mechanism of movement in both flagella is the same, the protein that accomplishes the movement is different.
D) The protein structure and the mechanism of movement in protist flagella are different from those of bacteria flagella.

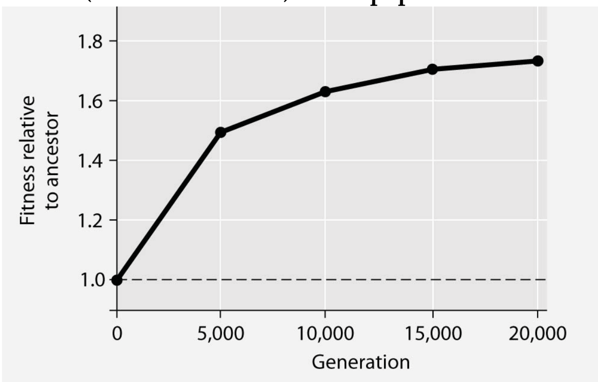
11) In a bacterium that possesses antibiotic resistance and the potential to persist through very adverse conditions, such as freezing, drying, or high temperatures, DNA should be located within, or be part of, which structures?
1. Nucleoid region.
2. Endospore.
3. Fimbriae.
4. Plasmids.
A) 1 and 2 only.
B) 1 and 4 only.
C) 2 and 4 only.
D) 1, 2, and 4.

12) If a bacterium regenerates from an endospore that did not possess any of the plasmids that were contained in its original parent cell, the regenerated bacterium will probably also _____.
A) Lack antibiotic-resistant genes.
B) Lack a cell wall.
C) Lack a chromosome.
D) Lack water in its cytoplasm.

13) Chloramphenicol is an antibiotic that targets prokaryotic (70S) ribosomes, but not eukaryotic (80S) ribosomes. Which of these questions stems from this observation, plus an understanding of eukaryotic origins?
A) Can chloramphenicol also be used to control human diseases that are caused by archaeans?
B) Can chloramphenicol pass through the capsules possessed by many cyanobacteria?
C) If chloramphenicol inhibits prokaryotic ribosomes, should it not also inhibit mitochondrial ribosomes?
D) Why aren't prokaryotic ribosomes identical to eukaryotic ribosomes?

The following question (14) refers to the figure below.

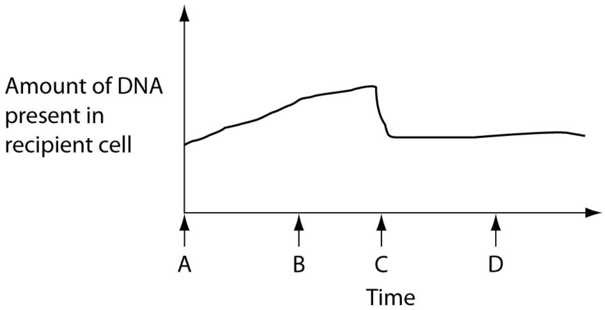
In this eight-year experiment, twelve populations of *E. coli*, each begun from a single cell, were grown in low-glucose conditions for 20,000 generations. Each culture was introduced to fresh growth medium every twenty-four hours. Occasionally, samples were removed from the populations, and their fitness in low-glucose conditions was tested against that of members sampled from the ancestral (common ancestor) *E. coli* population.



14) If new genetic variation in the experimental populations arose solely by spontaneous mutations, then the most effective process for subsequently increasing the prevalence of the beneficial mutations in the population over the course of generations is _____.
A) Transduction.
B) Binary fission.
C) Conjugation.
D) Transformation.

Use the following information and graph to answer the questions (15-18)) below.

The figure below depicts changes to the amount of DNA present in a recipient cell that is engaged in conjugation with an Hfr cell. Hfr cell DNA begins entering the recipient cell at Time A. Assume that reciprocal crossing over occurs (in other words, a fragment of the recipient's chromosome is exchanged for a homologous fragment from the Hfr cell's DNA).



15) What is occurring at Time C that is decreasing the DNA content?
A) Crossing over.
B) Cytokinesis.
C) Degradation of DNA that was not retained in the recipient's chromosome.
D) Reversal of the direction of conjugation.

16) How is the recipient cell different at Time D than it was at Time A?
A) It has a greater number of genes.
B) It has a greater mass of DNA.
C) It has a different sequence of base pairs.
D) It contains bacteriophage DNA.

17) Which two processes are responsible for the shape of the curve at Time B?
1. Transduction.
2. Entry of single-stranded Hfr DNA.
3. Rolling circle replication of single-stranded Hfr DNA.
4. Activation of DNA pumps in plasma membrane.
A) 1 and 2.
B) 1 and 4.
C) 2 and 3.
D) 3 and 4.

18) During which two times can the recipient accurately be described as "recombinant" due to the sequence of events portrayed in the figure?
A) During Times C and D.
B) During Times A and C.
C) During Times A and B.
D) During Times B and D.

Use the information in the following paragraph to answer the question(s) below.

A hypothetical bacterium swims among human intestinal contents until it finds a suitable location on the intestinal lining. It adheres to the intestinal lining using a feature that also protects it from phagocytes, bacteriophages, and dehydration. Fecal matter from a human in whose intestine this bacterium lives can spread the bacterium, even after being mixed with water and boiled. The bacterium is not susceptible to the penicillin family of antibiotics. It contains no plasmids and relatively little peptidoglycan.

19) The cell also lacks F factors and F plasmids. Upon its death, this bacterium should be able to participate in _____.
A) Conjugation.
B) Transduction.
C) Transformation.
D) Conjugation and transduction.

The following table depicts characteristics of five prokaryotic species (A -E). Use the information in the table to answer the question(s) below.

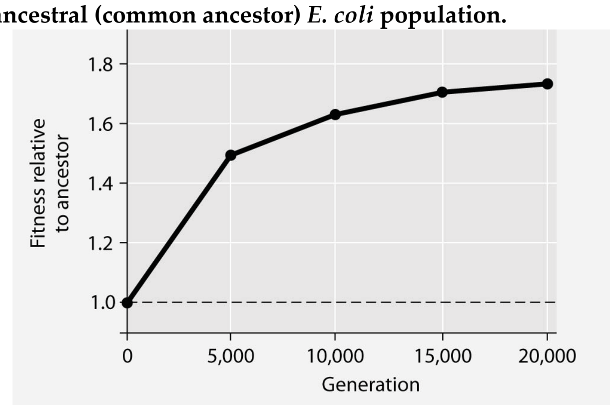
Trait	Species A	Species B	Species C	Species D	Species E
Plasmid	R	None	R	F	None
Gram Staining Results	Variable	Variable	Negative	Negative	Negative
Nutritional Mode	Chemo hetero-troph	Chemo auto-troph	Chemo hetero-troph	Chemo hetero-troph	Photo autotroph
Specialized Metabolic Pathways	Aerobic methanotroph (obtains carbon and energy from methane)	Anaerobic methanogen	Anaerobic butanolic fermentation	Anaerobic lactic acid fermentation	Anaerobic nitrogen fixation and aerobic photosystems I and II
Other Features	Fimbriae	Internal membranes	Flagellum	Pili	Thylakoids

20) Species D is pathogenic if it gains access to the human intestine. Which other species, if it coinhabited a human intestine along with species D, is most likely to result in a recombinant species that is both pathogenic and resistant to some antibiotics?
A) Species A.
B) Species B.
C) Species C.
D) Species E.

21) Which species might be able to include Hfr cells?
A) Species A.
B) Species B.
C) Species C.
D) Species D.

The following questions (22-24) refer to the figure below.

In this eight-year experiment, twelve populations of *E. coli*, each begun from a single cell, were grown in low-glucose conditions for 20,000 generations. Each culture was introduced to fresh growth medium every twenty-four hours. Occasionally, samples were removed from the populations, and their fitness in low-glucose conditions was tested against that of members sampled from the ancestral (common ancestor) *E. coli* population.



22) Which term best describes what has occurred among the experimental populations of cells over this eight-year period?
A) Microevolution.
B) Speciation.
C) Adaptive radiation.
D) Stabilizing selection.

23) Compare the bacteria in the figure above in generation 1 and generation 20,000. The bacteria in generation 1 have a greater _____.
A) Efficiency at exporting glucose from the cell to the environment.
B) Ability to survive on simple sugars, other than glucose.
C) Ability to synthesize glucose from amino acid precursors.
D) Reliance on glycolytic enzymes.

24) If the vertical axis of the figure above refers to relative fitness, then which of the following is the most valid and accurate measure of fitness?
A) Number of daughter cells produced per mother cell per generation.
B) Average swimming speed of cells through the growth medium.
C) Amount of glucose synthesized per unit time.
D) Number of generations per unit time.

25) *E. coli* cells typically make most of their ATP by metabolizing glucose. Under the conditions of this experiment, as time passed, E. coli generation times should be _____.
A) The same as in the typical environment.
B) Faster than in the typical environment.
C) Slower than in the typical environment.
D) None of them.

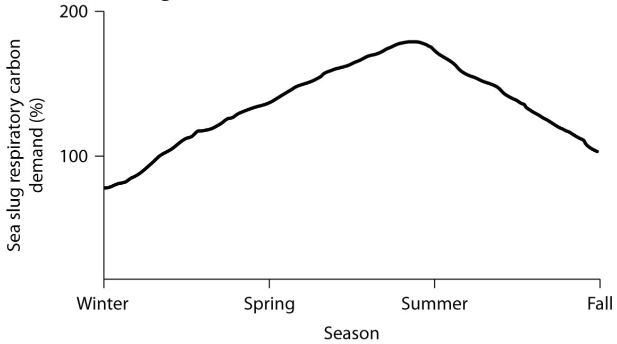
26) If the experimental population of *E. coli* lacks an F factor or F plasmid, and if bacteriophages are excluded from the bacterial cultures, then beneficial mutations might be transmitted horizontally to other *E. coli* cells via _____.
A) Sex pili.
B) Transduction.
C) Conjugation.
D) Transformation.

27) In a hypothetical situation, the genes for sex pilus construction and for tetracycline resistance are located on the same plasmid within a particular bacterium. If this bacterium readily performs conjugation involving a copy of this plasmid, then the result should be the _____.
A) Temporary possession by this bacterium of a completely diploid genome.
B) Rapid spread of tetracycline resistance to other bacteria in that habitat.
C) Subsequent loss of tetracycline resistance from this bacterium.
D) Production of endospores among the bacterium's progeny.

28) Which of the following is LEAST associated with the others?
A) Horizontal gene transfer.
B) Conjugation.
C) Transformation.
D) Binary fission.

Use the following information and figure to answer the questions (29-30) below.

The sea slug *Pteraeolidia ianthina (P. ianthina)* can harbor living dinoflagellates (photosynthetic protists) in its skin. These endosymbiotic dinoflagellates reproduce quickly enough to maintain their populations. Low populations do not affect the sea slugs very much, but high populations (> 5 x 10^5 cells/mg of sea slug protein) can promote sea slug survival.



Percent of sea slug respiratory carbon demand provided by indwelling dinoflagellates.

29) If we assume that carbon is the sole nutrient needed by sea slugs to drive their cellular respiration, then based on the graph, during which season(s) is it LEAST necessary for *P. ianthina* to act as a chemoheterotroph?

A) Winter.
B) Spring.
C) Summer.
D) Fall.

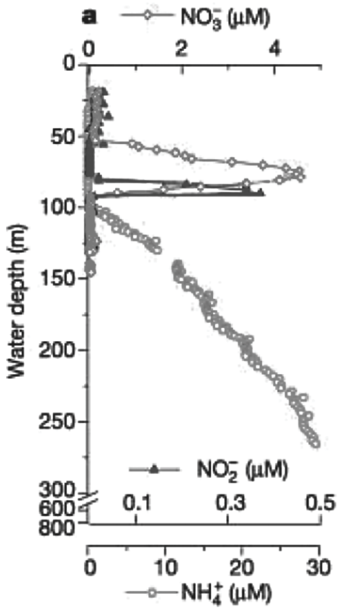
30) Use of synthetic fertilizers often leads to the contamination of groundwater with nitrates. Nitrate pollution is also a suspected cause of anoxic "dead zones" in the ocean. Which of the following might help reduce nitrate pollution?

A) Growing improved crop plants that have nitrogen-fixing enzymes.
B) Adding nitrifying bacteria to the soil.
C) Adding denitrifying bacteria to the soil.
D) Using ammonia instead of nitrate as a fertilizer.

31) Biologists sometimes divide living organisms into two groups: autotrophs and heterotrophs. These two groups differ in _____.

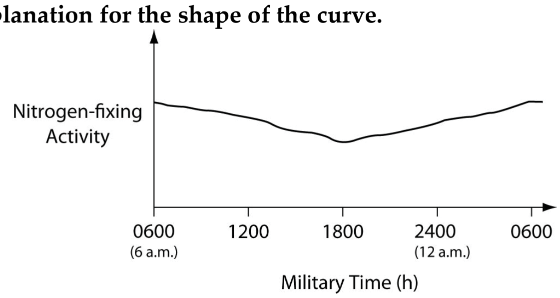
A) Their sources of energy.
B) Their electron acceptors.
C) Their mode of nutrition.
D) The way that they generate ATP.

32) Bacteria able to perform the $NH_4^+ + NO_2^- \rightarrow N_2 + 2H_2O$ reaction have been discovered in laboratory bioreactors and wastewater treatment systems. Researchers predicted that these bacteria should exist in oceans. They measured the concentration of NH_4^+, NO_2^-, NO_3^-, and O_2 in the Black Sea as a function of water depth (M. Kuypers et al. 2003 Anaerobic ammonium oxidation by anammox bacteria in the Black Sea. Nature 422:608—11) to determine where in the sea the bacteria might live. Analyzing data presented in the figure below, at what depth would you expect to find the bacteria? (Note: In the figure, different scales are used to show concentrations of NH_4^+, NO_2^-, NO_3^-, and O_2.)



A) In the top 50 meters.
B) At a depth of 75 meters.
C) At a depth of 92 meters.
D) Below 100 meters.

33) Data were collected from the heterocysts of a nitrogen-fixing cyanobacterium inhabiting equatorial ponds. Study the following graph and choose the most likely explanation for the shape of the curve.



A) Enough oxygen (O2) enters heterocysts during hours of peak photosynthesis to have a somewhat-inhibitory effect on nitrogen fixation.
B) Atmospheric nitrogen (N2) levels increase at night because plants are no longer metabolizing this gas, so they are not absorbing this gas through their stomata.
C) Heterocyst walls become less permeable to nitrogen (N2) influx during darkness.
D) The amount of fixed nitrogen that is dissolved in the pond water in which the cyanobacteria are growing peaks at the close of the photosynthetic day (1800 hours).

Use the information in the following paragraph to answer the question (34) below.

A hypothetical bacterium swims among human intestinal contents until it finds a suitable location on the intestinal lining. It adheres to the intestinal lining using a feature that also protects it from phagocytes, bacteriophages, and dehydration. Fecal matter from a human in whose intestine this bacterium lives can spread the bacterium, even after being mixed with water and boiled. The bacterium is not susceptible to the penicillin family of antibiotics. It contains no plasmids and relatively little peptidoglycan.

34) This bacterium derives nutrition by digesting human intestinal contents. Thus, this bacterium is an _____.
A) Aerobic chemoheterotroph.
B) Aerobic chemoautotroph.
C) Anaerobic chemoheterotroph.
D) Anaerobic chemoautotroph.

Use the following information to answer the question(s) below.

Nitrogenase, the enzyme that catalyzes nitrogen fixation, is inhibited whenever free oxygen (O_2) reaches a critical concentration. Consequently, nitrogen fixation cannot occur in cells wherein photosynthesis produces free O_2. Consider the colonial aquatic cyanobacterium, Anabaena, whose heterocytes are described as having "...a thickened cell wall that restricts entry of O_2 produced by neighboring cells. Intracellular connections allow heterocysts to transport fixed nitrogen to neighboring cells in exchange for carbohydrates."

35) Given that the enzymes that catalyze nitrogen fixation are inhibited by oxygen, what are two "strategies" that nitrogen-fixing prokaryotes might use to protect these enzymes from oxygen?
1. Couple them with photosystem II (the photosystem that splits water molecules).
2. Package them in membranes that are impermeable to all gases.
3. Be obligate anaerobes.
4. Be strict aerobes.
5. Package these enzymes in specialized cells or compartments that inhibit oxygen entry.
A) 1 and 4.
B) 2 and 4.
C) 3 and 4.
D) 3 and 5.

36) Think about this description of the colonial aquatic cyanobacterium, *Anabaena*. What two questions below are important for understanding how nitrogen (N2) enters heterocysts, and how oxygen (O_2) is kept out of heterocysts?
1. If carbohydrates can enter the heterocysts from neighboring cells via the "intracellular connections", how is it that O_2 doesn't also enter via this route?
2. If the cell walls of Anabaena photosynthetic cells are permeable to O_2 and carbon dioxide (CO_2), are they also permeable to N_2?
3. If the nuclei of the photosynthetic cells contain the genes that code for nitrogen fixation, how can these cells fail to perform nitrogen fixation?
4. If the nuclei of the heterocysts contain the genes that code for photosynthesis, how can these cells fail to perform photosynthesis?
5. If the cell walls of Anabaena heterocysts are permeable to N_2, how is it that N_2 doesn't diffuse out of the heterocysts before it can be fixed?
6. If the thick cell walls of Anabaena heterocysts exclude entry of oxygen gas, how is it that they don't also exclude the entry of nitrogen gas?
A) 1 and 3
B) 1 and 6
C) 2 and 5
D) 4 and 6.

The following table depicts characteristics of five prokaryotic species (A -E). Use the information in the table to answer the question(s) below.

Trait	Species A	Species B	Species C	Species D	Species E
Plasmid	R	None	R	F	None
Gram Staining Results	Variable	Variable	Negative	Negative	Negative
Nutritional Mode	Chemo hetero-troph	Chemo auto-troph	Chemo hetero-troph	Chemo hetero-troph	Photo autotroph
Specialized Metabolic Pathways	Aerobic methanotroph (obtains carbon and energy from methane)	Anaerobic methanogen	Anaerobic butanolic fermentation	Anaerobic lactic acid fermentation	Anaerobic nitrogen fixation and aerobic photosystems I and II
Other Features	Fimbriae	Internal membranes	Flagellum	Pili	Thylakoids

37) Which two species might be expected to cooperate metabolically, perhaps forming a biofilm wherein one species surrounds cells of the other species?
A) Species A and B.
B) Species A and C.
C) Species B and E.
D) Species C and D.

Use the following information to answer the question(s) below.

For several decades now, amphibian species worldwide have been in decline. A significant proportion of the decline seems to be due to the spread of the chytrid fungus, *Batrachochytrium dendrobatidis* (Bd). Chytrid sporangia reside within the epidermal cells of infected animals, animals that consequently show areas of sloughed skin. They can also be lethargic, which is expressed through failure to hide and failure to flee. The infection cycle typically takes four to five days, at the end of which zoospores are released from sporangia into the environment. In some amphibian species, mortality rates approach 100%; other species seem able to survive the infection.

38) If infection primarily involves the outermost layers of adult amphibian skin, and if the chytrids use the skin as their sole source of nutrition, then which term best applies to the chytrids?
A) Anaerobic chemoautotroph.
B) Aerobic chemoautotroph.
C) Anaerobic chemoheterotroph.
D) Aerobic chemoheterotroph.

39) While examining a rock surface, you have discovered an interesting new organism. Which of the following criteria will allow you to classify the organism as belonging to Bacteria but not Archaea or Eukarya?
A) Cell walls are made primarily of peptidoglycan.
B) The organism does not have nucleus.
C) The lipids in its plasma membrane consist of glycerol bonded to straight-chain fatty acids.
D) It can survive at a temperature over 100°C.

40) Which of the following describe all existing bacteria?
A) Pathogenic, omnipresent, morphologically diverse.
B) Extremophiles, tiny, abundant.
C) Tiny, ubiquitous, metabolically diverse.
D) Morphologically diverse, metabolically diverse, extremophiles.

41) You have found a new prokaryote. What line of evidence would support your hypothesis that the organism is a cyanobacterium?
A) It is able to form colonies and produce oxygen.
B) It is an endosymbiont.
C) It forms chains called mycelia.
D) It lacks cell walls.

42) Which statement about the domain Archaea is true?
A) Genetic prospecting has recently revealed the existence of many previously unknown archaean species.
B) The genomes of archaeans are unique, containing no genes that originated within bacteria.
C) No archaeans can inhabit solutions that are 30% salt.
D) No archaeans are adapted to waters with temperatures above the boiling point.

43) Which of the following traits do archaeans and bacteria share?
A) Composition of the cell wall.
B) Composition of the cell wall and lack of a nuclear envelope.
C) Lack of a nuclear envelope and presence of plasma membrane.
D) Presence of plasma membrane and composition of the cell wall.

44) Assuming that each of these possesses a cell wall, which prokaryotes should be expected to be most strongly resistant to plasmolysis in hypertonic environments?
A) Extreme halophiles.
B) Extreme thermophiles.
C) Methanogens.
D) Cyanobacteria.

45) The thermoacidophile *Sulfolobus acidocaldarius* lacks peptidoglycan, but still possesses a cell wall. What is likely to be true of this species?
1. It is a bacterium.
2. It is an archaean.
3. The optimal pH of its enzymes will lie above pH 7.
4. The optimal pH of its enzymes will lie below pH 7.
5. It could inhabit certain hydrothermal springs.
6. It could inhabit alkaline hot springs.
A) 1, 3, and 6.
B) 2, 4, and 5.
C) 1, 3, and 5.
D) 1, 4, and 5.

46) A fish that has been salt-cured subsequently develops a reddish color. You suspect that the fish has been contaminated by the extreme halophile *Halobacterium*. Which of these features of cells removed from the surface of the fish, if confirmed, would support your suspicion?
1. The presence of the same photosynthetic pigments found in cyanobacteria.
2. Cell walls that lack peptidoglycan.
3. Cells those are isotonic to conditions on the surface of the fish.
4. Cells unable to survive salt concentrations lower than 9%.
5. The presence of very large numbers of ion pumps in its plasma membrane.
A) 2 and 5.
B) 3 and 4.
C) 1, 4, and 5.
D) 2, 3, 4, and 5.

The following table depicts characteristics of five prokaryotic species (A -E). Use the information in the table to answer the question (47) below.

Trait	Species A	Species B	Species C	Species D	Species E
Plasmid	R	None	R	F	None
Gram Staining Results	Variable	Variable	Negative	Negative	Negative
Nutritional Mode	Chemo hetero-troph	Chemo auto-troph	Chemo hetero-troph	Chemo hetero-troph	Photo autotroph
Specialized Metabolic Pathways	Aerobic methanotroph (obtains carbon and energy from methane)	Anaerobic methanogen	Anaerobic butanolic fermentation	Anaerobic lactic acid fermentation	Anaerobic nitrogen fixation and aerobic photosystems I and II
Other Features	Fimbriae	Internal membranes	Flagellum	Pili	Thylakoids

47) Which species is most likely to be found in sewage treatment plants and in the guts of cattle?
A) Species A.
B) Species B.
C) Species C.
D) Species D.

48) Which of the following extremophiles might researchers most likely use as a model for the earliest organisms on Earth?
A) A bacterium found on another planet or moon.
B) An archaean capable of surviving in the polar ice caps.
C) An anaerobic archaean species.
D) A bacterium that thrives in a highly acidic environment.

49) Mitochondria are thought to be the descendants of certain alpha proteobacteria. They are, however, no longer able to lead independent lives because most genes originally present on their chromosomes have moved to the nuclear genome. Which phenomenon accounts for the movement of these genes?
A) Plasmolysis.
B) Conjugation.
C) Translation.
D) Horizontal gene transfer.

Use the following information to answer the question(s) below.

The most recently discovered phylum in the animal kingdom (1995) is the phylum Cycliophora. It includes three species of tiny organisms that live in large numbers on the outsides of the mouthparts and appendages of lobsters. The feeding stage permanently attaches to the lobster via an adhesive disk and collects scraps of food from its host's feeding by capturing the scraps in a current created by a ring of cilia. The body is sac-like and has a U-shaped intestine that brings the anus close to the mouth. Cycliophorans are coelomates, do not molt (though their host does), and their embryos undergo spiral cleavage.

50) If harboring large populations of cycliophorans neither helps nor harms their lobster hosts, then cycliophorans can be properly considered to be _____.
1. Parasites.
2. Mutualists.
3. Commensals.
4. Symbionts.
5. Endosymbionts.
A) 1 and 4.
B) 3 and 4.
C) 2 and 5.
D) 3 and 5.

51) Bacteria perform each of the following ecological roles. Which role typically does NOT involve symbiosis?
A) Skin commensalist.
B) Decomposer.
C) Aggregates with methane-consuming archaea.
D) Gut mutualist.

Use the following information to answer the question(s) below.

Healthy individuals of *Paramecium bursaria* contain photosynthetic algal endosymbionts of the genus *Chlorella*. When within their hosts, the algae are referred to as zoochlorellae. In aquaria with light coming from only one side, *P. bursaria* gather at the well-lit side, whereas other species of *Paramecium* gather at the opposite side. The zoochlorellae provide their hosts with glucose and oxygen, and *P. bursaria* provides its zoochlorellae with protection and motility. *P. bursaria* can lose its zoochlorellae in two ways: (1) if kept in darkness, the algae will die; and (2) if prey items (mostly bacteria) are absent from its habitat, *P. bursaria* will digest its zoochlorellae.

52) A *P. bursaria* cell that has lost its zoochlorellae is aposymbiotic. If aposymbiotic cells have population growth rates the same as those of healthy, zoochlorella-containing *P. bursaria* in well-lit environments with plenty of prey items, then such an observation would be consistent with which type of relationship?
A) Parasitic.
B) Commensalistic.
C) Toxic.
D) Mutualistic.

53) The termite gut protist *Mixotricha paradoxa* has at least two kinds of bacteria attached to its outer surface. One kind is a spirochete that propels its host through the termite gut. A second type of bacteria synthesizes adenosine triphosphate (ATP), some of which is used by the spirochetes. The locomotion provided by the spirochetes introduces the ATP-producing bacteria to new food sources. Which term(s) is (are) applicable to the relationship between the two kinds of bacteria?
1. Mutualism.
2. Parasitism.
3. Symbiosis.
4. Metabolic cooperation.
A) 1 and 2.
B) 2 and 3.
C) 1, 3, and 4.
D) 2, 3, and 4.

54) If all prokaryotes on Earth suddenly vanished, which of the following would be the most likely and most direct result?
A) Human populations would thrive in the absence of disease.
B) Bacteriophage numbers would dramatically increase.
C) The recycling of nutrients would be greatly reduced, at least initially.
D) There would be no more pathogens on Earth.

55) In a hypothetical situation, a bacterium lives on the surface of a leaf, where it obtains nutrition from the leaf's nonliving, waxy covering while inhibiting the growth of other microbes that are plant pathogens. If this bacterium gains access to the inside of a leaf, however, it causes a fatal disease in the plant. Once the plant dies, the bacterium and its offspring decompose the plant. What is the correct sequence of ecological roles played by the bacterium in the situation described here?
1. Nutrient recycler.
2. Mutualist.
3. Commensal.
4. Pathogen.
5. Primary producer.
A) 1, 3, 4
B) 2, 3, 4
C) 2, 4, 1
D) 1, 2, 5

56) What is the goal of bioremediation?
A) To improve human health with the help of living organisms such as bacteria.
B) To clean up areas polluted with toxic compounds by using bacteria.
C) To improve soil quality for plant growth by using bacteria.
D) To improve bacteria for production of useful chemicals.

57) Foods can be preserved in many ways by slowing or preventing bacterial growth. Which of these methods should be LEAST effective at inhibiting bacterial growth?
A) Refrigeration: slows bacterial metabolism and growth.
B) Closing previously opened containers: prevents more bacteria from entering, and excludes oxygen.
C) Pickling: creates a pH at which most bacterial enzymes cannot function.
D) Canning in heavy sugar syrup: creates osmotic conditions that remove water from most bacterial cells.

58) Broad-spectrum antibiotics inhibit the growth of most intestinal bacteria. Consequently, assuming that nothing is done to counter the reduction of intestinal bacteria, a hospital patient who is receiving broad-spectrum antibiotics is most likely to become _____.
A) Unable to fix carbon dioxide.
B) Antibiotic resistant.
C) Unable to synthesize peptidoglycan.
D) Deficient in certain vitamins and nutrients.

59) The pathogenic prokaryotes that cause cholera are _____.
A) Archaea that release an exotoxin.
B) Archaea that release an endotoxin.
C) Bacteria that release an exotoxin.
D) Bacteria that release an endotoxin.

60) The soil layer surrounding plant roots, called the rhizosphere, has been shown in some cases to _____.
A) Contain fungi that produce an endotoxin which inhibits the growth rhizomes.
B) Inhibit the growth of fungal pathogens.
C) Produce vitamins essential for the uptake of nitrogen.
D) Contain archaea which produce antibiotics that prevent the growth of most bacteria.

61) When a virus infects a bacterial cell, often new viruses are assembled and released when the host bacterial cell is lysed. If these new viruses go on to infect new bacterial cells the host cells may not be lysed. What is the most plausible explanation for this?
A) The bacterial cell must be resistant to infection by the virus.
B) The virus carries genes that confer resistance to the host bacterial cell.
C) The host bacterium couples the viral infection with transformation.
D) The virus has entered the genome of the bacterial cell and is in the lysogenic stage.

62) Sexual reproduction in eukaryotes increases genetic variation. In prokaryotes, transformation, transduction, and conjugation are mechanisms that increase genetic variation. A fundamental difference between the generations of genetic variation in the two domains is:

A) Eukaryotes are able to generate mutations in response to environmental stress while prokaryotes only generate random variation.

B) Eukaryotic variation occurs primarily within a single generation while prokaryotic variation occurs over many generations.

C) Crossing over is a major mechanism in creating genetic variation in prokaryotes while independent assortment is a major mechanism is eukaryotes.

D) Eukaryotic genetic variation occurs with vertical gene transfer while prokaryotic genetic variation occurs with horizontal gene transfer.

63) In prokaryotes new mutations accumulate quickly in populations, while in eukaryotes new mutations accumulate much more slowly. The primary reasons for this are

A) Prokaryotes have short generation times and large population sizes.

B) Prokaryotes have random mutations while eukaryotes can target genes for mutations; thus mutations may not accumulate as quickly in eukaryotes but they are more useful to the organism.

C) The DNA in prokaryotes is not as stable as eukaryotic DNA and is thus more likely to mutate.

D) Prokaryote mutations are less effective than eukaryote mutations in providing variation for evolution.

CHAPTER 28:

PROTISTS

1) According to the endosymbiotic theory, why was it adaptive for the larger (host) cell to keep the engulfed cell alive, rather than digesting it as food?
A) The engulfed cell provided the host cell with adenosine triphosphate (ATP).
B) The engulfed cell provided the host cell with carbon dioxide.
C) The engulfed cell allowed the host cell to metabolize glucose.
D) The host cell was able to survive anaerobic conditions with the engulfed cell alive.

2) The chloroplasts of land plants are thought to have been derived according to which evolutionary sequence?
A) Cyanobacteria → green algae → land plants.
B) Cyanobacteria → green algae → fungi → land plants.
C) Red algae → brown algae → green algae → land plants.
D) Cyanobacteria → red algae → green algae → land plants.

3) A particular species of protist has obtained a chloroplast via secondary endosymbiosis. You know this because the chloroplasts _____.
A) Have nuclear and cyanobacterial genes.
B) Are exceptionally small.
C) Have three or four membranes.
D) Have only a single pigment.

4) All protists are _____.
A) Unicellular.
B) Eukaryotic.
C) Symbionts.
D) Mixotrophic.

5) An individual mixotroph loses its plastids, yet continues to survive. Which of the following most likely accounts for its continued survival?
A) It relies on photosystems that float freely in its cytosol.
B) It must have gained extra mitochondria when it lost its plastids.
C) It engulfs organic material by phagocytosis or by absorption.
D) It has an endospore.

6) Which of the following have chloroplasts (or structures since evolved from chloroplasts) thought to be derived from ancestral green algae?
A) Stramenopiles.
B) Apicomplexans.
C) Dinoflagellates.
D) Chlorarachniophytes.

Use the following information to answer the question(s) below.

***Paulinella chromatophora* is one of the few cercozoans that is autotrophic, carrying out aerobic photosynthesis with its two elongated "chromatophores." The chromatophores are contained within vesicles of the host cell, and each is derived from a cyanobacterium, though not the same type of cyanobacterium that gave rise to the chloroplasts of algae and plants.**

7) The closest living relative of *P. chromatophora* is the heterotroph *P. ovalis*. *P. ovalis* uses threadlike pseudopods to capture its prey, which it digests internally. Which of the following, if observed, would be the best reason for relabeling *P. chromatophora* as a mixotroph instead of an autotroph?
A) A pigmented central vacuole, surrounded by a tonoplast.
B) A vacuole with food inside.
C) A secretory vesicle.
D) A contractile vacuole.

8) Which process could have allowed the nucleomorphs of chlorarachniophytes to be reduced, without the net loss of any genetic information?
A) Conjugation.
B) Horizontal gene transfer.
C) Phagocytosis.
D) Meiosis.

Use the following information to answer the questions (9-14) below.

***Giardia intestinalis* is an intestinal parasite of humans and other mammals that causes intestinal ailments in most people who ingest the cysts. Upon ingestion, each cyst releases two motile cells, called trophozoites. These attach to the small intestine's lining via a ventral adhesive disk. The trophozoites anaerobically metabolize glucose from the host's intestinal contents to produce ATP. Reproduction is completely asexual, occurring by longitudinal binary fission of trophozoites, with each daughter cell receiving two haploid nuclei ($n = 5$). A trophozoite will often encyst as it passes into the large intestine by secreting around itself a case that is resistant to cold, heat, and dehydration. Infection usually occurs by drinking untreated water that contains cysts.**

9) The cysts of *Giardia* function most like the _____.
A) Mitochondria of ancestral diplomonads.
B) Nuclei of archaeans.
C) Endospores of bacteria.
D) Capsids of viruses.

10) Consider the following data: (a) Most ancient eukaryotes are unicellular. (b) All eukaryotes alive today have a nucleus and cytoskeleton. (c) Most ancient eukaryotes lack a cell wall. Which of the following conclusions could reasonably follow the data presented? The first eukaryote may have been _____.
A) Very similar to a plant cell.
B) Anaerobic.
C) Capable of phagocytosis.
D) Photosynthetic.

11) *Giardia's* mitosome can be said to be "doubly degenerate," because it is a degenerate type of , an organelle that is itself a degenerate form of _____.
A) Nucleus; archaean.
B) Nucleus; bacterium.
C) Mitochondria; proteobacterium.
D) Mitochondria; spirochete.

12) The mitosome of *Giardia* has no DNA within it. If it did contain DNA, then what predictions should we be able to make about its DNA?
1. It is linear.
2. It is circular.
3. It has many introns.
4. It has few introns.
5. It is not associated with histone proteins.
6. It is complexed with histone proteins.
A) 1, 3, and 5.
B) 1, 4, and 5.
C) 2, 3, and 6.
D) 2, 4, and 5.

13) Given its mode of reproduction and internal structures, which of the following should be expected to occur in Giardia at some stage of its life cycle?
A) Separation (segregation) of daughter chromosomes.
B) Crossing over.
C) Meiosis.
D) None of them.

14) If the mitosomes of *Giardia* contain no DNA, yet are descendants of what were once free-living organisms, then where are we likely to find the genes that encode their structures, and what accounts for their current location there?
A) Plasmids; conjugation.
B) Plasmids; transformation.
C) Nucleus; horizontal gene transfer.
D) Nucleus; S phase.

15) When a mosquito infected with *Plasmodium* first bites a human, the *Plasmodium* _____.
A) Gametes fuse, forming an oocyst.
B) Cells infect the human liver cells.
C) Cells cause lysing of the human red blood cells.
D) Oocyst undergoes meiosis.

16) Which two genera have members that can evade the human immune system by frequently changing their surface proteins?
1. *Plasmodium.*
2. *Trichomonas.*
3. *Paramecium.*
4. *Trypanosoma.*
5. *Entamoeba.*
A) 1 and 4.
B) 2 and 3.
C) 2 and 4.
D) 4 and 5.

17) Which of the following pairs of protists and their characteristics is mismatched?
A) Apicomplexans - internal parasites.
B) Euglenozoans - unicellular flagellates.
C) Ciliates - red tide organisms.
D) Entamoebas - ingestive heterotrophs.

18) Dinoflagellates _____.
A) Possess two flagella.
B) Are all autotrophic.
C) Lack mitochondria.
D) Include species that cause malaria.

19) You are given an unknown organism to identify. It is unicellular and heterotrophic. It is motile, using many short extensions of the cytoplasm, each featuring the 9 + 2 filament pattern. It has well-developed organelles and two nuclei, one large and one small. This organism is most likely to be a _____.
A) Foraminiferan.
B) Radiolarian.
C) Ciliate.
D) Kinetoplastid.

20) Which of the following is characteristic of ciliates?
A) They use pseudopods as feeding structures.
B) They are often multinucleate.
C) They can exchange genetic material with other ciliates by the process of mitosis.
D) Most live as solitary autotrophs in fresh water.

21) Diatoms are mostly asexual members of the phytoplankton. Diatoms lack any organelles that might have the 9 + 2 pattern. They obtain their nutrition from functional chloroplasts, and each diatom is encased within two porous, glasslike valves. Which question would be most important for one interested in the day-to-day survival of individual diatoms?
A) How do diatoms get transported from one location on the water's surface layers to another location on the surface?
B) How do diatoms with their glasslike valves keep from sinking into poorly lit waters?
C) How do diatoms with their glasslike valves avoid being shattered by the action of waves?
D) How do diatom sperm cells locate diatom egg cells?

22) A large seaweed that floats freely on the surface of deep bodies of water would be expected to lack which of the following?
A) Thalli.
B) Bladders.
C) Holdfasts.
D) Gel-forming polysaccharides.

23) Reinforced, threadlike pseudopods that can perform phagocytosis are generally characteristic of _____.
A) Forams.
B) Water molds.
C) Dinoflagellates.
D) Oomycetes.

24) A snail-like, coiled, porous test (shell) of calcium carbonate is characteristic of _____.
A) Diatoms.
B) Foraminiferans.
C) Ciliates.
D) Water molds.

25) You are given the task of designing an aquatic protist that is a primary producer. It cannot swim on its own, yet must stay in well-lit surface waters. It must be resistant to physical damage from wave action. It should be most similar to a(n) _____.
A) Diatom.
B) Dinoflagellate.
C) Apicomplexan.
D) Red alga.

26) A gelatinous seaweed that grows in shallow, cold water and undergoes heteromorphic alternation of generations is most probably what type of alga?
A) Red.
B) Green.
C) Brown.
D) Yellow.

You are given four test tubes, each containing an unknown protist, and your task is to read the following description and match these four protists to the correct test tube in the following questions (27-28).

When light, especially red and blue light, is shone on the tubes, oxygen bubbles accumulate on the inside of test tubes 1 and 2. Chemical analysis of test tube 1 indicates the presence of a chemical that is toxic to fish and humans. Chemical analysis of test tube 2 indicates the presence of substantial amounts of silica. Microscopic analysis of organisms in test tubes 1, 3, and 4 reveals the presence of permanent, membrane-bounded sacs just under the plasma membrane. Microscopic analysis of organisms in test tube 3 reveals the presence of an apicoplast in each. Microscopic analysis of the contents in test tube 4 reveals the presence of one large nucleus and one small nucleus in each organism.

27) Test tube 3 contains _____.
A) *Paramecium.*
B) *Pfiesteria (dinoflagellate).*
C) *Entamoeba.*
D) *Plasmodium.*

28) Test tube 4 contains _____.
A) *Paramecium.*
B) *Pfiesteria (dinoflagellate).*
C) *Entamoeba.*
D) *Plasmodium.*

Use the following description and table to answer the question(s) below.

Diatoms are encased in petri-platelike cases (valves) made of translucent hydrated silica whose thickness can be varied. The material used to store excess calories can also be varied. At certain times, diatoms store excess calories in the form of the liquid polysaccharide, laminarin, and at other times as oil. The following are data concerning the density (specific gravity) of various components of diatoms, and of their environment.

Specific Gravities of Materials Relevant to Diatoms

Material	Specific Gravity (kg/m^3)
Pure water	1000
Seawater	1026
Hydrated silica	2250
Liquid laminarin	1500
Diatom oil	910

29) Water's density and, consequently, its buoyancy decrease at warmer temperatures. Based on this consideration and using data from the table above, at which time of year should one expect diatoms to be storing excess calories mostly as oil?
A) Mid-winter.
B) Early spring.
C) Late summer.
D) Late fall.

30) Water's density and, consequently, its buoyancy decrease at warmer temperatures. Considering the impact of temperature, and the table above, in which environment should diatoms sinking be slowest?
A) Cold fresh water.
B) Warm fresh water.
C) Cold seawater.
D) Warm seawater.

Use the following information to answer the question(s) below.

Healthy individuals of *Paramecium bursaria* contain photosynthetic algal endosymbionts of the genus *Chlorella*. When within their hosts, the algae are referred to as zoochlorellae. In aquaria with light coming from only one side, *P. bursaria* gather at the well-lit side, whereas other species of Paramecium gather at the opposite side. The zoochlorellae provide their hosts with glucose and oxygen, and *P. bursaria* provides its zoochlorellae with protection and motility. *P. bursaria* can lose its zoochlorellae in two ways: (1) if kept in darkness, the algae will die; and (2) if prey items (mostly bacteria) are absent from its habitat, *P. bursaria* will digest its zoochlorellae.

31) Which term best describes the symbiotic relationship of well-fed *P. bursaria* to their zoochlorellae?
A) Mutualistic.
B) Parasitic.
C) Predatory.
D) Pathogenic.

32) The motility that permits *P. bursaria* to move toward a light source is provided by _____.
A) Pseudopods.
B) A single flagellum featuring the 9 + 2 pattern.
C) Many cilia.
D) Contractile vacuoles.

33) A *P. bursaria* cell that has lost its zoochlorellae is said to be *aposymbiotic*. It might be able to replenish its contingent of zoochlorellae by ingesting them without subsequently digesting them. Which of the following situations would be most favorable to the reestablishment of resident zoochlorellae, assuming compatible *Chlorella* are present in *P. bursaria's* habitat?
A) Abundant light, no bacterial prey.
B) Abundant light, abundant bacterial prey.
C) No light, no bacterial prey.
D) No light, abundant bacterial prey.

34) A *P. bursaria* cell that has lost its zoochlorellae is *aposymbiotic*. If aposymbiotic cells have population growth rates the same as those of healthy, zoochlorella-containing *P. bursaria* in well-lit environments with plenty of prey items, then such an observation would be consistent with which type of relationship?
A) Parasitic.
B) Commensalistic.
C) Toxic.
D) Mutualistic.

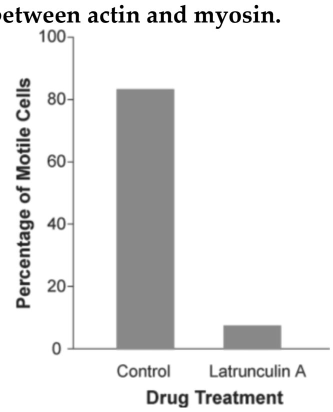
35) Theoretically, *P. bursaria* can obtain zoochlorella either vertically (via the asexual reproduction of its mother cell) or horizontally (by ingesting free-living *Chlorella* from its habitat). Consider a *P. bursaria* cell containing zoochlorellae but whose habitat lacks free-living *Chlorella*. If this cell subsequently undergoes many generations of asexual reproduction, if all of its daughter cells contain roughly the same number of zoochlorellae as it had originally contained, and if the zoochlorellae are all haploid and identical in appearance, then what is true? The zoochlorellae _____.
A) Also reproduced asexually, at an increasing rate over time.
B) Also reproduced asexually, at a decreasing rate over time.
C) Also reproduced asexually, at a fairly constant rate over time.
D) Reproduced sexually, undergoing heteromorphic alternation of generations.

36) Which process in Paramecium results in genetic recombination but no increase in population size?
A) Budding.
B) Meiotic division.
C) Conjugation.
D) Binary fission.

Use the following information to answer the question(s) below.

The mechanism of cell crawling in protist species is not well defined. Pseudopodia extension involves interactions between actin and myosin (the same molecules that are involved in vertebrate muscle contraction). However, prior to the study described below, no one had provided convincing data that actin and myosin were actually involved in cell crawling in protists. Anatomical studies had identified the cytoskeletal protein actin just below the surface of the cell membrane in several species of protist, but physiological studies had failed to show a functional link between actin, myosin, and cell crawling.

In a study by N. Poulsen et al. (Diatom gliding is the result of an actin-myosin motility system, *Cell Motility and the Cytoskeleton* 44 (1999):23-22), researchers tested whether motility in a particular species of diatom involves interactions between actin and myosin.



37) Refer to the study by Poulsen et al. and the figure above. Latrunculin A is a reversible toxin that disrupts the formation of actin fibers. A culture of a particular species of diatom was treated with this toxin diluted in a buffer, while another culture was treated only with the buffer (no toxin; control). The motility of cells in each culture was assessed by counting the number of cells that were moving during a defined period of time. Which of the following conclusions is reasonable based on the above figure?
A) Formation of actin fibers is not necessary for the movement in this species of diatom.
B) The buffer alone largely inhibited movement in this species of diatom.
C) In this species of diatom, fully formed actin fibers are necessary for movement.
D) None of them.

38) Refer to the study by Poulsen et al. and the figure above. The data graphed in the figure could be an artifact if latrunculin A kills this species of diatoms (that is, that may be why the cells are not moving). Which of the following would be the best evidence that latrunculin A is NOT killing the cells?
A) When the toxin was washed off the culture, the cells began to move again.
B) There were still a small percentage of motile cells in the culture treated with the toxin.
C) Most of the cells in the control were moving, indicating that they were alive.
D) When the toxin was applied to another species of diatom, 25% of them continued to move.

39) Refer to the study by Poulsen et al. and the figure above. Cultures of a species of diatom were treated with BDM, a reversible inhibitor of myosin function. Which of the following predictions is consistent with the hypothesis that an actin-myosin interaction is necessary for motility?
A) BDM will significantly decrease motility of the cells in culture.
B) BDM will not significantly alter motility of the cells in culture.
C) BDM will significantly increase motility of the cells in culture.
D) None of them.

Use the following information to answer the question(s) below.

***Paulinella chromatophora* is one of the few cercozoans that is autotrophic, carrying out aerobic photosynthesis with its two elongated "chromatophores." The chromatophores are contained within vesicles of the host cell, and each is derived from a cyanobacterium, though not the same type of cyanobacterium that gave rise to the chloroplasts of algae and plants.**

40) *P. chromatophora* secretes around itself a test, or case, of plates made of silica. Which of the following is another rhizarian that would be in competition with *P. chromatophora* for the silica needed to make these plates, assuming limited quantities of silica in the environment?
A) Radiolarians.
B) Foraminiferans.
C) Dinoflagellates.
D) Diatoms.

41) Including the membrane of the surrounding vesicle, how many phospholipid (NOT lipopolysaccharide) bilayers should be found around each *P. chromatophora's* chromatophore, and which one of these bilayers should have photosystems embedded in it?
A) Two; innermost.
B) Two; outermost.
C) Three; innermost.
D) Three; outermost.

42) If true, which of the following would be most important in determining whether *P. chromatophora's* chromatophore is still an endosymbiont, or is an organelle, as the term *chromatophore* implies?
A) If *P. chromatophora* is less fit without its chromatophore than with it.
B) If the chromatophore is less fit without the host cercozoan than with it.
C) If there is ongoing metabolic cooperation between the chromatophore and the host cercozoan.
D) If there has been movement of genes from the chromatophore genome to the nuclear genome, such that these genes are no longer present in the chromatophore genome.

43) The genome of modern chloroplasts is roughly 50% the size of the genome of the cyanobacterium from which it is thought to have been derived. In comparison, the genome of P. chromatophora's chromatophore is only slightly reduced relative to the size of the genome of the cyanobacterium from which it is thought to have been derived. What is a valid hypothesis that can be drawn from this comparison?
A) Lytic phage infections have targeted the chloroplast genome more often than the *P. chromatophora* genome.
B) *P. chromatophora's* chromatophore is the result of an evolutionarily recent endosymbiosis.
C) The genome of the chloroplast ancestor contained many more introns that could be lost without harm, compared to the chromatophore's genome.
D) The genome of the cyanobacteria was smaller than the genome of *P. chromatophora*.

44) A biologist discovers an alga that is marine, multicellular, and lives at a depth reached only by blue light. This alga is most likely a type of _____.
A) Red algae.
B) Brown algae.
C) Green algae.
D) Golden algae.

45) Green algae differ from land plants in that many green algae _____.
A) Are unicellular.
B) Have plastids.
C) Have alternation of generations.
D) Have cell walls containing cellulose.

46) You are given the task of designing an aerobic, mixotrophic protist that can perform photosynthesis in fairly deep water (for example, 250 meters deep) and can also crawl about and engulf small particles. With which two of the following structures would you provide your protist?
1. Hydrogenosome.
2. Apicoplast.
3. Pseudopods.
4. Chloroplast from red alga.
5. Chloroplast from green alga.
A) 1 and 2.
B) 2 and 3.
C) 3 and 4.
D) 4 and 5.

47) Similar to most amoebozoans, the forams and the radiolarians also have pseudopods, as do some of the white blood cells of animals (monocytes). If one were to erect a taxon that included all organisms that have cells with pseudopods, the taxon would _____.
A) Be polyphyletic.
B) Be paraphyletic.
C) Be monophyletic.
D) Include all eukaryotes.

48) Which of the following groups is matched with a correct anatomical feature?
A) Foraminifera → silicon-rich tests.
B) Dinoflagellata → holdfast.
C) Diatoms → tests made of cellulose.
D) Phaeophyta (brown algae) → blade.

49) Unikonta is a supergroup that includes all of the following except _____.
A) Fungi.
B) Protists.
C) Animals.
D) Plants.

50) Previously understood similarities that seemed to connect slime molds and fungi are now considered to be _____.
A) Homologies.
B) Examples of convergent evolution.
C) Variations of common ancestral traits.
D) Adaptations for much different functions.

51) Branching points at the root of the eukaryotic phylogenetic tree
A) Reveal that unikonts are derived from the SAR clade.
B) Suggest that Archaeplastids were the first eukaryotes.
C) Strongly suggest that fungi are more closely related to plants than animals.
D) Are presently unclear.

52) Super cells characteristic of plasmodial slime molds result when which one of the following common cellular processes does not occur?
A) Mitosis.
B) Cytokinesis.
C) Aerobic metabolism.
D) Endocytosis.

53) Which of the following is responsible for nearly 100,000 human deaths worldwide every year?
A) *Entamoeba histolytica.*
B) *Amoeba proteus.*
C) *Plasmodial slime molds.*
D) *Dictyostelium discoideum.*

54) Assume that some members of an aquatic species of motile, photosynthetic protists evolve to become parasitic to fish. They gain the ability to live in the fish gut, absorbing nutrients as the fish digests food. Over time, which of the following phenotypic changes would you expect to observe in this population of protists?
A) Loss of motility.
B) Loss of chloroplasts.
C) Gain of a rigid cell wall.
D) Gain of meiosis.

Use the following information to answer the question(s) below.

Healthy individuals of *Paramecium bursaria* contain photosynthetic algal endosymbionts of the genus *Chlorella*. When within their hosts, the algae are referred to as zoochlorellae. In aquaria with light coming from only one side, *P. bursaria* gather at the well-lit side, whereas other species of *Paramecium* gather at the opposite side. The zoochlorellae provide their hosts with glucose and oxygen, and *P. bursaria* provides its zoochlorellae with protection and motility. *P. bursaria* can lose its zoochlorellae in two ways: (1) if kept in darkness, the algae will die; and (2) if prey items (mostly bacteria) are absent from its habitat, *P. bursaria* will digest its zoochlorellae.

55) Which term most accurately describes the nutritional mode of healthy *P. bursaria?*
A) Photoautotroph.
B) Photoheterotroph.
C) Chemoautotroph.
D) Mixotroph.

56) Living diatoms contain brownish plastids. If global warming causes bloom of diatoms in the surface waters of Earth's oceans, how might this be harmful to the animals that build coral reefs?
A) The coral animals, which capture planktonic organisms, may be outcompeted by the diatoms.
B) The coral animals' endosymbiotic dinoflagellates may get "shaded out" by the diatoms.
C) The coral animals may die from overeating the plentiful diatoms with their cases of silica.
D) The diatoms' photosynthetic output may over-oxygenate the water.

57) Which of the following is a producer?
A) Kinetoplastid.
B) Apicomplexan.
C) Diatom.
D) Ciliates.

58) If we were to apply the most recent technique used to fight potato late blight to the fight against the malarial infection of humans, then we would _____.
A) Increase the dosage of the least-expensive antimalarial drug administered to humans.
B) Introduce a predator of the malarial parasite into infected humans.
C) Use a "cocktail" of at least three different pesticides against *Anopheles* mosquitoes.
D) Insert genes from a Plasmodium-resistant strain of mosquito into *Anopheles* Mosquitoes.

CHAPTER 29:

PLANT DIVERSITY I: HOW PLANTS COLONIZED LAND

1) According to the fossil record, plants colonized terrestrial habitats _____.
A) In conjunction with insects that pollinated them.
B) In conjunction with fungi that helped provide them with nutrients from the soil.
C) To escape abundant herbivores in the oceans.
D) Only about 150 million years ago.

2) The most direct ancestors of land plants were probably _____.
A) Kelp (brown alga) that formed large beds near the shorelines.
B) Green algae.
C) Photosynthesizing prokaryotes (cyanobacteria).
D) Liverworts and mosses.

3) About 450 million years ago, the terrestrial landscape on Earth would have _____.
A) Looked very similar to that of today, with flowers, grasses, shrubs, and trees
B) Been completely bare rock, with little pools that contained bacteria and cyanobacteria
C) Been covered with tall forests in swamps that became today's coal
D) Had non-vascular green plants similar to liverworts forming green mats on rock.

4) What evidence do paleobotanists look for that indicates the movement of plants from water to land?
A) Waxy cuticle to decrease evaporation from leaves.
B) Loss of structures that produce spores.
C) Sporopollenin to inhibit evaporation from leaves.
D) Remnants of chloroplasts from photosynthesizing cells.

5) Which of these time intervals, based on plant fossils, came last (most recently)?
A) Extensive growth of gymnosperm forests.
B) Colonization of land by early liverworts and mosses.
C) Rise and diversification of angiosperms.
D) Carboniferous swamps with giant horsetails and lycophytes.

6) Why have biologists hypothesized that the first land plants had a low, sprawling growth habit?
A) They were tied to the water for reproduction and thus needed to remain in close contact with the moist soil.
B) The ancestors of land plants, green algae, lack the structural support to stand erect in air.
C) Land animals of that period were small and could not pollinate tall plants.
D) There was less competition for space so they simply spread out flat.

7) Spores and seeds have basically the same function✢dispersal✢but are vastly different because _____.
A) Spores have a protective outer covering; seeds do not.
B) Spores have an embryo; seeds do not.
C) Spores have stored nutrition; seeds do not.
D) Spores are unicellular; seeds are not.

8) You find a green organism in a pond near your house and believe it is a plant, not an alga. The mystery organism is most likely a plant and not an alga if it _____.
A) Contains chloroplasts.
B) Is surrounded by a cuticle.
C) Does not contain vascular tissue.
D) Has cell walls that are comprised largely of cellulose.

9) Retaining the zygote on the living gametophyte of land plants _____.
A) Protects the zygote from herbivores.
B) Evolved concurrently with pollen.
C) Helps in dispersal of the zygote.
D) Allows it to be nourished by the parent plant.

10) The structural integrity of bacteria is to peptidoglycan as the structural integrity of plant spores is to _____.
A) Lignin.
B) Cellulose.
C) Secondary compounds.
D) Sporopollenin.

11) According to our current knowledge of plant evolution, which group of organisms should feature cell division most similar to that of land plants?
A) Unicellular green algae.
B) Cyanobacteria.
C) Charophytes.
D) Red algae.

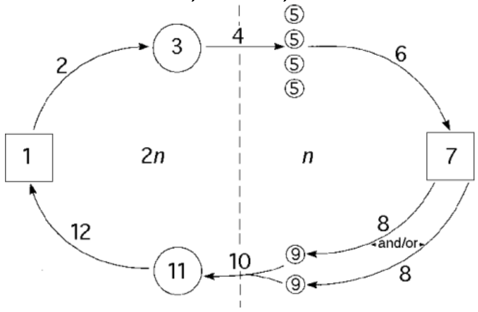
12) Which taxon is essentially equivalent to the "embryophytes"?
A) Plantae.
B) Pterophyta.
C) Bryophyta.
D) Charophycea.

13) If the kingdom Plantae is someday expanded to include the charophytes (stoneworts), then the shared derived characteristics of the kingdom will include _____.
1. Rings of cellulose-synthesizing complexes.
2. Chlorophylls *a* and *b*.
3. Alternation of generations.
4. Cell walls of cellulose.
5. Ability to synthesize sporopollenin.
A) 1 and 5.
B) 1, 2, and 3.
C) 1, 3, and 5.
D) 1, 2, 4, and 5.

14) Which of the following were probably factors that permitted early plants to successfully colonize land?
1. The relative number of potential predators (herbivores).
2. The relative number of competitors.
3. The relative availability of symbiotic partners.
4. Air's relative lack of support, compared to water's support.
A) 1 and 2.
B) 2 and 3.
C) 3 and 4.
D) 1, 2, and 3.

15) A student encounters a pondweed which appears to be a charophyte (stonewort). Which of the following features would help the student determine whether the sample comes from a charophyte or from some other type of green alga?
1. Molecular structure of enzymes inside the chloroplasts.
2. Structure of sperm cells.
3. Presence of phragmoplasts.
4. Rings of cellulose-synthesizing complexes.
A) 1 and 3.
B) 1 and 4.
C) 2 and 3.
D) 2, 3, and 4.

The following questions (16-22) refer to the generalized life cycle for land plants shown in the figure below. Each number within a circle or square represents a specific plant or plant part, and each number over an arrow represents either meiosis, mitosis, or fertilization.



16) In the figure above, which number represents the mature gametophyte?
A) 1.
B) 3.
C) 7.
D) 11.

17) In the figure above, which number represents an embryo?
A) 1.
B) 3.
C) 7.
D) 11.

18) In the figure above, meiosis is most likely to be represented by which number(s)?
A) 2.
B) 4.
C) 2 and 8.
D) 10 and 12.

19) Which number represents a megaspore mother cell in the figure above?
A) 1.
B) 3.
C) 5.
D) 7.

20) In the figure above, the process labeled "6" involves __.
A) Mitosis.
B) Meiosis.
C) Fertilization.
D) Binary fission.

21) The embryo sac of an angiosperm flower is best represented by which number in the figure above?
A) 1.
B) 3.
C) 7.
D) 11.

22) In angiosperms, which number in the figure above represents a component contributing to the formation of endosperm?
A) 1.
B) 7.
C) 9.
D) 11.

23) Stomata _____.
A) Occur in all land plants and define them as a monophyletic group.
B) Open to allow gas exchange and close to decrease water loss.
C) Occur in all land plants and are the same as pores.
D) Open to increase both water absorption and gas exchange.

24) Liverworts, hornworts, and mosses are grouped together as the Bryophytes. Besides not having vascular tissue, what do they all have in common?
A) They are all wind pollinated.
B) They are heterosporous.
C) They can reproduce asexually by producing gemmae.
D) They require water for reproduction.

25) Most moss gametophytes do not have a cuticle and are 1 -2 cells thick. What does this imply about moss gametophytes and their structure?
A) They use stomata for gas exchange regulation.
B) They can easily lose water to, and absorb water from, the atmosphere.
C) Photosynthesis occurs throughout the entire gametophyte surface.
D) They have branching veins in their leaves.

26) As you stroll through a moist forest, you are most likely to see a _____.
A) Zygote of a green alga.
B) Gametophyte of a moss.
C) Sporophyte of a liverwort.
D) Gametophyte of a fern.

27) Which of these are spore-producing structures?
A) Sporophyte (capsule) of a moss.
B) Antheridium of a moss or fern.
C) Archegonium of a moss or fern.
D) Gametophyte of a moss.

28) What is true about the genus Sphagnum?
A) It is an economically important liverwort.
B) It grows in extensive mats in grassland areas.
C) It accumulates to form coal and is burned as a fuel.
D) It is an important carbon sink, reducing atmospheric carbon dioxide.

29) How are the bryophytes and seedless vascular plants alike?
A) Plants in both groups have vascular tissue.
B) In both groups, sperm swim from antheridia to archegonia.
C) The dominant generation in both groups is the sporophyte.
D) Plants in both groups have true roots, stems, and leaves.

30) In general, liverworts have a cuticle and pores. However, some species do not have pores. What would you predict concerning the cuticle of these species and why?
A) The cuticle would be the same as in those species with pores.
B) The cuticle would be thicker than in those species with pores.
C) The cuticle would be thinner than in those species with pores.
D) None of them.

31) Archegonia _____.
A) Are the sites where male gametes are produced.
B) May contain sporophyte embryos.
C) Have the same function as sporangia.
D) Make asexual reproductive structures.

32) Which of the following is a true statement about plant reproduction?
A) Embryophytes are small plants in an early developmental stage.
B) Male and female bryophytes each produce a type of gametangia.
C) Eggs and sperm of most land plants swim toward one another.
D) Bryophytes are limited to asexual reproduction.

33) Assuming that they all belong to the same plant, arrange the following structures from largest to smallest.
1. Antheridia.
2. Gametes.
3. Gametophytes.
A) 2, 3, 1.
B) 2, 1, 3.
C) 3, 2, 1.
D) 3, 1, 2.

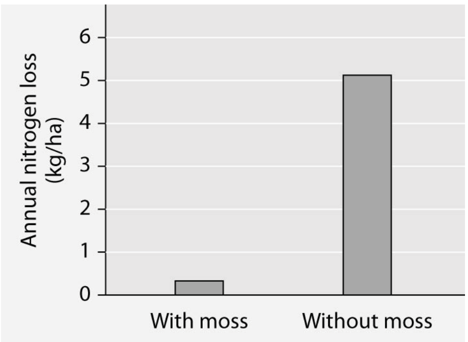
34) The leaflike appendages of moss gametophytes may be one to two cell layers thick. Consequently, which of the following is LEAST likely to be found associated with such appendages?
A) Cuticle.
B) Phenolics.
C) Stomata.
D) Peroxisomes.
D) None of them.

35) Which of the following is true of the life cycle of mosses?
A) The haploid generation grows on the sporophyte generation.
B) Spores are primarily distributed by water currents.
C) Antheridia and archegonia are produced by gametophytes.
D) The sporophyte generation is dominant.

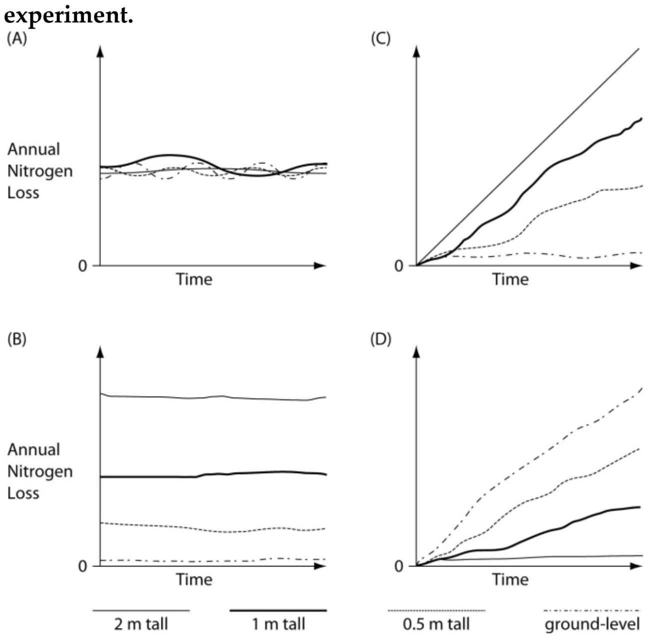
36) At some time during their life cycles, bryophytes make _____.
A) Microphylls.
B) True roots.
C) True leaves.
D) Sporangia.

37) Two small, poorly drained lakes lie close to each other in a northern forest. The basins of both lakes are composed of the same geologic substratum. One lake is surrounded by a dense *Sphagnum* mat; the other is not. Compared to the pond with *Sphagnum*, the pond lacking the moss mat should have _____.
A) Lower numbers of bacteria.
B) Reduced rates of decomposition.
C) Reduced oxygen content.
D) Less-acidic water.

Researchers tested nitrogen loss from soil where the moss *Dawsonia* was growing, and compared it to soil from which *Dawsonia* had been removed. The data are presented below.



Researchers decided to test the hypothesis that if the 1-meter-tall *Dawsonia* gametophyte-sporophyte plants had acted as a physical buffer, then they would have reduced water's ability to erode the soil and carry away its nitrogen. They began with four equal-sized areas where *Dawsonia* mosses grew to a height of 1 m above the soil surface. One of the four areas was not modified. In the second area, the mosses were trimmed to a height of 0.5 m above the soil surface. In the third area, the mosses were trimmed to a height of 0.25 m above the soil surface. In the fourth area, the mosses were trimmed all the way to the ground, leaving only the rhizoids. Water, simulating rainfall, was then added in a controlled fashion to all plots over the course of one year. The figure below presents four graphs that depict potential results of this experiment.



38) In the figure above, which graph of soil nitrogen loss over time most strongly supports the hypothesis that if the 1-m tall *Dawsonia* gametophyte-sporophyte plants had acted as a physical buffer, then they would have reduced water's ability to erode the soil and carry away its nitrogen?
A) A.
B) B.
C) C.
D) D.

39) Soil nitrogen is not solely contained within the rhizoids of the *Dawsonia* mosses because rhizoids _____.
A) Are associated with fungi that inhibit mineral transfer from soil to rhizoids.
B) Are not absorptive structures.
C) Lack direct attachment to the moss sporophytes.
D) Immediately transfer the nitrogen to the sporophyte.

40) The 1-m height attainable by *Dawsonia* moss is at the upper end of the size range reached by mosses. What accounts for the relative tallness of *Dawsonia*?
A) The cuticle that is found along the ridges of "leaves".
B) "Leaves" that are more than one cell layer thick.
C) Reduced size, mass, and persistence of the sporophytes, which allows gametophores to grow taller.
D) The presence of conducting tissues in the "stem".

41) Encouraging the growth (via nutrient fertilization) of photosynthetic protists in marine environments may help reduce global warming because _____.
A) Photosynthetic protists are primary consumers in many marine food chains.
B) Photosynthetic protists fix atmospheric carbon dioxide, decreasing atmospheric carbon dioxide levels.
C) The increased oxygen consumption by large populations of photosynthetic protists will increase photosynthesis in land plants.
D) Photosynthetic protists would release a lot of oxygen, and fertilizing them would increase levels of oxygen in the atmosphere.

42) Which set contains the most closely related terms?
A) Megasporangium, megaspore, pollen, ovule.
B) Microsporangium, microspore, egg, ovary.
C) Megasporangium, megaspore, egg, ovule.
D) Microsporangium, microspore, carpel, ovary.

43) How could you determine if a plant is heterosporous?
A) Male and female reproductive structures are located on separate plants.
B) It has vascular tissue.
C) It has multiple sporangia.
D) Its diploid sporophyte produces spores via meiosis.

44) A botanist discovers a new species of plant in a tropical rain forest. After observing its anatomy and life cycle, he notes the following characteristics: flagellated sperm, xylem with tracheids, separate gametophyte and sporophyte generations with the sporophyte dominant, and no seeds. This plant is probably most closely related to _____.
A) Mosses.
B) Ferns.
C) Gymnosperms.
D) Flowering plants.

45) You are hiking in a forest and come upon a mysterious plant, which you determine is either a lycophyte sporophyte or a Pterophyta sporophyte. Which of the following would be most helpful in determining the correct classification of the plant?
A) Whether or not it has true leaves.
B) Whether it has microphylls or megaphylls.
C) Whether or not it has seeds.
D) Its height.

46) Assuming that they all belong to the same plant, arrange the following structures from largest to smallest (or from most inclusive to least inclusive).
1. Spores.
2. Sporophylls.
3. Sporophytes.
4. Sporangia.
A) 2, 4, 3, 1.
B) 2, 3, 4, 1.
C) 3, 1, 4, 2.
D) 3, 2, 4, 1.

47) If humans had been present to build log structures during the Carboniferous period (they were not), which plant types would have been suitable sources of logs?
A) Horsetails and bryophytes.
B) Lycophytes and bryophytes.
C) Ferns, horsetails, and lycophytes.
D) Charophytes (stoneworts), bryophytes, and gymnosperms.

48) Arrange the following terms from most inclusive to least inclusive.
1. Embryophytes.
2. Green plants.
3. Seedless vascular plants.
4. Ferns.
5. Tracheophytes.
A) 1, 2, 5, 3, 4
B) 2, 1, 5, 3, 4
C) 2, 5, 1, 3, 4
D) 1, 4, 2, 5, 3

Use the following description to answer the questions (49-50) below.

A biology student hiking in a forest happens upon an erect, 15 -cm-tall plant that bears microphylls and a strobilus at its tallest point. When disturbed, the cone emits a dense cloud of brownish dust. A pocket magnifying glass reveals the dust to be composed of tiny spheres with a high oil content.

49) This student has probably found a _____.
A) Bryophyte sporophyte.
B) Fern sporophyte.
C) Horsetail gametophyte.
D) Lycophyte sporophyte.

50) Besides oil, what other chemical should be detected in substantial amounts upon chemical analysis of these small spheres?
A) Sporopollenins.
B) Phenolics.
C) Waxes.
D) Terpenes.

Use the following information to answer the questions (51-57) below.

Big Bend National Park in Texas is mostly Chihuahuan desert, where rainfall averages about 10 inches per year. Yet, it is not uncommon when hiking in this bone-dry desert to encounter mosses and ferns. One such plant is called "flower of stone." It is not a flowering plant, nor does it produce seeds. Under arid conditions, its leaflike structures curl up. However, when it rains, it unfurls its leaves, which form a bright green rosette on the desert floor. Consequently, it is sometimes called the "resurrection plant." At first glance; it could be a fern, a true moss, or a spike moss.

51) What feature of both true mosses and ferns makes it most surprising that they can survive for many generations in dry deserts?
A) Flagellated sperm.
B) Lack of vascular tissues.
C) Lack of true roots.
D) Lack of cuticle.

52) Which of the following features is most important for true mosses and ferns to reproduce in the desert?
A) That the sporophytes occupy only permanently shady, north-facing habitats.
B) That the sporophytes hug the ground, growing no taller than a couple of inches.
C) Either that their gametophytes grow close together, or that they are hermaphroditic.
D) That the sporophytes have highly lignified vascular tissues.

53) Which of the following characteristics is (are) possessed in common by true mosses, ferns, and spike mosses, and therefore becomes useless at helping to determine to which of these groups "flower of stone" belongs?
1. A sporophyte generation that is dominant.
2. True leaves and roots.
3. Flagellated sperm.
4. Strobili.
5. Alternation of generations.
A) 5 only
B) 1 and 5
C) 2 and 3
D) 3 and 5.

54) Upon closer inspection of the leaves of "flower of stone", one can observe tiny, cone-like structures. Each cone-like structure emits spores of two different sizes. Further investigation also reveals that the roots of "flower of stone" branch only at the growing tip of the root, forming a Y-shaped structure. Based on these additional observations, which of the following can be properly inferred about "flower of stone"?
1. It is heterosporous.
2. It is a fern.
3. The cone-like structures are sori.
4. It is a lycophyte.
5. It has separate male and female gametophytes.
A) 1 and 5.
B) 2 and 3.
C) 1, 2, and 3.
D) 1, 4, and 5.

55) Upon closer inspection of the leaves of "flower of stone," one can observe tiny, cone-like structures. Each cone-like structure emits spores of two different sizes. Further investigation also reveals that the roots of "flower of stone" branch only at the growing tip of the root, forming a Y-shaped structure. Consequently, which of the following is the closest living relative of "flower of stone"?
A) True moss.
B) Club moss.
C) Liverwort.
D) Fern.

56) Upon closer inspection of the leaves of "flower of stone," one can observe tiny, cone-like structures. Each cone-like structure emits spores of two different sizes. Further investigation also reveals that the roots of "flower of stone" branch only at the growing tip of the root, forming a Y-shaped structure. Consequently, "flower of stone" should be expected _____.
1. A gametophyte generation that is dominant.
2. Lignified vascular tissues.
3. Microphylls.
4. Filamentous rhizoids, but not true roots.
5. Spores those are diploid when mature.
A) 1 and 2.
B) 1 and 5.
C) 2 and 3.
D) 3, 4, and 5.

57) In which combination of locations would one who is searching for the gametophytes of "flower of stone" have the best chance of finding them?
1. Moist soil.
2. Underground, nourished there by symbiotic fungi.
3. South- or west-facing slopes.
4. Permanently shady places.
5. Far from any flower of stone sporophytes.
A) 1 only.
B) 1 and 2.
C) 1, 2, and 4.
D) 1, 2, and 5.

58) Suppose an efficient conducting system evolved in a moss that could transport water and other materials as high as a tall tree. Which of the following statements about "trees" of such a species would be FALSE?
A) Fertilization would probably be more difficult.
B) Spore dispersal distances would probably increase.
C) Females could produce only one Archegonium.
D) Individuals would probably compete more effectively for access to light.

59) Which of the following features of how seedless land plants get sperm to egg are the same as for some of their algal ancestors?
A) Conjugation tubes are formed between sperm and egg cells.
B) Packets of sperm are delivered by wind to the eggs.
C) Aquatic invertebrates carry sperm to eggs.
D) Flagellated sperm swim to the eggs in a water drop.

60) Increasing the number of stomata per unit surface area of a leaf when atmospheric carbon dioxide levels decline is most analogous to a human _____.
A) Breathing faster as atmospheric carbon dioxide levels increase.
B) Putting more red blood cells into circulation when atmospheric oxygen levels decline.
C) Breathing more slowly as atmospheric oxygen levels increase.
D) Increasing the volume of its lungs when atmospheric carbon dioxide levels increase.

CHAPTER 30:

PLANT DIVERSITY II: THE EVOLUTION OF SEED PLANTS

1) Which of these is a major trend in land plant evolution?
A) The trend toward smaller size.
B) The trend toward a gametophyte-dominated life cycle.
C) The trend toward a sporophyte-dominated life cycle.
D) The trend toward larger gametophytes.

2) Which of the following lines of evidence would best support your assertion that a particular plant is an angiosperm?
A) It produces seeds.
B) It retains its fertilized egg within its archaegonium.
C) It lacks gametangia.
D) It undergoes alternation of generations.

3) Which of the following is NOT functionally important in cells of the gametophytes of both angiosperms and gymnosperms?
A) Haploid nuclei.
B) Mitochondria.
C) Cell walls.
D) Chloroplasts.

4) In addition to seeds, which of the following characteristics is unique to the seed-producing plants?
A) Sporopollenin.
B) Lignin present in cell walls.
C) Pollen.
D) Megaphylls.

5) Suppose that the cells of seed plants, like the skin cells of humans, produce a pigment upon increased exposure to ultraviolet radiation. Rank the following cells, from greatest to least, in terms of the likelihood of producing this pigment.
1. Cells of sporangium.
2. Cells in the interior of a subterranean root.
3. Epidermal cells of sporophyte megaphylls.
4. Cells of a gametophyte.
A) 3, 4, 1, 2.
B) 3, 4, 2, 1.
C) 3, 1, 4, 2.
D) 3, 2, 1, 4.

6) Arrange the following in the correct sequence, from earliest to most recent, in which these plant traits originated.
1. Sporophyte dominance, gametophyte independence.
2. Sporophyte dominance, gametophyte dependence.
3. Gametophyte dominance, sporophyte dependence.
A) 1 → 2 → 3.
B) 2 → 1 → 3.
C) 3 → 2 → 1.
D) 3 → 1 → 2.

7) In seed plants, which of the following is part of a pollen grain and has a function most like that of the seed coat?
A) Sporophyll.
B) Sporopollenin.
C) Stigma.
D) Sporangium.

8) In terms of alternation of generations, the internal parts of the pollen grains of seed-producing plants are most similar to a _____.
A) Moss sporophyte.
B) Moss gametophyte bearing both male and female gametangia.
C) Fern sporophyte.
D) Fern gametophyte bearing only antheridia.

9) A researcher has developed two stains for use with seed plants. One stains sporophyte tissue blue; the other stains gametophyte tissue red. If the researcher exposes pollen grains to both stains, and then rinses away the excess stain, what should occur?
A) The pollen grains will be pure red.
B) The pollen grains will be pure blue.
C) The pollen grains will have red interiors and blue exteriors.
D) The pollen grains will have blue interiors and red exteriors.

10) Which of the following sex and generation combinations directly produces the pollen tube of angiosperms?
A) Male gametophyte.
B) Female gametophyte.
C) Male sporophyte.
D) Female sporophyte.

11) The closest relatives of the familiar pine and spruce trees are _____.
A) Ferns, horsetails, lycophytes, and club mosses.
B) Hornworts, liverworts, and mosses.
C) Gnetophytes, cycads, and ginkgos.
D) Elms, maples, and aspens.

12) Conifers and pines both have needlelike leaves, with the adaptive advantage of _____.
A) Increased surface area, increasing photosynthesis.
B) Increased surface area, increasing gas exchange.
C) Decreased surface area, reducing gas exchange.
D) Decreased surface area, reducing water loss.

13) Which of the following statements correctly describes a portion of the pine life cycle?
A) Female gametophytes use mitosis to produce eggs.
B) Seeds are produced in pollen-producing cones.
C) Pollen grains contains female gametophytes.
D) A pollen tube slowly digests its way through the triploid endosperm.

14) Which of the following statements is true of the pine life cycle?
A) The pine tree is a gametophyte.
B) Male and female gametophytes are in close proximity during gamete synthesis.
C) Conifer pollen grains contain male gametophytes.
D) Double fertilization is a relatively common phenomenon.

15) Within a gymnosperm megasporangium, what is the correct sequence in which the following should appear during development, assuming that fertilization occurs?
1. Sporophyte embryo.
2. Female gametophyte.
3. Egg cell.
4. Megaspore.
A) 4 → 2 → 3 → 1.
B) 4 → 1 → 2 → 3.
C) 1 → 4 → 3 → 2.
D) 1 → 4 → 2 → 3.

16) Arrange the following structures, which can be found on male pine trees, from the largest structure to the smallest structure (or from most inclusive to least inclusive).
1. Sporophyte.
2. Microspores.
3. Microsporangia.
4. Pollen cone.
5. Pollen nuclei.
A) 1, 4, 3, 2, 5.
B) 1, 2, 3, 5, 4.
C) 4, 1, 2, 3, 5.
D) 4, 3, 2, 5, 1.

17) Which of the following sex and generation combinations most directly produces the integument of a pine seed?
A) Male gametophyte.
B) Female gametophyte.
C) Male sporophyte.
D) Female sporophyte.

18) Which of the following sex and generation combinations directly produces the megasporangium of pine ovules?
A) Male gametophyte.
B) Female gametophyte.
C) Male sporophyte.
D) Female sporophyte.

Use the following description to answer the question (19) below.

The cycads, a mostly tropical phylum of gymnosperms, evolved about 300 million years ago and were dominant forms during the Age of the Dinosaurs. Though their sperm are flagellated, their ovules are pollinated by beetles. These beetles get nutrition (they eat pollen) and shelter from the microsporophylls. Upon visiting megasporophylls, the beetles transfer pollen to the exposed ovules. In cycads, pollen cones and seed cones are borne on different plants. Cycads synthesize neurotoxins, especially in the seeds, that are effective against most animals, including humans.

19) Which feature of cycads distinguishes them from most other gymnosperms?
1. They have exposed ovules.
2. They have flagellated sperm.
3. They are pollinated by animals.
A) 1 only.
B) 2 only.
C) 3 only.
D) 2 and 3.

20) On the Pacific island of Guam, large herbivorous bats called "flying foxes" commonly feed on cycad seeds, a potent source of neurotoxins. The flying foxes do not visit male cones. Consequently, what should be true?
A) The flying foxes are attracted to cycad fruit and eat the enclosed seeds only by accident.
B) Flying foxes are highly susceptible to the effects of the neurotoxins.
C) The flying foxes assist the beetles as important pollinating agents of the cycads.
D) Flying foxes can be dispersal agents of cycad seeds if the seeds sometimes get swallowed whole (in other words, without getting chewed).

Use the following information to answer the questions (21-24) below.

The Brazil nut tree, *Bertholletia excels* (*n* = 17), is native to tropical rain forests of South America. It is a hardwood tree that can grow to over 50 meters tall, is a source of high -quality lumber, and is a favorite nesting site for harpy eagles. As the rainy season ends, tough -walled fruits, each containing 8-25 seeds (Brazil nuts), fall to the forest floor. Brazil nuts are composed primarily of endosperm. About $50 million worth of nuts are harvested each year. Scientists have discovered that the pale yellow flowers of Brazil nut trees cannot fertilize themselves and admit only female orchid bees as pollinators. The agouti *(Dasyprocta spp.)*, a cat-sized rodent, is the only animal with teeth strong enough to crack the hard wall of Brazil nut fruits. It typically eats some of the seeds, buries others, and leaves still others inside the fruit, which moisture can now enter. The uneaten seeds may subsequently germinate.

21) Orchid bees are to Brazil nut trees as _____ are to pine trees.
A) Breezes.
B) Rain droplets.
C) Seed-eating birds.
D) Squirrels.

22) Many mammals have skins and mucous membranes that are sensitive to phenolic secretions of plants like poison oak *(Rhus)*. These secondary compounds are primarily adaptations that _____.
A) Favor pollination.
B) Foster seed dispersal.
C) Decrease competition.
D) Inhibit herbivory.

23) Entrepreneurs attempted, but failed, to harvest nuts from plantations grown in Southeast Asia. Attempts to grow Brazil nut trees in South American plantations also failed. In both cases, the trees grew vigorously, produced healthy flowers in profusion, but set no fruit. Consequently, what is the likely source of the problem?
A) Poor sporophyte viability
B) Failure to produce fertile ovules
C) Failure to produce pollen
D) Pollination failure.

24) The agouti is most directly involved with the Brazil nut tree's dispersal of .
A) Female gametophytes.
B) Sporophyte embryos.
C) Sporophyte megaspores.
D) Female gametes.

Use the following information to answer the questions (25-27) below.

Scarlet gilia *(Ipomopsis aggregata)* usually has red flowers in an inflorescence of up to 250 flowers. In certain populations in the Arizona Mountains, however, the flowers range from red to pink to white. In early summer, most of the flowers were red. Six to eight weeks later, the same individual plants were still present; the flowers ranged from pink to white, and few red flowers were present. The major pollinators early in the season were two species of hummingbirds active during the day; they emigrated to lower elevations, and the major pollinator later in the season was a hawk moth (a type of moth). The hawk moth was most active at sunset and later, and it preferred light pink to white flowers after dark. When hummingbirds were present, more red flowers than white flowers produced fruit. When only hawk moths were present, more white flowers produced fruit (K. N. Paige and T. G. Whitham. 1985. Individual and population shifts in flower color by scarlet gilia: A mechanism for pollinator tracking. Science 227:315-17).

25) Refer to the paragraph on scarlet gilia. What is the significance of measuring fruit production?
A) It is a measure of pollination success.
B) It is a measure of seed dispersal success.
C) It is easier than counting flowers.
D) It is an indication of predation on the seeds of the plants.

26) Refer to the paragraph on scarlet gilia. Late in the season, when only hawk moths were present, researchers painted the red flowers white. What would you expect?
A) Unpainted red flowers would produce more fruits than white flowers would.
B) Red flowers painted white would produce more fruits than red flowers would.
C) Red and white flowers would produce the same numbers of fruits.
D) None of them.

27) Refer to the paragraph on scarlet gilia. Some plants changed their flowers to lighter colors, and some retained the same darker color all season. Which plants do you expect produced more fruit?
A) Those that changed their color to a lighter shade.
B) Those that stayed darker.
C) The same numbers of fruit on both.
D) None of them.

28) Immature seed cones of conifers are usually green before pollination, and flowers of grasses are inconspicuously colored. What does this indicate about their pollination?
A) They self-fertilize and do not need pollen carried from one plant to another.
B) Their pollinating insects are color blind.
C) They are wind pollinated.
D) They probably attract pollinators using strong fragrances.

Use the following information to answer the questions (29-34) below.

In onions (Allium), cells of the sporophyte have 16 chromosomes within each nucleus. Match the number of chromosomes present in each of the following onion tissues.

29) How many chromosomes should be in a tube cell nucleus?
A) 8.
B) 16.
C) 24.
D) 32.

30) How many chromosomes should be in an endosperm nucleus?
A) 8.
B) 16.
C) 24.
D) 32.

31) How many chromosomes should be in a generative cell nucleus?
A) 8.
B) 16.
C) 24.
D) 32.

32) How many chromosomes should be in an embryo sac nucleus?
A) 8.
B) 16.
C) 24.
D) 32.

33) How many chromosomes should be in an embryo nucleus?
A) 8.
B) 16.
C) 24.
D) 32.

34) How many chromosomes should be in a megasporangium nucleus?
A) 8.
B) 16.
C) 24.
D) 32.

Use the following information to answer the question (15) below.

The Brazil nut tree, *Bertholletia excels (n = 17)*, is native to tropical rain forests of South America. It is a hardwood tree that can grow to over 50 meters tall, is a source of high -quality lumber, and is a favorite nesting site for harpy eagles. As the rainy season ends, tough -walled fruits, each containing 8-25 seeds (Brazil nuts), fall to the forest floor. Brazil nuts are composed primarily of endosperm. About $50 million worth of nuts are harvested each year. Scientists have discovered that the pale yellow flowers of Brazil nut trees cannot fertilize themselves and admit only female orchid bees as pollinators. The agouti *(Dasyprocta spp.)*, a cat-sized rodent, is the only animal with teeth strong enough to crack the hard wall of Brazil nut fruits. It typically eats some of the seeds, buries others, and leaves still others inside the fruit, which moisture can now enter. The uneaten seeds may subsequently germinate.

35) Animals that consume Brazil nuts derive nutrition mostly from tissue whose nuclei have how many chromosomes?
A) 17.
B) 34.
C) 51.
D) 68.

36) A botanist discovers a new species of land plant with a dominant sporophyte, chlorophylls a and b, and cell walls made of cellulose. In assigning this plant to a phylum, which of the following, if present, would be LEAST useful?
A) Endosperm.
B) Seeds.
C) Flowers.
D) Spores.

37) Which of the following sex and generation combinations directly produces the fruit of angiosperms?
A) Male gametophyte.
B) Female gametophyte.
C) Male sporophyte.
D) Female sporophyte.

38) Which of the following is a characteristic of all angiosperms?
A) Double internal fertilization.
B) Free-living gametophytes.
C) Carpels that contain microsporangia.
D) Ovules that are not contained within ovaries.

39) What adaptations should one expect of the seed coats of angiosperm species whose seeds are dispersed by frugivorous (fruit-eating) animals, as opposed to angiosperm species whose seeds are dispersed by other means?
1. The exterior of the seed coat should have barbs or hooks.
2. The seed coat should contain secondary compounds that irritate the lining of the animal's mouth.
3. The seed coat should be able to withstand low pH.
4. The seed coat, upon its complete digestion, should provide vitamins or nutrients to animals.
5. The seed coat should be resistant to the animals' digestive enzymes.
A) 4 only.
B) 1 and 2.
C) 2 and 3.
D) 3 and 5.

40) The generative cell of male angiosperm gametophytes is haploid. This cell divides to produce two haploid sperm cells. What type of cell division does the generative cell undergo to produce these sperm cells?
A) Binary fission.
B) Mitosis.
C) Meiosis.
D) Meiosis without subsequent cytokinesis.

41) Among plants known as legumes (beans, peas, alfalfa, clover, for example) the seeds are contained in a fruit that is itself called a legume, better known as a pod. Upon opening such pods, it is commonly observed that some ovules have become mature seeds, whereas other ovules have not. Thus, which of the following statements is (are) true?
1. The flowers that gave rise to such pods were not pollinated.
2. Pollen tubes did not enter all of the ovules in such pods.
3. There was apparently not enough endosperm to distribute to all of the ovules in such pods.
4. The ovules that failed to develop into seeds were derived from sterile floral parts.
5. Fruit can develop, even if all ovules within have not been fertilized.

A) 1 and 5.
B) 2 and 4.
C) 2 and 5.
D) 3 and 5.

42) Arrange the following structures from largest to smallest, assuming that they belong to two generations of the same angiosperm.
1. Ovary.
2. Ovule.
3. Egg.
4. Carpel.
5. Embryo sac.
A) 4, 5, 2, 1, 3.
B) 5, 4, 3, 1, 2.
C) 5, 1, 4, 2, 3.
D) 4, 1, 2, 5, 3.

43) The fruit of the mistletoe, a parasitic angiosperm, is a one-seeded berry. In members of the genus *Viscum*, the outside of the seed is viscous (sticky), which permits the seed to adhere to surfaces such as the branches of host plants or the beaks of birds. What should be expected of the fruit if the viscosity of *Viscum* seeds is primarily an adaptation for dispersal rather than an adaptation for infecting host plant tissues? It should _____.
A) Be drab in color.
B) Be colored so as to provide it with camouflage.
C) Be nutritious to the dispersing organisms.
D) Secrete enzymes that can digest bark.

Use the following description to answer the questions (44-46) below.

The cycads, a mostly tropical phylum of gymnosperms, evolved about 300 million years ago and were dominant forms during the Age of the Dinosaurs. Though their sperm are flagellated, their ovules are pollinated by beetles. These beetles get nutrition (they eat pollen) and shelter from the microsporophylls. Upon visiting megasporophylls, the beetles transfer pollen to the exposed ovules. In cycads, pollen cones and seed cones are borne on different plants. Cycads synthesize neurotoxins, especially in the seeds, that are effective against most animals, including humans.

44) Which feature of cycads makes them similar to many angiosperms?
1. They have exposed ovules.
2. They have flagellated sperm.
3. They are pollinated by animals.
A) 1 only.
B) 2 only.
C) 3 only.
D) 2 and 3.

45) If the beetles survive by consuming cycad pollen, then whether the beetles should be considered mutualists with, or parasites of, the cycads depends upon the _____.
A) Extent to which their overall activities affect cycad reproduction.
B) Extent to which the beetles are affected by the neurotoxins.
C) Extent to which the beetles damage the cycad flowers.
D) Distance the beetles must travel between cycad microsporophylls and cycad megasporophylls.

46) If one were to propose a new taxon of plants that included all plants that are pollinated by animals, and only plants that are pollinated by animals, then this new taxon would be _____.
A) Monophyletic.
B) Paraphyletic.
C) Polyphyletic.
D) Identical in composition to the phylum Anthophyta.

Use the following description to answer the questions (47-50) below.

Oviparous (egg-laying) animals have internal fertilization (sperm cells encounter eggs within the female's body). Yolk and/or albumen is (are) provided to the embryo, and a shell is then deposited around the embryo and its food source. Eggs are subsequently deposited in an environment that promotes their further development, or are incubated by one or both parents.

47) The yolk of an animal egg has what type of analog in angiosperms?
A) Endosperm.
B) Carpels.
C) Fruit.
D) Seed coat.

48) The shell of a fertilized animal egg has what type of analog in angiosperms?
A) Endosperm.
B) Carpels.
C) Fruit.
D) Seed coat.

49) The internal fertilization that occurs prior to shell deposition has what type of analog in angiosperms?
A) Endosperm proliferation.
B) Growth of pollen tube and delivery of sperm nuclei.
C) Fusion of carpels into a fruit.
D) Seed coat hardening.

50) The laying of eggs has what type of analog in angiosperms?
A) Endosperm breakdown.
B) Fusion of carpels into a fruit.
C) Fruit dispersal.
D) Seed coat hardening.

Use the following information to answer the questions (51-59) below.

The Brazil nut tree, *Bertholletia excels (n = 17)*, is native to tropical rain forests of South America. It is a hardwood tree that can grow to over 50 meters tall, is a source of high -quality lumber, and is a favorite nesting site for harpy eagles. As the rainy season ends, tough -walled fruits, each containing 8-25 seeds (Brazil nuts), fall to the forest floor. Brazil nuts are composed primarily of endosperm. About $50 million worth of nuts are harvested each year. Scientists have discovered that the pale yellow flowers of Brazil nut trees cannot fertilize themselves and admit only female orchid bees as pollinators. The agouti *(Dasyprocta spp.)*, a cat-sized rodent, is the only animal with teeth strong enough to crack the hard wall of Brazil nut fruits. It typically eats some of the seeds, buries others, and leaves still others inside the fruit, which moisture can now enter. The uneaten seeds may subsequently germinate.

51) If a female orchid bee has just left a Brazil nut tree with nectar in her stomach, and if she visits another flower on a different Brazil nut tree, what is the sequence in which the following events should occur?
1. Double fertilization.
2. Pollen tube emerges from pollen grain.
3. Pollen tube enters micropyle.
4. Pollination.
A) 4, 2, 3, 1.
B) 4, 3, 2, 1.
C) 2, 4, 3, 1.
D) 2, 4, 1, 3.

52) Stamens, sepals, petals, carpels, and pinecone scales are all _____.
A) Female reproductive parts.
B) Capable of photosynthesis.
C) Modified leaves.
D) Found on flowers.

53) The harpy eagle, *Harpia harpyja*, is the largest, most powerful raptor in the Americas. It nests only in trees taller than 25 meters. It is a "sloth specialist", but will also take agouti. Thus, if these eagles capture too many agoutis from a particular locale, they might contribute to their own demise by _____.
A) Having too many offspring.
B) Decreasing their habitat.
C) Decreasing atmospheric carbon dioxide.
D) Increasing the number of sloths.

54) Native peoples traditionally use Brazil nuts to treat stomach ache, inflammation, hypersensitivity, and hepatitis. Consequently, a scientist should be interested in promoting _____.
A) Better education for the native peoples so that they will overcome their old ways.
B) Clear-cutting forests containing Brazil nut trees to make way for crops with proven medical benefits.
C) An increase in the living standards of the native peoples so that they might be able to purchase modern pharmaceuticals.
D) The evaluation of Brazil nut chemicals for use as potential drugs.

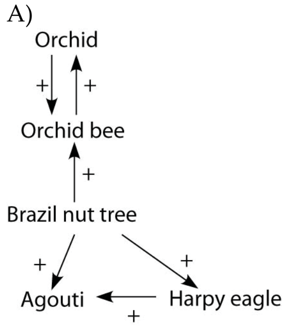
55) People who attempted to plant Brazil nuts in hopes of establishing plantations of Brazil nut trees played roles most similar to those of _____.
A) Agoutis.
B) Orchid bees.
C) Pollen tubes.
D) Harpy eagles.

56) The same bees that pollinate the flowers of the Brazil nut trees also pollinate orchids, which are epiphytes (in other words, plants that grow on other plants); however, orchids cannot grow on Brazil nut trees. These observations explain _____.
A) The coevolution of Brazil nut trees and orchids.
B) Why Brazil nut trees do not set fruit in monoculture plantations.
C) Why male orchid bees do not pollinate Brazil nut tree flowers.
D) Why male orchid bees are smaller than female orchid bees.

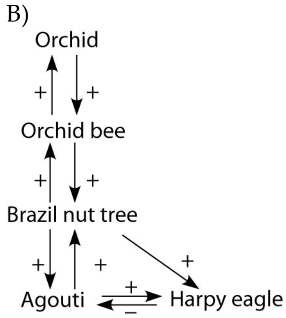
57) The taller a Brazil nut tree is _____.
1. The more valuable it is as a source of lumber.
2. The less useful it is to harpy eagles.
3. The greater its photosynthetic rate relative to neighboring plants.
A) 1 only.
B) 1 and 2.
C) 1 and 3.
D) 2 and 3.

58) Ecologists often build models to depict the relationships between organisms. In such models, an arrow is used to link two organisms in a relationship. The arrowhead is next to the organism that is affected. If the effect is positive, the arrow is labeled with (+), and if negative, then the label is (-). Which of the following models best illustrates the relationship of the Brazil nut tree and the other organisms associated with it?

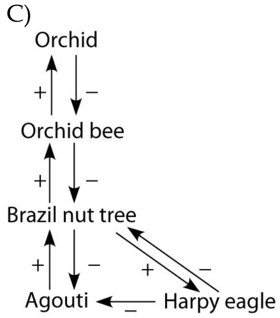
A)



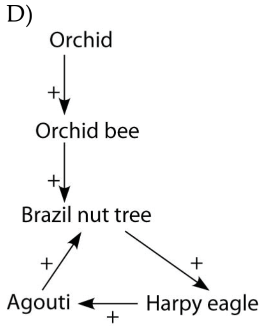
B)



C)

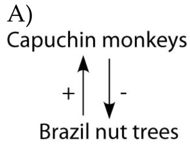


D)

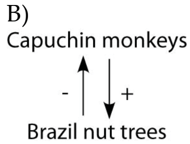


59) Ecologists often build models to depict the relationships between organisms. In such models, an arrow is used to link two organisms in a relationship. The arrowhead is next to the organism that is affected. If the effect is positive, the arrow is labeled with (+), and if negative, then the label is (-). Capuchin monkeys have been known to use rocks to smash open the fruits of Brazil nut trees. On the rare occasions when this has been observed, the monkeys consume all of the Brazil nuts. Thus, which of the following correctly depicts the relationship between capuchin monkeys and Brazil nut trees?

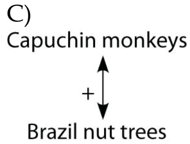
A)



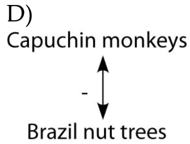
B)



C)



D)



Use the following information to answer the question (60) below.

Harold and Kumar are pre-med and pre-pharmacy students, respectively. They complain to their biology professor that they should not have to study about plants because plants have little relevance to their chosen professions.

60) Which adaptations of land plants are likely to provide Harold with future patients?
I) Sporophyte dominance.
II) Defenses against herbivory.
III) Adaptations related to wind dispersal of pollen.

A) I and II.
B) II and III.
C) I and III.
D) I, II, and III.

CHAPTER 31:

FUNGI

Use the following information to answer the questions (1-3) below.

In the United States and Canada, bats use one of two strategies to survive winter: They either migrate south, or they hibernate. Recently, those that hibernate seem to have come under attack by a fungus, *Geomyces destructans* (Gd), an attack that is occurring from Missouri to southern Canada. Many infected bats have a delicate, white filamentous mat on their muzzles, which is referred to as white-nose syndrome (WNS). The fungus invades the bat tissues, causes discomfort, and awakens the bat from its hibernation. The bat fidgets and wastes calories, using up its stored fat. The bat then behaves abnormally, leaving its cave during daytime in winter to search for food. Their food, primarily insects, is scarce during the winter, and the bats ultimately starve to death. Since 2007, it is estimated that up to one million bats have perished from WNS.

1) The Gd mat on the fur of the bats should be expected to consist of _____.
A) Hyphae.
B) Haustoria.
C) Yeasts.
D) Basidia.

2) What do fungi and arthropods have in common?
A) The haploid state is dominant in both groups.
B) Both groups are predominantly autotrophs that produce their own food.
C) Both groups use chitin for support.
D) Both groups have cell walls.

3) Fungi have an extremely high surface-area-to-volume ratio. What is the advantage of this to an organism that gets most of its nutrition through absorption?
A) The larger surface area allows for more material to be transported through the cell membrane.
B) The lower volume prevents the cells from drying out too quickly, which can interfere with absorption.
C) This high ratio creates more room inside the cells for additional organelles involved in absorption.
D) This high ratio means that fungi have a thick, fleshy structure that allows the fungi to store more of the food it absorbs.

Use the following information to answer the questions (4-5) below.

Suzanne Simard and colleagues knew that the same mycorrhizal fungal species could colonize multiple types of trees. They wondered if the same fungal individual would colonize different trees, forming an underground network that potentially could transport carbon and nutrients from one tree to another (S. Simard et al. 1997. Net transfer of carbon between mycorrhizal tree species in the field. *Nature* 388:579-82).

Figure A illustrates the team's experimental setup. Pots containing seedlings of three different tree species were set up and grown under natural conditions for three years; two of the three species (Douglas fir, birch) formed ectomycorrhizae and the other (cedar) formed arbuscular mycorrhizae. For the experiment, the researchers placed airtight bags over the Douglas fir and birch seedlings; into each bag, they injected either carbon dioxide made from carbon-13 or carbon-14 ($^{13}CO_2$ and $^{14}CO_2$, isotopes of carbon). As the seedlings photosynthesized, the radioactive carbon dioxide was converted into radioactively labeled sugars that could be tracked and measured by the researchers.

Figure A

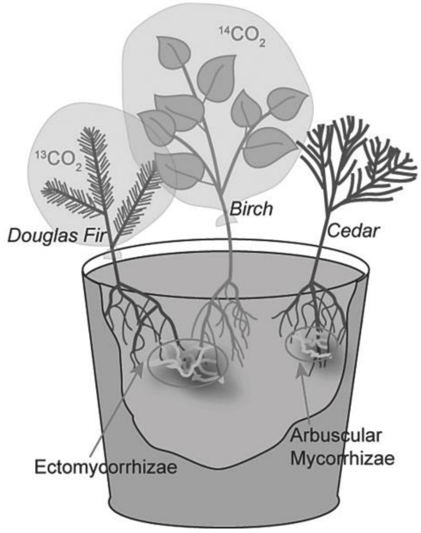


Figure B

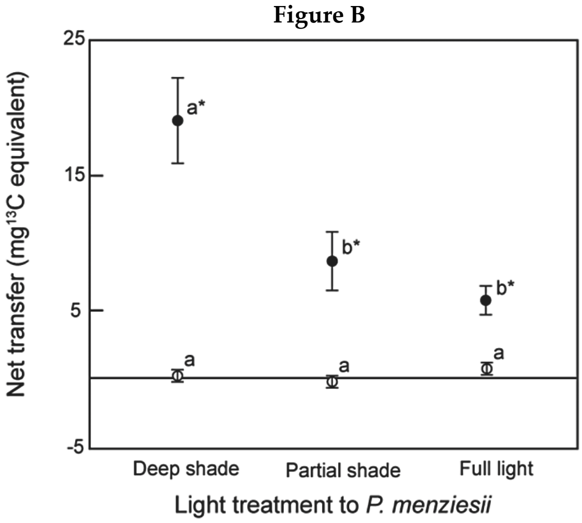


Figure 1 Mean (+/- s.e.) net transfer from *B. papyrifera* to *P. menziesii* in deep shade, partial shade and full ambient sunlight in the first (two-year-old seedlings, open circles) and second (three-year-old seedlings, filled circles) experiment years. Within years, means denoted by the same letter (a or b) do not differ significantly ($P<0.05$). Asterisks indicate treatments in which net transfer was greater than zero ($P<0.05$).

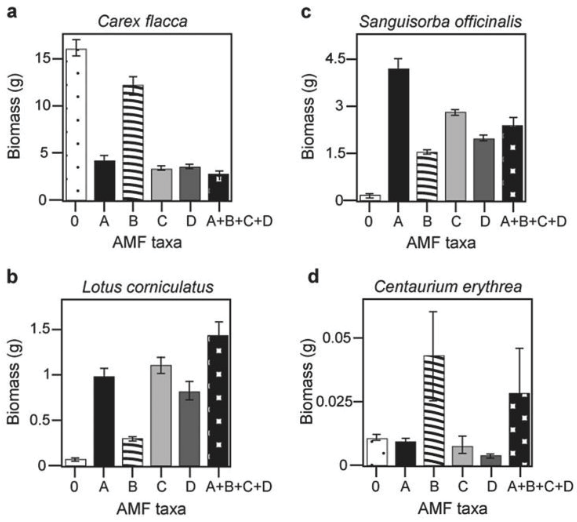
4) Refer to Figure A. Which of the following results would support Simard et al.'s (1997) hypothesis that fungi can move carbon from one plant to another? [Hypothesis: Sugars made by one plant during photosynthesis can travel through a mycorrhizal fungus and be incorporated into the tissues of another plant.]
A) Carbon-14 is found in the birch seedling's tissues and carbon-13 in the Douglas fir.
B) Carbon-14 is found in the Douglas fir seedling's tissues and carbon-13 in the birch.
C) Either carbon-13 or carbon-14 is found in the fungal tissues.
D) Either carbon-13 or carbon-14 is found in the cedar seedling's tissues.

5) Based on the idea that fungi have pores between their cell walls, which allow cytoplasm to move from one end of the mycelium to the other, which of the following hypotheses is the most plausible?
A) If a single mycorrhizal fungus formed symbiotic associations with more than one tree, carbon could travel from one plant to another.
B) Parasitic fungi steal nutrients from their hosts.
C) Predatory fungi capture their prey by encircling them with hyphae, and the flowing of the cytoplasm through the pores helps the hyphae to move around the prey.
D) Fungi function as part of the global carbon cycle not only by converting carbon from one form to another, but by physically moving it from one location to another.

Use the following information to answer the question(s) below.

There is much discussion in the media about protecting biodiversity. But does biodiversity really matter? Canadian and Swiss researchers wanted to know if the diversity of arbuscular mycorrhizal fungi (AMF) was important to the productivity of grasslands (M.G.A. van der Heijden, J. N. Klironomos, M. Ursic, P. Moutoglis, R. Streitwolf-Engel, T. Boler, A. Wiemken, and I. R. Sanders. 1998. Mycorrhizal fungal diversity determines plant biodiversity, ecosystem variability and productivity. *Nature* 396:69-72). Specifically, they wanted to know if it mattered which specific AMF species were present, or just that some type of AMF was present. They grew various plants in combination with one of four AMF species (A, B, C, and D), no AMF species (O), or all four AMF species together (A+B+C+D); and they measured plant growth under each set of conditions. All plant species were grown in each plot, so they always competed with each other with the only difference being which AMF species were present.

On the graphs below, the x-axis labels indicate the number and identity of AMF species (bar 0 = no fungi; bars A-D = individual AMF species; bar A+B+C+D = all AMF species together). The y-axis indicates the amount (grams) of plant biomass for the species shown in italics above each graph.



6) Based on the graphs in the figure above, which of the following plant species is most likely NOT to form mycorrhizal associations?
A) *Carex flacca* (graph a).
B) *Lotus corniculatus* (graph b).
C) *Sanguisorba officinalis* (graph c).
D) *Centaurium erythrea* (graph d).

7) If all fungi in an environment that perform decomposition were to suddenly die, then which group of organisms should benefit most, due to the fact that their fungal competitors have been removed?
A) Flowering plants.
B) Protists.
C) Prokaryotes.
D) Grasses.

8) When a mycelium infiltrates an unexploited source of dead organic matter, what are most likely to appear within the food source soon thereafter?
A) Fungal Haustoria.
B) Fungal enzymes.
C) Increased oxygen levels.
D) Larger bacterial populations.

9) The functional significance of porous septa in certain fungal hyphae is most similar to that represented by which pair of structures in animal cells and plant cells, respectively?
A) Desmosomes – tonoplasts.
B) Gap junctions – plasmodesmata.
C) Tight junctions – plastids.
D) Centrioles – plastids.

10) A fungal spore germinates; giving rise to a mycelium that grows outward into the soil surrounding the site where the spore originally landed. Which of the following accounts for the fungal movement, as described here?
A) Karyogamy.
B) Mycelial flagella.
C) Breezes distributing spores.
D) Cytoplasmic streaming in hyphae.

11) When pathogenic fungi are found growing on the roots of grape vines, grape farmers sometimes respond by covering the ground around their vines with plastic sheeting and pumping a gaseous fungicide into the soil. The most important concern of grape farmers who engage in this practice should be that the _____.
A) Fungicide might also kill the native yeasts residing on the surfaces of the grapes.
B) Lichens growing on the vines' branches are not harmed.
C) Fungicide might also kill mycorrhizae.
D) Sheeting is transparent so that photosynthesis can continue.

12) The adaptive advantage associated with the filamentous nature of fungal mycelia is primarily related to _____.
A) The ability to form haustoria and parasitize other organisms.
B) The potential to inhabit almost all terrestrial habitats.
C) The increased probability of contact between different mating types.
D) An extensive surface area well suited for invasive growth and absorptive nutrition.

13) Some fungal species can kill herbivores while feeding off of sugars from its plant host. What type of relationship does this fungus have with its host?
A) Parasitic.
B) Mutualistic.
C) Commensal.
D) Predatory.

Use the following information to answer the questions (14-16) below.

Suzanne Simard and colleagues knew that the same mycorrhizal fungal species could colonize multiple types of trees. They wondered if the same fungal individual would colonize different trees, forming an underground network that potentially could transport carbon and nutrients from one tree to another (S. Simard et al. 1997. Net transfer of carbon between mycorrhizal tree species in the field. *Nature* 388:579-82).

Figure A illustrates the team's experimental setup. Pots containing seedlings of three different tree species were set up and grown under natural conditions for three years; two of the three species (Douglas fir, birch) formed ectomycorrhizae and the other (cedar) formed arbuscular mycorrhizae. For the experiment, the researchers placed airtight bags over the Douglas fir and birch seedlings; into each bag, they injected either carbon dioxide made from carbon-13 or carbon-14 ($^{13}CO_2$ and $^{14}CO^2$, isotopes of carbon). As the seedlings photosynthesized, the radioactive carbon dioxide was converted into radioactively labeled sugars that could be tracked and measured by the researchers.

Figure A

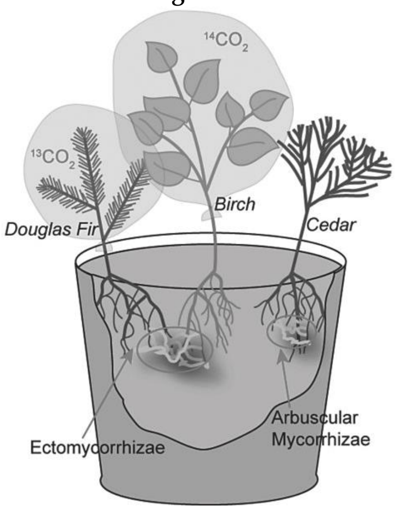
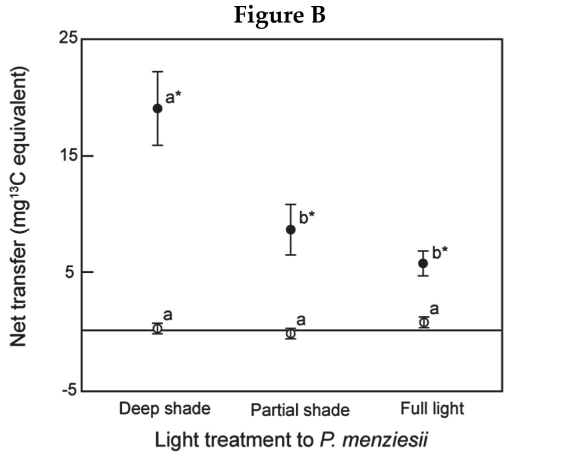


Figure B



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14) Referring to Simard et al. (1997), what is the result that would most strongly refute their hypothesis? [Hypothesis: Sugars made by one plant during photosynthesis can travel through a mycorrhizal fungus and be incorporated into the tissues of another plant.]
A) No movement: Carbon-14 is found in the birch seedling's tissues and carbon-13 in the Douglas fir.
B) Reciprocal exchange: Carbon-14 is found in the Douglas fir seedling's tissues and carbon-13 in the birch.
C) Either carbon-13 or carbon-14 is found in the fungal tissues.
D) Either carbon-13 or carbon-14 is found in the cedar seedling's tissues.

15) Referring to Simard et al. (1997), which design element is the control in this experiment and why?
A) The bags over the seedlings to contain the different types of carbon dioxide.
B) The fact that all the seedlings are different species.
C) The cedar seedling, because it is not bagged.
D) The cedar seedling, because it forms arbuscular mycorrhizae.

16) Simard et al. (1997) further hypothesized that if reciprocal transfer did occur, it would be a source-sink relationship driven by photosynthetic rates. That is, if one seedling is in full sun and the other in deep shade, there will be a net movement of carbon from the seedling in full sun to the one in deep shade. If a shade was placed over the birch seedlings and the cedar, and the Douglas fir was left in full sun, what result could Simard and colleagues expect?

A) More 13C would be found in the birch than in the Douglas fir.

B) More 13C would be found in the Douglas fir than in the birch.

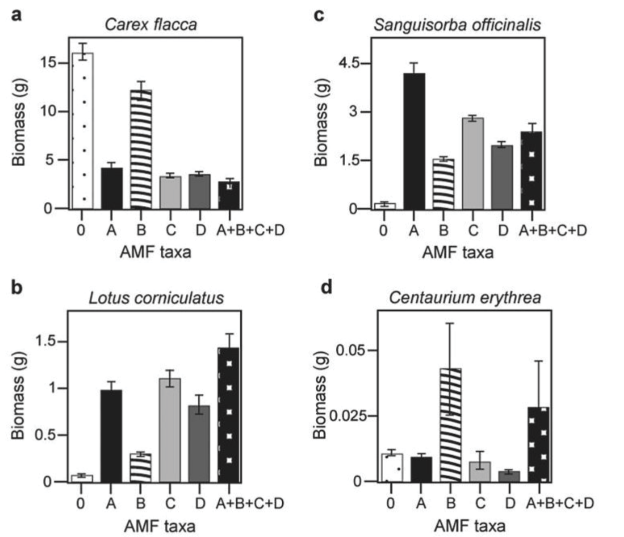
C) The most 13C would be found in the cedar.

D) The most 14C would be found in the cedar.

Use the following information to answer the question(s) below.

There is much discussion in the media about protecting biodiversity. But does biodiversity really matter? Canadian and Swiss researchers wanted to know if the diversity of arbuscular mycorrhizal fungi (AMF) was important to the productivity of grasslands (M.G.A. van der Heijden, J. N. Klironomos, M. Ursic, P. Moutoglis, R. Streitwolf-Engel, T. Boler, A. Wiemken, and I. R. Sanders. 1998. Mycorrhizal fungal diversity determines plant biodiversity, ecosystem variability and productivity. Nature 396:69-72). Specifically, they wanted to know if it mattered which specific AMF species were present, or just that some type of AMF was present. They grew various plants in combination with one of four AMF species (A, B, C, and D), no AMF species (O), or all four AMF species together (A+B+C+D); and they measured plant growth under each set of conditions. All plant species were grown in each plot, so they always competed with each other with the only difference being which AMF species were present.

On the graphs below, the x-axis labels indicate the number and identity of AMF species (bar 0 = no fungi; bars A-D = individual AMF species; bar A+B+C+D = all AMF species together). The y-axis indicates the amount (grams) of plant biomass for the species shown in italics above each graph.



17) Based on the van der Heijden et al. (1998) graphs in the figure above, which of the following is the best description of the data supporting the idea that a plant species did not form mycorrhizae with a fungus? Its biomass is greatest when _____.

A) No AMF are present.

B) AM fungus A is present.

C) AM fungus B is present.

D) AM fungus C is present.

18) In graph (b) in the figure above, which of the following best explains the data given about *Lotus corniculatus*?

A) This plant grows best when AMF taxa A or C are present.

B) Lotus corniculatus does not form mycorrhizal associations.

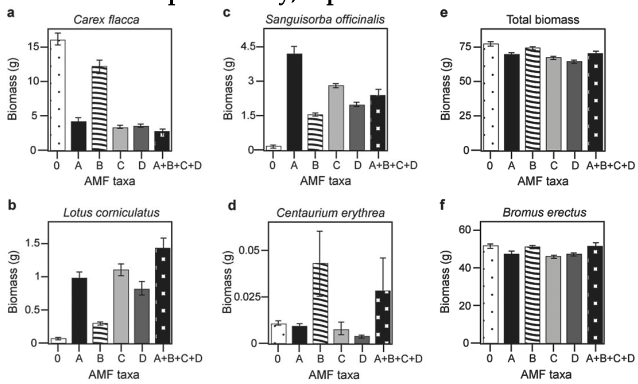
C) Mycorrhizal fungi parasitize the plant's roots when they are present, reducing its growth.

D) This plant forms multiple AMF associations, growing best with increased fungal diversity.

Use the following information to answer the question (19-21) below.

Canadian and Swiss researchers wanted to know if the diversity of arbuscular mycorrhizal fungi (AMF) was important to the productivity of grasslands (M.G.A. van der Heijden, J. N. Klironomos, M. Ursic, P. Moutoglis, R. Streitwolf-Engel, T. Boler, A. Wiemken, and I. R. Sanders. 1998. Mycorrhizal fungal diversity determines plant biodiversity, ecosystem variability and productivity. Nature 396:69-72). Specifically, they wanted to know if it mattered which specific AMF species were present, or just that some type of AMF was present. They grew various plants in combination with one of four AMF species (A, B, C, & D), no AMF species (O), or all four AMF species together (A +B+C+D); and they measured plant growth under each set of conditions. All plant species were grown in each plot, so they always competed with each other with the only difference being which AMF species were present.

On the graphs below, the x-axis labels indicate the number and identity of AMF species (bar 0 = no fungi; bars A-D = individual AMF species; bar A+B+C+D = all AMF species together). The y-axis indicates the amount (grams) of plant biomass for the species shown in italics above each graph. Graph (e) is the total biomass (grams) of all 11 plant species combined; graph (f) is the biomass of *Bromus erectus* plants only, separated from the total.



19) What is the major difference between *Bromus erectus* (graph f) and the other plant species (graphs a-d) included in the study?
A) *Bromus erectus* grows best with a diversity of fungal partners.
B) *Bromus erectus* is unaffected by AMF diversity.
C) *Bromus erectus* does not form mycorrhizal associations.
D) *Bromus erectus* produces very little biomass regardless of AMF.

20) Why does total biomass (graph e in the figure above) not vary with AMF diversity?
A) Plant growth is unaffected by fungal diversity.
B) Most of the plants in this system do not form mycorrhizal associations.
C) *Bromus erectus* is the dominant plant species.
D) *Lotus corniculatus* is a rare species.

21) Based on graphs (e) and (f) in the figure above, which is the most well -supported prediction for the effect on total plant biomass if AMF diversity were increased to eight species?
A) No effect is predicted, because the dominant species is unaffected by AMF diversity.
B) Total biomass for eight species would double in comparison to that for four species.
C) Rare species would produce more biomass compared to the case when fewer AMF are present.
D) No effect is predicted, because the dominant species is non -mycorrhizal.

22) At which stage of a basidiomycete's life cycle would reproduction be halted if an enzyme that prevented the fusion of hyphae was introduced?
A) Fertilization.
B) Karyogamy.
C) Plasmogamy.
D) Germination.

23) Deuteromycetes _____.
A) Represent the phylum in which all the fungal components of lichens are classified.
B) Are the groups of fungi that have, at present, no known sexual stage.
C) Are the group that includes molds, yeasts, and lichens.
D) Include the imperfect fungi that lack hyphae.

Use the following information to answer the question (24) below.

For several decades now, amphibian species worldwide have been in decline. A significant proportion of the decline seems to be due to the spread of the chytrid fungus, *Batrachochytrium dendrobatidis* (Bd). Chytrid sporangia reside within the epidermal cells of infected animals, animals that consequently show areas of sloughed skin. They can also be lethargic, which is expressed through failure to hide and failure to flee. The infection cycle typically takes four to five days, at the end of which zoospores are released from sporangia into the environment. In some amphibian species, mortality rates approach 100%; other species seem able to survive the infection.

24) Sexual reproduction has not been observed in Bd. If its morphology and genetics did not identify it as a chytridiomycete, then to which fungal group would Bd be assigned?
A) Zygomycetes.
B) Glomeromycetes.
C) Basidiomycetes.
D) Deuteromycetes.

25) Plasmogamy can directly result in which of the following?
1. Cells with a single haploid nucleus.
2. Heterokaryotic cells.
3. Dikaryotic cells.
4. Cells with two diploid nuclei.
A) 1 or 3.
B) 2 or 3.
C) 2 or 4.
D) 3 or 4.

26) After cytokinesis occurs in budding yeasts, the daughter cell has a _____.
A) Similar nucleus and more cytoplasm than mother cell.
B) Smaller nucleus and less cytoplasm than the mother cell.
C) Larger nucleus and less cytoplasm than the mother cell.
D) Similar nucleus and less cytoplasm than the mother cell.

27) In most fungi, karyogamy does not immediately follow plasmogamy, which consequently _____.
A) Means that sexual reproduction can occur in specialized structures.
B) Results in multiple diploid nuclei per cell.
C) Allows fungi to reproduce asexually most of the time.
D) Results in heterokaryotic or dikaryotic cells.

28) Asexual reproduction in yeasts occurs by budding. Due to unequal cytokinesis, the "bud" cell receives less cytoplasm than the parent cell. Which of the following should be true of the smaller cell until it reaches the size of the larger cell?
A) It should produce fewer fermentation products per unit time.
B) It should be transcriptionally less active.
C) It should have reduced motility.
D) It should have a smaller nucleus.

29) The microsporidian *Brachiola gambiae* parasitizes the mosquito *Anopheles gambiae*. Adult female mosquitoes must take blood meals for their eggs to develop, and it is while they take blood that they transmit malarial parasites to humans. Male mosquitoes drink flower nectar. If humans are to safely and effectively use *Brachiola gambiae* as a biological control to reduce human deaths from malaria, then how many of the following statements should be true?
1. *Brachiola* should kill the mosquitoes before the malarial parasite they carry reaches maturity.
2. The microsporidian should not be harmful to other insects.
3. Microsporidians should infect mosquito larvae, rather than mosquito adults.
4. The subsequent decline in anopheline mosquitoes should not significantly disrupt human food resources or other food webs.
5. *Brachiola* must be harmful to male mosquitoes, but not to female mosquitoes.
A) 2 and 5.
B) 1, 2, and 4.
C) 2, 3, 5.
D) 3 and 4.

30) Why are mycorrhizal fungi superior to plants at acquiring mineral nutrition from the soil?
A) Hyphae are one hundred to one thousand times larger than plant roots.
B) Hyphae have a smaller surface-area-to-volume ratio than do the hairs on a plant root.
C) Mycelia are able to grow in the direction of food.
D) Fungi secrete extracellular enzymes that can break down large molecules.

31) Fossil fungi date back to the origin and early evolution of plants. What combination of environmental and morphological change is similar in the evolution of both fungi and plants?
A) Presence of "coal forests" and change in mode of nutrition.
B) Periods of drought and presence of filamentous body shape.
C) Predominance in swamps and presence of cellulose in cell walls.
D) Colonization of land and loss of flagellated cells.

32) The multicellular condition of animals and fungi seems to have arisen _____.
A) Due to common ancestry.
B) By convergent evolution.
C) By inheritance of acquired traits.
D) By serial endosymbiosis.

33) Which feature seen in chytrids supports the hypothesis that they diverged earliest in fungal evolution?
A) The absence of chitin within the cell wall.
B) Coenocytic hyphae.
C) Flagellated spores.
D) Parasitic lifestyle.

34) Recent genetic studies of the structure of microsporidian genomes, as well as the sequences of their tubulin genes and the gene for RNA polymerase II, indicate that microsporidians are closely related to the fungi. Microsporidians lack flagella, centrioles, peroxisomes, and mitochondria (although they do have degenerate mitochondria, called mitosomes). They have the smallest genome of any eukaryote, and it is a genome that changes quickly. The genome is contained within two haploid nuclei. All microsporidians are obligate intracellular parasites. They use a unique organelle called a polar filament to gain access to the cells of their hosts. One species causes chronic diarrhea in AIDS patients. Another parasitizes *Anopheles gambiae*, the mosquito that transmits a fatal form of malaria to humans.
Given the eukaryotic structures they lack, it should be expected that microsporidians also lack _____.
A) The "9 + 2 pattern" of microtubules.
B) Centrosomes.
C) Lysosomes.
D) Nuclei.

35) It has been hypothesized that fungi and plants have a mutualistic relationship because plants make sugars available for the fungi's use. What is the best evidence in support of this hypothesis?
A) Fungi survive better when they are associated with plants.
B) Radioactively labeled sugars produced by plants eventually show up in the fungi with which they are associated.
C) Fungi associated with plants have the ability to undergo photosynthesis and produce their own sugars, while those not associated with plants do not produce their own sugars.
D) Radioactive labeling experiments show that plants pass crucial raw materials to the fungus for manufacturing sugars.

36) You observe the gametes of a fungal species under the microscope and realize that they resemble animal sperm. To which of the following group does the fungus belong?
A) Chytrids.
B) Zygomycetes.
C) Basidiomycota.
D) Ascomycota.

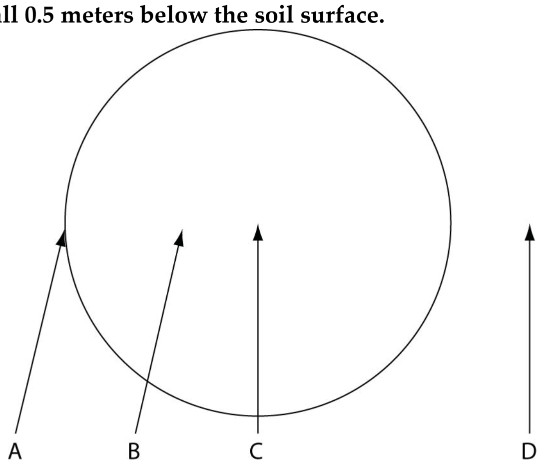
37) Which of the following has the LEAST affiliation with all of the others?
A) Glomeromycota.
B) Mycorrhizae.
C) Lichens.
D) Arbuscules.

38) Arrange the following in order from largest to smallest.
1. Ascospore.
2. Ascocarp.
3. Ascomycete.
4. Ascus.
A) 3 → 2 → 4 → 1.
B) 3 → 4 → 1 → 2.
C) 2 → 3 → 4 → 1.
D) 2 → 4 → 1 → 3.

39) Arrange the following in order from largest to smallest, assuming that they all come from the same fungus.
1. Basidiocarp.
2. Basidium.
3. Basidiospore.
4. Mycelium.
5. Gill.
A) 4 → 5 → 1 → 2 → 3.
B) 5 → 1 → 4 → 2 → 3.
C) 5 → 1 → 3 → 2 → 4.
D) 4 → 1 → 5 → 2 → 3.

Use the following information to answer the question (40-45) below.

The following figure depicts the outline of a large fairy ring that has appeared overnight in an open meadow, as viewed from above. The fairy ring represents the furthest advance of this mycelium through the soil. Locations A-D are all 0.5 meters below the soil surface.



40) What is the most probable location of the oldest portion of this mycelium?
A) A.
B) B.
C) C.
D) D.

41) Which location is nearest to basidiocarps?
A) A.
B) B.
C) C.
D) D.

42) At which location is the mycelium currently absorbing the most nutrients per unit surface area, per unit time?
A) A.
B) B.
C) C.
D) D.

43) At which location should one find the lowest concentration of fungal enzymes, assuming that the enzymes do not diffuse far from their source, and that no other fungi are present in this habitat?
A) A.
B) B.
C) C.
D) D.

44) Assume that all four locations are 0.5 meters above the surface. On a breezy day with prevailing winds blowing from left to right, where should one expect to find the highest concentration of free basidiospores in an air sample?
A) A.
B) B.
C) C.
D) D.

45) If the fungus that produced the fairy ring can also produce arbuscules, then which of the following is most likely to be buried at location "C"?
A) Tree stump.
B) Deceased animal.
C) Fire pit.
D) Cement-capped well.

Use the following information to answer the question (46) below.

For several decades now, amphibian species worldwide have been in decline. A significant proportion of the decline seems to be due to the spread of the chytrid fungus, *Batrachochytrium dendrobatidis* (Bd). Chytrid sporangia reside within the epidermal cells of infected animals, animals that consequently show areas of sloughed skin. They can also be lethargic, which is expressed through failure to hide and failure to flee. The infection cycle typically takes four to five days, at the end of which zoospores are released from sporangia into the environment. In some amphibian species, mortality rates approach 100%; other species seem able to survive the infection.

46) Apart from direct amphibian-to-amphibian contact, what is the most likely means by which the zoospores spread from one free-living amphibian to another?
A) By wind-blown spores.
B) By flagella.
C) By cilia.
D) By hyphae.

47) When adult amphibian skin harbors populations of the bacterium, *Janthinobacterium lividum* (Jl), chytrid infection seems to be inhibited. Which of the following represents the best experimental design to test whether this inhibition is real?
A) Inoculate uninfected amphibians with Jl, and determine whether the amphibians continue to remain uninfected by chytrids.
B) Inoculate infected amphibians with Jl, and determine whether the amphibians recover from infection by chytrids.
C) Take infected amphibians and assign them to two populations. Leave one population alone; inoculate the other with Jl. Measure the rate at which infection proceeds in both populations.
D) Take infected amphibians and assign them to two populations. Inoculate one population with a high dose of Jl; inoculate the other with a low dose of Jl. Measure the survival frequency in both populations.

48) A researcher took water in which a bacteria *Janthinobacterium lividum* (Jl), population had been thriving, filtered the water to remove all bacterial cells, and then applied the water to the skins of adult amphibians to see if there would subsequently be a reduced infection rate by Bd when frog skins were inoculated with Bd. For which of the following hypotheses is the procedure described a potential test?
A) The hypothesis that a toxin secreted by Jl cells kills Bd cells when both are present together on frog skin.
B) The hypothesis that Jl cells infect and kill Bd cells when both are present together on frog skin.
C) The hypothesis that Jl outcompetes Bd when both are present together on a frog's skin.
D) The hypothesis that the presence of Jl on frog skin causes a skin reaction that prevents attachment by Bd cells.

49) Basidiomycetes are the only fungal group capable of synthesizing lignin peroxidase. What advantage does this group of fungi have over other fungi because of this capability?
A) This is always the first group of fungi to begin any kind of plant decomposition.
B) This fungal group can break down the tough lignin, which cannot be harnessed for energy, to get to the more useful cellulose.
C) This is the only group of fungi that can use lignin for adenosine triphosphate (ATP) production.
D) This enzyme releases heat energy from the breakdown of lignin that is used to kill off competing fungi.

Use the following information to answer the question (50) below.

For several decades now, amphibian species worldwide have been in decline. A significant proportion of the decline seems to be due to the spread of the chytrid fungus, *Batrachochytrium dendrobatidis* (Bd). Chytrid sporangia reside within the epidermal cells of infected animals, animals that consequently show areas of sloughed skin. They can also be lethargic, which is expressed through failure to hide and failure to flee. The infection cycle typically takes four to five days, at the end of which zoospores are released from sporangia into the environment. In some amphibian species, mortality rates approach 100%; other species seem able to survive the infection.

50) If Bd cannot grow properly at temperatures above 28°C (82°F), then, assuming the amphibians can survive, in which time or place should the chytrid infection proceed most rapidly?
1. Cooler months.
2. Warmer months.
3. Lower altitudes.
4. Higher altitudes.
A) 1 or 3.
B) 1 or 4.
C) 2 or 3.
D) 2 or 4.

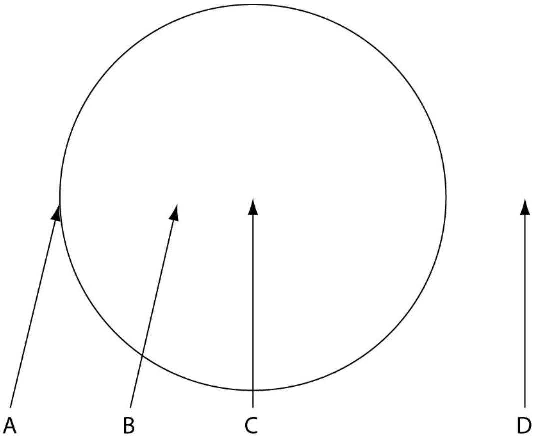
Use the following information to answer the question (51) below.

Diploid nuclei of the ascomycete, *Neurospora crassa*, contain 14 chromosomes. A single diploid cell in an ascus will undergo one round of meiosis, followed in each of the daughter cells by one round of mitosis, producing a total of eight ascospores.

51) If a single, diploid G2 nucleus in an ascus contains 400 nanograms (ng) of DNA, then a single ascospore nucleus of this species should contain how much DNA (ng), carried on how many chromosomes?
A) 100, carried on 7 chromosomes.
B) 100, carried on 14 chromosomes.
C) 200, carried on 7 chromosomes.
D) 200, carried on 14 chromosomes.

Use the following information to answer the question (52) below.

The following figure depicts the outline of a large fairy ring that has appeared overnight in an open meadow, as viewed from above. The fairy ring represents the furthest advance of this mycelium through the soil. Locations A-D are all 0.5 meters below the soil surface.



52) In which of the following human mycoses should one expect to find a growth pattern most similar to that of the mycelium that produced the fairy ring?
A) Skin mycoses.
B) Coccidiomycosis (lung infection).
C) Systemic (bloodborne) Candida infection.
D) Sporothrix infection of lymphatic vessels.

Use the following information to answer the question(s) below.

Rose-picker's disease is caused by the yeast *Sporothrix schenkii (S. schenkii)*. The yeast grows on the exteriors of rose-bush thorns. If a human gets pricked by such a thorn, the yeasts can be introduced under the skin. The yeasts then assume a hyphal morphology and grow along the interiors of lymphatic vessels until they reach a lymph node. This often results in the accumulation of pus in the lymph node, which subsequently ulcerates through the skin surface and then drains.

53) Humans have immune systems in which lymph nodes are important, because many phagocytes and lymphocytes reside there. Given that a successful infection by *S. schenkii* damages lymph nodes themselves, which of the following is most probable?
A) The hyphae secrete antibiotics, which increases the ability of the infected human to tolerate the fungus.
B) Their conversion from yeast to hyphal morphology allows such fast growth that the body's defenses are at least temporarily overwhelmed.
C) Defensive cells of humans cannot detect foreign cells that are covered with cell walls composed of cellulose.
D) Given that most fungal pathogens attack plants, human defenses are simply not adapted to seek out and destroy fungi.

54) If haustoria from the fungal partner were to appear within the photosynthetic partner of a lichen, and if the growth rate of the photosynthetic partner consequently slowed substantially, then this would support the claim that _____.
A) Algae and cyanobacteria are autotrophic.
B) Lichens are not purely mutualistic relationships.
C) Algae require maximal contact with the fungal partner in order to grow at optimal rates.
D) Soredia are asexual reproductive structures combining both the fungal and photosynthetic partners.

55) A billionaire buys a sterile volcanic island that recently emerged from the sea. To speed the arrival of conditions necessary for plant growth, the billionaire might be advised to aerially sow what over the island?
A) Basidiospores.
B) Spores of ectomycorrhizae.
C) Soredia.
D) Yeasts.

56) Orchid seeds are tiny, with virtually no endosperm and with miniscule cotyledons. If such seeds are deposited in a dark, moist environment, then which of the following represents the most likely means by which fungi might assist in seed germination, given what the seeds lack?
A) By transferring some chloroplasts to the embryo in each seed.
B) By providing the seeds with water and minerals.
C) By providing the embryos with some of the organic nutrients they have absorbed.
D) By strengthening the seed coat that surrounds each seed.

Use the following information to answer the questions (57-58) below.

Rose-picker's disease is caused by the yeast *Sporothrix schenkii (S. schenkii)*. The yeast grows on the exteriors of rose-bush thorns. If a human gets pricked by such a thorn, the yeasts can be introduced under the skin. The yeasts then assume a hyphal morphology and grow along the interiors of lymphatic vessels until they reach a lymph node. This often results in the accumulation of pus in the lymph node, which subsequently ulcerates through the skin surface and then drains.

57) The answer to which of these questions would be of most assistance to one who is attempting to assign the genus *Sporothrix* to the correct fungal phylum?
A) Do these yeasts perform fermentation while growing on the rose -bush thorns, or do they wait until inside a human host?
B) Does *S. schenkii* rely on animal infection to complete some part of its life cycle, or is the infection merely opportunistic?
C) Are the hyphae in lymphatic vessels septate, or are they coenocytic?
D) Is *S. schenkii* best described as a decomposer, parasite, pathogen, or mutualist of humans?

58) Suppose that *S. schenkii* had initially been classified as a deuteromycete. Asci were later discovered in the pus that oozed from an ulcerated lymph node, and the spores therein germinated, giving raises to *S. schenkii* yeasts. Which two of these are conclusions make sense on the basis of this information?
1. *S. schenkii* produces asexual spores within lymph nodes.
2. *S. schenkii* should be reclassified.
3. *S. schenkii* continues to have no known sexual stage.
4. The hyphae growing in lymphatic vessels probably belonged to a different fungal species.
5. *S. schenkii* yeasts belonging to two different mating strains were introduced by the same thorn prick.
A) 1 and 3
B) 1 and 5
C) 2 and 5
D) 4 and 5.

59) Which of the following best describes the physical relationship of the partners involved in lichens?
A) Fungal cells are enclosed within algal cells.
B) Lichen cells are enclosed within fungal cells.
C) Photosynthetic cells are surrounded by fungal hyphae.
D) The fungi grow on rocks and trees and are covered by algae.

60) Mycorrhizae are to the roots of vascular plants as endophytes are to vascular plants' _____.
A) Leaf mesophyll.
B) Stem apical meristems.
C) Root apical meristems.
D) Xylem.

CHAPTER 32:

AN OVERVIEW OF ANIMAL DIVERSITY

1) A researcher is trying to construct a molecular-based phylogeny of the entire animal kingdom. Assuming that none of the following genes is absolutely conserved, which of the following would be the best choice on which to base the phylogeny?
A) Genes involved in chitin synthesis.
B) Collagen genes.
C) β-catenin genes.
D) Genes involved in eye-lens synthesis.

2) Which of the following is (are) unique to animals?
A) The structural carbohydrate, chitin.
B) Nervous system signal conduction and muscular movement.
C) Heterotrophy.
D) Flagellated gametes.

3) The larvae of some insects are merely small versions of the adult, whereas the larvae of other insects look completely different from adults, eat different foods, and may live in different habitats. Which of the following is most directly involved in the evolution of these variations in metamorphosis?
A) Artificial selection of sexually immature forms of insects.
B) Changes in the homeobox genes governing early development.
C) The evolution of meiosis.
D) The origin of a brain.

4) As you are on the way to Tahiti for a vacation, your plane crash -lands on a previously undiscovered island. You soon find that the island is teeming with unfamiliar organisms, and you, as a student of biology, decide to survey them (with the aid of the Insta -Lab Portable Laboratory you brought along in your suitcase). You select three organisms and observe them in detail, making the notations found in the figure below. Which organism would you classify as an animal?

Organism	Appearance	Habitat/ Activity	Nutrient Acquisition	Reproduction
A	Microscopic, unicellular, with a flagellum	Swims around in freshwater pools	Envelops and consumes other microscopic organisms	Mates with others; young bud off
B	Shaped like a basketball, covered with purple filaments, multicellular	Rolls slowly across grassy fields	Thrives with access to only freshwater and sunlight	Mates with others; young emerge from hardened spherical structures
C	Hard and branched, multicellular, covered in a sticky coating	Attached to rocky surfaces	Traps insects in sticky coating and dissolves them	No mating; releases winged young that fly off and affix to bare rocks

A) Organism A.
B) Organism B.
C) Organism C.
D) None of them.

5) Both animals and fungi are heterotrophic. What distinguishes animal heterotrophy from fungal heterotrophy is that most animals derive their nutrition by _____.
A) Preying on animals.
B) Ingesting it.
C) Consuming living, rather than dead, prey.
D) Using enzymes to digest their food.

Use the following information to answer the question (6) below.

Trichoplax adhaerens **(Tp) is the only living species in the phylum Placozoa. Individuals are about 1 mm wide and only 27 μm high, are irregularly shaped, and consist of a total of about 2000 cells, which are diploid (2n = 12). There are four types of cells, none of which are nerve or muscle cells, and none of which have cell walls. They move using cilia, and any "edge" can lead. Tp feeds on marine microbes, mostly unicellular green algae, by crawling atop the algae and trapping it between its ventral surface and the substrate. Enzymes are then secreted onto the algae, and the resulting nutrients are absorbed. Tp sperm cells have never been observed, nor have embryos past the 64 -cell (blastula) stage.**

6) Which of the following Tp traits is different from all other known animals?
A) Tp is multicellular.
B) Tp lacks muscle and nerve cells.
C) Tp has cilia.
D) Tp lacks cell walls.

7) What do animals ranging from corals to monkeys have in common?
A) A mouth and an anus.
B) Number of embryonic tissue layers.
C) Type of body symmetry.
D) Presence of *Hox* genes.

8) In individual insects of some species, whole chromosomes that carry larval genes are eliminated from the genomes of somatic cells at the time of metamorphosis. A consequence of this occurrence is that _____.
A) We could not clone a larva from the somatic cells of such an adult insect.
B) Such species must reproduce only asexually.
C) The descendants of these adults do not include a larval stage.
D) Metamorphosis can no longer occur among the descendents of such adults.

9) Which of the following would you classify as something other than an animal?
A) Sponges.
B) Coral.
C) Jellyfish.
D) Choanoflagellates.

10) The evolution of animal species has been prolific (the estimates go into the millions and tens of millions). Much of this diversity is a result of the evolution of novel ways to _____.
A) Reproduce.
B) Arrange cells into tissues.
C) Sense, feed, and move.
D) Form an embryo and establish a basic body plan.

11) The last common ancestor of all animals was probably a _____.
A) Unicellular chytrid.
B) Multicellular algae.
C) Multicellular fungus.
D) Flagellated protist.

12) Evidence of which structure or characteristic would be most surprising to find among fossils of the Ediacaran fauna?
A) True tissues.
B) Hard parts.
C) Bilateral symmetry.
D) Embryos.

13) Which statement is most consistent with the hypothesis that the Cambrian explosion was caused by the rise of predator-prey relationships? The fossil record reveals an increased incidence of _____.
A) Worm burrows.
B) Larger animals.
C) Organic material.
D) Hard parts.

14) Which of the following genetic processes may be most helpful in accounting for the Cambrian explosion?
A) Binary fission.
B) Random segregation.
C) Gene duplication.
D) Chromosomal condensation.

15) Whatever its ultimate cause(s), the Cambrian explosion is a prime example of _____.
A) Mass extinction.
B) Evolutionary stasis.
C) Adaptive radiation.
D) A large meteor impact.

16) Arthropods invaded land about 100 million years before vertebrates did. This most clearly implies that _____.
A) Arthropods evolved before vertebrates did.
B) Extant terrestrial arthropods are better adapted to terrestrial life than are extant terrestrial vertebrates.
C) Vertebrates evolved from arthropods.
D) Arthropods have had more time to coevolve with land plants than have vertebrates.

17) Which tissue type, or organ, is NOT correctly matched with its germ layer tissue?
A) Nervous – mesoderm.
B) Muscular – mesoderm.
C) Stomach – endoderm.
D) Skin – ectoderm.

18) While looking at some seawater through your microscope, you spot the egg of an unknown animal. Which of the following tests could you use to determine whether the developing organism is a protostome or a deuterostome? See whether the embryo _____.
A) Develops germ layers
B) Exhibits spiral cleavage or radial cleavage
C) Develops a blastopore
D) Develops an archenteron.

19) In examining an unknown animal species during its embryonic development, how can you be sure what you are looking at is a protostome and not a deuterostome?
A) There is evidence of cephalization.
B) The animal is triploblastic.
C) The animal is clearly bilaterally symmetrical.
D) You see a mouth, but not an anus.

20) Which of the following is a feature of the tube-within-a-tube body plan in most animal phyla?
A) The outer tube consists of a hard exoskeleton.
B) The outer tube consists of digestive organs.
C) The mouth and anus form the ends of the inner tube.
D) The two "tubes" are separated by tissue that comes from embryonic endoderm.

21) If you think of the earthworm body plan as a drinking straw within a pipe, where would you expect to find most of the tissues that developed from endoderm?
A) Lining the straw.
B) Lining the space between the pipe and the straw.
C) Forming the outside of the pipe.
D) Forming the outside of the straw.

22) Among protostomes, which morphological trait has shown the most variation?
A) Type of symmetry (bilateral vs. radial vs. none).
B) Type of body cavity (coelom vs. pseudocoelom vs. no coelom).
C) Number of embryonic tissue types (diploblasty vs. triploblasty).
D) Type of development (protostome vs. deuterostome).

23) What do all deuterostomes have in common?
A) Adults are bilaterally symmetrical.
B) Embryos have pharyngeal pouches that may or may not form gill slits.
C) All have a spinal column.
D) The pore (blastopore) formed during gastrulation becomes the anus.

24) Soon after the coelom begins to form, a researcher injects a dye into the coelom of a deuterostome embryo. Initially, the dye should be able to flow directly into the _____.
A) Blastopore.
B) Blastocoel.
C) Archenteron.
D) Pseudocoelom.

25) You have before you a living organism, which you examine carefully. Which of the following should convince you that the organism is acoelomate?
A) It is triploblastic.
B) It has bilateral symmetry.
C) It possesses sensory structures at its anterior end.
D) Muscular activity of its digestive system distorts the body wall.

Use the following information to answer the question (26) below.

Trichoplax adhaerens **(Tp) is the only living species in the phylum Placozoa. Individuals are about 1 mm wide and only 27 µm high, are irregularly shaped, and consist of a total of about 2000 cells, which are diploid (2n = 12). There are four types of cells, none of which are nerve or muscle cells, and none of which have cell walls. They move using cilia, and any "edge" can lead. Tp feeds on marine microbes, mostly unicellular green algae, by crawling atop the algae and trapping it between its ventral surface and the substrate. Enzymes are then secreted onto the algae, and the resulting nutrients are absorbed. Tp sperm cells have never been observed, nor have embryos past the 64 -cell (blastula) stage.**

26) On the basis of information in the paragraph above, which of these should be able to be observed in Tp?
A) The act of fertilization.
B) The process of gastrulation.
C) Eggs.
D) All three of the listed responses are correct.

Use the following information to answer the questions (27-28) below.

A student encounters an animal embryo at the eight -cell stage. The four smaller cells that comprise one hemisphere of the embryo seem to be rotated 45 degrees and to lie in the grooves between larger, underlying cells.

27) This embryo may potentially develop into a _____.
A) Turtle.
B) Earthworm.
C) Sea star.
D) Sea urchin.

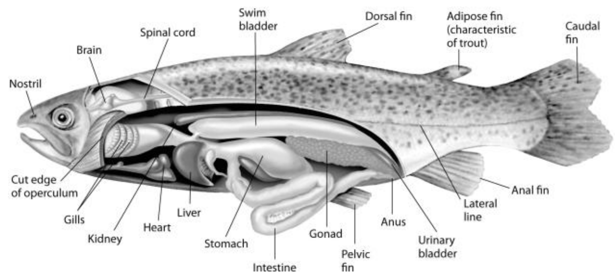
28) If we were to separate these eight cells and attempt to culture them individually, then what is most likely to happen?
A) All eight cells will die immediately.
B) Each cell may continue development, but only into a nonviable embryo that lacks many parts.
C) Each cell may develop into a full-sized, normal embryo.
D) Each cell may develop into a smaller-than-average, but otherwise normal, embryo.

Use the following information to answer the question (29) below.

The most recently discovered phylum in the animal kingdom (1995) is the phylum Cycliophora. It includes three species of tiny organisms that live in large numbers on the outsides of the mouthparts and appendages of lobsters. The feeding stage permanently attaches to the lobster via an adhesive disk and collects scraps of food from its host's feeding by capturing the scraps in a current created by a ring of cilia. The body is sac-like and has a U-shaped intestine that brings the anus close to the mouth. Cycliophorans are coelomates, do not molt (though their host does), and their embryos undergo spiral cleavage.

29) Which of these features is LEAST useful in assigning the phylum Cycliophora to a clade of animals?
A) Having a true coelom as a body cavity.
B) Having a body symmetry that permits a U-shaped intestine.
C) Having embryos with spiral cleavage.
D) Lacking ecdysis (molting).

Use the following figure and information to answer the question (30) below.



Fishes that have swim bladders can regulate their density and, thus, their buoyancy. There are two types of swim bladder: physostomous and physoclistous. The ancestral version is the physostomous version, in which the swim bladder is connected to the esophagus via a short tube (see the figure above). The fish fills this version by swimming to the surface, taking gulps of air, and directing them into the swim bladder. Air is removed from this version by "belching". The physoclistous version is more derived, and has lost its connection to the esophagus. Instead, gas enters and leaves the swim bladder via special circulatory mechanisms within the wall of the swim bladder.

30) We should expect the inner wall of the swim bladder to be lined with tissue that is derived from ______.
A) Ectoderm.
B) Endoderm.
C) Mesoderm.
D) Mesoglea.

31) What was an early selective advantage of a coelom in animals? A coelom _____.
A) Contributed to a hydrostatic skeleton, allowing greater range of motion.
B) Was a more efficient digestive system.
C) Allowed cephalization and the formation of a cerebral ganglion.
D) Allowed asexual and sexual reproduction.

32) The protostome developmental sequence arose just once in evolutionary history, resulting in two main subgroups: Lophotrochozoa and Ecdysozoa. What does this finding suggest?
A) These two subgroups have a common ancestor that was a deuterostome.
B) The protostomes are a polyphyletic group.
C) Division of these two groups occurred after the protostome developmental sequence appeared.
D) The lophotrochozoans are monophyletic.

33) Which of these, if true, would support the claim that the ancestral cnidarians had bilateral symmetry?
1. Cnidarian larvae possess anterior-posterior, left-right, and dorsal-ventral aspects.
2. Cnidarians have fewer *Hox* genes than bilaterians.
3. All extant cnidarians, including Nematostella, are diploblastic.
4. β-catenin turns out to be essential for gastrulation in all animals in which it occurs.
5. All cnidarians are acoelomate.
A) 1 and 4
B) 2 and 3
C) 2 and 4
D) 4 and 5.

34) An organism that exhibits cephalization probably also _____.
A) Is bilaterally symmetrical
B) Has a coelom
C) Is segmented
D) Is diploblastic.

35) Suppose a researcher for a pest-control company developed a chemical that inhibited the development of an embryonic mosquito's endodermal cell. Which of the following would be a likely mechanism by which this pesticide works?
A) The mosquito would develop a weakened exoskeleton that would make it vulnerable to trauma.
B) The mosquito would have trouble digesting food, due to impaired gut function.
C) The mosquito would have trouble with respiration and circulation, due to impaired muscle function.
D) The mosquito wouldn't be affected at all.

Use the following information to answer the question (36) below.

***Trichoplax adhaerens* (Tp) is the only living species in the phylum Placozoa. Individuals are about 1 mm wide and only 27 μm high, are irregularly shaped, and consist of a total of about 2000 cells, which are diploid (2n = 12). There are four types of cells, none of which are nerve or muscle cells, and none of which have cell walls. They move using cilia, and any "edge" can lead. Tp feeds on marine microbes, mostly unicellular green algae, by crawling atop the algae and trapping it between its ventral surface and the substrate. Enzymes are then secreted onto the algae, and the resulting nutrients are absorbed. Tp sperm cells have never been observed, nor have embryos past the 64 - cell (blastula) stage.**

36) Tp's body symmetry seems to be most like that of _____.
A) Most sponges.
B) Cnidarians.
C) Worms.
D) Tetrapods.

Use the following information to answer the question (37) below.

The most recently discovered phylum in the animal kingdom (1995) is the phylum Cycliophora. It includes three species of tiny organisms that live in large numbers on the outsides of the mouthparts and appendages of lobsters. The feeding stage permanently attaches to the lobster via an adhesive disk and collects scraps of food from its host's feeding by capturing the scraps in a current created by a ring of cilia. The body is sac-like and has a U-shaped intestine that brings the anus close to the mouth. Cycliophorans are coelomates, do not molt (though their host does), and their embryos undergo spiral cleavage.

37) Cycliophorans have two types of larvae. One type of larva is produced when the digestive system of a female is impregnated by a male. The digestive system then collapses and develops into a larva, which swims away in search of a new host after the surrounding female dies. Which is the embryonic tissue that is apparently most important in forming this type of larva?
A) Mesophyll.
B) Mesoderm.
C) Ectoderm.
D) Endoderm.

Use the following information to answer the question (38) below.

Nudibranchs, a type of predatory sea slug, can have various protuberances (that is, extensions) on their dorsal surfaces. Rhinophores are paired structures, located close to the head, which bear many chemoreceptors. Dorsal plummules, usually located posteriorly, perform respiratory gas exchange. Cerata usually cover much of the dorsal surface and contain nematocysts at their tips.

38) If nudibranch rhinophores are located at the anterior ends of these sea slugs, then they contribute to the sea slugs' _____.
A) Segmentation.
B) Lack of torsion.
C) Cephalization.
D) Identity as lophotrochozoans.

39) Fossil evidence indicates that the following events occurred in what sequence, from earliest to most recent?
1. Protostomes invade terrestrial environments.
2. Cambrian explosion occurs.
3. Deuterostomes invade terrestrial environments.
4. Vertebrates become top predators in the seas.
A) 2 → 1 → 4 → 3.
B) 2 → 4 → 1 → 3.
C) 2 → 3 → 1 → 4.
D) 2 → 1 → 3 → 4.

40) What is the probable sequence in which the following clades of animals originated, from earliest to most recent?
1. Tetrapods.
2. Vertebrates.
3. Deuterostomes.
4. Amniotes.
5. Bilaterians.
A) 5 → 3 → 2 → 1 → 4.
B) 5 → 3 → 4 → 2 → 1.
C) 3 → 5 → 4 → 2 → 1.
D) 3 → 5 → 2 → 1 → 4.

41) The most ancient branch point in animal phylogeny is the characteristic of having _____.
A) Radial or bilateral symmetry.
B) Diploblastic or triploblastic embryos.
C) True tissues or no tissues.
D) A body cavity or no body cavity.

42) You find a new species of worm and want to classify it. Which of the following lines of evidence would allow you to classify the worm as a nematode and not an annelid?
A) It is segmented.
B) It is triploblastic.
C) It has a coelom.
D) It sheds its external skeleton to grow.

Use the following information to answer the question (43) below.

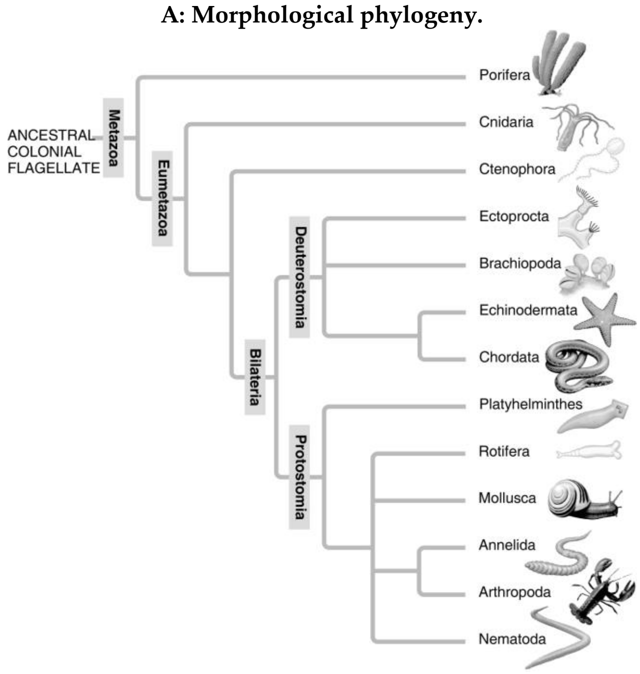
The most recently discovered phylum in the animal kingdom (1995) is the phylum Cycliophora. It includes three species of tiny organisms that live in large numbers on the outsides of the mouthparts and appendages of lobsters. The feeding stage permanently attaches to the lobster via an adhesive disk and collects scraps of food from its host's feeding by capturing the scraps in a current created by a ring of cilia. The body is sac-like and has a U-shaped intestine that brings the anus close to the mouth. Cycliophorans are coelomates, do not molt (though their host does), and their embryos undergo spiral cleavage.

43) Using similarities in embryonic development, body symmetry, and other anatomical features to assign an organism to a clade involves _____.
1. Cladistics based on body plan.
2. Molecular-based phylogeny.
3. Morphology-based phylogeny.
A) 1 only.
B) 2 only.
C) 3 only.
D) 1 and 3.

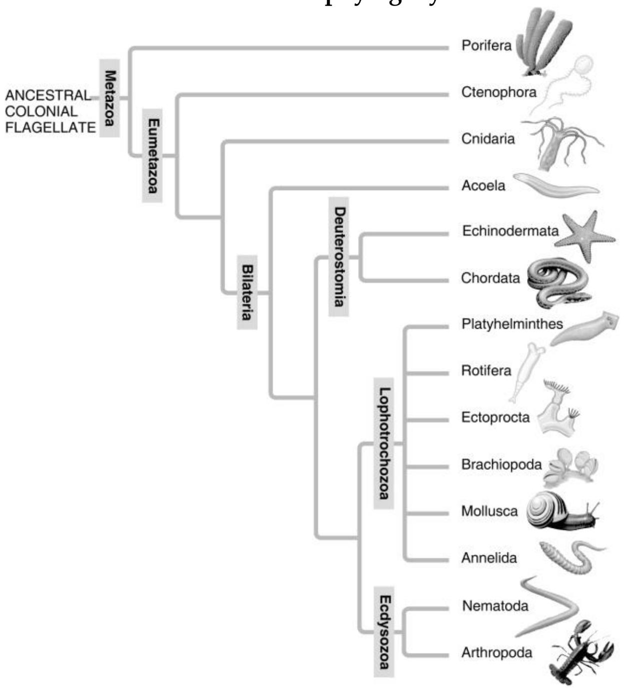
44) The common ancestor of the protostomes had a coelom. What does this suggest?
A) All lophotrochozoans have a coelom.
B) There are no pseudocoelomates within the protostomes.
C) There are no acoelomates within the protostomes.
D) The body cavity evolved before the lophophore.

Use the following information to answer the question (45) below.

A: Morphological phylogeny.



B: Molecular phylogeny.



45) In the traditional phylogeny (A), the phylum Platyhelminthes is depicted as a sister taxon to the rest of the protostome phyla and as having diverged earlier from the lineage that led to the rest of the protostomes. In the molecular phylogeny (B), Platyhelminthes is depicted as a lophotrochozoan phylum. What probably led to this change?
A) Platyhelminthes ceased to be recognized as true protostomes.
B) The removal of the acoel flatworms (Acoela) from the Platyhelminthes allowed the remaining flatworms to be a monophyletic clade clearly tied to the Lophotrochozoa.
C) All Platyhelminthes must have a well-developed lophophore as their feeding apparatus.
D) Platyhelminthes' close genetic ties to the arthropods became clear as their *Hox* gene sequences were studied.

46) The Hox genes came to regulate each of the following. From earliest to most recent, in what sequence did these events evolve?
1. Identity and position of paired appendages in protostome embryos.
2. Anterior-posterior orientation of segments in protostome embryos.
3. Positioning of tentacles in cnidarians.
4. Anterior-posterior orientation in vertebrate embryos.
A) 4 → 1 → 3 → 2.
B) 4 → 2 → 1 → 3.
C) 3 → 2 → 1 → 4.
D) 3 → 4 → 1 → 2.

47) The last common ancestor of all bilaterians is thought to have had four *Hox* genes. Most extant cnidarians have two *Hox* genes, except Nematostella (of β-catenin fame), which has three *Hox* genes. On the basis of these observations, some have proposed that the ancestral cnidarians were originally bilateral and, in stages, lost *Hox* genes from their genomes. If true, this would mean that _____.
A) The Radiata should be a true clade.
B) The radial symmetry of extant cnidarians is secondarily derived, rather than being an ancestral trait.
C) *Hox* genes play little actual role in coding for an animal's "body plan".
D) The Cnidaria may someday replace porifera as the basal bilaterians.

48) Some researchers claim that sponge genomes have homeotic genes, but no Hox genes. If true, this finding would _____.
A) Mean that sponges must no longer be classified as animals.
B) Confirm the identity of sponges as "basal animals".
C) Mean that extinct sponges must have been the last common ancestor of animals and fungi.
D) Require sponges to be reclassified as choanoflagellates.

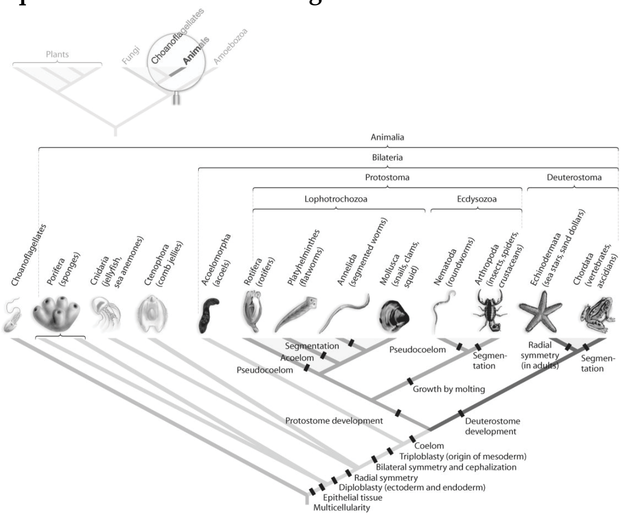
Use the following information to answer the question (49-50) below.

The most recently discovered phylum in the animal kingdom (1995) is the phylum Cycliophora. It includes three species of tiny organisms that live in large numbers on the outsides of the mouthparts and appendages of lobsters. The feeding stage permanently attaches to the lobster via an adhesive disk and collects scraps of food from its host's feeding by capturing the scraps in a current created by a ring of cilia. The body is sac-like and has a U-shaped intestine that brings the anus close to the mouth. Cycliophorans are coelomates, do not molt (though their host does), and their embryos undergo spiral cleavage.

49) Which of these, if discovered among cycliophorans, would cause the most confusion concerning our current understanding of cycliophoran taxonomy?
A) If the ciliated feeding ring is a lophophore.
B) If embryos are diploblastic.
C) If the body cavity is actually a pseudocoelom.
D) If the organisms show little apparent cephalization.

50) The feeding stage of cycliophorans _____.
1. Is autotrophic.
2. Is sessile.
3. Captures food in a manner similar to that of animals with lophophores.
4. Shows radial symmetry.
A) 1 and 2.
B) 1 and 3.
C) 2 and 4.
D) 1, 2, and 3.

51) Which morphological trait evolved more than once in animals, according to the phylogeny based on DNA sequence data found in the figure below?



A) Coelom.
B) Bilateral symmetry.
C) Segmentation.
D) Protostome development.

52) Why might researchers choose to use molecular data (such as ribosomal RNA sequences) rather than morphological data to study the evolutionary history of animals?
A) Molecular data can be gathered in the lab, while morphological data must be gathered in the field.
B) Sequence data can be gathered faster than morphological data, and morphological data provides a different perspective.
C) Morphological changes usually do not result from molecular changes.
D) Some phyla vary too widely in morphological characteristics to be classified accurately.

53) If in the future the current molecular evidence regarding animal origins is further substantiated, what will be true of any contrary evidence regarding the origin of animals derived from the fossil record?
A) The contrary fossil evidence will be seen as a hoax.
B) The fossil evidence will be understood to have been interpreted incorrectly because it is incomplete.
C) The fossil record will henceforth be ignored.
D) Phylogenies involving even the smallest bit of fossil evidence will need to be discarded.

Use the following information to answer the question(s) below.

The most recently discovered phylum in the animal kingdom (1995) is the phylum Cycliophora. It includes three species of tiny organisms that live in large numbers on the outsides of the mouthparts and appendages of lobsters. The feeding stage permanently attaches to the lobster via an adhesive disk and collects scraps of food from its host's feeding by capturing the scraps in a current created by a ring of cilia. The body is sac-like and has a U-shaped intestine that brings the anus close to the mouth. Cycliophorans are coelomates, do not molt (though their host does), and their embryos undergo spiral cleavage.

54) Basing your inferences on information in the paragraph above, to which clade(s) should cycliophorans belong?
1. Eumetazoa.
2. Deuterostomia.
3. Bilateria.
4. Ecdysozoa.
5. Lophotrochozoa.
A) 1 and 3.
B) 1, 3, and 5.
C) 2, 3, and 4.
D) 2, 3, and 5.

Use the following table to answer the question (55-57) below.

Last Common Ancestor of Bilateria	Last Common Ancestor of Insects and Vertebrates	Ancestral Vertebrates	Mammals
4	7	14	38-40

55) What conclusion is apparent from the data in the table above?
A) Land animals have more *Hox* genes than do those that live in water.
B) All bilaterian phyla have had the same degree of expansion in their numbers of *Hox* genes.
C) The expansion in number of Hox genes throughout vertebrate evolution cannot be explained merely by three duplications of the ancestral vertebrate *Hox* cluster.
D) Extant insects all have seven *Hox* genes.

56) All things being equal, which of these is the most parsimonious explanation for the change in the number of *Hox* genes from the last common ancestor of insects and vertebrates to ancestral vertebrates, as shown in the table above?
A) The occurrence of seven independent duplications of individual *Hox* genes.
B) The occurrence of two distinct duplications of the entire seven-gene cluster, followed by the loss of one cluster.
C) The occurrence of a single duplication of the entire seven-gene cluster.
D) None of them.

57) Two competing hypotheses to account for the increase in the number of *Hox* genes from the last common ancestor of bilaterians to the last common ancestor of insects and vertebrates are: (1) a single duplication of the entire four-gene cluster, followed by the loss of one gene, and (2) three independent duplications of individual Hox genes. To prefer the first hypothesis on the basis of parsimony requires the assumption that _____.
A) The duplication of a cluster of four *Hox* genes is equally likely as the duplication of a single *Hox* gene.
B) There is an actual process by which individual genes can be duplicated.
C) Genes can exist is spatial groupings called clusters.
D) Clusters of genes can undergo disruption, with individual genes moving to different chromosomes during evolution.

Use the following figure to answer the question (58-59) below.

Phylum	*Dll* Activity in Developing Appendages
Annelida	Yes
Arthropoda	Yes
Chordata	No
Echinodermata	No
Mollusca	Yes

58) In the experiment outlined in the figure above, what would you expect to happen if researchers supplied an enzyme that blocked the expression of the *Dll* gene?
A) The embryo would have organs in abnormal locations.
B) The origins of the embryo's appendages would fluoresce.
C) The developing embryo would have no appendages.
D) The embryo's appendages would be shorter than usual.

59) *Dll* is a gene known to direct limb development in the fruit fly. Researchers studying this gene have found that it is also expressed in developing appendages in animals from many other phyla, supporting the hypothesis that all animal appendages may be homologous. However, suppose researchers looking at *Dll* activity had instead found the results shown in the figure above. These results suggest instead that _____.
A) *Dll* is not actually involved in appendage development.
B) Appendages evolved separately in protostomes and deuterostomes.
C) Appendages coevolved with segmentation.
D) All animal appendages are homologous.

60) Which of the following statements concerning animal taxonomy is (are) true?
1. Animals are more closely related to plants than to fungi.
2. All animal clades based on body plan have been found to be incorrect.
3. Kingdom Animalia is monophyletic.
4. Animals only reproduce sexually.
5. Animals are thought to have evolved from flagellated protists similar to modern choanoflagellates.
A) 1 and 2.
B) 3 and 5.
C) 3, 4, and 5.
D) 2 and 4.

CHAPTER 33:

AN INTRODUCTION TO INVERTEBRATES

1) One should expect to find cilia associated with the feeding apparatus of _____.
A) Annelids.
B) Coral animals.
C) Tapeworms.
D) Sponges.

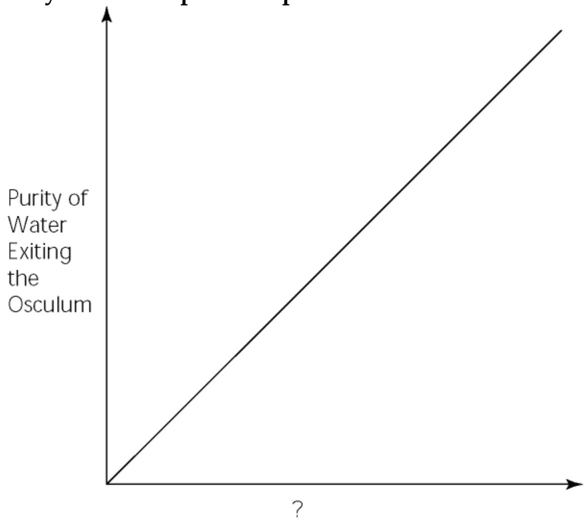
2) Sponges _____.
A) Have larvae which are motile and move via the motion of cilia.
B) Are the simplest diploblastic animals.
C) Have a nerve net but not a central nervous system.
D) Have feeding cells called dinoflagellates.

3) Which of the following is most likely to be aquatic?
A) Suspension feeder.
B) Mass feeder.
C) Deposit feeder.
D) Fluid feeder.

4) Comb jellies may not be the most familiar animal to you, but they are critical in the food chain because they make up a significant portion of the planktonic biomass. Their feeding strategy is predatory and involves adhesives or mucus on their tentacles or other body parts. What feeding tactic do these animals use?
A) Suspension feeder.
B) Fluid feeder.
C) Deposit feeder.
D) Food-mass feeder.

5) Which of the following can be found in the mesohyl of a sponge?
1. Amoebocytes.
2. Spicules.
3. Spongin.
4. Zygotes.
5. Dhoanocytes.
A) 1 and 2.
B) 2, 3, 4.
C) 1, 2, 3, and 4.
D) 1, 2, 3, 4, and 5.

6) Which of the following factors, when used to label the horizontal axis of the graph below, would account most directly for the shape of the plot?



A) Rate of cribrostatin synthesis (molecules/unit time).
B) Number of pores per sponge.
C) Number of spicules per sponge.
D) Number of choanocytes per sponge.

7) Most cnidarians are known to produce toxins. In fact, it has been claimed that one particular species produces the most deadly of all toxins on the planet. What feature of this group most likely evolved simultaneously with the evolution of these toxins?
A) The medusa body form.
B) Asexual reproduction.
C) A slow-moving or sessile lifestyle in the adult.
D) Diploblastic design.

8) Healthy corals are brightly colored because they _____.
A) Secrete colorful pigments to attract mates.
B) Host symbionts with colorful photosynthetic pigments.
C) Build their skeletons from colorful minerals.
D) Secrete colorful pigments to protect themselves from ultraviolet light.

9) In terms of food capture, which sponge cell is most similar to the cnidocyte of a cnidarian?
A) Amoebocyte.
B) Choanocyte.
C) Epidermal cell.
D) Pore cell.

10) The crown-of-thorns sea star, *Acanthaster planci*, preys on the flesh of live coral. If coral animals are attacked by these sea stars, then what actually provides nutrition to the sea star, and which chemical (besides the toxin within their nematocysts) do the corals rely on for protection?
A) Medusae; silica.
B) Exoskeleton; calcium carbonate.
C) Polyps; calcium carbonate.
D) Polyps; silica.

Use the following information to answer the question (11-14) below.

An elementary school science teacher decided to liven up the classroom with a saltwater aquarium. Knowing that saltwater aquaria can be quite a hassle, the teacher proceeded stepwise. First, the teacher conditioned the water. Next, the teacher decided to stock the tank with various marine invertebrates, including a polychaete, a siliceous sponge, several bivalves, a shrimp, several sea anemones of different types, a colonial hydra, a few coral species, an ectoproct, a sea star, and several herbivorous gastropod varieties. Lastly, she added some vertebrates: a parrotfish and a clownfish. She arranged for daily feedings of copepods and feeder fish.

11) One day, Tommy, a student in an undersupervised class of forty fifth graders, got the urge to pet Nemo (the clownfish), who was swimming among the waving petals of a pretty underwater "flower" that had a big hole in the midst of the petals. Tommy giggled upon finding that these petals felt sticky. A few hours later, Tommy was in the nurse's office with nausea and cramps. Microscopic examination of his fingers would probably have revealed the presence of _____.

A) Teeth marks.
B) Spines.
C) Spicules.
D) Nematocysts.

12) The teacher and class were especially saddened when the colonial hydrozoan died. They had watched it carefully, and the unfortunate creature never even got to produce offspring by budding. Yet, everyone was elated when one of the students noticed a small colonial hydrozoan growing in a part of the tank far from the location of the original colony. The teacher was apparently unaware that these hydrozoans exhibit _____.

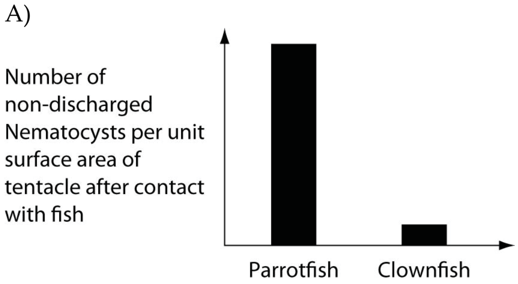
A) Spontaneous generation.
B) Abiogenesis.
C) Alternation of generations.
D) A medusa stage.

13) The sharp, inch-long thorns of the crown-of-thorns sea star are its spines. These spines, unlike those of most other sea stars, contain a potent toxin. If it were discovered that crown-of-thorns sea stars do not make this toxin themselves, then the most likely alternative would be that this toxin is _____.

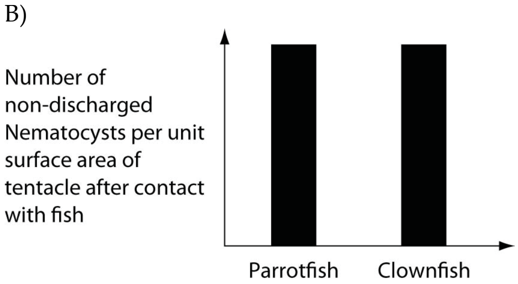
A) Derived from the nematocysts of its prey.
B) Absorbed from the surrounding seawater.
C) An endotoxin of cellulose-digesting bacteria that inhabit the sea star's digestive glands.
D) Injected into individual thorns by mutualistic corals which live on the aboral surfaces of these sea stars.

14) The clownfish readily swims among the tentacles of the sea anemones; the parrotfish avoids them. One hypothesis for the clownfish's apparent immunity is that they slowly build a tolerance to the sea anemone's toxin. A second hypothesis is that a chemical in the mucus that coats the clownfish prevents the nematocysts from being triggered. Which of the following graphs supports the second, but not the first, of these hypotheses?

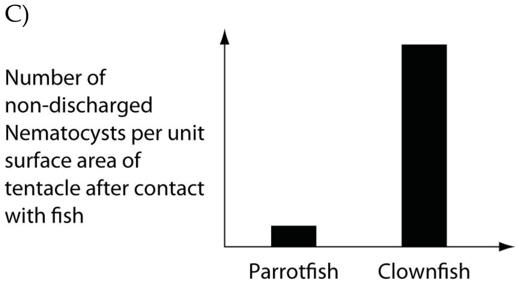
A)



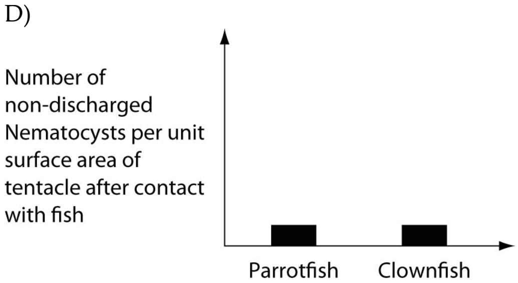
B)



C)



D)



15) The presence of a lophophore in a newly discovered species would suggest that the species _____.

A) Has an exoskeleton.
B) Grows by shedding its external covering.
C) Is motile.
D) Is a suspension feeder.

16) You find what you believe is a new species of animal. Which of the following characteristics would enable you to argue that it is more closely related to a flatworm than it is to a roundworm?

A) It is a suspension feeder.
B) It has no coelom.
C) It is shaped like a worm.
D) It has a mouth and an anus.

17) What would be the best anatomical feature to look for to distinguish a gastropod from a chiton?
A) Presence of a muscular foot.
B) Presence of a rasp-like feeding structure.
C) Production of eggs.
D) Number of shell plates.

18) Which of the following organisms would you expect to have the largest surface-area-to-volume ratio? Assume that all of the following are the same total length.
A) A mollusk.
B) An annelid.
C) An arthropod.
D) A platyhelminth.

19) Against which hard structure do the circular and longitudinal muscles of annelids work?
A) Cuticle.
B) Shell.
C) Endoskeleton.
D) Hydrostatic skeleton.

20) While sampling marine plankton in a lab, a student encounters large numbers of fertilized eggs. The student rears some of the eggs in the laboratory for further study and finds that the blastopore becomes the mouth. The embryo develops into a trochophore larva and eventually has a true coelom. These eggs probably belonged to a(n) _____.
A) Echinoderm.
B) Mollusk.
C) Nematode.
D) Arthropod.

This nudibranch, a type of sea slug, has many reddish cerata on its dorsal surface, as well as two white-tipped rhinophores located on the head.

The nontaxonomic term sea slug encompasses a wide variety of marine gastropods. One feature they share as adults is the lack of a shell. We might think, therefore, that they represent defenseless morsels for predators. In fact, sea slugs have multiple defenses. Some sea slugs prey on sponges and concentrate sponge toxins in their tissues. Others feed on cnidarians, digesting everything except the nematocysts, which they then transfer to their own skins. Whereas the most brightly colored sea slugs are often highly toxic, others are nontoxic and mimic the coloration of the toxic species. Their colors are mostly derived from pigments in their prey. There are also sea slugs that use their coloration to blend into their environments.

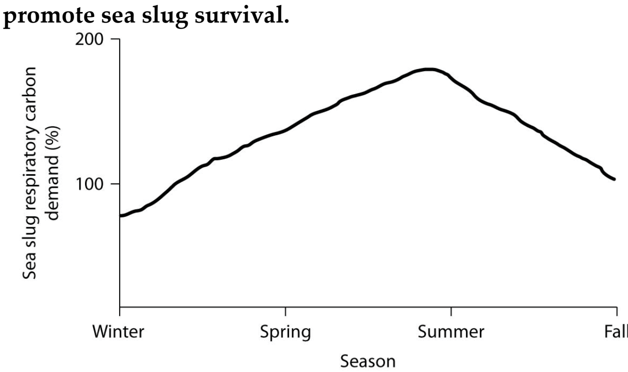
21) Which structure do sea slugs use to feed on their prey?
A) Nematocysts.
B) An incurrent siphon.
C) A radula.
D) A mantle cavity.

22) The nematocysts most likely reach the skin of sea slugs through branches of the _____.
A) Intestine.
B) Excurrent siphon.
C) Nephridium.
D) Pseudocoelom.

23) The nematocysts of sea slugs should be most effective at protecting individual sea slugs from predation if the predators _____.
A) Remove small bites of flesh from sea slugs and have long-term memory.
B) Remove small bites of flesh from sea slugs and have no long-term memory.
C) Consume entire sea slugs in one gulp and have no long-term memory.
D) Consume entire sea slugs in one gulp and have long-term memory.

Use the following information and figure to answer the question(s) below.

The sea slug *Pteraeolidia ianthina* (*P. ianthina*) can harbor living dinoflagellates (photosynthetic protists) in its skin. These endosymbiotic dinoflagellates reproduce quickly enough to maintain their populations. Low populations do not affect the sea slugs very much, but high populations (> 5 x 105 cells/mg of sea slug protein) can promote sea slug survival.



Percent of sea slug respiratory carbon demand provided by indwelling dinoflagellates.

24) If the dinoflagellate-containing sea slug *P. ianthina* preys on coral animals, then it would be LEAST surprising to find that _____.
A) *P. ianthina* has no tolerance to the toxin in the nematocysts of its prey.
B) *P. ianthina* can locate its coral prey by chemicals released into the water by corals.
C) The coral prey harbor dinoflagellates in their tissues.
D) The coral prey transform themselves into medusas to flee from approaching *P. ianthina*.

25) The sea slug *Elysia chorotica* has no nematocysts or dinoflagellates but, rather, has "naked" chloroplasts in its skin. The chloroplasts are all that remain of the seaweed (*Vaucheria sp.*) that *Elysia* feeds upon. The chloroplasts are transferred to the skin; consequently, this slug is green. It spends most of its time basking in shallow water on the surface of seaweeds. How should we expect its chloroplasts to benefit the *Elysia* sea slug?
1. Provide *Elysia* with fixed carbon dioxide.
2. Provide *Elysia* with fixed nitrogen.
3. Provide *Elysia* with protective coloration.
A) 1 only.
B) 2 only.
C) 3 only.
D) 1 and 3.

Use the following information to answer the question (26-27) below.

Nudibranchs, a type of predatory sea slug, can have various protuberances (that is, extensions) on their dorsal surfaces. Rhinophores are paired structures, located close to the head, which bear many chemoreceptors. Dorsal plummules, usually located posteriorly, perform respiratory gas exchange. Cerata usually cover much of the dorsal surface and contain nematocysts at their tips.

26) Nudibranchs usually have two rhinophores. However, if they had a single rhinophore, it could still carry out the function of two rhinophores, and with similar effectiveness, if this single rhinophore _____.
A) Had two branches, one directed to the left, and the other to the right.
B) Was located within the mantle cavity.
C) Was as long as two rhinophores placed end to end.
D) Had cilia whose power strokes directed water away from the surface of the slug.

27) A natural predator of the crown-of-thorns sea star is a mollusc called the Giant Triton, *Charonia tritonis*. If the triton uses a radula to saw into the sea star, then to which clade should the triton belong?
A) Chitons.
B) Bivalves.
C) Gastropods.
D) Cephalopods.

Use the following information to answer the question (28-30) below.

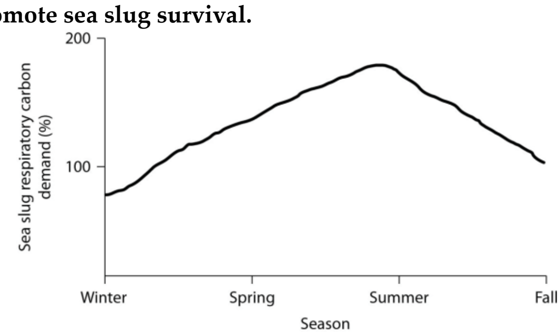
An elementary school science teacher decided to liven up the classroom with a saltwater aquarium. Knowing that saltwater aquaria can be quite a hassle, the teacher proceeded stepwise. First, the teacher conditioned the water. Next, the teacher decided to stock the tank with various marine invertebrates, including a polychaete, a siliceous sponge, several bivalves, a shrimp, several sea anemones of different types, a colonial hydra, a few coral species, an ectoproct, a sea star, and several herbivorous gastropod varieties. Lastly, she added some vertebrates: a parrotfish and a clownfish. She arranged for daily feedings of copepods and feeder fish.

28) If the teacher wanted to show the students what a lophophore is and how it works, the teacher would point out a feeding _____.
A) Hydra.
B) Sponge.
C) Gastropod.
D) Ectoproct.

29) The teacher was unaware of the difference between suspension feeding and predation. The teacher thought that providing live copepods (2 mm long) and feeder fish (2 cm long) would satisfy the dietary needs of all of the organisms. Consequently, which two organisms would have been among the first to starve to death (assuming they lack photosynthetic endosymbionts)?
A) Sponges and corals.
B) Sea stars and sponges.
C) Shrimp and bivalves.
D) Bivalves and sponges.

30) What would be the most effective method of reducing the incidence of blood flukes in a human population?
A) Reduce the mosquito population.
B) Reduce the population of the intermediate snail host.
C) Avoid contact with rodent droppings.
D) Carefully wash all raw fruits and vegetables.

Use the following information and figure to answer the question (31-33) below.
The sea slug *Pteraeolidia ianthina (P. ianthina)* can harbor living dinoflagellates (photosynthetic protists) in its skin. These endosymbiotic dinoflagellates reproduce quickly enough to maintain their populations. Low populations do not affect the sea slugs very much, but high populations (> 5 x 10^5 cells/mg of sea slug protein) can promote sea slug survival.



Percent of sea slug respiratory carbon demand provided by indwelling dinoflagellates.

31) According to the graph, during which season(s) of the year is the relationship between the sea slug and its dinoflagellates closest to being commensal?
A) Winter.
B) Spring.
C) Summer.
D) Spring and summer.

32) Planarians lack dedicated respiratory and circulatory systems because _____.
A) None of their cells are far removed from the gastrovascular cavity or from the external environment.
B) They lack mesoderm as embryos and, therefore, lack the adult tissues derived from mesoderm.
C) Their flame bulbs can carry out respiratory and circulatory functions.
D) Their body cavity, a pseudocoelom, carries out these functions.

33) Which one of these mollusk groups can be classified as suspension feeders?
A) Bivalves.
B) Gastropods.
C) Chitons.
D) Cephalopods.

34) Which characteristic is shared by cnidarians and flatworms?
A) Dorsoventrally flattened bodies.
B) Radial symmetry.
C) A digestive system with a single opening.
D) A distinct head.

35) If a lung were to be found in a mollusc, where would it be located?
A) Mantle cavity.
B) Incurrent siphon.
C) Visceral mass.
D) Excurrent siphon.

36) Parasitism is one of the most widespread life strategies ever to evolve. Which of the following is consistent with this finding?
A) Parasites almost always predigest their hosts' tissues and, therefore, spend less energy and require fewer structural adaptations.
B) Parasites, unlike predators, feed on almost all the tissues of their host.
C) Parasites do not generally kill their hosts; thus they can feed on the same host throughout the host's normal life span and do not have competition from decomposers.
D) Parasites generally kill their host and can feed for a very long time because they are much smaller than their host.

37) Nematodes and arthropods both _____.
A) Develop an anus from the blastopore (pore) formed in the gastrula stage.
B) Are suspension feeders.
C) Grow by shedding their exoskeleton.
D) Have ciliated larvae.

38) Arthropod exoskeletons and mollusk shells both _____.
A) Completely replace the hydrostatic skeleton.
B) Are secreted by the mantle.
C) Help retain moisture in terrestrial habitats.
D) Are comprised of the polysaccharide chitin.

39) You find a multi-legged animal in your garden and want to determine if it is a centipede or a millipede. You take the animal to a university where a myriapodologist quickly tells you that you have found a centipede. Which of the following may have allowed her to make this distinction?
A) Segmentation.
B) Poisonous fangs.
C) Egg-laying.
D) Molting.

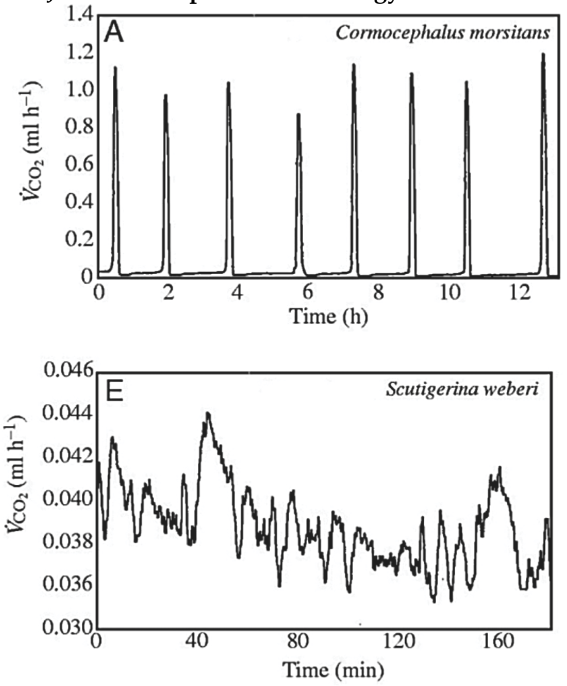
40) Whiteflies are common pest insects found on cotton, tomato, poinsettia, and many other plants. Nymphs are translucent and mostly sessile, feeding on their host plants' phloem (sap) from the undersides of leaves. They undergo incomplete metamorphosis into winged adults. Because whitefly nymphs cannot escape predation by moving, you hypothesize that their translucent bodies make them hard to spot by predators. How could you directly test this hypothesis?
A) Compare rates of predation on whitefly nymphs on plant leaves of different colors (for example, red vs. green poinsettia leaves).
B) Compare rates of predation on whitefly nymphs coated with a nontoxic dye vs. undyed whitefly nymphs.
C) Compare rates of predation on whitefly nymphs vs. whitefly adults.
D) Compare rates of predation on whitefly nymphs by predators that are translucent vs. predators that are not translucent.

41) All arthropods _____.
1) Undergo complete metamorphose.
2) Have jointed appendages.
3) Molt.
4) Have segmented bodies.
5) Have an exoskeleton or cuticle.
A) 1, 2, and 4.
B) 3 and 5.
C) 2, 3, 4, 5.
D) 1, 4, 5.

Use the following information when to answer the question(s) below.

Many terrestrial arthropods exchange gases with their environments by using tracheae, tubes that lead from openings (called spiracles) in the animal's exoskeleton or cuticle directly to the animal's tissues. Some arthropods can control whether their spiracles are opened or closed; opening the spiracles allows the carbon dioxide produced in the tissues to travel down the tracheae and be released outside the animal. Klok et al. measured the carbon dioxide emitted over time (represented by VCO2) by several species of centipedes. The figure below presents graphs of their results for two species,

***Cormocephalus morsitans* and *Scutigerina weberi*. (C. J. Klok, R. D. Mercer, and S. L. Chown. 2002. Discontinuous gas exchange in centipedes and its convergent evolution in tracheated arthropods. *Journal of Experimental Biology* 205:1019-29.) Copyright 2002. The Company of Biologists and the Journal of Experimental Biology.**



42) Look at the graph for *Cormocephalus morsitans* in the figure above. What is the best interpretation of these results?
A) The centipede had its spiracles open the entire time.
B) The centipede had its spiracles closed the entire time.
C) The centipede had its spiracles open when carbon dioxide (CO_2) emission peaked and closed when CO_2 emission was low.
D) The centipede had its spiracles closed when carbon dioxide (CO_2) emission peaked and open when CO_2 emission was low.

43) Look at the graph for *Scutigerina weberi* (note the scale of the y-axis) in the figure above. What is the best interpretation of these results?
A) The centipede had its spiracles open the entire time.
B) The centipede had its spiracles closed the entire time.
C) The centipede had its spiracles open when carbon dioxide (CO_2) emission peaked and closed when CO_2 emission was low.
D) The centipede had its spiracles closed when carbon dioxide (CO_2) emission peaked and open when CO_2 emission was low.

44) How would a terrestrial centipede most likely benefit from the ability to close its spiracles? Closing spiracles would _____.
A) Allow the centipede to move more quickly.
B) Allow the centipede to retain more moisture in its tissues.
C) Allow the centipede to stay warmer.
D) Allow more oxygen from the environment to reach the centipede's tissues.

45) Compare the graphs in the figure above of carbon dioxide (CO_2) emission for *Cormocephalus morsitans* and *Scutigerina weberi*. What hypothesis can you make about each centipede's habitat?
A) *C. morsitans* lives in a habitat that provides more carbon dioxide than does *S. weberi*.
B) *C. morsitans* lives in a habitat with more predators than does *S. weberi*.
C) *C. morsitans* lives in a colder habitat than does *S. weberi*.
D) *C. morsitans* lives in a drier habitat than does *S. weberi*.

46) What would be the most direct effect of removing or damaging an insect's antenna? The insect would have trouble _____.
A) Hearing.
B) Mating.
C) Seeing.
D) Smelling.

47) The heartworms that can accumulate within the hearts of dogs and other mammals have a pseudocoelom, an alimentary canal, and an outer covering that is occasionally shed. To which phylum does the heartworm belong?
A) Platyhelminthes.
B) Arthropoda.
C) Nematoda.
D) Annelida.

48) A terrestrial animal species is discovered with the following larval characteristics: exoskeleton, system of tubes for gas exchange, and modified segmentation. A knowledgeable zoologist should predict that the adults of this species would also feature _____.
A) Eight legs.
B) Two pairs of antennae.
C) A sessile lifestyle.
D) An open circulatory system.

49) In a tide pool, a student encounters an organism with a hard outer covering that contains much calcium carbonate, an open circulatory system, and gills. The organism could potentially be a crab, a shrimp, a barnacle, or a bivalve. The presence of which of the following structures would allow for the most certain identification of the organism?
A) A mantle.
B) A heart.
C) A body cavity.
D) A filter-feeding apparatus.

Use the following information to answer the question (50-53) below.

Nudibranchs, a type of predatory sea slug, can have various protuberances (that is, extensions) on their dorsal surfaces. Rhinophores are paired structures, located close to the head, which bear many chemoreceptors. Dorsal plummules, usually located posteriorly, perform respiratory gas exchange. Cerata usually cover much of the dorsal surface and contain nematocysts at their tips.

50) The claws (fangs) on the foremost trunk segment of centipedes have a function most similar to that of _____.
A) Rhinophores.
B) Dorsal plummules.
C) Cerata.
D) Chemoreceptors.

51) The stingers of honeybees have a function most similar to that of _____.
A) Rhinophores.
B) Dorsal plummules.
C) Cerata.
D) Chemoreceptors.

52) The spiracles and tracheae of insects have a function most similar to that of _____.
A) Rhinophores.
B) Dorsal plummules.
C) Cerata.
D) Chemoreceptors.

53) The antennae of insects have a function most similar to that of _____.
A) Rhinophores.
B) Dorsal plummules.
C) Cerata.
D) Chemoreceptors.

Use the following information to answer the question (54-59) below.

A farm pond, usually dry during winter, has plenty of water and aquatic pond life during the summer. One summer, Sarah returns to the family farm from college. Observing the pond, she is fascinated by some six-legged organisms that can crawl about on submerged surfaces or, when disturbed, seemingly "jet" through the water. Watching further, she is able to conclude that the "mystery organisms" are ambush predators, and their prey includes everything from insects to small fish and tadpoles.

54) If the pond organisms are larvae, rather than adults, Sarah should expect them to have all of the following structures, EXCEPT _____.
A) Antennae.
B) An open circulatory system.
C) An exoskeleton of chitin.
D) Sex organs.

55) Sarah observed that the mystery pond organisms never come up to the pond's surface. If she catches one of these organisms and observes closely, perhaps dissecting the organism, she should find _____.
A) Gills.
B) Spiracles.
C) Tracheae.
D) Book lungs.

56) As you are walking along a beach, you find an animal and believe that it belongs to the class Asteroidea. Which of the following characteristics would support your hypothesis that the animal is a sea star and not another type of echinoderm?
A) It is pentaradially symmetric.
B) It feeds on other animals.
C) It has a hydrostatic skeleton, formed from its water vascular system.
D) Its central region is not well-delineated from its appendages.

57) The water vascular system of echinoderms _____.
A) Functions as a circulatory system that distributes nutrients to body cells.
B) Functions in locomotion and feeding.
C) Is bilateral in organization, even though the adult animal is not bilaterally symmetrical.
D) Is analogous to the gastrovascular cavity of flatworms.

58) Which of the following combinations correctly matches a phylum to its description?
A) Echinodermata - bilateral symmetry as a larva, water vascular system.
B) Nematoda - segmented worms, closed circulatory system.
C) Cnidaria - flatworms, gastrovascular cavity, acoelomate.
D) Platyhelminthes - radial symmetry, polyp and medusa body forms.

59) Which of the following animal groups is entirely aquatic?
A) Mollusca.
B) Crustacea.
C) Echinodermata.
D) Nematoda.

Use the following information to answer the question (60) below.

An elementary school science teacher decided to liven up the classroom with a saltwater aquarium. Knowing that saltwater aquaria can be quite a hassle, the teacher proceeded stepwise. First, the teacher conditioned the water. Next, the teacher decided to stock the tank with various marine invertebrates, including a polychaete, a siliceous sponge, several bivalves, a shrimp, several sea anemones of different types, a colonial hydra, a few coral species, an ectoproct, a sea star, and several herbivorous gastropod varieties. Lastly, she added some vertebrates: a parrotfish and a clownfish. She arranged for daily feedings of copepods and feeder fish.

60) The bivalves started to die one by one; only the undamaged shells remained. To keep the remaining bivalves alive, the teacher would most likely need to remove the _____.

A) Sea anemones.
B) Sea star.
C) Gastropods.
D) Ectoprocts.

CHAPTER 34:

THE ORIGIN AND EVOLUTION OF VERTEBRATES

1) Which of the following is a characteristic of all chordates at some point during their life cycle?
A) Jaws.
B) Post-anal tail.
C) Four-chambered heart.
D) Vertebrae.

2) Why do adult urochordates (tunicates) lack notochords, even though larval urochordates have them? Larvae use notochords to _____.
A) Aid in swimming; adults are sessile and thus no longer propel themselves.
B) Stiffen their bodies; in adults, the notochord is replaced by a column of bone.
C) Induce tissue differentiation; in adults, tissue is already differentiated.
D) Organize their nervous systems; adults' nervous systems are fully developed and do not change.

3) If a tunicate's pharyngeal gill slits were suddenly blocked, the animal would have trouble _____.
A) Respiring.
B) Feeding.
C) Moving.
D) Respiring and feeding.

4) Chordate pharyngeal slits appear to have functioned first as _____.
A) The digestive system's opening.
B) Suspension-feeding devices.
C) Components of the jaw.
D) Sites of respiration.

5) Which of the following statements would be LEAST acceptable to most zoologists?
A) The first fossils resembling lancelets appeared in the fossil record around 530 million years ago.
B) Recent work in molecular systematics supports the hypothesis that lancelets are the basal clade of chordates.
C) The extant lancelets are the immediate ancestors of the fishes.
D) Lancelets display the same method of swimming as do fishes.

6) Which extant chordates are postulated to be most like the earliest chordates in appearance?
A) Lancelets.
B) Adult tunicates.
C) Amphibians.
D) Chondrichthyans.

7) Vertebrates and tunicates share _____.
A) Jaws adapted for feeding.
B) A high degree of cephalization.
C) The formation of structures from the neural crest.
D) A notochord and a dorsal, hollow nerve cord.

8) All chordates studied to date, except tunicates, share a set of _____.
A) 13 *Hox* genes.
B) 5 *Dlx* genes.
C) 9 *Otx* genes.
D) 7 *FOXP2* genes.

9) Which of the following characteristics is shared by a hagfish and a lamprey?
A) A rasping tongue.
B) Paired fins.
C) Jaws.
D) A well-developed notochord.

10) A new species of aquatic chordate is discovered that closely resembles an ancient form. It has the following characteristics: external armor of bony plates no paired lateral fins, and a suspension-feeding mode of nutrition. In addition to these, it will probably have which of the following characteristics?
A) Legs.
B) No jaws.
C) An amniotic egg.
D) Endothermy.

11) The earliest known mineralized structures in vertebrates are associated with _____.
A) Feeding.
B) Locomotion.
C) Defense.
D) Respiration.

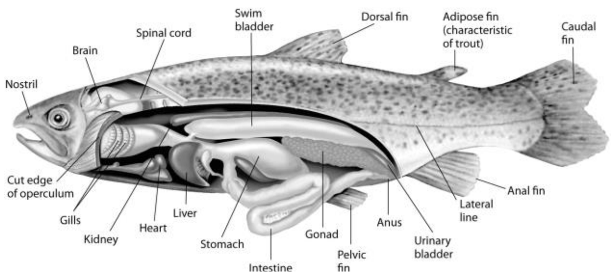
12) A team of researchers has developed a poison that has proven effective against lamprey larvae in freshwater cultures. The poison is ingested and causes paralysis by detaching segmental muscles from the skeletal elements. The team wants to test the poison's effectiveness in streams feeding Lake Michigan, but one critic worries about potential effects on lancelets, which are similar to lampreys in many ways. Why is this concern misplaced?
A) Lamprey larvae and lancelets have very different feeding mechanisms.
B) Lancelets do not have segmental muscles.
C) Lancelets live only in saltwater environments.
D) Lancelets and lamprey larvae eat different kinds of food.

13) To reproduce, many plants produce seeds–structures containing embryonic offspring along with nutrients inside a tough case. These offspring develop after being released by the parent plant. To which animal reproductive strategy is seed production most comparable?
A) Oviparous reproduction.
B) Ovoviviparous reproduction.
C) Viviparous reproduction.
D) None of them.

14) Why do skates and rays have flattened bodies, while sharks are torpedo shaped?
A) Sharks are more closely related to the tubelike lampreys than skates and rays are.
B) Skates and rays need enlarged pectoral fins to help them stay level in turbulent water, while sharks do not.
C) Skates and rays exchange gases across their skin and thus require a high surface-area-to-volume ratio, while sharks use gills to respire.
D) Sharks are streamlined for active swimming off the bottom, while skates move about mostly on the ocean bed.

15) Which of these statements accurately describes a similarity between sharks and ray-finned fishes?
A) They are equally able to exchange gases with the environment while stationary.
B) They are highly maneuverable due to their flexibility.
C) They have a lateral line that is sensitive to vibrations.
D) A swim bladder helps control buoyancy.

Use the following figure and information to answer the question (16-18) below.



Fishes that have swim bladders can regulate their density and, thus, their buoyancy. There are two types of swim bladder: physostomous and physoclistous. The ancestral version is the physostomous version, in which the swim bladder is connected to the esophagus via a short tube (see the figure above). The fish fills this version by swimming to the surface, taking gulps of air, and directing them into the swim bladder. Air is removed from this version by "belching". The physoclistous version is more derived, and has lost its connection to the esophagus. Instead, gas enters and leaves the swim bladder via special circulatory mechanisms within the wall of the swim bladder.

16) The presence of a swim bladder allows the typical ray-finned fish to stop swimming and still _____.
A) Effectively circulate its blood.
B) Use its lateral line system.
C) Use its swim bladder as a respiratory organ.
D) Not sink.

17) Which shark structure is closest in function to a swim bladder full of gas?
A) Its lateral line system.
B) Its spiral valve.
C) Its liver.
D) Its gills.

18) If a ray-finned fish is to both hover (remain stationary) in the water column and ventilate its gills effectively, then what other structure besides its swim bladder will it use?
A) Its pectoral fins.
B) Its lateral line system.
C) Its caudal (tail) fin.
D) Its opercula.

19) How did the evolution of the jaw contribute to diversification of early vertebrate lineages?
A) It allowed for smaller body size.
B) It was the first stage in the development of a bony skull.
C) It made additional food sources available.
D) It increased the surface area for respiration and feeding.

20) It is believed that the coelacanths and lungfish represent a crucial link between other fishes and tetrapods. What is the major feature in these fish in support of this hypothesis?
A) Like amphibians, they are tied to the water for reproduction.
B) Their fins have skeletal and muscular structures similar to amphibian limbs.
C) They have highly evolved nervous and circulatory systems.
D) They have lungs and are able to breathe air when water is scarce.

21) Jaws first occurred in which extant group of fishes?
A) Lampreys.
B) Chondrichthyans.
C) Ray-finned fishes.
D) Placoderms.

22) Which of these might have been observed in the common ancestor of chondrichthyans and osteichthyans?
A) A mineralized, bony skeleton.
B) Opercula.
C) A spiral valve intestine.
D) A swim bladder.

23) Arrange these groups in order from most inclusive (most general) to least inclusive (most specific).
1. Lobe-fins.
2. Amphibians.
3. Gnathostomes.
4. Osteichthyans.
5. Tetrapods.
A) 4, 3, 1, 5, 2.
B) 4, 3, 2, 5, 1.
C) 3, 4, 1, 5, 2.
D) 3, 4, 5, 1, 2.

24) Suppose, while out camping in a forest, you found a chordate with a long, slender, limbless body slithering across the ground near your tent. This critter could be ___.
A) A lampreys.
B) A mammal.
C) An amphibian.
D) A skate.

Use the following description to answer the questions (25-29) below.
While on an intersession course in tropical ecology, Kris pulls a large, snakelike organism from a burrow (the class was granted a collecting permit). The 1-meter-long organism has smooth skin, which appears to be segmented. It has two tiny eyes that are hard to see because they seem to be covered by skin. Kris brings it back to the lab at the field station, where it is a source of puzzlement to the class. Kris says that it is a giant oligochaete worm; Shaun suggests it is a legless amphibian; Kelly proposes it belongs to a snake species that is purely fossorial (lives in a burrow).

25) The class decided to humanely euthanize the organism and subsequently dissect it. Having decided that it was probably not a reptile, two of their original hypotheses regarding its identity remained. Which of the following, if observed, should help them arrive at a conclusive answer?
A) Presence of moist, highly vascularized skin.
B) Presence of lungs.
C) Presence of a nerve cord.
D) Presence of a digestive system with two openings.

26) The organism was found to have two lungs, but the left lung was much smaller than the right lung. Kelly added that the herpetology instructor had said that in most snakes, the same condition exists. If the size difference between the lungs in this organism is not a shared ancestral characteristic with its occurrence in snakes, then its existence in this organism is explained as:
1. A result of convergent evolution.
2. An example of homologous structures.
3. A similar adaptation to a shared lifestyle or body plan.
4. A result of having identical *Hox* genes.
A) 1 only
B) 1 and 3
C) 2 and 3
D) 3 and 4.

27) Which of the following could be considered the most recent common ancestor of living tetrapods?
A) A sturdy-finned, shallow-water lobe-fin whose appendages had skeletal supports similar to those of terrestrial vertebrates.
B) An armored, jawed placoderm with two pairs of appendages.
C) An early ray-finned fish that developed bony skeletal supports in its paired fins.
D) A salamander that had legs supported by a bony skeleton but moved with the side-to-side bending typical of fishes.

28) A trend first observed in the evolution of the earliest tetrapods was _____.
A) The appearance of jaws.
B) Feet with digits.
C) The mineralization of the endoskeleton.
D) The amniotic egg.

29) Fossils of the earliest tetrapods should _____.
A) Show evidence of internal fertilization.
B) Show evidence of having produced shelled eggs.
C) Indicate limited adaptation to life on land.
D) Feature the earliest indications of the appearance of jaws.

Use the following information to answer the question (30) below.

Terry catches a ray-finned fish from the ocean and notices that attached to its flank is an equally long, snakelike organism. The attached organism has no external segmentation, no scales, a round mouth surrounded by a sucker, and two small eyes. Terry thinks it might be a marine leech, a hagfish, or a lamprey.

30) Terry saved some of the tooth-like objects within the hagfish's round mouth to analyze their composition in his mentor's biochemistry research lab. Terry will find that they are composed of the same protein found in tetrapod_.
A) Skin.
B) Teeth.
C) Bones.
D) Cartilage.

31) What is believed to be the most significant result of the evolution of the amniotic egg?
A) Tetrapods were no longer tied to the water for reproduction.
B) Tetrapods can now function with just lungs.
C) Newborns are much less dependent on their parents.
D) Embryos are protected from predators.

32) Which structure of the amniotic egg most closely surrounds the embryo?
A) The chorion.
B) The yolk sac.
C) The allantois.
D) The amnion.

33) The evolution of similar insulating skin coverings such as fur, hair, and feathers in mammals and birds is a result of _____.
A) Shared ancestry.
B) Convergent evolution.
C) Homology.
D) Evolutionary divergence.

34) Which of the following characteristics evolved independently in mammals and birds?
A) Amniotic eggs.
B) Jaws.
C) Bone.
D) Endothermy.

35) Suppose you traveled back in time and located the first animals to have evolved feathers. You found that these animals were tree-dwelling ectotherms, able to run quickly but unable to fly. You also noticed that only males had feathers. Which hypothesis of feather evolution would these data most support? Feathers initially evolved in a role associated with _____.
A) Flight.
B) Insulation.
C) Sexual selection.
D) Gliding.

36) Mammals and birds eat more often than reptiles. Which of the following traits shared by mammals and birds best explains this habit?
A) Endothermy.
B) Ectothermy.
C) Amniotic egg.
D) Terrestrial.

37) Which characteristic is common to all the modern representatives of all major reptilian lineages (turtles, lepidosaurs, crocodilians, and birds)?
A) Presence of teeth.
B) Presence of four walking limbs.
C) Ectothermy.
D) Presence of a notochord.

38) Which of these are amniotes?
A) Amphibians.
B) Fishes.
C) Turtles.
D) Lungfish.

Use the following information to answer the question (39-40) below.

Due to its system of air sacs connected to the lungs, the respiratory system of birds is arguably the most effective respiratory system of all air-breathers. Upon inhalation, air first flows into posterior air sacs, then into the lungs, and then into anterior air sacs on the way to being exhaled. Thus, there is one-way flow of air through the lungs, along thousands of tubules called parabronchi.

39) If the inner lining of the air sacs is neither thin nor highly vascularized, then what can be inferred about the air sacs?
A) They must not belong to the respiratory system.
B) They cannot be derived from endoderm.
C) They are not efficient sites of gas exchange between air and blood.
D) They cannot effectively moisturize the air before it reaches the lungs.

40) The one-way flow of air along parabronchi makes what type of gas exchange mechanism possible, at least theoretically?
A) The same as that occurring in fish gills.
B) The same as that occurring in insect tracheae.
C) The same as that occurring in mammalian lungs.
D) The same as that occurring in echinoderm skin gills.

41) Which of these characteristics added most to vertebrate success in relatively dry environments?
A) The shelled, amniotic egg.
B) The ability to maintain a constant body temperature.
C) Two pairs of appendages.
D) A four-chambered heart.

42) Which of the following are the only extant animals that descended directly from dinosaurs?
A) Lizards.
B) Crocodiles.
C) Birds.
D) Tuataras.

43) During chordate evolution, what is the sequence (from earliest to most recent) in which the following structures arose?
1. Amniotic egg.
2. Paired fins.
3. Jaws.
4. Swim bladder.
5. Four-chambered heart.
A) 2, 3, 4, 1, 5.
B) 3, 2, 4, 5, 1.
C) 3, 2, 1, 4, 5.
D) 2, 1, 5, 3, 4.

44) Which clade does NOT include humans?
A) Lobe-fins.
B) Diapsids.
C) Craniates.
D) Osteichthyans.

45) Primate evolution and behavior, such as hunting skills, have been directed in part by the development of depth perception. What anatomical change made depth perception possible?
A) A larger brain.
B) The formation of compound eyes.
C) Movement of the eyes to the front of the head.
D) Diurnal activity.

46) What group of mammals have (a) embryos that spend more time feeding through the placenta than the mother's nipples, (b) young that feed on milk, and (c) a prolonged period of maternal care after leaving the placenta?
A) Eutheria.
B) Marsupiala.
C) Monotremata.
D) None of them.

47) Which of the following represents the strongest evidence that two of the three middle ear bones of mammals are homologous to certain reptilian jawbones?
A) They are similar in size to the reptilian jawbones.
B) They are similar in shape to the reptilian jawbones.
C) The mammalian jaw has fewer bones than does the reptilian jaw.
D) These bones can be observed to move from the evolving jaw to the evolving middle ear in mammalian embryos.

48) Which of the following is the most inclusive (most general) group in which all of the members have fully opposable thumbs?
A) Apes.
B) *Homo.*
C) Anthropoids.
D) Primates.

49) Which of these would a paleontologist most likely do to determine if a fossil represents a reptile or a mammal?
A) Look for the presence of milk-producing glands.
B) Look for the mammalian characteristics of a four-chambered heart and a diaphragm.
C) Use molecular analysis to look for the protein keratin.
D) Examine the teeth.

50) Female birds lay their eggs, thereby facilitating flight by reducing weight. Which "strategy" seems most likely for female bats to use to achieve the same goal?
A) Limit litters to a single embryo.
B) Refrain from flying throughout pregnancy (about six weeks long).
C) Give birth to underdeveloped young, and subsequently carry them in a pouch that has teats.
D) Feed multiple embryos internally using placentas.

51) Unlike eutherians, both monotremes and marsupials _____.
A) Lack nipples.
B) Have some embryonic development outside the uterus.
C) Lay eggs.
D) Are found in Australia and Africa.

52) On the back of your skull you can feel a small bump, below which is an opening where the spinal cord enters the skull. The location of this opening toward the bottom of the skull is significant in evolutionary biology for what reason?
A) It allowed for the hominin brain to grow much larger than other primates.
B) It provided greater protection for the spinal cord.
C) It occurred as a result of the change to a bipedal stance.
D) This change was necessary for the increase in size from prosimian forms to anthropoid forms.

Use the following information to answer the question (53-54) below.

Brown et al. and Morwood et al. reported in 2004 that they had found skeletal remains of a previously unknown type of hominin, now dubbed *Homo floresiensis*, on the Indonesian island of Flores. These hominins were small (approximately 1 meter tall) with small braincases (approximately 380 cubic centimeters) as compared with other hominins. The remains of *H. floresiensis* were found alongside handmade stone tools and the remains of dwarf elephants that also inhabited the island, suggesting that *H. floresiensis* was able both to make tools and to coordinate the hunting of animals much larger than itself. *H. floresiensis* is estimated to have lived at the site where the remains were found from at least 38,000 years ago to 18,000 years ago.

53) Refer to the paragraph on Brown et al. and Morwood et al. Which would be the most feasible method of figuring out to which other hominin species *H. floresiensis* was most closely related?
A) Compare the type of prey hunted by *H. floresiensis* to that hunted by each of the other hominin species.
B) Compare the average body size of *H. floresiensis* to that of each of the other hominin species.
C) Compare the skeletal morphology of *H. floresiensis* to that of each of the other hominin species.
D) Compare the estimated life span of *H. floresiensis* to that of each of the other hominin species.

54) In what respect do hominins differ from all other anthropoids?
A) Lack of a tail.
B) Eyes on the front of the face.
C) Bipedal posture.
D) Opposable thumbs.

55) The table below is a comparison of several characteristics of *H. floresiensis* to those of nine other hominin species (arranged roughly from oldest to most recent). What do these data suggest?

Species	Location of Fossils	Estimated Average Braincase Volume (cm^3)	Estimated Average Body Mass (kg)	Associated with Stone Tools?
Australopithecus afarensis	Africa	450	36	no
A. africanus	Africa	450	36	no
Paranthropus boisei	Africa	510	44	no
Homo habilis	Africa	550	34	yes
H. ergaster	Africa	850	58	yes
H. erectus	Africa, Asia	1000	57	yes
H. heidelbergensis	Africa, Europe	1200	62	yes
H. neanderthalensis	Middle East, Europe, Asia	1500	76	yes
H. floresiensis	Asia	380	16—36	yes
H. sapiens	Middle East, Europe, Asia	1350	53	yes

A) A large brain is not necessarily required for toolmaking.
B) Body mass and braincase volume are completely unrelated.
C) Hominins first evolved in and then radiated out from Asia.
D) *Homo floresiensis* is most closely related to *Australopithecus afarensis* or *A. africanus*.

56) Refer to the paragraph on Brown et al. and Morwood et al. It is speculated that *H. floresiensis* and *H. sapiens* may have lived on Flores concurrently. Suppose researchers obtained mitochondrial DNA samples from the *H. floresiensis* remains, amplified a 1000-base-pair sequence via PCR, and compared it to that of several currently living *H. sapiens* native to Indonesia, North Africa, and North America. Also suppose *H. floresiensis* were found to differ from the average Indonesian *H. sapiens* in 28 base pairs, from the average North African *H. sapiens* in 51 base pairs, and from the average North American *H. sapiens* in 53 base pairs, while two randomly selected *H. sapiens* differed from each other in an average of 21 base pairs. What would you surmise from these data?
A) *H. floresiensis* and *H. sapiens* probably did not live on Flores concurrently.
B) *H. floresiensis* and *H. sapiens* probably lived on Flores concurrently but did not interact.
C) *H. floresiensis* and *H. sapiens* probably lived on Flores concurrently, and *H. sapiens* killed and consumed *H. floresiensis*.
D) *H. floresiensis* and *H. sapiens* probably lived on Flores concurrently and interbred to some degree.

57) Arrange the following taxonomic terms in order from most inclusive (most general) to least inclusive (most specific).
1. Apes.
2. Hominins.
3. *Homo*.
4. Anthropoids.
5. Primates.
A) 5, 1, 4, 2, 3.
B) 5, 4, 1, 2, 3.
C) 5, 4, 2, 1, 3.
D) 5, 2, 1, 4, 3.

58) Which of these traits is most strongly associated with the adoption of bipedalism?
A) Enhanced depth perception.
B) Shortened hind limbs.
C) Opposable big toe.
D) Repositioning of foramen magnum.

59) Which of the following statements about human evolution is correct?
A) Modern humans are the only human species to have evolved on Earth.
B) Human ancestors were virtually identical to extant chimpanzees.
C) Human evolution has occurred within an unbranched lineage.
D) The upright posture and enlarged brain of humans evolved separately.

60) With which of the following statements would a biologist be most inclined to agree?
A) Humans and other apes represent divergent lines of evolution from a common ancestor.
B) Humans represent the pinnacle of evolution and have escaped from being affected by natural selection.
C) Humans evolved from chimpanzees.
D) Humans and other apes are the result of disruptive selection in a species of chimpanzee.

CHAPTER 35:

PLANT STRUCTURE, GROWTH, AND DEVELOPMENT

1) Which part of a plant absorbs most of the water and minerals taken up from the soil?
A) Root cap.
B) Root hairs.
C) The thick parts of the roots near the base of the stem.
D) Storage roots.

2) What is the primary function of stems?
A) Facilitation of gas exchange.
B) Water absorption and movement.
C) Maximization of photosynthesis by leaves.
D) Reproduction.

3) When you eat Brussels sprouts, you are eating _____.
A) Immature flowers.
B) Large axillary buds.
C) Petioles.
D) Storage leaves.

4) Some of the largest leaves in the world can be found on plants near the forest floor of dense tropical rain forests. Which of the following precursors for photosynthesis is most likely limited in these large leaves?
A) Oxygen.
B) Carbon dioxide.
C) Glucose.
D) Light.

5) Leaf thickness represents a trade-off between _____.
A) Light collection and carbon dioxide absorption.
B) Water retention and carbon dioxide absorption.
C) Water retention and oxygen absorption.
D) Light collection and oxygen absorption.

6) One important difference between the anatomy of roots and the anatomy of leaves is that _____.
A) Only leaves have phloem and only roots have xylem.
B) Root cells have cell walls and leaf cells do not.
C) A waxy cuticle covers leaves but is absent from roots.
D) Vascular tissue is found in roots but is absent in leaves.

7) Which of the following was a challenge to the survival of the first land plants?
A) Too much sunlight.
B) A shortage of carbon dioxide.
C) Desiccation.
D) Animal predation.

8) Trichomes _____.
A) Absorb sunlight, increasing the temperature of leaves.
B) Open and close for gas exchange.
C) Repel or trap insects.
D) Increase water loss from leaves.

9) Which structure is correctly paired with its tissue system?
A) Root hair - vascular tissue.
B) Guard cell - vascular tissue.
C) Companion cell - ground tissue.
D) Tracheid - vascular tissue.

10) The main source of water necessary for photosynthesis to occur in the leaf mesophyll is _____.
A) Soil via the xylem.
B) Soil via the phloem.
C) The atmosphere through the cuticle and stomata.
D) All of the listed responses.

11) The vascular bundle in the shape of a single central cylinder in a root is called the _____.
A) Cortex.
B) Stele.
C) Periderm.
D) Pith.

12) Which of the following cell types retains the ability to undergo cell division?
A) A parenchyma cell near the root tip.
B) A functional sieve tube element.
C) A tracheid.
D) A stem fiber.

13) Which of these is NOT an example of a parenchyma cell?
A) Cells that can form clones in tissue culture of plants.
B) Support cells near the outside of nonwoody stems.
C) Edible cells in fruits and vegetables.
D) Tissue in leaves that photosynthesizes.

14) Which of the following have unevenly thickened primary walls that support young, growing parts of plant?
A) Parenchyma cells.
B) Collenchyma cells.
C) Sclerenchyma cells.
D) Tracheids and vessel elements.

15) Which of the following is correctly paired with its structure and function?
A) Sclerenchyma - supporting cells with thick secondary walls.
B) Ground meristem - protective coat of woody stems and roots.
C) Guard cells - waterproof ring of cells surrounding the central stele in roots.
D) Periderm - parenchyma cells functioning in photosynthesis in leaves.

16) Which of the following occurs in vascular land plants but not charophytes (stoneworts)?
A) Sporopollenin.
B) Lignin.
C) Chlorophyll a.
D) Cellulose.

17) Which of the following are water-conducting cells that are dead at functional maturity?
A) Parenchyma cells.
B) Collenchyma cells.
C) Tracheids and vessel elements.
D) Sieve-tube elements.

18) Which of the following cells transport sugars over long distances?
A) Parenchyma cells.
B) Sclerenchyma cells.
C) Tracheids and vessel elements.
D) Sieve-tube elements.

19) Plant meristematic cells _____.
A) Are distributed evenly in all tissues throughout the plant.
B) Are undifferentiated cells that produce new cells.
C) Increase the surface area of dermal tissue by developing root hairs.
D) Subdivide into three distinct cell types named parenchyma, ground meristem, and procambium.

20) Which of the following arise, directly or indirectly, from meristematic activity?
A) Secondary xylem.
B) Leaves.
C) Dermal tissue.
D) Secondary xylem, leaves, dermal tissue, and tubers.

21) Compared to most animals, the growth of most plant structure is best described as _____.
A) Perennial.
B) Weedy.
C) Indeterminate.
D) Primary.

22) What is present in a shoot apical meristem region?
I) The region of cell division.
II) Immature buds and leaves.
III) Cells that will give rise to the protoderm, ground meristem, and procambium.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

23) Shoot elongation in a growing bud is due primarily to _____.
A) Cell division at the shoot apical meristem.
B) Cell elongation directly below the shoot apical meristem.
C) Cell elongation localized in each internode.
D) Cell division at the shoot apical meristem and cell elongation directly below the shoot apical meristem.

24) Apical meristems of dicots are at the tips of stems. Apical meristems of grasses are at ground level or slightly below, concealed by the leaves. The leaves also have an intercalary meristem at their bases. What does this mean when considering care of a lawn or soccer field?
A) If you mow right at ground level, the leaves can keep growing with no problem.
B) Grass mowed two inches above ground level grows at a slower rate compared to grass mowed three inches above the ground level.
C) If you mow two inches above ground level, most apical meristems will be cut down.
D) If you mow two inches above ground level, both the apical and intercalary meristems can keep producing new cells.

25) In a meristematic region, the cell plate during mitosis is perpendicular to the side of the stem. In what direction will the stem grow?
A) Laterally in width.
B) Vertically in height.
C) At a 45-degree angle from the ground.
D) Away from the sun.

26) Which of the following cells or tissues arise from lateral meristem activity?
A) Secondary xylem.
B) Leaves.
C) Trichomes.
D) Tubers.

27) Cells produced by lateral meristems are known as _____.
A) Dermal and ground tissue.
B) Lateral tissues.
C) Pith.
D) Secondary tissues.

28) Which of the following can be used to determine a twig's age?
A) Number of apical bud scar rings.
B) Number of leaf scars.
C) Number and arrangement of axillary buds.
D) Length of internodes.

29) A plant that grows one year, dies back, and then grows again the following year, produces flowers and then dies would be considered _____.
A) Annual.
B) Biennial.
C) Perennial.
D) Not very fit.

30) Which of the following is the correct sequence of the zones in the primary growth of a root, moving from the root cap inward?
A) Zone of cell division, zone of elongation, zone of differentiation.
B) Zone of differentiation, zone of elongation, zone of cell division.
C) Zone of elongation, zone of cell division, zone of differentiation.
D) Zone of cell division, zone of differentiation, zone of elongation.

31) The driving force that pushes the root tip through the soil is primarily _____.
A) Continuous cell division in the root cap at the tip of root.
B) Continuous cell division just behind the root cap in the center of the apical meristem.
C) Elongation of cells behind the root apical meristem.
D) Continuous cell division of root cap cells.

32) Mitotic activity by the apical meristem of a root makes which of the following more possible?
A) Increased delivery of water to the aboveground stem.
B) Decreased absorption of mineral nutrients.
C) Increased absorption of carbon dioxide.
D) Effective lateral growth of the stem.

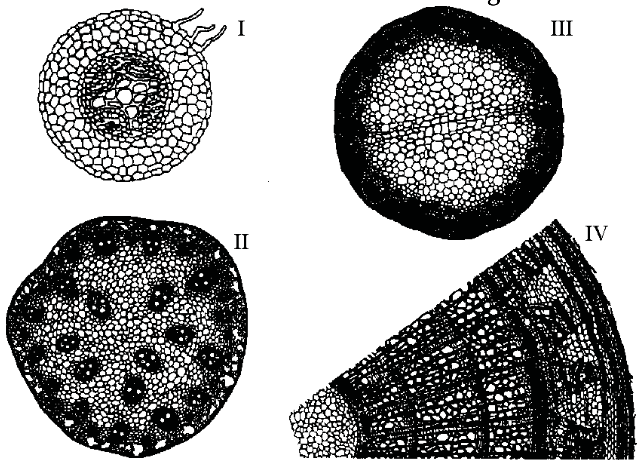
33) Which of the following root tissues gives rise to lateral roots?
A) Endodermis.
B) Phloem.
C) Epidermis.
D) Pericycle.

34) As a youngster, you drive a nail in the trunk of a young tree that is 3 meters tall. The nail is about 1.5 meters from the ground. Fifteen years later, you return and discover that the tree has grown to a height of 30 meters. About how many meters above the ground is the nail?
A) 0.5.
B) 1.5.
C) 3.0.
D) 15.0.

35) You find a plant unfamiliar to you and observe that it has vascular bundles scattered throughout the stem cross section. What do you conclude about the plant?
A) It is probably an herbaceous eudicot.
B) It will probably get annual rings of wood.
C) It is probably a monocot.
D) It could be either a young eudicot or a monocot.

36) Monocot vascular bundles do not have a vascular cambium between the xylem and phloem. This means that monocots _____.
A) Are much less efficient at conducting water and sugars.
B) Have very thin stems.
C) Do not produce wood in annual rings.
D) Cannot produce lateral shoots.

The following questions (37-39) are based on the drawings of root or stem cross sections shown in the figure below.



37) Refer to the figure above. A monocot stem is represented by _____.
A) I only.
B) II only.
C) III only.
D) IV only.

38) Refer to the figure above. A woody eudicot is represented by _____.
A) II only.
B) III only.
C) IV only.
D) I and III.

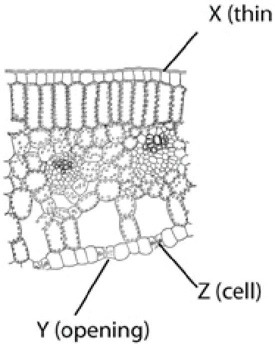
39) Refer to the figure above. A plant that is at least three years old is represented by _____.
A) I only.
B) II only.
C) III only.
D) IV only.

40) Canada thistle is a dicot that spreads via growth from lateral roots. You want to use a root miner insect for weed control. What would you need to observe in the underground growth to verify that this weed spreads via lateral roots and not by underground stems?
A) An epidermis at the periphery.
B) Vascular bundles in a ring around the outside of a cross section.
C) A vascular bundle in the center surrounded by parenchyma tissue.
D) Meristematic tissue at the tips of the branches.

41) A student examining leaf cross sections under a microscope finds many loosely packed cells with relatively thin cell walls. The cells have numerous chloroplasts. What type of cells are they?
A) Parenchyma.
B) Endodermis.
C) Collenchyma.
D) Sclerenchyma.

42) The veins of leaves are _____.
I) Composed of xylem and phloem.
II) Continuous with vascular bundles in the stem and roots.
III) Finely branched to be in close contact with photosynthesizing cells.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

The following diagram is of a cross section of a plant leaf. Use the diagram to answer the question (43-44) below.



43) The main function associated with structure X is _____.
A) Absorption of carbon dioxide.
B) Retention of water.
C) Collection of light.
D) Release of carbon dioxide.

44) The main function associated with structure Y is _____.
A) Absorption of carbon dioxide.
B) Retention of water.
C) Collection of light.
D) Release of carbon dioxide.

45) Increasing the number of stomata per unit surface area of a leaf when atmospheric carbon dioxide levels decline is most analogous to a human _____.
A) Breathing faster as atmospheric carbon dioxide levels increase.
B) Putting more red blood cells into circulation when atmospheric oxygen levels decline.
C) Removing red blood cells from circulation when atmospheric oxygen levels increase.
D) Increasing the volume of its lungs when atmospheric carbon dioxide levels increase.

46) Where is primary growth occurring in an old tree?
A) Nowhere; trees more than a year old have only secondary growth.
B) Closest to ground level at the base of the tree.
C) In young branches where leaves are forming.
D) Where the vascular cambium and cork cambium are located.

47) What tissue makes up most of the wood of a tree?
A) Primary xylem.
B) Secondary xylem.
C) Secondary phloem.
D) Vascular cambium.

48) A plant has the following characteristics: a taproot system, several growth rings evident in a cross section of the stem, and a layer of bark around the outside. Which of the following best describes the plant?
A) Herbaceous eudicot.
B) Woody eudicot.
C) Woody monocot.
D) Herbaceous monocot.

49) If you were able to walk into an opening cut into the center of a large redwood tree, when you exited from the middle of the trunk (stem) outward, you would cross, in order, _____.
A) The annual rings, new xylem, vascular cambium, phloem, and bark.
B) The secondary xylem, cork cambium, phloem, and periderm.
C) The vascular cambium, oldest xylem, and newest xylem.
D) The secondary xylem, secondary phloem, and vascular cambium.

50) Heartwood and sapwood consist of _____.
A) Periderm.
B) Secondary xylem.
C) Secondary phloem.
D) Cork.

51) Two examples of lateral meristems in plants are _____.
A) Vascular cambium, producing cork; cork cambium, producing secondary phloem.
B) Vascular cambium, producing secondary xylem; cork cambium, producing secondary phloem.
C) Vascular cambium, producing secondary xylem; cork cambium, producing cork.
D) Vascular cambium, producing secondary phloem; cork cambium, producing secondary xylem.

52) Additional vascular tissue produced as secondary growth in a root originates from which cells?
A) Vascular cambium.
B) Apical meristem.
C) Endodermis.
D) Xylem.

53) Girdling is a procedure to kill unwanted tress by cutting a groove into the bark of the tree. The groove must completely encircle the trunk and should penetrate into the wood to a depth of at least 1-1*11\10-inch on small trees, and 1-1*11\10 inches on larger trees. Why does this procedure cause tree death?
A) No water can be transported from the roots to the leaves.
B) No water can be transported from the leaves to the roots.
C) No sugars can be transported from the leaves to the roots.
D) Both water and sugars are prevented from being transported.

54) Where are the youngest wood and the youngest bark in a tree trunk?
A) Youngest wood is in the center of a tree; youngest bark is the outside of the bark.
B) Youngest wood is in the center of a tree; youngest bark is the inner part, next to the vascular cambium.
C) Youngest wood is toward the outside, near the vascular cambium; youngest bark is the outside of the bark.
D) Youngest wood is toward the outside, near the vascular cambium; youngest bark is the inner part, next to the vascular cambium.

55) The polarity of a plant is established when _____.
A) Cotyledons form at the shoot end of the embryo.
B) The shoot-root axis is established in the embryo.
C) The primary root breaks through the seed coat.
D) The shoot first breaks through the soil into the light as the seed germinates.

56) Growth and development of plant parts involves ___.
I) Cell division to produce new cells.
II) Enlargement and elongation of cells.
III) Specialization of cells into tissues.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

57) Totipotency is a term used to describe a cell's ability to give rise to a complete new organism. In plants, this means that _____.
A) Plant development is not under genetic control.
B) The cells of shoots and the cells of roots have different genes.
C) Cell differentiation depends largely on the control of gene expression.
D) A cell's environment has no effect on its differentiation.

58) Which of the following statements is true?
A) A band of ribosomes determines where a cell plate will form in a dividing plant cell.
B) The way in which a plant cell differentiates is determined by the position of the nucleus in the developing plant cell.
C) Homeotic genes often control morphogenesis.
D) Plant cells differentiate because the cytoskeleton determines which genes will be turned "on" and "off".

59) The phase change of an apical meristem from the juvenile to the mature vegetative phase is often revealed by _____.
A) A change in the morphology of the leaves produced.
B) The initiation of secondary growth.
C) A change in the orientation of preprophase bands and cytoplasmic microtubules in lateral meristems.
D) The activation of floral meristem identity genes.

CHAPTER 36:

RESOURCE ACQUISITION AND TRANSPORT IN VASCULAR PLANTS

1) Given that early land plants most likely share a common ancestor with green algae, the earliest land plants were most likely _____.
A) Nonvascular plants that grew leafless, photosynthetic shoots.
B) Species that did not exhibit alternation of generations.
C) Vascular plants with well-defined root systems.
D) Plants with well-developed leaves.

2) A fellow student brought in a leaf to be examined. The leaf was dark green, thin, had stoma on the lower surface only, and had a total surface area of ten square meters. Where is the most likely environment where this leaf was growing?
A) A large, still pond.
B) A tropical rain forest.
C) An oasis within grassland.
D) The floor of a deciduous forest.

3) Why do most angiosperms have alternate phyllotaxy, with leaf emergence at an angle of 137.5° compared to leaves above and below?
A) To allow maximum exposure to light.
B) To promote a leaf area index above 8.
C) To reduce shading of lower leaves.
D) To allow maximum exposure to light and to reduce shading of lower leaves.

4) A plant developed a mineral deficiency after being treated with a fungicide. What is the most probable cause of the deficiency?
A) Mineral receptor proteins in the plant membrane were not functioning.
B) Mycorrhizal fungi were killed.
C) Active transport of minerals was inhibited.
D) The genes for the synthesis of transport proteins were destroyed.

5) Which structure or compartment is separate from the apoplastic route?
A) The lumen of a xylem vessel.
B) The lumen of a sieve tube.
C) The cell wall of a mesophyll cell.
D) The cell wall of a root hair.

6) The _____ is the most efficient route of water movement in plants, while the _____ is the most select.
A) Apoplast: symplast.
B) Apoplast: transmembrane.
C) Symplast apoplast.
D) Transmembrane: symplast.

7) Active transport of amino acids in plants at the cellular level requires _____.
A) NADP and channel proteins.
B) Xylem membranes and channel proteins.
C) Sodium/potassium pumps and xylem membranes.
D) ATP, transport proteins, and a proton gradient.

8) What is the function of proton pumps localized in the plant plasma membrane?
A) To transfer phosphorus groups from ATP to proteins.
B) To transfer metal ions across the plasma membrane.
C) To transfer anions across the plasma membrane.
D) To create a membrane potential.

9) Which of the following would be LEAST likely to affect osmosis in plants?
A) A difference in solute concentrations.
B) Receptor proteins in the membrane.
C) Aquaporins.
D) A difference in water potential.

10) The movement of water across biological membranes can best be predicted by _____.
A) Prevailing weather conditions.
B) Aquaporins.
C) Level of active transport.
D) Water potentials.

11) If isolated plant cells with a water potential averaging -0.5 MPa are placed into a solution with a water potential of -0.3 MPa, which of the following would be the most likely outcome?
A) The pressure potential of the cells would increase.
B) Water would move out of the cells.
C) The cell walls would rupture, killing the cells.
D) Solutes would move out of the cells.

12) The value for Ψ in root tissue was found to be -0.15 MPa. If you take the root tissue and place it in a 0.1 *M* solution of sucrose (Ψ = -0.23 MPa), the net water flow would _____.
A) Be from the tissue into the sucrose solution.
B) Be from the sucrose solution into the tissue.
C) Be in both directions and the concentration of water would remain equal.
D) Be impossible to determine from the values given here.

13) When an animal cell is placed in a hypotonic solution and water enters the cell via osmosis, the volume of the cell increases until it bursts. This does not happen to plant cells, because _____.
A) They have large central vacuoles, which provide abundant space for storage of incoming water.
B) They have cell walls, which prevent the entry of water by osmosis.
C) They have cell walls, which provide pressure to counteract the pressure of the incoming water.
D) Certain gated channel proteins embedded in their plasma membranes open as osmotic pressure decreases, allowing excess water to leave the cell.

14) How does a flaccid cell differ from a turgid cell?
A) A flaccid cell has higher pressure potential.
B) A flaccid cell has lower pressure potential.
C) A flaccid cell has higher solute potential.
D) A flaccid cell has lower solute potential.

15) Compared to a cell with few aquaporins in its membrane, a cell containing many aquaporins wills _____.
A) Have a faster rate of osmosis.
B) Have a lower water potential.
C) Have a higher water potential.
D) Have a faster rate of active transport.

16) Which of the following statements about bulk flow are correct?
I) Bulk flow is driven primarily by pressure potential.
II) Bulk flow depends on a difference in pressure potential at the source and sink.
III) Bulk flow depends on the force of gravity on a column of water.
IV) Bulk flow may be the result of either positive or negative pressure potential.
A) I and III.
B) II and III.
C) I, II, and IV.
D) I, II, III, and IV.

17) Which of the following are important components of the long-distance transport process in plants?
I) The cohesion of water molecules.
II) A negative water potential.
III) The root parenchyma.
IV) The active transport of solutes.
V) Bulk flow from source to sink.
A) II, III, IV, and V.
B) I, III, IV, and V.
C) I, II, IV, and V.
D) I, II, III, and V.

18) Loss of water from the aerial parts of plants is called _____.
A) Dehydration.
B) Respiration.
C) Gas exchange.
D) Transpiration.

19) Which of the following contribute to the surface area available for water absorption from the soil by a plant root system?
I) Root hairs.
II) Endodermis.
III) Mycorrhizae.
IV) Fibrous arrangement of the roots.
A) II and III.
B) I, III, and IV.
C) I, II, and IV.
D) I, II, III, and IV.

20) What is the overall charge on the cytoplamsic side of a plant cell plasma membrane?
A) Positive.
B) Negative.
C) Neutral.
D) None of them.

21) A water molecule could move all the way through a plant from soil to root to leaf to air and pass through a living cell only once. This living cell would be a part of which structure?
A) A guard cell.
B) The root epidermis.
C) The endodermis.
D) The root cortex.

22) In plant roots, the Casparian strip _____.
A) Aids in the uptake of nutrients.
B) Provides energy for the active transport of minerals into the stele from the cortex.
C) Ensures that all minerals are absorbed from the soil in equal amounts.
D) Ensures that all water and dissolved substances must pass through a cell membrane before entering the stele.

23) Which of the following observations provides the strongest evidence against root pressure being the principal mechanism of water transport in the xylem?
A) Not all soils have high concentrations of ions.
B) Root pressure requires movement of water into the xylem from surrounding cells in the roots.
C) Over long distances, the force of root pressure is not enough to overcome the force of gravity.
D) There is no water potential gradient between roots and shoots.

24) Most likely to see guttation in small plants when the _.
A) Transpiration rates are high.
B) Root pressure exceeds transpiration pull.
C) The preceding evening was hot, windy, and dry.
D) Roots are not absorbing minerals from the soil.

25) Most of the water taken up by a plant is _____.
A) Used as a solvent.
B) Used as a hydrogen source in photosynthesis.
C) Lost during transpiration.
D) Used to keep cells turgid.

26) Transpiration in plants requires _____.
I) Adhesion of water molecules to cellulose.
II) Cohesion between water molecules.
III) Evaporation of water molecules.
IV) Active transport through xylem cells.
V) Transport through tracheids.
A) I, III, IV, and V.
B) I, II, IV, and V.
C) I, II, III, and V.
D) I, II, III, and IV.

27) What is the main force by which most of the water within xylem vessels moves toward the top of a tree?
A) Active transport of ions into the stele.
B) Evaporation of water through stoma.
C) The force of root pressure.
D) Osmosis in the root.

28) Water potential is generally most negative in which of the following parts of a plant?
A) Mesophyll cells of the leaf.
B) Xylem vessels in leaves.
C) Xylem vessels in roots.
D) Cells of the root cortex.

29) As an undergraduate research assistant, your duties involve measuring water potential in experimental soil-plant-atmosphere systems. Assume you make a series of measurements in a system under normal daylight conditions, with stomata open and photosynthesis occurring. Which of the following correctly depicts the trend your measurement data should follow if the cohesion-tension mechanism is operating?
A) ψ soil < ψ roots = ψ leaves < ψ atmosphere.
B) ψ atmosphere < ψ leaves = ψ roots < ψ soil.
C) ψ soil < ψ roots < ψ leaves < ψ atmosphere.
D) ψ atmosphere < ψ leaves < ψ roots < ψ soil.

30) Formation of menisci, such as occurs in a tube filled with water, is an important factor in plant water movement. A meniscus is created by _____.
A) The upward pull of gravity on the water column in the tube.
B) Downward pressure from the atmosphere on the topmost layer of water molecules.
C) The water molecules being pulled upward by adhesion to the air.
D) The topmost layer of water molecules being pulled downward by the hydrogen bonds to the water molecules below.

31) Which mechanism of water transport in xylem can contribute to recovery from cavitation?
A) Cohesion-tension.
B) Capillarity.
C) Pressure flow.
D) Root pressure.

32) Which of the following primarily enters a plant somewhere other than through the roots?
A) Carbon dioxide.
B) Nitrogen.
C) Potassium.
D) Water.

33) The opening of stomata is thought to involve _____.
A) An increase in the solute concentration of the guard cells.
B) Active transport of water out of the guard cells.
C) Decreased turgor pressure in guard cells.
D) Movement of K^+ from the guard cells.

34) Ignoring all other factors, what kind of day would result in the fastest delivery of water and minerals to the leaves of an oak tree?
A) Cool, dry day.
B) Very hot, dry, windy day.
C) Warm, humid day.
D) Cool, humid day.

35) Photosynthesis ceases when leaves wilt, mainly because _____.
A) The chlorophyll in wilting leaves is degraded.
B) Flaccid mesophyll cells are incapable of photosynthesis.
C) Stomata close, preventing carbon dioxide from entering the leaf.
D) Accumulation of carbon dioxide in the leaf inhibits enzymes.

36) The water lost during transpiration is a side effect of the plant's exchange of gases. However, the plant derives some benefit from this water loss in the form of _____.
A) Increased turgor and increased growth.
B) Mineral transport and increased growth.
C) Evaporative cooling and increased turgor.
D) Evaporative cooling and mineral transport.

37) Which of the following experimental procedures would most likely reduce transpiration while allowing the normal growth of a plant?
A) Subjecting the leaves of the plant to a partial vacuum.
B) Increasing the level of carbon dioxide around the plant.
C) Putting the plant in drier soil.
D) Decreasing the relative humidity around the plant.

38) Several tomato plants are growing in a small garden plot. If soil water potential were to drop significantly on a hot, summer afternoon, which of the following would most likely occur?
A) Size of stomatal openings would decrease.
B) Transpiration would increase.
C) The leaves would become more turgid.
D) The uptake of carbon dioxide would be enhanced.

39) What is the advantage of having small, needlelike leaves?
A) Increased transpiration rate.
B) Decreased transpiration rate.
C) Increased efficiency of light capture.
D) Decreased efficiency of light capture.

40) Which of the following is a net sugar source for a deciduous angiosperm tree?
A) New leaves in early spring.
B) Fruits in summer.
C) Roots in early spring.
D) Roots in early autumn.

41) Phloem transport is described as being from source to sink. Which of the following would most accurately complete this statement about phloem transport as applied to most plants in the late spring?
Phloem transports _____ from the _____ source to the _____ sink.
A) Amino acids; root; mycorrhizae.
B) Sugars; leaf; apical meristem.
C) Proteins; root; leaf.
D) Sugars; woody stem; root.

42) Arrange the following five events in an order that explains the mass flow of materials in the phloem.
1. Water diffuses into the sieve tubes.
2. Leaf cells produce sugar by photosynthesis.
3. Solutes are actively transported into sieve tubes.
4. Sugar is transported from cell to cell in the leaf.
5. Sugar moves down the stem.
A) 1, 2, 3, 4, 5.
B) 2, 4, 3, 1, 5.
C) 4, 2, 1, 3, 5.
D) 2, 4, 1, 3, 5.

43) Water flows into the source end of a sieve tube because _____.
A) Sucrose has been actively transported into the sieve tube, making it hypertonic.
B) Water pressure outside the sieve tube forces in water.
C) The companion cell of a sieve tube actively pumps in water.
D) Sucrose has been transported out of the sieve tube by active transport.

44) Which of the following supports the finding that sugar translocation in phloem is an active (energy-requiring) process?
A) Sucrose occurs in higher concentrations in companion cells than in the mesophyll cells where it is produced.
B) Movement of water occurs from xylem to phloem and back again.
C) Strong pH differences exist between the cytoplasm of the companion cell and the mesophyll cell.
D) ATPases are abundant in the plasma membranes of the mesophyll cells.

45) Which one of the following statements about transport of nutrients in phloem is correct?
A) Solute particles are actively transported from phloem at the source.
B) Companion cells control the rate and direction of movement of phloem sap.
C) Differences in osmotic concentration at the source and sink cause a hydrostatic pressure gradient to be formed.
D) A sink is the part of a plant where a particular solute is produced.

46) In the pressure-flow mechanism, loading of sucrose from companion cells to sieve-tube elements takes place through _____.
A) Plasmodesmata.
B) Facilitated diffusion.
C) Sucrose-H^+ symporters.
D) Sucrose-H_+ antiporters.

47) Which of the following is a correct statement about sugar movement in phloem?
A) Diffusion can account for the observed rates of transport.
B) Movement can occur both upward and downward in the plant.
C) Sugar is translocated from sinks to sources.
D) Only phloem cells with nuclei can perform sugar movement.

48) Plants do not have a circulatory system like that of some animals. If a water molecule in a plant did "circulate" (that is, go from one point in a plant to another and back in the same day), it would require the activity of _____.
A) Only the xylem.
B) Only the phloem.
C) Only the endodermis.
D) Both the xylem and the phloem.

49) As an undergraduate research assistant, you are assisting with a radioisotope tracer experiment. You expose a mature leaf on one side of the lower shoot of a sugar beet plant to $^{14}CO_2$ and then track the movement of the ^{14}C atoms by radiography. Where are you LEAST likely to detect ^{14}C?
A) The treated leaf.
B) The shoot apical meristem.
C) The roots.
D) A mature upper leaf on the opposite side of the plant from the treated leaf.

50) Which of the following is a similarity between xylem and phloem transport?
A) Many cells in both tissues have sieve plates.
B) Expenditure of energy from ATP is required.
C) Transpiration is required for both processes.
D) Bulk flow of water is involved.

51) Some botanists argue that the entire plant should be considered as a single unit rather than a composite of many individual cells. Which of the following cellular structures best supports this view?
A) Cell wall.
B) Cell membrane.
C) Vacuole.
D) Plasmodesmata.

52) Plasmodesmata can change in number, and when dilated can provide a passageway for _____.
A) Macromolecules such as ribonucleic acid (RNA) and proteins.
B) Ribosomes.
C) Chloroplasts.
D) Mitochondria.

53) The symplastic route can transport _____.
A) Sugars, mRNA, and mitochondria.
B) mRNA, mitochondria, and proteins.
C) Mitochondria, mRNA, and viruses.
D) Viruses, sugars, and mRNA.

CHAPTER 37:

SOIL AND PLANT NUTRITION

1) What soil composition would be best for availability of nutrients, water, and root development?
A) Equal amounts of sand, clay, and humus.
B) Higher proportion of humus, lower amounts of clay and sand.
C) Higher proportion of clay, lower amounts of humus and sand.
D) Higher proportion of sand, lower amount of humus and clay.

2) The highest amount of oxygen will be found in soils containing large amounts of _____.
A) Clay.
B) Sand.
C) Gravel.
D) Silt.

3) A group of ten tomato plants are germinated and maintained in a large tray with no drainage. After several weeks they all begin to wilt and die despite repeated watering and fertilization. The most likely cause of this die-off is _____.
A) Competition for resources.
B) A lack of oxygen for the roots.
C) Organic nutrient depletion.
D) No room left for root growth.

4) There are several properties that are characteristic of a soil in which typical plants would grow well. Of the following, which would be the least conducive to plant growth?
A) Abundant humus.
B) Numerous soil organisms.
C) Compacted soil.
D) High cation exchange capacity.

5) Clay in soils represents a trade-off in nutrient availability, such that _____.
A) Anions are less likely to leach out of soil and are difficult for plants to extract.
B) Cations are less likely to leach out of soil and are difficult for plants to extract.
C) Nitrogen levels are exceptionally high, but much of the nitrogen leaches away.
D) Oxygen levels are exceptionally high, but much of the nitrogen leaches away.

6) Which of the following soil minerals is most likely leached away during a hard rain?
A) Na^+.
B) K^+.
C) Ca^{++}.
D) NO_3^-.

7) Which of the following are problems associated with intensive irrigation?
I) Mineral runoff.
II) Over fertilization.
III) Aquifer depletion.
IV) Soil salinization.
A) Only I and II.
B) Only I, III, and IV.
C) Only III and IV.
D) I, II, III, and IV.

8) A young farmer purchases some land in a relatively arid area and is interested in earning a reasonable profit for many years. Which of the following strategies would best allow the farmer to achieve such a goal?
A) Establishing an extensive irrigation system.
B) Using plenty of the best fertilizers.
C) Finding a way to sell all parts of crop plants.
D) Selecting crops adapted to arid areas.

9) The NPK percentages on a package of fertilizer refer to the _____.
A) Percentages of manure collected from different types of animals.
B) Relative percentages of organic and inorganic nutrients in the fertilizer.
C) Percentages of three important mineral nutrients.
D) Proportions of three different nitrogen sources.

10) Which of the following would inhibit the growth of most plants?
A) Abundant humus.
B) Air spaces.
C) Good drainage.
D) A pH above 10.0.

11) Which of the following contributed to the dust bowl in the American southwest during the 1930s?
I) Overgrazing by cattle.
II) Clear-cutting of forest trees.
III) Plowing of native grasses.
IV) Lack of soil moisture.
A) I and II.
B) II, III, and IV.
C) I, III, and IV.
D) I, II, III, IV.

12) Why does planting a cover crop help conserve soil?
I) Controls weed populations.
II) Reduces soil loss due to wind and water erosion.
III) Increases soil nutrients when plowed under and allowed to decompose.
A) Only I.
B) I and II.
C) II and III.
D) I, II, and III.

13) Which of the following would be the most effective strategy to remove toxic heavy metals from a soil?
A) Heavy irrigation to leach out the heavy metals.
B) Application of sulfur to lower the soil pH and precipitate the heavy metals.
C) Adding plant species that have the ability to take up and accumulate heavy metals.
D) Inoculating soil with mycorrhizae to avoid heavy-metal uptake.

14) When comparing root systems of corn plants growing on a square foot in a cornfield with the root systems of natural prairie plants growing in a meadow, what result do you expect to see?
A) The overall mass of roots will be identical in a cornfield and on a meadow.
B) The mass and length of roots in a cornfield will be higher because soil is fertilized.
C) The mass and length of roots on a meadow will be higher because soil is poorer and there is higher diversity of plants.
D) The overall mass of roots on a meadow will be lower, but the length will be higher since roots need to grow deeper to reach nutrients.

15) Most of the dry mass of a plant is the result of uptake of _____.
A) Water and minerals through root hairs.
B) Water and minerals through mycorrhizae.
C) Carbon dioxide through stoma.
D) Carbon dioxide and oxygen through stomata in leaves.

16) How would you expect the root system of a plant grown by hydroponics to compare to the root system of a plant grown in soil? The root system of a plant grown by hydroponics would be _____.
A) More developed.
B) Less developed.
C) About the same.
D) Absent.

17) Which of the following statements about essential nutrients are true? Essential nutrients _____.
I) Are necessary for plant growth and reproduction.
II) Are required for a specific structure or metabolic function.
III) Cannot be synthesized by a plant.
IV) Are produced by symbiotic bacteria.
A) I and IV.
B) II, III, and IV.
C) I, II, and III.
D) I, II, III, and IV.

18) Which of the following experiments is the best way to determine if an element is essential for plant growth?
A) Measure the amount of the element stored in plant tissues.
B) Measure the amount of the element in the soil after plant growth.
C) Measure the weight of the plant and soil before and after plant growth.
D) Grow a plant using hydroponics with and without the element.

19) Which criteria allow biologists to divide chemicals into macronutrients and micronutrients?
A) Molecular weight of the element or compound.
B) The quantities of each required by plants.
C) How they are used in metabolism.
D) Whether or not they are essential for plant growth.

20) Which elements are most often the limiting nutrients for plant growth?
A) Nitrogen, potassium, phosphorus.
B) Nitrogen, oxygen, hydrogen.
C) Carbon, sodium, chlorine.
D) Carbon, nitrogen, oxygen.

21) Synthesis of which of the following compounds in a mature leaf would be least impacted by a temporary soil nitrogen deficiency?
A) DNA.
B) RNA.
C) Amino acids.
D) Cellulose.

22) A major function of magnesium in plants is to be _____.
A) Required to regenerate phosphoenolpyruvate in C_4 and CAM plants.
B) A component of DNA and RNA.
C) A component of chlorophyll.
D) Active in amino acid formation.

23) Micronutrients are needed in very small amounts because _____.
A) Most of them are mobile in the plant.
B) Most serve mainly as cofactors of enzymes.
C) They play only a minor role in the growth and health of the plant.
D) Only the most actively growing regions of the plants require micronutrients.

24) Two groups of tomatoes were grown under laboratory conditions, one with humus added to the soil and one a control without humus. The leaves of the plants grown without humus were yellowish (less green) compared with those of the plants grown in the humus-enriched soil. The best explanation for this difference is that _____.
A) The healthy plants used the food in the decomposing leaves of the humus for energy to make chlorophyll.
B) The humus made the soil more loosely packed, so water penetrated more easily to the roots.
C) The humus contained minerals such as magnesium and iron, needed for the synthesis of chlorophyll.
D) The heat released by the decomposing leaves of the humus caused more rapid growth and chlorophyll synthesis

25) Soil leaching can cause nutrient deficiencies in the soil. Which of the following are symptoms of nutrient deficiency in plants?
I) Chlorosis.
II) Death of meristems.
III) Excess storage of chlorophyll.
IV) Small internodes.
A) I, II, and III.
B) II, III, and IV.
C) I, II, and IV.
D) I, II, III, and IV.

26) If an African violet has chlorosis, which of the following elements might be a useful addition to the soil?
A) Molybdenum.
B) Copper.
C) Iodine.
D) Magnesium.

27) Which of the following are characteristic of both rhizobia and mycorrhizae?
I) They both benefit by receiving sugars from the plant.
II) They both become parasitic in nutrient-rich environments.
III) They both enhance the growth of most plants.
IV) They both are found in most ecosystems of the world.
A) Only I and II.
B) Only I, III, and IV.
C) Only III and IV.
D) I, II, III, and IV.

28) Which of the following are functions of rhizobacteria?
I) Produce hormones that stimulate plant growth.
II) Produce antibiotics that protect roots from disease.
III) Absorb toxic metals.
IV) Carry out nitrogen fixation.
A) Only II and IV.
B) Only I, II, and III.
C) Only III and IV.
D) I, II, III, and IV.

29) Nitrogen fixation is a process that _____.
A) Recycles nitrogen compounds from dead and decaying materials.
B) Converts ammonia to ammonium.
C) Releases nitrate from the rock substrate.
D) Converts nitrogen gas into ammonia.

30) Why is nitrogen fixation an essential process?
A) Nitrogen fixation can only be done by certain prokaryotes.
B) Fixed nitrogen is often the limiting factor in plant growth.
C) Nitrogen fixation is very expensive in terms of metabolic energy.
D) Nitrogen fixers are sometimes symbiotic with legumes.

31) You are weeding your garden when you accidentally expose some roots of your pea plants. You notice swellings (root nodules) on the roots and there is a reddish tinge to the ones you accidentally damaged. Most likely your pea plants _____.
A) Suffer from a mineral deficiency.
B) Are infected with a parasite.
C) Are benefiting from a mutualistic bacterium.
D) Are developing offshoots from the root.

32) The specific relationship between a legume and its mutualistic *Rhizobium* strain probably depends on _____.
A) Each legume having a chemical dialogue with a fungus.
B) Each *Rhizobium* strain having a form of nitrogenase that works only in the appropriate legume host.
C) Each legume being found where the soil has only the *Rhizobium* specific to that legume.
D) Specific recognition between the chemical signals and signal receptors of the *Rhizobium* strain and legume species.

33) Rhizobia, actinomycetes, and cyanobacteria all share the common feature that they can _____.
A) Increase water uptake in plants.
B) Kill parasites in the soil.
C) Exist in extreme environments.
D) Fix atmospheric nitrogen.

34) The earliest vascular plants on land had underground stems (rhizomes) but no roots. Water and mineral nutrients were most likely obtained by _____.
A) Diffusion through stomata.
B) Absorption by mycorrhizae.
C) Osmosis through the root hairs.
D) Diffusion across the cuticle of the rhizome.

35) What major benefits do plants and mycorrhizal fungi receive from their symbiotic relationship?
A) Plants receive enzymes, and fungi receive nitrogen and phosphorus.
B) Plants receive increased root surface area, and fungi receive digestive enzymes.
C) Fungi receive photosynthetic products in exchange for living in plant root nodules.
D) Plants receive nitrogen and phosphorus, and fungi receive photosynthetic products.

36) Hyphae form a covering over roots. These hyphae create a large surface area that helps to do which of the following?
A) Aid in absorbing minerals and ions.
B) Maintain cell shape.
C) Increase cellular respiration.
D) Anchor a plant.

37) A plant developed a mineral deficiency after being treated with a fungicide. What is the most probable cause of the deficiency?
A) Mineral receptor proteins in the plant membrane were not functioning.
B) Mycorrhizal fungi were killed.
C) Active transport of minerals was inhibited.
D) Proton pumps reversed the membrane potential.

38) We would expect the greatest difference in plant health between two groups of plants of the same species, one group with mycorrhizae and one group without mycorrhizae, in an environment _____.
A) Where nitrogen-fixing bacteria are abundant.
B) That has soil with poor drainage.
C) In which the soil is relatively deficient in mineral nutrients.
D) That is near a body of water, such as a pond or river.

39) Which of the following is a primary difference between ectomycorrhizae and endomycorrhizae?
A) Endomycorrhizae has thicker, shorter hyphae than ectomycorrhizae.
B) Ectomycorrhizae does not penetrate root cells, whereas endomycorrhizae grow into invaginations of the root cell membranes.
C) Endomycorrhizae is more common than ectomycorrhizae.
D) There are no significant differences between ectomycorrhizae and endomycorrhizae.

40) Carnivorous plants have evolved mechanisms that trap and digest small animals. The products of this digestion are used to supplement the plant's supply of _____.
A) Energy.
B) Carbohydrates.
C) Lipids and steroids.
D) Nitrogen and other minerals.

41) Epiphytes are _____.
A) Aerial vines common in tropical regions.
B) Plants that live in poor soil and digest insects to obtain nitrogen.
C) Plants that grow on other plants but do not obtain nutrients from their hosts.
D) Plants that have a symbiotic relationship with fungi.

42) While hiking in a forest, you notice an unusual plant growing on the branches of a tree. What will help you to determine if this plant is epiphytic or parasitic?
A) If the plant is green, it is epiphytic; if not, then it is parasitic.
B) The root of an epiphytic plant will be in the soil, but a parasitic plant will grow from the trunk of a tree.
C) The roots of a parasitic plant will penetrate under the bark into the tree xylem, and the roots of epiphytic plant will not.
D) The epiphytic plant will have large water collecting leaves, and the parasitic plant will not.

CHAPTER 38:

ANGIOSPERM REPRODUCTION AND BIOTECHNOLOGY

1) Which of the following is the correct order of floral organs from the outside to the inside of a complete flower?
A) Petals → sepals → stamens → carpels.
B) Sepals → stamens → petals → carpels.
C) Spores → gametes → zygote → embryo.
D) Sepals → petals → stamens → carpels.

2) Arrange the following structures from largest to smallest, assuming that they belong to two generations of the same angiosperm.
1. Ovary.
2. Ovule.
3. Egg.
4. Carpel.
5. Embryo sac.
A) 4, 2, 1, 5, 3.
B) 5, 4, 3, 1, 2.
C) 5, 1, 4, 2, 3.
D) 4, 1, 2, 5, 3.

3) A summer occupation in the Corn Belt states is de-tasseling the corn: removing unwanted male flowers so that female flowers on the same plant are pollinated by the desired pollen for the hybrid corn. What does this tell you about corn? The flowers are _____.
A) Perfect and the plant are dioecious.
B) Perfect and the plant are monoecious.
C) Imperfect and the plant are dioecious.
D) Imperfect and the plant are monoecious.

4) During the alternation of generations in plants, _____.
A) Meiosis produces gametes.
B) Mitosis produces gametes.
C) Fertilization produces spores.
D) Fertilization produces gametes.

5) Which of these is a major trend in land plant evolution?
A) The trend toward smaller size.
B) The trend toward a gametophyte-dominated life cycle.
C) The trend toward a sporophyte-dominated life cycle.
D) The trend toward larger gametophytes.

6) Retaining the zygote on the living gametophyte of land plants _____.
A) Protects the zygote from herbivores.
B) Evolved concurrently with pollen.
C) Helps in dispersal of the zygote.
D) Allows it to be nourished by the parent plant.

7) Sperm cells are formed in plants by _____.
A) Meiosis in pollen grains.
B) Meiosis in anthers.
C) Mitosis in male gametophyte.
D) Mitosis in the micropyle.

8) A researcher has developed two stains for use with seed plants. One stains sporophyte tissue blue; the other stains gametophyte tissue red. If the researcher exposes pollen grains to both stains, and then rinses away the excess stain, what should occur?
A) The pollen grains will be pure red.
B) The pollen grains will be pure blue.
C) The pollen grains will have red interiors and blue exteriors.
D) The pollen grains will have blue interiors and red exteriors.

9) In which of the following pairs are the two terms equivalents?
A) Ovule – egg.
B) Embryo sac - female gametophyte.
C) Seed – zygote.
D) Microspore - pollen grain.

10) The generative cell of male angiosperm gametophytes is haploid. This cell divides to produce two haploid sperm cells. What type of cell division does the generative cell undergo to produce these sperm cells?
A) Mitosis.
B) Meiosis.
C) Mitosis without subsequent cytokinesis.
D) Meiosis without subsequent cytokinesis.

11) Which of the following statements regarding flowering plants is correct?
A) The gametophyte is the dominant generation.
B) Female gametophytes develop from megaspores within the anthers.
C) Pollination is the delivery of pollen to the stigma of a carpel.
D) The food-storing endosperm is derived from the cell that contains one polar nucleus and two sperm nuclei.

12) In a typical angiosperm, what is the sequence of structures encountered by the tip of a growing pollen tube on its way to the egg?
1. Micropyle.
2. Style.
3. Ovary.
4. Stigma.
A) 4 →2 → 3 → 1.
B) 4 → 3 → 2 → 1.
C) 1 → 3 → 4 → 2.
D) 3 → 2 → 4 → 1.

13) If an ovary contains 50 ovules, what is the minimum number of pollen grains that must land to form 50 mature seeds?
A) 25.
B) 50.
C) 100.
D) 500.

14) Double fertilization means that _____.
A) Flowers must be pollinated twice to yield fruits and seeds.
B) One sperm is needed to fertilize the egg, and a second sperm is needed to fertilize the polar nuclei.
C) The egg of the embryo sac is diploid.
D) Every sperm has two nuclei.

15) What is typically the result of double fertilization in angiosperms?
A) The endosperm develops into a diploid nutrient tissue.
B) A triploid zygote is formed.
C) Both a diploid embryo and triploid endosperm are formed.
D) Two embryos develop in every seed.

16) Suppose that 100 pollen grains land on a stigma, and 50 mature seeds are formed in the fruit. What does this indicate about the pollination process and success?
A) 50% success: 100 pollen grains grew to 50 ovules, and double fertilization occurred.
B) 50% success: evidently, only 50 sperm pollinated 50 anthers.
C) 50% success: 50 sperm fertilized 50 eggs, and 50 sperm fused with 50 polar nuclei.
D) 50% success: 50 sperm fertilized 50 eggs, and 50 sperm fused with 100 polar nuclei.

17) Which of the following flower parts develops into a seed?
A) Ovule.
B) Ovary.
C) Stamen.
D) Carpel.

18) The vast number and variety of flower species is probably related to various kinds of _____.
A) Seed dispersal agents.
B) Pollinators.
C) Herbivores.
D) Climatic conditions.

19) It is estimated that animal-pollinated or insect-pollinated plants produce 1000 pollen grains for each ovule; wind-pollinated plants produce 1,000,000 pollen grains for each ovule. What does that indicate about pollination systems?
A) Wind-pollinated plants rarely produce seeds.
B) Wind pollination is more efficient than animal-assisted pollination.
C) Wind pollination is less efficient than animal-assisted pollination.
D) Wind pollination is costlier to the plant than animal-assisted pollination.

20) Cottonwood, aspen, and willow trees have beige flowers, with no petals, that appear before the tree's leaves are out in the spring; and they are dioecious. What does this indicate about these trees?
A) Their insect pollinators are specialists.
B) Early emerging insects are probably the pollinators.
C) Their pollen is dispersed by wind.
D) The trees are self-pollinating.

21) Which of the following occur during the formation of an embryo from a zygote in angiosperms?
I) The root and shoot systems emerge from the seed.
II) Basal cells form a connection between the parent plant and the developing embryo.
III) Meiosis produces a mass of cells that become the young embryo.
IV) Cells differentiate to form the basic plant tissue types.
V) The early root-shoot axis is formed.
A) I, II, and III.
B) II, IV, and V.
C) II, III, IV, and V.
D) III, IV, and V.

22) The egg of a plant has a haploid chromosome number of 12 ($n = 12$). What is true about the number of chromosomes in the cells of other tissues of this plant?
A) The sperm has 6 chromosomes.
B) The leaves and stems have 12 chromosomes.
C) The zygote has 12 chromosomes.
D) The endosperm has 36 chromosomes.

23) What adaptations should one expect of the seed coats of angiosperm species whose seeds are dispersed by frugivorous (fruit-eating) animals, as opposed to angiosperm species whose seeds are dispersed by other means?
1. The exterior of the seed coat should have barbs or hooks.
2. The seed coat should contain secondary compounds that irritate the lining of the animal's mouth.
3. The seed coat should be able to withstand low pHs.
4. The seed coat, upon its complete digestion, should provide vitamins or nutrients to animals.
5. The seed coat should be resistant to the animals' digestive enzymes.
A) 4 only.
B) 1 and 2.
C) 3 and 5.
D) 3, 4, and 5.

24) Which of these events occurs first in seed germination?
A) Cell division occurs in the embryo and growth starts.
B) Mitochondria multiply and provide energy for growth processes.
C) Water is taken up.
D) Oxygen is produced and proteins are synthesized.

25) Before plowing a field, a farmer thought the bare field looked weed-free. Three days after plowing and turning over the soil, he was amazed to see thousands of tiny seedlings. What is the most likely reason for the mass germination of seeds?
A) Large seeds that needed soil disturbance to germinate.
B) Small seeds that need light to germinate.
C) Small seeds that were scarified by exposure to plow.
D) Large seeds that needed exposure to higher levels of oxygen to germinate.

26) Angiosperms are the most successful terrestrial plants. Which of the following features is unique to them and helps account for their success?
A) Wind pollination.
B) Dominant gametophytes.
C) Fruits enclosing seeds.
D) Sperm cells without flagella.

27) Which of the following flower parts develops into the pulp of a fleshy fruit?
A) Stigma.
B) Style.
C) Ovule.
D) Ovary.

28) The black dots that cover strawberries are actually individual fruits. The fleshy and tasty portion of a strawberry derives from the receptacle of a flower with many separate carpels. Therefore, a strawberry is _____.
A) Both a multiple fruit and an aggregate fruit.
B) Both a multiple fruit and an accessory fruit.
C) Both an aggregate fruit and an accessory fruit.
D) A simple fruit with many seeds.

29) Among plants known as legumes (beans, peas, alfalfa, clover, for example) the seeds are contained in a fruit that is itself called a legume, better known as a pod. Upon opening such pods, it is commonly observed that some ovules have become mature seeds, whereas other ovules have not. Thus, which of the following statements is (are) true?
1. The flowers that gave rise to such pods were not pollinated.
2. Pollen tubes did not enter all of the ovules in such pods.
3. There was apparently not enough endosperm to distribute to all of the ovules in such pods.
4. The ovules that failed to develop into seeds were derived from sterile floral parts.
5. Fruit can develop, even if all ovules within have not been fertilized.
A) 1 only.
B) 1 and 5.
C) 2 and 5.
D) 3 and 5.

30) Unripe fruits protect seeds from predation and early germination. What is the major function of ripe fruits?
A) Attracting pollinators.
B) Dispersing seed.
C) Releasing nutrients to seeds.
D) Keeping the seed hydrated before germination.

Use the following information to answer the questions (31-36) below.

The Brazil nut tree, *Bertholletia excels* (*n* = 17), is native to tropical rain forests of South America. It is a hardwood tree that can grow to over 50 meters tall, is a source of high -quality lumber, and is a favorite nesting site for harpy eagles. As the rainy season ends, tough -walled fruits, each containing 8-25 seeds (Brazil nuts), fall to the forest floor. Brazil nuts are composed primarily of endosperm. About $50 million worth of nuts are harvested each year. Scientists have discovered that the pale yellow flowers of Brazil nut trees cannot fertilize themselves and admit only female orchid bees as pollinators. The agouti (*Dasyprocta spp.*), a cat-sized rodent, is the only animal with teeth strong enough to crack the hard wall of Brazil nut fruits. It typically eats some of the seeds, buries others, and leaves still others inside the fruit, which moisture can now enter. The uneaten seeds may subsequently germinate.

31) If a female orchid bee has just left a Brazil nut tree with nectar in her stomach, and if she visits another flower on a different Brazil nut tree, what is the sequence in which the following events should occur?
1. Double fertilization.
2. Pollen tube emerges from pollen grain.
3. Pollen tube enters micropyle.
4. Pollination.
A) 4, 2, 3, 1.
B) 4, 3, 2, 1.
C) 2, 4, 3, 1.
D) 2, 4, 1, 3.

32) Orchid bees are to Brazil nut trees as _ are to pine trees.
A) Breezes.
B) Rain droplets.
C) Seed-eating birds.
D) Squirrels.

33) Entrepreneurs attempted, but failed, to harvest nuts from plantations grown in Southeast Asia. Attempts to grow Brazil nut trees in South American plantations also failed. In both cases, the trees grew vigorously, produced healthy flowers in profusion, but set no fruit. Consequently, what is the likely source of the problem?
A) Poor sporophyte fertility.
B) Failure to produce fertile ovules.
C) Failure to produce pollen.
D) Pollination failure.

34) The same bees that pollinate the flowers of the Brazil nut trees also pollinate orchids, which are epiphytes (in other words, plants that grow on other plants); however, orchids cannot grow on Brazil nut trees. These observations explain _____.
A) The coevolution of Brazil nut trees and orchids.
B) Why Brazil nut trees do not set fruit in monoculture plantations.
C) Why male orchid bees do not pollinate Brazil nut tree flowers.
D) Why male orchid bees are smaller than female orchid bees.

35) The agouti is most directly involved with the Brazil nut tree's dispersal of _____.
A) Male gametophytes.
B) Female gametophytes.
C) Sporophyte embryos.
D) Sporophyte megaspores.

36) Animals that consume Brazil nuts derive nutrition mostly from tissue whose nuclei have how many chromosomes?
A) 17.
B) 34.
C) 51.
D) 68.

37) Which of the following could be considered an evolutionary advantage of asexual reproduction in plants?
A) Increased success of progeny in a stable environment.
B) Increased agricultural productivity in a rapidly changing environment.
C) Maintenance and expansion of a large genome.
D) Increased ability to adapt to a change in the environment.

38) Plants produce more seeds when they reproduce asexually than sexually. Yet most plants reproduce sexually in nature. What is the probable explanation for the prevalence of sexual reproduction? Sexual reproduction _____.
A) Is more energy efficient than asexual reproduction.
B) Ensures genetic continuity from parents to offspring.
C) Mixes up alleles contributing to variation in a species.
D) Is not dependent on other agents of pollination.

39) Which of the following is a true statement about asexual reproduction in plants?
A) Clones of plants do not occur naturally.
B) Cloning, although achieved in animals, has not been demonstrated in plants.
C) Making cuttings of ornamental plants is a form of fragmentation.
D) Reproduction of plants by cloning may be either sexual or asexual.

40) While looking at a flower in your garden, you notice that it has carpels with very long styles, and stamens with very short filaments. This plant is most likely to reproduce by _____.
A) Cross-pollination.
B) Selfing.
C) Asexual reproduction.
D) None of them.

41) Which of the following types of plants are incapable of self-pollination?
A) Dioecious.
B) Monoecious.
C) Wind-pollinated.
D) Insect-pollinated.

42) Which of the following is a potential advantage of introducing apomixis into hybrid crop species?
A) Cultivars would be better able to cope with a rapidly changing environment.
B) They would have a larger potential genome than inbred crops.
C) All of the desirable traits of the cultivar would be passed on to offspring.
D) They would benefit from positive mutations in their DNA.

43) Pollen from a plant with the S_1S_2 genotype is recognized and allowed to germinate on the stigma of the same plant with the S_1S_2 genotype. According to the S-system hypothesis, this indicates that the plant is _____.
A) Self-compatible and can self-pollinate.
B) Self-compatible and must cross-pollinate.
C) Self-incompatible and can self-pollinate.
D) Self-incompatible and must cross-pollinate.

44) Over human history, which process has been most important in improving the features of plants that have long been used by humans as staple foods?
A) Genetic engineering.
B) Artificial selection.
C) Sexual selection.
D) Pesticide and herbicide application.

45) Which of these activities is part of the development of crop plants from wild relatives?
I) People planting seeds of the plants with the characteristic wanted.
II) People making observations of desired plant characteristics.
III) People eating products from only the plants with desired characteristics.
IV) People developing several varieties of crops from a wild relative.
A) I and II.
B) I and IV.
C) I, III, and IV.
D) I, II, and IV.

46) Regardless of where in the world a vineyard is located, for the winery to produce a Burgundy, it must use varietal grapes that originated in Burgundy, France. The most effective way for a new California grower to plant a vineyard to produce Burgundy is to _____.
A) Plant seeds obtained from French many Burgundy grape.
B) Transplant varietal Burgundy plants from France.
C) Acquire a tissue culture of varietal Burgundy grapes from France.
D) Graft varietal Burgundy grape scions onto native (Californian) root stocks.

47) The most immediate potential benefits of introducing genetically modified crops include _____.
I) Creating crops that can grow on land previously unsuitable for agriculture.
II) Creating crops with better potential for biofuel production.
III) Creating crops with better nutritional attributes.
IV) Increasing crop yield.
V) Decreasing the mutation rate of certain genes.
A) Only II, III, and IV.
B) Only I, II, III, and IV.
C) Only III, IV, and V.
D) I, II, III, IV, and V.

48) "Golden Rice" _____.
A) Is resistant to various herbicides, making it practical to weed rice fields with those herbicides.
B) Includes bacterial genes that produce a toxin that reduces damage from insect pests.
C) Produces larger, golden grains that increase crop yields.
D) Contains daffodil genes that increase vitamin A content.

49) Which of the following is a scientific concern related to creating genetically modified crops?
A) Herbicide resistance may spread to weedy species.
B) Genetically modified crops cannot survive without the addition of great amounts of fertilizer to the soil.
C) The monetary costs of growing genetically modified plants are significantly greater than traditional breeding techniques.
D) Genetically modified plants are less stable and may revert back to parental genotypes.

50) Which of the following would be the most problematic for the natural environment in the development of genetically engineered crops?
A) The introduction of male sterility into crops.
B) The creation of transgenic crops with apomictic seeds.
C) The creation of crops with flowers that develop normally, but fail to open.
D) The creation of transgenic crops that hybridize more easily.

51) Fruit ripening represents an example of positive feedback. Which one of the following statements accurately justifies why the process of fruit ripening involves positive feedback?
A) Once seeds have reached maturity, chemical signals increase enzymatic activity in fruit to convert sugars into starches, thicken pulp, and maintain color in the fruit.
B) Once seeds have reached maturity, chemical signals block enzymatic steps that normally convert sugars into starches and sugars accumulate from photosynthetic activity within the fruit.
C) Chemical signals initiate a process that triggers enzymatic activity, which involves converting starches into sugars, softening of pulp, and color change of fruit.
D) Chemical signals shut down enzymatic activity within the fruit, which results in breakdown of starches into sugars, softening of pulp, and color change of fruit.

52) In order for an ovule (egg cell) in a flower to be fertilized and form a viable seed, pollination must occur. In this process, a sperm cell is delivered to the ovule when the pollen grain lands on the stigma and grows a tube, which enters the ovary and discharges the sperm cell to form a diploid zygote when it fuses with the egg cell. Although it only takes one pollen grain to successfully deliver sperm to the egg, numerous pollen grains are generally transferred to the stigma during insect pollination of flowering plants. Which phenotypic traits of pollen would you predict to be selected upon to promote survival and fitness within an insect-pollinated flowering plant?
A) High pollen tube growth rate and ability to detect chemicals from cells surrounding the egg.
B) Ability to produce the most cells during mitotic growth of the pollen tube.
C) Elaborate and striking UV "nectary guides" on the petals to guide an insect to the stigma.
D) Larger pollen in order to carry the tube that is necessary for delivery of the sperm cells.

53) Many flowering plants coevolve with specific pollinators. The Madagascar orchid has a 12-inch floral tube and is a reliable nectar source for the hawkmoth, which has a correspondingly long proboscis (tongue). Which statement most accurately describes how coevolution might have occurred for the hawkmoth and Madagascar orchid shown here?

A) The hawkmoths that expended the most effort to reach the nectar would be the most fit, and pass the longer tongue phenotype to their offspring.

B) Natural selection would favor orchids with nectar tubes just long enough to for an insect with pollen to make contact. Hawkmoths whose tongue could reach the deep tubes would be more fit.

C) Hawkmoths whose tongue was just long enough to obtain nectar, but not able to pick up pollen would become the most fit in the population.

D) It is most likely that mutations that resulted in both the length of the orchid floral tube and the length of the hawkmoth tongue occurred abruptly and simultaneously.

CHAPTER 39:

PLANT RESPONSES TO INTERNAL AND EXTERNAL SIGNALS

1) The detector of light during de-etiolation (greening) of a tomato plant is (are) _____.
A) Carotenoids.
B) Xanthophylls.
C) Phytochrome.
D) Auxin.

2) Plant hormones _____.
A) In plant cells naturally exist in very large amounts.
B) Change their shape in response to stimulus.
C) Are unable to move from one cell to another.
D) Affect only cells with the appropriate receptor.

3) Which of the following can function in signal transduction in plants ?
I) Calcium ions.
II) Nonrandom mutations.
III) Receptor proteins.
IV) Autochrome.
V) Secondary messengers.
A) Only I, III, and IV.
B) Only I, II, and V.
C) Only I, III, and V.
D) Only II, III, and V.

4) Plant hormones produce their effects by _____.
I) Altering the expression of genes.
II) Modifying the permeability of the plasma membrane.
III) Modifying the structure of the nuclear envelope membrane.
A) Only I.
B) Only II.
C) Only III.
D) Only I and II.

5) Plant hormonal regulation differs from animal hormonal regulation in that _____.
A) There are no dedicated hormone-producing organs in plants as there are in animals.
B) All production of hormones is local in plants with little long-distance transport.
C) Only animal hormone concentrations are developmentally regulated.
D) Only animal hormones may have either external or internal receptors.

6) Which of the following plant growth responses is primarily due to the action of auxins?
A) Leaf abscission.
B) Cell division.
C) The detection of photoperiod.
D) Phototropism.

7) Auxins in plants are known to affect which of the following processes?
I) Gravitropism of shoots.
II) Maintenance of seed dormancy.
III) Phototropism of shoots.
IV) Inhibition of lateral buds.
V) Apical dominance.
A) Only I and II.
B) Only I, III and V.
C) Only I, III, IV and V.
D) Only II, III, IV and V.

8) Experiments on the positive phototropic response of plants indicate that _____.
A) Light destroys auxin.
B) Auxin moves down the plant apoplastically.
C) Auxin is synthesized in the area where the stem bends.
D) Auxin can move to the shady side of the stem.

9) Which of the following statements best summarizes the acid -growth hypothesis in an actively growing shoot?
A) Auxin stimulates proton pumps in the plasma membrane and tonoplast.
B) Auxin-activated proton pumps lower the pH of the cell wall, which breaks bonds and makes the walls more flexible.
C) Auxins and gibberellins together act as a lubricant to help stretch cellulose microfibrils.
D) Auxins activate aquaporins that increase turgor pressure in the cells.

10) Which of the following conclusions is supported by the research of both Went and Charles and Francis Darwin on shoot responses to light?
A) When shoots are exposed to light, a chemical substance migrates toward the light.
B) A chemical substance involved in shoot bending is produced in shoot tips.
C) Once shoot tips have been cut, normal growth cannot be induced.
D) Light stimulates the synthesis of a plant hormone that responds to light.

11) An eccentric millionaire botanist has offered a $25,000 scholarship to anyone who can successfully get a plant to grow through a vertical maze in complete darkness. The maze is not in a box; the maze is simply drawn on the wall, and the contestants must get their plant to grow in a pattern that matches the path through the maze. You need the money and feel confident that you can accomplish this task. Which of the following techniques will help you succeed?
A) Apply auxin directly to the shoot tip on the side to which you want the tip to bend.
B) Apply auxin directly to the part of the stem just below the tip opposite from the direction you want the stem to bend.
C) Inject compounds that block auxin receptors into the part of the stem opposite from the direction you want the stem to bend.
D) Plant the roots in two different pots, and apply auxin to the root bucket that is on the same side as the direction you want the plant to bend.

12) You have a small tree in your yard that is the height that you want it, but does not have as many branches as you want. How can you prune it to trigger it to increase the number of branches?
A) Cut off the leaves at the ends of several branches.
B) Cut off the tips of the main shoots.
C) Cut off lower branches.
D) Cut off the leaves at the base of most of the branches.

13) As cytokinins are primarily produced in roots, what route would they travel to influence lateral shoot formation in a recently topped tree?
A) Symplastic.
B) Trachieds/vessels.
C) Phloem.
D) Apoplastic.

14) Who might be interested in using cytokinins?
A) Grocers, to spray on fruit to enhance ripening in the store.
B) Consumers, to spray on fruit before eating to enhance taste.
C) Florists, to dip stems in to keep leaves green longer.
D) Farmers, to spray on fruit after picking to stall ripening.

15) If a farmer wanted more loosely packed clusters of grapes, he would most likely spray the immature bunches with _____.
A) Auxin.
B) Gibberellins.
C) Cytokinins.
D) Abscisic acid.

16) A researcher found a beautiful plant while traveling in Alaska and collected its seeds. When she came back to Florida, she soaked some seeds in pure water and some in water with a hormone. When she put the seeds in soil to grow, only the seeds that had been soaked with the hormone germinated. The hormone most likely was _____.
A) Gibberellin.
B) Abscisic acid (ABA).
C) Auxin.
D) Ethylene.

17) _____ prevents seeds from germinating until conditions are favorable for the growth of the plant.
A) Ethylene.
B) Zeaxanthin.
C) Gibberellin
D) Abscisic acid.

18) A population of plants experiences several years of severe drought. Much of the population dies due to lack of water, but a few individuals survive. You set out to discover the physiological basis for their adaptation to such an extreme environmental change. You hypothesize that the survivors have the ability to synthesize higher levels of _____ than their siblings do.
A) Auxin.
B) Gibberellin.
C) Cytokinins.
D) Abscisic acid.

19) If you were shipping green bananas to a supermarket thousands of miles away, which of the following chemicals would you want to eliminate from the plants' environment?
A) Carbon dioxide.
B) Cytokinins.
C) Ethylene.
D) Auxin.

20) In the fall, the leaves of some trees change color. This happens because chlorophyll breaks down and the accessory pigments become visible. What hormone is responsible for this?
A) Phototropin.
B) Abscisic acid.
C) Cytokinins.
D) Ethylene.

21) Which type of mutant would be most likely to produce a bushier phenotype?
A) Auxin overproducer.
B) Strigolactone overproducer.
C) Cytokinin underproducer.
D) Strigolactone underproducer.

22) Plant growth regulators _____.
A) Only act by altering gene expression.
B) Often have a multiplicity of effects.
C) Function independently of other hormones.
D) Affect the division and elongation, but not the differentiation, of cells.

23) Vines in tropical rain forests must grow toward large trees before being able to grow toward the sun. To reach a large tree, the most useful kind of growth movement for a tropical vine presumably would be _____.
A) Negative thigmotropism.
B) Negative phototropism.
C) Negative gravitropism.
D) The opposite of circadian rhythms.

24) In lettuce seeds, blue light initiates germination. If you measured hormone levels within the seed, which hormone would be produced upon exposure to blue light?
A) Gibberellin.
B) Ethylene.
C) Abscisic acid.
D) Cytokinins.

25) Upon exposure to blue light, plants not only begin to grow toward the light, but move their chloroplasts to the sunny side of each cell. The adaptive advantage of moving chloroplasts to the sunny side of each cell _____.
A) Maximizes light absorption by the chloroplasts for photosynthesis.
B) Increases production of phototropic hormones.
C) Maximizes heat absorption by the chloroplasts for cellular respiration.
D) Increases adenosine triphosphate (ATP) production during the light-independent reactions.

26) Mammalian eyes sense light because the photoreceptor cells have molecules called opsins, which change structure when exposed to light. Which of the following plant molecules would be analogous to mammalian opsins in their light -sensing ability?
A) Auxin and Phytochrome.
B) Auxin and Pfr.
C) Pfr and Phytochrome.
D) Cytokinins and phototropins.

27) Seed packets give a recommended planting depth for the enclosed seeds. The most likely reason some seeds are to be covered with only 1\4 -inch of soil is that the _____.
A) Seedlings do not have an etiolation response.
B) Seeds require light to germinate.
C) Seeds require a higher temperature to germinate.
D) Seeds are very sensitive to waterlogging.

28) Suppose a plant had a photosynthetic pigment that absorbed far-red wavelengths of light. In which of the following environments could that plant thrive?
A) On the surface of a lake.
B) On the forest floor, beneath a canopy of taller plants.
C) On the ocean floor, in very deep waters.
D) On mountaintops, closer to the Sun.

29) The biological clock controlling circadian rhythms must ultimately _____.
A) Depend on environmental cues.
B) Affect gene transcription.
C) Stabilize on a 24-hour cycle.
D) Speed up or slow down with increasing or decreasing temperature.

30) Many plants flower in response to day-length cues. Which of the following statements best summarizes this phenomenon?
A) As a rule, short-day plants flower in the summer.
B) As a rule, long-day plants flower in the spring or fall.
C) Long-day plants flower in response to long days, not short nights.
D) Flowering in short-day and long-day plants is controlled by phytochrome.

31) Plants often use changes in day length (photoperiod) to trigger events such as dormancy and flowering. It is logical that plants have evolved this mechanism because photoperiod changes _____.
A) Are more predictable than air temperature changes.
B) Predict moisture availability.
C) Are modified by soil temperature changes.
D) Can reset the biological clock.

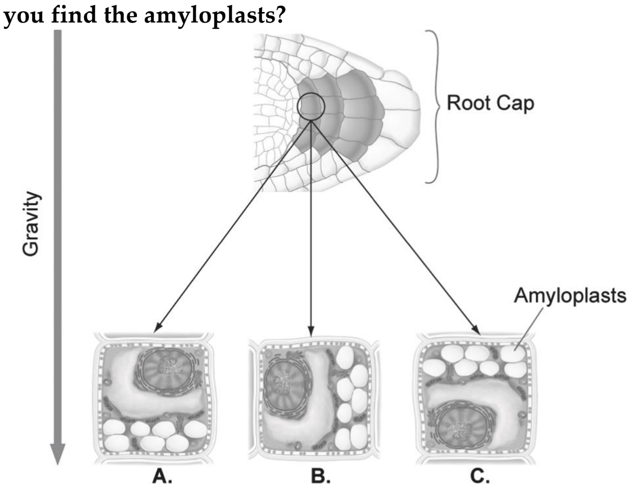
32) A gardener in Canada wants to surprise his mother on her birthday and make her favorite hibiscus bush flower in May instead of at the end of June. The bush is growing in the greenhouse. Which of the following might make the hibiscus bush flower early?
A) Grafting leaves of a hibiscus that was exposed to long night.
B) Grafting leaves of a hibiscus that was exposed to short night.
C) Exposing flower buds of the hibiscus bush to long nights.
D) Exposing flower buds of the hibiscus bush to short nights.

33) Which of the following can be sensed by plants?
I) Gravity.
II) Pathogens.
III) Wind.
IV) Light.
A) Only I and III.
B) Only I, II, and IV.
C) Only II, III, and IV.
D) I, II, III, and IV.

34) You have discovered a previously unidentified plant, and you cultivate it in your lab. You notice that its flowers close when people are talking, yet are open when the lab is relatively quiet. You suspect that this plant may have the ability to hear! Which of the following hypotheses is (are) the most reasonable to explain this phenomenon?
I) There is a cell-surface protein on the epidermal cells that becomes phosphorylated in response to vibration by sound waves.
II) There are tiny hairs on epidermal cells that bend in response to the vibration of sound waves, triggering an action potential in epidermal cells.
III) There is a cell-surface receptor on root cells that becomes phosphorylated when the soil vibrates in response to sound waves.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

35) Shoots that grow vertically toward the sun can be characterized as _____.
A) Positive for phototropism and negative for gravitropism.
B) Neutral for phototropism and positive for gravitropism.
C) Negative for phototropism and positive for gravitropism.
D) Positive for phototropism and neutral for gravitropism.

36) Suppose you laid a seedling on its side so that the root was parallel to the ground as shown in the figure below. Several hours after the change in position, where in the root cells (position A, B, or C in the figure above) would you find the amyloplasts?



A) A.
B) B.
C) C.
D) None of them.

37) The rapid leaf movements resulting from a response to touch (thigmotropism) primarily involve _____.
A) Potassium channels.
B) Nervous tissue.
C) Aquaporins.
D) Stress proteins.

38) In extremely cold regions, woody species may survive freezing temperatures by _____.
A) Emptying water from the vacuoles to prevent freezing.
B) Decreasing the numbers of phospholipids in cell membranes.
C) Decreasing the fluidity of all cellular membranes.
D) Increasing cytoplasmic levels of specific solute concentrations, such as sugars.

39) Most scientists agree that global warming is underway; thus, it is important to know how plants respond to heat stress. Which of the following would be a useful line of inquiry to try and improve plant response and survival to heat stress?
A) The production of heat-stable carbohydrates.
B) Increased production of heat-shock proteins.
C) The opening of stomata to increase evaporational heat loss.
D) Protoplast fusion experiments with xerophytic plants.

40) When an arborist prunes a limb off a valuable tree, he or she may paint the cut surface. The primary purpose of the paint is to _____.
A) Minimize water loss by evaporation from the cut surface.
B) Improve the appearance of the cut surface.
C) Stimulate growth of the cork cambium to "heal" the wound.
D) Block entry of pathogens through the wound.

41) You are out working in your garden, and you notice that one of your favorite flowering plants has black, dead spots on the leaves. You immediately suspect that the plant has been invaded by a pathogen and has initiated a(n) _____.
A) Avirulence response.
B) Hypersensitive response.
C) Resistance response.
D) Virulence response.

42) Generalized defense responses in organs distant from the infection site are called _____.
A) Hyperactive responses.
B) Systemic acquired resistance.
C) Pleiotropy.
D) Hyperplasia.

43) A certain bacterium infects a plant's upper leaves. A few days later, bacteria of the same species attempt to infect the same plant's roots but are unsuccessful. What process is responsible for the plant's ability to prevent this infection?
A) Antivirulence response.
B) Pathogenesis resistance.
C) Systemic acquired resistance.
D) Sequential immunity.

44) A particular species of virus carries a gene for salicylate hydroxylase, an enzyme that breaks down salicylic acid. Will this virus be more or less virulent to plants than other viruses?
A) More.
B) Less.
C) This will not make the virus more or less virulent.
D) None of them.

45) Which event during the evolution of land plants favored the synthesis of secondary compounds?
A) The greenhouse effect throughout the Devonian period.
B) The reverse-greenhouse effect during the Carboniferous period.
C) The association of the roots of land plants and fungi.
D) The rise of herbivory.

46) The major function of the medicinal compounds in plants is to _____.
A) Attract pollinators for seed dispersal.
B) Attract insects and birds to spread seeds and fruits.
C) Defend the plant against herbivores.
D) Defend the plant against microbes.

47) For a plant to initiate chemical responses to herbivory, before it is directly affected by herbivores, _____.
A) A plant must have already flowered at least once.
B) Volatile "signal" compounds must be perceived.
C) Gene-for-gene intraspecific recognition must occur.
D) Phytoalexins must be released.

Use the following information to answer the questions (48-49) below.

Some plants have continually produced secondary defense compounds. Other plants are induced to form secondary defense compounds when they are injured. Corn seedling leaves that are chewed on by the caterpillars of a type of cutworm moth emit immediate volatile chemicals (LOX products), and after six hours large amounts of terpenoid compounds are released into the air. The terpenoids are released not only from the leaf being chewed, but from all leaves of the plant. The terpenoid compounds attract a parasitoid wasp female that lays her eggs on the caterpillar. When the wasp larvae hatch, they eat and kill the moth caterpillar. (T.C.J. Turlings, J. H. Loughrin, P. J. McCall, U. S. R. Rose, W. J. Lewis, and J. H. Tumlinson. 1995. How caterpillar-damaged plants protect themselves by attracting parasitic wasps. *Proceedings of the National Academy of Sciences* 92:4169-74.)

48) Refer to the paragraph on how caterpillar-damaged plants protect themselves by attracting parasitic wasps. What can you conclude based only on the information in the preceding paragraph?
A) The attracting terpenoid compounds are always present in the corn seedling.
B) Physical injury by the caterpillar mouthparts results in the immediate release of terpenoids.
C) Chemical signals from the caterpillar saliva attract the parasitic wasp.
D) The parasitoid wasp is attracted by compounds produced by an injured corn plant.

49) Refer to the paragraph on how caterpillar-damaged plants protect themselves by attracting parasitic wasps. To test the hypothesis that caterpillar saliva and a wound are both necessary to attract the parasitoid wasps you would need to include which of the following in your experiment?
I) Slash the corn leaves with a razor blade.
II) Put caterpillar saliva on a leaf wound.
III) Put caterpillar saliva on an intact leaf.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

50) You may have observed plants rotate towards a light source, thereby increasing the plant 's ability to intercept light energy and increase photosynthesis. You, however, are given the task of preventing grass seedlings from rotating toward the light. Using your knowledge of phototropism, which of the following experimental procedures would you use to complete your task?
A) Cover the growing tip of the grass seedling with black paper.
B) Supply the seedlings with very dim light (red light does not induce a bend).
C) Cover the portion of the seedling below the tip with a black shield.
D) Supply the seedling with nutrient-rich fertilizer solution.

51) A plant scientist was hired by a greenhouse operator to devise a way to force iris plants to bloom in the short days of winter. Iris normally blooms as long-day (short-night) plants. Which of the following has the best chance of creating iris blooms in winter?
A) Artificially increase the period of darkness in the greenhouse.
B) Increase the temperature to more closely follow summer temperatures.
C) Alternate four hours of darkness with four hours of light repeatedly over each 24-hour period.
D) Interrupt the long winter nights with a brief period of light.

52) An individual plant was discovered that could not grow towards light. After some research, it was determined that the reason was a defective gene that did not allow for the level of cell elongation necessary for a phototropic response. This mutation greatly reduces the fitness of the individual plant. Which reason best describes the reason for the loss of fitness?

A) The plant was too short to attract insects for pollination.

B) The plant could not adjust to directional light, which reduced photosynthetic activity and therefore energy available for reproduction.

C) Because the plant grew much taller and straighter, resources that could be used for reproduction were used for growth.

D) The loss of a phototropic response meant that the plant's seeds could not germinate so reproduction would be unsuccessful.

CHAPTER 40:

BASIC PRINCIPLES OF ANIMAL FORM AND FUNCTION

1) Penguins, seals, and tuna have body forms that permit rapid swimming, because _____.
A) All share a recent common ancestor.
B) All of their bodies have been compressed since birth by intensive underwater pressures.
C) The shape is a convergent evolutionary solution, which reduces drag while swimming.
D) This is the only shape that will allow them to maintain a constant body temperature in water.

2) As the size of some animals has evolved to greater sizes, the effectiveness of their adaptations that promote exchanges with the environment have also increased. For example, in many larger organisms, evolution has favored lungs and a digestive tract with _____.
A) More branching or folds.
B) Increased thickness.
C) Larger cells.
D) Decreased blood supply.

3) Much of the coordination of vertebrate body functions via chemical signals is accomplished by the _____.
A) Respiratory system.
B) Endocrine system.
C) Integumentary system.
D) Excretory system.

4) Compared with a smaller cell, a larger cell of the same shape has _____.
A) Less surface area.
B) Less surface area per unit of volume.
C) A smaller average distance between its mitochondria and the external source of oxygen.
D) A smaller cytoplasm-to-nucleus ratio.

5) If you were to view a sample of animal tissue under a light microscope and notice an extensive extracellular matrix surrounding a tissue, which tissue type would you most suspect?
A) Connective.
B) Epithelial.
C) Nervous.
D) Striated muscle.

6) Some animals have no gills when young, but then develop gills that grow larger as the animal grows larger. What is the reason for this increase in gill size?
A) The young of these animals are much more active than the adult, which leads to a higher BMR (basal metabolic rate) and, therefore, a higher need for oxygen.
B) Relative to their volume, the young have more surface area across which they can transport all the oxygen they need.
C) The young have a higher basal metabolic rate.
D) Relative to their surface area, the young have more body volume in which they can store oxygen for long periods of time.

7) Evolutionary adaptations that help diverse animals directly exchange matter between cells and the environment include _____.
A) A gastrovascular activity, a two-layered body, and a torpedo -like body shape.
B) An external respiratory surface, a small body size, and a two -cell-layered body.
C) A large body volume, a long, tubular body, and a set of wings.
D) An unbranched internal surface, a small body size, and thick covering.

8) All animals, whether large or small, have _____.
A) An external body surface that is dry.
B) A basic body plan that resembles a two-layered sac.
C) A body surface covered with hair to keep them warm.
D) Most of their cells in contact with an aqueous medium.

9) Interstitial fluid is _____.
A) The internal environment inside animal cells.
B) Identical to the composition of blood.
C) A common site of exchange between blood and body cells.
D) Found only in the lumen of the small intestine.

10) Of the following choices, the epithelium with the shortest diffusion distance is _____.
A) Simple squamous epithelium.
B) Simple columnar epithelium.
C) Pseudostratified ciliated columnar epithelium.
D) Stratified squamous epithelium.

11) Most of the exchange surfaces of multicellular animals are lined with _____.
A) Connective tissue.
B) Smooth muscle cells.
C) Neural tissue.
D) Epithelial tissue.

12) Connective tissues typically have _____.
A) Little space between the membranes of adjacent cells.
B) The ability to transmit electrochemical impulses.
C) The ability to shorten upon stimulation.
D) Relatively few cells and a large amount of extracellular matrix.

13) If you gently bend your ear, and then let go, the shape of your ear will return because the cartilage of your ear contains _____.
A) Collagenous fibers.
B) Elastic fibers.
C) Reticular fibers.
D) Adipose tissue.

14) Blood is best classified as connective tissue because _____.
A) Its cells are separated from each other by an extracellular matrix.
B) It contains more than one type of cell.
C) Its cells can move from place to place.
D) It is found within all the organs of the body.

15) Most types of communication between cells utilize _____.
A) The exchange of cytosol between the cells.
B) The movement of the cells.
C) Chemical or electrical signals.
D) The exchange of DNA between the cells.

16) All types of muscle tissue have _____.
A) Striated banding patterns seen under the microscope.
B) Cells that lengthen when appropriately stimulated.
C) A response that can be consciously controlled.
D) Interactions between actin and myosin.

17) Cardiac muscle cells are both _____.
A) Striated and interconnected by intercalated disks.
B) Smooth and under voluntary control.
C) Striated and under voluntary control.
D) Smooth and under involuntary control.

18) The type of muscle tissue surrounding the intestines and blood vessels is _____.
A) Skeletal muscle.
B) Cardiac muscle.
C) Intercalated cells.
D) Smooth muscle.

19) Food moves along the digestive tract as the result of contractions by _____.
A) Cardiac muscle.
B) Smooth muscle.
C) Striated muscle.
D) Skeletal muscle.

20) The cells lining the air sacs in the lungs make up a _____.
A) Simple squamous epithelium.
B) Stratified squamous epithelium.
C) Pseudostratified ciliated columnar epithelium.
D) Simple columnar epithelium.

21) The body tissue that consists largely of material located outside of cells is _____.
A) Epithelial tissue.
B) Connective tissue.
C) Skeletal muscle.
D) Nervous tissue.

22) You are looking through a microscope at a slide of animal tissue and see a single layer of flat, closely packed cells that cover a surface. This specific tissue is most likely _____.
A) Adipose.
B) A tendon.
C) Epithelial.
D) A neuron.

23) Environmental influences appear to contribute to cellular mutations that lead to tumor growth. For example, certain diets lead to higher incidence of colon cancers, and overexposure to sunlight leads to higher incidence of skin cancers. The tissues in closest contact with a carcinogen or mutagen (anything that causes genetic mutations) are obviously the ones most likely to develop tumors. Carcinomas and melanomas account for well over half of all cancers. What type of tissue would you guess the term *carcinoma* and *melanoma* is most closely associated with?
A) Connective.
B) Muscle.
C) Epithelial.
D) Nervous.

24) Which of the following is a true statement about body size and physiology?
A) The amount of food and oxygen an animal requires and the amount of heat and waste it produces are inversely proportional to its mass.
B) The rate at which an animal uses nutrients and produces waste products is independent of its volume.
C) Small and large animals face different physiological challenges because an animal's body mass increases cubically while its surface area increases as a squared function.
D) The wastes produced by an animal double as its volume doubles and triple as its surface area triples.

25) An elephant and a mouse are running in full sunlight, and both overheat by the same amount above their normal body temperatures. When they move into the shade and rest, which animal will cool down faster?
A) The elephant will because it has the higher surface-area-to-volume ratio.
B) The elephant will because it has the lower surface-area-to-volume ratio.
C) The mouse will because it has the higher surface-area-to-volume ratio.
D) They will cool at the same rate because they overheated by the same amount.

26) You have a cube of modeling clay in your hands. Which of the following changes to the shape of this cube of clay will decrease its surface area relative to its volume?
A) Pinch the edges of the cube into small folds.
B) Flatten the cube into a pancake shape.
C) Round the clay up into a sphere.
D) Stretch the cube into a long, shoebox shape.

27) If an organism was discovered that had no epithelial tissues, it would require adaptations to maintain homeostasis in which of the following areas? The organism would require adaptations _____.
A) In its skeleton for structure.
B) In its nervous system for sensing external stimuli.
C) That would prevent water loss from the body in a terrestrial environment.
D) In its muscular system for movement.

Use the following information to answer the question (28) below.

The crucian carp (*Carassius carassius*) is a Northern European freshwater fish often inhabiting ponds that become hypoxic (have reduced oxygen levels) and even anoxic (have no oxygen) when the surface freezes during the winter. Surprisingly, when oxygen levels are normal, these fish lack the lamellae that provide a large surface area for gas exchange between water and blood: their gills are smooth. Yet when the level of oxygen in the water falls, the gill morphology undergoes a change: packing cells stop dividing and programmed cell death is induced, exposing gill lamellae that were buried in other tissue. With lamellae exposed, the gills have increased surface area for gas exchange. These changes in gill lamellar profile are reversible: investigators observed that the gills return to their normal structure within seven days after returning the fish to well-oxygenated water. (Jørund Sollid, Paula De Angelis, Kristian Gundersen, and Göran E. Nilsson. 2003. Hypoxia induces adaptive and reversible gross morphological changes in crucian carp gills. *Journal of Experimental Biology* 206:3667-73.)

28) Refer to the paragraph on crucian carp. Gills serve multiple functions in fish in addition to gas exchange. Given the large surface area of gills with lamellae, what is the most likely explanation for why crucian carp cover protruding lamellae in their gills when levels of oxygen are normal?
A) To prevent loss of heat to the surrounding water.
B) To prevent loss of ions to the surrounding water.
C) To prevent protein loss to the surrounding water.
D) To prevent loss of oxygen to the surrounding water.

29) Once labor begins in childbirth, contractions increase in intensity and frequency, causing more contractions to occur until delivery. The increasing labor contractions of childbirth are an example of which type of regulation?
A) Positive feedback.
B) Negative feedback.
C) Feedback inhibition.
D) Enzymatic catalysis.

30) When the body's blood glucose level rises, the pancreas secretes insulin and, as a result, the blood glucose level declines. When the blood glucose level is low, the pancreas secretes glucagon and, as a result, the blood glucose level rises. Such regulation of the blood glucose level is the result of _____.
A) Catalytic feedback.
B) Positive feedback.
C) Negative feedback.
D) Protein-protein interactions.

31) The body's automatic tendency to maintain a constant and optimal internal environment is termed _____.
A) Balanced equilibrium.
B) Physiological chance.
C) Homeostasis.
D) Static equilibrium.

32) An example of a properly functioning homeostatic control system is seen when _____.
A) The core body temperature of a runner rises gradually from 37°C to 45°C.
B) The kidneys excrete salt into the urine when dietary salt levels rise.
C) A blood cell shrinks when placed in a solution of salt and water.
D) The blood pressure increases in response to an increase in blood volume.

33) Positive feedback differs from negative feedback in that _____.
A) Positive feedback benefits the organism, whereas negative feedback is detrimental.
B) The positive feedback's effector responses are in the same direction as the initiating stimulus rather than opposite of it.
C) The effector's response increases some parameter (such as body temperature), whereas in negative feedback it can only decrease the parameter.
D) Positive feedback systems have only effectors, whereas negative feedback systems have only receptors.

34) Which of the following is an example of negative feedback?
A) During birthing contractions, oxytocin (a hormone) is released and acts to stimulate further contractions.
B) When a baby is nursing, suckling leads to the production of more milk and a subsequent increase in the secretion of prolactin (a hormone that stimulates lactation).
C) After a blood vessel is damaged, signals are released by the damaged tissues that activate platelets in the blood. These activated platelets release chemicals that activate more platelets.
D) When the level of glucose in the blood increases, the pancreas produces and releases the hormone insulin. Insulin acts to decrease blood glucose. As blood glucose decreases, the rate of production and release of insulin decreases as blood glucose decreases.

35) You discover a new species of bacteria that grows in aquatic environments with high salt levels. While studying these bacteria, you note that their internal environment is similar to the salt concentrations in their surroundings. You also discover that the internal salt concentrations of the bacteria change as the salt concentration in their environment changes. The new species can tolerate small changes in this way, but dies from large changes because it has no mechanism for altering its own internal salt levels. What type of homeostatic mechanism is this species using to regulate its internal salt levels?
A) Conformation.
B) Regulation.
C) Integration.
D) Assimilation.

36) Chum salmon (*Oncorhynchus keta*) are born in freshwater environments and then migrate to the sea. Near the end of their lives, they return to the freshwater stream where they were born to spawn. In freshwater, water constantly diffuses into the body and ions are lost from the body. In salt water, body water diffuses out of the body and excess ions are gained from the water. A salmon's gills have special cells to pump salt in or out of the body to maintain homeostasis. In response to the salmon's moves between freshwater and salt water, some cells in the gills are produced and others are destroyed. These changes made in the cells of the gills during the lifetime of an individual salmon are an example of which of the following?
A) Evolution.
B) Trade-off.
C) Acclimatization.
D) Adaptation.

37) To prepare flight muscles for use on a cool morning, hawkmoths _____.
A) Relax the muscles completely until after they launch themselves into the air.
B) Decrease their standard metabolic rate.
C) Rapidly contract and relax these muscles to generate metabolic warmth.
D) Reduce the metabolic rate of the muscles to rest them before flight.

38) In a cool environment, an ectotherm is more likely to survive an extended period of food deprivation than would an equally sized endotherm because the ectotherm _____.
A) Maintains a higher basal metabolic rate.
B) Expends more energy per kilogram of body mass than does the endotherm.
C) Invests little energy in temperature regulation.
D) Has greater insulation on its body surface.

39) Sweating allows a person to lose heat through the process of _____.
A) Conduction.
B) Convection.
C) Radiation.
D) Evaporation.

40) An example of an ectothermic organism that has few or no behavioral options when it comes to its ability to adjust its body temperature is a _____.
A) Sea star living deep in the ocean.
B) Bass living in a farm pond.
C) Hummingbird flying through a prairie.
D) Honeybee in a hive on a rural farm.

41) The panting responses that are observed in overheated birds and mammals dissipate excess heat by _____.
A) Countercurrent exchange.
B) Acclimation.
C) Vasoconstriction.
D) Evaporation.

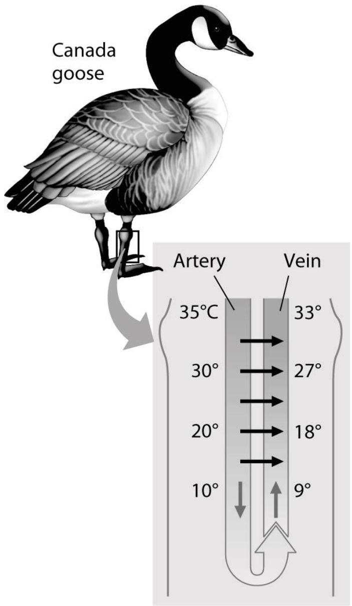
42) Most land-dwelling invertebrates and all of the amphibians _____.
A) Are ectothermic organisms with variable body temperatures.
B) Alter their metabolic rates to maintain a constant body temperature of 37°C.
C) Are endotherms but become thermoconformers when they are in water.
D) Become more active when environmental temperatures drop below 15°C.

43) The temperature-regulating center of vertebrate animals is located in the _____.
A) Thyroid gland.
B) Hypothalamus.
C) Subcutaneous layer of the skin.
D) Liver.

44) The metabolic breakdown of specialized brown fat depots in certain animals is substantially increased during _____.
A) Acclimatization.
B) Torpor.
C) Nonshivering thermogenesis.
D) Shivering thermogenesis.

45) A moth preparing for flight on a cold morning may warm its flight muscles via _____.
A) Acclimatization.
B) Torpor.
C) Evaporative cooling.
D) Shivering thermogenesis.

46) The thin horizontal arrows in the figure below shows that the _____.



A) Warmer arterial blood can bypass the legs as needed, when the legs are too cold to function well.
B) Warmer venous blood transfers heat to the cooler arterial blood.
C) Warmer arterial blood transfers heat to the cooler venous blood.
D) Arterial blood is always cooler in the abdomen, compared to the temperature of the venous blood in the feet of the goose.

47) Examine the figure above. Near a goose's abdomen, the countercurrent arrangement of the arterial and venous blood vessels causes the _____.
A) Temperature difference between the contents of the two sets of vessels to be minimized.
B) Venous blood to be as cold near the abdomen as it is near the feet.
C) Blood in the feet to be as warm as the blood in the abdomen.
D) Loss of the maximum possible amount of heat to the environment.

48) Which of the following would increase the rate of heat exchange between an animal and its environment?
A) Feathers or fur.
B) Vasoconstriction.
C) Wind blowing across the body surface.
D) Blubber or fat layer.

49) You are studying a large tropical reptile that has a high and relatively stable body temperature. How would you determine whether this animal is an endotherm or an ectotherm?
A) You know from its high and stable body temperature that it must be an endotherm.
B) You know that it is an ectotherm because it is not a bird or mammal.
C) You subject this reptile to various temperatures in the lab and find that its body temperature and metabolic rate change with the ambient temperature. You conclude that it is an ectotherm.
D) You note that its environment has a high and stable temperature. Because its body temperature matches the environmental temperature, you conclude that it is an ectotherm.

50) A woman standing and watching the stars on a cool, calm night will lose most of her body heat by _____.
A) Radiation.
B) Convection.
C) Conduction.
D) Evaporation.

51) There are advantages and disadvantages to adaptations. Animals that are endothermic are likely to be at the greatest disadvantage in _____.
A) Very cold environments.
B) Very hot environments.
C) Environments with a constant food source.
D) Environments with variable and limited food sources.

52) Which principle of heat exchange is the most important explanation for why birds look larger in colder weather because they fluff their feathers?
A) Fluffing feathers results in less cooling by radiation because feathers emit less infrared radiation than other tissues do.
B) Fluffing decreases the amount of heat lost by conduction when the bird makes contact with cold objects in its environment.
C) Fluffing creates a pocket of air near the bird that acts as insulation.
D) Fluffing decreases the surface-area-to-volume ratio, thus decreasing the amount of heat lost to the environment.

53) Snake behavior in Wisconsin changes throughout the year. For example, a snake is _____.
A) Less active in winter because the food supply is decreased.
B) Less active in winter because it does not need to avoid predators.
C) More active in summer because that is the period for mating.
D) More active in summer because it can gain body heat by conduction.

54) Standard metabolic rate (SMR) and basal metabolic rate (BMR) are _____.
A) Used differently: SMR is measured during exercise, whereas BMR is measured at rest.
B) Used to compare metabolic rates during feeding and other active conditions.
C) Both measured across a wide range of temperatures for a given species.
D) Both measured in animals in a resting and fasting state.

55) Independent of whether an organism is an endotherm or ectoderm, the LEAST reliable indicator of an animal's metabolic rate is the amount of _____.
A) Food eaten in one day.
B) Heat generated in one day.
C) Oxygen used in mitochondria in one day.
D) Water consumed in one day.

56) Consider the energy budgets for a human, an elephant, a penguin, a mouse, and a snake. The _____ would have the highest total annual energy expenditure, and the _____ would have the highest energy expenditure per unit mass.
A) Elephant; mouse.
B) Elephant; human.
C) Human; penguin.
D) Mouse; snake.

57) An animal's inputs of energy and materials would exceed its outputs if _____.
A) The animal is an endotherm, which must always take in more energy because of its high metabolic rate.
B) It is actively foraging for food.
C) It is hibernating.
D) It is growing and increasing its mass.

58) Which of the following animals most likely uses the largest percentage of its energy budget for homeostatic regulation?
A) A marine jelly (an invertebrate) living deep in the ocean.
B) A snake in a tropical forest.
C) A shark swimming in the open ocean.
D) A bird living year round in a desert.

59) A researcher is setting up an experiment to measure basal metabolic rate in prairie voles (*Microtus ochrogaster*–a small rodent). Which of the following would be the best set of conditions for the voles immediately before and during the measurement?
A) House the animals in a cage with plenty of food and water to avoid stress; conduct measurements in a warmer room than the room where housed.
B) House the animals in a cage with plenty of food and water to avoid stress; conduct measurements in a room the same temperature as the room where housed.
C) House the animals in a cage with no food for a few hours before measurement; conduct measurements in a colder room than the room where housed, and exercise the voles.
D) House the animals in a cage with no food for a few hours before measurement; conduct measurements in a room the same temperature as the room where housed.

60) Hummingbirds are small birds that require a regular food supply. When hummingbirds are faced with a situation that decreases their food supply, such as a storm, which of the following adaptations would be most useful for the bird to survive such an unpredictable and short-term absence of food resources?
A) Shivering.
B) Torpor.
C) Hibernation.
D) Burrowing into soil.

61) Organisms maintain dynamic homeostasis (internal balance) through behavioral and physiological mechanisms. Which of the following statements is an accurate explanation of a negative feedback mechanism used by animals to regulate body temperature?
A) Squirrels are able to cool themselves during warmer months by producing more brown fat, which contains abundant mitochondria and a rich blood supply.
B) Desert jackrabbits have unusually large ears that serve as solar heat collectors to enable them to maintain their body temperatures.
C) A ground squirrel’s hypothalamus detects changes in environmental temperatures and responds by activating or suppressing metabolic heat production.
D) A goldfish slows its movements when the water temperature is lower.

CHAPTER 41:

ANIMAL NUTRITION

1) The following table shows the contents of a multivitamin supplement and its percentage of recommended daily values (%DV).
The most likely reason that some of the vitamins and minerals in this supplement are found at less than 100% is that _____.

Dietary Supplement	% DV
Vitamin A	70
Vitamin C	100
Vitamin D	100
Vitamin E	150
Vitamin K	13
Vitamin B1	100
Vitamin B2	100
Folic acid	100
Vitamin B12	41.7
Calcium	20
Phosphorus	5
Iodine	100
Magnesium	25
Zinc	100
Copper	100
Chromium	125
Molybdenum	100
Iron	0

A) It would be chemically impossible to add more.
B) These vitamins and minerals are too large in size to reach 100%.
C) It is too easy to overdose on minerals such as phosphorus and calcium.
D) It is dangerous to overdose on fat-soluble vitamins such as A and K.

2) In a well-fed human eating a Western diet, the richest source of stored chemical energy in the body is _____.
A) Fat in adipose tissue.
B) Glucose in the blood.
C) Protein in muscle cells.
D) Glycogen in muscle cells.

3) Animals that migrate great distances would obtain the greatest energetic benefit of storing chemical energy as _____.
A) Proteins.
B) Minerals.
C) Carbohydrates.
D) Fats.

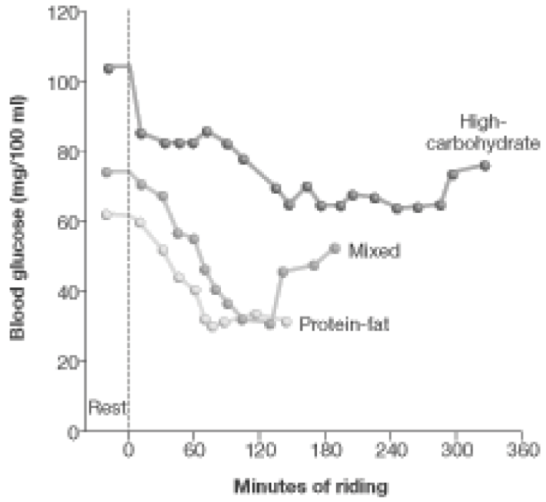
4) Certain nutrients are considered "essential" in the diets of some animals because _____.
A) Only those animals use those nutrients.
B) These animals are not able to synthesize these nutrients.
C) The nutrients are necessary coenzymes.
D) Only certain foods contain them.

5) Which pair correctly associates a physiological process with the appropriate vitamin?
A) Blood clotting - vitamin C.
B) Normal vision - vitamin A.
C) Synthesis of cell membranes - vitamin D.
D) Production of white blood cells - vitamin K.

6) Folic acid supplements have become especially important for pregnant women because _____.
A) Folic acid supplies vitamins that only pregnant women can use.
B) The fetus makes high levels of folic acid.
C) Folic acid deprivation is associated with neural tube abnormalities in a fetus.
D) Folic acid deprivation is a cause of heart abnormalities in a newborn.

7) What is the importance of consuming an adequate amount of proteins in the diet?
A) They are most commonly used to meet energy demands of cells.
B) Proteins serve a variety of functions, and the body does not store excess quantities of protein.
C) They are used as cofactors for metabolic reactions and are required in minute quantities.
D) Proteins are necessary to produce urea and other important metabolites.

8) Three groups of cyclists consumed three different types of diets: high-carbohydrate; a diet mixed in carbohydrates, fat, and protein; or a diet higher in protein and fat. The average time each group could spend cycling over a six-hour period is shown in the above graph. What conclusion from the data would help an athlete or trainer improve performance?



A) Endurance is entirely related to diet.
B) Maintaining elevated blood sugar improves performance.
C) An early 50 percent drop in blood glucose is associated with improved endurance.
D) Diet is not at all related to endurance.

9) Which of the following animals is correctly paired with its feeding mechanism?
A) Baleen whale - fluid feeder.
B) Aphid - suspension feeder.
C) Clam - substrate feeder.
D) Snake - bulk feeder.

10) The process of obtaining food is known as _____ and requires specialized mouthparts.
A) Ingestion.
B) Digestion.
C) Absorption.
D) Excretion.

11) In marine sponges, intracellular digestion of peptides is usually immediately preceded by _____.
A) Hydrolysis.
B) Phagocytosis.
C) Absorption.
D) Secretion.

12) An advantage of a complete digestive system over a gastrovascular cavity is that the complete system _____.
A) Excludes the need for extracellular digestion.
B) Allows for specialized regions with specialized functions.
C) Allows extensive branching.
D) Facilitates intracellular digestion.

13) Because the foods eaten by animals are often composed largely of macromolecules, animals need to have mechanisms for _____.
A) Dehydration synthesis.
B) Enzymatic hydrolysis.
C) Regurgitation.
D) Demineralization.

14) Fat digestion yields fatty acids and glycerol, whereas protein digestion yields amino acids; both digestive processes _____.
A) Are catalyzed by the same enzyme.
B) Use water molecules when breaking bonds (hydrolysis).
C) Require the presence of hydrochloric acid to lower the pH.
D) Require adenosine triphosphate (ATP) as an energy source.

15) Ingested dietary substances must cross cell membranes to be used by the body. This process is known as _____.
A) Ingestion
B) Digestion
C) Hydrolysis
D) Absorption.

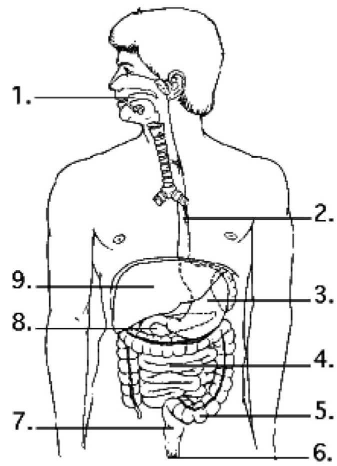
16) The function of mechanical digestion is to break down large chunks of food into smaller pieces. Why is this important? Smaller pieces of food _____.
A) Do not taste as good as larger pieces of food.
B) Have more surface area for chemical digestion than do larger pieces of food.
C) Are easier to excrete than are larger pieces of food.
D) Are more easily stored in the stomach than are larger pieces of food.

17) The large surface area in the gut directly facilitates _____.
A) Secretion.
B) Absorption.
C) Filtration.
D) Temperature regulation.

18) In the digestive system, peristalsis is_____.
A) A process of fat emulsification in the small intestine.
B) Voluntary control of the rectal sphincters regulating defecation.
C) The transport of nutrients to the liver through the hepatic portal vessel.
D) Smooth muscle contractions that move food along the esophagus.

19) Among mammals, it is generally true that_____.
A) All types of foods begin their enzymatic digestion in the mouth.
B) After leaving the oral cavity, the bolus enters the larynx.
C) The epiglottis prevents swallowed food from entering the trachea.
D) The trachea leads to the esophagus and then to the stomach.

Use the following figure to answer the questions (20-23) below.



20) Examine the digestive system structures in the figure above. The agents that help emulsify fats are produced in location _____.
A) 1.
B) 3.
C) 8.
D) 9.

21) Examine the digestive system structures in the figure above. The highest rate of nutrient absorption occurs at location _____.
A) 1.
B) 4.
C) 5.
D) 8.

22) Examine the digestive system structures in the figure above. Most of digestion of fats occurs in structure(s) ___.
A) 3 only.
B) 4 only.
C) 1 and 4.
D) 3 and 4.

23) Examine the digestive system structures in the figure above. Bacteria that produce vitamins are found in the greatest concentration in location _____.
A) 3.
B) 4.
C) 5.
D) 8.

24) Mammalian trachea and esophagus connect to the ___.
A) Stomach.
B) Pharynx.
C) Rectum.
D) Epiglottis.

25) Which of the following organs is correctly paired with its function?
A) Stomach - protein digestion.
B) Large intestine - bile production.
C) Small intestine - starch digestion.
D) Pancreas - starch digestion.

26) Stomach cells are moderately well adapted to the acidity and protein-digesting activities in the stomach by having _____.
A) A sufficient colony of *H. pylori.*
B) A thick, mucous secretion and active mitosis of epithelial cells.
C) A high level of secretion of enzymes by chief cells.
D) A cell wall impermeable to acid.

27) Villi and microvilli in the small intestine _____.
A) Neutralize stomach acid.
B) Activate trypsinogen.
C) Increase the surface area to increase the efficiency of nutrient absorption.
D) Emulsify lipid molecules.

28) Upon activation by stomach acidity, the secretions of the parietal cells _____.
A) Initiate the chemical digestion of protein in the stomach.
B) Initiate the mechanical digestion of lipids in the stomach.
C) Initiate the chemical digestion of lipids in the stomach.
D) Delay digestion until the food arrives in the small intestine.

29) Historically inaccurate diagnosis of acid reflux disorders and gastric ulcers has been improved by _____.
A) pH monitoring.
B) X-ray technology.
C) Screening for *H. pylori* infections.
D) Sonography.

30) What is the importance of the mucins that are released by salivary glands?
A) They aid in degradation of triglycerides to fatty acids and monoglycerides.
B) They are beginning the process of starch digestion.
C) They are hormonal molecules that stimulate the release of gastric juice by the stomach in anticipation of receipt of the contents of the mouth.
D) They are glycoproteins that make food slippery enough to slide easily through the esophagus.

31) Jahasz-Pocsine and co-workers found a correlation between gastric bypass surgery and neurological complications. Surgeons performed gastric bypass surgery on 150 patients at the University of Arkansas for Medical Sciences Neurology Clinic. Of the 150 patients, 26 experienced neurological complications related to the surgery. What is the most likely cause for the neurological complications?
A) Sudden weight loss and caloric deficiency interfering with neurological function.
B) Nutrient (for example, vitamin and mineral) deficiencies.
C) Sloppy surgical technique of physicians performing the bypass surgery.
D) Infections following surgical intervention.

32) Why did scientists originally hypothesize that proteolytic enzymes such as pepsin and trypsin are secreted in inactive form?
A) These proteolytic enzymes, in active form, would digest the very tissues that synthesize them.
B) They identified the hormone that activates pepsin and trypsin.
C) The stomach is too acidic to maintain these enzymes in their active form.
D) Pepsin and trypsin have never been isolated in their fully activated form.

33) Over-the-counter medications for acid reflux or heartburn block the production of stomach acid. Which of the following cells are directly affected by this medication?
A) Goblet cells.
B) Chief cells.
C) Parietal cells.
D) Smooth muscle cells.

34) *Helicobacter pylori* is a bacterial organism that causes ulcers and digestive disturbances. How might they survive the acid pH of the stomach?
A) They secrete buffers to neutralize acid.
B) They burrow under the mucus layer that covers the stomach epithelium.
C) They take over the parietal cells.
D) They release chemicals that decrease acid production in the stomach.

35) The active ingredient orlistat acts to decrease the amount of fat that is absorbed by attaching to enzymes that digest fat. Which of the following are potential targets of orlistat?
A) Salivary amylase.
B) Pepsidase.
C) Pancreatic lipase.
D) Secretin.

36) The pancreas is involved in the digestion of _____.
I) Protein.
II) Fat.
III) Nucleic acids.
IV) Carbohydrates.
A) I and III.
B) I, II, and IV.
C) II, III, and IV.
D) I, II, III, and IV.

37) Digestive secretions with a pH of 2 are characteristic of the _____.
A) Small intestine.
B) Stomach.
C) Pancreas.
D) Liver.

38) The bile salts _____.
A) Are enzymes.
B) Are manufactured by the pancreas.
C) Emulsify fats in the duodenum.
D) Are normally an ingredient of gastric juice.

39) The absorption of fats differs from that of carbohydrates in that _____.
A) Fat absorption primarily occurs in the stomach, whereas carbohydrates are absorbed from the small intestine.
B) Carbohydrates need to be emulsified before they can be digested, whereas fats do not.
C) Most absorbed fat first enters the lymphatic system, whereas carbohydrates directly enter the blood.
D) Fats, but not carbohydrates, are digested by bacteria before absorption.

40) Constipation can result from the consumption of a substance that _____.
A) Promotes water reabsorption in the large intestine.
B) Speeds up movement of material in the large intestine.
C) Decreases water reabsorption in the small intestine.
D) Stimulates peristalsis.

41) A significant contribution of intestinal bacteria to human nutrition is the benefit of bacterial _____.
A) Production of vitamins A and C.
B) Absorption of organic materials.
C) Production of vitamin K.
D) Recovery of water from fecal matter.

42) After surgical removal of an infected gallbladder, a person must be especially careful to restrict dietary intake of _____.
A) Protein.
B) Sugar.
C) Fat.
D) Water.

43) When a woman has her gallbladder removed, she should probably reduce her consumption of _____.
A) Proteins.
B) Carbohydrates.
C) Fats.
D) Proteins and carbohydrates.

44) If you place a small piece of a cracker on your tongue, what would you expect to happen?
A) The vitamins in the cracker are immediately absorbed.
B) Salivary amylase degrades the starch from the cracker into glucose.
C) The proteins in the cracker begin to be digested.
D) The flavor becomes less noticeable because the sugars are digested.

45) A relatively long cecum is characteristic of animals that are _____.
A) Carnivores.
B) Herbivores.
C) Autotrophs.
D) Omnivores.

46) Cattle are able to survive on a diet consisting almost entirely of plant material because cattle _____.
A) Are autotrophic.
B) Re-ingest their feces.
C) Manufacture all fifteen amino acids out of sugars in liver.
D) Have cellulose-digesting, symbiotic microorganisms in chambers of their stomachs.

47) A zoologist analyzes the jawbones of an extinct mammal and concludes that it was an herbivore. The zoologist most likely came to this conclusion based upon_.
A) The position of muscle attachment sites.
B) The shape of the teeth.
C) The size of the mouth opening.
D) The angle of the teeth in the mouth.

48) An enlarged cecum is typical of _____.
A) Rabbits, horses, and herbivorous bears.
B) Carnivorous animals.
C) Tubeworms that digest via symbionts.
D) Humans and other primates.

49) Coprophagy is important for the nutritional balance of _____.
A) Ruminants such as cows.
B) Insects and arthropods.
C) Rabbits and their relatives.
D) Squirrels and some rodents.

50) If you found a vertebrate skull in the woods and the teeth were sharp and scissor -like, what type of food would you expect this animal to eat?
A) Grass.
B) Flesh of another animal.
C) Nectar.
D) Blood.

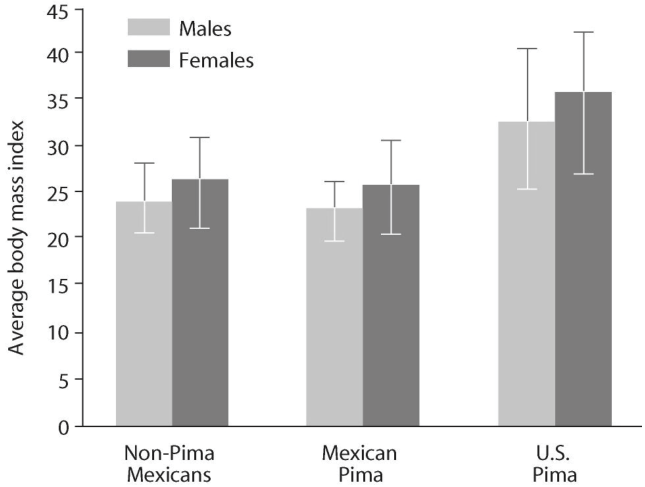
51) You are most likely to observe coprophagy in _____.
A) Carnivores.
B) Herbivores.
C) Fluid feeders.
D) Suspension feeders.

52) Obesity in humans is most clearly linked to _____.
A) Type 1 diabetes and prostate cancer.
B) Type 2 diabetes and muscle hypertrophy.
C) Type 2 diabetes and cardiovascular disease.
D) Type 2 diabetes and decreased appetite.

53) If you were to jog one kilometer a few hours after lunch, which stored fuel would you probably tap?
A) Muscle proteins.
B) Liver glycogen and muscle glycogen.
C) Fat stored in adipose tissue.
D) Blood proteins.

54) Food being digested in the stomach is in a highly acidic environment. When the food is released from the stomach into the small intestine, why is the environment no longer acidic?
A) Secretin increases the flow of bicarbonate ions from the pancreas into the small intestine to neutralize the stomach acid.
B) Enterokinase activates trypsinogen, thus neutralizing the stomach acid.
C) Bile salts from the gallbladder neutralize the stomach acid.
D) When pepsinogen activates pepsin, one result is the neutralization of stomach acid in the stomach.

55) The Pimas are a group of people living in the southwestern United States and Mexico. Although Pima Indians living in the United States and Mexico have a similar genetic background, a five-fold increase in the incidence of type II diabetes mellitus among U.S. Pima has been reported. The body mass index of Mexicans of non-Pima descent, Mexicans of Pima descent, and Pimas living in the United States is shown in the figure below. Based on this information and the graph below, what can you infer about the incidence of type II diabetes mellitus?



A) Obesity is a risk factor for development of type II diabetes mellitus.
B) If you maintain a normal body weight (body mass index less than 25), you will not get type II diabetes mellitus.
C) The incidence of type II diabetes mellitus has increased in the past ten years.
D) People who develop type II diabetes mellitus are typically diagnosed in childhood or adolescence.

56) In a healthy person, after a large meal, the production of _____ will increase. After fasting, the production of _____ will increase.
A) Trypsin; trypsinogen.
B) Glucagon; glucose.
C) Glucagon; pepsinogen.
D) Insulin; glucagon.

57) If there is a strong genetic link for type II diabetes mellitus in your family, how might you minimize your risk of developing the disorder?
A) Monitor your blood glucose levels daily.
B) Take oral insulin daily.
C) Maintain a healthy weight, eat a balanced direct, and exercise.
D) Eat complex carbohydrates like starch instead of sweets.

58) When the digestion and absorption of organic carbohydrates results in more energy -rich molecules than are immediately required by an animal, the excess is _____.
A) Eliminated in the feces.
B) Stored as starch in the liver.
C) Stored as glycogen in the liver and muscles.
D) Oxidized and converted to ATP.

59) A fasting animal whose energy needs exceed those provided in its diet draws on its stored resources in which order?
A) Fat, then glycogen, then protein.
B) Glycogen, then protein, then fat.
C) Liver glycogen, then muscle glycogen, then fat.
D) Muscle glycogen, then fat, then liver glycogen.

CHAPTER 42:

CIRCULATION AND GAS EXCHANGE

1) The circulatory systems of bony fishes, rays, and sharks are most similar to _____.
A) Those of birds, with a four-chambered heart.
B) The portal systems of mammals, where two capillary beds occur sequentially, without passage of blood through a pumping chamber.
C) Those of sponges, where gas exchange in all cells occurs directly with the external environment.
D) Those of humans, where there are four pumping chambers to drive blood flow.

2) Organisms with a circulating body fluid that is distinct from the fluid that directly surrounds the body's cells are likely to have _____.
A) An open circulatory system.
B) A closed circulatory system.
C) A gastrovascular cavity.
D) Branched tracheae.

3) In which of the following organisms does blood flow from the pulmocutaneous circulation to the heart before circulating through the rest of the body?
A) Annelids.
B) Fishes.
C) Frogs.
D) Insects.

4) The only vertebrates in which blood flows directly from respiratory organs to body tissues without first returning to the heart are the _____.
A) Amphibians.
B) Fishes.
C) Mammals.
D) Reptiles.

5) To adjust blood pressure independently in the capillaries of the gas-exchange surface and in the capillaries of the general body circulation, an organism would need a(n) _____.
A) Open circulatory system.
B) Hemocoel.
C) Two-chambered heart.
D) Four-chambered heart.

6) An anthropologist discovers the fossilized heart of an extinct animal. The evidence indicates that the organism's heart was large, was well-formed, and had four chambers, with no connection between the right and left sides. A reasonable conclusion supported by these observations is that the _____.
A) Animal had evolved from birds.
B) Animal was endothermic and had a high metabolic rate.
C) Animal was most closely related to crocodiles.
D) Species had little to no need to regulate blood pressure.

7) In an open circulatory system, blood is _____.
A) Always inside of vessels and is under higher pressure than in closed circulatory systems.
B) Not always confined to blood vessels and is under higher pressure than in closed circulatory systems.
C) Always inside of vessels and is under lower pressure than in closed circulatory systems.
D) Not always confined to blood vessels and is under lower pressure than in closed circulatory systems.

8) Circulatory systems compensate for _____.
A) Temperature differences between the lungs and the active tissue.
B) The slow rate at which diffusion occurs over large distances.
C) The problem of communication systems involving only the nervous system.
D) The need to cushion animals from trauma.

9) Which of the following develops the greatest pressure on the blood in the mammalian aorta?
A) Systole of the left atrium.
B) Diastole of the right ventricle.
C) Systole of the left ventricle.
D) Diastole of the right atrium.

10) Which of the following is the correct sequence of blood flow in birds and mammals?
A) Left ventricle → aorta → lungs → systemic circulation.
B) Vena cava → right atrium → right ventricle → pulmonary vein.
C) Pulmonary vein → left atrium → left ventricle → pulmonary circuit.
D) Vena cava → right atrium → right ventricle → pulmonary artery.

11) A patient with a blood pressure of 120/75, a pulse rate of 70 beats/minute, a stroke volume of 70 mL/beat (milliliters per beat), and a respiratory rate of 25 breaths/minute will have a cardiac output of _____.
A) 1,000 mL/minute.
B) 1,750 mL/minute.
C) 2,800 mL/minute.
D) 4,900 mL/minute.

12) Damage to the sinoatrial node in humans _____.
A) Would block conductance between the bundle branches and the Purkinje fibers.
B) Would have a negative effect on peripheral resistance.
C) Would disrupt the rate and timing of cardiac muscle contractions.
D) Would have a direct effect on blood pressure monitors in the aorta.

13) A stroke volume of 70 mL/cycle in a heart with a pulse of 72 cycles per minute results in a cardiac output of about _____.
A) 5 liters per minute.
B) 50 milliliters per minute.
C) 0.5 liters per minute.
D) 50 liters per minute.

14) Atria contract _____.
A) Just prior to the beginning of diastole.
B) During diastole.
C) Immediately after systole.
D) During systole.

15) The greatest difference in the concentration of respiratory gases is found in which of the following pairs of mammalian blood vessels?
A) The pulmonary vein and the jugular vein.
B) The veins from the right and left legs.
C) The pulmonary artery and the inferior vena cava.
D) The pulmonary vein and the aorta.

16) A human red blood cell in an artery of the left arm is on its way to deliver oxygen to a cell in the thumb. To travel from the artery to the thumb and then back to the left ventricle, this red blood cell must pass through _____.
A) One capillary bed.
B) Two capillary beds.
C) Three capillary beds.
D) Four capillary beds.

17) If a molecule of carbon dioxide released into the blood in your left toe is exhaled from your nose, it must pass through all of the following EXCEPT _____.
A) The pulmonary vein.
B) An alveolus.
C) The trachea.
D) The right atrium.

18) Among the following choices, which organism likely has the highest systolic pressure?
A) Mouse.
B) Human.
C) Hippopotamus.
D) Giraffe.

19) The velocity of blood flow is the lowest in capillaries because _____.
A) The capillaries have internal valves that slow the flow of blood.
B) The diastolic blood pressure is too low to deliver blood to the capillaries at a high flow rate.
C) The systemic capillaries are supplied by the left ventricle, which has a lower cardiac output than the right ventricle.
D) The total cross-sectional area of the capillaries is greater than the total cross-sectional area of the arteries or any other part of the circulatory system.

20) A species that has a normal resting systolic blood pressure of greater than 260 mm Hg is likely to be _____.
A) An animal that is small and compact, without the need to pump blood very far from the heart.
B) A species that has very wide diameter veins.
C) An animal that has a very long distance between its heart and its brain.
D) An animal that makes frequent, quick motions.

21) Small swollen areas in the neck, groin, and axillary region are associated with _____.
A) Increased activity of the immune system.
B) Blood sugar that is abnormally high.
C) Dehydration.
D) Sodium depletion.

22) What will be the long-term effect of blocking the lymphatic vessels associated with a capillary bed?
A) More fluid entering the venous capillaries.
B) An increase in the blood pressure in the capillary bed.
C) The accumulation of more fluid in the interstitial areas.
D) The area of the blockage becoming abnormally small.

23) Which of the following conditions would most likely be due to high blood pressure in a mammal?
A) Bursting of blood vessels in capillary beds.
B) Inability of the right ventricle to contract.
C) Reversal of normal blood flow direction in arteries.
D) Destruction of red blood cells.

24) Which of the following mechanisms are used to regulate blood pressure in the closed circulatory system of vertebrates?
I) Changing the force of heart contraction.
II) Constricting and relaxing sphincters in the walls of arterioles.
III) Adjusting the volume of blood contained in the veins.
A) Only I and II.
B) Only I and III.
C) Only II and III.
D) I, II, and III.

25) Blood is pumped at high pressures in arteries from the heart to ensure that all parts of the body receive adequate blood flow. Capillary beds, however, would hemorrhage under direct arterial pressures. How does the design of the circulatory network contribute to reducing blood pressure to avoid this scenario?
A) Blood flow through the capillaries is essentially frictionless, and this reduces the amount of pressure on their walls.
B) The total cross-sectional diameter of the arterial circulation increases with progression from artery to arteriole to capillary, leading to a reduced blood pressure.
C) Fluid loss from the arteries is high enough that pressure drops off significantly by the time blood reaches the capillaries.
D) Capillary beds have the thickest walls of any blood vessel to resist these high pressures.

Use the following description to answer the question (26) below.
Lymph hearts are pumping structures that drive lymph through the lymphatic system, returning it to the circulatory system at the large veins entering the heart. Researchers examined rate and strength of pumping of lymph hearts in two species of amphibians, a toad (*Bufo marinus*) and a frog (*Rana catesbiana*). During hemorrhage or dehydration, the volume of blood in the circulatory system falls. (E.A. DeGrauw and S. S. Hillman. 2004. General function and endocrine control of the posterior lymph hearts in *Bufo marinus* and *Rana catesbiana*. *Physiological and Biochemical Zoology* 77(4):594-600.)
26) Refer to the paragraph on lymph hearts. What effect would increasing lymph heart pressure have first?
A) Blood volume would increase.
B) Blood volume would decrease.
C) Hemorrhage would increase.
D) Hemorrhage would decrease.

27) If, during protein starvation, the osmotic pressure on the venous side of capillary beds drops below the hydrostatic pressure, then _____.
A) Hemoglobin will not release oxygen.
B) Fluids will tend to accumulate in tissues.
C) The pH of the interstitial fluids will increase.
D) Plasma proteins will escape through the endothelium of the capillaries.

28) Large proteins such as albumin remain in capillaries rather than diffusing out, resulting in the _____.
A) Loss of osmotic pressure in the capillaries.
B) Development of an osmotic pressure difference across capillary walls.
C) Loss of fluid from capillaries.
D) Increased diffusion of hemoglobin.

29) The production of red blood cells is stimulated by ___.
A) Low-density lipoproteins.
B) Immunoglobulins.
C) Erythropoietin.
D) Epinephrine.

30) To become bound to hemoglobin for transport in a mammal, atmospheric molecules of oxygen must cross ___.
A) One membrane - that of the lining in the lungs - and then bind directly to hemoglobin, a protein dissolved in the plasma of the blood.
B) Two membranes - in and out of the cell lining the lung - and then bind directly to hemoglobin, a protein dissolved in the plasma of the blood.
C) Four membranes - in and out of the cell lining the lung, in and out of the endothelial cell lining an alveolar capillary - and then bind directly to hemoglobin, a protein dissolved in the plasma of the blood.
D) Five membranes - in and out of the cell lining the lung, in and out of the endothelial cell lining an alveolar capillary, and into the red blood cell - to bind with hemoglobin.

31) The diagnosis of hypertension in adults is based on the _____.
A) Measurement of fatty deposits on the endothelium of arteries.
B) Measurement of the LDL/HDL ratio in peripheral blood.
C) Percentage of blood volume made up of platelets.
D) Blood pressure being greater than 140 mm Hg systolic and/or greater than 90 mm Hg diastolic.

32) Cyanide poisons mitochondria by blocking the final step in the electron transport chain. Human red blood cells placed in an isotonic solution containing cyanide are likely to _____.
A) Retain the normal cell shape, but the mitochondria will be poisoned.
B) Lyse as the cyanide concentration increases inside the cell.
C) Switch to anaerobic metabolism.
D) Be unaffected.

33) A normal event in the process of blood clotting is the _.
A) Production of erythropoietin.
B) Conversion of fibrin to fibrinogen.
C) Activation of prothrombin to thrombin.
D) Synthesis of hemoglobin.

34) You cut your finger, and after putting pressure on the wound for several minutes, you notice that it is still bleeding profusely. What may be the problem?
A) Platelets are not functioning properly, or there are too few to be effective.
B) Mast cells are not releasing their chemical messengers.
C) There are too many antigens to allow clotting.
D) Hemoglobin levels are too high to allow clotting.

35) The hormone that stimulates the production of red blood cells, and the organ where this hormone is synthesized, are _____.
A) Growth hormone and pancreas, respectively.
B) Erythropoietin and kidney, respectively.
C) Cortisol and adrenal gland, respectively.
D) Acetylcholine and bone marrow, respectively.

36) Countercurrent exchange is evident in the flow of ___.
A) Water across the gills of a fish and the blood within those gills.
B) Blood in the dorsal vessel of an insect and that of air within its tracheae.
C) Air within the primary bronchi of a human and the blood within the pulmonary veins.
D) Water across the skin of a frog and the blood flow within the ventricle of its heart.

37) Countercurrent exchange in the fish gill helps to maximize _____.
A) Blood pressure.
B) Diffusion.
C) Active transport.
D) Osmosis.

38) Which of the following statements comparing respiration in fish and in mammals is correct?
A) The respiratory medium for fish carries more oxygen than the respiratory medium of mammals.
B) A countercurrent exchange mechanism between the respiratory medium and blood flow is seen in mammals but not in fish.
C) The movement of the respiratory medium in mammals is bidirectional, but in fish it is unidirectional.
D) In blood, oxygen is primarily transported by plasma in fish, but by red blood cells in mammals.

39) Flying insects typically _____.
A) Decrease metabolism as much as 200-fold during flight.
B) Switch from diffusion of tracheal gases to active transport during flight.
C) Utilize high numbers of mitochondria in flight muscles.
D) Generate fuel molecules from catabolism of carbon dioxide.

40) When the air in a testing chamber is specially mixed so that its oxygen content is 10 percent and its overall air pressure is 400 mm Hg, then PO_2 is _____.
A) 400 mm Hg.
B) 82 mm Hg.
C) 40 mm Hg.
D) 4 mm Hg.

41) The sun shining on a tidal pool during a hot day heats the water. As some water evaporates, the pool becomes saltier, causing _____.
A) An increase in its carbon dioxide content.
B) A decrease in its oxygen content.
C) An increase in its ability to sustain aerobic organisms.
D) A decrease in the water's density.

42) An oil-water mixture works as an insecticidal spray against mosquitoes and other insects because it _____.
A) Blocks the openings into the tracheal system.
B) Interferes with gas exchange across the capillaries.
C) Clogs their bronchi.
D) Prevents gases from leaving the atmosphere.

43) Atmospheric pressure at sea level is equal to a column of 760 mm Hg. Oxygen makes up 21 percent of the atmosphere by volume. The partial pressure of oxygen (PO_2) in such conditions is _____.
A) 160 mm Hg.
B) 16 mm Hg.
C) 21/760.
D) 760/21.

44) Some human infants, especially those born prematurely, suffer serious respiratory failure because of _____.
A) The sudden change from the uterine environment to the air.
B) The overproduction of surfactants.
C) Lung collapse due to inadequate production of surfactant.
D) Mutations in the genes involved in lung formation.

45) At the summit of a high mountain, the atmospheric pressure is 380 mm Hg. If the atmosphere is still composed of 21percent oxygen, then the partial pressure of oxygen at this altitude is about _____.
A) 80 mm Hg.
B) 160 mm Hg.
C) 380 mm Hg.
D) 760 mm Hg.

46) Compared with the interstitial fluid that bathes active muscle cells, blood reaching these muscle cells in arteries has a _____.
A) Higher PO_2.
B) Greater bicarbonate concentration.
C) Lower pH.
D) Lower osmotic pressure.

47) A rabbit taken from a meadow near sea level and moved to a meadow high on a mountainside would have some trouble breathing. Why?
A) The percentage of oxygen in the air at high elevations is lower than at sea level.
B) The percentage of oxygen in the air at high elevations is higher than at sea level.
C) The partial pressure of oxygen in the air at high elevations is lower than at sea level.
D) The partial pressure of oxygen in the air at high elevations is higher than at sea level.

48) What would be the consequences if we were to reverse the direction of water flow over the gills of a fish, moving water inward past the operculum, past the gills, the out the mouth? This reversal of water flow would _____.
A) Reduce efficiency of gas exchange.
B) Change the exchange of gases in the body from carbon dioxide out and oxyhemoglobin in to carbon dioxide in and oxygen out.
C) Increase the efficiency of gas exchange.
D) None of them.

49) Under identical atmospheric conditions, freshwater _____.
A) Has more oxygen than seawater.
B) Has less oxygen than seawater.
C) Can hold 10-40 times more carbon dioxide than air.
D) Can hold 10-40 times more oxygen than air.

50) Which of the following statements comparing respiration in fish and in mammals is correct?
A) The respiratory medium for fish carries more oxygen than the respiratory medium of mammals.
B) A countercurrent exchange mechanism between the respiratory medium and blood flow is seen in mammals but not in fish.
C) The movement of the respiratory medium in mammals is bidirectional, but in fish it is unidirectional.
D) In blood, oxygen is primarily transported by plasma in fish, but by red blood cells in mammals.

51) How has the avian lung adapted to the metabolic demands of flight?
A) Airflow through the avian lung is bidirectional like in mammals.
B) There is more dead space within the avian lung so that oxygen can be stored for future use.
C) Countercurrent circulation is present in the avian lung.
D) Gas exchange occurs during both inhalation and exhalation.

52) Carbon dioxide levels in the blood and cerebrospinal fluid affect pH. This enables the organism to sense a disturbance in gas levels as _____.
A) The brain directly measures and monitors oxygen levels and causes breathing changes accordingly.
B) The medulla oblongata, which is in contact with cerebrospinal fluid, monitors pH and uses this measure to control breathing.
C) The brain alters the pH of the cerebrospinal fluid to force the animal to retain more or less carbon dioxide.
D) Stretch receptors in the lungs cause the medulla oblongata to speed up or slow breathing.

53) A person with a tidal volume of 450 mL (milliliters), a vital capacity of 4000 mL, and a residual volume of 1000 mL would have a potential total lung capacity of _____.
A) 1450 mL.
B) 4000 mL.
C) 4450 mL.
D) 5000 mL.

54) During most daily activities, the human respiration rate is most closely linked to the blood levels of _____.
A) Nitrogen.
B) Oxygen.
C) Carbon dioxide.
D) Carbon monoxide.

55) An decrease from pH 7.4 to pH 7.2 causes hemoglobin to _____.
A) Release all bound carbon dioxide molecules.
B) Bind more oxygen molecules.
C) Increase its binding of H^+.
D) Give up more of its oxygen molecules.

56) The Bohr shift on the oxygen-hemoglobin dissociation curve is produced by changes in _____.
A) The partial pressure of oxygen.
B) Hemoglobin concentration.
C) Temperature.
D) pH.

57) Most of the carbon dioxide produced by humans is _____.
A) Converted to bicarbonate ions by an enzyme in red blood cells.
B) Bound to hemoglobin.
C) Transported in the erythrocytes as carbonic acid.
D) Simply dissolved in the plasma.

58) Which of the following events would be predicted by the Bohr shift effect as the amount of carbon dioxide released from your tissues into the blood capillaries increases? The amount of oxygen in _____.
A) Arterial blood would increase.
B) Arterial blood would decrease.
C) Venous blood would increase.
D) Venous blood would decrease.

59) You are a physician, and you are seeing a patient who complains of abnormal fatigue during exercise. You find that the immediate problem is a buildup of carbon dioxide in the tissues. What is the most likely cause?
A) Abnormally shaped platelets.
B) Abnormal carbonic anhydrase.
C) Abnormal hemoglobin.
D) Not enough hemoglobin.

60) Carbonic anhydrase inhibitors cause less movement of carbonic acid toward carbon dioxide production and are used as a prophylactic treatment of altitude sickness. Altitude sickness occurs when a hiker ascends to altitudes where the density of oxygen is low. How does this decrease the symptoms of high altitude sickness?
A) The excess hydrogen ions are excreted in the urine and the resulting loss of acidity increases respiration rate.
B) The excess bicarbonate ions are excreted in the urine and the resulting loss of blood pressure increases respiration rate.
C) The excess bicarbonate ions are excreted in the urine and the resulting increase in blood acidity leads to an increase in ventilation.
D) The excess bicarbonate ions in the blood increase the affinity of hemoglobin for oxygen.

CHAPTER 43:

THE IMMUNE SYSTEM

1) Innate immunity _____.
A) Is activated immediately upon infection.
B) Depends on an infected animal's previous exposure to the same pathogen.
C) Is based on recognition of antigens that are specific to different pathogens.
D) Is found only in vertebrate animals.

2) A fruit fly, internally infected by a potentially pathogenic fungus, is protected by its _____.
A) Immunoglobulins.
B) Antibodies.
C) Antimicrobial peptides.
D) B cells.

3) Engulfing-phagocytic cells of innate immunity of vertebrates include _____.
I) Neutrophils.
II) Macrophages.
III) Dendritic cells.
IV) Natural killer cells.
A) I and III.
B) II and IV.
C) I and IV.
D) I, II, and III.

4) The cells and signaling molecules involved in the initial stages of the inflammatory response are _____.
A) Phagocytes and chemokines.
B) Dendritic cells and interferons.
C) Mast cells and histamines.
D) Lymphocytes and interferons.

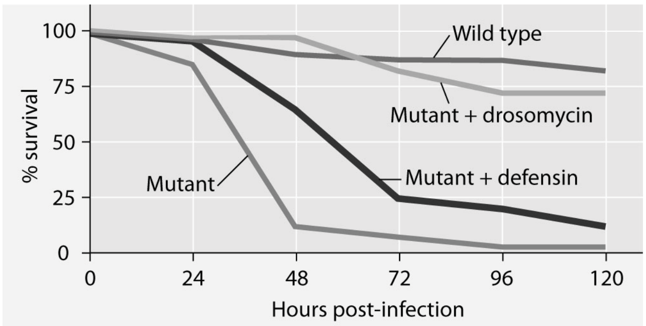
5) Inflammatory responses typically include _____.
A) Increased activity of phagocytes in an inflamed area.
B) Reduced permeability of blood vessels to conserve plasma.
C) Release of substances to decrease the blood supply to an inflamed area.
D) Inhibiting the release of white blood cells from bone marrow.

6) Mammals have Toll-like receptors (TLRs) that can recognize a kind of macromolecule that is absent from vertebrates but present in or on certain groups of pathogens, such as viral _____.
A) Double-stranded DNA.
B) Double-stranded RNA.
C) Glycoproteins.
D) Phospholipids.

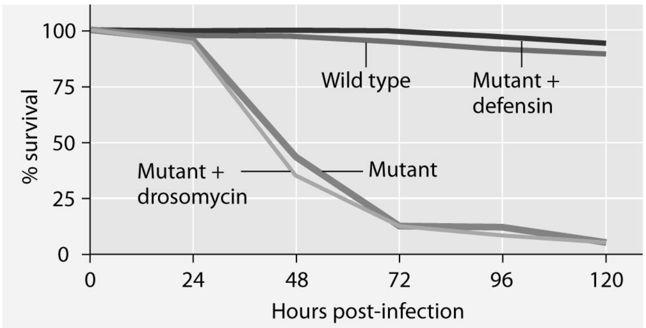
7) Septic shock, a systemic response including high fever and low blood pressure, is a response to _____.
A) Certain bacterial infections.
B) Specific forms of viruses.
C) The presence of natural killer cells.
D) Increased production of neutrophils.

8) The cells involved in innate immunity, whose absence increases the chances of developing malignant tumors, are _____.
A) Cytotoxic T cells.
B) Natural killer cells.
C) Macrophages.
D) B cells.

9) Mutant fruit flies that make only one antimicrobial peptide were tested for survival after infection with *Neurospora crassa* fungi or with *Micrococcus luteus* bacteria.



Fruit fly survival after infection by *N. crassa* fungi



Fruit fly survival after infection by *M. luteus* bacteria

The results shown in the graphs support the hypothesis that _____.
A) Adding the defensin gene to such mutants protects them from death by fungal infection.
B) Adding the drosomycin gene to such mutants protects them from death by fungal infection.
C) Wild-type flies with the full set of genes for antimicrobial peptides are highly susceptible to these infective agents.
D) The presence of any single antimicrobial peptide protects against both infective agents.

10) You and a friend were in line for a movie when you noticed the woman in front of you sneezing and coughing. Both of you were equally exposed to the woman's virus, but over the next few days, only your friend acquired flu-like symptoms and was ill for almost a week before recovering. Which one of the following is a logical explanation for this?
A) Your friend had antibodies to that virus.
B) You had an adaptive immunity to that virus.
C) Your friend had an autoimmune disorder.
D) Your friend had allergies.

11) A boy falls while riding his bike. A scrape on his hand almost immediately begins to bleed and becomes red, warm, and swollen. What response is occurring?
A) Inflammatory response.
B) Lytic response.
C) Adaptive immune response.
D) Autoimmune response.

12) Acidity in human sweat is an example of _____.
A) Cell-mediated immune responses.
B) Acquired immunity.
C) Adaptive immunity.
D) Innate immunity.

13) The eyes and the respiratory tract are both protected against infections by _____.
A) Interferons produced by immune cells.
B) The secretion of complement proteins.
C) The release of slightly alkaline secretions.
D) The secretion of lysozyme onto their surfaces.

14) The complement system is _____.
A) A set of proteins involved in innate but not acquired immunity.
B) A group of proteins that includes interferons and interleukins.
C) A group of antimicrobial proteins that act together in a cascade fashion.
D) A set of proteins that act individually to attack and lyse microbes.

15) Bacteria entering the body through a small cut in the skin _____.
A) Inactivate the erythrocytes.
B) Stimulate apoptosis of nearby body cells.
C) Stimulate release of interferons.
D) Activate a group of proteins called complement.

16) Mucus occurs in both the respiratory and digestive tracts. What is its main immunological function?
A) Sweeping away debris.
B) Physically trapping pathogens.
C) Destruction of pathogens because it is acidic.
D) Increasing oxygen absorption.

17) Within a differentiated B cell, the rearrangement of DNA sequences between variable regions and joining regions is accomplished by a(n) _____.
A) RNA polymerase.
B) Reverse transcriptase.
C) Epitopase.
D) Recombinase.

18) Clonal selection and differentiation of B cells activated by antigen exposure leads to the production of _____.
A) Large quantities of the antigen initially recognized.
B) Vast numbers of B cells with random antigen-recognition receptors.
C) Long-lived erythrocytes that can later secrete antibodies for the antigen.
D) Short-lived plasma cells that secrete antibodies for the antigen.

19) A newborn that is accidentally given a drug that destroys the thymus would most likely _____.
A) Lack innate immunity.
B) Be unable to genetically rearrange antigen receptors.
C) Be unable to differentiate and mature T cells.
D) Have a reduced number of B cells and be unable to form antibodies.

20) Clonal selection is an explanation for how _____.
A) *V*, *J*, and *C* gene segments are rearranged.
B) An antigen can provoke production of high levels of specific antibodies.
C) HIV (human immunodeficiency virus) can disrupt the immune system.
D) Macrophages can recognize specific T cells and B cells.

21) An immunoglobulin (Ig) molecule, of whatever class, with regions symbolized as C or V, H or L, has a light chain made up of _____.
A) One C region and one V region.
B) One H region and one L region.
C) Three H regions and one L region.
D) Two C regions and two V regions.

22) Immunological memory accounts for _____.
A) The human body's ability to distinguish self from non-self.
B) The observation that some strains of the pathogen that causes dengue fever cause more severe disease than others.
C) The ability of a helper T cell to signal B cells via cytokines.
D) The ancient observation that someone who had recovered from the plague could safely care for those newly diseased.

Use the following information to answer the question (23) below.

An otherwise healthy student in your class is infected with EBV, the virus that causes infectious mononucleosis. The same student had already been infected when she was a child, at which time she had merely experienced a mild sore throat and swollen lymph nodes in her neck. This time, though infected, she does not get sick.

23) The EBV antigen fragments will be presented by the virus-infected cells along with _____.
A) A complement.
B) Antibodies.
C) Class I MHC molecules.
D) Class II MHC molecules.

24) Vaccination increases the number of _____.
A) Lymphocytes with receptors that can bind to the pathogen.
B) Epitopes that the immune system can recognize.
C) Macrophages specific for a pathogen.
D) Major Histocompatability (MHC) molecules that can present an antigen.

25) If a patient is missing B and T cells, what would be absent from the immune response?
A) Memory.
B) Lysozymes.
C) Cytokines.
D) Defense against bacteria.

26) Lymphocytes mature in the _____.
I) Spleen.
II) Thymus.
III) Bone marrow.
A) Only I and III.
B) Only I and II.
C) Only II and III.
D) I, II, and III.

27) Which of the following statements are fundamental to the clonal-selection theory of how the adaptive immune system functions?
I) Each lymphocyte has a unique membrane receptor that recognizes one antigen.
II) When the lymphocyte binds an antigen, it is activated and begins dividing to form many identical copies of itself.
III) Cloned lymphocytes have slight differences and are selected by the spleen for removal if they do not bind an antigen.
IV) Cloned cells descend from an activated lymphocyte and persist even after the pathogen is eliminated.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

28) What major advantage is conveyed by having a system of adaptive immunity?
A) It enables a rapid defense against an antigen that has been previously encountered.
B) It enables an animal to counter most pathogens almost instantly the first time they are encountered.
C) It results in effector cells with specificity for a large number of antigens.
D) It allows for the destruction of antibodies.

29) Which of the following is a difference between B cells and T cells?
A) One has a major role in antibody production, while the other has a major role in cytotoxicity.
B) One binds a receptor called BCR (B-cell receptor), while the other recognizes a receptor called TCR (T-cell receptor).
C) B cells are activated by free-floating antigens in the blood or lymph. T cells are activated by membrane-bound antigens.
D) T cells are produced in the thymus and B cells are produced in the bone marrow.

30) A certain cell type has existed in the blood and tissue of its vertebrate host's immune system for over twenty years. One day, it recognizes a newly arrived antigen and binds to it, subsequently triggering a secondary immune response in the body. Which of the following cell types most accurately describes this cell?
A) Plasma cell.
B) Thyroid cell.
C) Memory cell.
D) Macrophage.

31) Which of the following statements about epitopes are correct?
I) B-cell receptors bind to epitopes.
II) T-cell receptors bind to epitopes.
III) There can be 10 or more different epitopes on each antigen.
IV) There is a one-to-one correspondence between antigen and epitope.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and III.
D) Only II, III, and IV.

32) Which of the following pairs of proteins shares the most overall similarity in structure?
A) B-cell receptors and T-cell receptors.
B) B-cell receptors and antibodies.
C) T-cell receptors and antibodies.
D) Antibodies and antigens.

33) What type of immunity is associated with breast feeding?
A) Innate immunity.
B) Active immunity.
C) Passive immunity.
D) Cell-mediated immunity.

34) Select the pathway that would lead to the activation of cytotoxic T cells.
A) B cell contact antigen → helper T cell is activated → clonal selection occurs.
B) Body cell becomes infected with a virus → new viral proteins appear → class I MHC molecule-antigen complex displayed on cell surface.
C) Complement is secreted → B cell contacts antigen → helper T cell activated → cytokines released.
D) Cytotoxic T cells → class II MHC molecule-antigen complex displayed → cytokines released → cell lysis.

35) Arrange in the correct sequence these components of the mammalian immune system as it first responds to a pathogen.
I) Pathogen is destroyed.
II) Lymphocytes secrete antibodies.
III) Antigenic determinants from pathogen bind to antigen receptors on lymphocytes.
IV) Lymphocytes specific to antigenic determinants from pathogen become numerous.
V) Only memory cells remain.
A) I → III → II → IV → V.
B) II → I → IV → III → V.
C) IV → II → III → I → V.
D) III → IV → II → I → V.

36) A nonfunctional CD4 protein on a helper T cell would result in the helper T cell being unable to _____.
A) Respond to T-independent antigens.
B) Lyse tumor cells.
C) Stimulate a cytotoxic T cell.
D) Interact with a class II MHC-antigen complex.

37) CD4 and CD8 are _____.
A) Proteins secreted by antigen-presenting cells.
B) Receptors present on the surface of natural killer cells.
C) Molecules present on the surface of T cells where they interact with major histocompatability (MHC) molecules.
D) Molecules on the surface of antigen-presenting cells where they enhance B cell activity.

38) T cells of the immune system include _____.
A) CD4, CD8, and plasma cells.
B) Cytotoxic and helper cells.
C) Plasma, antigen-presenting, and memory cells.
D) Lymphocytes, macrophages, and dendritic cells.

39) B cells interacting with helper T cells are stimulated to differentiate when _____.
A) B cells produce IgE antibodies.
B) B cells release cytokines.
C) Cytotoxic T cells present the class II MHC molecule-antigen complex on their surface.
D) Helper T cells release cytokines.

40) When antibodies bind antigens, the clumping of antigens results from _____.
A) The antibody having at least two binding regions.
B) Disulfide bridges between the antigens.
C) Bonds between class I and class II MHC molecules.
D) Denaturation of the antibodies.

41) Phagocytosis of microbes by macrophages is enhanced by _____.
I) The binding of antibodies to the surface of microbes.
II) Antibody-mediated agglutination of microbes.
III) The release of cytokines by activated B cells.
A) Only I and II.
B) Only II and III.
C) Only I and III.
D) I, II, and, III.

42) Naturally acquired passive immunity can result from the _____.
A) Injection of vaccine.
B) Ingestion of interferon.
C) Placental transfer of antibodies.
D) Absorption of pathogens through mucous membranes.

43) Jenner's successful use of cowpox virus as a vaccine against the smallpox virus was due to the fact that _____.
A) The immune system responds nonspecifically to antigens.
B) The cowpox virus made antibodies in response to the presence of smallpox.
C) There are some epitopes (antigenic determinants) common to both pox viruses.
D) Cowpox and smallpox are caused by the same virus.

44) An individual who has been bitten by a poisonous snake that has a fast -acting toxin would likely benefit from _____.
A) Vaccination with a weakened form of the toxin.
B) Injection of antibodies to the toxin.
C) Injection of interleukin-1.
D) Injection of interferon.

45) For the successful development of a vaccine to be used against a pathogen, it is necessary that _____.
A) The surface antigens of the pathogen stay the same.
B) All of the surface antigens on the pathogen be identified.
C) The pathogen has only one epitope.
D) The major histocompatability (MHC) molecules are heterozygous.

46) The switch of one B cell from producing one class of antibody to another class of antibody that is responsive to the same antigen is due to _____.
A) The rearrangement of V region genes in that clone of responsive B cells.
B) A switch in the kind of antigen-presenting cell that is involved in the immune response.
C) A patient's reaction to the first kind of antibody made by the plasma cells.
D) The rearrangement of immunoglobulin heavy-chain C region DNA.

47) The number of major histocompatability (MHC) protein combinations possible in a given population is enormous. However, an individual in that diverse population has a far more limited array of MHC molecules because _____.
A) The MHC proteins are made from several different gene regions that are capable of rearranging in a number of ways.
B) MHC proteins from one individual can only be of class I or class II.
C) Each of the MHC genes has a large number of alleles, but each individual only inherits two for each gene.
D) Once a B cell has matured in the bone marrow, it is limited to two MHC response categories.

48) A bone marrow transplant may not be appropriate from a given donor (Jane) to a given recipient (Jane's cousin Bob), even though Jane has previously given blood for one of Bob's needed transfusions, because _____.
A) Even though Jane's blood type is a match to Bob's, her major histocompatability (MHC) proteins may not be a match.
B) A blood type match is less stringent than a match required for transplant because blood is more tolerant of change.
C) For each gene, there is only one blood allele but many tissue alleles.
D) Jane's MHC class II genes are not expressed in bone marrow.

49) An immune response to a tissue graft will differ from an immune response to a bacterium because _____.
A) MHC molecules of the donor may stimulate rejection of the graft tissue, but bacteria lack MHC molecules.
B) The tissue graft, unlike the bacterium, is isolated from the circulation and will not enter into an immune response.
C) A bacterium cannot escape the immune system by replicating inside normal body cells.
D) The graft will stimulate an autoimmune response in the recipient.

Use the following information to answer the question (50) below.

An otherwise healthy student in your class is infected with EBV, the virus that causes infectious mononucleosis. The same student had already been infected when she was a child, at which time she had merely experienced a mild sore throat and swollen lymph nodes in her neck. This time, though infected, she does not get sick.

50) Her immune system's recognition of the second infection involves memory _____.
A) Helper T cells.
B) Natural killer cells.
C) Plasma cells.
D) Cytotoxic T cells.

51) Which of the following should be the same in identical twins?
A) The set of antibodies produced.
B) The set of major histocompatability (MHC) molecules produced.
C) The set of T cell antigen receptors produced.
D) The susceptibility to a particular virus.

52) Which of the following is crucial to activation of the adaptive immune response?
A) Memory cells.
B) Presentation of MHC (major histocompatibility complex)-antigen complex on a cell surface.
C) Somatic hypermutation.
D) Phagocytosis of antibody-antigen complex by macrophages in the blood (the humoral response).

53) Which of the following components of the immune system destroys bacteria in a way similar to an antitank weapon destroying armored military tanks by punching holes in the wall of the bacteria?
A) Complement protein.
B) Macrophages.
C) Plasma cells.
D) Major histocompatibility complex proteins.

54) Yearly vaccination of humans for influenza viruses is necessary because _____.
A) Of an increase in immunodeficiency diseases.
B) The flu can generate anaphylactic shock.
C) Surviving the flu one year exhausts the immune system to nonresponsiveness the second year.
D) Rapid mutation in flu viruses alters the surface proteins in infected host cells.

55) A patient who has a high level of mast cell activity, dilation of blood vessels, and acute drop in blood pressure is likely suffering from _____.
A) An autoimmune disease.
B) A typical skin allergy (contact dermatitis) that can be treated by antihistamines.
C) An organ transplant, such as a skin graft.
D) Anaphylactic shock immediately following exposure to an allergen.

56) The ability of some viruses to remain inactive (latent) for a period of time is exemplified by _____.
A) Influenza, a particular strain of which returns every 10-20 years.
B) Herpes simplex viruses (oral or genital) whose reproduction is triggered by physiological or emotional stress in the host.
C) Kaposi's sarcoma, which causes a skin cancer in people with AIDS but rarely in those not infected by HIV.
D) The virus that causes a form of the common cold, which recurs in patients many times in their lives.

57) A patient complaining of watery, itchy eyes and sneezing after being given a flower bouquet as a birthday gift should first be treated with _____.

A) A vaccine.
B) Sterile pollen.
C) Antihistamines.
D) Monoclonal antibodies.

58) Which of the following would help a virus avoid triggering an effective adaptive immune response?
I) Having frequent mutations in genes for surface proteins.
II) Building the viral shell from host proteins.
III) Producing proteins very similar to those of other viruses.
IV) Infecting and killing helper T cells.

A) Only I and III.
B) Only I, II, and IV.
C) Only I, II, and III.
D) Only II, III, and IV.

59) Which of the following is the best definition of autoimmune disease?

A) A condition in which B cells and T cells respond independently to antigens and do not interact correctly.
B) A condition in which the adaptive immune system fails to recognize the second infection by the same antigen.
C) A condition in which self-molecules are treated as non–self.
D) A condition in which the immune system creates random antibodies without being triggered by an antigen.

60) Which of the following would prevent allergic attacks?

A) Blocking the attachment of the IgE antibodies to the mast cells.
B) Blocking the antigenic determinants of the IgM antibodies.
C) Reducing the number of helper T cells in the body.
D) Reducing the number of cytotoxic cells.

61) In a humoral or antibody-mediated immune response, specific B cells are stimulated by Helper T cells to transform into plasma cells that secrete antibodies. What would be an important feature added to B cells in this transition process?

A) Duplication of specific gene sequences for the appropriate antibody.
B) Increased rough endoplasmic reticulum in order to have the surface area needed for antibody production.
C) Duplication of lysosomes in order to store the antibodies before transport.
D) None of them.

CHAPTER 44:

OSMOREGULATION AND EXCRETION

1) The force driving simple diffusion is _____, while the energy source for active transport is _____.
A) The concentration gradient; ADP.
B) The concentration gradient; ATP.
C) Transmembrane pumps; electron transport.
D) Phosphorylated protein carriers; ATP.

2) To maintain homeostasis freshwater fish must _____.
A) Excrete large quantities of electrolytes.
B) Consume large quantities of water.
C) Excrete large quantities of water.
D) Take in electrolytes through simple diffusion.

3) Single-celled *Paramecium* lives in pond water (a hypotonic environment). They have a structural feature, a contractile vacuole, which enables them to osmoregulate. If you observed them in the following solutions, at which sucrose concentration (in millimolars, mM) would you expect the contractile vacuole to be most active?
A) 0.0 mM sucrose.
B) 0.05 mM saline.
C) 0.08 mM sucrose.
D) 1.0 mM saline.

4) Sharks live in seawater. Their tissues are isotonic to seawater, but their concentrations of sodium ions, potassium ions, and chloride ions in cells and extracellular fluids are similar to those of freshwater fishes. How is that possible?
A) Urea and trimethylamine oxide contribute to intra- and extracellular osmolarity in shark tissues.
B) Metabolic intermediates of sharks tie up intracellular chloride and potassium ions.
C) Their blood is hypotonic to their tissues.
D) They excrete large quantities of electrolytes.

5) Hagfish (*Eptatretus cirrhauts*) are a jawless marine vertebrate that are isotonic with their environment and are considered to be osmoconformers. How might this interesting adaptation limit the habitat that the hagfish can tolerate?
A) Hagfish are not limited by salinity.
B) Osmoconformers do not face the same pressures as osmoregulators and can live in any marine environment.
C) Individual hagfish will adapt to different salinities over their lifetime and, therefore, can inhabit any marine environment.
D) Hagfish habitat is limited by the salinity of the environment.

6) Tissues of sharks are isotonic to seawater, but their concentrations of sodium ions, potassium ions, and chloride ions in cells and extracellular fluids are similar to those of freshwater fishes. What can you infer about the movement of sodium and chloride in these animals?
A) To maintain homeostasis of sodium and chloride levels, the shark must take up additional sodium and chloride from seawater.
B) Sodium and chloride will diffuse into shark gills from seawater down their concentration gradient.
C) Sharks conserve sodium and chloride, limiting excretion.
D) Sodium and chloride must be eliminated through the gills.

7) What role do chloride cells play in osmoregulation of marine fish with bony skeletons?
A) They actively transport chloride into the gills.
B) They mediate the movement of salt from seawater through their gills.
C) They are involved in excretion of excess salt.
D) They actively transport salt across the basolateral membrane of the rectal gland.

8) Salmon eggs hatch in freshwater. The fish then migrate to the ocean (a hypertonic solution) and, after several years of feeding and growing, return to freshwater to breed. How can these organisms make the transition from freshwater to ocean water and back to freshwater?
A) The rectal gland functions in the ocean water and chloride cells function in freshwater.
B) Different gill cells are involved in osmoregulation in freshwater than in salt water.
C) Salmon in freshwater excrete dilute urine, and salmon in salt water secrete concentrated urine.
D) Their metabolism changes in salt water to degrade electrolytes.

9) Terrestrial organisms lose water through evaporation. In what ecosystem might an entomologist find a good study organism to examine the prevention of water loss?
A) Wet rain forest.
B) Desert.
C) Prairie.
D) Chaparral.

10) A necropsy (postmortem analysis) of a marine sea star that died after it was mistakenly placed in fresh water would likely show that it died because _____.
A) It was stressed and needed more time to acclimate to the new conditions.
B) It was so hypertonic to the fresh water that it could not osmoregulate.
C) Its contractile vacuoles ruptured.
D) Its cells dehydrated and lost the ability to metabolize.

11) The body fluids of an osmoconformer would be _____ with its _____ environment.
A) Isoosmotic; freshwater.
B) Hyperosmotic; saltwater.
C) Isoosmotic; saltwater.
D) Hypoosmotic; saltwater.

12) Compared to the seawater around them, most marine invertebrates are _____.
A) Hyperosmotic.
B) Hypoosmotic.
C) Isosmotic.
D) Hyperosmotic and isosmotic.

13) The fluid with the highest osmolarity is _____.
A) Distilled water.
B) Plasma in birds.
C) Plasma in mammals.
D) Seawater in a tidal pool.

14) A human who has no access to fresh water but is forced to drink seawater instead will _____.
A) Thrive under such conditions, as long as he has lived at the ocean most of his life.
B) Excrete more water molecules than taken in, because of the high load of ion ingestion.
C) Develop structural changes in the kidneys to accommodate the salt overload.
D) Risk becoming overhydrated within twelve hours.

15) Unlike most bony fishes, sharks maintain body fluids that are isoosmotic to seawater, so they are considered by many to be osmoconformers. Nonetheless, these sharks osmoregulate at least partially by _____.
A) Using their gills and kidneys to rid themselves of sea salts.
B) Monitoring dehydration at the cellular level with special gated aquaporins.
C) Tolerating high urea concentrations that are balanced with internal salt concentrations to seawater osmolarity.
D) Synthesizing trimethylamine oxide, a chemical that binds and precipitates salts inside cells.

16) The necropsy (postmortem analysis) of a freshwater fish that died after being placed accidentally in saltwater would likely show that _____.
A) Loss of water by osmosis from cells in vital organs resulted in cell death and organ failure.
B) High amounts of salt had diffused into the fish's cells, causing them to swell and lyse.
C) The kidneys were not able to keep up with the water removal necessary in this hyperosmotic environment, creating an irrevocable loss of homeostasis.
D) The gills became encrusted with salt, resulting in inadequate gas exchange and a resulting asphyxiation.

17) Which of the following animals generally has the lowest volume of urine production?
A) A vampire bat.
B) A salmon in fresh water.
C) A marine bony fish.
D) A shark inhabiting the Mississippi River.

18) One of the waste products that accumulates during cellular functions is carbon dioxide. It is removed via the respiratory system. What is another waste product that accumulates during normal physiological functions in vertebrates?
I) Ammonia.
II) Uric acid.
III) Urea.
A) Only I and III.
B) Only II and III.
C) Only I and II.
D) I, II, and III.

19) Urea is produced in the _____.
A) Liver from NH_3 and carbon dioxide.
B) Liver from glycogen.
C) Kidneys from glycerol and fatty acids.
D) Bladder from uric acid and water.

20) Urea is _____.
A) Insoluble in water.
B) The primary nitrogenous waste product of humans.
C) The primary nitrogenous waste product of most birds.
D) The primary nitrogenous waste product of most aquatic invertebrates.

21) Which nitrogenous waste has the greatest number of nitrogen atoms?
A) Ammonia.
B) Ammonium ions.
C) Urea.
D) Uric acid.

22) Ammonia is likely to be the primary nitrogenous waste in living conditions that include _____.
A) Lots of fresh water flowing across the gills of a fish.
B) Lots of seawater, such as a bird living in a marine environment.
C) A terrestrial environment, such as that supporting crickets.
D) A moist system of burrows, such as those of naked mole rats.

23) Excessive formation of uric acid crystals in humans leads to _____.
A) A condition called diabetes, where excessive urine formation occurs.
B) A condition of insatiable thirst and excessive urine formation.
C) Gout, a painful inflammatory disease that primarily affects the joints.
D) Osteoarthritis, an inevitable consequence of aging.

24) Ammonia _____.
A) Is soluble in water.
B) Has low toxicity relative to urea.
C) Is metabolically more expensive to synthesize than urea.
D) Is the major nitrogenous waste excreted by insects.

25) The advantage of excreting nitrogenous wastes as urea rather than as ammonia is that _____.
A) Urea can be exchanged for Na^+.
B) Urea is less toxic than ammonia.
C) Urea does not affect the osmolar gradient.
D) Less nitrogen is removed from the body.

26) In animals, nitrogenous wastes are produced mostly from the catabolism of _____.
A) Starch and cellulose.
B) Triglycerides and steroids.
C) Proteins and nucleic acids.
D) Phospholipids and glycolipids.

27) Birds secrete uric acid as their nitrogenous waste because uric acid _____.
A) Is readily soluble in water.
B) Is metabolically less expensive to synthesize than other excretory products.
C) Requires little water for nitrogenous waste disposal, thus reducing body mass.
D) Can be reused by birds as a protein source.

28) Among the following choices, the most concentrated urine is excreted by _____.
A) Frogs.
B) Kangaroo rats.
C) Humans.
D) Freshwater bass.

29) African lungfish, which are often found in small, stagnant pools of fresh water, produce urea as a nitrogenous waste. What is the advantage of this adaptation?
A) Urea takes less energy to synthesize than ammonia.
B) Small, stagnant pools do not provide enough water to dilute the toxic ammonia.
C) The highly toxic urea makes the pool uninhabitable to potential competitors.
D) Urea makes lungfish tissue hypoosmotic to the pool.

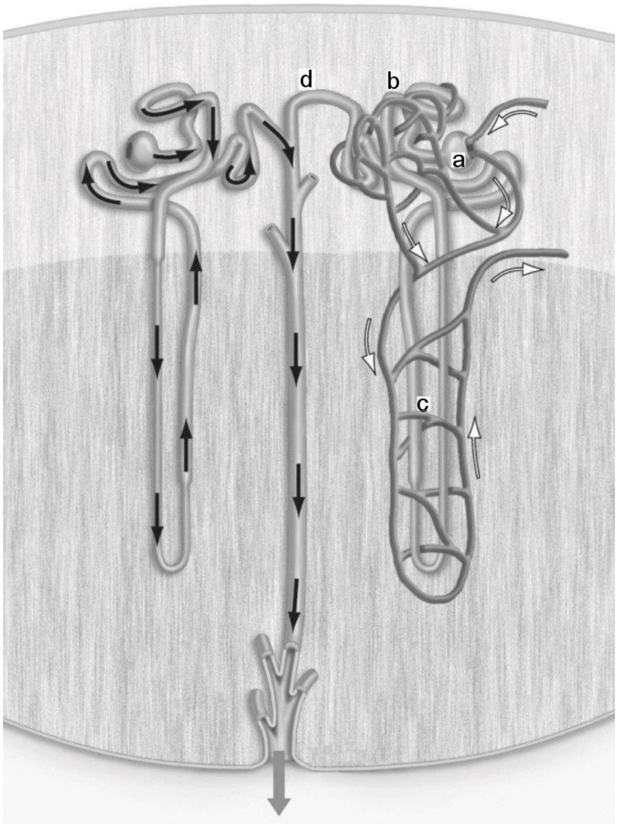
30) Which of the following most accurately describes selective permeability?
A) An input of energy is required for transport.
B) Lipid-soluble molecules pass through a membrane.
C) There must be a concentration gradient for molecules to pass through a membrane.
D) Only certain molecules can cross a cell membrane.

31) Through studies of insect Malpighian tubules, researchers found that K+ accumulated on the inner face of the tubule, against its concentration gradient. What can you infer about the mechanism of transport?
A) Potassium transport is a passive process.
B) Movement of potassium into the lumen of the Malpighian tubules is an energy-requiring process.
C) Potassium moves out of the tubules at a faster rate than it moves into the lumen of the tubules.
D) Sodium ions will follow potassium ions.

32) A potassium ion gradient is set up in insect Malpighian tubules through an active transport process. As a result, potassium concentration is higher in the lumen of the tubules than in hemolymph. How would the potassium gradient affect water movement?
A) Water would be forced out of the lumen of the Malpighian tubules through an osmotic gradient.
B) The potassium gradient would have no effect on water movement.
C) There would be a net movement of water into the lumen of the tubules.
D) Water would be conserved, forming a hypertonic solution in the Malpighian tubules.

33) Why are the renal artery and vein critical to the process of osmoregulation in vertebrates?
A) The kidneys require constant and abnormally high oxygen supply to function.
B) The renal artery delivers blood with nitrogenous waste to the kidney and the renal vein brings blood with less nitrogenous wastes away from the kidneys.
C) The kidneys require higher than normal levels of hormones.
D) The renal artery and vein are the main pathways regulating how much is produced by the kidneys.

34) The figure below shows a nephron. Filtration takes place in the structure labeled _____.



A) a.
B) b.
C) c.
D) d.

35) The osmoregulatory/excretory system of a freshwater flatworm is based on the operation of _____.
A) Protonephridia.
B) Metanephridia.
C) Malpighian tubules.
D) Nephrons.

36) Materials are returned to the blood from the filtrate by which of the following processes?
A) Filtration.
B) Selective reabsorption.
C) Secretion.
D) Excretion.

37) Excretory organs known as Malpighian tubules are present in _____.
A) Flatworms.
B) Insects.
C) Jellyfish.
D) Sea stars.

38) Osmoregulatory process called secretion refers to the _.
A) Reabsorption of nutrients from a filtrate.
B) Selective elimination of excess ions and toxins from body fluids.
C) Formation of an osmotic gradient along an excretory structure.
D) Expulsion of urine from the body.

39) The osmoregulatory/excretory system of an earthworm is based on the operation of _____.
A) Protonephridia.
B) Metanephridia.
C) Malpighian tubules.
D) Nephrons.

40) Choose a pair that correctly associates the mechanism for osmoregulation or nitrogen removal with the appropriate animal.
A) Metanephridium – flatworm.
B) Malpighian tubule – frog.
C) Flame bulb – snake.
D) Exchange across the body surface - marine invertebrate.

41) An excretory system that is partly based on the filtration of fluid under high hydrostatic pressure is the _.
A) Flame bulb system of flatworms.
B) Protonephridia of rotifers.
C) Malpighian tubules of insects.
D) Kidneys of vertebrates.

42) The transfer of fluid from the glomerulus to Bowman's capsule _____.
A) Results from active transport.
B) Transfers large molecules as easily as small ones.
C) Is very selective as to which subprotein-sized molecules are transferred.
D) Is mainly a consequence of blood pressure in the capillaries of the glomerulus.

43) Within a normally functioning kidney, blood can be found in _____.
A) The vasa recta.
B) Bowman's capsule.
C) The proximal tubule.
D) The collecting duct.

44) A primary reason that the kidneys have one of the highest metabolic rates of all body organs is that _____.
A) They have membranes of varying permeability to water.
B) They operate extensive set of active-transport ion pumps.
C) They are the body's only means of shedding excess nutrients.
D) They have an abundance of myogenic smooth muscle.

45) Which process in the nephron is LEAST selective?
A) Filtration.
B) Reabsorption.
C) Active transport.
D) Secretion.

46) What is the function of the osmotic gradient found in the kidney? The osmotic gradient allows for _____.
A) Electrolytes to move from low to high concentrations in the absence of ATP.
B) The precise control of the retention of water and electrolytes.
C) The loop of Henle to deliver water to the renal vein.
D) The filtration of large cells at the glomerulus.

47) The loop of Henle dips into the renal cortex. This is an important feature of osmoregulation in terrestrial vertebrates because _____.
A) Absorptive processes taking place in the loop of Henle are hormonally regulated.
B) Differential permeabilities of ascending and descending limbs of the loop of Henle are important in establishing an osmotic gradient.
C) The loop of Henle plays an important role in detoxification.
D) Additional filtration takes place along the loop of Henle.

48) Low selectivity of solute movement is a characteristic of _____.
A) H^+ pumping to control pH.
B) Reabsorption mechanisms along the proximal tubule.
C) Filtration from the glomerular capillaries.
D) Secretion along the distal tubule.

49) If ATP production in a human kidney was suddenly halted, urine production would _____.
A) Decrease, and the urine would be hypoosmotic compared to plasma.
B) Increase, and the urine would be isoosmotic compared to plasma.
C) Increase, and the urine would be hyperosmotic compared to plasma.
D) Decrease, and the urine would be isoosmotic compared to plasma.

50) Compared to wetland mammals, water conservation in mammals of arid regions is enhanced by having more _____.
A) Juxtamedullary nephrons.
B) Urinary bladders.
C) Ureters.
D) Podocytes.

51) Processing of filtrate in the proximal and distal tubules _____.
A) Achieves the conversion of toxic ammonia to less toxic urea.
B) Maintains homeostasis of pH in body fluids.
C) Regulates the speed of blood flow through the nephrons.
D) Reabsorbs urea to maintain osmotic balance.

52) In humans, the transport epithelial cells in the ascending loop of Henle _____.
A) Are the largest epithelial cells in the body.
B) Are not in contact with interstitial fluid.
C) Have plasma membranes of low permeability to water.
D) Are not affected by high levels of nitrogenous wastes.

53) The high osmolarity of the renal medulla is maintained by all of the following EXCEPT _____.
A) Active transport of salt from the upper region of the ascending limb.
B) The spatial arrangement of juxtamedullary nephrons.
C) Diffusion of urea from the collecting duct.
D) Diffusion of salt from the descending limb of the loop of Henle.

54) Natural selection should favor the highest proportion of juxtamedullary nephrons in which of the following species?
A) A river otter.
B) A mouse species living in a tropical rain forest.
C) A mouse species living in a temperate broadleaf forest.
D) A mouse species living in a desert.

55) If you are hiking through the desert for several days, one would pack which of the following to ensure proper hydration?
A) A drink with a combination of water and electrolytes.
B) Caffeinated beverages.
C) Bottled water kept at room temperature.
D) Bottled water that had been frozen to ensure that it would be as cold as possible.

56) Increased antidiuretic hormone (ADH) secretion is likely after _____.
A) Drinking lots of pure water.
B) Sweating-induced dehydration increases plasma osmolarity.
C) Eating a small sugary snack.
D) Blood pressure becomes abnormally high.

57) After blood flow is artificially reduced at one kidney, you would expect that kidney to secrete more of the hormone known as _____.
A) Angiotensinogen.
B) Renin.
C) Antidiuretic hormone.
D) Atrial natriuretic peptide.

58) After drinking alcoholic beverages, increased urine excretion is the result of _____.
A) Increased aldosterone production.
B) Increased blood pressure.
C) Inhibited secretion of antidiuretic hormone (ADH).
D) Increased reabsorption of water in the proximal tubule.

59) Osmoregulatory adjustment via the renin-angiotensin-aldosterone system can be triggered by _____.
A) Sleeping for one hour.
B) Severe sweating on a hot day.
C) Eating a pizza with olives and pepperoni.
D) Drinking several glasses of water.

60) Antidiuretic hormone (ADH) and the renin-angiotensin-aldosterone system (the RAAS) work together in maintaining osmoregulatory homeostasis through which of the following ways?
A) ADH regulates the osmolarity of the blood by altering renal reabsorption of water, and the RAAS maintains the osmolarity of the blood by stimulating Na^+ and water reabsorption.
B) ADH and the RAAS work antagonistically; ADH stimulates water reabsorption during dehydration and the RAAS causes increased excretion of water when it is in excess in body fluids.
C) Both stimulate the adrenal gland to secrete aldosterone, which increases both blood volume and pressure via its receptors in the urinary bladder.
D) ADH and the RAAS combine at the receptor sites of proximal tubule cells, where reabsorption of essential nutrients takes place.

CHAPTER 45:

HORMONES AND THE ENDOCRINE SYSTEM

1) You are dissecting a fish in your biology laboratory section. Your teaching assistant points out a long oval structure and tells you it is an endocrine gland. Which of the following would you then know is a true statement about this structure?
A) It secretes a product that is released through a series of ducts.
B) The gland's product will only interact with receptors on the cell membrane.
C) The gland's product is lipid soluble.
D) The gland produces and secretes its product into the blood.

2) In experiments where researchers suspect that a hormone may be responsible for a certain physiological effect, they may cut the neurons leading to the organ where the effect being studied occurs. What is the purpose of cutting these neurons?
A) To make sure that the effect is not occurring through actions in the nervous system.
B) To make sure that the organ being affected cannot function unless the researchers stimulate it with an external electrical probe.
C) To impair the normal functions of the organ so that the hormonal effect can be more easily studied.
D) To numb the organ so that it can be probed without inducing pain in the lab animal.

3) What is the only type of chemical signal that does not alter the physiology of the animal producing that signal?
A) Neural.
B) Paracrine.
C) Neuroendocrine.
D) Pheromones.

4) Testosterone is an example of a chemical signal that affects the very cells that synthesize it, the neighboring cells in the testis, along with distant cells outside the gonads. Thus, testosterone is an example of _____.
I) An autocrine signal.
II) A paracrine signal.
III) An endocrine signal.
A) Only I and II.
B) Only II and III.
C) Only I and III.
D) I, II, and III.

5) Prostaglandins are local regulators whose chemical structure is derived from _____.
A) Oligosaccharides.
B) Fatty acids.
C) Steroids.
D) Amino acids.

6) Aspirin and ibuprofen both _____.
A) Inhibit the synthesis of prostaglandins.
B) Inhibit the release of nitric oxide, a potent vasodilator.
C) Activate the paracrine signaling pathways that form blood clots.
D) Stimulate vasoconstriction in the kidneys.

7) A cell with membrane-bound proteins that selectively bind a specific hormone is called that hormone's _____.
A) Secretory cell.
B) Endocrine cell.
C) Target cell.
D) Regulatory cell.

8) The reason that the steroid hormone aldosterone affects only a small number of cells in the body is that _____.
A) Only its target cells get exposed to aldosterone.
B) Only its target cells contain aldosterone receptors.
C) It is unable to enter nontarget cells.
D) Nontarget cells destroy aldosterone before it can produce any effect.

9) Different body cells can respond differently to the same peptide hormones because _____.
A) Different target cells have different sets of genes.
B) A target cell's response is determined by the components of its signal transduction pathways.
C) The circulatory system regulates responses to hormones by routing the hormones to specific targets.
D) The hormone is chemically altered in different ways as it travels through the circulatory system.

10) Hormone X activates the cAMP second messenger system in its target cells. The greatest response by a cell would come from _____.
A) Applying a molecule of hormone X to the extracellular fluid surrounding the cell.
B) Injecting a molecule of hormone X into the cytoplasm of the cell.
C) Applying a molecule of cAMP to the extracellular fluid surrounding the cell.
D) Injecting a molecule of activated, cAMP-dependent protein kinase into the cytoplasm of the cell.

11) When a steroid hormone and a peptide hormone exert similar effects on a population of target cells, then _____.
A) The steroid and peptide hormones must use the same biochemical mechanisms.
B) The steroid and peptide hormones must bind to the same receptor protein.
C) The steroid hormones affect the synthesis of effector proteins, whereas peptide hormones activate effector proteins already present in the cell.
D) The steroid hormones affect the activity of certain proteins within the cell, whereas peptide hormones directly affect the processing of mRNA.

12) Based on their effects, which pair below would NOT be expected to be active at the same time and place?
A) Prostaglandin and nitric oxide.
B) Endocrine and exocrine glands.
C) Hormones and target cells.
D) Neurosecretory cells and neurotransmitters.

13) The steroid hormone that coordinates molting in arthropods is _____.
A) Ecdysteroid.
B) Glucagon.
C) Thyroxine.
D) Growth hormone.

14) Growth factors are local regulators that _____.
A) Are modified fatty acids that stimulate bone and cartilage growth.
B) Are found on the surface of cancer cells and stimulate abnormal cell division.
C) Bind to cell-surface receptors and stimulate growth and development of target cells.
D) Convey messages between nerve cells.

15) Steroid and peptide hormones typically have in common _____.
A) The building blocks from which they are synthesized.
B) Their solubility in cell membranes.
C) Their requirement for travel through the bloodstream.
D) Their reliance on signal transduction in the cell.

16) A cluster of tumor cells that produces and secretes growth factors to induce surrounding cells to grow and divide is showing which type of cell-to-cell signaling?
A) Autocrine.
B) Paracrine.
C) Endocrine.
D) Neuroendocrine.

17) If a portion of the pancreas is surgically removed from a rat and the rat subsequently loses its appetite, one explanation is that the removed portion contains cells that secrete a chemical signal that somehow stimulates appetite. Given this scenario, what type of chemical signaling is occurring?
A) Autocrine.
B) Paracrine.
C) Endocrine.
D) Neuroendocrine.

18) If a biochemist discovers a new molecule, which of the following pieces of data would allow her to draw the conclusion that the molecule is a steroid hormone?
I) The molecule is lipid soluble.
II) The molecule is derived from a series of steps beginning with cholesterol.
III) The molecule acts at a target tissue some distance from where it is produced.
IV) The molecule uses a carrier protein when in an aqueous solution such as blood.
A) Only I and III.
B) Only II and IV.
C) Only I, III, and IV.
D) I, II, III, and IV.

19) Which of the following are similar in structure to cholesterol?
A) Leptin and serotonin.
B) Luteinizing hormone and insulin.
C) Melanocyte-stimulating hormone and vasopressin.
D) Testosterone, estradiol, and cortisol.

20) Polypeptides can have which of the following types of effects?
I) Autocrine.
II) Paracrine.
III) Endocrine.
A) Only I and III.
B) Only II and III.
C) Only I and II.
D) I, II, and III.

21) What property of steroid hormones allows them to cross the phospholipid bilayer?
A) Steroid hormones are lipid soluble and easily cross the phospholipid bilayer.
B) Steroid hormones can act in very small concentrations and very few molecules of steroids need to cross the lipid bilayer.
C) Steroid hormones act on cells close to where they were produced and very few molecules are required to travel such a short distance to cross the lipid bilayer.
D) Steroid hormones act on the same cells in which they are produced and, therefore, are within the cell they are acting upon.

22) Tadpoles must undergo a major metamorphosis to become frogs. This change includes reabsorption of the tail, growth of limbs, calcification of the skeleton, increase in rhodopsin in the eye, development of lungs, change in hemoglobin structure, and reformation of the gut from the long gut of an herbivore to the short gut of a carnivore. Amazingly, all of these changes are induced by thyroxine. What is the most likely explanation for such a wide array of effects of thyroxine?
A) There are many different forms of thyroxine, each specific to a different tissue.
B) Different tissues have thyroxine receptors that activate different signal transduction pathways.
C) Some tissues have membrane receptors for thyroxine, while other tissues have thyroxine receptors within the nucleus.
D) Different releasing hormones release thyroxine to different tissues.

23) When adenylyl cyclase is activated _____.
A) cAMP is created.
B) cAMP is destroyed.
C) G proteins bind to cAMP.
D) Steroid hormones pass through the lipid bilayer.

24) Insect hormones and their receptors _____.
A) Act independently of each other.
B) Are a focus in pest control research.
C) Utilize cell-surface receptors only.
D) Are active independently of environmental cues.

25) During mammalian labor and delivery, the contraction of uterine muscles is enhanced by oxytocin. This is an example of _____.
A) A negative feedback system.
B) A hormone that acts in an antagonistic way with another hormone.
C) A hormone that is involved in a positive feedback loop.
D) Signal transduction immediately changing gene expression in its target cells.

26) Which of the following has both endocrine and exocrine activity?
A) The pituitary gland.
B) Parathyroid glands.
C) Salivary glands.
D) The pancreas.

27) Analysis of a blood sample from a fasting individual who had not eaten for twenty -four hours would be expected to reveal high levels of _____.
A) Insulin.
B) Glucagon.
C) Gastrin.
D) Glucose.

28) When the beta cells of the pancreas release insulin into the blood, _____.
A) The skeletal muscles and the adipose cells take up glucose at a faster rate.
B) The liver catabolizes glycogen.
C) The alpha cells of the pancreas release glucose into the blood.
D) The kidneys begin gluconeogenesis.

29) The steroid hormone that coordinates molting in arthropods is _____.
A) Ecdysteroid.
B) Glucagon.
C) Thyroxine.
D) Growth hormone.

30) Which of the following statements are correct?
I) Hormones often regulate homeostasis through antagonistic functions.
II) Hormones of the same chemical class usually have the same function.
III) Hormones are secreted by specialized cells usually located in exocrine glands.
IV) Hormones are often regulated through feedback loops.
A) Only II and III.
B) Only I and III.
C) Only III and IV.
D) Only I and IV.

31) An example of antagonistic hormones controlling homeostasis is _____.
A) Thyroxine and parathyroid hormone in calcium balance.
B) Insulin and glucagon in glucose metabolism.
C) Progestins and estrogens in sexual differentiation.
D) Epinephrine and norepinephrine in fight-or-flight responses.

32) The relationship between the insect hormones ecdysteroid and PTTH is an example of _____.
A) An interaction of the endocrine and nervous systems.
B) Competitive inhibition of a hormone receptor.
C) How peptide-derived hormones have more widespread effects than steroid hormones.
D) Homeostasis maintained by antagonistic hormones.

33) Hormones secreted by the posterior pituitary gland are made in the _____.
A) Cerebellum.
B) Thalamus.
C) Hypothalamus.
D) Medulla oblongata.

34) Injury localized to the hypothalamus would most likely disrupt _____.
A) Short-term memory.
B) Coordination during locomotion.
C) Executive functions, such as decision making.
D) Regulation of body temperature.

35) The interrelationships between the endocrine and the nervous systems are especially apparent in a _____.
A) Steroid-producing cell in the adrenal cortex.
B) Neurosecretory cell in the hypothalamus.
C) Brain cell in the cerebral cortex.
D) Cell in the pancreas that produces digestive enzymes.

36) Portal blood vessels connect two capillary beds found in the _____.
A) Hypothalamus and thalamus.
B) Anterior pituitary and posterior pituitary.
C) Hypothalamus and anterior pituitary.
D) Posterior pituitary and thyroid gland.

37) If a person loses a large amount of water in a short period of time, he or she may die from dehydration. Antidiuretic hormone (ADH) can help reduce water loss through its interaction with its target cells in the _____.
A) Anterior pituitary.
B) Posterior pituitary.
C) Bladder.
D) Kidney.

38) A product of the anterior pituitary gland that causes color changes in its target cells is _____.
A) Follicle-stimulating hormone (FSH).
B) Luteinizing hormone (LH).
C) Thyroid-stimulating hormone (TSH).
D) Melanocyte-stimulating hormone (MSH).

39) In a lactating mammal, the two hormones that promote milk synthesis and milk release, respectively, are _____.
A) Prolactin and calcitonin.
B) Prolactin and oxytocin.
C) Follicle-stimulating hormone and luteinizing hormone.
D) Luteinizing hormone and oxytocin.

40) Oxytocin and antidiuretic hormone (ADH) are synthesized in the _____.
A) Hypothalamus.
B) Adenohypophysis.
C) Anterior pituitary.
D) Posterior pituitary.

41) Which endocrine disorder is correctly matched with the malfunctioning gland?
A) Dwarfism - the adrenal cortex.
B) Gigantism - the anterior pituitary gland.
C) Goiter - the adrenal medulla.
D) Diabetes mellitus - the parathyroid glands.

42) The body's reaction to PTH (parathyroid hormone), raising plasma levels of calcium, can be opposed by _____.
A) Thyroxine.
B) Epinephrine.
C) Growth hormone.
D) Calcitonin.

43) Which of the following is the most likely explanation for hypothyroidism in a patient whose iodine level is normal?
A) Greater production of T3 than of T4.
B) Hyposecretion of thyroid-stimulating hormone (TSH).
C) Hypersecretion of thyroid-stimulating hormone (TSH).
D) A decrease in the thyroid secretion of calcitonin.

44) When a person drinks alcohol, the rate of urination increases. This suggests that antidiuretic hormone (ADH) may be affected by alcohol consumption in some way. Which of the following best accounts for the increase in urination?
A) Alcohol stimulates the release of ADH.
B) Alcohol inhibits the release of ADH.
C) Alcohol inhibits the binding of ADH to receptors in the nephron.
D) Alcohol could inhibit ADH release or the binding of ADH to receptors in the nephron.

45) Removing which of the following glands would have the most wide -reaching effect on bodily functions of an adult human?
A) Adrenal glands.
B) Pituitary gland.
C) Thyroid gland.
D) Ovaries (in female) or testes (in male).

46) Glucocorticoids do which of the following?
A) Promote the immune response.
B) Promote the release of fatty acids.
C) Increase blood glucose levels.
D) Increase insulin production.

47) Nitric oxide and epinephrine _____.
A) Both function as neurotransmitters.
B) Both function as steroid hormones.
C) Bind the same receptors.
D) Both cause a reduction in the blood levels of glucose.

48) Fight-or-flight reactions include activation of the ___.
A) Parathyroid glands, leading to increased metabolic rate.
B) Anterior pituitary gland, leading to cessation of gonadal function.
C) Adrenal medulla, leading to increased secretion of epinephrine.
D) Pancreas, leading to a reduction in the blood sugar concentration.

49) The amino acid tyrosine is a starting substrate for the synthesis of _____.
A) Epinephrine.
B) Steroid hormones.
C) Parathyroid hormone.
D) Acetylcholine.

50) A disease that destroys the adrenal cortex should lead to an increase in the plasma levels of _____.
A) Glucocorticoid hormones.
B) Epinephrine.
C) Adrenocorticotropic hormone (ACTH).
D) Acetylcholine.

51) During a stressful interval, _____.
A) Thyroid-stimulating hormone (TSH) stimulates the adrenal cortex and medulla to secrete acetylcholine.
B) The alpha cells of islets secrete insulin and simultaneously the beta cells of the islets secrete glucagon.
C) Adrenocorticotropic hormone (ACTH) stimulates the adrenal cortex, and neurons of the sympathetic nervous system stimulate the adrenal medulla.
D) The calcium levels in the blood are increased due to actions of two antagonistic hormones, epinephrine and norepinephrine.

52) In response to stress, the adrenal gland promotes the synthesis of glucose from non-carbohydrate substrates via the action of the steroid hormone _____.
A) Glucagon.
B) Cortisol.
C) Thyroxine.
D) Adrenocorticotropic hormone (ACTH).

53) Melatonin is secreted by the _____.
A) Hypothalamus during the day.
B) Pineal gland during the night.
C) Autonomic nervous system during the winter.
D) Posterior pituitary gland during the day.

54) DES is called an "endocrine disrupting chemical" because it structurally resembles, and interferes with, the endocrine secretions of the _____.
A) Thyroid gland.
B) Adrenal medulla.
C) Ovaries.
D) Hypothalamus.

55) Epinephrine is an example of _____.
A) An androgen.
B) An estrogen.
C) A catecholamine.
D) A glucocorticoid.

56) In an experiment, rats' ovaries were removed immediately after impregnation and then the rats were divided into two groups. Treatments and results are summarized in the table below. The results most likely occurred because progesterone exerts an effect on the ____.

	Group 1	Group 2
Daily injections of progesterone (mg)	0.25	20
Percentage of rats carrying fetus to birth	0	100

A) General health of the rat.
B) Metabolism of the uterus.
C) Gestation period of rats.
D) Number of eggs fertilized.

57) People with type II diabetes mellitus have defective insulin receptors that cannot respond to insulin properly. Relative to normal individuals, what would be the effect on blood glucose levels under conditions of chronic stress that kept blood cortisol levels high? There would be ___.
A) A greater increase in blood glucose levels in individuals with type II diabetes mellitus than in normal individuals.
B) Less increase in blood glucose levels in individuals with type II diabetes mellitus than in normal individuals.
C) Be a greater decrease in blood glucose levels in individuals with type II diabetes mellitus than in normal individuals.
D) Less decrease in blood glucose levels in individuals with type II diabetes mellitus than in normal individuals.

58) Osteoporosis is a condition in which the density of bones is decreased so much that the individual is at a higher risk of fractures. The more calcium in the bones, the better the bone density. Which of the following would produce the greatest increase in bone calcium levels?
A) Calcitonin injection.
B) Calcitonin receptor blocker.
C) Parathyroid hormone injection.
D) Glucagon receptor blocker.

59) Predict the effects of a drug that increases adrenocorticotropic hormone (ACTH) synthesis.
A) Increase in glucocorticoid production.
B) Increase in release of corticotropin-releasing hormone (CRH).
C) Decrease in cortisol release.
D) Decrease in release of corticotropin-releasing hormone (CRH).

60) Of the following types of molecules, which can function as both neurotransmitters and hormones?
A) Glucocorticoids.
B) Second messengers.
C) Catecholamines.
D) Adipocytes.

61) Correct and appropriate signal transduction processes are generally under strong selective pressure and are determined by the properties of the molecules involved, the concentrations of signal and receptor molecules, and the binding affinities between signal and receptor. Therefore, a hormone action is very specific in a species at any one point in time. However, there are examples of very diverse functions of a specific hormone between groups of organisms. For example, thyroxin, which is produced in all vertebrates and many invertebrates, can trigger growth, differentiation, metamorphosis, maturation, reproduction, behavior, temperature tolerance, osmoregulation, or seasonal adaptation depending on the organism in which it is produced. What is the most logical explanation for such different responses triggered by thyroxin in organisms?
A) The concentration of thyroxin varies in different organisms. Invertebrate organisms do not have as much thyroxin as vertebrate organisms.
B) Thyroxin and its receptor molecules have a different binding affinity in different organisms.
C) Receptor molecules for thyroxin are located on different tissues in different organisms.
D) The structure of thyroxin is substantially different in different organisms.

CHAPTER 46:

ANIMAL REPRODUCTION

1) Regeneration normally follows _____.
A) All types of asexual reproduction.
B) Fission.
C) Fragmentation.
D) Parthenogenesis.

2) Asexual reproduction results in offspring that are genetically identical to their parent. What type of cell process occurs to generate this type of offspring?
A) Mitosis.
B) Meiosis.
C) Cell fusion.
D) None of them.

3) What makes sexually reproduced offspring genetically different from their parents?
A) Genetic recombination during meiosis.
B) Genetic recombination during mitosis.
C) Crossing over during mitosis.
D) Sexual reproduction does not produce genetically different offspring.

4) Which of the following aspects of eukaryotic reproduction are found only among invertebrate animals?
A) Sexual and asexual reproduction.
B) External and internal fertilization.
C) Hermaphroditism and parthenogenesis.
D) Fission and budding.

5) On a submarine expedition to the ocean bottom, you discover a population of fish that are only female. What type of reproduction does this fish most likely use?
A) Sexual.
B) Budding.
C) Cloning.
D) Parthenogenesis.

6) Which of the following is most true of sexual reproduction?
A) Only half of the offspring from sexually reproducing females are also females.
B) Asexual reproduction produces offspring of greater genetic variety.
C) Sexual reproduction is completed more rapidly than asexual reproduction.
D) Asexual reproduction is better suited to environments with extremely varying conditions.

7) Environmental cues that influence the timing of reproduction generally do so by _____.
A) Increasing the body temperature.
B) Providing access to water for external fertilization.
C) Increasing ambient temperature most favorable for sex.
D) Direct effects on hormonal control mechanisms.

8) Evidence that parthenogenic whiptail lizards are derived from sexually reproducing ancestors includes _____.
A) The requirement for male-like behaviors in some females before their partners will ovulate.
B) The development and then regression of testes prior to sexual maturation.
C) The observation that all of the offspring are haploid.
D) The persistence of a vestigial penis among some of the females.

9) In an animal that switches between sexual and asexual reproduction, when is sexual reproduction more likely to occur?
A) When conditions for survival are favorable.
B) When conditions for survival are unfavorable.
C) When males and females find each other.
D) What conditions favor sexual over asexual remains a complete mystery.

10) Genetic mutations in asexually reproducing organisms lead to more evolutionary change than do genetic mutations in sexually reproducing ones because _____.
A) Asexually reproducing organisms, but not sexually reproducing organisms, pass all mutations on to their offspring.
B) Sexually reproducing organisms can produce more offspring in a given time than can asexually reproducing organisms.
C) More genetic variation is present in organisms that reproduce asexually than is present in those that reproduce sexually.
D) Asexually reproducing organisms have more dominant genes than organisms that reproduce sexually.

11) Asexual reproduction results in greater reproductive success than does sexual reproduction when _____.
A) Pathogens are rapidly diversifying.
B) There is some potential for rapid overpopulation.
C) A species is expanding into diverse geographic settings.
D) A species is in stable and favorable environments.

12) Sexual reproduction _____.
A) Allows animals to conserve resources and reproduce only during optimal conditions.
B) Can produce diverse phenotypes that may enhance survival of a population in a changing environment.
C) Enables males and females to remain isolated from each other while rapidly colonizing habitats.
D) Guarantees that both parents will care for each offspring.

13) For water fleas of the genus *Daphnia*, switching from a pattern of asexual reproduction to sexual reproduction coincides with _____.
A) Environmental conditions becoming more favorable for offspring.
B) Greater abundance of food resources for offspring.
C) Periods of temperature or food stresses on adults.
D) Exhaustion of an individual's supply of eggs.

14) Among non-mammalian vertebrates, the cloaca is an anatomical structure that functions as _____.
A) A specialized sperm-transfer device produced only by males.
B) A shared pathway for the digestive, excretory, and reproductive systems.
C) A source of nutrients for developing sperm in the testes.
D) A gland that secretes mucus to lubricate the vaginal opening.

15) Females of many insect species, including honeybee queens, can store gametes shed by their mating partners in _____.
A) Their nests.
B) The abdominal tract.
C) The uterus.
D) The spermathecal.

16) Animals that have external fertilization are most likely to reproduce in which of the following areas?
A) Sand dune.
B) Polar ice sheet.
C) Shallow lake.
D) Tallgrass prairie.

17) In close comparisons, external fertilization often yields more offspring than does internal fertilization. However, internal fertilization typically offers the advantage that _____.
A) It requires less time and energy to be devoted to reproduction.
B) The smaller number of offspring produced often receive a greater amount of parental investment.
C) It permits the most rapid population increase.
D) It requires expression of fewer genes and maximizes genetic stability.

18) You decide to study two species of birds, both of which form monogamous pairs (one male and one female). In species 1, you find that the eggs in a pair's nest are in fact almost always the offspring of that pair. In species 2, you are surprised to find that many of the eggs in a nest were actually fathered by males of neighboring pairs. Apparently, mating outside of monogamous pairings is widespread in species 2. Given this information, what would be the logical prediction to make before comparing testes size of males of the two species?
A) Testes of species 1 are larger than testes of species 2.
B) Testes of species 2 are larger than testes of species 1.
C) There is no relationship between this observation and the size of testes.
D) None of them.

19) Which of the following is a challenge to the hypothesis that bats, which produce an unusually large number of sperm, increase the probability of conception when females engage in multiple matings?
I) Female bats can eject the sperm of certain males after mating.
II) A female can actively choose which male mates with her last, thus increasing the odds of conception with that male's sperm.
A) Only I is correct.
B) Only II is correct.
C) Both I and II are correct.
D) None of them.

20) Which of the following structures in females is analogous in function to the vas deferens in males?
A) Urethra.
B) Oviduct.
C) Uterus.
D) Vagina.

21) In humans, the follicular cells that remain behind in the ovary following ovulation become _____.
A) The ovarian endometrium that is shed at the time of the menses.
B) A steroid-hormone synthesizing structure called the corpus luteum.
C) The thickened portion of the uterine wall.
D) The placenta, which secretes cervical mucus.

22) At the time of fertilization, the maturation of the human oogonium has resulted in _____.
A) One secondary oocyte.
B) Two primary oocytes.
C) Four secondary oocytes.
D) Four zygotes.

23) In vertebrate animals, spermatogenesis and oogenesis differ in that _____.
A) Oogenesis begins at the onset of sexual maturity, whereas spermatogenesis begins during embryonic development.
B) Oogenesis produces four functional haploid cells, whereas spermatogenesis produces only one functional spermatozoon.
C) Cytokinesis is unequal in oogenesis, whereas it is equal in spermatogenesis.
D) Oogenesis ends at menopause, whereas spermatogenesis is finished before birth.

24) Mature human sperm and ova are similar in that they _____.
A) Both have the same number of chromosomes.
B) Are approximately the same size.
C) Each have a flagellum that provides motility.
D) Are produced from puberty until death.

25) Which of the following correctly describes a difference between spermatogenesis and oogenesis?
A) Spermatogenesis results in four mature sperm cells, while oogenesis results in one mature egg cell.
B) Spermatogenesis results in one mature sperm cell, while oogenesis results in four mature egg cells.
C) In spermatogenesis, mitosis occurs twice and meiosis once, while in oogenesis, mitosis occurs once and meiosis twice.
D) Spermatogenesis results in four mature sperm cells, while oogenesis results in one mature egg cell. In spermatogenesis, mitosis occurs twice and meiosis once, while in oogenesis, mitosis occurs once and meiosis twice.

26) Among mammals, the male and female genital structures that consist mostly of erectile tissue include the _____.
A) Penis and clitoris.
B) Vas deferens and oviduct.
C) Testes and ovaries.
D) Prostate and ovaries.

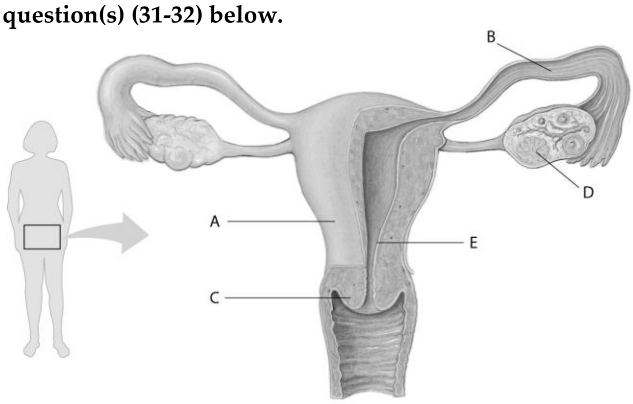
27) Among human males, both semen and urine normally travel along the _____.
A) Vas deferens.
B) Seminal vesicle.
C) Urethra.
D) Ureter.

28) Human sperm cells first arise in the _____.
A) Prostate gland.
B) Vas deferens.
C) Seminiferous tubules.
D) Epididymis.

29) The surgical removal of the seminal vesicles would likely _____.
A) Cause sterility because sperm would not be produced.
B) Cause sterility because sperm would not be able to exit the body.
C) Greatly reduce the volume of semen.
D) Cause the testes to migrate back into the abdominal cavity.

30) Increasing the temperature of the human scrotum by 2°C (that is, near the normal body core temperature) and holding it there would most likely _____.
A) Reduce the fertility of the man by impairing the production of gonadal steroid hormones.
B) Reduce the fertility of the man by impairing spermatogenesis.
C) Reduce the man's sexual interest.
D) Increase the fertility of the affected man by enhancing the rate of steroidogenesis.

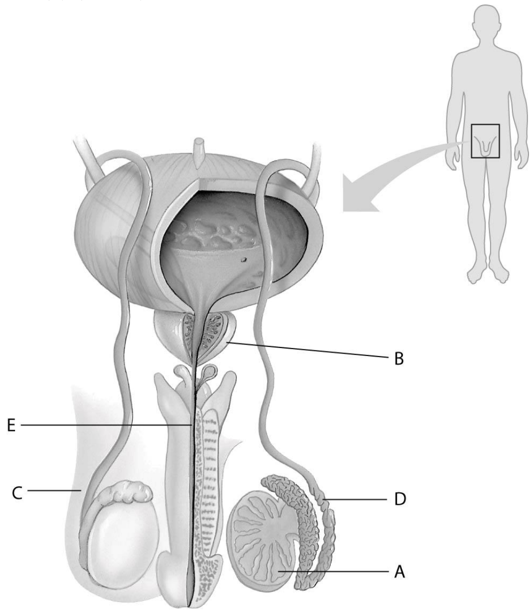
Refer to the following figure, which diagrams the reproductive anatomy of the human female, to answer the question(s) (31-32) below.



31) In the above figure, which letter points to the corpus luteum?
A) A.
B) B.
C) C.
D) D.

32) In the above figure, which letter points to the cervix?
A) A.
B) B.
C) C.
D) D.

Refer to the following figure, which diagrams the reproductive anatomy of the human male, to answer the question(s) (33-34) below.



33) In the above figure, which letter points to the prostate gland?
A) A
B) B
C) C
D) D.

34) In the above figure, which letter points to the vas deferens?
A) A.
B) B.
C) C.
D) D.

35) For which of the following is the number the same in spermatogenesis and oogenesis?
A) Timing of meiotic divisions.
B) Functional gametes produced by meiosis.
C) Meiotic divisions required to produce each gamete.
D) Different cell types produced by meiosis.

36) Which statement about human reproduction is correct?
A) Fertilization occurs in the uterus.
B) In humans, spermatogenesis and oogenesis function best at normal, core body temperatures.
C) A human oocyte completes meiosis after a sperm penetrates it.
D) The earliest stages of spermatogenesis occur closest to the lumen of the seminiferous tubules.

37) A physician finds that a nine-year-old male patient is entering puberty much earlier than is usual. Such a condition is most likely the result of a tumor in the _____.
A) Hypothalamus, producing elevated levels of testosterone.
B) Anterior pituitary, producing elevated levels of testosterone.
C) Testes, producing elevated levels of estrogen.
D) Anterior pituitary, producing elevated levels of gonadotropin-stimulating hormone.

38) A male's "primary" sex characteristics include _____.
A) Deepening of the voice at puberty.
B) Development of the seminal vesicles and associated ducts.
C) Elongation of the skeleton prior to puberty.
D) Onset of growth of facial hair at puberty.

39) The primary difference between estrous and menstrual cycles is that _____.
A) The endometrium shed by the uterus during the estrous cycle is reabsorbed with no extensive fluid flow out of the body, whereas the shed endometrium of menstrual cycles is excreted from the body.
B) Behavioral changes during estrous cycles are much less apparent than those of menstrual cycles.
C) Season and climate have less pronounced effects on estrous cycles than they do on menstrual cycles.
D) Copulation normally occurs across the estrous cycle, whereas in menstrual cycles copulation only occurs during the period surrounding ovulation.

40) In correct chronological order, the three phases of the human ovarian cycle are _____.
A) Follicular → luteal → secretory.
B) Menstrual → proliferative → secretory.
C) Follicular → ovulation → luteal.
D) Proliferative → luteal → ovulation.

41) In correct chronological order, the three phases of the human uterine cycle are _____.
A) Follicular → luteal → secretory.
B) Menstrual → proliferative → secretory.
C) Follicular → ovulation → luteal.
D) Proliferative → luteal → ovulation.

42) An inactivating mutation in the progesterone receptor gene would likely result in _____.
A) The absence of secondary sex characteristics.
B) The inability of the uterus to support pregnancy.
C) Enlarged and hyperactive uterine endometrium.
D) The absence of mammary gland development.

43) A primary response by the Leydig cells in the testes to the presence of luteinizing hormone is an increase in the synthesis and secretion of _____.
A) Inhibin.
B) Testosterone.
C) Oxytocin.
D) Progesterone.

44) A reproductive hormone that is secreted directly from a structure in the brain is _____.
A) Estradiol.
B) Progesterone.
C) Follicle-stimulating hormone (FSH).
D) Gonadotropin-releasing hormone (GnRH).

45) The primary function of the corpus luteum is to _____.
A) Nourish and protect the egg cell.
B) Maintain progesterone and estrogen synthesis after ovulation has occurred.
C) Stimulate the development of the mammary glands.
D) Support pregnancy in the second and third trimesters.

46) Menopause is characterized by _____.
A) The loss of responsiveness by the ovaries to follicle-stimulating hormone (FSH) and luteinizing hormone (LH).
B) A decline in production of the gonadotropin hormones by the anterior pituitary gland.
C) Wearing away of the uterine endometrium.
D) A halt in the synthesis of gonadotropin-releasing hormone by the brain.

Use the following information to answer the questions (47-50) below.

For your internship at the local zoo, you have been assigned to help with the new orangutan-breeding program. Little is known about orangutan reproductive hormones, but hormone feedback cycles are often the same in closely related animals. You have been asked to use your knowledge of the interactions of human reproductive hormones to recommend injections to promote ovulation in a female orangutan when a visiting male arrives for a brief breeding visit.

47) Refer to the paragraph on the orangutan breeding program. Which of the following hormones would you use if you want to induce ovulation right away?
A) Estradiol (estrogen).
B) Progesterone.
C) Luteinizing hormone (LH).
D) Human chorionic gonadotropin (HCG).

48) Refer to the paragraph on the orangutan breeding program. Which of the following hormones would you use if you want to induce ovulation in a few days?
A) Estradiol (estrogen).
B) Progesterone.
C) Luteinizing hormone (LH).
D) Human chorionic gonadotropin (HCG).

49) Labor contractions can be increased by the medical use of a synthetic drug that mimics the action of _____.
A) Inhibin.
B) Luteinizing hormone.
C) Oxytocin.
D) Prolactin.

50) In excreted urine, a reliable "marker" that a pregnancy has initiated is _____.
A) Progesterone.
B) Estrogen.
C) Follicle-stimulating hormone.
D) Human chorionic gonadotropin (hCG).

51) Two contraceptive methods that are generally irreversible and which block the gametes from moving to a site where fertilization can occur are _____.
A) The male condom and female condom.
B) The male condom and oral contraceptives.
C) Vasectomy and tubal ligation.
D) The diaphragm and subcutaneous progesterone implant.

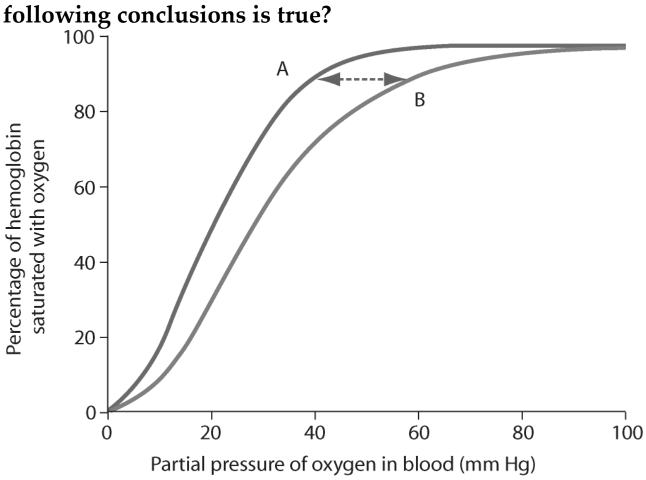
52) Tubal ligation _____.
A) Reduces the incidence of ovulation.
B) Prevents fertilization by preventing sperm from entering the uterus.
C) Prevents sperm from exiting the male urethra.
D) Prevents oocytes from entering the uterus.

53) A vasectomy _____.
A) Eliminates spermatogenesis.
B) Eliminates testosterone synthesis.
C) Prevents implantation of an embryo.
D) Prevents sperm from exiting the male urethra.

54) Imagine that a woman is in the final week of her pregnancy. Her doctor gives her an injection of oxytocin. The likely result of this is that the pregnant woman would ____.
A) Stop secreting prostaglandins from the placenta.
B) Undergo vigorous contractions of her uterine muscles.
C) Increase the synthesis and secretion of progesterone.
D) Be prevented from lactation.

55) A pregnant woman comes into the hospital past her due date. The doctor decides it is time for the baby to be delivered. Before performing a cesarean section, the doctor wants to try to induce labor. Which of the following would she most likely inject?
A) Progesterone.
B) Luteinizing hormone (LH).
C) Follicle-stimulating hormone (FSH).
D) Oxytocin.

56) Based on the figure below, showing fetal (A) and adult (B) oxygen -hemoglobin saturation curves, which of the following conclusions is true?



A) The mother binds oxygen with greater affinity than the fetus.
B) Adult saturation occurs at lower partial pressures of oxygen than fetal saturation does.
C) At 50 percent saturation, fetal blood will have a higher affinity for oxygen than adult blood will.
D) As the partial pressure of oxygen increases, adult hemoglobin approaches saturation faster than fetal hemoglobin does.

57) An ectopic pregnancy occurs when a fertilized egg implants somewhere other than in the lining of the uterus. Usually it implants in the oviduct. Which of the following would be the most likely explanation for such a pregnancy being unsuccessful?
A) The orientation of the baby would be sideways.
B) Human chorionic gonadotropin (HCG) cannot be produced in the oviduct.
C) The lining of the oviduct is unable to support the developing fetus.
D) The necessary hormones cannot reach the developing fetus in the oviduct.

Use the following information to answer the question(s) below.

Estrogens found in the environment have raised concerns about effects on reproductive health of animals. Researchers studied the effects that estrogens in the water have on sexual differentiation in zebrafish. They exposed embryo-larval (0-21 days post-hatching), juvenile (21-42 days post-hatching), and adult (over 200 days post-hatching) fish to three concentrations of 17β-estradiol (5, 25, and 100 nanograms/liter) that are within the range of concentrations found in water leaving sewage treatment plants in different countries. They then examined the proportion of males and females when the fish exposed at embryo-larval and juvenile stages reached adulthood. Embryo-larval stage fish that had been exposed to 100 ng/l 17β-estradiol resulted in adult populations that had substantially more females than males compared to control groups. Embryo -larval fish that had been exposed to 5 and 25 ng/l of 17β -estradiol did not show a statistically significant shift in the proportion of females. (Brion, F., C. R. Tyler, X. Palazzi, B. Laillet, J. M. Porcher, J. Garric, and P. Flammarion. 2004. Impacts of 17β -estradiol, including environmentally relevant concentrations, on reproduction after exposure during embryo-larval-, juvenile-, and adult-life stages in zebrafish *(Danio rerio)*. *Aquatic Toxicology* 68:193-217.)

58) Refer to the paragraph on the effects of estrogens in the environment. What is the significance of using the concentrations of 5, 25, and 100 ng/l of 17β -estradiol for the dose in this experiment?
A) These concentrations are similar to those found in many animals.
B) These concentrations are found in the environment.
C) These concentrations are effective, yet not lethal to the fish.
D) These concentrations are standard in toxicology assays.

59) Refer to the paragraph on the effects of estrogens in the environment. You are assigned to write the report to the Environmental Protection Agency, which needs to decide what level of 17β-estradiol to permit in sewage output. You do not want to make the level any lower than necessary, because it requires substantial additional money for the extra treatment of sewage. Given the data presented above, what level of 17β-estradiol would you suggest is safe to prevent feminization of fish?
A) 2.5 ng/l.
B) 12.5 ng/l.
C) 25 ng/l.
D) 100 ng/l.

60) Which of the following is the most likely explanation for the lack of a filter blocking the passage of alcohol between the maternal and fetal circulations in humans?
A) There has not been enough time to evolve such a barrier.
B) Such a barrier would probably also block important molecules that need to be passed to the fetus.
C) The maternal and fetal blood mix directly together in an area with many villi, so a barrier is impossible.
D) Alcohol has some positive effects on the fetus, so evolution has resulted in an intermediate level of filtering that blocks all but the worst abuses of alcohol.

CHAPTER 47:

ANIMAL DEVELOPMENT

1) Even in the absence of sperm, metabolic activity in an egg can be artificially activated by _____.
A) Abnormally high levels of carbonic acid in the cytosol.
B) Abnormally low levels of extracellular oxygen.
C) Injection of calcium ions into the cytosol.
D) Depletion of its ATP supplies.

2) The formation of the fertilization envelope requires an increase in the availability of _____.
A) Calcium ions.
B) Hydrogen ions.
C) Potassium ions.
D) Sodium ions.

3) Contact of a sea urchin egg with signal molecules on sperm causes the egg to undergo a brief _____.
A) Mitosis.
B) Membrane depolarization.
C) Vitellogenesis.
D) Acrosomal reaction.

4) The plasma membrane of the sea urchin egg _____.
A) Is outside of the fertilization membrane.
B) Releases calcium, which initiates the cortical reaction.
C) Has receptor molecules that are specific for binding acrosomal proteins.
D) Is a mesh of proteins crossing through the cytosol of the egg.

5) Fertilization of an egg without activation is most like _____.
A) Placing the key in the ignition of a car but not starting the engine.
B) Resting during halftime of a basketball game.
C) Preparing a pie from scratch and baking it in the oven.
D) Walking to the cafeteria and eating lunch.

6) A reproductive difference between sea urchins and humans is _____.
A) The sea urchin egg completes meiosis prior to fertilization, but meiosis in humans is completed after fertilization.
B) Sea urchin eggs and sperm are of equal size, but human eggs are much bigger than human sperm.
C) Sea urchins, but not humans, have a need to block polyspermy, because only in sea urchins can there be more than one source of sperm to fertilize the eggs.
D) Sea urchin zygotes get their mitochondria from the sperm, but human zygotes get their mitochondria from the egg.

7) During fertilization, the acrosomal contents _____.
A) Block polyspermy.
B) Help propel more sperm toward the egg.
C) Digest the protective jelly coat on the surface of the egg.
D) Trigger the completion of meiosis by the sperm.

8) In a newly fertilized egg, the vitelline layer _____.
A) Lifts away from the egg and hardens to form a fertilization envelope.
B) Secretes hormones that enhance steroidogenesis by the ovary.
C) Reduces the loss of water from the egg and prevents desiccation.
D) Provides most of the nutrients used by the zygote.

9) In sea urchins, the "fast block" and the longer lasting "slow block" to polyspermy, respectively, are _____.
A) The acrosomal reaction and the formation of egg white.
B) The cortical reaction and the formation of yolk protein.
C) The jelly coat of the egg and the vitelline membrane.
D) Membrane depolarization and the cortical reaction.

10) In an egg cell treated with a chemical that binds calcium and magnesium ions, the _____.
A) Acrosomal reaction would be blocked.
B) Fusion of sperm and egg nuclei would be blocked.
C) Fast block to polyspermy would not occur.
D) Fertilization envelope would not be formed.

11) In mammalian eggs, the receptors for sperm are found in the _____.
A) Fertilization membrane.
B) Egg plasma membrane.
C) Cytosol of the egg.
D) Mitochondria of the egg.

12) A human blastomere is _____.
A) An embryonic cell that is smaller than the ovum.
B) An embryonic structure that includes a fluid-filled cavity.
C) That part of the acrosome that opens the egg's membrane.
D) A cell that contains a (degenerating) second polar body.

13) At the moment of sperm penetration, human eggs _____.
A) Have used flagellar propulsion to move from the ovary to the oviduct.
B) Are still located within the ovary.
C) Have a paper-thin cell of calcium carbonate that prevents desiccation.
D) Are still surrounded by follicular cells.

14) In a developing frog embryo, most of the yolk is _____.
A) Located near the animal pole.
B) Located near the vegetal pole.
C) Found within the cleavage furrow.
D) Distributed equally throughout the embryo.

15) Among these choices, the largest cell involved in frog reproduction is _____.
A) An egg.
B) A blastomere in the vegetal pole.
C) A blastomere in the animal pole.
D) One of the products of the first cleavage.

16) The pattern of embryonic development in which only the cells lacking yolk subsequently undergo cleavage is called _____.
A) Holoblastic development, which is typical of marsupial mammals.
B) Meroblastic development, which is typical of humans.
C) Holoblastic development, which is typical of amphibians.
D) Meroblastic development, which is typical of birds.

17) As cleavage continues during frog development, the size of the blastomeres _____.
A) Increases as the number of the blastomeres decreases.
B) Increases as the number of the blastomeres increases.
C) Decreases as the number of the blastomeres increases.
D) Decreases as the number of the blastomeres decreases.

18) The vegetal pole of a frog zygote differs from the animal pole in that _____.
A) The vegetal pole has a higher concentration of yolk.
B) The blastomeres originate only in the vegetal pole.
C) The vegetal pole cells undergo mitosis, but not cytokinesis.
D) The polar bodies bud from this region.

19) Meroblastic cleavage occurs in _____.
A) Sea urchins, but not in humans or birds.
B) Humans, but not in sea urchins or birds.
C) Birds, but not in sea urchins or humans.
D) Both sea urchins and birds, but not in humans.

20) Which of the following correctly displays the sequence of developmental milestones?
A) Blastula → gastrula → cleavage.
B) Cleavage → gastrula → blastula.
C) Cleavage → blastula → gastrula.
D) Gastrula → blastula → cleavage.

21) The first cavity formed during frog development is the _____.
A) Blastopore.
B) Mouth.
C) Blastocoel.
D) Anus.

22) In some rare salamander species, all individuals are females. Reproduction relies on those females having access to sperm from males of another species. However, the resulting embryos receive no genetic contribution from the males. In this case, the sperm appear to be used only for _____.
A) Morphogenesis.
B) Egg activation.
C) Cell differentiation.
D) The creation of a diploid cell.

23) The cortical reaction of sea urchin eggs functions directly in _____.
A) The formation of a fertilization envelope.
B) The production of a fast block to polyspermy.
C) The release of hydrolytic enzymes from the sperm.
D) The generation of an electrical impulse by the egg.

24) The structure of the *Drosophila* gene, called Tinman, is similar to a gene in humans that also _____.
A) Promotes ear development.
B) Specifies the location of the heart.
C) Determines structures in the eyes.
D) Specifies limb elongation points.

25) From earliest to latest, the overall sequence of early development proceeds in which of the following sequences?
A) First cell division → synthesis of embryo's DNA begins → acrosomal reaction → cortical reaction.
B) Cortical reaction → synthesis of embryo's DNA begins → acrosomal reaction → first cell division.
C) Cortical reaction → acrosomal reaction → first cell division → synthesis of embryo's DNA begins.
D) Acrosomal reaction → cortical reaction → synthesis of embryo's DNA begins → first cell division.

26) An embryo with meroblastic cleavage, extraembryonic membranes, and a primitive streak must be that of _____.
A) An insect.
B) An amphibian.
C) A bird.
D) A sea urchin.

27) Cells move to new positions as an embryo establishes its three germ-tissue layers during _____.
A) Determination.
B) Cleavage.
C) Induction.
D) Gastrulation.

28) The outer-to-inner sequence of tissue layers in a post-gastrulation vertebrate embryo is _____.
A) Endoderm → ectoderm → mesoderm.
B) Mesoderm → endoderm → ectoderm.
C) Ectoderm → mesoderm → endoderm.
D) Ectoderm → endoderm → mesoderm.

29) If gastrulation was blocked by an environmental toxin, then _____.
A) Cleavage would not occur in the zygote.
B) Embryonic germ layers would not form.
C) The blastula would not be formed.
D) The blastopore would form above the gray crescent in the animal pole.

30) The archenteron of the developing sea urchin eventually develops into the _____.
A) Blastocoel.
B) Heart and lungs.
C) Digestive tract.
D) Brain and spinal cord.

31) In a frog embryo, gastrulation _____.
A) Produces a blastocoel displaced into the animal hemisphere.
B) Occurs along the primitive streak in the animal hemisphere.
C) Proceeds by involution as cells roll over the lip of the blastopore.
D) Occurs within the inner cell mass that is embedded in the large amount of yolk.

32) Which of the following is a correct description of the fate of the germ layers?
A) The mesoderm gives rise to the notochord.
B) The endoderm gives rise to the hair follicles.
C) The ectoderm gives rise to the liver.
D) The mesoderm gives rise to the lungs.

33) The primitive streak in a bird is the functional equivalent of _____.
A) The lip of the blastopore in the frog.
B) The archenteron in a frog.
C) The notochord in a mammal.
D) Neural crest cells in a mammal.

34) In all vertebrate animals, development requires _____.
A) A large supply of yolk.
B) An aqueous environment.
C) Extraembryonic membranes.
D) A primitive streak.

35) The least amount of yolk would be found in the egg of a _____.
A) Bird.
B) Frog.
C) Eutherian mammal.
D) Reptile.

36) At the time of implantation, the human embryo is called a _____.
A) Blastocyst.
B) Gastrula.
C) Fetus.
D) Zygote.

37) Uterine implantation due to enzymatic digestion of the endometrium is initiated by the _____.
A) Inner cell mass.
B) Endoderm.
C) Mesoderm.
D) Trophoblast.

38) Thalidomide, now banned for use as a sedative in pregnancy, was used in the early 1960s by many women in their first trimester of pregnancy. Some of these women gave birth to children with arm and leg deformities, suggesting that the drug most likely influenced _____.
A) Early cleavage divisions.
B) Differentiation of bone tissue.
C) Morphogenesis.
D) Organogenesis.

39) The migratory neural crest cells _____.
A) Form most of the central nervous system.
B) Form the spinal cord in the frog.
C) Form a variety of neural and non-neural structures.
D) Form the lining of the lungs and of the digestive tract.

40) From earliest to latest, the overall sequence of early development proceeds in which of the following sequences?
A) Gastrulation → organogenesis → cleavage.
B) Cleavage → gastrulation → organogenesis.
C) Gastrulation → blastulation → neurulation.
D) Preformation → morphogenesis → neurulation.

41) Changes in cell position occur extensively during _____.
A) Organogenesis, but not during gastrulation or cleavage.
B) Cleavage, but not during gastrulation or organogenesis.
C) Gastrulation and cleavage.
D) Gastrulation.

42) Changes in the shape of a cell usually involve a reorganization of the _____.
A) Nucleus.
B) Cytoskeleton.
C) Extracellular matrix.
D) Transport proteins.

43) When we compare animal development to plant development, we find that _____.
A) Plant cells, but not animal cells, migrate during morphogenesis.
B) Animal cells, but not plant cells, migrate during morphogenesis.
C) Plant cells and animal cells migrate extensively during morphogenesis.
D) Neither plant cells nor animal cells migrate during morphogenesis.

44) Select the choice that correctly associates the organ with its embryonic sources.
A) Anterior pituitary gland - mesoderm and endoderm.
B) Thyroid gland - mesoderm and ectoderm.
C) Adrenal gland - ectoderm and mesoderm.
D) Skin - endoderm and mesoderm.

45) The embryonic precursor to the human spinal cord is the _____.
A) Notochord.
B) Neural tube.
C) Mesoderm.
D) Archenteron.

46) During metamorphosis, a tadpoles tail is reduced in size by the process of _____.
A) Regeneration.
B) Apoptosis.
C) Oxidative phosphorylation.
D) Re-differentiation.

47) The term applied to a morphogenetic process whereby cells extend themselves, making the mass of the cells narrower and wider, is _____.
A) Convergent extension.
B) Induction.
C) Elongational streaming.
D) Bi-axial elongation.

48) Which of the following is common to the development of birds and mammals?
A) The formation of an embryonic epiblast and hypoblast.
B) The formation of an embryonic trophoblast.
C) The formation of an embryonic yolk plug.
D) The formation of an embryonic gray crescent.

49) The archenteron of a frog develops into the _____.
A) Blastocoel.
B) Endoderm.
C) Placenta.
D) Lumen of the digestive tract.

50) What structural adaptation in chickens allows them to lay their eggs in arid environments rather than in water?
A) Amnion.
B) Yolk.
C) Gastrulation.
D) Development of the brain from ectoderm.

51) If an amphibian zygote is manipulated so that the first cleavage plane fails to divide the gray crescent, then _____.
A) The daughter cell with the entire gray crescent will die.
B) Both daughter cells will develop normally, because amphibians are totipotent at this stage.
C) Only the daughter cell with the gray crescent will develop normally.
D) Both daughter cells will develop abnormally.

52) Hans Spemann and colleagues developed the concept of the organizer in amphibian embryos while studying the _____.
A) Medial cells between the optic cups.
B) Anterior terminus of the notochord.
C) Lateral margins of the neural tube.
D) Dorsal lip of the blastopore.

53) Which of the following is an adult organism that has fewer than 1,000 cells?
A) Chickens, *Gallus domesticus*.
B) African clawed frogs, *Xenopus laevis*.
C) Fruit flies, *Drosophila melanogaster*.
D) Nematodes, *Caenorhabditis elegans*.

54) The developmental precursors to the gonadal tissues of *Caenorhabditis elegans* uniquely contain _____.
A) Proteins of maternal origin.
B) High concentrations of potassium ions.
C) T tubules for the propagation of action potentials.
D) P granules of mRNA and protein.

55) Two primary factors in shaping the polarity of the body axes in chick embryos are _____.
A) Light and temperature.
B) Salt gradients and membrane potentials.
C) Gravity and pH.
D) Moisture and mucus.

56) The arrangement of organs and tissues in their characteristic places in 3-D space defines _____.
A) Pattern formation.
B) Differentiation.
C) Determination.
D) Organogenesis.

57) If the apical ectodermal ridge is surgically removed from an embryo, it will lose _____.
A) Positional information for limb-bud pattern formation.
B) Guidance signals needed for correct gastrulation.
C) Unequal cytokinesis of blastomeres.
D) The developmental substrate for the kidneys.

58) The nematode *Caenorhabditis elegans* _____.
A) Is composed of about 1000 cells, in which the developmental origin of each cell has been mapped.
B) Has only a single chromosome, which has been fully sequenced.
C) As about 1000 genes, each of which has been fully sequenced.
D) Uniquely, among animals, utilizes programmed cell death during normal development.

59) In humans, identical twins are possible because _____.
A) Cytoplasmic determinants are distributed unevenly in unfertilized eggs.
B) Extraembryonic cells interact with the zygote nucleus.
C) Early blastomeres can form a complete embryo if isolated.
D) The gray crescent divides the dorsal-ventral axis into new cells.

60) Cells transplanted from the neural tube of a frog embryo to the ventral part of another embryo develop into nervous system tissues. This result indicates that the transplanted cells were _____.
A) Totipotent.
B) Determined.
C) Differentiated.
D) Mesenchymal.

61) Embryonic induction, the influence of one group of cells on another group of cells, plays a critical role in embryonic development. In 1924, Hans Spemann and Hilde Mangold transplanted a piece of tissue that influences the formation of the notochord and neural tube, from the dorsal lip of an amphibian embryo to the ventral side of another amphibian embryo. If embryonic induction occurred, which of the following observations justifies the claim of embryonic induction?
A) The transplanted tissue induced multiple limbs to develop on the ventral side of the recipient embryo.
B) The transplanted tissue inhibited normal cell division on the dorsal side of the recipient embryo that lead to its death.
C) The transplanted tissue had no effect on either the ventral or dorsal side of the recipient embryo so it continued to develop normally.
D) The transplanted tissue induced the formation of a second notochord and neural tube on the ventral side of the developing embryo.

CHAPTER 48:

NEURONS, SYNAPSES, AND SIGNALING

1) A simple nervous system _____.
A) Must include chemical senses, mechanoreception, and vision.
B) Includes a minimum of twelve effector neurons.
C) Has information flow in only one direction: away from an integrating center.
D) Includes sensory information, an integrating center, and effectors.

2) Most of the neurons in the human brain are _____.
A) Sensory neurons.
B) Motor neurons.
C) Interneurons.
D) Peripheral neurons.

3) The motor (somatic nervous) system can alter the activities of its targets, the skeletal muscle fibers, because _____.
A) It is electrically coupled by gap junctions to the muscles.
B) Its signals bind to receptor proteins on the muscles.
C) Its signals reach the muscles via the blood.
D) It is connected to the internal neural network of the muscles.

4) The point of connection between two communicating neurons is called the _____.
A) Axon hillock.
B) Dendrite.
C) Synapse.
D) Cell body.

5) In a simple synapse, neurotransmitter chemicals are released by _____.
A) The presynaptic membrane.
B) Axon hillocks.
C) Cell bodies.
D) Ducts on the smooth endoplasmic reticulum.

6) In a simple synapse, neurotransmitter chemicals are received by _____.
A) The postsynaptic membrane.
B) The presynaptic membrane.
C) Axon hillocks.
D) Cell bodies.

7) Although the membrane of a "resting" neuron is highly permeable to potassium ions, its membrane potential does not exactly match the equilibrium potential for potassium because the neuronal membrane is also _____.
A) Slightly permeable to sodium ions.
B) Fully permeable to calcium ions.
C) Impermeable to sodium ions.
D) Highly permeable to chloride ions.

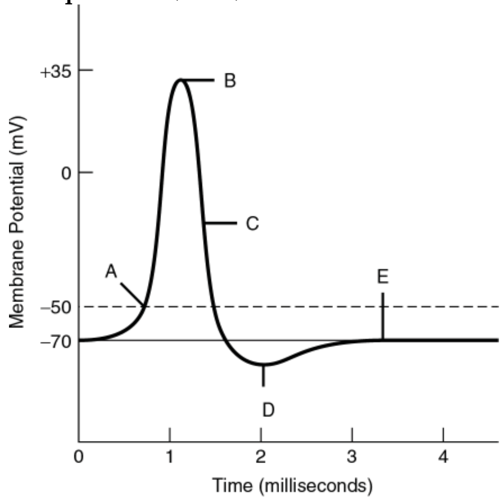
8) The operation of the sodium-potassium "pump" moves _____.
A) Sodium and potassium ions into the cell.
B) Sodium and potassium ions out of the cell.
C) Sodium ions into the cell and potassium ions out of the cell.
D) Sodium ions out of the cell and potassium ions into the cell.

9) In a resting potential, an example of a cation that is more abundant as a solute in the cytosol of a neuron than it is in the interstitial fluid outside the neuron is _____.
A) Cl^-.
B) Ca^{++}.
C) Na^+.
D) K^+.

10) The membrane potential in which there is no net movement of the ion across the membrane is called the _____.
A) Graded potential.
B) Threshold potential.
C) Equilibrium potential.
D) Action potential.

11) Two fundamental concepts about the ion channels of a "resting" neuron are that the channels _____.
A) Are always open, but the concentration gradients of ions frequently change.
B) Are always closed, but ions move closer to the channels during excitation.
C) Open and close depending on stimuli, and are specific as to which ion can traverse them.
D) Open in response to stimuli, and then close simultaneously, in unison.

Refer to the following graph of an action potential to answer the questions (12-16) below.



12) The membrane potential is closest to the equilibrium potential for potassium at label _____.
A) A.
B) B.
C) C.
D) D.

13) The membrane's permeability to sodium ions is at its maximum at label _____.
A) A.
B) B.
C) C.
D) D.

14) The minimum graded depolarization needed to operate the voltage -gated sodium and potassium channels is indicated by the label _____.
A) A.
B) B.
C) D.
D) E.

15) The cell is not hyperpolarized; however, repolarization is in progress, as the sodium channels are closing or closed, and many potassium channels have opened at label _____.
A) B.
B) C.
C) D.
D) E.

16) The neuronal membrane is at its resting potential at label _____.
A) A.
B) B.
C) D.
D) E.

17) If you experimentally increase the concentration of Na^+ outside a cell while maintaining other ion concentrations as they were, what would happen to the cell's membrane potential?
A) The membrane potential would decrease.
B) The membrane potential would increase.
C) The membrane potential would be unaffected.
D) The answer depends on the thermodynamic potential.

18) The concentrations of ions are very different inside and outside a nerve cell due to _____.
A) Osmosis.
B) Diffusion.
C) Sodium-potassium pumps.
D) Symports and antiports.

19) Which of the following ions is most likely to cross the plasma membrane of a resting neuron?
A) K^+.
B) Na^+.
C) Ca^{2+}.
D) Cl-.

20) The Nernst equation specifies the equilibrium potential for a particular ion. This equilibrium potential is a function of _____.
A) Hydrostatic pressure.
B) Ion concentration gradient.
C) Osmotic gradient.
D) Temperature (thermal) gradient.

21) For a neuron with an initial membrane potential at -70 mV, an increase in the movement of potassium ions out of that neuron's cytoplasm would result in the _____.
A) Depolarization of the neuron.
B) Hyperpolarization of the neuron.
C) Replacement of potassium ions with sodium ions.
D) Replacement of potassium ions with calcium ions.

22) Opening all of the sodium channels on an otherwise typical neuron, with all other ion channels closed (which is an admittedly artificial setting), should move its membrane potential to _____.
A) -90 mV.
B) 0 mV.
C) +30 mV.
D) +62 mV.

23) A graded hyperpolarization of a membrane can be induced by _____.
A) Increasing its membrane's permeability to Na^+.
B) Decreasing its membrane's permeability to Cl^-.
C) Increasing its membrane's permeability to Ca^{++}.
D) Increasing its membrane's permeability to K^+.

24) Self-propagation and refractory periods (states) are typical of _____.
A) Action potentials.
B) Graded hyperpolarizations.
C) Excitatory postsynaptic potentials.
D) Threshold potentials.

25) The "threshold" potential of a membrane is the _____.
A) Lowest frequency of action potentials a neuron can produce.
B) Minimum hyperpolarization needed to prevent the occurrence of action potentials.
C) Minimum depolarization needed to operate the voltage-gated sodium and potassium channels.
D) Peak amount of depolarization seen in action potential.

26) Action potentials move along axons _____.
A) More slowly in axons of large than in small diameter.
B) By activating the sodium-potassium "pump" at each point along the axonal membrane.
C) More rapidly in myelinated than in non-myelinated axons.
D) By reversing the concentration gradients for sodium and potassium ions.

27) A toxin that binds specifically to voltage-gated sodium channels in axons would be expected to _____.
A) Prevent the hyperpolarization phase of the action potential.
B) Prevent the depolarization phase of the action potential.
C) Prevent graded potentials.
D) Increase the release of neurotransmitter molecules.

28) After the depolarization phase of an action potential, the resting potential is restored by _____.
A) The opening of voltage-gated potassium channels and the closing of sodium channels.
B) A decrease in the membrane's permeability to potassium and chloride ions.
C) A brief inhibition of the sodium-potassium pump.
D) The opening of more voltage-gated sodium channels.

29) The "undershoot" phase of after-hyperpolarization is due to _____.
A) Slow opening of voltage-gated sodium channels.
B) Sustained opening of voltage-gated potassium channels.
C) Rapid opening of voltage-gated calcium channels.
D) Slow restorative actions of the sodium-potassium ATPase.

30) The fastest possible conduction velocity of action potentials is observed in _____.
A) Thin, non-myelinated neurons.
B) Thin, myelinated neurons.
C) Thick, non-myelinated neurons.
D) Thick, myelinated neurons.

31) Action potentials are normally carried in only one direction: from the axon hillock toward the axon terminals. If you experimentally depolarize the middle of the axon to threshold, using an electronic probe, then _____.
A) No action potential will be initiated.
B) An action potential will be initiated and proceed only in the normal direction toward the axon terminal.
C) An action potential will be initiated and proceed only back toward the axon hillock.
D) Two action potentials will be initiated, one going toward the axon terminal and one going back toward the hillock.

32) Why are action potentials usually conducted in one direction?
A) The nodes of Ranvier conduct potentials in one direction.
B) The brief refractory period prevents reopening of voltage-gated Na^+ channels.
C) The axon hillock has a higher membrane potential than the terminals of the axon.
D) Voltage-gated channels for both Na^+ and K^+ open in only one direction.

33) If you experimentally increase the concentration of K^+ inside a cell while maintaining other ion concentrations as they were, what would happen to the cell's membrane potential?
A) The membrane potential would become more negative.
B) The membrane potential would become less negative.
C) The membrane potential would remain the same.
D) None of them.

34) Which of the following statements about action potentials is correct?
A) Action potentials for a given neuron vary in magnitude.
B) Action potentials for a given neuron vary in duration.
C) Action potentials are propagated down the length of the axon.
D) Movement of ions during the action potential occurs mostly through the sodium pump.

35) Why do Na^+ ions enter the cell when voltage-gated Na^+ channels are opened in neurons?
A) Because the Na^+ concentration is much lower outside the cell than it is inside.
B) Because the Na^+ ions are actively transported by the sodium-potassium pump into the cell.
C) Because the Na^+ concentration is much higher outside the cell than it is inside, and the Na^+ ions are attracted to the negatively charged interior.
D) Because the Na^+ concentration is much higher outside the cell than it is inside, and the Na^+ ions are actively transported by the sodium-potassium pump into the cell.

36) What would probably happen if a long neuron had one continuous myelin sheath down the length of the axon with no nodes of Ranvier?
A) The action potential would be propagated nearly instantaneously to the synapse.
B) There could be no action potential generated at the axon hillock.
C) The signal would fade because it is not renewed by the opening of more sodium channels.
D) None of them.

37) A neurophysiologist is investigating nerve reflexes in two different animals: a crab and a fish. Action potentials are found to pass more rapidly along the fish's neurons. What is the most likely explanation?
A) The fish's axons are smaller in diameter; small axons transmit action potentials faster than large axons do.
B) Unlike the crab, the fish's axons are wrapped in myelin.
C) There are more ion channels in the axons of the crab compared with fish axons.
D) Unlike the crab, the fish's axons are wrapped in myelin and the fish's axons are smaller in diameter; small axons transmit action potentials faster than large axons do.

38) Tetrodotoxin blocks voltage-gated sodium channels and ouabain blocks sodium-potassium pumps. If you added both tetrodotoxin and ouabain to a solution containing neural tissue, what responses would you expect?
A) Immediate loss of resting potential.
B) Immediate loss of action potential with gradual loss of resting potential.
C) Slow decrease of resting potential and action potential amplitudes.
D) No effect; the substances counteract each other.

39) Which of the following will increase the speed of an action potential moving down an axon?
I) Action potentials move faster in larger diameter axons.
II) Action potentials move faster in axons lacking potassium ion channels.
III) Action potentials move faster in myelinated axons.
A) Only I and II.
B) Only II and III.
C) Only I and III.
D) I, II, and III.

40) In multiple sclerosis the myelin sheaths around the axons of the brain and spinal cord are damaged and demyelination results. How does this disease manifest at the level of the action potential?
I) Action potentials move in the opposite direction on the axon.
II) Action potentials move more slowly along the axon.
III) No action potentials are transmitted.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

41) Neurotransmitters are released from axon terminals via _____.
A) Osmosis.
B) Active transport.
C) Diffusion.
D) Exocytosis.

42) Acetylcholine released into the junction between a motor neuron and a skeletal muscle binds to a sodium channel and opens it. This is an example of _____.
A) A voltage-gated potassium channel.
B) A ligand-gated sodium channel.
C) A second-messenger-gated sodium channel.
D) A chemical that inhibits action potentials.

43) An inhibitory postsynaptic potential (IPSP) occurs in a membrane made more permeable to _____.
A) Potassium ions.
B) Sodium ions.
C) ATP.
D) All neurotransmitter molecules.

44) The following steps refer to various stages in transmission at a chemical synapse.
1. Neurotransmitter binds with receptors associated with the postsynaptic membrane.
2. Calcium ions rush into neuron's cytoplasm.
3. An action potential depolarizes the membrane of the presynaptic axon terminal.
4. The ligand-gated ion channels open.
5. The synaptic vesicles release neurotransmitter into the synaptic cleft.
Which sequence of events is correct?
A) 1 → 2 → 3 → 4 → 5.
B) 2 → 3 → 5 → 4 → 1.
C) 3 → 2 → 5 → 1 → 4.
D) 4 → 3 → 1 → 2 → 5.

45) The activity of acetylcholine in a synapse is terminated by its _____.
A) Diffusion across the presynaptic membrane.
B) Active transport across the postsynaptic membrane.
C) Diffusion across the postsynaptic membrane.
D) Degradation on the postsynaptic membrane.

46) Ionotropic receptors found at synapses are operated via _____.
A) Ligand-gated ion channels.
B) Electrical synapses.
C) Inhibitory, but not excitatory, synapses.
D) Excitatory, but not inhibitory, synapses.

47) An example of ligand-gated ion channels is _____.
A) The spreading of action potentials in the heart.
B) Acetylcholine receptors at the neuromuscular junction.
C) cAMP-dependent protein kinases.
D) Action potentials on the axon.

48) Neurotransmitters categorized as inhibitory are expected to _____.
A) Act independently of their receptor proteins.
B) Close potassium channels.
C) Open sodium channels.
D) Hyperpolarize the membrane.

49) Excitatory postsynaptic potentials (EPSPs) produced nearly simultaneously by different synapses on the same postsynaptic neuron can also add together, creating _____.
A) A temporal summation.
B) A spatial summation.
C) A tetanus.
D) The refractory state.

50) When two excitatory postsynaptic potentials (EPSPs) occur at a single synapse so rapidly in succession that the postsynaptic neuron's membrane potential has not returned to the resting potential before the second EPSP arrives, the EPSPs add together producing _____
A) Temporal summation.
B) Spatial summation.
C) Tetanus.
D) The refractory state.

51) Receptors for neurotransmitters are of primary functional importance in assuring one-way synaptic transmission because they are mostly found on the _____.
A) Axonal membrane.
B) Axon hillock.
C) Postsynaptic membrane.
D) Presynaptic membrane.

52) Neurotransmitters affect postsynaptic cells by _____.
I) Initiating signal transduction pathways in the cells.
II) Causing molecular changes in the cells.
III) Affecting ion-channel proteins.
IV) Altering the permeability of the cells.
A) I and III.
B) II and IV.
C) III and IV.
D) I, II, III, and IV.

53) The amino acid that operates at most inhibitory synapses in the brain is _____.
A) Acetylcholine.
B) Endorphin.
C) Nitric oxide.
D) Gamma-aminobutyric acid (GABA).

54) The botulinum toxin, which causes botulism, reduces the synaptic release of _____.
A) Acetylcholine.
B) Endorphin.
C) Nitric oxide.
D) Gamma-aminobutyric acid (GABA).

55) The heart rate decreases in response to the arrival of _____.
A) Acetylcholine.
B) Endorphin.
C) Nitric oxide.
D) Gamma-aminobutyric acid (GABA).

56) A chemical that affects neuronal function but is not stored in presynaptic vesicles is _____.
A) Acetylcholine.
B) Epinephrine.
C) Nitric oxide.
D) Gamma-aminobutyric acid (GABA).

57) Which of the following is a *direct* result of depolarizing the presynaptic membrane of an axon terminal?
A) Voltage-gated calcium channels in the membrane open.
B) Synaptic vesicles fuse with the membrane.
C) The postsynaptic cell produces an action potential.
D) Ligand-gated channels open, allowing neurotransmitters to enter the synaptic cleft.

58) How could you increase the magnitude of inhibitory postsynaptic potentials (IPSPs) generated at a synapse?
A) Increase sodium-potassium pump activity.
B) Increase K^+ permeability.
C) Increase the influx of calcium.
D) All of the listed responses are correct.

59) What happens if twice as many inhibitory postsynaptic potentials (IPSPs) as excitatory postsynaptic potentials (EPSPs) arrive at a postsynaptic neuron in close proximity?
A) A stronger action potential results.
B) A weaker action potential results.
C) No action potential results.
D) None of them.

60) Motor neurons release the neurotransmitter acetylcholine (ACh) and acetylcholinesterase degrades ACh in the synapse. If a neurophysiologist applies onchidal (a naturally occurring acetylcholinesterase inhibitor produced by the mollusc *Onchidella binneyi*) to a synapse, what would you expect to happen?
A) Paralysis of muscle tissue.
B) Convulsions due to constant muscle stimulation.
C) Decrease in the frequency of action potentials.
D) No effect.

61) The myelin sheath plays an important role in neuron structure and function. However, when the myelin sheath is missing or not fully intact, there are consequences. There are many conditions that cause demyelination of neurons, some are autoimmune disorders, such as multiple schlerosis, and others are hereditary. The symptoms of these conditions vary, but often include speech impairment and difficulty coordinating movement. Which of the following correctly connects the symptoms of demyelination with the process of nerve impulse transmission?
A) Demyelination prevents the formation of an action potential in sensory neurons that transmit signals from the environment to the central nervous system.
B) Demyelination slows nerve impulse transmission.
C) Demyelination prevents the uptake of neurotransmitters needed to propagate a message to the next neuron.
D) Demyelination targets the central nervous system.

62) What happens if a neuron is stimulated enough midway in an axon to trigger an action potential?
A) The nerve impulse would go both directions from the stimulus point, but only the axon end could transfer the message through neurotransmitters to another neuron.
B) Since neuron transmission is one-way, the nerve impulse would only be transmitted to the end of the axon and then through neurotransmitters to the next neuron.
C) The nerve impulse could not be transmitted because it must be initiated at the dendrite end of a neuron.
D) The nerve impulse would go both directions and the dendrite end would be stimulated to send a second message through this neuron.

63) *C.elegans* is a model organism and was the first eukaryotic organism to have its genome sequenced. The free-living nematode is often used in laboratories investigating reproduction, particularly egg-laying. There are 16 muscles, 2 types of neurons, and multiple receptors involved in the process of laying eggs in *C.elegans*, and there are mutations in all of those structures for the study of the process. One particular mutation that prevents the laying of eggs by the worm is rescued by the neurotransmitter, serotonin. That rescue suggests that this mutation is most likely in which of the following?

A) A post-synaptic neuron involved in egg-laying.
B) A pre-synaptic neuron involved in egg-laying.
C) A receptor for serotonin on cells needed for egg-laying.
D) One of the muscles needed for egg-laying.

CHAPTER 49:

NERVOUS SYSTEMS

1) An organism that lacks integration centers _____.
A) Cannot receive stimuli.
B) Will not have a nervous system.
C) Will not be able to interpret stimuli.
D) Can be expected to lack myelinated neurons.

2) In the human knee-jerk reflex of a seated individual, as the calf is raised from a vertical position to a horizontal position, the muscles of the quadriceps (on the front of the thighs) and the muscles of the hamstring (on the back side of the thighs) are _____.
A) Both excited and contracting.
B) Both inhibited and relaxed.
C) Excited and inhibited, respectively.
D) Inhibited and excited, respectively.

3) The stretch sensors of the sensory neurons in the human knee -jerk reflex are located in the _____.
A) Cartilage of the knee.
B) Quadriceps muscles on the front side of the thighs.
C) Hamstring muscles on the back side of the thighs.
D) Brain, the sensorimotor relay.

4) Choose the correct match of glial cell type and function.
A) Astrocytes - metabolize neurotransmitters and modulate synaptic effectiveness.
B) Oligodendrocytes - produce the myelin sheaths of myelinated neurons in the peripheral nervous system.
C) Radial glia - the source of immunoprotection against pathogens.
D) Schwann cells - provide nutritional support to non-myelinated neurons.

5) The cerebrospinal fluid is _____.
A) A filtrate of the blood.
B) A secretion of glial cells.
C) Cytosol secreted from ependymal cells.
D) Secreted by the hypothalamus.

6) The human knee-jerk reflex requires an intact _____.
A) Spinal cord.
B) Corpus callosum.
C) Cerebellum.
D) Medulla.

7) Myelinated neurons are especially abundant in the ___.
A) Gray matter of the brain and the white matter of the spinal cord.
B) White matter of the brain and the gray matter of the spinal cord.
C) Gray matter of the brain and the gray matter of the spinal cord.
D) White matter in the brain and the white matter in the spinal cord.

8) Cerebrospinal fluid can be described as which of the following?
I) Functioning in transport of nutrients and hormones through the brain.
II) A product of the filtration of blood in the brain.
III) Functioning to cushion the brain.
IV) Filling spaces between glial cells and neurons in the gray matter.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and III.
D) Only II, III, and IV.

9) The divisions of the nervous system that have antagonistic, or opposing, actions are _____.
A) Motor and sensory systems.
B) Sympathetic and parasympathetic systems.
C) Presynaptic and postsynaptic membranes.
D) Central nervous system and peripheral nervous system.

10) Preparation for the fight-or-flight response includes activation of the _____ nervous system.
A) Sympathetic.
B) Somatic.
C) Central.
D) Parasympathetic.

11) Exercise and emergency reactions include _____.
A) Increased activity in all parts of the peripheral nervous system.
B) Increased activity in the sympathetic and decreased activity in the parasympathetic divisions.
C) Decreased activity in the sympathetic and increased activity in the parasympathetic divisions.
D) Increased activity in the enteric nervous system.

12) Increased activity in the sympathetic nervous system leads to _____.
A) Decreased heart rate.
B) Increased secretion by the pancreas.
C) Increased contractions of the stomach.
D) Relaxation of the airways in the lungs.

13) The activation of the parasympathetic branch of the autonomic nervous system is associated with _____.
A) Resting and digesting.
B) Release of epinephrine into the blood.
C) Increased metabolic rate.
D) Intensive aerobic exercise.

14) In a cephalized invertebrate, the system that transmits "efferent" impulses from the anterior ganglion to distal segments is the _____.
A) Central nervous system.
B) Peripheral nervous system.
C) Autonomic nervous system.
D) Parasympathetic nervous system.

15) Imagine you are resting comfortably on a sofa after dinner. This could be described as a state with _____.
A) Increased activity in the sympathetic, parasympathetic, and enteric nervous systems.
B) Decreased activity in the sympathetic, parasympathetic, and enteric nervous systems.
C) Decreased activity in the sympathetic nervous system, and increased activity in the parasympathetic and enteric nervous systems.
D) Increased activity in the sympathetic nervous system, and decreased activity in the parasympathetic and enteric nervous systems.

16) If a doctor attempts to trigger the patellar tendon reflex and a lack of response occurs, what are potential regions where pathology might exist?
I) The brain.
II) The knee.
III) The spinal cord.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

17) Upon witnessing a robber hold up a convenience store at gunpoint, which of the following reactions would your nervous system initiate?
A) Increased heartbeat.
B) Constriction of airways.
C) Constriction of pupils.
D) Decreased heartbeat.

18) After eating a large meal, which nerves are most active in your digestive system?
I) Parasympathetic nerves
II) Somatic (motor) nerves
III) Sympathetic nerves
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

19) The central nervous system is lacking in animals that have _____.
A) A complete gut.
B) Bilateral symmetry.
C) Radial symmetry.
D) A closed circulatory system.

20) Cephalization, the clustering of neurons and interneurons in the anterior part of the animal, is apparent in _____.
A) Cnidarians.
B) *Planaria.*
C) Sea stars.
D) Invertebrate animals with radial symmetry.

21) Which of the following structures or regions is correctly paired with its function?
A) Limbic system - motor control of speech.
B) Medulla oblongata - emotional memory.
C) Cerebellum - homeostatic control.
D) Corpus callosum - communication between the left and right cerebral cortices.

22) Calculation, contemplation, and cognition are human activities associated with increased activity in the _____.
A) Hypothalamus.
B) Cerebrum.
C) Cerebellum.
D) Spinal cord.

23) Central coordination of vertebrate biological rhythms in physiology and behavior reside in the _____.
A) Pituitary gland.
B) Hypothalamus.
C) Cerebrum.
D) Thalamus.

24) Biological rhythms in animals isolated from light and dark cues _____.
A) Continue to have cycles of exactly twenty-four hours' duration.
B) Continue to have cycles of approximately twenty-four hours duration; some more rapid, some slower.
C) Synchronize activity with whatever lighting cycle is imposed on them.
D) Cease having any rhythms.

25) Bottlenose dolphins breathe air but can sleep in the ocean because _____.
A) They sleep for only thirty minutes at a time, which is the maximum interval they can cease breathing.
B) They fill their swim bladder with air to keep their blowholes above the surface of the water while they sleep.
C) They move to shallow water to sleep, so they do not need to swim to keep their blowholes above the surface of the water.
D) They alternate which half of their brain is asleep and which half is awake.

26) The limbic system in the central nervous system sustains many vegetative functions in mammals and is closely associated with structures that process cues about _____.
A) Olfaction.
B) Vision.
C) Audition.
D) Mechanosensation.

27) Increases and decreases of the heart rate result from changes in the activity of the _____.
A) Medulla oblongata.
B) Thalamus.
C) Pituitary.
D) Cerebellum.

28) The unconscious control of respiration and circulation are associated with the _____.
A) Thalamus.
B) Cerebellum.
C) Medulla oblongata.
D) Cerebrum.

29) Which of the following structures are correctly paired?
A) Forebrain – cerebellum.
B) Midbrain – cerebrum.
C) Hindbrain – cerebellum.
D) Brainstem - anterior pituitary gland.

30) Hormones secreted by the posterior pituitary gland are made in the_____.
A) Cerebrum.
B) Cerebellum.
C) Thalamus.
D) Hypothalamus.

31) The coordination of groups of skeletal muscles is driven by activity in the_____.
A) Cerebrum.
B) Cerebellum.
C) Thalamus.
D) Medulla oblongata.

32) Body temperature is regulated by _____.
A) Cerebrum.
B) Cerebellum.
C) Thalamus.
D) Hypothalamus.

33) The regulatory centers for the respiratory and circulatory systems are found in the _____.
A) Cerebrum.
B) Cerebellum.
C) Thalamus.
D) Medulla oblongata.

34) Food and water appetites are under the regulatory influence of the _____.
A) Cerebrum.
B) Thalamus.
C) Hypothalamus.
D) Medulla oblongata.

35) The suprachiasmatic nuclei are found in the _____.
A) Hypothalamus.
B) Epithalamus.
C) Amygdala.
D) Broca's area.

36) Wakefulness is regulated by the reticular formation, which is present in the _____.
A) Basal nuclei.
B) Cerebral cortex.
C) Brainstem.
D) Limbic system.

37) If a patient has an injury in the brain stem, which of the following would be observed?
A) Auditory hallucinations.
B) Visual hallucinations.
C) An inability to regulate body temperature.
D) An inability to regulate heart function.

38) The telencephalon region of the developing brain of a mammal _____.
A) Divides further into the metencephalon and myelencephalon.
B) Develops from the midbrain.
C) Is the brain region most like that of ancestral vertebrates.
D) Gives rise to the cerebrum.

39) The motor cortex is part of the _____.
A) Cerebrum.
B) Cerebellum.
C) Spinal cord.
D) Medulla oblongata.

40) In mammals, advanced cognition is usually correlated with a large and very convoluted neocortex, but birds are capable of sophisticated cognition because they have _____.
A) A more advanced cerebellum.
B) A cerebellum with several flat layers.
C) A pallium with neurons clustered into nuclei.
D) Microvilli to increase the brain's surface area.

41) Wernicke's and Broca's regions of the brain affect _____.
A) Olfaction.
B) Vision.
C) Speech.
D) Hearing.

42) Which of the following shows a brain structure correctly paired with one of its primary functions?
A) Frontal lobe - decision making.
B) Occipital lobe - control of skeletal muscles.
C) Temporal lobe - visual processing.
D) Occipital lobe - speech production.

43) If you were writing an essay, the part(s) of your brain that would be actively involved in this task is/are the _____.
A) Frontal lobes.
B) Parietal lobe.
C) Broca's area.
D) Occipital lobe.

44) Wernicke's area _____.
A) Is active when speech is heard and comprehended.
B) Is active during the generation of speech.
C) Coordinates the response to olfactory sensation.
D) Is found on the left side of the brain.

45) When Phineas Gage had a metal rod driven into his frontal lobe, or when someone had a frontal lobotomy, they would _____.
A) Lose their sense of balance.
B) Lose all short-term memory.
C) Have greatly altered emotional responses.
D) Have greatly increased long-term memory.

46) Patients with damage to Wernicke's area have difficulty _____.
A) Generating speech.
B) Recognizing faces.
C) Understanding language.
D) Experiencing emotion.

47) After suffering a stroke, a patient can see objects anywhere in front of him, but pays attention only to objects in his right field of vision. When asked to describe these objects, he has difficulty judging their size and distance. What part of the brain was likely damaged by the stroke?
A) The left frontal lobe.
B) The right frontal lobe.
C) The left parietal lobe.
D) The right parietal lobe.

48) An injury to the occipital lobe will likely impair the function of the _____.
A) Primary visual cortex.
B) Thalamus.
C) Sense of taste.
D) Sense of touch.

49) Short-term and long-term memory are related but have important differences. Short-term memory _____.
A) Involves temporary links formed in the cerebral cortex while long-term memory involves permanent connections within the hippocampus.
B) And long-term memory store information in the cerebellum but use different neurotransmitters.
C) Is essential for acquiring and retaining long-term memories.
D) Is essential for acquiring new long-term memories but not for maintaining them.

50) One of the fundamental processes by which memories are stored and learning takes place _____.
A) Is related to changes in degree of myelination of axons.
B) Results in an increase in the diameter of axons.
C) Results in a shift from aerobic to anaerobic respiration in neurons.
D) Involves two types of glutamate receptors.

51) The point of connection between two communicating neurons is called the _____.
A) Axon hillock.
B) Dendrite.
C) Synapse.
D) Glia.

52) Short-term memory information processing usually causes changes in the _____.
A) Brainstem.
B) Medulla.
C) Hypothalamus.
D) Hippocampus.

53) Forming new long-term memories is strikingly disrupted after damage to the _____.
A) Thalamus.
B) Cerebral cortex.
C) Somatosensory cortex.
D) Primary motor cortex.

54) In a simple synapse, neurotransmitter chemicals are released by _____.
A) The dendritic membrane.
B) The presynaptic membrane.
C) Axon hillocks.
D) Cell bodies.

55) In a simple synapse, neurotransmitter chemicals are received by _____.
A) The postsynaptic membrane.
B) The presynaptic membrane.
C) Axon hillocks.
D) Cell bodies.

56) Our understanding of mental illness has been most advanced by discoveries involving the _____.
A) Degree of convolutions in the brain's surface.
B) Sequence of developmental specialization.
C) Chemicals involved in brain communications.
D) Nature of the blood-brain barrier.

57) Bipolar disorder differs from schizophrenia in that __.
A) Schizophrenia typically involves hallucinations.
B) Schizophrenia typically involves manic and depressive states.
C) Bipolar disorder involves both genes and environment.
D) Bipolar disorder increases biogenic amines.

58) One of the complications of Alzheimer's disease is an interference with learning and memory. This disease would most likely involve _____.
A) Changes in the concentration of ions in the extracellular fluid surrounding neurons.
B) Changes in myelination of axons.
C) Molecular and structural changes at synapses.
D) Structural changes to ion channels in axons.

59) Stem cell transplants may someday be used to treat Parkinson's disease. Researchers are hopeful that these cells would alleviate the symptoms of Parkinson's disease by _____.
A) Preventing temporal lobe seizures.
B) Repairing sites of traumatic brain injury.
C) Replenishing missing ion channels.
D) Secreting the neurotransmitter dopamine.

60) The brain reward system _____.
A) Represents an emergent brain property that has arisen independent of natural selection.
B) Is a reflex of the peripheral nervous primarily under autonomic control.
C) Is housed in the thalamus and primarily regulates the enteric division of the autonomic nervous system.
D) Utilizes the neurotransmitter dopamine and is affected by drug addiction.

61) *C.elegans* is a model organism and was the first eukaryotic organism to have its genome sequenced. The free-living nematode is often used in laboratories investigating nervous system development with all 302 of its neurons and their effectors categorized. Interesting control mechanisms have been investigated with reproduction, particularly with egg-laying in the nematode. There are 16 muscles, 2 types of neurons, and multiple receptors involved in the process of laying eggs in *C.elegans*, and there are mutations in all of those structures for the study of the process. One particular mutation that prevents the laying of eggs by the worm is rescued by the neurotransmitter, serotonin. That rescue suggests that the mutation is most likely in which of the following?
A) A post-synaptic neuron involved in egg-laying.
B) A pre-synaptic neuron involved in egg-laying.
C) A receptor for serotonin on cells needed for egg-laying.
D) One of the muscles needed for egg-laying.

CHAPTER 50:

SENSORY AND MOTOR MECHANISMS

1) When the mammalian brain compares the actual temperature of the body to the preferred temperature of the body, which general component is being used?
A) Sensor.
B) Effector.
C) Integrator.
D) None of them.

2) A behavioral physiologist is studying the homeostatic control of blood pH. In a trial, a lizard runs on a treadmill for a set amount of time and the blood pH is measured. The blood pH drops as carbon dioxide is released into the bloodstream. Which component of the homeostatic feedback system is responsible for deciding if the blood pH is far enough from normal that a response is necessary?
A) Effector.
B) Sensor.
C) Integrator.
D) Assimilator.

3) The eleven pairs of appendages projecting from the rostral area of star-nosed moles are _____.
A) Chemosensory structures.
B) Tactile structures.
C) Olfactory structures.
D) Gustatory structures.

4) The correct sequence of sensory processing is _____.
A) Sensory adaptation → stimulus reception → sensory transduction → sensory perception.
B) Stimulus reception → sensory transduction → sensory perception → sensory adaptation.
C) Sensory perception → stimulus reception → sensory transduction → sensory adaptation.
D) Stimulus reception → sensory perception → sensory adaptation → sensory transduction.

5) Artificial electrical stimulation of human's capsaicin-sensitive neurons would likely produce the sensation of _____.
A) Cold temperature.
B) Hot temperature.
C) Tactile stimulus.
D) Deep pressure.

6) Artificial electrical stimulation of human's menthol-sensitive neurons would likely produce the sensation of _____.
A) Cold temperature.
B) Hot temperature.
C) Odor of pepper.
D) Deep pressure.

7) Stimuli alter the activity of excitable sensory cells and generate action potentials via _____.
A) Integration.
B) Transmission.
C) Transduction.
D) Amplification.

8) Immediately after putting on a shirt, your skin might feel itchy. However, this perception soon fades due to ___.
A) Sensory adaptation.
B) Accommodation.
C) Reduced motor unit recruitment.
D) Reduced receptor amplification.

9) A given photon of light may trigger an action potential with thousands of times more energy because the signal strength is amplified by _____.
A) The receptor.
B) A G protein.
C) A signal transduction pathway.
D) Triggering several receptors at once.

10) Although some sharks close their eyes just before they bite, their bites are on target. Researchers have noted that sharks often misdirect their bites at metal objects and that they can find batteries buried under sand. This evidence suggests that sharks keep track of their prey during the split second before they bite in the same way that a _____.
A) Rattlesnake finds a mouse in its burrow.
B) Male silkworm moth locates a mate.
C) Bat finds moths in the dark.
D) Platypus locates its prey in a muddy river.

11) Which type of receptor would you expect to be most abundant in the antennae of a moth?
A) Thermoreceptors.
B) Mechanoreceptors.
C) Chemoreceptors.
D) Electroreceptors.

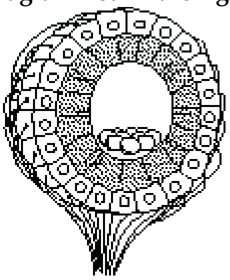
12) The middle ear converts _____.
A) Air pressure waves to fluid pressure waves.
B) Fluid pressure waves to air pressure waves.
C) Air pressure waves to nerve impulses.
D) Fluid pressure waves to nerve impulses.

13) Statocysts contain cells that are _____.
A) Mechanoreceptors used to detect orientation relative to gravity.
B) Chemoreceptors used in selecting migration routes.
C) Photoreceptors used in setting biological rhythms.
D) Thermoreceptors used in prey detection.

14) During an auditory transduction, ion flow varies across the _____.
A) Tectorial membrane.
B) Round-window membrane.
C) Hair cell membrane.
D) Basilar membrane.

15) Dizziness is a perceived sensation that can occur when _____.
A) The hair cells in the cochlea move more than their normal limits.
B) Moving fluid in the semicircular canals encounters a stationary cupula.
C) Rods and cones provide information that does not correspond with information received by cochlear hair cells.
D) The basilar membrane makes physical contact with the tectorial membrane.

16) The structure diagrammed in the figure is the _____.



A) Neuromast.
B) Statocyst.
C) Ommatidium.
D) Olfactory bulb.

17) A person able to hear only high-frequency sounds would probably have which of the following structural problems in the ear?
A) The tympanum is damaged because of chronic ear infections.
B) The basilar membrane is stiffened along its entire length.
C) The ear ossicles are abnormally thickened.
D) All of these problems could result in inability to detect low-frequency sound.

18) Partial or complete loss of hearing (deafness) can be caused by damage to the _____.
I) Axons of the neurons associated with each hair cell that carries information to the brain.
II) Hair cells (the sensory receptors) in the cochlea.
III) Tympanic membrane, or eardrum.
A) Only II.
B) Only III.
C) Only I and II.
D) I, II, and III.

19) The cochlea _____.
I) Amplifies sound vibrations.
II) Collects sound pressure waves.
III) Detects the frequency of sounds.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

20) Elephants hear sounds that are too low for humans to hear. This sensitivity is primarily due to the differences in the _____.
A) Arrangement and shape of the ossicles.
B) Flexibility of the basilar membrane in the cochlea.
C) Size and flexibility of the tympanic membrane (eardrum).
D) Size and shape of the outer ear.

21) Hair cells in the vertebrate ear are responsible for transducing sound pressure waves. Ion channels in the hair cell membrane open when _____.
A) A chemical ligand binds to the ion channel.
B) Light is absorbed by a molecule in the membrane.
C) The cell membrane reaches a threshold voltage.
D) The membrane is distorted mechanically.

22) If you experimentally reduce the concentration of K^+ in the extracellular fluid surrounding hair cells in the inner ear, the result would be like which of the following?
A) Decreasing the volume of sound reaching the hair cells.
B) Increasing the volume of sound reaching the hair cells.
C) Decreasing the frequency of sound reaching the hair cells.
D) Increasing the frequency of sound reaching the hair cells.

23) It can be very difficult to select an angle for sneaking up to a grasshopper to catch it because grasshoppers have _____.
A) Excellent hearing for detecting predators.
B) Compound eyes with multiple ommatidia.
C) Eyes with multiple fovea.
D) A camera-like eye with multiple fovea.

24) Compared to viewing a distant object, a human viewing an object held within five centimeters of the eye requires a lens that _____.
A) Has been flattened, as a result of contraction of the ciliary muscles.
B) Has been made more spherical, as a result of contraction of the ciliary muscles.
C) Has been flattened, as a result of relaxation of the ciliary muscles.
D) Has been made more spherical, as a result of relaxation of the ciliary muscles.

25) Sensory transduction of light in the vertebrate retina is accomplished by _____.
A) Ganglion cells.
B) Amacrine cells.
C) Bipolar cells.
D) Rods and cones.

26) Lateral inhibition via amacrine cells in the mammalian retina _____.
A) Underlies habituation of vision.
B) Enhances visual contrast.
C) Prevents bleaching in bright light.
D) Recycles neurotransmitter molecules.

27) The blind spot in the human retina is the location that has the collected axons of _____.
A) Ganglion cells.
B) Bipolar cells.
C) Primary visual cortex.
D) Lateral geniculate nuclei.

28) Corneal surgery is now routinely performed to change the shape of the cornea and improve vision. This surgery is beneficial because it _____.
A) Improves the circulation of nutrients to the eye.
B) Improves the focusing of light onto the retina.
C) Decreases the amount of light entering the eye.
D) Increases the sensitivity of the photoreceptors.

29) Rods exposed to light will _____.
A) Depolarize due to the opening of sodium channels.
B) Hyperpolarize due to the closing of sodium channels.
C) Depolarize due to the opening of potassium channels.
D) Hyperpolarize due to the closing of potassium channels.

30) What structures would neurobiologists look for if they are interested in determining if an animal can see in color?
A) Opsins.
B) Electroreceptors.
C) Pupil.
D) Lens.

31) How could you genetically modify an animal so that it would distinguish more shades of green?
A) Induce genes to produce a greater number of cone cells in the fovea.
B) Introduce genes for different opsins that respond in the green region of the spectrum.
C) Introduce genes to produce green fluid in the eyeball, because green fluids will not absorb green light.
D) Induce increased production of cGMP to increase opening of cGMP-gated sodium channels.

32) Which sensory distinction is NOT encoded by a difference in neuron identity?
A) Red and green.
B) Loud and faint.
C) Salty and sweet.
D) Spicy and cool.

33) Tastes and smells are distinct kinds of environmental information in that _____.
A) Neural projections from taste receptors reach different parts of the brain than the neural projections from olfactory receptors.
B) The single area of the cerebral cortex that receives smell and taste signals can distinguish tastes and smells by the pattern of action potentials received.
C) Tastant molecules are airborne, whereas odorant molecules are dissolved in fluids.
D) Distinguishing tastant molecules requires learning, whereas smell discrimination is an innate process.

34) Most of the chemosensory neurons arising in the nasal cavity have axonal projections that terminate in the _____.
A) Gustatory complex.
B) Olfactory bulb.
C) Occipital lobe.
D) Posterior pituitary gland.

35) Umami perception would be stimulated by _____.
A) Chocolate milk.
B) A slice or roast beef.
C) Acidic orange juice.
D) Salt water.

36) The olfactory bulbs are located in the _____.
A) Nasal cavity.
B) Anterior pituitary gland.
C) Brain.
D) Brainstem.

37) Which of the following sensory receptors is correctly paired with its category?
A) Hair cell - mechanoreceptor.
B) Muscle spindle - electromagnetic receptor.
C) Taste receptor - mechanoreceptor.
D) Rod - chemoreceptor.

38) The umami receptor in the sense of taste detects _____.
A) Glucose.
B) Potassium ions.
C) Hydrogen ions.
D) Monosodium glutamate.

39) Experiments with genetically altered mice showed that the mice would consume abnormally high amounts of bitter-tasting compounds in water after their _____.
A) Hormone receptors for digestive hormones were reduced or eliminated, showing that bitter tastes are reinforced by digestive responses.
B) Salt-taste cells were altered to express receptors for bitter tastants, suggesting that animals have unregulated salt appetites.
C) Visual sense was reduced or eliminated, suggesting that mice learn visual cues about bitter tastes.
D) Sweet-taste cells were altered to express receptors for bitter tastants, suggesting that the sensation of taste depends only on which taste cell is stimulated.

40) Two students studying physiology taste a known "bitter" substance, and both report sensing bitterness. They then sample another substance. Student A reports sensing both a bitter taste and a salty taste, but student B reports only a salty taste. What is the most logical explanation?
A) Student A had an allergic reaction to the food, causing him to perceive the food as being bitter.
B) Student A has normal "bitter" taste buds; student B has defective "bitter" taste buds that result in lower sensitivity to bitterness.
C) Student A has a protein receptor capable of detecting a bitter molecule found in that substance, whereas student B lacks that particular protein receptor.
D) Student A has normal saliva, whereas student B's saliva is more alkaline than normal.

41) Which of the following are present in high densities in both smooth and skeletal muscle cells?
I) Cilia.
II) Mitochondria.
III) Nuclei.
IV) Endoplasmic reticulum.
A) Only I and II.
B) Only II and IV.
C) Only III and IV.
D) Only I, II, and III.

42) The contraction of skeletal muscles is based on _____.
A) Myosin filaments coiling up to become shorter.
B) Actin and myosin filaments both coiling up to become shorter.
C) Actin cross-bridges binding to myosin and transitioning from a high -energy to a low-energy state.
D) Myosin cross-bridges binding to actin and transitioning from a high-energy to a low-energy state.

43) Compared to oxidative skeletal muscle fibers, those classified as glycolytic typically have _____.
A) A higher concentration of myoglobin.
B) A higher density of mitochondria.
C) A smaller diameter.
D) Less resistance to fatigue.

44) Myasthenia gravis is a form of muscle paralysis in which _____.
A) Motor neurons lose their myelination and the ability to rapidly fire action potentials.
B) Acetylcholine receptors are destroyed by an overactive immune system.
C) ATP production becomes uncoupled from mitochondrial electron transport.
D) Troponin molecules become unable to bind calcium ions.

45) A skeletal muscle deprived of adequate ATP supplies will _____.
A) Immediately relax.
B) Enter a state where actin and myosin are unable to separate.
C) Fire many more action potentials than usual and enter a state of "rigor".
D) Sequester all free calcium ions into the sarcoplasmic reticulum.

46) Most of the ATP supplies for a skeletal muscle undergoing one hour of sustained exercise come from _____.
A) Creatine phosphate.
B) Glycolysis.
C) Substrate phosphorylation.
D) Oxidative phosphorylation.

47) The "motor unit" in vertebrate skeletal muscle refers to _____.
A) One actin binding site and its myosin partner.
B) One sarcomere and all of its actin and myosin filaments.
C) One myofibril and all of its sarcomeres.
D) One motor neuron and all of the muscle fibers on which it has synapses.

48) The muscles of a recently deceased human can remain in a contracted state, termed *rigor mortis*, for several hours, due to the lack of _____.
A) ATP needed to break actin-myosin bonds.
B) Calcium ions needed to bind to troponin.
C) Oxygen supplies needed for myoglobin.
D) Sodium ions needed to fire action potentials.

49) Which of the following is the correct sequence that describes the excitation and contraction of a skeletal muscle fiber?
1. Tropomyosin shifts and unblocks the cross-bridge binding sites.
2. Calcium is released and binds to the troponin complex.
3. Transverse tubules depolarize the sarcoplasmic reticulum.
4. The thin filaments are ratcheted across the thick filaments by the heads of the myosin molecules using energy from ATP.
5. An action potential in a motor neuron causes the axon to release acetylcholine, which depolarizes the muscle cell membrane.
A) 1 → 2 → 3 → 4 → 5.
B) 2 → 1 → 3 → 5 → 4.
C) 2 → 3 → 4 → 1 → 5.
D) 5 → 3 → 2 → 1 → 4.

50) Action potentials in the heart move from one contractile cell to the next via _____.
A) Chemical synapses using acetylcholine.
B) Chemical synapses using norepinephrine.
C) Electrical synapses using gap junctions.
D) Non-myelinated motor neurons.

51) What would happen to people exposed to a chemical warfare agent that blocked acetylcholine from binding to muscle receptors?
A) Action potentials would be continuously generated, causing convulsive muscle contractions.
B) Muscle contractions would be prevented, causing paralysis.
C) Muscle contractions could still occur, but relaxation of the muscle would be impaired.
D) Action potentials would be continuously generated, causing convulsive muscle contractions; muscle contractions would then be prevented, causing paralysis.

52) When an action potential from a motor neuron arrives at the neuromuscular junction (NMJ), a series of events occurs that leads to muscle contraction. Which of the following events will occur last (that is, after all of the others)?
A) Acetylcholine (ACh) release.
B) Conformational change in troponin.
C) Depolarization of the muscle cell.
D) Release of Ca^{2+} from the sarcoplasmic reticulum.

53) A patient is hospitalized with muscle spasms caused by failure of back muscles to relax after contraction. Which of the following would be most likely to help?
A) Inject calcium into the muscle cell, because it is not being released from the sarcoplasmic reticulum.
B) Induce tropomycin and troponin to bind to the myosin binding sites on actin.
C) Increase the amount of acetylcholine at the synapses between motor neurons and muscle cells.
D) Depolarize the motor neurons to send an action potential to the muscle cells.

Use the following information to answer the question(s) below.

"Marine cone snails from the genus *Conus* are estimated to consist of up to 700 species. These predatory molluscs have devised an efficient venom apparatus that allows them to successfully capture polychaete worms, other molluscs, or in some cases fish as their primary food sources. … conotoxins from Australian species of *Conus* … have the capacity to inhibit specifically the nicotinic acetylcholine receptors in higher animals." (B. G. Livett, K. R. Gayler, and Z. Khalil. 2004. Drugs from the sea: Conopeptides as potential therapeutics. *Current Medicinal Chemistry* 11:1715-23.)

54) Refer to the paragraph above on the venom of marine core snails. This particular conotoxin inhibits acetylcholine receptors that are located _____.
A) Along the motor neuron axon.
B) On motor neuron dendrites.
C) On the presynaptic membrane of the neuromuscular junction.
D) On the postsynaptic membrane, on the muscle cell.

55) Refer to the paragraph above on the venom of marine core snails. What is the adaptive value of this toxin?
I) It would cause muscle spasms in the prey.
II) It would result in paralysis of the skeletal muscle of the prey.
III) It would stimulate digestive tract smooth muscle to cause nausea and vomiting of the prey.
A) Only I.
B) Only II.
C) Only III.
D) Only I and II.

56) An endoskeleton is the primary body support for the _____.
A) Annelids, including earthworms.
B) Insects, including beetles.
C) Cartilaginous fishes, including sharks.
D) Bivalves, including clams.

57) A ball-and-socket joint connects _____.
A) The radius to the ulna.
B) The radius to the humerus.
C) The ulna to the humerus.
D) The humerus to the scapula.

58) Among these choices, the most energetically efficient locomotion per unit mass is likely _____.
A) Running by a 50-gram rodent.
B) Running by a 40-kilogram ungulate.
C) Flying by a 100-gram bird.
D) Swimming by a 100-kilogram tuna (bony fish).

59) The hydrostatic skeleton of the earthworm allows it to move around in its environment by _____.
A) Walking on its limbs.
B) Swimming with its setae.
C) Using peristaltic contractions of its circular and longitudinal muscles.
D) Alternating contractions and relaxations of its flagella.

60) Chitin is a major component of the _____.
A) Skeleton of mammals.
B) Hydrostatic skeletons of earthworms.
C) Exoskeleton of insects.
D) Body hairs of mammals.

CHAPTER 51:

ANIMAL BEHAVIOR

1) What type of signal is long-lasting and works at night?
A) Olfactory.
B) Visual.
C) Auditory.
D) Electrical.

2) What type of signal is brief and can work among obstructions at night?
A) Olfactory.
B) Visual.
C) Auditory.
D) Magnetic.

3) What type of signal is fast and requires daylight with no obstructions?
A) Olfactory.
B) Visual.
C) Auditory.
D) Tactile.

4) Circannual rhythms in birds are influenced by _____.
A) Periods of food availability.
B) Periods of daylight and darkness.
C) Magnetic fields.
D) Lunar cycles.

5) Upon returning to its hive, a European honeybee communicates to other worker bees the presence of a nearby food source it has discovered by _____.
A) Vibrating its wings at varying frequencies.
B) Performing a round dance.
C) Performing a waggle dance.
D) Visual cues.

6) Displays of nocturnal mammals are usually _____.
A) Visual and auditory.
B) Tactile and visual.
C) Olfactory and auditory.
D) Visual and olfactory.

7) A cage containing male mosquitoes has a small earphone placed on top, through which the sound of a female mosquito is played. All the males immediately fly to the earphone and go through all of the steps of copulation. What is the best explanation for this behavior?
A) Copulation is a fixed action pattern, and the female flight sound is a sign stimulus that initiates it.
B) The sound from the earphone irritates the male mosquitoes, causing them to attempt to sting it.
C) The reproductive drive is so strong that when males are deprived of females, they will attempt to mate with anything that has even the slightest female characteristic.
D) Through classical conditioning, the male mosquitoes have associated the inappropriate stimulus from the earphone with the normal response of copulation.

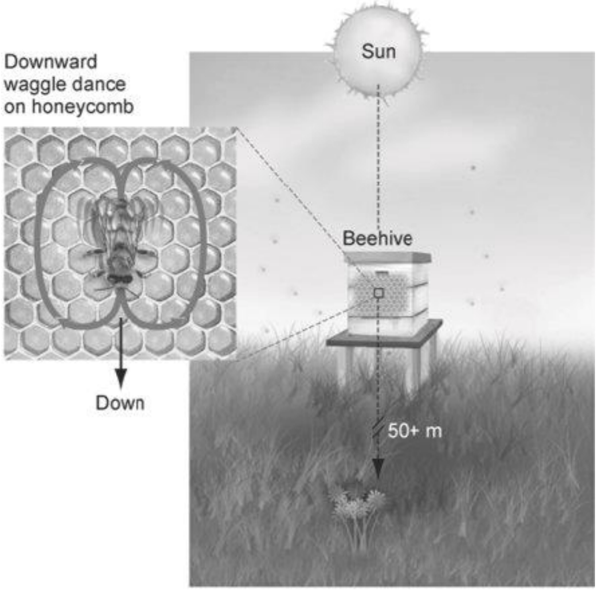
8) A stickleback fish will attack a fish model as long as the model has red coloring. What animal behavior idea is manifested by this observation?
A) Sign stimulus.
B) Cognition.
C) Imprinting.
D) Classical conditioning.

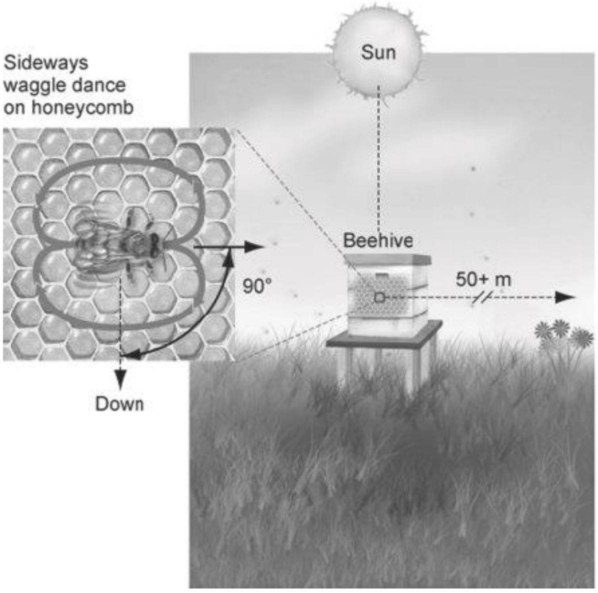
9) Which of the following experiments best addresses the hypothesis that moths stop flying in response to high-intensity bat sounds?
A) Isolate and characterize the neurons that control flight muscle.
B) Play prerecorded high-intensity bat sounds to flying moths.
C) Observe responses of moths to bats in nature.
D) Put bats and moths in an enclosure and make detailed observations of predator-prey interactions.

10) A lizard's bobbing dewlap (a colorful flap of skin hanging from an *Anolis* lizard's throat) is an example of a(n) _____.
A) Stimulus.
B) Reflex.
C) Signal.
D) Innate releasing mechanism.

11) What was the main reason the honeybees switched from the "round dance" to the "waggle dance"?
A) The waggle dance communicates the presence of nectar.
B) The preferred food source was farther away.
C) The round dance uses too much energy.
D) The round dance did not communicate the quality of the food.

12) From the figure below, what can we determine about the location of the food source?

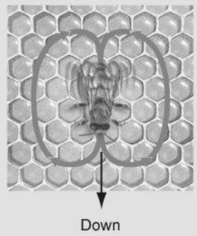




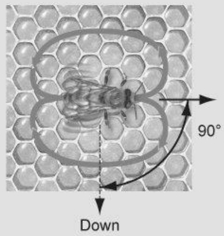
A) The waggle dance in the top figure indicates that the food is directly under the hive.
B) The waggle dance in the bottom figure indicates that the food is to the west of the hive.
C) The waggle dance in the top figure indicates that the food is close to the hive.
D) The waggle dance in the bottom figure indicates that the food is 90 degrees to the right of the Sun.

13) If the figure below shows the dances of bees in a hive at twelve noon on March 21 in the northern hemisphere, which dance is communicating that the food is to the south of the hive?

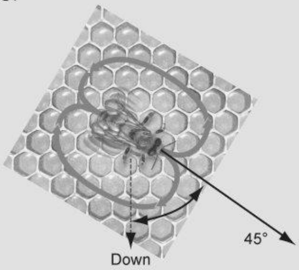
A.



B.



C.



A) Dance A.
B) Dance B.
C) Dance C.
D) It is not possible to tell if any of the dances indicate the food is to the south of the hive.

14) Scientists believe that the direction birds go when migrating is guided in part by _____.
I) The stars in the night sky.
II) The Sun during the day.
III) The magnetic field of the Earth.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

15) Which of the following examples describes a behavioral pattern that results from a proximate cause?
A) A cat kills a mouse to obtain nutrition.
B) A male sheep fights with another male because it helps to improve its social position.
C) A female bird lays its eggs because the amount of daylight is decreasing slightly each day.
D) A goose squats and freezes motionless to escape a predator.

16) The proximate causes of behavior are interactions with the environment, but behavior is ultimately shaped by _____.
A) Hormones.
B) Evolution.
C) Pheromones.
D) The nervous system.

17) During a field trip, an instructor touched a moth resting on a tree trunk. The moth raised its forewings to reveal large eyespots on its hind wings. The instructor asked why the moth lifted its wings. One student answered that sensory receptors had fired and triggered a neuronal reflex culminating in the contraction of certain muscles. A second student responded that the behavior might frighten predators. Which statement best describes these explanations?
A) The first explanation is correct, but the second is incorrect.
B) The first explanation refers to proximate causation, whereas the second refers to ultimate causation.
C) The first explanation is testable as a scientific hypothesis, whereas the second is not.
D) Both explanations are reasonable and simply represent a difference of opinion.

18) Which of the following is required for a behavioral trait to evolve by natural selection?
A) The behavior is determined entirely by genes.
B) The behavior is the same in all individuals in the population.
C) An individual's reproductive success depends in part on how the behavior is performed.
D) The behavior is not genetically inherited.

19) In testing a hypothesis that "territorial defense in European robins is a fixed action pattern that is released by the sight of orange feathers", researchers found that robins defended their territory by attacking anything that was of similar size and had an orange patch. What experiment would you perform next to determine that the color initiated the defense response?
A) Repeat the experiment using a blue patch instead of an orange patch.
B) Repeat the experiment by removing the patch completely.
C) Repeat the experiment by using a model of a robin that was twice the size of a normal robin but with a small orange patch.
D) Repeat the experiment by using a model of a robin that had an orange patch that was twice the size of a normal patch.

Listed below are several examples of types of animal behavior. Choose the letter of the correct term (A-E) that matches each example in the following questions (20-22).

A. Operant conditioning.
B. Agonistic behavior.
C. Innate behavior.
D. Imprinting.
E. Altruistic behavior.

20) Through trial and error, a rat learns to run a maze without mistakes to receive a food reward.
A) A.
B) B.
C) C.
D) D.

21) A human baby performs a sucking behavior perfectly when it is put in the presence of the nipple of its mother's breast.
A) A.
B) B.
C) C.
D) D.

22) A mother goat can recognize its own kid by smell.
A) A.
B) B.
C) C.
D) D.

23) Every morning at the same time, John went into the den to feed his new tropical fish. After a few weeks, he noticed that the fish swam to the top of the tank when he entered the room. This is an example of ____.
A) Cognition.
B) Imprinting.
C) Classical conditioning.
D) Operant conditioning.

24) Some dogs love attention, and Frodo the beagle learns that if he barks, he gets attention. Which of the following might you use to describe this behavior?
A) The dog is displaying an instinctive fixed action pattern.
B) The dog is trying to protect its territory.
C) The dog has been classically conditioned.
D) The dog's behavior is a result of operant conditioning.

25) Scientists have tried raising endangered whooping cranes in captivity by using sandhill cranes as foster parents. This strategy is no longer used because _____.
A) The fostered whooping cranes' critical period was variable such that different chicks imprinted on different "mothers".
B) Sandhill crane parents rejected their fostered whooping crane chicks soon after incubation.
C) None of the fostered whooping cranes formed a mating pair -bond with another whooping crane.
D) Sandhill crane parents did not properly incubate whooping crane eggs.

26) White-crowned sparrows can only learn the "crystallized" song for their species by _____.
A) Listening to adult sparrow songs during a sensitive period as a fledgling, followed by a practice period until the juvenile matches its melody to its memorized fledgling song.
B) Listening to the song of its own species during a critical period so that it will imprint to its own species song and not the songs of other songbird species.
C) Performing the crystallized song as adults when they become sexually mature, as the song is programmed into the innate behavior for the species.
D) Observing and practicing after receiving social confirmation from other adults at a critical period during their first episode of courtship behavior.

27) One way to understand how an early environment influences behavior in similar species is through the "cross-fostering" experimental technique. Suppose that the curly-whiskered mud rat differs from the bald mud rat in several ways, including being much more aggressive. How would you set up a cross-fostering experiment to determine if environment plays a role in the curly-whiskered mud rat's aggression?
A) You would cross curly-whiskered mud rats and bald mud rats and hand -rear the offspring to see if any grew up to be aggressive.
B) You would place newborn curly-whiskered mud rats with bald mud rat parents and place newborn bald mud rats with curly-whiskered mud rat parents. Finally, let some mud rats of both species be raised by their own species. Then you would compare the outcomes.
C) You would remove the offspring of curly-whiskered mud rats and bald mud rats from their parents, raise them in the same environment but without parents, and then compare the outcomes.
D) You would replace normal newborn mud rats with deformed newborn mud rats to see if it triggered an altruistic response.

28) Which of the following is true of innate behaviors? Innate behaviors _____.
A) Are only weakly influenced by genes.
B) Occur in invertebrates and some vertebrates but not mammals.
C) Are limited to invertebrate animals.
D) Are expressed in most individuals in a population.

29) A region of the canary forebrain shrinks during the nonbreeding season and enlarges when breeding season begins. This change is probably associated with the annual _____.
A) Addition of new syllables to a canary's song repertoire.
B) Crystallization of subsong into adult songs.
C) Renewal of mating and nest-building behaviors.
D) Elimination of the memorized template for songs sung the previous year.

30) Although many chimpanzees live in environments containing oil palm nuts, members of only a few populations use stones to crack open the nuts. The likely explanation is that _____.
A) The behavioral difference is caused by genetic differences between populations.
B) Members of different populations have different nutritional requirements.
C) The cultural tradition of using stones to crack nuts has arisen in only some populations.
D) Members of different populations differ in learning ability.

31) You observe a species of bird that, upon hatching, has contact with its parents only while being fed. You also never hear the parents sing during the feeding process. What would you propose about song learning in this species of bird?
A) Song learning in this species is most likely learned.
B) The period of imprinting is likely later in the bird's life.
C) The males will learn song when they congregate with other males of their species during the winter.
D) Song learning in this species is most likely innate.

32) Learning has the most influence on behavior when _____.
A) Making mistakes does not result in death.
B) Animals reproduce asexually.
C) Animals have enormous cognitive ability.
D) Making mistakes result in death.

33) You have captured a number of rats from a wild population and quickly surmise with tests that they are very good at avoiding food with poisons. What would best explain this observation?
A) Rats are probably just intelligent enough to avoid poison.
B) Rats may experience a large variety of toxins in their environment and learn to avoid them.
C) Rats are taught by their parents to test small bits of food first and then return later if the food seems safe.
D) Rats may be able to tolerate large amounts of poison.

34) You observe scrub jays hiding food and notice that one particular individual only pretends to hide food. What kind of experiment could you perform to test whether this behavior was random or in response to another signal?
A) Observe more of these behaviors in the wild and try to determine if the behavior is random.
B) Hypothesize a set of signals that could produce this behavior and try to match the behaviors with the signals.
C) Attempt to reproduce the behavior in captivity by using bird models and a computer simulation.
D) None of them.

35) You observe scrub jays hiding food and notice that one particular individual only pretends to hide food. Your experiments associate the presence of other individuals with the frequency of pretending to cache food. A colleague shows you animals of the same species that do not perform this pretend caching. How does this information affect your conclusions about this behavior?
A) It suggests that this behavior might be learned.
B) It prevents you from making conclusions.
C) It suggests that your experimental design is flawed.
D) It does not change your initial conclusions.

36) You discover a rare new bird species, but you are unable to observe its mating behavior. You see that the male is large and ornamental compared with the female. On this basis, you can probably conclude that the species is _____.
A) Polygamous.
B) Monogamous.
C) Polyandrous.
D) Agonistic.

37) Fred and Joe, two unrelated, mature male gorillas, encounter one another. Fred is courting a female. Fred grunts as Joe comes near. As Joe continues to advance, Fred begins drumming (pounding his chest) and bares his teeth. Joe then rolls on the ground on his back, gets up, and quickly leaves. This behavioral pattern is repeated several times during the mating season. Choose the most specific behavior described by this example.
A) Agonistic behavior.
B) Territorial behavior.
C) Learned behavior.
D) Fixed action pattern.

38) Female spotted sandpipers aggressively court males and, after mating and egg-laying, leave the clutch of young for the male to incubate. This sequence may be repeated several times with different males until no available males remain, forcing the female to incubate her last clutch. Which of the following terms best describes this behavior?
A) Monogamy.
B) Polygyny.
C) Polyandry.
D) Promiscuity.

39) Feeding behavior with a high energy intake -to-expenditure ratio is called _____.
A) Autotrophy.
B) Heterotrophy.
C) Search scavenging.
D) Optimal foraging.

40) Which of the following might affect the foraging behavior of an animal in the context of optimal foraging?
I) Risk of predation.
II) Prey size.
III) Prey defenses.
IV) Prey density.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and III.
D) I, II, III, and IV.

41) Which of the following is most likely associated with the evolution of mating systems?
A) Population density.
B) Territoriality.
C) Certainty of paternity.
D) Sexual dimorphism.

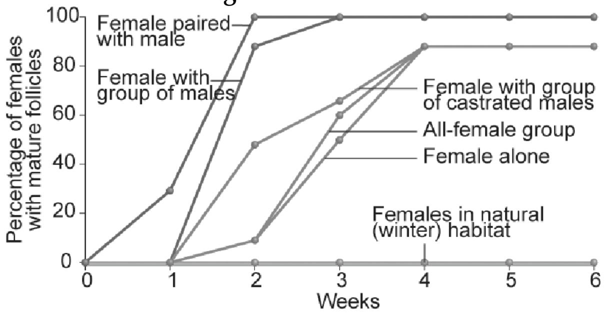
42) Females are typically larger and more ornamented than males where _____ occurs.
A) Monogamy.
B) Polyandry.
C) Polygamy.
D) Polygyny.

43) Which of the following statements is true about certainty of paternity?
A) Certainty of paternity is high in most species with internal fertilization because the acts of mating and birth are separated by time.
B) Certainty of paternity is low when males guard females they have mated.
C) Certainty of paternity is low when egg laying and mating occur together, as in external fertilization.
D) Paternal behavior exists because it has been reinforced over generations by natural selection.

44) Which of the following best describes "game theory" as it applies to animal behavior?
A) The fitness of a particular behavior is influenced by other behavioral phenotypes in a population.
B) The total of all of the behavioral displays, both male and female, is related to courtship.
C) The play behavior performed by juveniles allows them to perfect adult behaviors that are needed for survival, such as hunting, courtship, and so on.
D) The evolutionary "game" is played between predator and prey. The prey evolves a behavior in response to the nature of the predatory behavior.

45) The color of throats of males in a population of side-blotched lizards is determined by _____.
A) Ambient temperature: blue = cold; orange = normal; yellow = hot.
B) Stage of development/maturity.
C) Their receptiveness to mate.
D) The success of the mating behavior of each of the throat color phenotypes.

46) In the figure below, which of the following conclusions is most logical based on the data?



A) Females produce more eggs more quickly when exposed to breeding males.
B) Females produce eggs more quickly when exposed to many males than females paired with a male.
C) All non-isolated females do just as well as isolated females.
D) After four weeks together, females with males produce mature follicles to the same extent as females without males.

Listed below are several examples of types of animal behavior. Choose the letter of the correct term (A-E) that matches each example in the following questions (47).

A. Operant conditioning.
B. Agonistic behavior.
C. Innate behavior.
D. Imprinting.
E. Altruistic behavior.

47) Upon observing a golden eagle flying overhead, a sentry prairie dog gives a warning call to other foraging members of the prairie dog community.
A) B.
B) C.
C) D.
D) E.

48) The *fru* gene in fruit flies _____.
I) Controls sex-specific development in the fruit fly.
II) Is a master regulatory gene that directs expression of many other genes.
III) Can be genetically manipulated in females so that they will perform male sex behaviors.
IV) Programs males for appropriate courtship behaviors.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) I, II, III, and IV.

49) Pair-bonding in a population of prairie voles can be prevented by _____.
A) The ensuing confusion caused by introducing meadow voles
B) Administering a drug that inhibits the brain receptor for vasopressin in the central nervous system (CNS) of males
C) Dying the coat color from brown to blond in either male or female prairie voles
D) Allowing the population size to reach critically low levels

50) Which of the following statements about evolution of behavior is correct?
A) Natural selection will favor behavior that enhances survival and reproduction.
B) An animal may show behavior that minimizes reproductive fitness.
C) If a behavior is less than optimal, it will eventually become optimal through natural selection.
D) Innate behaviors cannot be altered by natural selection.

51) How do altruistic behaviors arise through natural selection?
A) By his/her actions, the altruist increases the likelihood that some of its genes will be passed on to the next generation.
B) The altruist is appreciated by other members of the population because its survivability has been enhanced by virtue of its risky behavior.
C) Animals that perform altruistic acts are allowed by their population to breed more, thereby passing on their behavior genes to future generations.
D) Altruistic behaviors lower stress in populations, which increases the survivability of all the members of the population.

52) Which of the following has a coefficient of relatedness of 0.25?
A) A father to his daughter.
B) An uncle to his nephew.
C) A brother to his brother.
D) A sister to her brother.

53) Animals that help other animals of the same species _.
A) Have excess energy reserves.
B) Are bigger and stronger than the other animals.
C) Are usually related to the other animals helped.
D) Are always male.

54) The presence of altruistic behavior is most likely due to kin selection, a theory maintaining that _____.
A) Genes enhance survival of copies of themselves by directing organisms to assist others who share those genes.
B) Companionship is advantageous to animals because in the future they can help each other.
C) Critical thinking abilities are normal traits for animals and they have arisen, like other traits, through natural selection.
D) Natural selection has generally favored the evolution of exaggerated aggressive and submissive behaviors to resolve conflict without grave harm to participants.

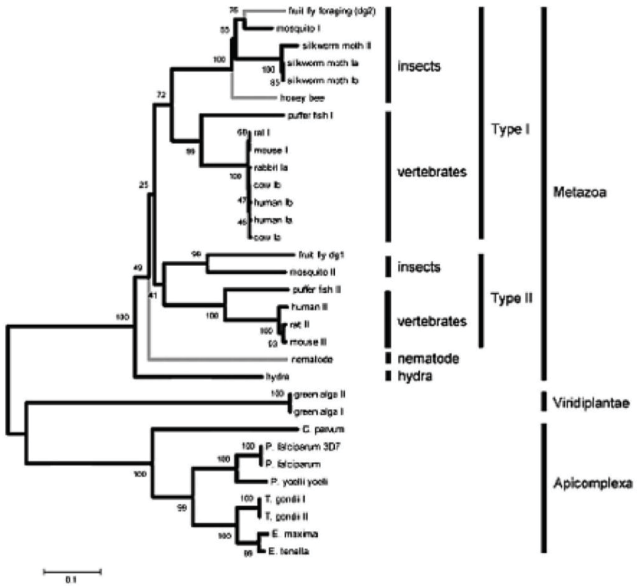
55) If a prairie dog had the opportunity to perform an altruistic act (that is, give an alarm call) to help its relatives, which combination of the following relatives would the prairie dog be most likely to help (base your answer solely on the genetic relationships)?
A) Two nieces, two cousins, and one half-brother.
B) Two half-sisters and two nieces.
C) One son, one niece, and one half-sister.
D) The prairie dog would be equally likely to act altruistically to each of the combinations described.

56) How would you classify the genetic basis for most behavioral traits in the animal kingdom?
A) One gene typically codes for one behavior.
B) One gene typically codes for many behaviors.
C) Many genes typically code for one behavior.
D) Behaviors are learned, not coded by genes.

Use the following information to answer the questions (57-58) below.

The following are an abstract and figure from a paper that explores the evolutionary relationship between a protein kinase and behavior (M. Fitzpatrick and M. Sokolowski. 2004. In search of food: Exploring the evolutionary link between cGMP-dependent protein kinase (PKG) and behavior. *Integrative and Comparative Biology* 44:28-36).

Abstract:
Despite an immense amount of variation in organisms throughout the animal kingdom, many of their genes show substantial conservation in DNA sequence and protein function. Here we explore the potential for a conserved evolutionary relationship between genes and their behavioral phenotypes. We investigate the evolutionary history of cGMP-dependent protein kinase (PKG) and its possible conserved function in food-related behaviors. First identified for its role in the foraging behavior of fruit flies, the PKG encoded by the *foraging* gene had since been associated with the maturation of behavior (from nurse to forager) in honey bees and the roaming and dwelling food -related locomotion in nematodes. These parallels encouraged us to construct protein phylogenies using 32 PKG sequences that include 19 species. Our analyses suggest five possible evolutionary histories that can explain the apparent conserved link between PKG and behavior in fruit flies, honey bees, and nematodes. Three of these raise the hypothesis that PKG influences the food -related behaviors of a wide variety of animals including vertebrates. Moreover it appears that the PKG gene was duplicated sometime between the evolution of nematodes and a common ancestor of vertebrates and insects whereby current evidence suggests only the *for*-like PKG might be associated with food-related behavior. Neighbor joining trees depicting the evolutionary relationships of 32 PKG kinase domain and C-terminal amino acid sequences spanning 19 species of protozoans and metazoans. Values at the nodes represent the results of 5000 bootstrap replications. Lineages with known behavioral links with PKG are indicated by gray branches.



57) Look at the evolutionary relationship of protein kinases in the figure above. Knowing that there is evidence that this particular protein kinase is linked to food-related behaviors in the animals studied, what conclusions can you draw?
A) There is enormous variation in amino acid sequences between taxa.
B) There is likely a conserved evolutionary relationship between these genes and their behavioral phenotype.
C) PKG influences food-related behaviors in a wide variety of animals, and there is enormous variation in amino acid sequences between taxa.
D) PKG influences food-related behaviors in a wide variety of animals, and there is likely a conserved evolutionary relationship between these genes and their behavioral phenotype.

58) Using the figure above and the accompanying paragraph, and knowing that the PKG encoded by the *foraging* gene has recently been associated with the maturation of out-of-nest behavior in honeybees, what would be a logical explanation for this relationship? As animals mature, they _____.
A) Require more food; therefore, PKG levels must increase.
B) Are more likely to forage; therefore, PKG levels must increase.
C) Respond to pheromones from the queen, which increases PKG levels.
D) Are able to fly, an activity that is connected to the increase in PKG levels.

59) What probably explains why coastal and inland garter snakes react differently to banana slug prey?
A) Ancestors of coastal snakes that could eat the abundant banana slugs had increased fitness. No such selection occurred inland, where banana slugs were absent.
B) Banana slugs are camouflaged, and inland snakes, which have poorer vision than coastal snakes, are less able to see them.
C) Garter snakes learns about prey from other garter snakes. Inland garter snakes have fewer types of prey because they are less social.
D) Inland banana slugs are distasteful, so inland snakes learn to avoid them. Coastal banana slugs are palatable to garter snakes.

60) Behaviors are diverse and important for survival and reproduction. Some behaviors are learned, such as the species-specific song of a yellow warbler which is different from the song of a blue-winged warbler. Other behaviors are innate, such as a female cat in heat urinating more often and in many places to attract a mate or honeybees do a "dance" that indicates the distance and direction of a food source when they return to their hive. Which of the following statements supports the idea that behaviors are important in survival and therefore affect natural selection?
A) Learned behaviors may not necessarily increase fitness. Baby warblers can learn the song of another species.
B) Innate behaviors are the result of selection for individual survival and reproductive success.
C) All behaviors are survival mechanisms that increase reproductive fitness by increasing mutation rates.
D) Both innate and learned behaviors are entirely based on genes inherited from parents.

61) The graph indicates that males leave the area where they were born while females stay in the area. What is the most likely reason for the evolution of this behavior?
A) Movement of males out of the territory reduces competition for food among the males.
B) The females are not as strong as the male ground squirrels and therefore stay closer to their birthplace.
C) Females reproduce by parthenogenesis, which means they produce offspring from unfertilized eggs so the males are not needed.
D) Within the ground squirrel population, males leave the area of their birth and are replaced by new males, thus maintaining genetic diversity in the population.

62) A male stickleback fish will attack other male sticklebacks that invade its nesting territory. It will only attack male fish, which display the red belly characteristic of the species. Why has natural selection favored this behavior?
A) The behavior reduces interspecific competition, which gives the male stickleback access to more food.
B) The behavior allows the male stickleback to attract females with its aggressive display.
C) The behavior allows the male to establish a defined space for breeding with female sticklebacks.
D) The behavior is a mechanism to reduce predation and resource competition.

CHAPTER 52:

AN INTRODUCTION TO ECOLOGY AND THE BIOSPHERE

1) Which level of ecological study focuses the most on abiotic factors?
A) Speciation ecology.
B) Population ecology.
C) Community ecology.
D) Ecosystem ecology.

2) What would happen to the seasons if the Earth were tilted 35 degrees off its orbital plane instead of the usual 23.5 degrees?
A) The seasons would disappear.
B) Winters and summers would be more severe.
C) Winters and summers would be less severe.
D) The seasons would be shorter.

3) Which of the following might be an investigation of microclimate?
A) The effect of ambient temperature on the onset of caribou migration.
B) The seasonal population fluctuation of nurse sharks in coral reef communities.
C) Competitive interactions between various species of songbirds during spring migration.
D) The effect of sunlight intensity on species composition in a decaying rat carcass.

4) Which of the following choices includes all of the others in creating global terrestrial climates?
A) Differential heating of Earth's surface.
B) Ocean currents.
C) Global wind patterns.
D) Earth's rotation on its axis.

5) Why is the climate drier on the leeward (downwind) side of mountain ranges that are subjected to prevailing winds?
A) Deserts create dry conditions on the leeward side of mountain ranges.
B) The sun illuminates the leeward side of mountain ranges at a more direct angle, converting to heat energy, which evaporates most of the water present.
C) Pushed by the prevailing winds on the windward side, air is forced to rise, cool, condense, and drop its precipitation, leaving drier air to descend the leeward side.
D) Air masses pushed by the prevailing winds are stopped by mountain ranges and the moisture is used up in the stagnant air masses on the leeward side.

6) What would be the effect on climate in the temperate latitudes if Earth were to slow its rate of rotation from a 24-hour period of rotation to a 48-hour period of rotation?
A) Seasons would be longer and more distinct (colder winters and warmer summers).
B) Large-scale weather events such as tornadoes and hurricanes would no longer be a part of regional climates.
C) Winter seasons in both the northern and southern hemispheres would have more abundant and frequent precipitation events.
D) There often would be a larger range between daytime high and nighttime low temperatures.

7) Subtropical plants are commonplace in Land's End, England, whose latitude is the equivalent of Labrador in coastal Canada, where the local flora is subarctic. Which statement best explains why this apparent anomaly exists between North America and Europe?
A) Labrador does not get enough rainfall to support the subtropical flora found in Land's End.
B) Warm ocean currents interact with England, whereas cold ocean currents interact with Labrador.
C) Rainfall fluctuates greatly in England; rainfall is consistently high in Labrador.
D) Labrador receives sunlight of lower duration and intensity than does Land's End.

8) In mountainous areas of western North America, north-facing slopes would be expected to _____.
A) Receive more sunlight than similar southern exposures.
B) Be warmer and drier than comparable southern exposed slopes.
C) Support biological communities similar to those found at lower elevations on similar south-facing slopes.
D) Support biological communities similar to those found at higher elevations on similar south-facing slopes.

9) Which of the following events might you predict to occur if the tilt of Earth's axis relative to its plane of orbit was increased to 33.5 degrees?
A) Summers and winters in the United States would likely become warmer and colder, respectively.
B) Seasonal variation at the equator might decrease.
C) Both northern and southern hemispheres would experience summer and winter at the same time.
D) Both poles would experience massive ice melts.

10) Imagine some cosmic catastrophe jolts Earth so that its axis is perpendicular to the orbital plane between Earth and the sun. The most obvious effect of this change would be _____.
A) The elimination of tides.
B) An increase in the length of a year.
C) A decrease in temperature at the equator.
D) The elimination of seasonal variation.

11) The main reason Polar Regions are cooler than the equator is that _____.
A) Sunlight strikes the poles at a lower angle.
B) The poles are farther from the sun.
C) The polar atmosphere is thinner and contains fewer greenhouse gases.
D) The poles are permanently tilted away from the sun.

12) The success of plants extending their range northward following glacial retreat is best determined by _____.
A) Whether there is simultaneous migration of herbivores.
B) Their tolerance to shade.
C) Their seed dispersal rate.
D) Their size.

13) As climate changes because of global warming, plant species' ranges in the northern hemisphere may move northward. The trees that are most likely to avoid extinction in such an environment are those that _____.
A) Have seeds that are easily dispersed by wind or animals.
B) Produce well-provisioned seeds.
C) Have seeds that become viable only after a forest fire.
D) Disperse many seeds in close proximity to the parent tree.

14) Generalized global air circulation and precipitation patterns are caused by _____.
A) Rising, warm, moist air masses that cool and release precipitation as they rise and then, at high altitude, cool and sink back to the surface as dry air masses after moving north or south of the tropics.
B) Air masses that are dried and heated over continental areas that rise, cool aloft, and descend over oceanic areas followed by a return flow of moist air from ocean to land, delivering high amounts of precipitation to coastal areas.
C) Polar, cool, moist high-pressure air masses from the poles that move along the surface, releasing precipitation along the way to the equator, where they are heated and dried.
D) The revolution of Earth around the sun.

15) Air masses formed over the Pacific Ocean are moved by prevailing westerlies where they encounter extensive north-south mountain ranges, such as the Sierra Nevada and the Cascades. Which statement best describes the outcome of this encounter between a landform and an air mass?
A) The cool, moist Pacific air heats up as it rises, releasing its precipitation as it passes the tops of the mountains. This warm, now dry air cools as it descends on the leeward side of the range.
B) The warm, moist Pacific air rises and cools, releasing precipitation as it moves up the windward side of the range. This cool, now dry air mass heats up as it descends on the leeward side of the range.
C) The cool, dry Pacific air heats up and picks up moisture from evaporation of the snowcapped peaks of the mountain range, releasing this moisture as precipitation when the air cools while descending on the leeward side of the range.
D) These air masses are blocked by the mountain ranges, producing high annual amounts of precipitation on the windward sides of these mountain ranges.

16) Coral reefs can be found on the southeast coast of the United States but not at similar latitudes on the southwest coast. Differences in which of the following most likely account for this?
A) Precipitation.
B) Day length.
C) Ocean currents.
D) Salinity.

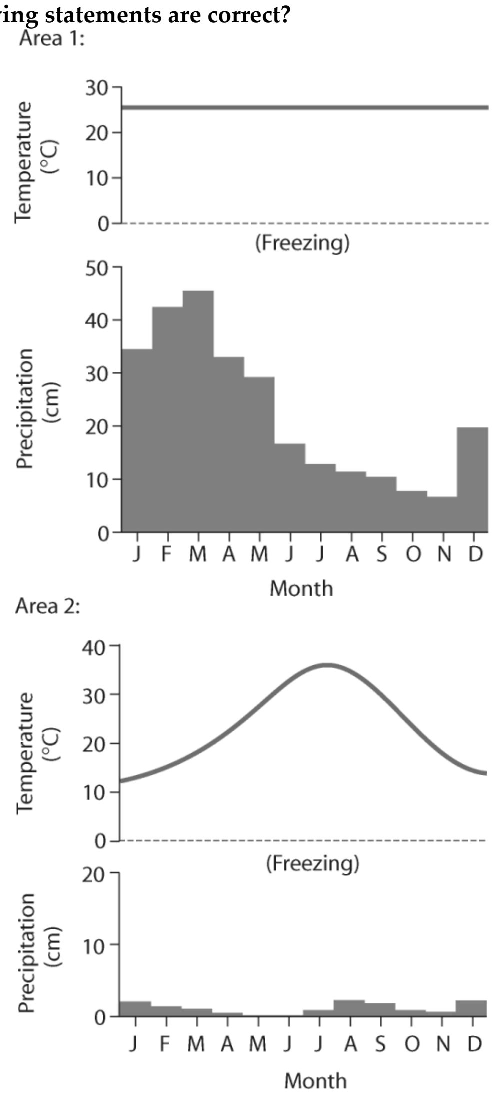
17) Which of the following investigations would shed the most light on the future distribution of organisms in temperate regions that are faced with climate change?
A) Remove, to the mineral soil, all of the organisms from an experimental plot and monitor the colonization of the area over time in terms of both species diversity and abundance.
B) Look at the climatic changes that occurred since the last Ice Age and how species redistributed as glaciers melted, then make predictions on future distribution in species based on past trends.
C) Compare and contrast the flora and fauna of warm/cold/dry/wet climates to shed light on how they evolved to be suited to their present -day environment.
D) Quantify the impact of man's activities on present-day populations of threatened and endangered species to assess the rate of extirpation and extinction.

18) Generally speaking, deserts are located in places where air masses are usually _____.
A) Tropical.
B) Humid.
C) Rising.
D) Descending.

19) When climbing a mountain, we can observe transitions in biological communities that are analogous to the changes _____.
A) In biomes at different latitudes.
B) In a community through different seasons.
C) In an ecosystem as it evolves over time.
D) Across the United States from east to west.

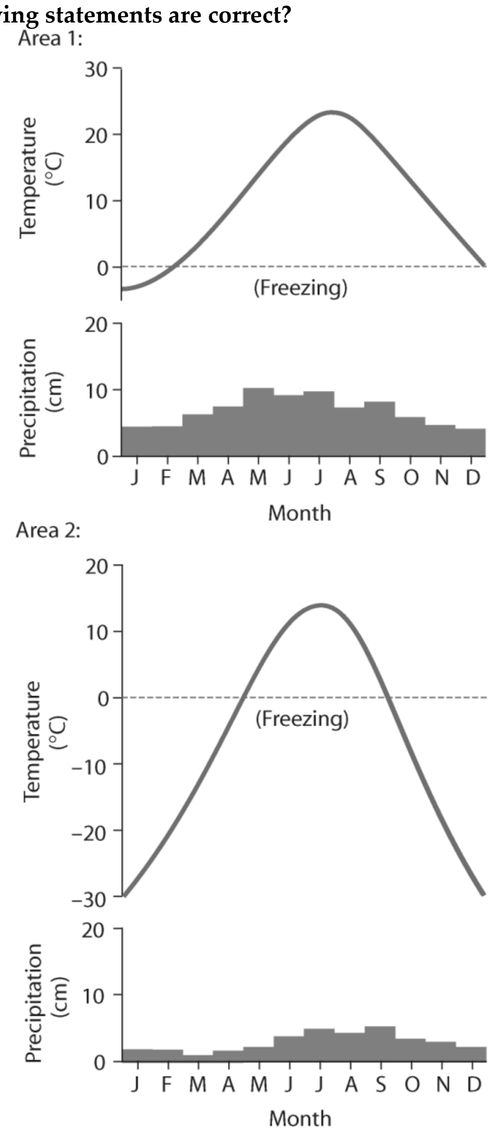
20) If the direction of Earth's rotation reversed, the most predictable effect would be _____.
A) A big change in the length of the year.
B) Winds blowing from west to east along the equator.
C) A loss of seasonal variation at high latitudes.
D) The elimination of ocean currents..

21) Based on the data in the figure below, which of the following statements are correct?



I) Area 1 would be considered a desert because of its high average temperature.
II) Area 1 has more average precipitation than Area 2.
III) Area 2 would be considered a desert because of its low average precipitation.
IV) Area 2 has a larger annual temperature variation.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

22) Based on the data in the figure below, which of the following statements are correct?



I) Area 1 has more average precipitation than Area 2.
II) Area 1 has a higher average temperature than Area 2.
III) Both areas have low variation in monthly precipitation.
IV) Area 2 has a lower annual temperature variation compared to Area 1.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) Only I, II, and III.

23) Besides sunlight, which would be the next most important climatic factors for terrestrial plants?
A) Wind and fire.
B) Moisture and wind.
C) Temperature and wind.
D) Temperature and moisture.

24) Which statement describes how climate might change if Earth was 75 percent land and 25 percent water?
A) Terrestrial ecosystems would likely experience more precipitation.
B) Earth's daytime temperatures would be higher and nighttime temperatures lower.
C) Summers would be longer and winters shorter at midlatitude locations.
D) Earth would experience an unprecedented global warming.

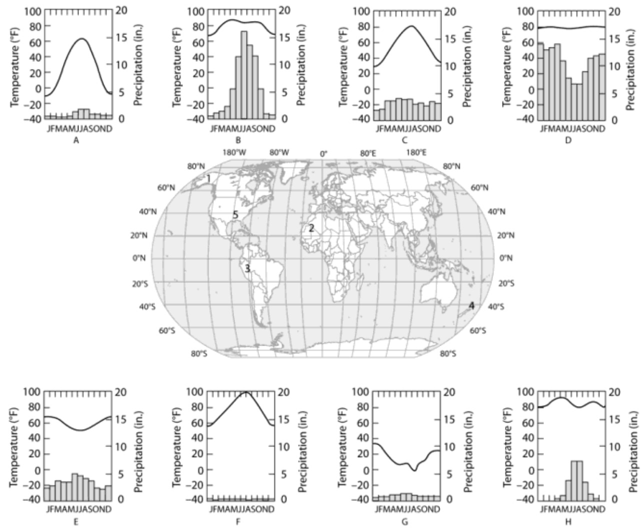
25) If global warming continues at its present rate, which biomes will likely take the place of the coniferous forest (taiga)?
A) Temperate broadleaf forest and grassland.
B) Desert and chaparral.
C) Tropical forest and savanna.
D) Chaparral and temperate broadleaf forest.

26) Fire suppression by humans _____.
A) Can change the species composition within biological communities.
B) Will result ultimately in sustainable production of increased amounts of forest products for human use.
C) Is necessary for the protection of threatened and endangered forest species.
D) Is a management goal of conservation biologists to maintain the healthy condition of forest communities.

27) Which of the following statements best describes the interaction between fire and ecosystems?
A) The likelihood of a wildfire occurring in a given ecosystem is highly predictable over the short term.
B) Many kinds of plants and plant communities have adapted to frequent fires.
C) The suppression of forest fires by man has prevented certain communities, such as grasslands, from reaching their climax stage.
D) Chaparral communities have evolved to the extent that they rarely burn.

Use the information and figure given below to answer the following questions (28-32).

The eight climographs below show yearly temperature (line graph and left vertical axis) and precipitation (bar graph and right vertical axis) averages for each month for some locations on Earth.



28) Which climograph shows the climate for location 1?
A) A.
B) C.
C) E.
D) H.

29) Which climograph shows the climate for location 2?
A) C.
B) D.
C) F.
D) H.

30) Which climograph shows the climate for location 3?
A) B.
B) C.
C) D.
D) E.

31) Which climograph shows the climate for location 4?
A) A.
B) C.
C) E.
D) G.

32) Which climograph shows the climate for location 5?
A) A.
B) C.
C) D.
D) H.

33) Which of the following best substantiates why location 3 is an equatorial (tropical) climate?
A) It has a monsoon season during the winter months.
B) The temperature is high for each monthly average.
C) The temperatures reach 100°F during some months.
D) The temperatures are lower in June, July, and August.

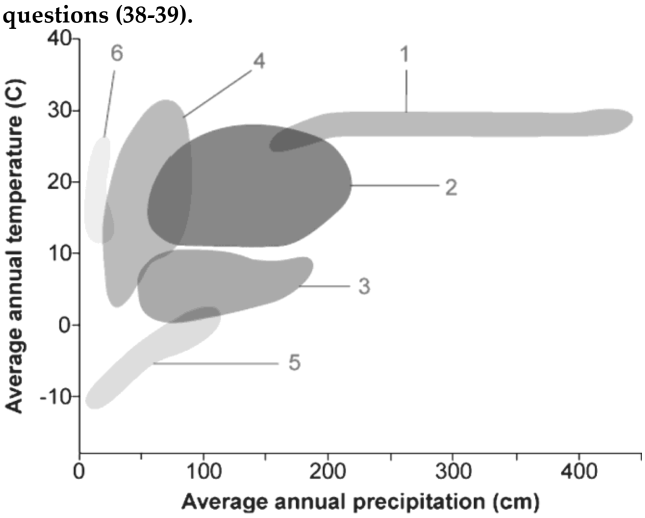
34) In areas of permafrost, stands of black spruce are frequently observed in the landscape, while other tree species are noticeably absent. Often these stands are referred to as "drunken forests" because many of the black spruce are displaced from their normal vertical alignment. What is the most likely explanation for the unusual growth of these forests in this marginal habitat?
A) Branches are adapted to absorb more carbon dioxide with this displaced alignment.
B) Taproot formation is impossible, so trees developed shallow root beds.
C) Trees are tilted so snow prevents them from breaking or tipping over.
D) Trees tip so that they do not compete with each other for sunlight.

35) Which of the following is an important feature of most terrestrial biomes?
A) Annual average rainfall in excess of 250 centimeters.
B) A distribution predicted almost entirely by rock and soil patterns.
C) Clear boundaries between adjacent biomes.
D) Vegetation demonstrating vertical layering.

36) Suppose that the number of bird species is determined mainly by the number of vertical strata found in the environment. If so, in which of the following biomes would you find the greatest number of bird species?
A) Tropical rain forest.
B) Savanna.
C) Temperate broadleaf forest.
D) Temperate grassland.

37) Two plant species live in the same biome but on different continents. Although the two species are not at all closely related, they may appear quite similar as a result of _____.
A) Convergent evolution.
B) Allopatric speciation.
C) Introgression.
D) Gene flow.

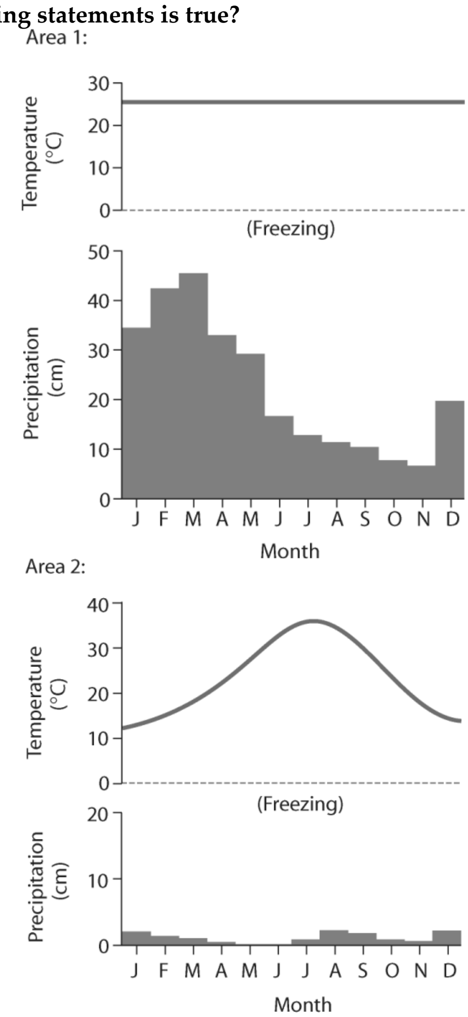
Use the figure given below to answer the following questions (38-39).



38) In the figure above, which number would designate the arctic tundra biome?
A) 2.
B) 3.
C) 4.
D) 5.

39) In the figure above, which number would designate the biome with the highest variation in annual precipitation?
A) 1.
B) 2.
C) 3.
D) 4.

40) Based on the data in the figure below, which of the following statements is true?



A) Area 1 could be called a boreal forest/taiga.
B) Area 2 could be called temperate grassland.
C) Area 1 could be called a tropical wet/rain forest.
D) Area 2 could be tundra.

41) In deep water, which of the following abiotic factors would most limit productivity?
A) Temperature.
B) Light availability.
C) Solute concentration.
D) None of them.

42) Wetlands are standing bodies of freshwater, just like lakes and ponds. However, wetlands are different from lakes and ponds because wetlands have _____.
A) Emergent vegetation.
B) Oxygen-poor water.
C) Shallow water and emergent vegetation.
D) Emergent vegetation and oxygen-poor water.

43) Which of the following statements regarding turnover in a lake is correct?
A) In fall turnover, dense water at 4°C sinks and disturbs sediments in the benthic zone.
B) In fall turnover, dense water at 4°C rises and disturbs sediments in the benthic zone.
C) The surface water gets to 4°C only by cooling.
D) Fall turnovers and spring turnovers are exactly the same.

44) A fish swimming into an estuary from a river would have which of the following as its greatest physiological challenge?
A) The high water flow would make the fish expend more energy.
B) The low oxygen content would give the fish difficulty in swimming aerobically.
C) The temperature change would stress the fish by denaturing its proteins.
D) The change in water solute content would challenge the osmotic balance of the fish.

45) Which of the following types of organisms is likely to have the widest geographic distribution?
A) Bacteria.
B) Songbirds.
C) Bears.
D) Lizards.

46) Which of the following can be said about light in aquatic environments?
A) Water selectively reflects and absorbs certain wavelengths of light.
B) Longer wavelengths penetrate to greater depths.
C) Light penetration seldom limits the distribution of photosynthetic species.
D) Most photosynthetic organisms avoid the surface where the light is too intense.

47) Turnover of water in temperate lakes during the spring and fall is made possible by which of the following?
A) Warm, less dense water layered at the top.
B) Cold, more dense water layered at the bottom.
C) A distinct thermocline between less dense, warm water and cold, dense water.
D) The changes in the density of water as seasonal temperatures change.

48) Imagine that a deep temperate zone lake did not turn over during the spring and fall seasons. Based on the physical and biological properties of limnetic ecosystems, what would be the difference from normal seasonal turnover?
A) The lake would fail to freeze over in winter.
B) An algal bloom of algae would result every spring.
C) Lakes would suffer a nutrient depletion in surface layers.
D) The pH of the lake would become increasingly alkaline.

49) If you are interested in observing a relatively simple community structure in a clear water lake, you would do well to choose diving into _____.
A) An oligotrophic lake.
B) A eutrophic lake.
C) A relatively shallow lake.
D) A nutrient-rich lake.

50) If a meteor impact or volcanic eruption injected a lot of dust into the atmosphere and reduced the sunlight reaching Earth's surface by 70 percent for one year, which of the following marine communities most likely would be least affected?
A) Deep-sea vent.
B) Coral reef.
C) Intertidal.
D) Estuary.

51) The oceans affect the biosphere by _____.
I) Producing a substantial amount of the biosphere's oxygen.
II) Adding carbon dioxide to the atmosphere.
III) Being the source of most of Earth's rainfall.
IV) Regulating the pH of freshwater biomes and terrestrial groundwater.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only I, II, and III.

52) A fish species known for its success in the aphotic zone may have which of the following characteristics?
I) Symbioses with photosynthetic organisms.
II) Highly developed chemoreception.
III) Adaptations for burrowing.
IV) Adaptations for sit-and-wait predation.
A) Only I and III.
B) Only I, II, and IV.
C) Only II, III, and IV.
D) Only I, II, and III.

53) Which of the following is responsible for the differences in summer and winter temperature stratification of deep temperate zone lakes?
A) Water is densest at 4°C.
B) Oxygen is most abundant in deeper waters.
C) Winter ice sinks in the summer.
D) Stratification is caused by a thermocline.

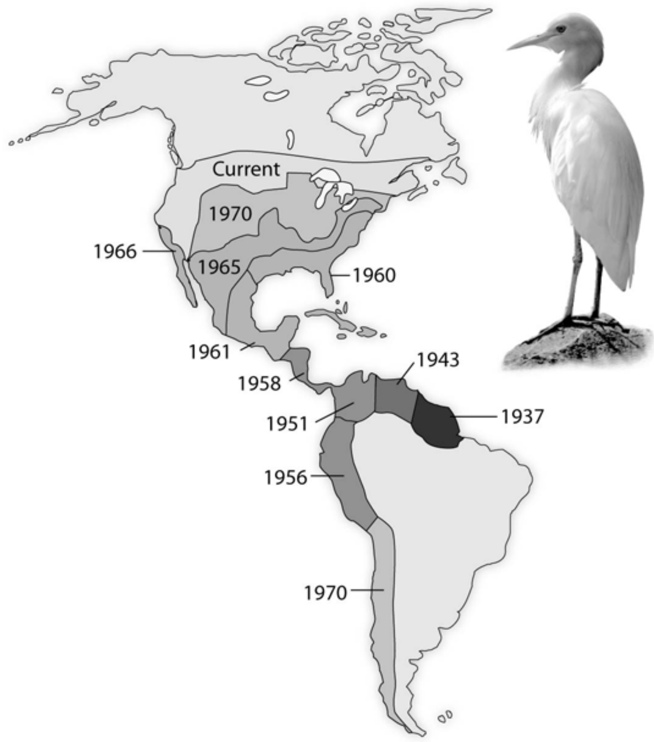
54) A certain species of pine tree survives only in scattered locations at elevations above 2800 meters in the western United States. To understand why this tree grows only in these specific places, an ecologist should _____.
A) Study the anatomy and physiology of this species.
B) Investigate the various biotic and abiotic factors that are unique to high altitude.
C) Analyze the soils found in the vicinity of these trees, looking for unique chemicals that may support their growth.
D) Collect data on temperature, wind, and precipitation at several of these locations for a year.

55) Which of the following statements best describes the effect of climate on biome distribution?
A) Average annual temperature and precipitation are sufficient to predict which biome will be found in an area.
B) Seasonal fluctuation of temperature is not a limiting factor in biome distribution if areas have the same annual temperature and precipitation means.
C) The average climate and pattern of climate are important in determining biome distribution.
D) Correlation of climate with biome distribution is sufficient to determine the cause of biome patterns.

56) In the development of terrestrial biomes, which factor is most dependent on all the others?
A) The species of colonizing animals.
B) Prevailing rainfall.
C) Mineral nutrient availability.
D) Soil structure.

57) Studying species transplants is a way that ecologists _____.
A) Determine the distribution of a species in a specified area.
B) Develop mathematical models for distribution and abundance of organisms.
C) Determine if dispersal is a key factor in limiting distribution of organisms.
D) Consolidate a landscape region into a single ecosystem.

Use the following diagram showing the spread of the cattle egret, *Bubulcus ibis*, since its arrival in the New World, to answer the question (58) below.



58) The range of cattle egrets has expanded between 1937 and today. How would an ecologist likely explain the expansion of the cattle egret?
A) Climatic factors, such as temperature and precipitation, provide a suitable habitat for cattle egrets.
B) There are no predators for cattle egrets in the New World, so they continue to expand their range.
C) A habitat left unoccupied by native herons and egrets met the biotic and abiotic requirements of the cattle egret transplants and their descendants.
D) The first egrets to colonize South America evolved into a new species capable of competing with the native species of herons and egrets.

59) Which statements about dispersal are correct?
I) Dispersal is a common component of the life cycles of plants and animals.
II) Colonization of devastated areas after floods or volcanic eruptions primarily depends upon climate.
III) Seeds are important dispersal stages in the life cycles of most flowering plants.
IV) Dispersal occurs only on an evolutionary time scale.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

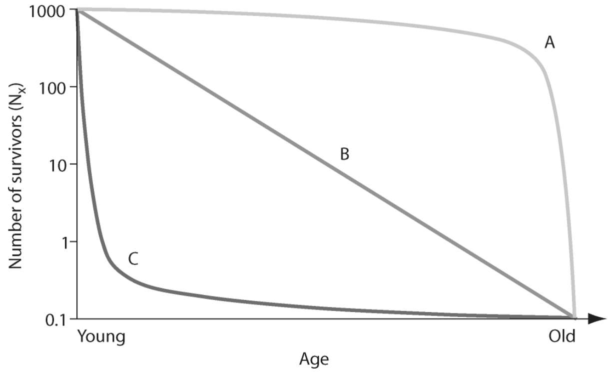
60) Which of the following examples of an ecological effect leading to an evolutionary effect is most correct?
A) When seeds are not plentiful, trees produce more seeds.
B) A few individuals with denser fur survive the coldest days of an ice age, and the reproducing survivors of the ice age will likely have more dense fur.
C) Fish that swim the fastest in running water catch the most prey and more easily escape predation.
D) The insects that spend the most time exposed to sunlight have the most mutations.

61) If carbon dioxide levels continue to increase and climate change continues over the next century, which of the following would best predict the directional migration of the North American ecosystems from the biomes shown in this climograph?
A) The ecosystems will shift to the south due to decreasing transpiration rates.
B) The ecosystems will move to both the eastern and western coastlines as these areas will be more moderate.
C) The ecosystems will move down mountains as the temperatures warm.
D) The ecosystems will shift to the north as temperatures warm.

CHAPTER 53:

POPULATION ECOLOGY

Use the following figure to answer the questions (1-2) below.



1) In the figure above, which of the following survivorship curves implies that an animal may lay many eggs, of which a regular number die each year on a logarithmic scale?
A) Curve A.
B) Curve B.
C) Curve C.
D) None of them.

2) In the figure above, which of the following survivorship curves most applies to humans living in developed countries?
A) Curve A.
B) Curve B.
C) Curve C.
D) Curve A or curve B.

Use the following table to answer the questions (3-4) below.

Life Table for *Lacerta vivipara* in the Netherlands

Year	Number alive	Survivorship	Fecundity	Survivorship x Fecundity = Average Number of Offspring Produced per Female of Age *x*
0	1000	1.000	0.00	0.00
1	763	0.763	1.70	1.30
2	308	0.308	2.94	0.91
3	158	0.158	4.13	0.65
4	57	0.057	4.88	0.28
5	10	0.010	6.50	0.07
6	7	0.007	6.40	0.04
7	2	0.002	6.30	0.01

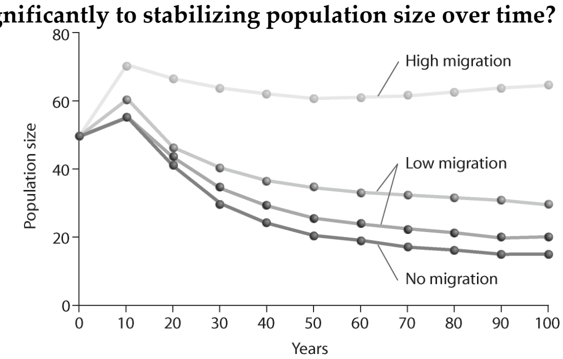
Data are from Strijbosch and Creemers, 1988.

3) Using the table above, how would you describe the population dynamics of *L. vivipara*?
A) The population is increasing.
B) The population is decreasing.
C) The population is stable.
D) The figure does not provide this information.

4) Using the table above, determine which age class year would hurt the population growth most if it were wiped out by disease.
A) Age class year 1.
B) Age class year 2.
C) Age class year 3.
D) Age class year 4.

5) Suppose researchers marked 800 turtles and later were able to trap a total of 300 individuals in that population, of which 150 were marked. What is the estimate for total population size?
A) 200.
B) 1050.
C) 1600.
D) 2100.

6) Looking at the figure below, what is contributing significantly to stabilizing population size over time?

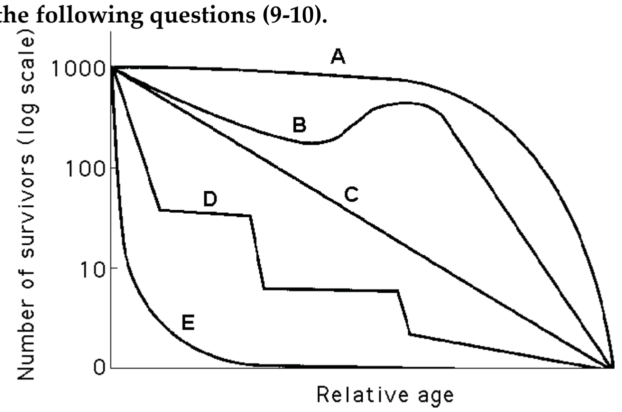


I) No migration.
II) Low migration.
III) High migration.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

7) Which of the following assumptions have to be made regarding the mark-recapture estimate of population size?
I) Marked and unmarked individuals have the same probability of being trapped.
II) The marked individuals have thoroughly mixed with the population after being marked.
III) No individuals have entered or left the population by immigration or emigration, and no individuals have been added by birth or eliminated by death during the course of the estimate.
A) I only.
B) II only.
C) I and II only
D) I, II, and III.

8) Which of the following is the most important assumption for the mark-recapture method to estimate the size of wildlife populations?
A) More individuals emigrate from, as opposed to immigrate into, a population.
B) Over 50% of the marked individuals need to be trapped during the recapture phase.
C) There is a 50:50 ratio of males to females in the population before and after trapping and recapture.
D) Marked individuals have the same probability of being recaptured as unmarked individuals during the recapture phase.

Use the survivorship curves in the figure below to answer the following questions (9-10).



9) Refer to the figure above. Which curve best describes survivorship in marine molluscs?
A) A.
B) B.
C) C.
D) E.

10) Refer to the figure above. Which curve best describes survivorship in elephants?
A) A.
B) B.
C) C.
D) E.

11) To measure the population of lake trout in a 250-hectare lake, 400 individual trout were netted and marked with a fin clip, then returned to the lake. The next week, the lake was netted again, and out of the 200 lake trout that were caught, 50 had fin clips. Using the mark-recapture estimate, the lake trout population size could be closest to which of the following?
A) 200.
B) 400.
C) 1,600.
D) 80,000.

12) Long-term studies of Belding's ground squirrels show that immigrants move nearly 2 kilometers from where they are born and become 1%-8% of the males and 0.7% - 6% of the females in other populations. On an evolutionary scale, why is this significant?
A) These immigrants make up for the deaths of individuals, keeping the other populations' size stable.
B) These immigrants provide a source of genetic diversity for the other populations.
C) Those individuals that emigrate to these new populations are looking for less crowded conditions with more resources.
D) Gradually, the populations of ground squirrels will move from a clumped to a uniform population pattern of dispersion.

13) A population is *correctly* defined as having which of the following characteristics?
I) Inhabiting the same general area.
II) Belonging to the same species.
III) Possessing a constant and uniform density and dispersion.
A) III only.
B) I and II only.
C) II and III only.
D) I, II, and III.

14) An ecologist recorded twelve white-tailed deer, *Odocoileus virginianus*, per square kilometer in one woodlot and twenty per square kilometer in another woodlot. What was the ecologist comparing?
A) Density.
B) Dispersion.
C) Carrying capacity.
D) Range.

15) Uniform spacing patterns in plants such as the creosote bush are most often associated with _____.
A) Patterns of high humidity.
B) The random distribution of seeds.
C) Competitive interaction between individuals of the same population.
D) The concentration of nutrients within the population's range.

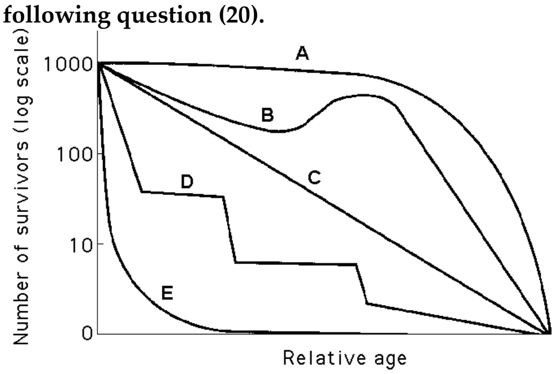
16) Which of the following groups would be most likely to exhibit uniform dispersion?
A) Red squirrels, which actively defend territories.
B) Cattails, which grow primarily at edges of lakes and streams.
C) Dwarf mistletoes, which parasitize particular species of forest tree.
D) Lake trout, which seek out cold, deep water high in dissolved oxygen.

17) Which of the following examples would most accurately measure the density of the population being studied?
A) Counting the number of times a one-kilometer transect is intersected by tracks of red squirrels after a snowfall.
B) Counting the number of coyote droppings per hectare.
C) Counting the number of moss plants in one-square-meter quadrants.
D) Counting the number of zebras from airplane census observations.

18) Which of the following scenarios would provide the most relevant data on population density?
A) Count the number of nests of a particular species of songbird and multiply this by a factor that extrapolates these data to actual animals.
B) Count the number of pine trees in several randomly selected 10-meter-square plots and extrapolate this number to the fraction of the study area these plots represent.
C) Use the mark-recapture method to estimate the size of the population.
D) Calculate the difference between all of the immigrants and emigrants to see if the population is growing or shrinking.

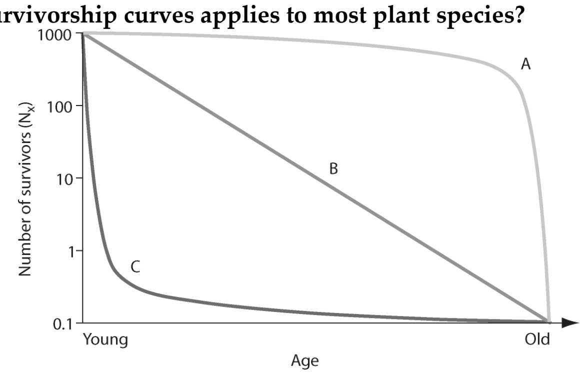
19) Which of the following is the best natural example of uniform distribution?
A) Bees collecting pollen in a wildflower meadow.
B) Snails in an intertidal zone at low tide.
C) Territorial songbirds in a mature forest during mating season.
D) Mushrooms growing on the floor of an old growth forest.

Use the survivorship curves in the figure below to answer the following question (20).



20) Refer to the figure above. Which statement best explains survivorship curve B?
A) It is likely a species that provides little postnatal care, but lots of care for offspring during midlife as indicated by increased survivorship.
B) This curve is likely for a species that produces lots of offspring, only a few of which are expected to survive.
C) It is likely a species where no individuals in the cohort die when they are at 60 -70% relative age.
D) Survivorship can only decrease; therefore, this curve could not happen in nature.

21) In the figure below, which of the following survivorship curves applies to most plant species?

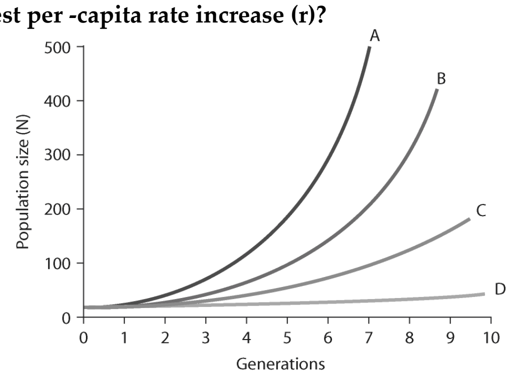


A) Curve A.
B) Curve B.
C) Curve C.
D) None of them.

22) In July 2008, the United States had a population of approximately 302,000,000 people. How many Americans were there in July 2009, if the estimated 2008 growth rate was 0.88%?
A) 5,500,000.
B) 303,000,000.
C) 304,000,000.
D) 2,710,800,000.

23) In 2008, the population of New Zealand was approximately 4,275,000 people. If the birth rate was 14 births for every 1000 people, approximately how many births occurred in New Zealand in 2008?
A) 6,000.
B) 42,275.
C) 60,000.
D) 140,000.

24) In the figure below, which of the lines represents the highest per -capita rate increase (r)?



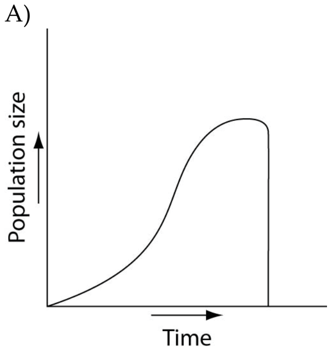
A) Line A.
B) Line B.
C) Line C.
D) Line D.

25) A population of ground squirrels has an annual per capita birth rate of 0.06 and an annual per capita death rate of 0.02. Calculate an estimate of the total number of individuals added to (or lost from) a population of 1000 individuals in one year.
A) 120 individuals added.
B) 40 individuals added.
C) 20 individuals added.
D) 400 individuals added.

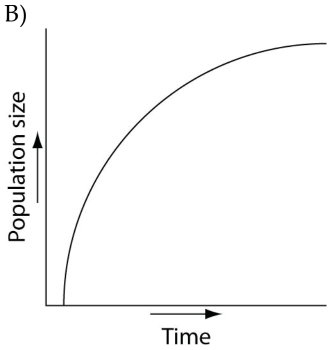
26) Starting from a single individual, what is the size of a population of bacteria at the end of a two-hour time period if they reproduce by binary fission every twenty minutes? (Assume unlimited resources and no mortality)
A) 16.
B) 32.
C) 64.
D) 128.

27) Which of the following graphs illustrates the population growth curve of single bacterium growing in a flask of ideal medium at optimum temperature over a two-hour period?

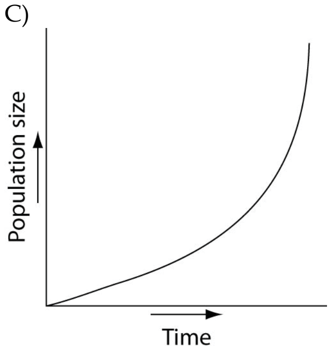
A)



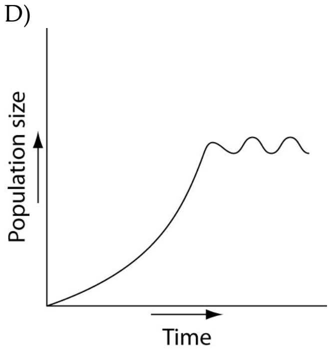
B)



C)



D)



28) During exponential growth, a population always _____.

A) Grows at its maximum per capita rate.
B) Quickly reaches its carrying capacity.
C) Cycles through time.
D) Loses some individuals to emigration.

29) Consider two old-growth forests: one is undisturbed while the other is being logged. In which region are species likely to experience exponential growth, and why?

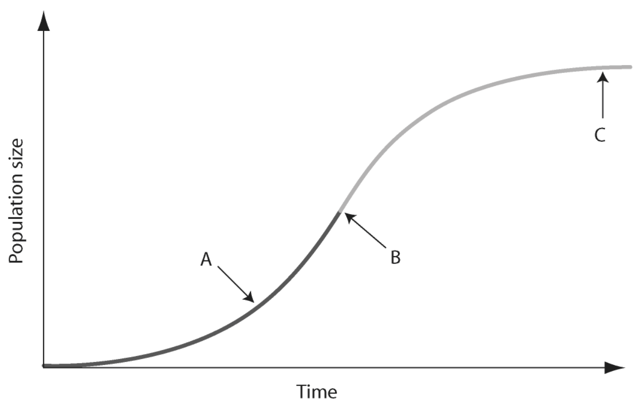
A) Old growth, because of stable conditions that would favor exponential growth of all species in the forest.
B) Old growth, because each of the species is well established and can produce many offspring.
C) Logged, because the disturbed forest affords more resources for increased specific populations to grow.
D) Logged, because the various populations are stimulated to a higher reproductive potential.

30) Imagine that you are managing a large game ranch. You know from historical accounts that a species of deer used to live there, but they have been extirpated. After doing some research to determine what might be an appropriately sized founding population, you reintroduce them. You then watch the population increase for several generations, and graph the number of individuals (vertical axis) against the number of generations (horizontal axis). With no natural predators impacting the population, the graph will likely appear as _____.

A) A diagonal line, getting higher with each generation.
B) An "S" that ends with a vertical line.
C) An upside-down "U".
D) A "J" increasing with each generation.

31) In the figure below, which of the arrows represents the carrying capacity?

(a) Density dependence: Growth rate slows at high density.



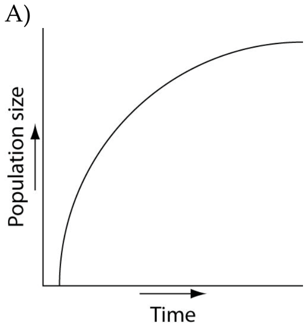
A) Arrow A.
B) Arrow B.
C) Arrow C.
D) Carrying capacity cannot be found in the figure because species under density-dependent control never reach carrying capacity.

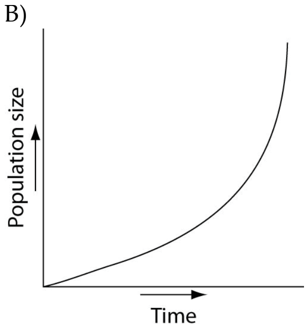
32) Which statements about *K* are correct?
I) *K* varies among populations.
II) *K* varies in space.
III) *K* varies in time.
IV) *K* is constant for any given species.

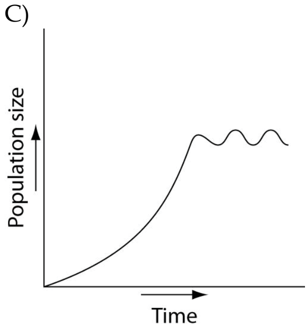
A) Only I and III.
B) Only II and IV.
C) Only I, II, and III.
D) Only II, III, and IV.

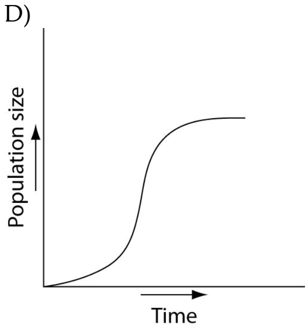
33) As *N* approaches *K* for a certain population, which of the following is predicted by the logistic equation?
A) The growth rate will not change.
B) The growth rate will approach zero.
C) The population will increase exponentially.
D) The carrying capacity of the environment will increase.

34) Which of the following causes populations to shift most quickly from an exponential to a logistic population growth?
A) Favorable climatic conditions.
B) Removal of predators.
C) Decreased death rate.
D) Competition for resources.

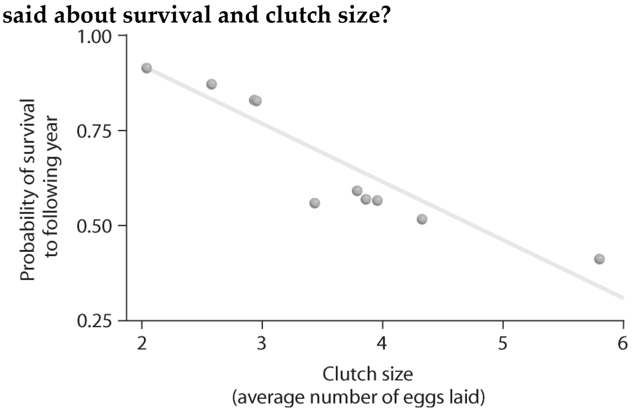
35) Which of the following graphs best illustrates the growth curve of a small population of rodents that has increased to a static carrying capacity?
A)


B)


C)


D)


36) According to the logistic growth equation,
***dN/dt* = r_{max}*N**(*K-N*)/*K* _____.**
A) The number of individuals added per unit time is greatest when *N* is close to zero.
B) The per capita growth rate (*r*) increases as *N* approaches *K*.
C) Population growth is zero when *N* equals *K*.
D) The population grows exponentially when K is small.

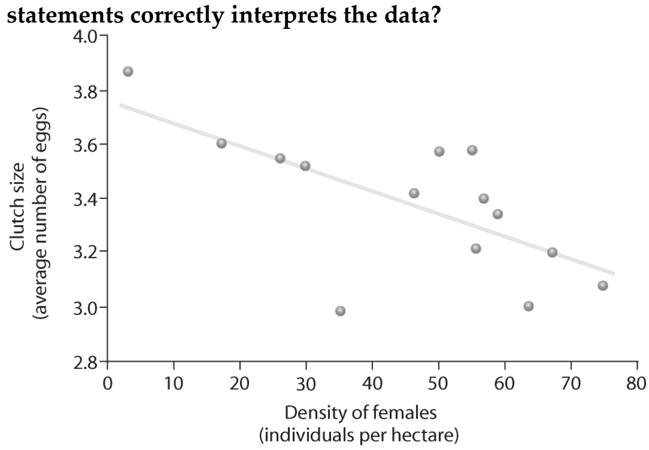
37) Looking at the data in the figure below, what can be said about survival and clutch size?


A) Animals with low survival tend to have smaller clutch sizes.
B) Large clutch size correlates with low survival.
C) Animals with high survival tend to have larger clutch sizes.
D) Probability of survivorship does not correlate with clutch size.

38) What is the primary limiting factor that determines why no female animal can produce a very large number of very large eggs?
A) Time is limited.
B) There are energy constraints.
C) Temperature constraints will prevent females from carrying too many eggs.
D) There will be an increase in predation pressure if the females carry too many large eggs.

39) You observe two female fish of the same species breeding. One female lays 100 eggs and the other female lays 1000 eggs. Which one of the following is LEAST likely given the limits of fitness trade-offs?
A) The female laying 100 eggs breeds more often than the female laying 1000 eggs.
B) The female laying 100 eggs lives longer than the female laying 1000 eggs.
C) The eggs from the female laying 1000 eggs have larger yolks than the yolks of the eggs from the female laying 100 eggs.
D) The female laying 1000 eggs is larger than the female laying 100 eggs.

40) Based on the figure below, which of the following statements correctly interprets the data?



A) As female density increases, clutch size increases.
B) As female density increases, survivorship decreases.
C) Clutch size decreases as female density increases.
D) None of them.

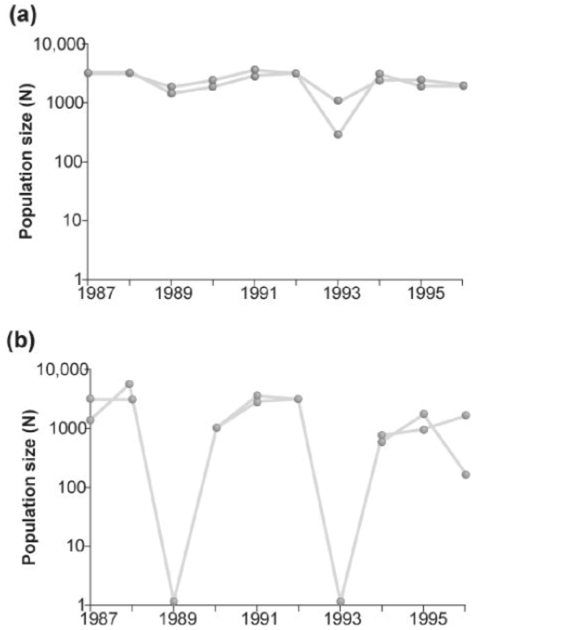
41) Which pair of terms most accurately describes life history traits for a stable population of wolves?
A) Semelparous; *r*-selected.
B) Semelparous; *K*-selected.
C) Iteroparous; *r*-selected.
D) Iteroparous; *K*-selected.

42) Natural selection involves energetic trade-offs between ______.
A) Choosing how many offspring to produce over the course of a lifetime and how long to live.
B) Producing large numbers of gametes when employing internal fertilization versus fewer numbers of gametes when employing external fertilization.
C) Increasing the number of individuals produced during each reproductive episode and a corresponding decrease in parental care.
D) High survival rates of offspring and the cost of parental care.

43) Which of the following is characteristic of *K*-selected populations?
A) Offspring with good chances of survival.
B) Many offspring per reproductive episode.
C) Small offspring.
D) A high intrinsic rate of increase.

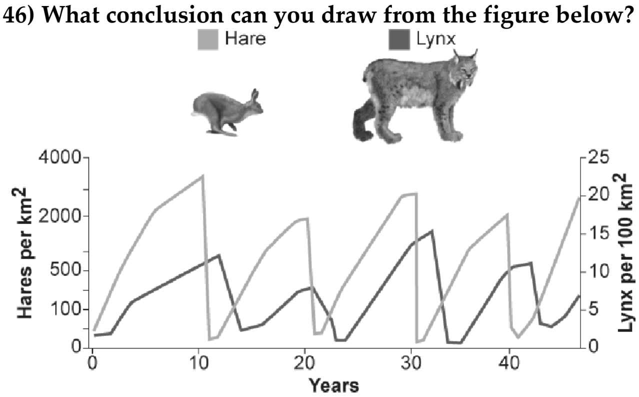
44) In which of the following situations would you expect to find the largest number of *K*-selected individuals?
A) A recently abandoned agricultural field in Ohio.
B) A shifting sand dune community of south Lake Michigan.
C) An old-growth forest.
D) South Florida after a hurricane.

45) Graph (b) in the figure below shows the normal fluctuations of a population of grouse. Assuming graph (a) in the figure above is the result of some experimental treatment in the grouse population, what can be concluded?



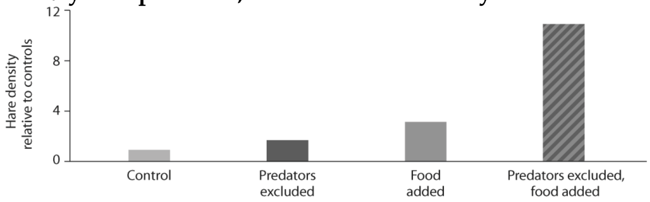
A) The experimental treatment exacerbated the population cycling.
B) The experimental treatment did not affect population cycling in this species.
C) The experimental treatment has most likely identified the cause of population cycling.
D) None of the other responses is true.

46) What conclusion can you draw from the figure below?



A) Hares control lynx population size.
B) Lynx control hare population size.
C) Lynx and hare populations are independent of each other.
D) The relationship between the populations cannot be determined only from this graph.

47) Looking at the data in the figure below from the hare/lynx experiment, what conclusion can you draw?



I) Food is a factor in controlling hare population size.
II) Excluding lynx is a factor in controlling hare population size.
III) The effect of excluding predators and adding food in the same experiment is greater than the sum of excluding lynx alone plus adding food alone.

A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

48) Often the growth cycle of one population has an effect on the cycle of another. As moose populations increase, for example, wolf populations also increase. Thus, if we are considering the logistic equation for the wolf population:

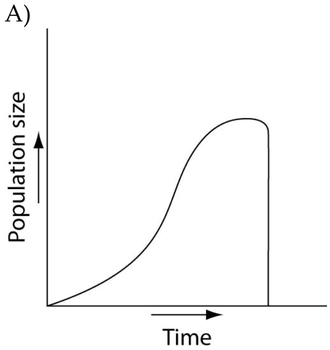
$dN/dt = rN^*(K\text{-}N)/K$ _____.

Which of the factors accounts for the effect of the moose population?

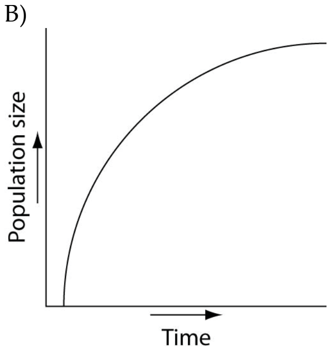
A) r.
B) N.
C) rN.
D) K.

49) Which of the following graphs illustrates the growth over several seasons of a population of snowshoe hares that were introduced to an appropriate habitat also inhabited by predators in northern Canada?

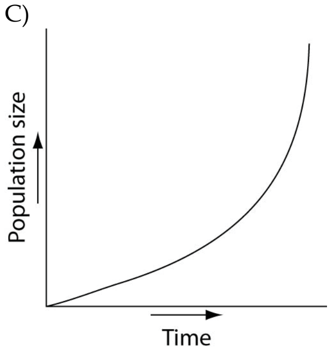
A)



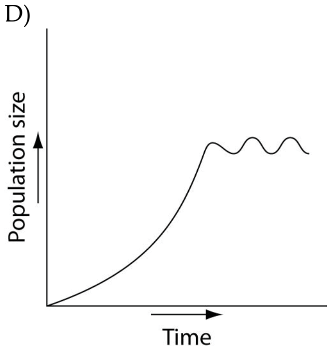
B)



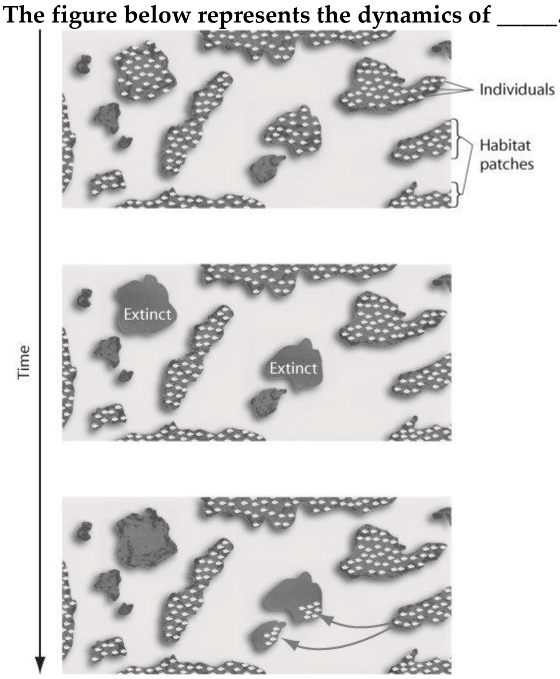
C)



D)



50) The figure below represents the dynamics of _____.



A) Metapopulations.
B) Extinction.
C) Emigration.
D) Both extinction and emigration.

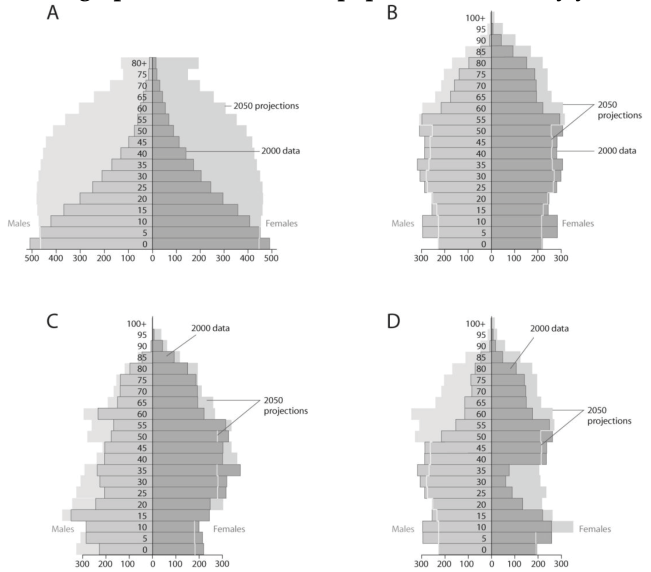
51) Use the following abstract from *Theoretical Population Biology* to answer the question.

Abstract:
We derive measures for assessing the value of an individual habitat fragment for the dynamics and persistence of a metapopulation living in a network of many fragments. We demonstrate that the most appropriate measure of fragment value depends on the question asked. Specifically, we analyze four alternative measures: the contribution of a fragment to the metapopulation capacity of the network, to the equilibrium metapopulation size, to the expected time to metapopulation extinction and the long-term contribution of a fragment to colonization events in the network. The latter measure is comparable to density-dependent measures in general matrix population theory; though some differences are introduced by the fact that "density dependence" is spatially localized in the metapopulation context. We show that the value of a fragment depends not only on the properties of the landscape but also on the properties of the species. Most importantly, variation in fragment values between the habitat fragments is greatest in the case of rare species that occur close to the extinction threshold, as these species are likely to be restricted to the most favorable parts of the landscape. We expect that the measures of habitat fragment described and analyzed here have applications in landscape ecology and in conservation biology. Copyright © 2003 Elsevier Inc. All rights reserved. (Otso Ovaskainen and Ilkka Hanski. 2003. How much does an individual habitat fragment contribute to metapopulation dynamics and persistence? *Theoretical Population Biology* 64:481-95.)

One measure for the value of the patch was given by the long -term contribution of a fragment to colonization events in the network. How do the properties of a landscape and the properties of a species affect the value of a patch? The value of the fragment depends _____.
A) On the properties of the landscape and the properties of the species.
B) Only on the properties of the landscape and not on the properties of the species.
C) Not on the properties of the landscape but only on the properties of the species.
D) On neither the properties of the landscape nor on the properties of the species.

52) A population of white-footed mice becomes severely overpopulated in a habitat that has been disturbed by human activity. Sometimes intrinsic factors cause the population to increase in mortality and lower reproduction rates to occur in reaction to the stress of overpopulation. Which of the following is an example of intrinsic population control?
A) Owl populations frequent the area more often because of increased hunting success.
B) Females undergo hormonal changes that delay sexual maturation, and many individuals suffer depressed immune systems and die due to the stress of overpopulation.
C) Clumped dispersion of the population leads to increased spread of disease and parasites, resulting in a population crash.
D) All of the resources (food and shelter) are used up by overpopulation, and much of the population dies of exposure and/or starvation.

53) Based on the diagrams in the figure above and on the large population of baby boomers in the United States, which graph best reflects U.S. population in twenty years?

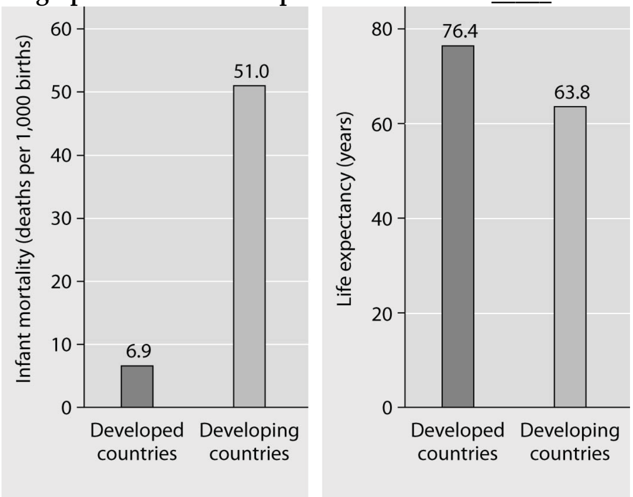


A) A.
B) B.
C) C.
D) D.

54) Which of the following statements regarding the future of populations in developing countries are correct?
I) The fecundity is predicted to increase.
II) Survivorship will increase.
III) Overall population size will increase dramatically.
IV) The number of offspring each year is predicted to remain high.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) Only I, II, and III.

55) Why does the 2009 U.S. population continue to grow even though the United States has essentially established a zero population growth (ZPG)?
A) Emigration.
B) Immigration.
C) Baby boomer reproduction.
D) The 2007-2009 economic recessions.

56) What is a logical conclusion that can be drawn from the graphs below? Developed countries have _____.



Infant mortality and life expectancy at birth in developed and developing countries (data as of 2005).

A) Lower infant mortality rates and lower life expectancy than developing countries.
B) Higher infant mortality rates and lower life expectancy than developing countries.
C) Lower infant mortality rates and higher life expectancy than developing countries.
D) Higher infant mortality rates and higher life expectancy than developing countries.

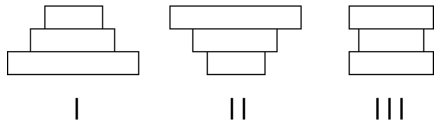
57) A recent study of ecological footprints concluded that _____.

A) Earth's carrying capacity would increase if per capita meat consumption increased.
B) Current demand by industrialized countries for resources is much smaller than the ecological footprint of those countries.
C) It is not possible for technological improvements to increase Earth's carrying capacity for humans.
D) The ecological footprint of the United States is large because per capita resource use is high.

58) Which of the following statements about human population in industrialized countries are correct?
I) Life history is *r*-selected.
II) The population has undergone the demographic transition.
III) The survivorship curve is Type III.
IV) Age distribution is relatively uniform.

A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

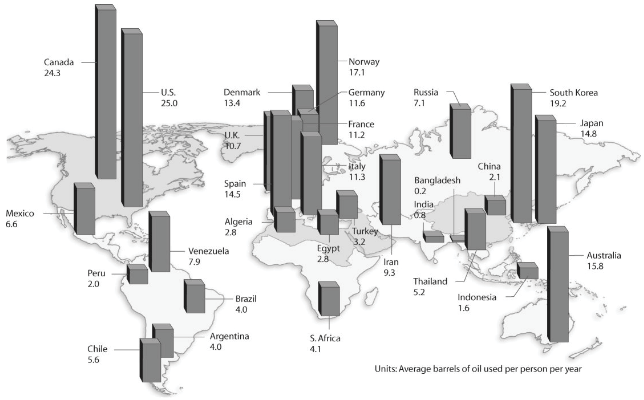
The following question (59) refers to the figure below, which depicts the age structure of three populations.



59) Which population(s) appears to be stable?

A) I.
B) III.
C) I and II.
D) II and III.

60) Based on the figure below and given the populations of the following countries, which country uses the most oil overall?



A) United States (population = 320 million).
B) Canada (population = 36 million).
C) China (population = 1.33 billion).
D) Russia (population = 144 million).

61) A population of squirrels on an island has a carrying capacity of 350 individuals. If the maximum rate of increase is 1.0 per individual per year and the population size is 275, determine the population growth rate (Round to the nearest whole number).

A) 59 squirrels per year.
B) -34 squirrels per year.
C) 75 squirrels per year.
D) 15 squirrels per year.

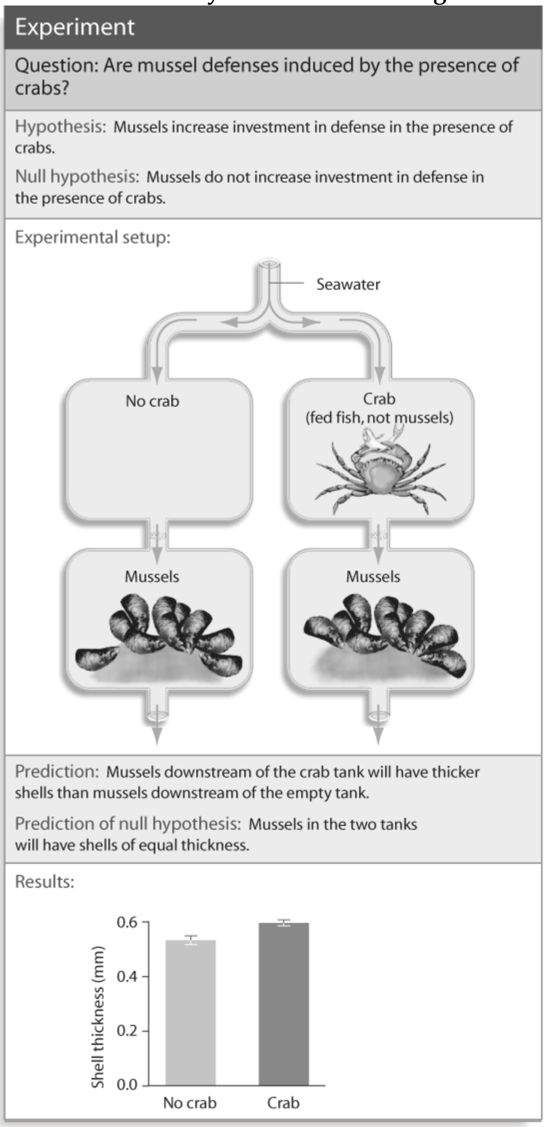
CHAPTER 54:

COMMUNITY ECOLOGY

1) Some birds follow moving swarms of army ants in the tropics. As the ants march along the forest floor hunting insects and small vertebrates, birds follow and pick off any insects or small vertebrates that fly or jump out of the way of the ants. This situation is an example of what kind of species interaction between the birds and the ants?
A) Consumption.
B) Commensalism.
C) Parasitism.
D) Mutualism.

2) In the hypothesis that *C. stellatus* (a species of barnacle) is competitively excluded from the lower intertidal zone by *B. balanoides* (another species of barnacle), what could be concluded about the two species?
A) The fundamental and realized niches of *B. balanoides* and *C. stellatus* are identical.
B) The fundamental and realized niches of *B. balanoides* and *C. stellatus* are different.
C) The fundamental and realized niches of *B. balanoides* are different, but the fundamental and realized niches of *C. stellatus* are identical.
D) The fundamental and realized niches of *B. balanoides* are identical, but the fundamental and realized niches of *C. stellatus* are different.

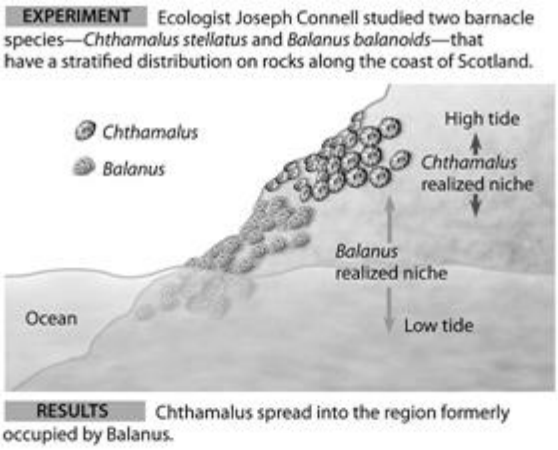
3) What conclusion can you draw from the figure below?



A) Without direct contact, mussels can sense the presence of crabs.
B) Mussels can sense the presence of crabs only visually.
C) Mussels are increasing their shell thickness in response to water current.
D) Crabs hunt for mussels by focusing on the chemicals they emit into the water.

4) As you study two closely related predatory insect species, the two -spot and the three-spot avenger beetles, you notice that each species seeks prey at dawn in areas without the other species. However, where their ranges overlap, the two-spot avenger beetle hunts at night and the three-spot hunts in the morning. When you bring them into the laboratory and isolate the two different species, you discover that the offspring of both species are found to be nocturnal. You have discovered an example of _____.
A) Mutualism.
B) Character displacement.
C) Batesian mimicry.
D) Resource partitioning.

Use the following diagram to answer the questions (5-6) below.



In this experiment, *Balanus* was removed from the habitat shown on the EXPERIMENT.

5) Which of the following statements is a valid conclusion of this experiment?
A) *Balanus* can survive only in the lower intertidal zone because it is unable to resist desiccation.
B) *Balanus* is inferior to *Chthamalus* in competing for space on rocks lower in the intertidal zone.
C) The two species of barnacles do not compete with each other because they feed at different times of day.
D) The removal of *Balanus* shows that the realized niche of *Chthamalus* is smaller than its fundamental niche.

6) Connell conducted this experiment to learn more about _____.
A) Character displacement in the color of barnacles.
B) Habitat preference in two different species of barnacles.
C) How sea-level changes affect barnacle distribution.
D) Competitive exclusion and distribution of barnacle species.

Use the following information to answer the questions (7-8).

The symbols +, -, and o are to be used to show the results of interactions between individuals and groups of individuals in the examples that follow. The symbol + denotes a positive interaction, - denotes a negative interaction, and o denotes where individuals are not affected by interacting. The first symbol refers to the first organism mentioned.

7) What interactions exist between a lion pride and a hyena pack?
A) +/+.
B) +/-.
C) o/o.
D) -/-.

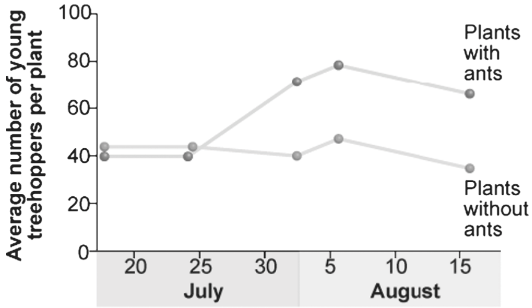
8) What interactions exist between a tick on a dog and the dog?
A) +/o.
B) +/-.
C) o/o.
D) -/-.

9) Which of the following statements is consistent with the principle of competitive exclusion?
A) The random distribution of one competing species will have a positive impact on the population growth of the other competing species.
B) Two species with the same fundamental niche will exclude other competing species.
C) Even a slight reproductive advantage will eventually lead to the elimination of the less well adapted of two competing species.
D) Natural selection tends to increase competition between related species.

10) If two species are close competitors, and one species is experimentally removed from the community, the remaining species would be expected to _____.
A) Change its fundamental niche.
B) Decline in abundance.
C) Become the target of specialized parasites.
D) Expand its realized niche.

11) Which of the following is an example of a commensalism?
A) Fungi residing in plant roots, such as endomycorrhizae.
B) Bacteria fixing nitrogen in plants.
C) Rancher ants that protect aphids in exchange for sugar-rich honeydew.
D) Cattle egrets eating insects stirred up by grazing bison.

12) Treehoppers (a type of insect) produce honeydew, which ants use for food. Treehoppers have a major predator, the jumping spider. Researchers hypothesized that the ants would protect the treehoppers from the spiders. In an experiment, researchers followed study plots with ants removed from the system and compared them to a control plot. In the figure below, what can you conclude?



A) Ants do somehow protect the treehoppers from spiders.
B) Ants eat the honeydew produced by treehoppers.
C) Ants reduce the numbers of treehoppers.
D) No specific conclusions can be drawn from this figure.

13) During a one-year study, researchers found no difference in treehopper populations in any of their control and experimental groups. What could they measure during the second year to gain information about why this might have occurred?
A) Measure the number of ant females.
B) Measure the relative sizes of the treehoppers.
C) Measure the relative abundance of jumping spiders.
D) Measure the relative sizes of different ant species.

14) Resource partitioning would be most likely to occur between _____.
A) Sympatric populations of species with similar ecological niches.
B) Sympatric populations of a flowering plant and its specialized insect pollinator.
C) Allopatric populations of the same animal species.
D) Allopatric populations of species with similar ecological niches.

15) Which of the following is an example of cryptic coloration?
A) Bands on a coral snake.
B) Brown or gray color of tree bark.
C) Markings of a viceroy butterfly's wings.
D) A "walking stick" insect that resembles a twig.

16) Which of the following is an example of Müllerian mimicry?
A) Two species of unpalatable butterfly that have the same color pattern.
B) A day-flying hawkmoth that looks like a wasp.
C) A chameleon that changes its color to look like a dead leaf.
D) Two species of rattlesnakes that both rattle their tails.

17) Which of the following is an example of Batesian mimicry?
A) A butterfly that resembles a leaf.
B) A nonvenomous snake that looks like a venomous snake.
C) A fawn with fur coloring that camouflages it in the forest environment.
D) A snapping turtle that uses its tongue to mimic a worm, thus attracting fish.

18) Which of the following is an example of aposematic coloration?
A) The brightly colored patterns of poison dart frogs.
B) Eye color in humans.
C) Green color of a plant.
D) A katydid whose wings look like a dead leaf.

19) Dwarf mistletoes are flowering plants that grow on certain forest trees. They obtain nutrients and water from the vascular tissues of the trees. The trees derive no known benefits from the dwarf mistletoes. Which of the following best describes the interactions between dwarf mistletoes and trees?
A) Mutualism.
B) Parasitism.
C) Competition.
D) Facilitation.

20) In some circumstances, grasses benefit from being grazed. Which of the following terms would best describe such a plant-herbivore interaction?
A) Mutualism.
B) Commensalism.
C) Parasitism.
D) Predation.

21) Which of the following would be most significant in understanding the structure of an ecological community?
I) Determining how many species are present overall.
II) Determining which particular species are present.
III) Determining the kinds of interactions that occur among organisms of different species.
IV) Determining the relative abundance of species
A) Only I and III.
B) Only II and IV.
C) Only I, II, and III.
D) I, II, III, and IV.

22) Which of the following studies would a community ecologist undertake to learn about competitive interactions?
I) Selectivity of nest sites among cavity-nesting songbirds.
II) The grass species preferred by grazing pronghorn antelope and bison.
III) Stomach analysis of brown trout and brook trout in streams where they coexist.
A) Only I and II.
B) Only I and III.
C) Only II and III.
D) I, II, and III.

23) How might an ecologist test whether a species is occupying its entire fundamental niche or only a portion of it?
A) Study the temperature range and humidity requirements of the species.
B) Observe if the niche size changes after the introduction of a similar non-native species.
C) Measure the change in reproductive success when the species is subjected to environmental stress.
D) Observe if the species expands its range after the removal of a competitor.

Use the following information to answer the questions (24-25).

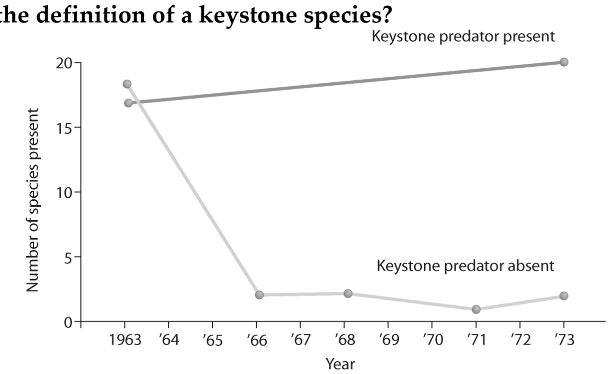
The symbols +, -, and o are to be used to show the results of interactions between individuals and groups of individuals in the examples that follow. The symbol + denotes a positive interaction, - denotes a negative interaction, and o denotes where individuals are not affected by interacting. The first symbol refers to the first organism mentioned.

24) What interactions exist between cellulose-digesting organisms in the gut of a termite and the termite?
A) +/+.
B) +/o.
C) +/-.
D) o/o.

25) What interactions exist between mycorrhizae and evergreen tree roots?
A) +/+.
B) +/o.
C) +/-.
D) o/o.

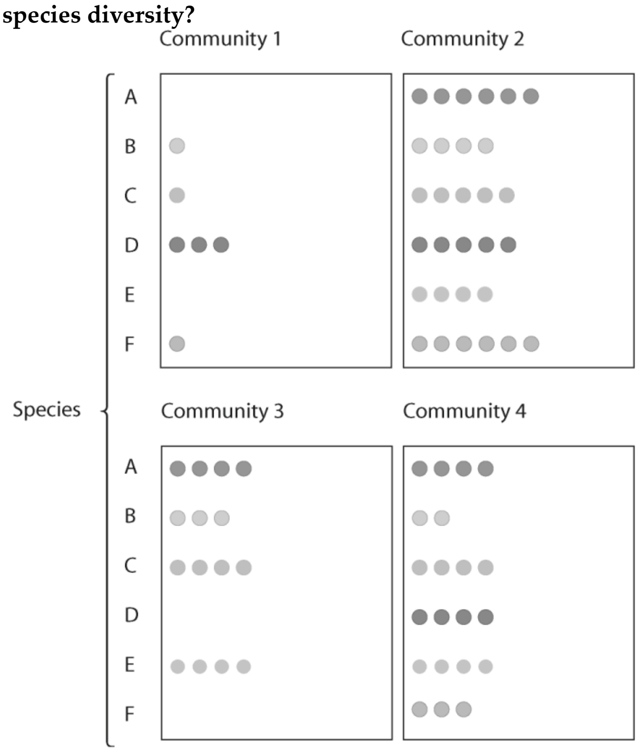
26) Bouchard and Brooks studied the effect of insect flight on dispersal and speciation in rain forest insects. They sampled all of the insects in the study area and found that 60 insect species are flightless and 19 are macropterous (able to fly). What can you conclude so far about this study? (P. Bouchard and D. R. Brooks. 2004. Effect of vagility potential on dispersal and speciation in rainforest insects. *Journal of Evolutionary Biology* 17:994—1006.)
A) Flightless insects have a greater dispersal potential from this study area.
B) Flightless insects are more numerous in the study area.
C) Flightless insects have a higher richness in the study area.
D) Flightless insects are better suited for the tropics.

27) What does the graph in the figure below tell you about the definition of a keystone species?



A) A keystone species has little interaction with other species in an environment.
B) Removing a keystone species from the community drastically reduces diversity.
C) Adding a keystone species to the community will make it more diverse.
D) Removing a keystone species from the community will eventually allow for the invasion of a new species.

28) In the figure below, which community has the highest species diversity?



A) Community 1.
B) Community 2.
C) Community 1 and community 3 have the highest species diversity.
D) Community 4.

29) Approximately how many kilograms (kg) of carnivore (secondary consumer) biomass can be supported by a field plot containing 1000 kg of plant material?
A) 1000.
B) 100.
C) 10.
D) 1.

30) In a tide pool, fifteen species of invertebrates were reduced to eight after one species was removed. The species removed was likely a(n) _____.
A) Pathogen.
B) Keystone species.
C) Herbivore.
D) Resource partitioner.

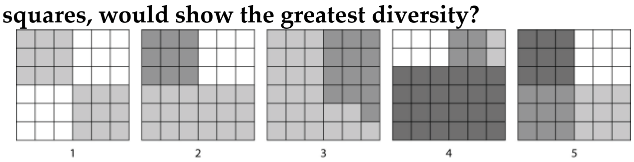
31) Elephants are not the most dominant species in African grasslands, yet they influence community structure. The grasslands contain scattered woody plants, but they are kept in check by the uprooting activities of the elephants. Take away the elephants, and the grasslands convert to forests or to shrublands. The newly growing forests support fewer species than the previous grasslands. Which of the following describes why elephants are the keystone species in this scenario?
A) Elephants exhibit a disproportionate influence on the structure of the community relative to their abundance.
B) Grazing animals depend upon the elephants to convert forests to grassland.
C) Elephants are the biggest herbivore in this community.
D) Elephants help other populations survive by keeping out many of the large African predators.

32) According to bottom-up and top-down control models of community organization, which of the following expressions would imply that an increase in the size of a carnivore (C) population would negatively impact its prey (P) population, but not vice versa?
A) P ← C.
B) P → C.
C) C ↔ P.
D) P ← C → P.

33) Which of the following is a likely explanation for why invasive species take over communities into which they have been introduced?
A) Invasive species are less efficient than native species in competing for the limited resources of the environment.
B) Invasive species are not held in check by the predators and agents of disease that have always been in place for native species.
C) Invasive species have a higher reproductive potential than native species.
D) Invasive species come from geographically isolated regions, so when they are introduced to regions where there is more competition, they thrive.

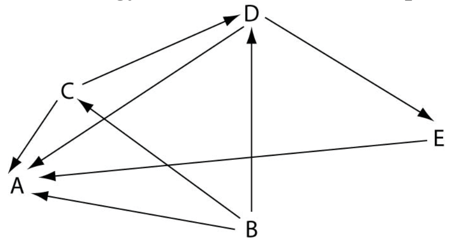
34) Imagine five forest communities, each with one hundred individuals distributed among four different tree species (W, X, Y, and Z). Which forest community would be most diverse?
A) 25W, 25X, 25Y, 25Z.
B) 40W, 30X, 20Y, 10Z.
C) 50W, 25X, 15Y, 10Z.
D) 70W, 10X, 10Y, 10Z.

35) According to the Shannon Diversity Index, which of the five blocks below, with each containing thirty-six squares, would show the greatest diversity?



A) 1.
B) 2.
C) 4.
D) 5.

Use the following diagram of a hypothetical food web to answer the questions (36-37) below. The arrows represent the transfer of energy between the various trophic levels.



36) Which letter represents an organism that could be a producer?
A) A.
B) B.
C) D.
D) E.

37) Which letter represents an organism that could only be a primary consumer?
A) A.
B) B.
C) C.
D) D.

38) Keystone predators can maintain species diversity in a community if they _____.
A) Competitively exclude other predators.
B) Prey on the community's dominant species.
C) Allow immigration of other predators.
D) Prey only on the least abundant species in the community.

39) Food chains are sometimes short because _____.
A) Only a single species of herbivore feeds on each plant species.
B) Local extinction of a species causes extinction of the other species in its food chain.
C) Most of the energy in a trophic level is lost as it passes to the next higher level.
D) Predator species tend to be less diverse and less abundant than prey species.

40) Which of the following could qualify as a top-down control on a grassland community?
A) Limitation of plant biomass by rainfall amount.
B) Influence of temperature on competition among plants.
C) Influence of soil nutrients on the abundance of grasses versus wildflowers.
D) Effect of grazing intensity by bison on plant species diversity.

41) Recall that Clements's view of biological communities is that of a highly predictable and interrelated structure, while Gleason's view of biological communities is that individual species operate independently. If we set up many identical sterilized ponds in the same area and allowed them to be colonized, what should we predict if we wished to test Gleason's hypothesis?
A) Identical plankton communities will develop in all ponds.
B) Similar plankton communities will develop in all ponds.
C) Different plankton communities will develop in all ponds.
D) Limited plankton communities will develop in all ponds.

42) What is the main advantage of controlled burnings of forested areas? Controlled burnings _____.
A) Eliminate the possibility of forest fires.
B) Clear forested areas for farmland.
C) Prevent the overgrowth of the underbrush.
D) Allow new species to form.

43) According to the nonequilibrium model of community diversity, _____.
A) Community structure remains stable in the absence of interspecific competition.
B) Communities are assemblages of closely linked species that are irreparably changed by disturbance.
C) Interspecific interactions induce changes in community composition over time.
D) Communities are constantly changing after being influenced by disturbances.

44) Why do moderate levels of disturbance result in an increase in community diversity?
A) Habitats are opened up for less competitive species.
B) Competitively dominant species infrequently exclude less competitive species after a moderate disturbance.
C) The resulting uniform habitat supports stability, which in turn supports diversity.
D) Less-competitive species evolve strategies to compete with dominant species.

45) Based on the intermediate disturbance hypothesis, a community's species diversity is increased by _____.
A) Frequent massive disturbance.
B) Stable conditions with no disturbance.
C) Moderate levels of disturbance.
D) Intensive disturbance by humans.

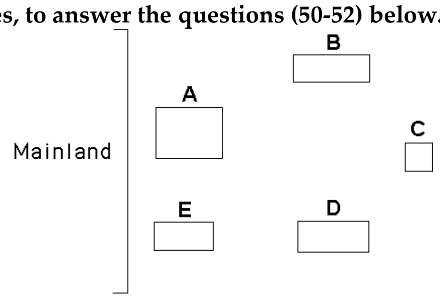
46) In a particular case of secondary succession, three species of wild grass all invaded a field. By the second season, a single species dominated the field. A possible factor in this secondary succession was _____.
A) Equilibrium.
B) Immigration.
C) Inhibition.
D) Parasitism.

47) There are more species in tropical areas than in places more distant from the equator. This is probably a result of _____.
A) Fewer predators.
B) More intense annual solar radiation.
C) More frequent ecological disturbances.
D) Fewer agents of disease.

48) Which of the following is a widely supported explanation for the tendency of tropical communities to have greater species diversity than temperate or polar communities?
A) There are fewer parasites to negatively affect the health of tropical communities.
B) Tropical communities are low in altitude, whereas temperate and polar communities are high in altitude.
C) Tropical communities are generally older than temperate and polar communities.
D) More competitive dominant species have evolved in temperate and polar communities.

49) Which of the following is a correct statement about the MacArthur/Wilson Island Biogeography Model?
A) As the number of species on an island increases, the emigration rate decreases.
B) Competitive exclusion is less likely on an island that has large numbers of species.
C) Small islands receive few new immigrant species.
D) Islands closer to the mainland have higher extinction rates.

Use the following diagram of five islands formed at around the same time near a particular mainland, as well as MacArthur and Wilson's island biogeography principles, to answer the questions (50-52) below.



50) Which island would likely have the greatest species diversity?
A) A.
B) C.
C) D.
D) E.

51) Which island would likely exhibit the most impoverished species diversity?
A) A.
B) B.
C) C.
D) E.

52) Which island would likely have the lowest extinction rate?
A) A.
B) C.
C) D.
D) E.

53) According to the equilibrium model of island biogeography, species richness would be greatest on an island that is _____.
A) Large and close to a mainland.
B) Large and remote.
C) Small and remote.
D) Small and close to a mainland.

54) The most plausible hypothesis to explain why species richness is higher in tropical than in temperate regions is that _____.
A) Tropical communities are younger.
B) Tropical regions generally have more available water and higher levels of solar radiation.
C) Higher temperatures cause more rapid speciation.
D) Tropical regions have very high rates of immigration and very low rates of extinction.

55) Zoonotic disease _____.
A) Is caused by suborganismal pathogens such as viruses, viroids, and prions only.
B) Is caused by pathogens that are transferred from other animals to humans by direct contact or by means of a vector.
C) Can only be spread from animals to humans through direct contact.
D) Can only be transferred from animals to humans by means of an intermediate host.

56) Why is a pathogen generally more virulent in a new habitat?
A) Intermediate host species are more motile and transport pathogens to new areas.
B) Pathogens evolve more efficient forms of reproduction in new environments.
C) Hosts in new environments have not had a chance to become resistant to the pathogen through natural selection.
D) New environments are almost always smaller in area so that transmission of pathogens is easily accomplished between hosts.

57) Which of the following best describes the consequences of white-band disease in Caribbean coral reefs?
A) Staghorn coral is decimated by the pathogen, and Elkhorn coral takes its place.
B) Key habitat for lobsters, snappers, and other reef fishes improves.
C) Algal species take the place of the dead coral, and the fish community is dominated by herbivores.
D) Algal species take over and the overall reef diversity increases due to increases in primary productivity.

58) Which of the following studies would shed light on the mechanism of spread of H5N1 virus from Asia?
A) Perform cloacal or saliva smears of migrating waterfowl to monitor whether any infected birds show up in Alaska.
B) Test fecal samples for H5N1 in Asian waterfowl that live near domestic poultry farms in Asia.
C) Test for the presence of H5N1 in poultry used for human consumption worldwide.
D) Locate and destroy birds infected with H5N1 in Asian open-air poultry markets.

59) In terms of community ecology, why are pathogens often more virulent now than in the past?
A) More new pathogens have recently evolved.
B) Host organisms have become more susceptible because of weakened immune systems.
C) Human activities are transporting pathogens into new habitats (or communities) at an unprecedented rate.
D) Medicines for treating pathogenic disease are in short supply.

60) The oak tree fungal pathogen, *Phytophthora ramorum*, has migrated eight hundred kilometers in fifteen years. West Nile virus spread from New York State to forty -six other states in five years. The difference in the rate of spread is probably related to _____.
A) The lethality of each pathogen.
B) The mobility of their hosts.
C) The fact that viruses are very small.
D) Innate resistance.

61) Scientists interested in how populations interact within communities are attempting to determine the species diversity of an island under study. What kind of data would be most helpful to the scientists in determining diversity?
A) The number of different species on the island and the size of the population of each species.
B) The number of species on the island that are consumers, producers, and decomposers.
C) The relative biomass of each species on the island separated by trophic level.
D) The number of trophic levels on the island and the niche of each species.

62) Red-cheeked salamanders are partially protected from predators because of cardiac glycosides they produce from glands on their back. When ingested, cardiac glycosides disrupt normal heart rhythms. A different salamander species, the imitator salamander, also has red cheek patches, but does not produce cardiac glycosides. It does gain protection from predators that have learned to avoid red cheeked salamanders. How does this relationship affect the population dynamics of both species?
A) Both species are negatively affected.
B) Both species are positively affected.
C) The red cheeked salamander is positively affected, the imitator is negatively affected.
D) The red cheeked salamander is negatively affected, the imitator is positively affected.

CHAPTER 55:

ECOSYSTEMS AND RESTORATION ECOLOGY

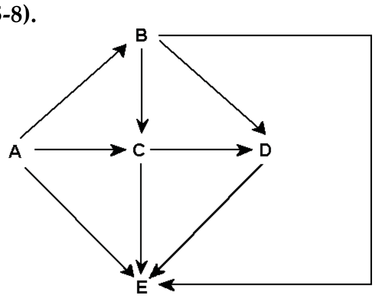
1) If the sun were to suddenly stop providing energy to Earth, most ecosystems would vanish. Which of the following ecosystems would likely survive the longest after this hypothetical disaster?
A) Tropical rain forest.
B) Tundra.
C) Benthic ocean.
D) Desert.

2) Which of the following terms encompasses all of the others?
A) Heterotrophs.
B) Herbivores.
C) Carnivores.
D) Primary consumers.

3) To recycle nutrients, an ecosystem must have, at a minimum, _____.
A) Producers.
B) Producers and decomposers.
C) Producers, primary consumers, and decomposers.
D) Producers, primary consumers, secondary consumers, and decomposers.

4) Which of the following is an example of an ecosystem?
A) All of the brook trout in a 500-square-hectare river drainage system.
B) The plants, animals, and decomposers that inhabit an alpine meadow.
C) The intricate interactions of the various plant and animal species on a savanna during a drought.
D) All of the organisms and their physical environment in a tropical rain forest.

Use the figure given below to answer the following questions (5-8).



Food web for a particular terrestrial ecosystem (arrows represent energy flow and letters represent species)

5) Examine this food web for a particular terrestrial ecosystem. Which species is autotrophic?
A) A.
B) C.
C) D.
D) E.

6) Examine this food web for a particular terrestrial ecosystem. Which species is most likely a decomposer on this food web?
A) A.
B) B.
C) C.
D) E.

7) Examine this food web for a particular terrestrial ecosystem. Species C is toxic to predators. Which species is most likely to benefit from being a mimic of C?
A) A.
B) B.
C) C.
D) E.

8) Examine this food web for a particular terrestrial ecosystem. Which pair of species could be omnivores?
A) A and D.
B) B and C.
C) C and D.
D) C and E.

9) If the figure above represents a marine food web, the smallest organism might be _____.

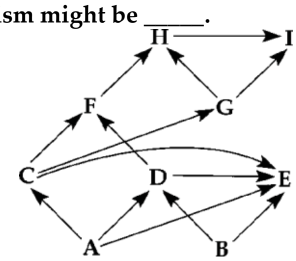


Diagram of a food web (arrows represent energy flow and letters represent species)

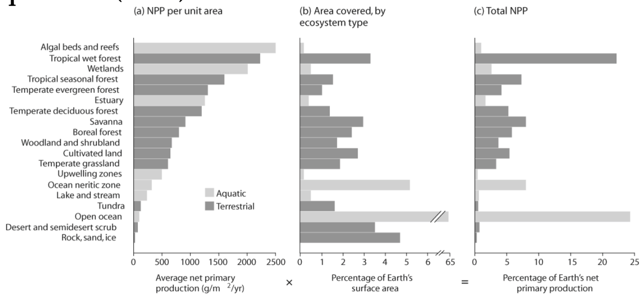
A) A.
B) C.
C) I.
D) E.

10) Which of the following organisms is INCORRECTLY paired with its trophic level?
A) Cyanobacterium - primary producer.
B) Grasshopper - primary consumer.
C) Zooplankton - primary producer.
D) Fungus - detritivore.

11) Which of the following has the greatest effect on the rate of chemical cycling in an ecosystem?
A) The ecosystem's rate of primary production.
B) The production efficiency of the ecosystem's consumers.
C) The rate of decomposition in the ecosystem.
D) The trophic efficiency of the ecosystem.

12) Matter is gained or lost in ecosystems. How it occurs?
A) Chemoautotrophic organisms can convert matter to energy.
B) Matter can be moved from one ecosystem to another.
C) Photosynthetic organisms convert solar energy to sugars.
D) Heterotrophs convert heat to energy.

Use the figure given below to answer the following questions (13-17).



13) Which habitat types in the figure below cover the largest area?

A) Tropical wet forest plus the ocean neritic zone.
B) Open ocean.
C) Algal beds and reefs plus the ocean neritic zone.
D) Wetlands plus the ocean neritic zone.

14) Which habitat type in the figure below makes available the most new tissue to consumers?

A) Tropical wet forest.
B) Open ocean.
C) Algal beds and reefs.
D) Wetlands.

15) Which category in the figure above makes available the highest productivity per square meter?

A) Tropical wet forest.
B) Open ocean.
C) Algal beds and reefs.
D) Wetlands.

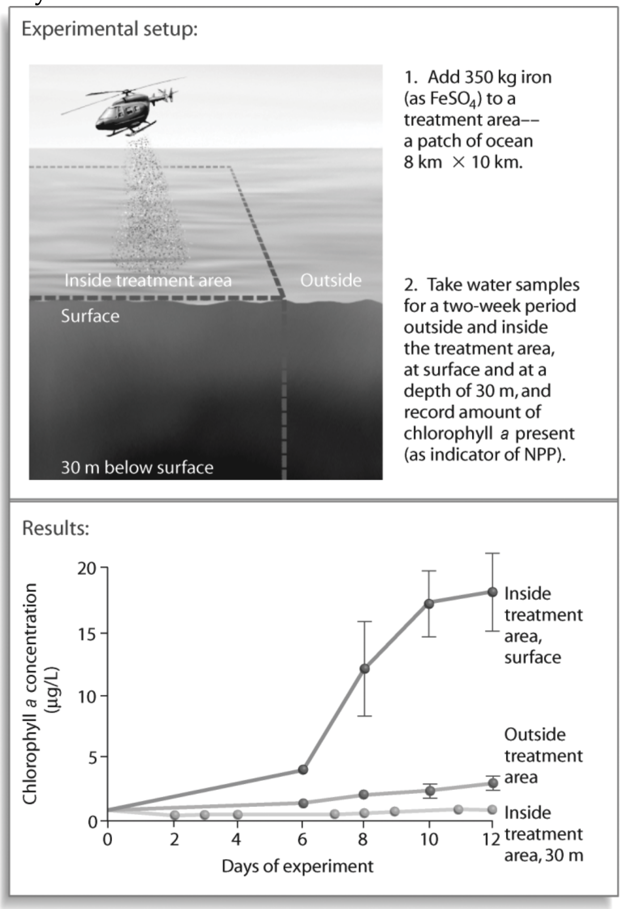
16) Considering its total area covered, which ecosystem type represented in the figure above has a very low level of economic impact on Earth's ecosystem?

A) Tropical wet forest.
B) Rock, sand, and ice.
C) Tropical seasonal forest.
D) Ocean neritic zone.

17) Why is terrestrial productivity higher in equatorial climates?

A) Productivity increases with temperature.
B) Productivity increases with water availability.
C) Productivity increases with available sunlight.
D) The answer is most likely a combination of the other responses.

18) After looking at the experiment in the figure below, what can be said about productivity in marine ecosystems?



A) Nothing can be said based on this information.
B) Marine organisms break down iron for energy and thus for productivity.
C) Iron can be a limiting nutrient in productivity.
D) Productivity increases when chlorophyll a is added.

19) During a year, plants never use 100% of the incoming solar radiation for photosynthesis. What is a reasonable explanation for this?
I) Plants cannot photosynthesize as well during winter (in cold winter climates).
II) Plants cannot photosynthesize as well on cloudy days.
III) The pigments that drive photosynthesis respond to only a fraction of the wavelengths that are available.

A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

20) You own three hundred acres of patchy temperate forest. Which one of the following actions would increase the net primary productivity of the area the most?

A) Adding fertilizer to the entire area.
B) Introducing one hundred rabbits into the area.
C) Planting five hundred new trees.
D) Relocating all of the deer found in the area.

Use the following information to answer the questions (21-22) below.

Abstract:
Increased radiative forcing is an inevitable part of global climate change, yet little is known of its potential effects on the energy fluxes in natural ecosystems. To simulate the conditions of global warming, we exposed peat monoliths (depth, 0.6 cm; surface area, 2.1 m 2) from a bog and fen in northern Minnesota, USA, to three infrared (IR) loading (ambient, +45, and +90 W m-2) and three water table (-16, -20, and -29 cm in bog and -1, -10 and -18 cm in fen) treatments, each replicated in three mesocosm plots. Net radiation (Rn) and soil energy fluxes at the top, bottom, and sides of the mesocosms were measured in 1999, five years after the treatments had begun. Soil heat flux (G) increased proportionately with IR loading, comprising about 3%-8% of Rn. In the fen, the effect of IR loading on G was modulated by water table depth, whereas in the bog, it was not. Energy dissipation from the mesocosms occurred mainly via vertical exchange with air, as well as the deeper soil layers through the bottom of the mesocosms, whereas lateral fluxes were 10- to 20-fold smaller and independent of IR loading and water table depth. The exchange with deeper soil layers was sensitive to water table depth, in contrast to G, which responded primarily to IR loading. The qualitative responses in the bog and fen were similar, but the fen displayed wider seasonal variations and greater extremes in soil energy fluxes. The differences of G in the bog and fen are attributed to differences in the reflectance in the long waveband as a function of vegetation type, whereas the differences in soil heat storage may also depend on different soil properties and different water table depth at comparable treatments. These data suggest that the ecosystem-dependent controls over soil energy fluxes may provide an important constraint on biotic response to climate change. Copyright © 2004 Springer-Verlag (A. Noormets et al. 2004. The effects of infrared loading and water table on soil energy fluxes in northern peatlands. Ecosystems 7:573-582.)

21) The Noormets et al. study (2004) shows that there was an ecosystem -specific control over soil energy fluxes, and this constrained the biotic response to climate change. How do you think radiative heat would affect the water table in a wetland versus a temperate forest?
A) The wetland would likely absorb less heat than the temperate forest and, therefore, not significantly change water table depth.
B) The wetland would likely absorb more heat than the temperate forest and significantly change water table depth.
C) The temperate forest would likely absorb more heat than the wetland and significantly change water table depth.
D) Both areas would absorb similar amounts of radiative heat and, therefore, affect the water table equally.

22) Once heat is transferred to the soil, where does it go next (reference the study by Noormets et al. 2004)?
I) The heat is emitted back to the atmosphere.
II) The heat is transferred to other soil layers.
III) The heat is stored in the soil.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

23) Suppose you are studying the nitrogen cycling in a pond ecosystem over the course of a month. While you are collecting data, a flock of one hundred Canada geese lands and spends the night during a fall migration. What could you do to eliminate error in your study as a result of this event?
A) Find out how much nitrogen is consumed in plant material by a Canada goose over about a twelve-hour period, multiply this number by 100, and add that amount to the total nitrogen in the ecosystem.
B) Find out how much nitrogen is eliminated by a Canada goose over about a twelve-hour period, multiply this number by 100, and subtract that amount from the total nitrogen in the ecosystem.
C) Find out how much nitrogen is consumed and eliminated by a Canada goose over about a twelve-hour period and multiply this number by 100; enter this +/- value into the nitrogen budget of the ecosystem.
D) Put a net over the pond so that no more migrating flocks can land on the pond and alter the nitrogen balance of the pond.

24) As big as it is, the ocean is nutrient-limited. If you wanted to investigate this phenomenon, one reasonable approach would be to ______.
A) Observe Antarctic Ocean productivity from year to year to see if it changes.
B) Experimentally enrich some areas of the ocean and compare their productivity to that of untreated areas.
C) Compare nutrient concentrations between the photic zone and the benthic zone in various marine locations.
D) Contrast nutrient uptake by autotrophs in marine locations that are different temperatures.

25) If you applied a fungicide to a cornfield, what would you expect to happen to the rate of decomposition and net ecosystem production (NEP)?
A) Both decomposition rate and NEP would decrease.
B) Both decomposition rate and NEP would increase.
C) Decomposition rate would increase and NEP would decrease.
D) Decomposition rate would decrease and NEP would increase.

26) Which of the following is a true statement regarding mineral nutrients in soils and their implication for primary productivity?
A) Globally, phosphorous availability is most limiting to primary productivity.
B) Adding a nonlimiting nutrient will stimulate primary productivity.
C) Phosphorous is sometimes unavailable to producers due to leaching.
D) Alkaline soils are more productive than acidic soils.

27) Why is net primary production (NPP) a more useful measurement to an ecosystem ecologist than gross primary production (GPP)?
A) NPP can be expressed in energy/unit of area/unit of time.
B) NPP can be expressed in terms of carbon fixed by photosynthesis for an entire ecosystem.
C) NPP represents the stored chemical energy that is available to consumers in the ecosystem.
D) NPP shows the rate at which the standing crop is utilized by consumers.

28) How is net ecosystem production (NEP) typically estimated in ecosystems?
A) The amount of heat energy released by the ecosystem.
B) The net flux of carbon dioxide or oxygen in or out of an ecosystem.
C) The rate of decomposition by detritivores.
D) The annual total of incoming solar radiation per unit of area.

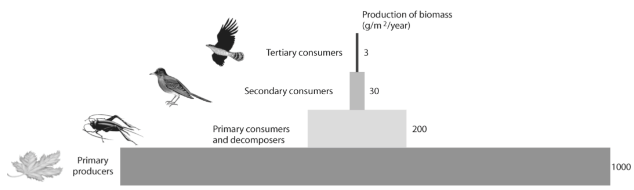
29) Which of the following ecosystems would likely have the largest net primary productivity per hectare and why?
A) Open ocean, because of the total biomass of photosynthetic autotrophs.
B) Grassland, because of the small standing crop biomass that results from consumption by herbivores and rapid decomposition.
C) Tundra, because of the incredibly rapid period of growth during the summer season.
D) Cave, due to the lack of photosynthetic autotrophs.

30) How is it that satellites can detect differences in primary productivity on Earth?
A) Satellite instruments can detect reflectance patterns of the photosynthetic organisms of different ecosystems.
B) Sensitive satellite instruments can measure the amount of NADPH (nicotinamide adenine dinucleotide phosphate) produced in the summative light reactions of different ecosystems.
C) Satellites compare the wavelengths of light captured and reflected by photoautotrophs in different ecosystems.
D) Satellites detect differences by measuring the amount of water vapor emitted by transpiring producers.

31) Which one of the following correctly ranks these organisms in order from lowest to highest percent in production efficiency?
A) Mammals, fish, insects.
B) Insects, fish, mammals.
C) Fish, insects, mammals.
D) Mammals, insects, fish.

32) Owls eat rats, mice, shrews, and small birds. Assume that, over a period of time, an owl consumes 5000 J of animal material. The owl loses 2300 J in feces and owl pellets and uses 2500 J for cellular respiration. What is the production efficiency of this owl?
A) 0.02%.
B) 0.2%.
C) 4%.
D) 40%.

33) After looking at the figure below, what can be said about productivity in this ecosystem?



A) Nothing can be said based on this information.
B) Between 80% and 90% of the energy is lost between most trophic levels.
C) Between 10% and 20% of the energy is lost between most trophic levels.
D) Productivity increases with each trophic level.

34) How does inefficient transfer of energy among trophic levels result in the typically high endangerment status of many top-level predators?
A) Top-level predators are destined to have small populations that are sparsely distributed.
B) Predators have relatively large population sizes.
C) Predators are more disease-prone than animals at lower trophic levels.
D) Top-level predators are more likely to be stricken with parasites.

35) Why does a vegetarian leave a smaller ecological footprint than an omnivore?
A) Fewer animals are slaughtered for human consumption.
B) There is an excess of plant biomass in all terrestrial ecosystems.
C) Vegetarians need to ingest less chemical energy than omnivores.
D) Eating meat is an inefficient way of acquiring photosynthetic productivity.

36) For most terrestrial ecosystems, pyramids composed of species abundances, biomass, and energy are similar in that they have a broad base and a narrow top. The primary reason for this pattern is that _____.
A) Secondary consumers and top carnivores require less energy than producers.
B) At each step, energy is lost from the system.
C) Biomagnification of toxic materials limits the secondary consumers and top carnivores.
D) Top carnivores and secondary consumers have a more general diet than primary producers.

37) Which of the following is primarily responsible for limiting the number of trophic levels in most ecosystems?
A) Many primary and higher-order consumers are opportunistic feeders.
B) Decomposers compete with higher-order consumers for nutrients and energy.
C) Nutrient cycling rates tend to be limited by decomposition.
D) Energy transfer between trophic levels is usually less than 20 percent efficient.

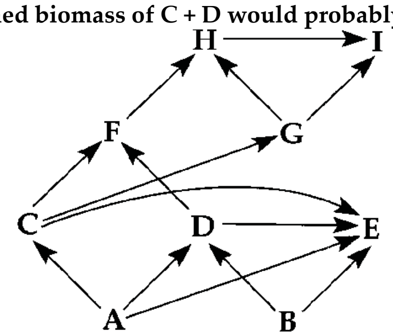
38) Which trophic level is most vulnerable to extinction?
A) Producer level.
B) Primary consumer level.
C) Secondary consumer level.
D) Tertiary consumer level.

39) Which statement best describes what ultimately happens to the chemical energy that is not converted to new biomass in the process of energy transfer between trophic levels in an ecosystem?
A) It is undigested and winds up in the feces and is not passed on to higher trophic levels.
B) It is used by organisms to maintain their life processes through the reactions of cellular respiration.
C) Heat produced by cellular respiration is used by heterotrophs for thermoregulation.
D) It is eliminated as feces or is dissipated into space as heat, consistent with the second law of thermodynamics.

40) Consider the food chain of grass → grasshopper → mouse → snake → hawk. About how much of the chemical energy fixed by photosynthesis of the grass (100 percent) is available to the hawk?
A) 0.01%.
B) 0.1%.
C) 1%.
D) 10%.

41) If the flow of energy in an arctic ecosystem goes through a simple food chain, perhaps involving humans, starting from phytoplankton to zooplankton to fish to seals to polar bears, then which of the following could be true?
A) Polar bears can provide more food for humans than seals can.
B) The total biomass of the fish is lower than that of the seals.
C) Seal populations are larger than fish populations.
D) Fish can potentially provide more food for humans than seal meat.

42) If the figure below represents a terrestrial food web, the combined biomass of C + D would probably be _____.

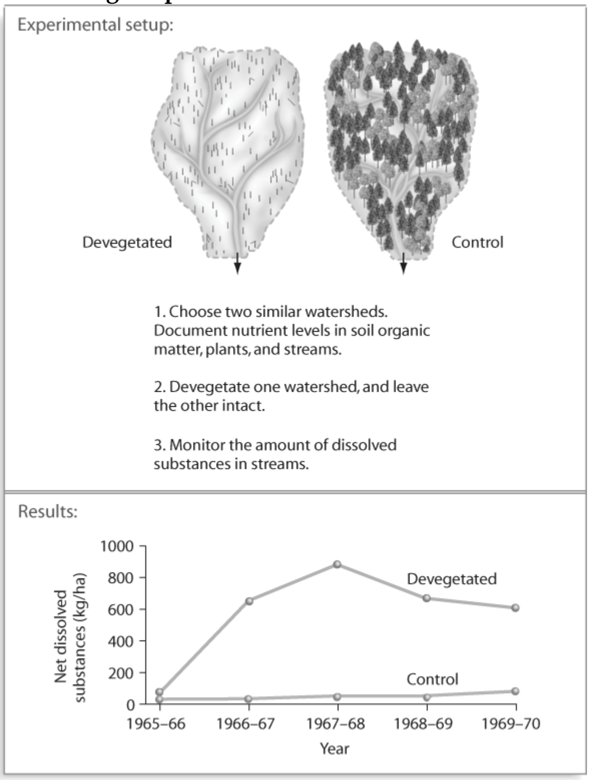


A) Greater than the biomass of A.
B) Greater than the biomass of B.
C) Less than the biomass of A+B.
D) Less than the biomass of E.

43) A porcupine eats 3000 J of plant material. Of this, 2100 J is indigestible and is eliminated as feces, 800 J are used in cellular respiration, and 100 J are used for growth and reproduction. What is the approximate production efficiency of this animal?
A) 0.03%.
B) 3%.
C) 10%.
D) 33%.

44) What do researchers typically focus on when they study a particular biogeochemical cycle?
I) The nature and size of the reservoirs.
II) The rate of element movement between reservoirs.
III) Interaction of the current cycle with other cycles.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

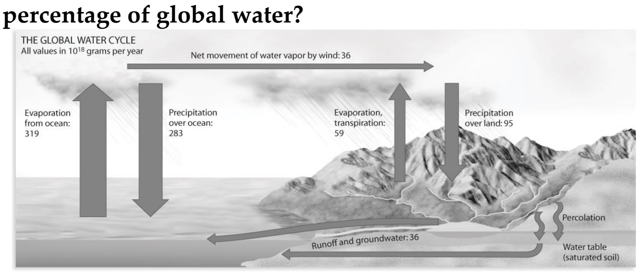
45) Based on the experiment in the figure below, which of the following are plausible reasons for the result?



I) No nutrients evaporate now that vegetation is absent.
II) Nutrients dissolve in the water running through the watershed.
III) Nutrients are attached to small particles of sand or clay that leave the watershed.
IV) Plant roots that held soil particles in place are no longer there.

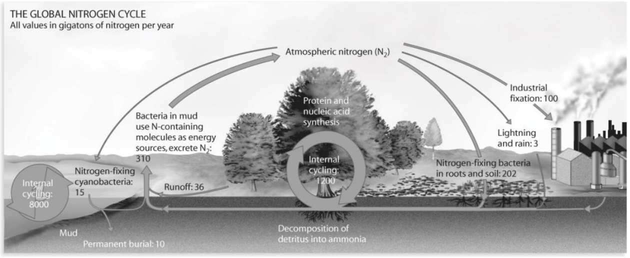
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

46) Consider the global water cycle depicted in the figure below. Which one of the reserves contains the smallest percentage of global water?



A) Rivers and lakes.
B) Polar ice caps.
C) Glaciers.
D) Atmosphere.

47) Consider the global nitrogen cycle depicted in the figure below. What is the limiting portion of the cycle for plants?



A) Industrial nitrogen fixation.
B) Nitrogen lost to the atmosphere.
C) Internal nitrogen cycling in the oceans.
D) Nitrogen fixation by bacteria.

48) Consider the global nitrogen cycle depicted in the figure above. How are humans altering this cycle?

A) Industrial nitrogen fixation
B) Nitrogen lost to the atmosphere
C) Reduction of nitrogen available to terrestrial ecosystems
D) Reduction of nitrogen fixation by bacteria.

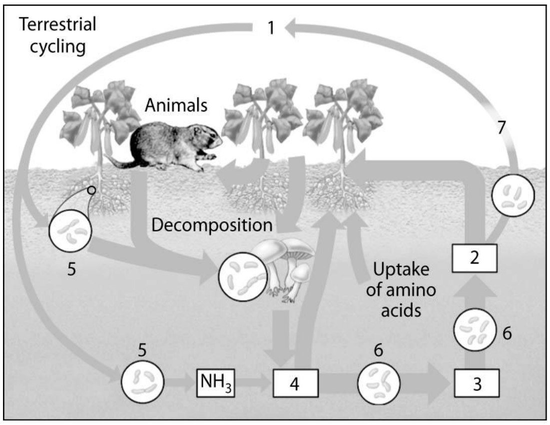
49) Which of the following locations are major reservoirs for carbon for the carbon cycle?
I) Atmosphere.
II) Sediments and sedimentary rocks.
III) Fossilized plant and animal remains (coal, oil, and natural gas).
IV) Plant and animal biomass.

A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) I, II, III, and IV.

50) Which of the following statements is correct about biogeochemical cycling?

A) The phosphorus cycle involves the recycling of atmospheric phosphorus.
B) The phosphorus cycle involves the weathering of rocks.
C) The carbon cycle has maintained a constant atmospheric concentration of carbon dioxide for the past million years.
D) The nitrogen cycle involves movement of diatomic nitrogen between the biotic and abiotic components of the ecosystem.

Use the following diagram to answer the questions (51-53) below.



51) On the diagram of the nitrogen cycle, which number represents nitrite (NO_2)?
A) 1.
B) 2.
C) 3.
D) 4.

52) On the diagram of the nitrogen cycle, which number represents the ammonium ion (NH_4^+)?
A) 1.
B) 2.
C) 3.
D) 4.

53) Nitrifying bacteria participate in the nitrogen cycle mainly by _____.
A) Converting nitrogen gas to ammonia.
B) Releasing ammonium from organic compounds, thus returning it to the soil.
C) Converting ammonium to nitrate, which plants absorb.
D) Incorporating nitrogen into amino acids and organic compounds.

54) The Hubbard Brook watershed deforestation experiment revealed that _____.
I) Deforestation increased water runoff.
II) Nitrate concentration in waters draining the deforested area became dangerously high.
III) Calcium levels remained high in the soil of deforested areas.
A) Only I.
B) Only II.
C) Only III.
D) Only I and II.

55) Why do logged tropical rain forest soils typically have nutrient-poor soils?
A) Tropical bedrock contains little phosphorous.
B) Logging results in soil temperatures that are lethal to nitrogen-fixing bacteria.
C) Most of the nutrients in the ecosystem are removed in the harvested timber.
D) The cation exchange capacity of the soil is reversed as a result of logging.

56) How can biodiversity affect the way we decontaminate industrial sites?
I) Bacteria have been found to be able to detoxify certain chemicals; perhaps there are more.
II) Trees produce sawdust, which can be used to soak up chemicals.
III) Species evolving in contaminated areas could adapt and detoxify the area.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

57) The first step in ecosystem restoration is to _____.
A) Restore the physical structure.
B) Restore native species that have been extirpated due to disturbance.
C) Remove competitive invasive species.
D) Remove toxic pollutants.

58) The goal of restoration ecology is to _____.
A) Replace a ruined ecosystem with a more suitable ecosystem for that area.
B) Return degraded ecosystems to a more natural state.
C) Manage competition between species in human-altered ecosystems.
D) Prevent further degradation by protecting an area with park status.

59) The discipline that applies ecological principles to returning degraded ecosystems to a more natural state is known as _____.
A) Landscape ecology.
B) Conservation ecology.
C) Restoration ecology.
D) Resource conservation.

60) Which of the following would be considered an example of bioremediation?
A) Adding nitrogen-fixing microorganisms to a degraded ecosystem to increase nitrogen availability.
B) Using a bulldozer to regrade a strip mine.
C) Dredging a river bottom to remove contaminated sediments.
D) Adding seeds of a chromium-accumulating plant to soil contaminated by chromium.

61) Acid precipitation lowered the pH of soil in a terrestrial ecosystem that supported a diverse community of plants and animals. The decrease in pH eliminated all nitrogen fixing bacteria populations in the area. Which prediction have on the community?

A) Since phosphorus can replace nitrogen as an essential nutrient, the impact will be minimal.

B) Plants can obtain the nitrogen necessary for growth from the atmosphere, but bacterial communities will be negatively impacted.

C) Primary producers will suffer from nitrogen deficiency and the entire community will experience a decrease in carrying capacity.

D) The decrease in pH actually increases the availability of soil nutrients, so other nutrients that were less available cause an increase in primary production and an increase in biomass at other trophic levels.

CHAPTER 56:

CONSERVATION BIOLOGY AND GLOBAL CHANGE

1) Philippe Bouchet and colleagues conducted a massive survey of marine molluscs on the west coast of New Caledonia. Twenty percent of the species found were represented by a single specimen. What does that suggest about the diversity of molluscs in this area?
A) The west coast of New Caledonia is not an appropriate habitat for molluscs.
B) Many of the species from this 20 percent are probably rare.
C) They were not sampling uniformly throughout the area.
D) Many of the species from this 20 percent are most likely just dispersing through the area.

2) If all individuals in the last remaining population of a particular bird species were all highly related, which type of diversity would be of greatest concern when planning to keep the species from going extinct?
I) Genetic diversity.
II) Species diversity.
III) Ecosystem diversity.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

3) What is the biological significance of genetic diversity between populations?
A) Genes for traits conferring an advantage to local conditions make microevolution possible.
B) The population that is most fit would survive by competitive exclusion.
C) Genetic diversity allows for species stability by preventing speciation.
D) Diseases and parasites are not spread between separated populations.

4) With regard to the destruction of tropical forests, the focus is often on biodiversity and the impact to these ecosystems. What is a direct benefit to humans that helps explain why these forests need to be preserved?
A) This diversity could contain undocumented insect species.
B) Natural and undisturbed areas are important wildlife habitats.
C) The diversity could contain novel drugs for consumers.
D) The plant diversity provides shade, which lowers global warming.

5) Ecosystem services include processes that increase the quality of the abiotic environment. Which of the following processes would fall under this category?
I) Keystone predators have a marked effect on species diversity.
II) Green plants produce the oxygen we breathe.
III) The presence of land plants builds soil.
IV) The presence of diverse wetlands helps in flood control.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

6) During the inventory of bacterial genes present in the Sargasso Sea in the middle of the Atlantic Ocean, a research team concluded that at least 1800 bacterial species were discovered. Based on what you know about this area, what would you expect to see in coral reef waters?
A) Slightly greater genetic diversity.
B) Slightly smaller genetic diversity.
C) Markedly greater genetic diversity.
D) Markedly smaller genetic diversity.

7) Erwin and Scott used an insecticidal fog to knock down insects from the top of a *L. seemannii* tree. The researchers identified over 900 species of beetles among the individuals that fell. Erwin also projected that this entire tree is host to about 600 arthropod species unique to this tree species. There are approximately 50,000 species of tropical trees. Although it could not be entirely accurate, what would be the best way to estimate the total number of arthropod species?
A) Estimate the species density and then multiply by 50,000.
B) Multiply 900 by 50,000.
C) Multiply 600 by 50,000.
D) Divide 900 by 600 and then multiply by 50,000.

8) Which of the following statements regarding extinction is (are) correct?
I) Only a small percentage of species is immune from extinction.
II) Extinction occurs whether humans interfere or not.
III) Extinctions can even be caused indirectly by humans.
A) Only I.
B) Only II.
C) Only III.
D) Only II and III.

9) Which of the following ecological locations has the greatest species diversity?
A) Deciduous forests.
B) Tropical rain forest.
C) Grasslands.
D) Islands.

10) Which of the following provides the best evidence of a biodiversity crisis?
A) The incursion of a non-native species.
B) Increasing pollution levels.
C) High rate of extinction.
D) Climate change.

11) Which of the following terms includes all of the others?
A) Species diversity.
B) Biodiversity.
C) Genetic diversity.
D) Ecosystem diversity.

12) To better comprehend the magnitude of current extinctions, it will be necessary to _____.
A) Differentiate between plant extinction and animal extinction numbers.
B) Focus on identifying more species of mammals and birds.
C) Identify more of the yet unknown species of organisms on Earth.
D) Use the average extinction rates of vertebrates as a baseline.

13) We should care about loss in biodiversity in other species because of _____.
I) Potential loss of medicines and other products yet undiscovered from threatened species.
II) Potential loss of genes, some of which may code for proteins useful to humans.
III) The risk to global ecological stability.
A) Only I.
B) Only II.
C) Only II and III.
D) I, II, and III.

14) Which of the following is the most direct threat to biodiversity?
A) Increased levels of atmospheric carbon dioxide.
B) The depletion of the ozone layer.
C) Overexploitation of selected species.
D) Habitat destruction.

15) Introduced species can have deleterious effects on biological communities by _____.
I) Preying on native species.
II) Competing with native species for food or light.
III) Displacing native species.
IV) Competing with native species for space or breeding/nesting habitat.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) I, II, III and IV.

16) Overharvesting encourages extinction and is most likely to affect _____.
A) Animals that occupy a broad ecological niche.
B) Large animals with low intrinsic reproductive rates.
C) Most organisms that live in the oceans.
D) Edge-adapted species.

17) Of the following ecosystem types, which have been impacted the most by humans?
A) Wetland and riparian.
B) Desert and high alpine.
C) Taiga and second-growth forests.
D) Tundra and arctic.

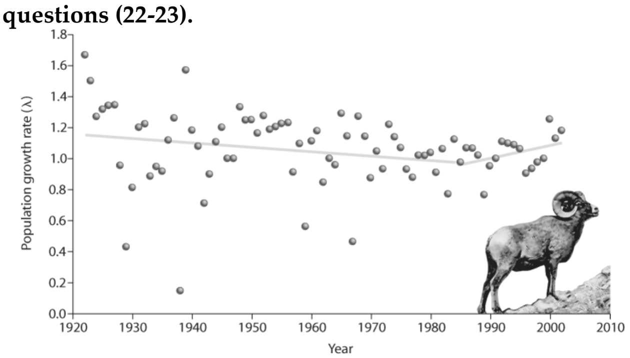
18) Which of the following is a type of work a *conservation* biologist would be involved?
I) Reestablishing whooping cranes in their former breeding grounds in North Dakota.
II) Studying species diversity and interaction in the Florida Everglades, past and present.
III) Studying population ecology of grizzly bears in Yellowstone National Park.
IV) Determining the effects of hunting white-tailed deer in Vermont.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) I, II, III and IV.

19) Burning fossil fuels releases oxides of sulfur and nitrogen. These air pollutants can be responsible for _____.
I) The death of fish in lakes.
II) Precipitation with a pH as low as 3.0.
III) Eutrophication of lakes.
A) Only I.
B) Only II.
C) Only III.
D) Only I and II.

20) Suppose you attend a town meeting at which some experts tell the audience that they have performed a cost-benefit analysis of a proposed transit system that would probably reduce overall air pollution and fossil fuel consumption. The analysis, however, reveals that ticket prices will not cover the cost of operating the system when fuel, wages, and equipment are taken into account. As a biologist, you know that if ecosystem services had been included in the analysis, the experts might have arrived at a different answer. Why are ecosystem services rarely included in economic analyses?
A) Their cost is difficult to estimate and people take them for granted.
B) They are not worth much and are usually not cost effective.
C) Ecosystem services only take into account abiotic factors that affect local environments.
D) Federal laws of the United States exclude their inclusion in any cost benefit analysis.

21) Researchers have been studying a rare population of eighty -seven voles in an isolated area. Ten voles from a larger population were added to this isolated population. Besides having ten additional animals, what benefits are there to importing individuals?
A) Additional animals from a distant population will likely bring genetic diversity and reduce inbreeding depression.
B) Additional animals will bring additional competition and could hurt the population.
C) Additional animals would increase beneficial genetic drift.
D) There is no benefit other than increasing the overall population size.

Use the figure given below to answer the following questions (22-23).



22) Five new individuals were added to this small population in 1986 and ten more were added between 1990-1994. According to the figure above, what occurred in this population after these additions?
A) The population increased exponentially.
B) The population increased in overall numbers.
C) The population growth rate increased.
D) The population continued to decline.

23) According to the figure above, what is the LEAST likely explanation for the data after 1985?
A) Emigration.
B) Immigration.
C) Introduction of new alleles into the population.
D) Increased resources in the area.

24) On Easter Island, data show that it was once covered by massive palm trees. How can an ecosystem collapse from removal of just one species of large tree?
I) Without large trees, soil erosion increases and reduces productivity.
II) Species of plants needing shade no longer have it.
III) Large trees are habitats for many species.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

25) Which of the following criteria have to be met for a species to qualify as invasive?
A) Endemic to the area, spreads rapidly, and displaces foreign species.
B) Introduced to a new area, spreads rapidly, and displaces native species.
C) Introduced to a new area, spreads rapidly, and displaces other invasive species.
D) Endemic to the area, spreads slowly, and displaces native species.

26) Which of the following conditions is the most likely indicator of a population in an extinction vortex?
A) The species in question is found only in small, stable pockets of its former range.
B) The effective population size of the species falls below 1000.
C) Genetic measurements indicate a loss of genetic variation over time.
D) The population is connected only by corridors.

27) What strategy was used to rescue Illinois prairie chickens from a recent extinction vortex?
A) Determining the minimum viable population size by taking into account the effective population size.
B) Establishing a nature reserve to protect its habitat.
C) Introducing individuals from other populations to increase genetic variation.
D) Reducing the population size of its predators and competitors.

28) If the sex ratio in a population is significantly different from 50:50, then which of the following will always be true?
A) The population will enter the extinction vortex.
B) The genetic variation in the population will increase over time.
C) The effective population size will be greater than the actual population size.
D) The effective population size will be less than the actual population size.

29) Which of the following life history traits can potentially influence effective population size?
I) Age of sexual maturation.
II) Genetic relatedness among individuals in a population.
III) Family size and population size.
IV) Gene flow between geographically separated populations.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) I, II, III and IV.

30) The primary difference between the small-population approach (S-PA) and the declining-population approach (D-PA) to biodiversity recovery is _____.

A) S-PA is interested in bolstering the genetic diversity of a threatened population rather than the environmental factors that caused the population's decline.
B) S-PA applies for conservation biologists when population numbers fall below 500.
C) D-PA would likely involve bringing together individuals from scattered small populations to interbreed in order to promote genetic diversity.
D) S-PA would investigate and eliminate all of the human impacts on the habitat of the species being studied for recovery.

31) What problem with red-cockaded woodpecker populations in the southeastern United States was addressed by habitat intervention?

A) The only habitat that can support their recovery is large tracts of mature oak forest.
B) The mature pine forests in which they live cannot ever be subjected to forest fire.
C) All of the appropriate red-cockaded woodpecker habitat has already been logged or converted to agricultural land.
D) The social organization of the red-cockaded woodpecker precluded the dispersal of reproductive individuals.

32) Managing southeastern forests specifically for the red-cockaded woodpecker _____.

A) Required the growth of a dense understory of trees and shrubs.
B) Contributed to greater abundance and diversity of other forest bird species.
C) Caused other species of songbird to decline.
D) Involved strict fire suppression measures.

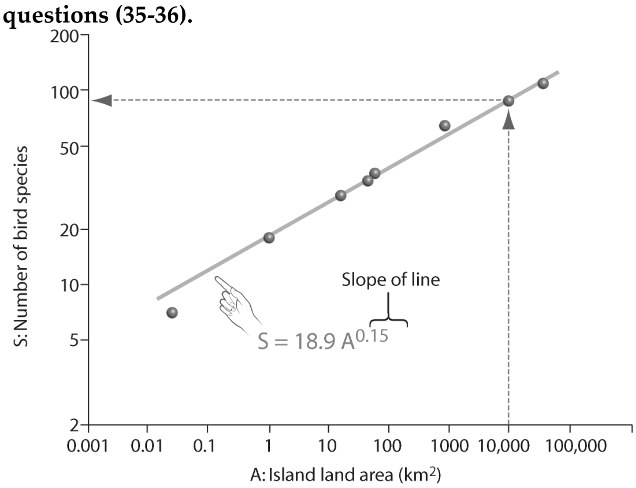
33) Which of the following strategies would most rapidly increase the genetic diversity of a population in an extinction vortex?

A) Capture all remaining individuals in the population for captive breeding followed by reintroduction to the wild.
B) Establish a reserve that protects the population's habitat.
C) Introduce new individuals transported from other populations of the same species.
D) Sterilize the least fit individuals in the population.

34) Which one of the following is LEAST likely to be a hotspot for birds?

A) Amazon River basin.
B) East Africa.
C) Southwest China.
D) Greenland.

Use the figure given below to answer the following questions (35-36).



35) In looking at the species—area plot in the figure above, what can be concluded?

A) Number of bird species increases linearly with island area.
B) Number of bird species remains the same on these various islands.
C) Diversity is independent from island area.
D) Number of bird species increases exponentially with island area.

36) Based on the species-area plot in the figure above, if habitable area on an island were reduced from 10,000 km2 (square kilometers) to 1000 km2, roughly what percentage of the species would disappear?

A) 0.3 percent.
B) 3 percent.
C) 30 percent.
D) 60 percent.

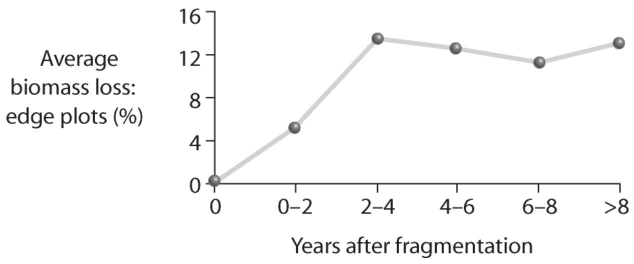
37) The main purpose of wildlife (movement) corridors is to _____.

A) Slow down the introduction of new individuals of a species.
B) Slowly introduce a species to a new preserve.
C) Create more edge habitat.
D) Connect two otherwise isolated populations.

38) A land developer and several ecologists are discussing how a parcel of private land should be developed while saving twenty hectares as natural habitat. The land developer suggests that the twenty hectares be divided into twenty separate one -hectare areas. The ecologists suggest that it would be better to have one intact parcel of twenty hectares. What is the significance of these different arrangements of the twenty hectares?

A) There really is no difference; they should both work equally well.
B) The isolated hectare plots increase the ability of individuals to disperse from one habitat to another.
C) The isolated plots are more vulnerable to edge effects.
D) The large plot will create more inbreeding in many species.

39) Looking at the figure below, what can be said about edge effects?



A) Biomass declines along edges of forest fragments.
B) Biomass increases along the edges of forest fragments.
C) Species diversity decreases along the edges of forest fragments.
D) Fragmentation does not affect biomass.

40) Which of the following is generally true about the current research regarding forest fragmentation?
A) Fragmented forests promote biodiversity because they result in the combination of forest-edge species and forest-interior species.
B) In fragmented forests, the number of forested-adapted species tends to decline and the number of edge species tends to increase.
C) Fragmented forests are the goal of conservation biologists who design wildlife preserves.
D) The disturbance of timber extraction causes the species diversity to increase because of the new habitats created.

41) Brown-headed cowbirds utilize fragmented forests effectively by _____.
A) Feeding on the fruits of shrubs that tend to grow at the forest/open -field interface.
B) Parasitizing the nests of forest birds and feeding on open-field insects.
C) Roosting in forest trees and nesting in grassy fields.
D) Outcompeting other songbird species for access to nesting holes in old-growth trees.

42) Movement (wildlife) corridors can be harmful to certain species because they _____.
A) Increase inbreeding.
B) Spread disease and parasites.
C) Increase genetic diversity.
D) Allow seasonal migration.

43) Locating new nature preserves in biodiversity hot spots may not necessarily the best choice because _____.
A) Hot spots are situated in remote areas not accessible to the public.
B) Their ecological importance makes land purchase very expensive.
C) A hot spot helps conserve only a few species.
D) Changing environmental conditions may shift the location of the communities.

44) What is the biggest problem with selecting a site for a preserve?
A) Making a proper selection is difficult because currently the environmental conditions of almost any site can change quickly.
B) Keystone species are difficult to identify in potential preserve sites.
C) Only lands that are not useful to human activities are available for preserves.
D) Most of the best sites are inaccessible by land transportation, so making roads to them is often prohibitively expensive.

45) What is a critical load?
A) The amount of nutrient augmentation necessary to bring a depleted habitat back to its former level.
B) The level of a given toxin in an ecosystem that is lethal to 50 percent of the species present.
C) The amount of added nutrient that can be absorbed by plants without damaging ecosystem integrity.
D) The number of predators an ecosystem can support that effectively culls prey populations to healthy levels.

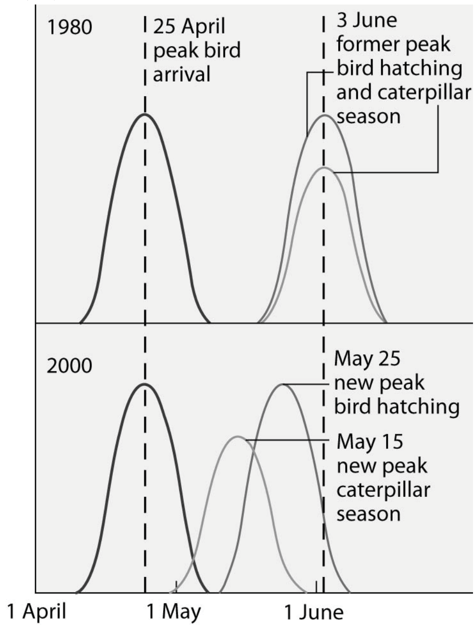
46) Agricultural lands frequently require nutrient augmentation because _____.
A) Nitrogen-fixing bacteria are not as plentiful in agricultural soils because of the use of pesticides.
B) The nutrients that become the biomass of plants are not cycled back to the soil on lands where they are harvested.
C) Land that is available for agriculture tends to be nutrient-poor.
D) Cultivation of agricultural land inhibits the decomposition of organic matter.

To answer the following questions (47-48), choose the term that matches the description.

47) This is caused by excessive nutrient runoff into aquatic ecosystems.
A) Depletion of ozone layer.
B) Acid precipitation.
C) Biological magnification.
D) Eutrophication.

48) This causes extremely high levels of toxic chemicals in fish-eating birds.
A) Acid precipitation.
B) Biological magnification.
C) Greenhouse effect.
D) Eutrophication.

Use the following graph and information to answer the question (49) below.



Flycatcher birds that migrate from Africa to Europe feed their nestlings a diet that is almost exclusively moth caterpillars. The graph below shows the mean dates of arrival, bird hatching, and peak caterpillar season for the years 1980 and 2000.

49) The shift in the peak of caterpillar season is most likely due to _____.
A) Earlier migration returns of flycatchers.
B) An innate change in the biological clock of the caterpillars.
C) Global warming.
D) Acid precipitation in Europe.

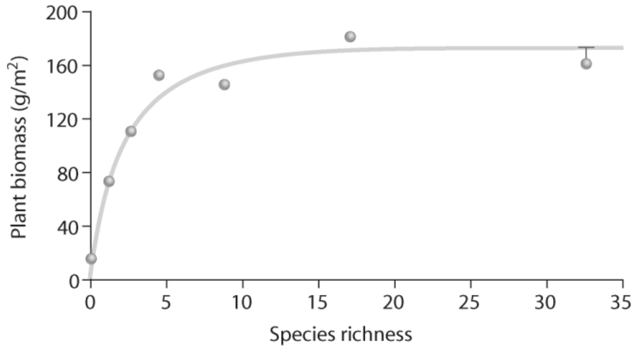
50) Your friend is wary of environmentalists' claims that global warming could lead to major biological change on Earth. Which of the following statements can you use in response to your friend's suspicions?
I) We know that atmospheric carbon dioxide has increased over the past 150 years.
II) Through measurements and observations, we know that carbon dioxide levels and temperature fluctuations are directly correlated, even in prehistoric times.
III) Global warming could have significant effects on agriculture in the United States.
A) Only I.
B) Only II.
C) Only III.
D) I, II, and III.

51) The main cause of the increase in the amount of carbon dioxide in Earth's atmosphere over the past 150 years is __.
A) Increased worldwide primary production.
B) An increase in the amount of infrared radiation absorbed by the atmosphere.
C) The burning of larger amounts of wood and fossil fuels.
D) Additional respiration by the rapidly growing human population.

52) Which of the following is a consequence of biological magnification (biomagnification)?
A) Toxic chemicals in the environment pose greater risk to top-level predators than to primary consumers.
B) Populations of top-level predators are generally smaller than populations of primary consumers.
C) Only a small portion of the energy captured by producers is transferred to consumers.
D) The amount of biomass in the producer level of an ecosystem decreases if the producer turnover time increases.

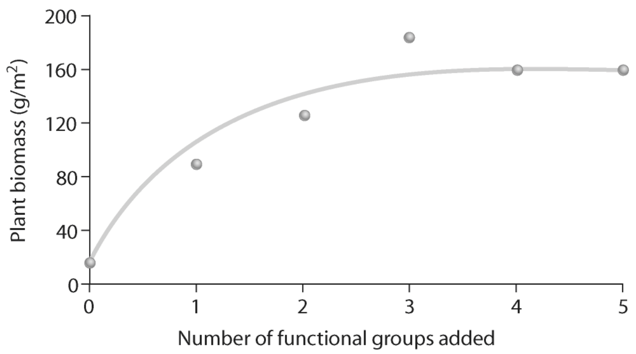
53) Why are changes in the global carbon cycle important?
I) Burning reduces available carbon for primary producers and, therefore, primary consumers.
II) Deforestation and suburbanization reduce an area's net primary productivity.
III) Increasing atmospheric concentrations of carbon dioxide could alter Earth's climate.
IV) By using fossil fuels we are destroying a nonrenewable resource.
A) Only I and III.
B) Only II and IV.
C) Only II, III, and IV.
D) Only I, II, III, and IV.

54) Looking at the figure below, what can you conclude about the data?



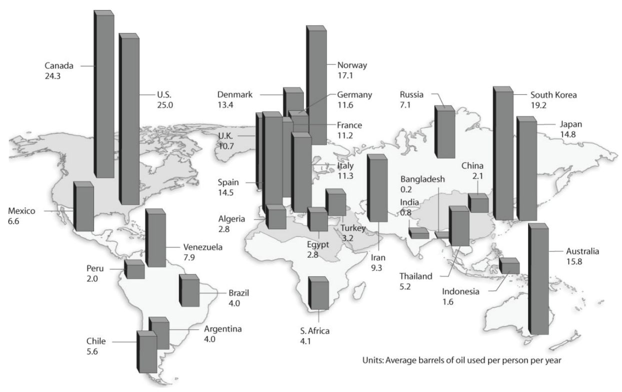
A) As species richness changes, plant biomass remains consistent.
B) As species richness increases, plant biomass increases.
C) As species richness increases, plant biomass increases and then levels off.
D) As species richness decreases, plant biomass increases.

55) Examine the figure below and consider this hypothesis: Plant biomass increases with species richness. In looking at the data in the figure below, how would you relate it to this hypothesis? The hypothesis is _____.



A) Partially supported.
B) Supported.
C) Rejected.
D) None of them.

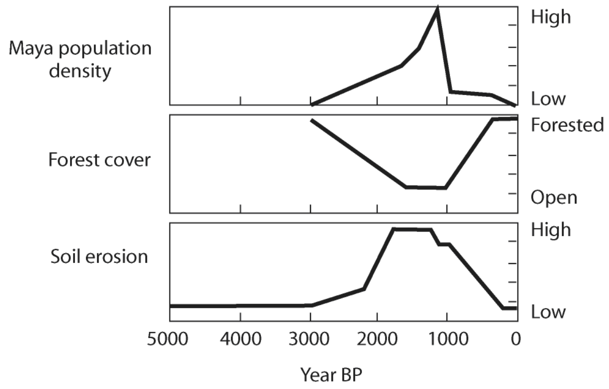
56) Examine the figure below, which notes the average barrels of oil used per person per year in different countries. What can be concluded?



A) Residents in warmer climates use more energy per person.
B) Residents of industrialized countries use more energy per person.
C) Residents of more populated countries use more energy per person.
D) English-speaking countries tend to use more energy per person.

57) The main goal of sustainable development is to _____.
A) Involve more countries in conservation efforts.
B) Use only natural resources in the construction of new buildings.
C) Use natural resources such that they do not decline over time.
D) Reevaluating and re-implementing management plans over time.

58) Based on the figure below, what can you conclude about the history of land use in the southern Yucatán?



A) Massive soil erosion caused the Mayan population to crash.
B) Reduction in forest cover caused the Mayan population to crash.
C) As Mayan population increased, deforestation increased, probably leading to increased soil erosion.
D) This Mayan population practiced sustainable development.

59) Which of the following nations has become a world leader in the establishment of zoned reserves?
A) Costa Rica.
B) China.
C) United States.
D) Mexico.

60) Which of the following statements about protected areas that have been established to preserve biodiversity are correct?
I) About 25 percent of Earth's land area is now protected.
II) National parks are one of many types of protected areas.
III) Management of a protected area should be coordinated with management of the land surrounding the area.
IV) It is especially important to protect biodiversity hot spots.
A) Only I and III.
B) Only II and IV.
C) Only I, II, and IV.
D) Only II, III, and IV.

61) Eutrophication is often caused by excess limiting nutrient runoff from agricultural fields into aquatic ecosystems. These results in massive algal blooms, which eventually die and decompose, ultimately depleting the dissolved oxygen and killing large numbers of fish and other aquatic organism. Predict which of the following human actions would best address the problem of eutrophication near agricultural areas?
A) After each eutrophication event, remove the dead fish and invertebrates to place on agricultural fields instead of fertilizer.
B) Determine which limiting nutrient is responsible for the algal bloom and use other fertilizers to apply to crops.
C) Remove the algae before it dies and decomposes to prevent eutrophication from occurring.
D) Determine critical nutrient loads required for certain crops and do not exceed this amount during fertilizer application.

62) Elevated carbon dioxide levels have been shown to contribute to the greenhouse effect, resulting in an increase in mean global temperature. Ecosystems where the largest warming has already occurred include snow-covered northern coniferous forests, tundra, and arctic sea ice habitats. Which statement best explains how the elimination of ice-covered ecosystems affects the rise or fall in global temperature?
A) Melting ice releases dissolved ozone gas, which adds to the greenhouse effect.
B) More reflective surfaces of ice are replaced with darker, more absorptive surfaces, thereby contributing to the warming trend.
C) Large-scale ice melts actually contribute toward lowering global temperatures by decreasing salinity of the oceans.
D) Carbon dioxide levels are lowered as a result of greater volume of water to accommodate greater dissolved gas.

63) A parasitic fungus, *Geomyces destructans*, has decimated millions of bats in the United States since it was first observed in upstate New York in 2006. The disease has been named White-nose syndrome because of the white fungal hyphae that cover the bat upon infection. It is believed that this fungus was introduced from Europe by into caves with hibernating bat populations. Which prediction most likely reflects changes that will occur in natural communities as a result of massive bat mortality?
A) Increased animal populations as a result of niche availability.
B) Increased rodent populations as a result of an increase in flying insect populations.
C) Increased flying insect populations and decreased populations of bat-pollinated plants.
D) Decreased bird populations as the spread of the fungus infects other closely.

ANSWER KEY

CHAPTER 1			
1	D	41	
2	C	42	
3	C	43	
4	C	44	
5	B	45	
6	C	46	
7	B	47	
8	D	48	
9	A	49	
10	B	50	
11	A	51	
12	D	52	
13	A	53	
14	D	54	
15	C	55	
16	C	56	
17	C	57	
18	C	58	
19	C	59	
20	C	60	
21	D	61	
22	C	62	
23	B	63	
24	C	64	
25	D	65	
26	C	66	
27	C	67	
28	D	68	
29	D	69	
30	C	70	
31	B	71	
32	C	72	
33	C	73	
34	B	74	
35	A	75	
36	A	76	
37	B	77	
38	A	78	
39		79	
40		80	

CHAPTER 2			
1	D	41	B
2	B	42	D
3	C	43	D
4	D	44	A
5	C	45	A
6	C	46	C
7	A	47	C
8	C	48	A
9	B	49	A
10	B	50	D
11	D	51	C
12	A	52	A
13	B	53	A
14	B	54	
15	C	55	
16	A	56	
17	C	57	
18	A	58	
19	D	59	
20	D	60	
21	C	61	
22	D	62	
23	C	63	
24	B	64	
25	C	65	
26	B	66	
27	A	67	
28	C	68	
29	B	69	
30	A	70	
31	B	71	
32	B	72	
33	B	73	
34	A	74	
35	C	75	
36	A	76	
37	C	77	
38	D	78	
39	C	79	
40	C	80	

CHAPTER 3			
1	C	41	A
2	B	42	D
3	B	43	D
4	D	44	C
5	A	45	D
6	C	46	D
7	B	47	A
8	B	48	B
9	D	49	C
10	B	50	B
11	B	51	
12	B	52	
13	C	53	
14	C	54	
15	C	55	
16	A	56	
17	D	57	
18	B	58	
19	D	59	
20	D	60	
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22	C	62	
23	B	63	
24	B	64	
25	C	65	
26	D	66	
27	C	67	
28	B	68	
29	A	69	
30	D	70	
31	A	71	
32	D	72	
33	A	73	
34	A	74	
35	C	75	
36	C	76	
37	C	77	
38	D	78	
39	C	79	
40	A	80	

CHAPTER 4			
1	C	41	C
2	A	42	D
3	D	43	C
4	B	44	A
5	B	45	B
6	D	46	A
7	D	47	C
8	D	48	C
9	C	49	B
10	C	50	A
11	B	51	B
12	A	52	D
13	B	53	A
14	B	54	D
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26	C	66	
27	B	67	
28	B	68	
29	C	69	
30	C	70	
31	C	71	
32	C	72	
33	D	73	
34	B	74	
35	D	75	
36	A	76	
37	D	77	
38	D	78	
39	B	79	
40	B	80	

CHAPTER 5			
1	A	41	B
2	A	42	A
3	C	43	C
4	A	44	A
5	B	45	B
6	C	46	B
7	D	47	B
8	A	48	C
9	A	49	B
10	D	50	C
11	C	51	B
12	A	52	B
13	A	53	D
14	A	54	D
15	B	55	D
16	A	56	B
17	C	57	B
18	C	58	B
19	A	59	C
20	B	60	B
21	A	61	B
22	A	62	
23	B	63	
24	B	64	
25	C	65	
26	B	66	
27	A	67	
28	C	68	
29	A	69	
30	D	70	
31	C	71	
32	C	72	
33	B	73	
34	B	74	
35	B	75	
36	B	76	
37	D	77	
38	D	78	
39	D	79	
40	B	80	

CHAPTER 6			
1	A	41	A
2	C	42	D
3	B	43	C
4	C	44	B
5	A	45	C
6	B	46	A
7	D	47	D
8	C	48	D
9	B	49	A
10	C	50	D
11	B	51	A
12	C	52	C
13	B	53	C
14	C	54	C
15	C	55	D
16	A	56	B
17	C	57	D
18	C	58	A
19	C	59	A
20	C	60	C
21	B	61	C
22	B	62	B
23	B	63	
24	B	64	
25	D	65	
26	B	66	
27	A	67	
28	C	68	
29	B	69	
30	C	70	
31	C	71	
32	D	72	
33	C	73	
34	B	74	
35	A	75	
36	C	76	
37	D	77	
38	A	78	
39	C	79	
40	C	80	

CHAPTER 7			
1	C	41	B
2	A	42	B
3	A	43	B
4	A	44	B
5	B	45	D
6	C	46	B
7	B	47	D
8	C	48	A
9	D	49	D
10	D	50	A
11	D	51	D
12	A	52	B
13	C	53	C
14	B	54	A
15	C	55	A
16	B	56	C
17	A	57	C
18	B	58	A
19	D	59	B
20	A	60	B
21	A	61	
22	D	62	
23	C	63	
24	C	64	
25	C	65	
26	A	66	
27	C	67	
28	B	68	
29	B	69	
30	B	70	
31	B	71	
32	C	72	
33	C	73	
34	A	74	
35	B	75	
36	B	76	
37	C	77	
38	D	78	
39	B	79	
40	D	80	

CHAPTER 8			
1	C	41	C
2	C	42	B
3	B	43	C
4	A	44	B
5	A	45	D
6	B	46	A
7	A	47	B
8	B	48	D
9	C	49	B
10	C	50	C
11	C	51	A
12	A	52	D
13	A	53	B
14	C	54	A
15	D	55	B
16	B	56	B
17	A	57	C
18	D	58	C
19	B	59	A
20	A	60	B
21	B	61	B
22	D	62	C
23	C	63	
24	B	64	
25	D	65	
26	C	66	
27	C	67	
28	C	68	
29	A	69	
30	C	70	
31	B	71	
32	B	72	
33	D	73	
34	B	74	
35	B	75	
36	B	76	
37	C	77	
38	C	78	
39	C	79	
40	B	80	

CHAPTER 9

1	C	41	C
2	B	42	A
3	A	43	A
4	A	44	A
5	B	45	D
6	C	46	A
7	A	47	A
8	B	48	A
9	C	49	C
10	B	50	D
11	D	51	B
12	C	52	A
13	B	53	D
14	B	54	B
15	C	55	B
16	D	56	C
17	D	57	A
18	B	58	C
19	B	59	D
20	B	60	D
21	C	61	B
22	B	62	
23	B	63	
24	C	64	
25	C	65	
26	C	66	
27	C	67	
28	A	68	
29	A	69	
30	A	70	
31	B	71	
32	D	72	
33	B	73	
34	C	74	
35	C	75	
36	C	76	
37	C	77	
38	D	78	
39	B	79	
40	C	80	

CHAPTER 10

1	B	41	B
2	D	42	D
3	C	43	D
4	A	44	A
5	D	45	D
6	D	46	C
7	A	47	B
8	C	48	C
9	C	49	B
10	A	50	B
11	B	51	D
12	C	52	B
13	C	53	B
14	B	54	A
15	C	55	C
16	B	56	C
17	B	57	A
18	A	58	A
19	D	59	B
20	C	60	A
21	C	61	D
22	C	62	
23	B	63	
24	A	64	
25	B	65	
26	C	66	
27	C	67	
28	C	68	
29	C	69	
30	B	70	
31	B	71	
32	C	72	
33	D	73	
34	B	74	
35	D	75	
36	B	76	
37	A	77	
38	B	78	
39	B	79	
40	D	80	

CHAPTER 11

1	C	41	D
2	A	42	C
3	B	43	D
4	C	44	A
5	D	45	D
6	A	46	D
7	C	47	C
8	D	48	C
9	C	49	D
10	B	50	B
11	A	51	C
12	B	52	C
13	A	53	B
14	D	54	C
15	B	55	B
16	B	56	B
17	D	57	A
18	C	58	A
19	B	59	C
20	A	60	D
21	B	61	C
22	B	62	C
23	D	63	
24	C	64	
25	C	65	
26	A	66	
27	B	67	
28	C	68	
29	A	69	
30	C	70	
31	A	71	
32	A	72	
33	C	73	
34	B	74	
35	C	75	
36	D	76	
37	C	77	
38	C	78	
39	A	79	
40	C	80	

CHAPTER 12

1	C	41	B
2	A	42	D
3	C	43	D
4	B	44	C
5	D	45	C
6	A	46	D
7	B	47	A
8	B	48	C
9	D	49	D
10	A	50	B
11	B	51	D
12	C	52	C
13	B	53	B
14	A	54	B
15	D	55	B
16	A	56	C
17	A	57	D
18	B	58	C
19	C	59	C
20	C	60	C
21	D	61	
22	C	62	
23	B	63	
24	C	64	
25	A	65	
26	D	66	
27	C	67	
28	A	68	
29	B	69	
30	C	70	
31	A	71	
32	C	72	
33	D	73	
34	B	74	
35	B	75	
36	B	76	
37	B	77	
38	D	78	
39	C	79	
40	C	80	

CHAPTER 13

1	C	41	B
2	D	42	C
3	B	43	D
4	D	44	C
5	B	45	A
6	B	46	B
7	C	47	B
8	B	48	B
9	B	49	C
10	B	50	D
11	D	51	A
12	B	52	D
13	A	53	A
14	C	54	C
15	A	55	A
16	C	56	C
17	B	57	C
18	A	58	C
19	B	59	A
20	D	60	C
21	C	61	D
22	B	62	
23	A	63	
24	C	64	
25	D	65	
26	C	66	
27	A	67	
28	D	68	
29	B	69	
30	C	70	
31	B	71	
32	D	72	
33	A	73	
34	D	74	
35	A	75	
36	B	76	
37	A	77	
38	D	78	
39	C	79	
40	D	80	

CHAPTER 14

1	B	41	C
2	B	42	B
3	B	43	A
4	C	44	D
5	B	45	D
6	A	46	C
7	D	47	C
8	C	48	A
9	D	49	C
10	D	50	A
11	C	51	A
12	D	52	C
13	C	53	C
14	A	54	B
15	B	55	C
16	D	56	B
17	C	57	C
18	C	58	B
19	C	59	D
20	A	60	D
21	B	61	A
22	C	62	B
23	C	63	
24	D	64	
25	D	65	
26	B	66	
27	B	67	
28	D	68	
29	C	69	
30	A	70	
31	D	71	
32	C	72	
33	D	73	
34	B	74	
35	C	75	
36	D	76	
37	D	77	
38	B	78	
39	C	79	
40	D	80	

CHAPTER 15

1	B	41	C
2	B	42	D
3	D	43	A
4	C	44	A
5	C	45	B
6	D	46	D
7	D	47	C
8	A	48	A
9	A	49	B
10	B	50	C
11	C	51	A
12	C	52	C
13	A	53	B
14	A	54	C
15	B	55	A
16	C	56	A
17	D	57	D
18	B	58	A
19	A	59	
20	C	60	
21	A	61	
22	B	62	
23	A	63	
24	A	64	
25	A	65	
26	B	66	
27	D	67	
28	C	68	
29	A	69	
30	D	70	
31	C	71	
32	C	72	
33	A	73	
34	A	74	
35	C	75	
36	B	76	
37	A	77	
38	B	78	
39	C	79	
40	C	80	

CHAPTER 16

1	A	41	C
2	D	42	B
3	D	43	B
4	B	44	D
5	C	45	B
6	A	46	B
7	A	47	A
8	C	48	C
9	D	49	A
10	D	50	D
11	C	51	B
12	B	52	B
13	B	53	A
14	D	54	A
15	B	55	C
16	C	56	D
17	C	57	C
18	C	58	B
19	D	59	A
20	C	60	
21	D	61	
22	C	62	
23	A	63	
24	A	64	
25	D	65	
26	A	66	
27	B	67	
28	D	68	
29	D	69	
30	D	70	
31	D	71	
32	B	72	
33	C	73	
34	A	74	
35	B	75	
36	C	76	
37	A	77	
38	B	78	
39	B	79	
40	D	80	

CHAPTER 17			
1	D	41	D
2	A	42	A
3	A	43	B
4	B	44	C
5	C	45	C
6	C	46	A
7	D	47	C
8	D	48	D
9	B	49	B
10	D	50	D
11	C	51	B
12	B	52	B
13	B	53	C
14	D	54	A
15	B	55	C
16	C	56	C
17	D	57	A
18	B	58	D
19	B	59	B
20	B	60	B
21	C	61	
22	A	62	
23	C	63	
24	B	64	
25	B	65	
26	C	66	
27	D	67	
28	C	68	
29	B	69	
30	B	70	
31	D	71	
32	D	72	
33	B	73	
34	B	74	
35	D	75	
36	C	76	
37	A	77	
38	B	78	
39	A	79	
40	B	80	

CHAPTER 18			
1	D	41	D
2	D	42	D
3	A	43	B
4	A	44	C
5	A	45	C
6	C	46	C
7	B	47	B
8	C	48	C
9	C	49	B
10	B	50	A
11	C	51	C
12	D	52	C
13	D	53	A
14	D	54	C
15	C	55	A
16	A	56	D
17	B	57	A
18	D	58	B
19	D	59	
20	C	60	
21	C	61	
22	C	62	
23	D	63	
24	C	64	
25	D	65	
26	B	66	
27	C	67	
28	A	68	
29	D	69	
30	A	70	
31	B	71	
32	D	72	
33	D	73	
34	A	74	
35	C	75	
36	B	76	
37	A	77	
38	B	78	
39	B	79	
40	A	80	

CHAPTER 19			
1	D	41	D
2	D	42	B
3	A	43	B
4	B	44	A
5	C	45	C
6	D	46	A
7	C	47	D
8	C	48	A
9	C	49	C
10	C	50	
11	D	51	
12	A	52	
13	C	53	
14	D	54	
15	C	55	
16	D	56	
17	B	57	
18	C	58	
19	D	59	
20	C	60	
21	B	61	
22	A	62	
23	B	63	
24	B	64	
25	C	65	
26	B	66	
27	A	67	
28	B	68	
29	B	69	
30	B	70	
31	B	71	
32	B	72	
33	B	73	
34	B	74	
35	A	75	
36	D	76	
37	A	77	
38	A	78	
39	D	79	
40	C	80	

CHAPTER 20			
1	A	41	C
2	B	42	A
3	C	43	A
4	D	44	A
5	C	45	D
6	A	46	C
7	B	47	D
8	C	48	D
9	A	49	C
10	B	50	D
11	B	51	A
12	C	52	C
13	D	53	
14	B	54	
15	A	55	
16	D	56	
17	D	57	
18	C	58	
19	B	59	
20	D	60	
21	C	61	
22	A	62	
23	A	63	
24	C	64	
25	B	65	
26	C	66	
27	A	67	
28	B	68	
29	D	69	
30	C	70	
31	C	71	
32	B	72	
33	B	73	
34	A	74	
35	D	75	
36	D	76	
37	A	77	
38	D	78	
39	B	79	
40	A	80	

CHAPTER 21			
1	D	41	C
2	A	42	A
3	C	43	C
4	B	44	
5	A	45	
6	A	46	
7	D	47	
8	C	48	
9	A	49	
10	D	50	
11	A	51	
12	D	52	
13	C	53	
14	D	54	
15	C	55	
16	C	56	
17	C	57	
18	B	58	
19	D	59	
20	A	60	
21	B	61	
22	A	62	
23	A	63	
24	D	64	
25	B	65	
26	C	66	
27	D	67	
28	A	68	
29	D	69	
30	A	70	
31	C	71	
32	C	72	
33	D	73	
34	B	74	
35	D	75	
36	A	76	
37	B	77	
38	D	78	
39	A	79	
40	C	80	

CHAPTER 22			
1	C	41	A
2	B	42	C
3	D	43	B
4	A	44	B
5	A	45	B
6	A	46	D
7	B	47	A
8	C	48	A
9	B	49	B
10	B	50	D
11	C	51	C
12	A	52	A
13	D	53	B
14	C	54	B
15	D	55	A
16	A	56	A
17	C	57	D
18	A	58	
19	C	59	
20	D	60	
21	A	61	
22	B	62	
23	A	63	
24	D	64	
25	A	65	
26	C	66	
27	A	67	
28	C	68	
29	B	69	
30	C	70	
31	D	71	
32	C	72	
33	C	73	
34	D	74	
35	D	75	
36	A	76	
37	B	77	
38	C	78	
39	B	79	
40	D	80	

CHAPTER 23			
1	A	41	A
2	A	42	A
3	B	43	D
4	D	44	C
5	B	45	B
6	C	46	C
7	D	47	C
8	D	48	C
9	C	49	C
10	C	50	A
11	D	51	D
12	C	52	C
13	D	53	A
14	D	54	B
15	C	55	D
16	C	56	D
17	C	57	B
18	D	58	B
19	A	59	D
20	B	60	C
21	D	61	
22	A	62	
23	D	63	
24	D	64	
25	D	65	
26	D	66	
27	A	67	
28	B	68	
29	C	69	
30	A	70	
31	D	71	
32	D	72	
33	D	73	
34	D	74	
35	A	75	
36	D	76	
37	D	77	
38	D	78	
39	B	79	
40	D	80	

CHAPTER 24			
1	D	41	C
2	A	42	C
3	C	43	C
4	C	44	C
5	D	45	D
6	D	46	B
7	A	47	C
8	C	48	B
9	B	49	A
10	C	50	B
11	D	51	C
12	D	52	D
13	C	53	A
14	B	54	D
15	A	55	D
16	C	56	C
17	D	57	C
18	C	58	D
19	C	59	D
20	B	60	A
21	B	61	
22	D	62	
23	D	63	
24	D	64	
25	C	65	
26	A	66	
27	C	67	
28	A	68	
29	A	69	
30	A	70	
31	A	71	
32	D	72	
33	B	73	
34	C	74	
35	D	75	
36	A	76	
37	C	77	
38	D	78	
39	C	79	
40	A	80	

CHAPTER 25

1	C	41	C
2	B	42	C
3	C	43	C
4	B	44	B
5	D	45	D
6	A	46	C
7	B	47	C
8	D	48	A
9	B	49	D
10	D	50	A
11	D	51	B
12	D	52	A
13	B	53	B
14	D	54	C
15	C	55	A
16	B	56	C
17	C	57	D
18	B	58	B
19	A	59	C
20	A	60	B
21	A	61	C
22	A	62	D
23	A	63	
24	A	64	
25	C	65	
26	A	66	
27	C	67	
28	A	68	
29	D	69	
30	C	70	
31	D	71	
32	C	72	
33	B	73	
34	C	74	
35	C	75	
36	D	76	
37	C	77	
38	C	78	
39	D	79	
40	C	80	

CHAPTER 26

1	A	41	D
2	C	42	A
3	D	43	D
4	C	44	C
5	C	45	B
6	B	46	D
7	D	47	D
8	C	48	D
9	C	49	C
10	D	50	D
11	D	51	D
12	A	52	C
13	A	53	D
14	B	54	A
15	A	55	B
16	C	56	B
17	D	57	B
18	C	58	B
19	A	59	A
20	D	60	D
21	D	61	
22	C	62	
23	D	63	
24	A	64	
25	B	65	
26	A	66	
27	D	67	
28	B	68	
29	D	69	
30	D	70	
31	D	71	
32	A	72	
33	D	73	
34	B	74	
35	D	75	
36	D	76	
37	A	77	
38	C	78	
39	C	79	
40	B	80	

CHAPTER 27

1	D	41	A
2	A	42	A
3	D	43	C
4	C	44	A
5	C	45	B
6	D	46	D
7	C	47	D
8	C	48	C
9	C	49	D
10	D	50	B
11	D	51	B
12	A	52	B
13	C	53	C
14	B	54	C
15	C	55	C
16	C	56	B
17	C	57	B
18	A	58	D
19	C	59	C
20	C	60	B
21	D	61	D
22	A	62	D
23	D	63	A
24	D	64	
25	C	65	
26	D	66	
27	B	67	
28	D	68	
29	C	69	
30	A	70	
31	C	71	
32	C	72	
33	A	73	
34	C	74	
35	D	75	
36	B	76	
37	A	77	
38	D	78	
39	A	79	
40	C	80	

CHAPTER 28

1	A	41	A
2	A	42	D
3	C	43	B
4	B	44	A
5	C	45	A
6	D	46	C
7	B	47	A
8	B	48	D
9	C	49	D
10	C	50	B
11	C	51	D
12	D	52	B
13	A	53	A
14	C	54	B
15	B	55	D
16	A	56	B
17	C	57	C
18	A	58	D
19	C	59	
20	B	60	
21	B	61	
22	C	62	
23	A	63	
24	B	64	
25	A	65	
26	C	66	
27	D	67	
28	A	68	
29	C	69	
30	C	70	
31	A	71	
32	C	72	
33	B	73	
34	B	74	
35	C	75	
36	C	76	
37	C	77	
38	A	78	
39	A	79	
40	A	80	

CHAPTER 29			
1	B	41	B
2	B	42	C
3	D	43	A
4	A	44	B
5	C	45	B
6	B	46	D
7	D	47	C
8	B	48	B
9	D	49	D
10	D	50	A
11	C	51	A
12	A	52	C
13	A	53	D
14	D	54	D
15	D	55	B
16	C	56	C
17	D	57	C
18	B	58	C
19	B	59	D
20	A	60	B
21	C	61	
22	C	62	
23	B	63	
24	D	64	
25	B	65	
26	B	66	
27	A	67	
28	D	68	
29	B	69	
30	C	70	
31	B	71	
32	B	72	
33	D	73	
34	C	74	
35	C	75	
36	D	76	
37	D	77	
38	D	78	
39	B	79	
40	D	80	

CHAPTER 30			
1	C	41	C
2	C	42	D
3	D	43	C
4	C	44	C
5	C	45	A
6	D	46	C
7	B	47	A
8	D	48	D
9	C	49	B
10	A	50	C
11	C	51	A
12	D	52	C
13	A	53	B
14	C	54	D
15	A	55	A
16	A	56	B
17	D	57	C
18	D	58	B
19	D	59	A
20	D	60	B
21	A	61	
22	D	62	
23	D	63	
24	B	64	
25	A	65	
26	B	66	
27	A	67	
28	C	68	
29	A	69	
30	C	70	
31	A	71	
32	A	72	
33	B	73	
34	B	74	
35	C	75	
36	D	76	
37	D	77	
38	A	78	
39	D	79	
40	B	80	

CHAPTER 31			
1	A	41	A
2	C	42	A
3	A	43	D
4	B	44	D
5	A	45	A
6	A	46	B
7	C	47	C
8	B	48	A
9	B	49	B
10	D	50	B
11	C	51	A
12	D	52	A
13	B	53	B
14	D	54	B
15	D	55	C
16	A	56	C
17	A	57	B
18	D	58	C
19	B	59	C
20	C	60	A
21	A	61	
22	C	62	
23	B	63	
24	D	64	
25	B	65	
26	D	66	
27	D	67	
28	A	68	
29	B	69	
30	D	70	
31	D	71	
32	B	72	
33	C	73	
34	A	74	
35	B	75	
36	A	76	
37	C	77	
38	A	78	
39	D	79	
40	C	80	

CHAPTER 32			
1	B	41	C
2	B	42	D
3	B	43	D
4	C	44	D
5	B	45	B
6	B	46	C
7	D	47	B
8	A	48	B
9	D	49	B
10	C	50	D
11	D	51	C
12	B	52	B
13	D	53	B
14	C	54	B
15	C	55	C
16	D	56	C
17	A	57	A
18	B	58	C
19	D	59	B
20	C	60	B
21	A	61	
22	B	62	
23	D	63	
24	C	64	
25	D	65	
26	C	66	
27	B	67	
28	B	68	
29	A	69	
30	B	70	
31	A	71	
32	C	72	
33	A	73	
34	A	74	
35	B	75	
36	A	76	
37	D	77	
38	C	78	
39	B	79	
40	A	80	

CHAPTER 33			
1	D	41	C
2	A	42	C
3	A	43	A
4	A	44	B
5	C	45	D
6	D	46	D
7	C	47	C
8	B	48	D
9	B	49	A
10	C	50	C
11	D	51	C
12	D	52	B
13	A	53	A
14	C	54	D
15	D	55	A
16	B	56	D
17	D	57	B
18	D	58	A
19	D	59	C
20	B	60	B
21	C	61	
22	A	62	
23	A	63	
24	C	64	
25	D	65	
26	A	66	
27	C	67	
28	D	68	
29	D	69	
30	B	70	
31	A	71	
32	A	72	
33	A	73	
34	C	74	
35	A	75	
36	C	76	
37	C	77	
38	C	78	
39	B	79	
40	B	80	

CHAPTER 34			
1	B	41	A
2	A	42	C
3	D	43	A
4	B	44	B
5	C	45	C
6	A	46	A
7	D	47	D
8	A	48	C
9	D	49	D
10	B	50	A
11	A	51	B
12	C	52	C
13	A	53	C
14	D	54	C
15	C	55	A
16	D	56	D
17	C	57	B
18	D	58	D
19	C	59	D
20	B	60	A
21	B	61	
22	A	62	
23	C	63	
24	C	64	
25	B	65	
26	B	66	
27	A	67	
28	B	68	
29	C	69	
30	A	70	
31	A	71	
32	D	72	
33	B	73	
34	D	74	
35	C	75	
36	A	76	
37	D	77	
38	C	78	
39	C	79	
40	A	80	

CHAPTER 35			
1	B	41	A
2	C	42	D
3	B	43	B
4	D	44	A
5	B	45	B
6	C	46	C
7	C	47	B
8	C	48	B
9	D	49	A
10	A	50	B
11	B	51	C
12	A	52	A
13	B	53	C
14	B	54	D
15	A	55	B
16	B	56	D
17	C	57	C
18	D	58	C
19	B	59	A
20	D	60	
21	C	61	
22	D	62	
23	C	63	
24	D	64	
25	B	65	
26	A	66	
27	D	67	
28	A	68	
29	A	69	
30	A	70	
31	C	71	
32	A	72	
33	D	73	
34	B	74	
35	C	75	
36	C	76	
37	B	77	
38	C	78	
39	D	79	
40	C	80	

CHAPTER 36			
1	A	41	B
2	B	42	B
3	D	43	A
4	B	44	A
5	B	45	C
6	B	46	A
7	D	47	B
8	D	48	D
9	B	49	D
10	D	50	D
11	A	51	D
12	A	52	A
13	C	53	D
14	B	54	
15	A	55	
16	C	56	
17	C	57	
18	D	58	
19	B	59	
20	B	60	
21	C	61	
22	D	62	
23	C	63	
24	B	64	
25	C	65	
26	C	66	
27	B	67	
28	A	68	
29	D	69	
30	D	70	
31	D	71	
32	A	72	
33	A	73	
34	B	74	
35	C	75	
36	D	76	
37	B	77	
38	A	78	
39	B	79	
40	C	80	

CHAPTER 37

1	B	41	C
2	C	42	C
3	B	43	
4	C	44	
5	B	45	
6	D	46	
7	B	47	
8	D	48	
9	C	49	
10	D	50	
11	C	51	
12	C	52	
13	C	53	
14	C	54	
15	C	55	
16	B	56	
17	C	57	
18	D	58	
19	B	59	
20	A	60	
21	D	61	
22	C	62	
23	B	63	
24	C	64	
25	C	65	
26	D	66	
27	B	67	
28	D	68	
29	D	69	
30	B	70	
31	C	71	
32	D	72	
33	D	73	
34	B	74	
35	D	75	
36	A	76	
37	B	77	
38	C	78	
39	B	79	
40	D	80	

CHAPTER 38

1	D	41	A
2	D	42	C
3	D	43	A
4	B	44	B
5	C	45	D
6	D	46	D
7	C	47	B
8	C	48	D
9	B	49	A
10	A	50	D
11	C	51	C
12	A	52	A
13	B	53	B
14	B	54	
15	C	55	
16	D	56	
17	A	57	
18	B	58	
19	C	59	
20	C	60	
21	B	61	
22	D	62	
23	C	63	
24	C	64	
25	B	65	
26	C	66	
27	D	67	
28	C	68	
29	C	69	
30	B	70	
31	A	71	
32	A	72	
33	D	73	
34	B	74	
35	C	75	
36	C	76	
37	A	77	
38	C	78	
39	C	79	
40	A	80	

CHAPTER 39

1	C	41	B
2	D	42	B
3	C	43	C
4	D	44	A
5	A	45	D
6	D	46	C
7	B	47	B
8	D	48	D
9	B	49	D
10	B	50	A
11	B	51	D
12	B	52	B
13	D	53	
14	B	54	
15	B	55	
16	A	56	
17	D	57	
18	D	58	
19	C	59	
20	D	60	
21	D	61	
22	B	62	
23	B	63	
24	A	64	
25	A	65	
26	C	66	
27	B	67	
28	B	68	
29	B	69	
30	D	70	
31	A	71	
32	B	72	
33	D	73	
34	D	74	
35	A	75	
36	A	76	
37	A	77	
38	D	78	
39	B	79	
40	D	80	

CHAPTER 40

1	C	41	D
2	A	42	A
3	B	43	B
4	B	44	C
5	A	45	D
6	B	46	C
7	B	47	A
8	D	48	C
9	C	49	C
10	A	50	A
11	D	51	D
12	D	52	C
13	B	53	D
14	A	54	D
15	C	55	D
16	D	56	A
17	A	57	D
18	D	58	D
19	B	59	D
20	A	60	B
21	B	61	C
22	C	62	
23	C	63	
24	C	64	
25	C	65	
26	C	66	
27	C	67	
28	B	68	
29	A	69	
30	C	70	
31	C	71	
32	B	72	
33	B	73	
34	D	74	
35	A	75	
36	C	76	
37	C	77	
38	C	78	
39	D	79	
40	A	80	

CHAPTER 41			
1	D	41	C
2	A	42	C
3	D	43	C
4	B	44	B
5	B	45	B
6	C	46	D
7	B	47	B
8	B	48	A
9	D	49	C
10	A	50	B
11	B	51	B
12	B	52	C
13	B	53	B
14	B	54	A
15	D	55	A
16	B	56	D
17	B	57	C
18	D	58	C
19	C	59	C
20	D	60	
21	B	61	
22	B	62	
23	C	63	
24	B	64	
25	A	65	
26	B	66	
27	C	67	
28	A	68	
29	C	69	
30	D	70	
31	B	71	
32	A	72	
33	C	73	
34	B	74	
35	C	75	
36	D	76	
37	B	77	
38	C	78	
39	C	79	
40	A	80	

CHAPTER 42			
1	B	41	B
2	B	42	A
3	C	43	A
4	B	44	C
5	D	45	A
6	B	46	A
7	D	47	C
8	B	48	A
9	C	49	A
10	D	50	C
11	D	51	D
12	C	52	B
13	A	53	D
14	D	54	C
15	A	55	C
16	B	56	D
17	A	57	A
18	D	58	D
19	D	59	B
20	C	60	C
21	A	61	
22	C	62	
23	A	63	
24	D	64	
25	B	65	
26	A	66	
27	B	67	
28	B	68	
29	C	69	
30	D	70	
31	D	71	
32	D	72	
33	C	73	
34	A	74	
35	B	75	
36	A	76	
37	B	77	
38	C	78	
39	C	79	
40	C	80	

CHAPTER 43			
1	A	41	A
2	C	42	C
3	D	43	C
4	C	44	B
5	A	45	A
6	B	46	D
7	A	47	C
8	B	48	A
9	B	49	A
10	B	50	D
11	A	51	B
12	D	52	B
13	D	53	A
14	C	54	D
15	D	55	D
16	B	56	B
17	D	57	C
18	D	58	B
19	C	59	C
20	B	60	A
21	A	61	B
22	D	62	
23	C	63	
24	A	64	
25	A	65	
26	C	66	
27	C	67	
28	A	68	
29	A	69	
30	C	70	
31	C	71	
32	B	72	
33	C	73	
34	B	74	
35	D	75	
36	D	76	
37	C	77	
38	B	78	
39	D	79	
40	A	80	

CHAPTER 44			
1	B	41	D
2	C	42	D
3	A	43	A
4	A	44	B
5	D	45	A
6	B	46	B
7	C	47	B
8	B	48	C
9	B	49	B
10	B	50	A
11	C	51	B
12	C	52	C
13	D	53	D
14	B	54	D
15	C	55	A
16	A	56	B
17	C	57	B
18	D	58	C
19	A	59	B
20	B	60	A
21	D	61	
22	A	62	
23	C	63	
24	A	64	
25	B	65	
26	C	66	
27	C	67	
28	B	68	
29	B	69	
30	D	70	
31	B	71	
32	C	72	
33	B	73	
34	A	74	
35	A	75	
36	B	76	
37	B	77	
38	B	78	
39	B	79	
40	D	80	

CHAPTER 45

1	D	41	B
2	A	42	D
3	D	43	B
4	D	44	B
5	B	45	B
6	A	46	C
7	C	47	A
8	B	48	C
9	B	49	A
10	A	50	C
11	C	51	C
12	A	52	B
13	A	53	B
14	C	54	C
15	C	55	C
16	B	56	B
17	C	57	A
18	D	58	A
19	D	59	A
20	D	60	C
21	A	61	D
22	B	62	
23	A	63	
24	B	64	
25	C	65	
26	D	66	
27	B	67	
28	A	68	
29	A	69	
30	D	70	
31	B	71	
32	A	72	
33	C	73	
34	D	74	
35	B	75	
36	C	76	
37	D	77	
38	D	78	
39	B	79	
40	A	80	

CHAPTER 46

1	C	41	B
2	A	42	B
3	A	43	B
4	D	44	D
5	D	45	B
6	A	46	A
7	D	47	C
8	A	48	A
9	B	49	C
10	A	50	D
11	D	51	C
12	B	52	D
13	C	53	D
14	B	54	B
15	D	55	D
16	C	56	C
17	B	57	C
18	B	58	B
19	C	59	C
20	B	60	B
21	B	61	
22	A	62	
23	C	63	
24	A	64	
25	A	65	
26	A	66	
27	C	67	
28	C	68	
29	C	69	
30	B	70	
31	D	71	
32	C	72	
33	B	73	
34	D	74	
35	C	75	
36	C	76	
37	D	77	
38	B	78	
39	A	79	
40	C	80	

CHAPTER 47

1	C	41	D
2	A	42	B
3	B	43	B
4	C	44	C
5	A	45	B
6	A	46	B
7	C	47	A
8	A	48	A
9	D	49	D
10	D	50	A
11	B	51	C
12	A	52	D
13	D	53	D
14	B	54	D
15	A	55	C
16	D	56	A
17	C	57	A
18	A	58	A
19	C	59	C
20	C	60	B
21	C	61	D
22	B	62	
23	A	63	
24	B	64	
25	D	65	
26	C	66	
27	D	67	
28	C	68	
29	B	69	
30	C	70	
31	C	71	
32	A	72	
33	A	73	
34	B	74	
35	C	75	
36	A	76	
37	D	77	
38	C	78	
39	C	79	
40	B	80	

CHAPTER 48

1	D	41	D
2	C	42	B
3	B	43	A
4	C	44	C
5	A	45	D
6	A	46	A
7	A	47	B
8	D	48	D
9	D	49	B
10	C	50	A
11	C	51	C
12	D	52	D
13	B	53	D
14	A	54	A
15	B	55	A
16	D	56	C
17	B	57	A
18	C	58	B
19	A	59	C
20	B	60	B
21	B	61	B
22	D	62	A
23	D	63	A
24	A	64	
25	C	65	
26	C	66	
27	B	67	
28	A	68	
29	B	69	
30	D	70	
31	D	71	
32	B	72	
33	B	73	
34	C	74	
35	C	75	
36	C	76	
37	B	77	
38	B	78	
39	C	79	
40	B	80	

CHAPTER 49			
1	C	41	C
2	C	42	A
3	B	43	A
4	A	44	A
5	A	45	C
6	A	46	C
7	D	47	D
8	C	48	A
9	B	49	D
10	A	50	D
11	B	51	C
12	D	52	D
13	A	53	B
14	B	54	B
15	C	55	A
16	D	56	C
17	A	57	A
18	A	58	C
19	C	59	D
20	B	60	D
21	D	61	B
22	B	62	
23	B	63	
24	B	64	
25	D	65	
26	A	66	
27	A	67	
28	C	68	
29	C	69	
30	D	70	
31	B	71	
32	D	72	
33	D	73	
34	C	74	
35	A	75	
36	C	76	
37	D	77	
38	D	78	
39	A	79	
40	C	80	

CHAPTER 50			
1	C	41	B
2	C	42	D
3	B	43	D
4	B	44	B
5	B	45	B
6	A	46	D
7	C	47	D
8	A	48	A
9	C	49	D
10	D	50	C
11	C	51	B
12	A	52	B
13	A	53	B
14	C	54	D
15	B	55	B
16	B	56	C
17	B	57	D
18	D	58	D
19	D	59	C
20	B	60	C
21	D	61	
22	A	62	
23	B	63	
24	B	64	
25	D	65	
26	B	66	
27	A	67	
28	B	68	
29	B	69	
30	A	70	
31	B	71	
32	B	72	
33	A	73	
34	B	74	
35	B	75	
36	C	76	
37	A	77	
38	D	78	
39	D	79	
40	C	80	

CHAPTER 51			
1	A	41	D
2	C	42	B
3	B	43	D
4	B	44	A
5	B	45	D
6	C	46	A
7	A	47	D
8	A	48	D
9	B	49	B
10	C	50	A
11	B	51	A
12	D	52	B
13	D	53	C
14	D	54	A
15	C	55	D
16	B	56	C
17	B	57	D
18	C	58	B
19	A	59	A
20	A	60	B
21	C	61	D
22	D	62	C
23	C	63	
24	D	64	
25	C	65	
26	A	66	
27	B	67	
28	D	68	
29	A	69	
30	C	70	
31	D	71	
32	A	72	
33	B	73	
34	B	74	
35	A	75	
36	A	76	
37	A	77	
38	C	78	
39	D	79	
40	D	80	

CHAPTER 52			
1	D	41	B
2	B	42	C
3	D	43	A
4	A	44	D
5	C	45	A
6	D	46	A
7	B	47	D
8	D	48	C
9	A	49	A
10	D	50	A
11	A	51	A
12	C	52	C
13	A	53	A
14	A	54	B
15	B	55	C
16	C	56	A
17	B	57	C
18	D	58	C
19	A	59	A
20	B	60	B
21	D	61	D
22	D	62	
23	D	63	
24	B	64	
25	A	65	
26	A	66	
27	B	67	
28	A	68	
29	C	69	
30	C	70	
31	C	71	
32	B	72	
33	B	73	
34	B	74	
35	D	75	
36	A	76	
37	A	77	
38	D	78	
39	A	79	
40	C	80	

CHAPTER 53			
1	B	41	D
2	A	42	D
3	A	43	A
4	A	44	C
5	C	45	C
6	C	46	D
7	D	47	D
8	D	48	D
9	D	49	D
10	A	50	A
11	C	51	A
12	B	52	B
13	B	53	B
14	A	54	C
15	C	55	B
16	A	56	C
17	C	57	D
18	B	58	B
19	C	59	B
20	D	60	A
21	C	61	A
22	C	62	
23	C	63	
24	A	64	
25	B	65	
26	C	66	
27	C	67	
28	A	68	
29	C	69	
30	D	70	
31	C	71	
32	C	72	
33	B	73	
34	D	74	
35	D	75	
36	C	76	
37	B	77	
38	B	78	
39	C	79	
40	C	80	

CHAPTER 54			
1	B	41	C
2	D	42	C
3	A	43	D
4	D	44	A
5	D	45	C
6	D	46	C
7	D	47	B
8	B	48	C
9	C	49	C
10	D	50	A
11	D	51	C
12	A	52	A
13	C	53	A
14	A	54	B
15	D	55	B
16	A	56	C
17	B	57	C
18	A	58	A
19	B	59	C
20	A	60	B
21	D	61	A
22	D	62	D
23	D	63	
24	A	64	
25	A	65	
26	C	66	
27	B	67	
28	B	68	
29	C	69	
30	B	70	
31	A	71	
32	A	72	
33	B	73	
34	A	74	
35	D	75	
36	B	76	
37	C	77	
38	B	78	
39	C	79	
40	D	80	

CHAPTER 55			
1	C	41	D
2	A	42	C
3	B	43	B
4	D	44	D
5	A	45	D
6	D	46	D
7	B	47	D
8	D	48	A
9	A	49	D
10	C	50	B
11	C	51	C
12	B	52	D
13	B	53	C
14	B	54	D
15	C	55	C
16	B	56	A
17	D	57	A
18	C	58	B
19	D	59	C
20	C	60	A
21	B	61	C
22	D	62	
23	C	63	
24	B	64	
25	D	65	
26	C	66	
27	C	67	
28	B	68	
29	B	69	
30	C	70	
31	A	71	
32	C	72	
33	B	73	
34	A	74	
35	D	75	
36	B	76	
37	D	77	
38	D	78	
39	D	79	
40	A	80	

CHAPTER 56			
1	B	41	B
2	A	42	B
3	A	43	D
4	C	44	A
5	D	45	C
6	C	46	B
7	C	47	D
8	D	48	B
9	B	49	C
10	C	50	D
11	B	51	C
12	C	52	A
13	D	53	D
14	D	54	C
15	D	55	A
16	B	56	B
17	A	57	C
18	D	58	C
19	D	59	A
20	A	60	D
21	A	61	D
22	C	62	B
23	A	63	C
24	D	64	
25	B	65	
26	C	66	
27	C	67	
28	D	68	
29	D	69	
30	A	70	
31	D	71	
32	B	72	
33	C	73	
34	D	74	
35	A	75	
36	C	76	
37	D	77	
38	C	78	
39	A	79	
40	B	80	